université PARIS-SACLAY

Nuclear Data Uncertainty Quantification for PWR Vessel Aging with a ^{56}Fe Elastic-Scattering Evaluation

Quantification des incertitudes dues aux données nucléaires sur le vieillissement des cuves REP avec une évaluation de la diffusion élastique du ^{56}Fe

Thèse de doctorat de l'université Paris-Saclay

École doctorale n° 576 PHENIICS: Particules, hadrons, énergie et noyau: instrumentation, image, cosmos et simulation
Spécialité de doctorat: Physique nucléaire
Graduate School : Physique. Référent : Faculté des Sciences d'Orsay

Thèse préparée dans l'unité de recherche **ASNR/PSN-RES/SNC/LN**, sous la direction d'**Éric DUMONTEIL**, chercheur HDR (CEA), le co-encadrement de **Mariya BROVCHENKO**, chercheuse (ASNR), et le co-encadrement de **Dimitri ROCHMAN**, chercheur (Nagra).

Thèse soutenue à Paris-Saclay, le 1er octobre 2026, par

Juan Antonio MONLEÓN DE LA LLUVIA MAZAGATOS

Composition du jury

Membres du jury avec voix délibérative

Nuria GARCÍA HERRANZ Professeure Universidad Politécnica de Madrid	Rapporteure & Examinatrice
David BERNARD Directeur de recherche, CEA Université Aix-Marseille	Rapporteur & Examinateur
Cyrille DE SAINT JEAN Directeur de recherche, CEA Université Paris-Saclay	Président & Examinateur
Ivan KODELI Chercheur senior UKAEA / Institut Jožef Stefan	Examinateur
Alexander VASILIEV Chercheur senior Paul Scherrer Institute (PSI)	Examinateur

THESE DE DOCTORAT

NNT : 2026UPASP088

université PARIS-SACLAY

ÉCOLE DOCTORALE
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Titre : Quantification des incertitudes dues aux données nucléaires sur le vieillissement des cuves REP avec une évaluation de la diffusion élastique du ^{56}Fe

Mots clés : données nucléaires, quantification d'incertitudes, analyse de sensibilité, transport neutronique Monte-Carlo, fluence cuve, distribution angulaire du ^{56}Fe

Résumé : La fluence neutronique rapide reçue par la cuve d'un réacteur à eau sous pression fixe la limite de sa durée d'exploitation. En s'accumulant dans l'acier de la cuve, les neutrons rapides le fragilisent, et c'est le flux au-dessus de 1 MeV intégré sur la vie du réacteur qui gouverne ce dommage. Cette fluence ne peut être mesurée dans l'acier ; elle est calculée par transport neutronique, ancrée sur quelques dosimètres de surveillance, et toute incertitude de calcul se reporte aux marges de sûreté. À mesure que les codes de transport et les modèles de réacteur ont mûri, les données nucléaires en sont devenues la source dominante, et les sections efficaces du fer en portent la plus grande part du terme de transport, car les neutrons n'atteignent la cuve qu'après avoir traversé l'acier du baffle, de l'enveloppe de cœur et de la protection thermique.

Cette thèse quantifie l'incertitude des données nucléaires sur la fluence neutronique rapide d'une cuve française de 900 MWe. Un cadre logiciel, `kika-nd`, propage les covariances évaluées par théorie des perturbations et par échantillonnage Monte-Carlo et décompose la variance de fluence par isotope et par réaction. Un canal est décrit de façon inégale par les bibliothèques évaluées, la distribution angulaire de la diffusion élastique du ^{56}Fe, qui fixe la direction dans laquelle chaque neutron diffusé poursuit sa route vers la cuve. Plusieurs n'en fournissent aucune covariance, et celles qui en donnent la traitent de façon très différente. Elle est ici réévaluée directement à partir des mesures différentielles de la base EXFOR sur 847 keV–4 MeV, ce qui produit une covariance `MF34` indépendante, dont les valeurs centrales restent proches des bibliothèques existantes mais dont l'incertitude est répartie de façon lisse et cohérente avec les données.

La méthodologie est testée sur les benchmarks ASPIS-88 et VENUS-3, puis appliquée à la cuve REP900. Les sections efficaces de transport, dominées par le fer, contribuent pour environ 5,9% à l'incertitude de fluence neutronique rapide à la cuve. La forme du spectre de fission émis, perturbée ici à partir de sa covariance évaluée à travers le même modèle de transport, y contribue pour environ 6,1%, et le budget de données nucléaires atteint environ 8,6% à la cuve. La distribution angulaire qui a motivé une grande partie de ce travail se révèle un contributeur mineur à la cuve pour chaque bibliothèque : elle augmente l'incertitude totale de moins de 0,1% en valeur absolue avec l'évaluation partagée et la nouvelle évaluation, et d'environ 0,5% au plus avec JENDL-5, ce qui répond à une question restée ouverte en dosimétrie de cuve.

université PARIS-SACLAY

ÉCOLE DOCTORALE
Particules, hadrons, énergie et noyau : instrumentation, imagerie, cosmos et simulation (PHENIICS)

Title: Nuclear Data Uncertainty Quantification for PWR Vessel Aging with a ^{56}Fe Elastic-Scattering Evaluation

Keywords: nuclear data, uncertainty quantification, sensitivity analysis, Monte Carlo neutron transport, vessel fluence, ^{56}Fe angular distributions

Abstract: The fast-neutron fluence received by a pressurized-water-reactor vessel sets the limit on how long the reactor can operate. As fast neutrons accumulate in the vessel steel they embrittle it, and the flux above 1 MeV integrated over the operating life drives that damage. This fluence cannot be measured inside the steel, so it is obtained from neutron transport calculations anchored to a few surveillance dosimeters, and every uncertainty in the calculation must be carried as a safety margin. As the transport codes and reactor models have matured, the nuclear data have become the dominant source of that uncertainty, and the iron cross sections carry the largest share of the transport term, since the neutrons reach the vessel only after crossing the steel of the baffle, core barrel, and neutron pad.

This thesis quantifies the nuclear-data uncertainty on the fast-neutron fluence of a French 900 MWe vessel. A software framework, `kika-nd`, propagates the evaluated covariances through perturbation theory and Monte Carlo sampling and decomposes the fluence variance by isotope and reaction. One channel is described unevenly by the evaluated libraries, the angular distribution of ^{56}Fe elastic scattering, which sets the direction in which each scattered neutron continues toward the vessel. Several give no covariance for it at all, and those that do treat it very differently. It is re-evaluated here directly from the differential-scattering measurements of the EXFOR database over 847 keV–4 MeV, producing an independent `MF34` covariance whose central values stay close to the existing libraries but whose uncertainty is spread smoothly and consistently with the data.

The methodology is tested on the ASPIS-88 iron benchmark and the reactor-like VENUS-3 benchmark, then applied to the PWR900 vessel. The transport cross sections, dominated by iron, contribute about 5.9% to the fast-neutron fluence uncertainty at the vessel. The shape of the emitted fission spectrum, perturbed here from its evaluated covariance through the same transport model, contributes about 6.1%, and the nuclear-data budget totals about 8.6% at the vessel. The angular distribution that motivated much of the work proves to be a minor contributor at the vessel for every library, raising the fluence uncertainty by less than 0.1% in absolute terms with the shared and the new evaluations and by at most about 0.5% with JENDL-5, which settles a question that had remained open in vessel dosimetry.

A mis abuelos, Juan y Consuelo.

Acknowledgements

Mariya Brovchenko supervised this work day to day, and most of what I learned in these years I learned from her: to look at the details, to ask what physics is behind a result, and to question everything, my own results included. I hope it has shaped the way I do research for good. I also thank her for her kindness, and for caring as much as she did.

I thank Dimitri Rochman for always being available and for his feedback. From our conversations, often far from the task at hand, I learned more than from a week of reading papers, probably more than he realises.

I thank my thesis director, Éric Dumonteil, for his feedback on the papers and on this manuscript, and for guiding me through the administrative side of the doctorate.

I am grateful to the members of the jury: to Nuria García Herranz and David Bernard, who reviewed this manuscript, and to Cyrille De Saint Jean, Ivan Kodeli and Alexander Vasiliev, for the time they devoted to this work.

Outside the laboratory, I thank the JEFF and INDEN community and colleagues at EPFL, PSI, Uppsala University, the JRC, SCK CEN, the CEA and the UPM, in particular Mathieu Hursin (EPFL), Antonio and Anastasia (NEA), Roberto Capote (INDEN) and Georgios (JRC). I also thank the fellow PhD students whose paths crossed mine so often over these years: Daniele, the two Federicos, Dan, Íñigo and Alejandro.

I also thank Óscar, Nuria and Gonzalo, and the other professors at the UPM. It was in their classes that I got hooked on nuclear science, and that is why I went on to the master's and ended up here. Je remercie l'ensemble du Laboratoire de Neutronique (LN) et du Service de Neutronique et de Criticité (SNC) de l'ASNR pour leur accueil, et en particulier Vaibhav et Raphaëlle. Merci à Pierre et à Aitor, mes collègues de bureau, pour toutes ces discussions, sur le travail et sur le reste, qui m'ont aidé à rester sain d'esprit pendant toutes ces heures passées au labo. Merci à Flavien Bazon pour son travail de stage sur la propagation des incertitudes des données nucléaires dans les calculs du terme source du PWR900. Merci enfin à Marine et à toute l'équipe de l'association des doctorants de l'ASNR, avec qui j'ai partagé la charge de la thèse et organisé plein de choses chouettes.

A mis amigas de toda la vida, Eli, Ana, Susana y Elena, por seguir ahí todos estos años.

A Aurora, que ha estado a mi lado durante casi todo el doctorado y a quien más le ha tocado aguantar el estrés, las dudas y también las alegrías. Gracias por escucharme siempre y por apoyarme en todo.

Y a mi familia, en especial a mi madre, por su apoyo de siempre, antes y durante el doctorado; por hacer posible que viviera aquí sin que me faltara de nada; y por estar ahí en todo lo que he necesitado.

List of contents

List of acronyms **xvi**

Introduction **1**

I General Context **5**

1 Neutron Transport . . . 6
1.1 The Boltzmann Transport Equation . . . 7
1.2 Computational Neutron Transport . . . 9
1.2.1 Deterministic Methods . . . 10
1.2.2 Stochastic Methods . . . 12
1.2.3 Hybrid Methods . . . 13
2 Variance Reduction . . . 14
2.1 Statistical Reliability and Convergence Diagnostics . . . 15
2.1.1 The Relative Error (R) . . . 15
2.1.2 Figure of Merit (FOM) . . . 15
2.1.3 Variance of the Variance (VOV) . . . 16
2.1.4 Empirical Slope of the Score Distribution . . . 16
2.2 Particle Weights and Importance Sampling . . . 17
2.2.1 Fundamental Principles: Weight and Importance . . . 17
2.2.2 Common Weight Control Mechanisms . . . 18
2.2.3 Weight Windows . . . 19
2.3 Adjoint Transport Theory . . . 21
2.4 Adjoint-Driven Importance Sampling . . . 22
2.4.1 Consistent Adjoint Driven Importance Sampling (CADIS) . . . 23
2.4.2 Forward-Weighted CADIS (FW-CADIS) . . . 24
3 Codes . . . 24
3.1 MCNP6.3 . . . 24
3.2 ADVANTG 3.2.0 . . . 25
3.3 Serpent 2.2 . . . 26
3.4 CASMO5/CMSLINK5/SIMULATE5 . . . 26
4 Nuclear Data . . . 27
4.1 Nuclear Data Cycle . . . 28
4.1.1 EXFOR . . . 28
4.1.2 Evaluated Nuclear Data Libraries . . . 30
4.1.3 Benchmarking and Validation . . . 31
4.2 ENDF-6 Files . . . 31
4.2.1 Hierarchical Organization . . . 31
4.2.2 File Formatting . . . 34
4.2.3 Elastic-Scattering Angular Distributions . . . 35
4.3 ACE Files . . . 37

4.4 Nuclear Data Processing . . . 38
4.5 Covariance Data . . . 40
4.5.1 Data Representations . . . 42
4.5.2 The Correlation Matrix . . . 42
4.5.3 Angular-Distribution Covariances (MF34) . . . 44
4.5.4 Covariance Data Formats . . . 44
4.5.5 Uncertainty Propagation . . . 45
5 VACS: Vessel Aging Calculation Scheme . . . 45
5.1 Displacement Damage Cross Sections . . . 46
5.2 Step 1: Fission Neutron Source Term . . . 46
5.3 Step 2: Neutron Transport to the Vessel . . . 47
6 Summary . . . 48

II Sensitivity Analysis and Uncertainty Quantification 49
1 Introduction and Scope . . . 50
1.1 Definitions . . . 51
1.2 Classification of Uncertainties . . . 51
1.3 Responses of Interest . . . 52
1.4 Input Parameters . . . 53
2 Sensitivity Analysis Framework . . . 53
2.1 Local Sensitivity Analysis . . . 53
2.1.1 Direct Perturbation . . . 54
2.1.2 Perturbation Theory . . . 55
2.1.3 Generalized Perturbation Theory (GPT) . . . 58
2.1.4 Differential Operator Sampling (DOS) . . . 65
2.1.5 Correlated Sampling (CS) . . . 70
2.1.6 Scope of the local sensitivity framework . . . 72
3 Uncertainty Quantification Framework . . . 72
3.1 First-Order Second-Moment Propagation (FOSM) . . . 72
3.1.1 Response Linearization and the Sandwich Rule . . . 73
3.1.2 Relative Form and Connection to Sensitivity . . . 73
3.1.3 Applicability and Limitations . . . 74
3.2 Monte Carlo Sampling (MCS) for Uncertainty Quantification . . . 74
3.2.1 Sampling Techniques . . . 74
3.2.2 Matrix Decomposition and Correlated Sampling . . . 75
3.2.3 Decoupling Parametric and Statistical Uncertainties . . . 77
4 The KIKA Computational Framework . . . 81
4.1 Motivation . . . 81
4.2 Architecture Overview . . . 82
4.3 Supported Formats and External Codes . . . 82
4.4 Multigroup Collapse of Angular Distribution Covariances . . . 83
4.5 Nuclear Data Perturbation Pipelines . . . 84
4.5.1 Cross Section Perturbation . . . 84
4.5.2 Angular Distribution Perturbation . . . 84
4.5.3 Combined Perturbation . . . 86
5 Methodology Verification: The PWR Sphere Toy Model . . . 87
5.1 Model Description . . . 87
5.2 Sensitivity Analysis and FOSM Propagation . . . 90

5.2.1 Sensitivity Profiles . . . 90
5.2.2 First-Order Applicability . . . 91
5.2.3 Energy-Integrated Sensitivity . . . 92
5.2.4 Uncertainty Propagation . . . 94
5.3 Monte Carlo Sampling Propagation . . . 94
5.3.1 Sampling and Decomposition . . . 94
5.3.2 Simulation Sets and Response Distribution . . . 95
5.3.3 Convergence and Efficiency . . . 97
5.4 Comparison of Methods and Discussion . . . 100
5.5 Angular Distribution Sensitivity and Uncertainty . . . 101
5.5.1 Sensitivity . . . 101
5.5.2 Uncertainty Quantification . . . 102
6 Conclusions . . . 103

III Data-Driven Evaluation of ^{56}Fe Elastic-Scattering Angular Distributions and Covariances 105

1 Introduction . . . 106
1.1 Role of Angular Distributions in RPV Fluence . . . 106
1.2 Status of ^{56}Fe Angular Distributions in Major Libraries . . . 107
1.3 Scope of the Evaluation . . . 110
2 Mathematical Formalism for Uncertainty Quantification . . . 111
2.1 Propagation of MF34 Uncertainties to the Differential Cross Section . . . 111
2.1.1 General first-order propagation . . . 111
2.1.2 Simplification adopted in this work (shape-only bands) . . . 112
2.2 Physical Constraints and Cross-Section Coupling . . . 112
3 Experimental Database and Comparison Methodology . . . 113
3.1 Experimental Configurations and Measurement Extractions . . . 113
3.2 Energy Resolution Folding and Normalization Bias . . . 115
3.3 Treatment of Experimental Uncertainties . . . 117
3.4 Goodness-of-Fit Metric . . . 119
3.5 Preliminary Library Comparison . . . 121
4 Evaluation Pipeline: Fitting and Sampling Methodology . . . 122
4.1 Pipeline Overview . . . 122
4.2 Data Selection and Energy Grid . . . 125
4.3 Nominal Fit . . . 125
4.3.1 Block-correlated generalized least squares . . . 125
4.3.2 Per-point weights for experimental heterogeneity . . . 126
4.3.3 Automatic degree selection via AIC . . . 126
4.3.4 Angular-band discrepancy model . . . 127
4.3.5 Model averaging and degree sampling . . . 128
4.4 Per-Bin Variance Pass — Monte Carlo Pass 1 . . . 129
4.4.1 Perturbation model . . . 130
4.4.2 Refit with frozen c_0 and per-sample degree . . . 130
4.4.3 Positivity enforcement . . . 131
4.5 Kernel-Weight Correlation Pass — Monte Carlo Pass 2 . . . 131
4.5.1 Overlap weights from TOF resolution . . . 132
4.5.2 Perturbation model . . . 132
4.6 Covariance Assembly . . . 133

4.6.1 Sample covariance and block structure . . . 134
4.6.2 Near-zero regularization . . . 134
4.6.3 Adaptive multigroup collapse . . . 135
4.7 Pipeline Output: Use and Limitations . . . 135
5 Results and Comparison . . . 136
5.1 Pipeline Output: Use and Limitations . . . 137
5.2 Legendre Coefficient Comparison Across Evaluations . . . 137
5.3 Uncertainty Band Comparison at Selected Energies . . . 143
5.4 Goodness-of-Fit Analysis . . . 145
5.5 Integral Cross-Check: Criticality Benchmarks . . . 149
6 Conclusions . . . 152

IV Integral Validation and Application to PWR Vessel Fluence 155
1 Introduction . . . 156
2 The ASPIS-88 Iron Benchmark . . . 157
2.1 Benchmark Description . . . 157
2.2 MCNP Model and Variance Reduction . . . 157
2.3 C/E Comparison with Nuclear-Data Uncertainties . . . 159
2.3.1 C/E Comparison . . . 159
2.3.2 Propagated Nuclear-Data Uncertainties . . . 161
2.3.3 Integral Effect of the Chapter-3 Angular Evaluation . . . 162
2.4 Sensitivity Analysis and the Origin of Library Differences . . . 163
2.4.1 Sensitivity Profiles . . . 163
2.4.2 Isolating ^{56}Fe by a Single-Isotope Library Swap . . . 163
3 The VENUS-3 Benchmark . . . 166
3.1 Benchmark Description . . . 166
3.2 MCNP Model and Uncertainty Propagation . . . 166
3.3 C/E Comparison . . . 167
3.4 Propagated Uncertainties and the Processing of the ^{16}O Covariances . . . 168
4 Reactor Model, Tallies, and Variance Reduction . . . 171
5 Nuclear Data Uncertainty in the Source Term . . . 175
5.1 Calculation Setup . . . 176
5.2 Outputs of the Core Calculation . . . 177
5.3 Perturbation of the Emitted Fission Spectrum . . . 178
5.4 Propagation to the Vessel . . . 178
5.5 Summary . . . 180
6 Nuclear Data Uncertainty in the Vessel Fluence and Capsule Responses . . . 182
6.1 Calculation Setup . . . 182
6.2 Fast Neutron Flux at the Vessel . . . 182
6.3 Vessel Atomic-Displacement Damage . . . 184
6.4 Capsule Dosimetry . . . 184
6.5 Cross Section and Angular-Distribution Contributions . . . 185
6.6 Comparison of ^{56}Fe Angular Evaluations . . . 186
6.7 Per-Isotope Decomposition of the Cross Section Channel . . . 187
6.8 Cross-Check: MCS Against FOSM for Cross-Section Propagation . . . 189
7 Comparison with Published Studies . . . 191
8 Conclusions . . . 194

Conclusion 197

Appendices **201**

A Global Sensitivity Analysis **203**
A.1 Variance-Based Sensitivity Analysis . . . 203
A.2 Estimation Methods . . . 204
A.3 Practical Limitations for Nuclear Data Applications . . . 205
A.4 Relevance to This Work . . . 206

B The KIKA Desktop Application **207**
B.1 Overview . . . 207
B.2 Data Management and Visualization . . . 208
B.3 Analysis Tools . . . 212
B.4 Transport Code Integration . . . 214
B.5 Distribution and Availability . . . 215

C KIKA-ND Implementation Details **217**
C.1 Design Principles . . . 217
C.2 Supported Formats and Capabilities . . . 218
C.2.1 ENDF-6 . . . 218
C.2.2 ACE . . . 218
C.2.3 Covariance Data . . . 219
C.2.4 NJOY Processing . . . 219
C.2.5 Transport Code Integration . . . 220
C.2.6 EXFOR . . . 220
C.2.7 Materials . . . 220
C.3 Sampling and Perturbation . . . 220
C.4 Covariance Repair . . . 221

D NJOY Perturbation Processing Template **223**

E Supplementary Material for Chapter III **225**
E.1 Energy-Resolution Folding: Magnitude and Shape Contributions . . . 225
E.2 Per-Experiment Uncertainty Cases . . . 228
E.2.1 Generic Handling Cases . . . 228
E.2.2 Per-Experiment Special Cases . . . 229
E.2.3 The 5% Default Systematic . . . 230
E.2.4 Coverage Summary . . . 230
E.2.5 Largest Contributors to the Fit . . . 230
E.3 Statistical-Uncertainty Floor Sensitivity . . . 231
E.4 Information-Criterion Comparison: AIC, AICc, BIC . . . 231
E.4.1 Criteria . . . 231
E.4.2 Selected-Order Distribution . . . 232
E.4.3 Per-Bin Reduced Chi-Squared . . . 232
E.4.4 Choice Rationale . . . 232
E.5 Positivity Enforcement . . . 233
E.6 Single-Source Correlation Matrix and PSD . . . 234
E.7 Near-Zero Regularization . . . 234
E.8 MF3–MF4 Cross-Correlations Under EXFOR-Only Methodology . . . 235
E.9 Strong Negative Adjacent-Bin Correlations . . . 236

E.10 Validity of the Gaussian Covariance Representation 236
E.11 Pipeline Diagnostics 237
E.12 Criticality Benchmark Cross-Check: Full-Suite Figure 239

F Supplementary Material for Chapter IV **241**
F.1 C/E with Nuclear-Data Uncertainties for the Remaining Foils 241
F.2 Single-Isotope Library Swap 243
F.3 Chapter-3 Angular Evaluation Swap 245
F.4 Per-Foil Uncertainty Tables 245
F.5 Perturbed Isotopes in the VENUS-3 Propagation 252
F.6 Reference Tally Values Across Libraries 254
F.7 Per-Assembly Distributions of νR_f 254
F.8 Spatial Contribution to Fission Neutron Production 256
F.9 Fast Neutron Flux Above 0.1 MeV 256
F.10 Azimuth-by-Height Uncertainty Maps 257
F.11 Highest-Uncertainty Tallies and Largest Library Spreads 259
F.12 Nuclear-Data Budget at Every Location 260

G Synthèse en français **263**
G.1 Contexte et problématique 263
G.2 Méthodologie : analyse de sensibilité et quantification d'incertitudes 264
G.3 Évaluation des distributions angulaires du ^{56}Fe 266
G.4 Confrontation aux benchmarks intégraux : ASPIS-88 et VENUS-3 269
G.5 Application : budget d'incertitude de la cuve REP900 271
G.6 Conclusion et perspectives 273

Bibliography **275**

List of figures **291**

List of tables **299**

List of acronyms

ACE A Compact ENDF

AIC Akaike Information Criterion

ASNR Autorité de Sûreté Nucléaire et de Radioprotection

C/E Calculated-to-Experimental ratio

CADIS Consistent Adjoint Driven Importance Sampling

CEA Commissariat à l'Énergie Atomique et aux énergies alternatives

DCS Differential Cross Section

DOS Differential Operator Sampling

dpa Displacements per Atom

EALF Energy of Average Lethargy causing Fission

EDF Électricité De France

ENDF Evaluated Nuclear Data File

FOSM First-Order Second-Moment

FW-CADIS Forward-Weighted CADIS

GNDS Generalised Nuclear Data Structure

GPT Generalized Perturbation Theory

GRS Gesellschaft für Anlagen- und Reaktorsicherheit (uncertainty-propagation method)

IRSN Institut de Radioprotection et de Sûreté Nucléaire

JEFF Joint Evaluated Fission and Fusion library

JENDL Japanese Evaluated Nuclear Data Library

MCS Monte Carlo Sampling

MF Magnetic File, the ENDF file number (data type)

MT Magnetic Tape, the ENDF reaction identifier (section number)

NEA Nuclear Energy Agency

NNDC National Nuclear Data Center

OECD Organisation for Economic Co-operation and Development

PFNS Prompt Fission Neutron Spectrum

PWR Pressurized Water Reactor

REP Réacteur à Eau sous Pression (French for PWR)

RPV Reactor Pressure Vessel

SA Sensitivity Analysis

SINBAD Shielding Integral Benchmark Archive and Database

SVD Singular Value Decomposition

TENDL TALYS-based Evaluated Nuclear Data Library

TMC Total Monte Carlo

UQ Uncertainty Quantification

VACS Vessel Aging Calculation Scheme

WPEC Working Party on international nuclear data Evaluation Cooperation

Introduction

Nuclear power supplies close to a tenth of the world's electricity and is one of its largest sources of low-carbon generation. Most of that electricity comes from reactors built decades ago. Of the roughly four hundred power reactors in operation today, the average age is about thirty-two years, and more than a third have already run beyond the forty-year period for which they were originally designed.[1] Few new plants come online each year, so the near-term contribution of nuclear energy to electricity supply and to decarbonization depends less on new construction than on whether these existing reactors can be operated safely for longer than first planned. Long-term operation has become a central question for operators and safety regulators worldwide.

Nowhere is this clearer than in France, which draws about 70% of its electricity from nuclear power, the highest share of any country. Its fleet of fifty-eight reactors, all operated by EDF, was built largely between the late 1970s and the early 1990s and licensed initially for forty years (Figure 1). Rather than retire this fleet, France has chosen to extend and renew it. The safety authority has cleared the thirty-two 900 MWe reactors to operate up to fifty years, subject to extensive upgrades, and is already setting the conditions for operation beyond fifty,[2] while EDF carries out a refurbishment program of close to fifty billion euros and the government has launched the construction of a new generation of reactors. The reactor studied in this thesis is one of these French 900 MWe units.

Other countries have followed a different path. Spain, which obtains about a fifth of its electricity from seven reactors, agreed in 2019 with its utilities and its waste-management agency to close them one by one between 2027 and 2035, beginning with the two units at Almaraz. That schedule has since been partially revised: in August 2026, the operating licences for both Almaraz units were extended until 8 June 2030, while the planned closure of the Spanish nuclear fleet by 2035 remains unchanged.[3] Whether a country extends the life of its reactors or retires them on schedule, the same technical question sits underneath the decision: for how long can the steel of the reactor pressure vessel withstand the neutron irradiation it receives?

The pressure vessel is the one large component of a reactor that cannot be replaced. It holds the core and the primary coolant, and it is the main structural barrier against the release of radioactive material. Throughout the life of the plant, the steel of the vessel is exposed to the fast neutrons, with energies above 1 MeV, that escape the core. These neutrons displace atoms in the steel lattice, and the accumulated damage gradually embrittles the material, raising the temperature at which it shifts from ductile to brittle behavior (Figure 2a). Because this

[1] International Atomic Energy Agency, *Nuclear Power Reactors in the World*, Reference Data Series No. 2, 2025 edition. https://www-pub.iaea.org/MTCD/publications/PDF/RDS-2-45_web.pdf

[2] World Nuclear Association, "Nuclear Power in France". https://world-nuclear.org/information-library/country-profiles/countries-a-f/france. The favorable opinion on fifty-year operation of the 900 MWe reactors was issued by the French regulator (ASN, now ASNR) in 2021.

[3] Spanish Ministry for the Ecological Transition and the Demographic Challenge, Order TED/864/2026 of 12 August 2026, extending the operating licence of Almaraz Units I and II until 8 June 2030.

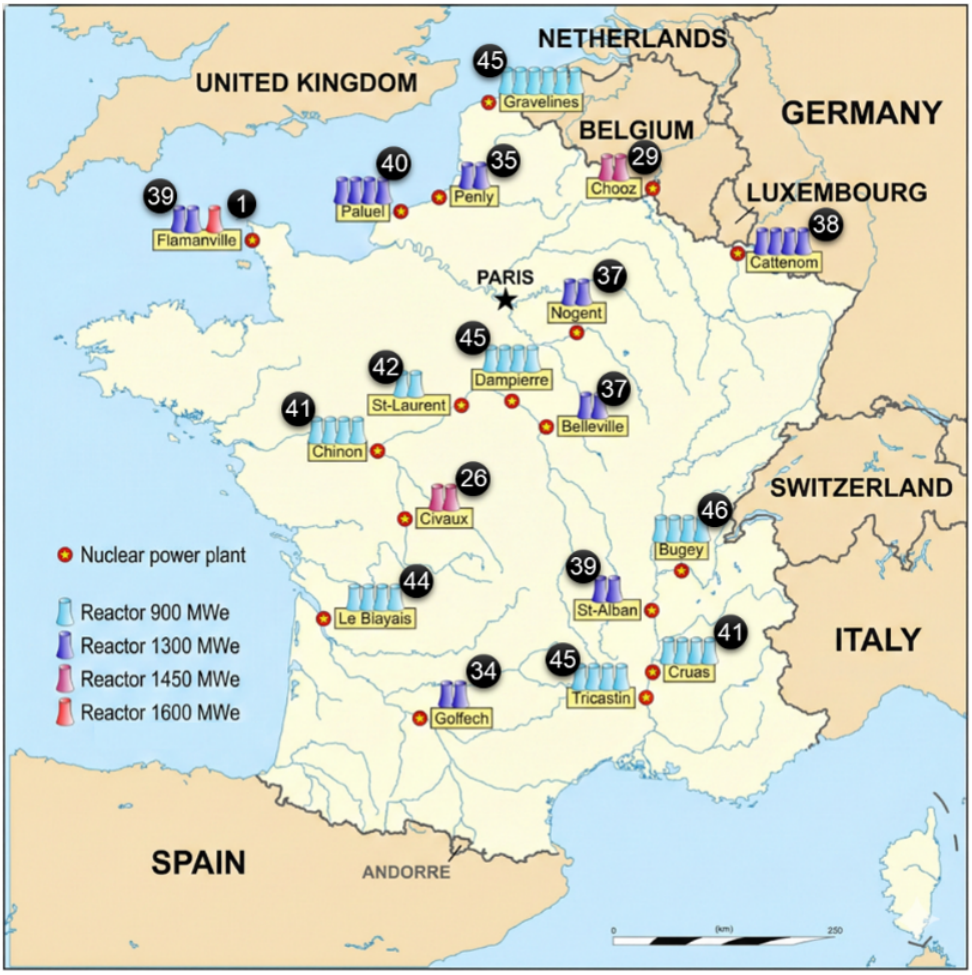
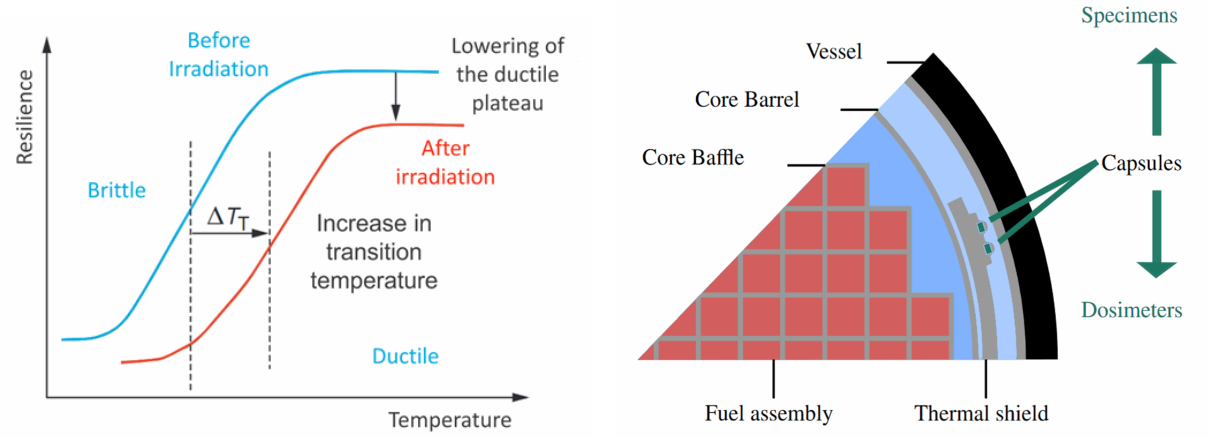


Figure 1: Location of the French nuclear power plants operated by EDF, colored by the age of each site.

embrittlement is irreversible and the vessel cannot be changed, it sets the ultimate limit on how long a reactor may safely operate.



(a) Irradiation shifts the ductile-to-brittle transition of the vessel steel toward higher temperature and lowers the upper-shelf toughness.

(b) Surveillance capsules, holding steel specimens and activation dosimeters, sit between the core barrel and the vessel, shown here in an eighth of the core.

Figure 2: Embrittlement of the reactor pressure vessel and its surveillance.

The quantity that drives the embrittlement correlations is the fast neutron fluence at the vessel wall, the flux above 1 MeV integrated over the operating life. This fluence cannot be measured directly inside the vessel steel. It is obtained instead from neutron transport calculations, anchored by the activation dosimeters of a few surveillance capsules placed inside the reactor (Figure 2b), which sample only a handful of positions and only part of the vessel life. The calculation therefore carries the burden of extending a sparse set of measurements across the whole vessel in space, energy, and time. Any uncertainty in the calculated fluence becomes a safety margin: the less confident the prediction, the larger the margin that must be carried, with a direct cost in operating flexibility [1, 2]. Quantifying that uncertainty, rather than hiding it inside a conservative margin, is the purpose of this work.

The uncertainty of a vessel fluence calculation has three broad sources: the numerical method, the geometric and technological description of the reactor, and the nuclear data that describe how neutrons interact with matter. As transport codes and reactor models have matured, the nuclear data have become the leading contributor, and within them the covariances that quantify how well each cross section is known [1]. The fast neutrons reach the vessel only after crossing the steel of the baffle, core barrel, and neutron pad, where the iron largely governs how they are scattered and slowed; the iron cross sections are therefore the largest single contributor to the nuclear-data budget. A central part of this thesis characterizes that share directly. The cross section covariances are propagated through a high-fidelity Monte Carlo model of the reactor, and the resulting fluence variance is decomposed by isotope.

The calculation depends on more than the size of the cross sections. The angle through which a neutron elastically scatters off iron sets the direction in which it continues toward the vessel. This elastic channel, the (n, el) reaction, is the one of interest here, and the uncertainty on its angular distribution is held in a separate part of the evaluated data files. That information is sparse and inconsistent: several modern libraries provide no angular covariance for ^{56}Fe elastic scattering, and those that do disagree on its size and structure. Rather than propagate these covariances as

published, this thesis looks inside them. The differential scattering measurements collected in the EXFOR database are re-analyzed, the quality of the released covariances is judged against them, and an independent evaluation of the ^{56}Fe elastic angular distribution and its covariance is produced, of a kind that the major evaluated libraries do not provide. Propagating this evaluation answers a question that has remained open in vessel dosimetry: how much the angular distribution of scattered neutrons adds to the fast neutron fluence uncertainty [3]. Chapter IV finds this contribution small at the vessel wall; establishing that bound is itself a result. The fission neutron spectrum also shapes the source. Its energy-shape uncertainty is propagated here from the evaluated covariance, through the same Monte Carlo model as the transport data.

These threads are brought together in a nuclear data uncertainty budget for the fast neutron fluence and the surveillance dosimetry of a French 900 MWe vessel, propagated through the two-step vessel aging calculation scheme used in France [4, 2]. The document is organized as follows. Chapter I sets out the general context: the neutron transport equation and the methods used to solve it, the variance reduction techniques and computational tools, the nuclear data and their covariances, and the vessel aging calculation scheme that frames the application. Chapter II develops the sensitivity-analysis and uncertainty-quantification framework and verifies it on a simplified PWR model. Chapter III presents the data-driven evaluation of ^{56}Fe elastic-scattering angular distributions and their covariances, built from the EXFOR database and compared against the JEFF-4.0 and JENDL-5 libraries. Chapter IV brings these threads together. It first confronts the methodology with experiment, mainly on the ASPIS-88 shielding benchmark and with additional results on the VENUS-3 benchmark, and then applies the validated method to the PWR900 vessel, propagating the nuclear data uncertainties through the vessel aging calculation scheme to its fast neutron fluence and surveillance dosimetry. The reactor and benchmark models come from previous ASNR studies, while the processing, perturbation, and analysis chain applied to them was written for this thesis.

Chapter I

General Context

Contents

1 Neutron Transport 6
1.1 The Boltzmann Transport Equation 7
1.2 Computational Neutron Transport 9
1.2.1 Deterministic Methods 10
1.2.2 Stochastic Methods 12
1.2.3 Hybrid Methods 13
2 Variance Reduction 14
2.1 Statistical Reliability and Convergence Diagnostics 15
2.1.1 The Relative Error (R) 15
2.1.2 Figure of Merit (FOM) 15
2.1.3 Variance of the Variance (VOV) 16
2.1.4 Empirical Slope of the Score Distribution 16
2.2 Particle Weights and Importance Sampling 17
2.2.1 Fundamental Principles: Weight and Importance . . . 17
2.2.2 Common Weight Control Mechanisms 18
2.2.3 Weight Windows 19
2.3 Adjoint Transport Theory 21
2.4 Adjoint-Driven Importance Sampling 22
2.4.1 Consistent Adjoint Driven Importance Sampling (CADIS) 23
2.4.2 Forward-Weighted CADIS (FW-CADIS) 24
3 Codes 24
3.1 MCNP6.3 24
3.2 ADVANTG 3.2.0 25
3.3 Serpent 2.2 26
3.4 CASMO5/CMSLINK5/SIMULATE5 26
4 Nuclear Data 27
4.1 Nuclear Data Cycle 28

4.1.1 EXFOR . . . 28
4.1.2 Evaluated Nuclear Data Libraries . . . 30
4.1.3 Benchmarking and Validation . . . 31
4.2 ENDF-6 Files . . . 31
4.2.1 Hierarchical Organization . . . 31
4.2.2 File Formatting . . . 34
4.2.3 Elastic-Scattering Angular Distributions . . . 35
4.3 ACE Files . . . 37
4.4 Nuclear Data Processing . . . 38
4.5 Covariance Data . . . 40
4.5.1 Data Representations . . . 42
4.5.2 The Correlation Matrix . . . 42
4.5.3 Angular-Distribution Covariances (MF34) . . . 44
4.5.4 Covariance Data Formats . . . 44
4.5.5 Uncertainty Propagation . . . 45
5 VACS: Vessel Aging Calculation Scheme . . . 45
5.1 Displacement Damage Cross Sections . . . 46
5.2 Step 1: Fission Neutron Source Term . . . 46
5.3 Step 2: Neutron Transport to the Vessel . . . 47
6 Summary . . . 48

The fast neutron fluence at the vessel is obtained from a chain of physics, numerical methods, and nuclear data, each link of which carries part of the uncertainty this thesis sets out to quantify. This chapter assembles that chain and collects the background on which the rest of the document builds. Section 1 presents the Boltzmann transport equation and the deterministic and Monte Carlo paradigms used to solve it. Section 2 develops the variance reduction techniques that make the deep-penetration vessel problem tractable, since the fast neutrons reaching the vessel form a rare tail of the source spectrum. Section 3 describes the transport and depletion codes used throughout the work. Section 4 turns to the nuclear data, from the evaluated files and their processing to the covariance information whose propagation is the central concern of this thesis. Finally, Section 5 sets out the Vessel Aging Calculation Scheme (VACS) that frames the application in Chapter IV.

1. Neutron Transport

The analysis of nuclear systems relies fundamentally on our ability to predict the distribution and behavior of neutrons as they interact with matter. In a nuclear reactor or a radiation shield, each neutron undergoes a sequence of physical events, including emission, free flight, scattering, and absorption, within a high-dimensional phase space of position, direction, and energy. This section establishes that framework, beginning with the governing equation of particle balance and concluding with the computational paradigms used to solve it in practical applications.

1.1. The Boltzmann Transport Equation

At the heart of neutron transport theory lies the Boltzmann Transport Equation (BTE), often referred to as the linear Boltzmann equation in the context of reactor physics. It provides a mathematical description of the expected or mean neutron population within a medium by treating the particles as a continuous distribution in a high-dimensional phase space [5].

To describe *"where the neutrons are and what they are doing"*, we must track it across seven independent variables: three spatial components ($\mathbf{r}$), two angles defining the direction of flight ($\mathbf{\Omega}$), kinetic energy (E), and time (t) [5]. For this discussion we assume steady state, so the explicit time dependence is dropped and the problem becomes six-dimensional. There is also an integral formulation of the transport equation, which describes particle propagation through the medium from a trajectory-based perspective. This formulation can be useful both conceptually and numerically, but it is not developed here; the interested reader is referred to [6, 7].

The fundamental logic of the BTE is a simple conservation balance: in any small element of phase space, the rate at which neutrons leave that element must be balanced by the rate at which neutrons enter it (plus any independent sources). We can categorize these physical processes into two groups:

- **Losses:** These occur via leakage (neutrons physically streaming out of a spatial volume) and collisions (neutrons being absorbed or scattered out of their current energy or direction).
- **Gains:** These occur through in-scattering (neutrons from other states redirecting into the current one), fission (the birth of new neutrons), and external sources.

In its most abstract operator notation, the equation is written as:

$$\mathcal{L}\psi = \mathcal{S}\psi + \mathcal{F}\psi + Q_{ext} \tag{I.1}$$

where $\mathcal{L}$ collects the loss mechanism, $\mathcal{S}$ is the (linear) scattering-in operator, $\mathcal{F}$ is the (linear) fission production operator, Q_{ext} is an external source (also denoted in this work as q) and the unknown variable is the angular flux, $\psi(\mathbf{r}, \mathbf{\Omega}, E)$. It is often convenient to rewrite this balance in terms of a single transport operator. Defining the Boltzmann (or transport) operator as $\mathcal{H} \equiv \mathcal{L} - \mathcal{S}$, the steady-state transport equation can be written as:

$$\mathcal{H}\psi = \mathcal{F}\psi + Q_{ext} \tag{I.2}$$

In many fixed-source shielding problems, fission production is absent or negligible and one may set $\mathcal{F}\psi \approx 0$, recovering the compact fixed-source form. Neutron-multiplying reactions such as (n,2n) are not part of the fission operator; they are included in the scattering operator $\mathcal{S}$ through the appropriate multiplicity in the scattering kernel [8]. The fixed-source form is then:

$$\mathcal{H}\psi = Q_{ext} \tag{I.3}$$

The operator $\mathcal{H}$ therefore represents the full physics of neutron transport associated with streaming, removal, and collision processes (losses) together with the subtraction of in-scattering gains. Fission production, when present, is represented separately through the operator $\mathcal{F}\psi$. This form is particularly useful for defining the adjoint transport equation and for expressing response functionals through bilinear identities.

Having introduced the transport operator in abstract form, we now look at each of its terms, beginning with the physical mechanisms that contribute to the loss operator $\mathcal{L}$. The most intuitive loss is leakage. Consider a small spatial volume dV. Neutrons moving in direction $\mathbf{\Omega}$ cross the boundaries of this volume; the net outflow is the divergence of the directional flow. Because the directional current is defined as $\mathbf{\Omega}\psi$, the leakage contribution is $\nabla \cdot (\mathbf{\Omega}\psi)$. Since the direction $\mathbf{\Omega}$ is an independent coordinate and does not change with position $\mathbf{r}$ during free flight, the term simplifies to:

$$\mathbf{\Omega} \cdot \nabla\psi(\mathbf{r}, \mathbf{\Omega}, E) \tag{I.4}$$

This gradient term represents the mathematical "streaming" of neutrons through space.

Next, consider collisional removal. Interactions are local-in-space: they do not depend on particles crossing a spatial boundary, so they appear as local rates proportional to ψ, not as derivatives. The macroscopic total cross section $\Sigma_t(\mathbf{r}, E)$ gives the total interaction rate per unit path length, so the collision loss term is simply:

$$\Sigma_t(\mathbf{r}, E)\psi(\mathbf{r}, \mathbf{\Omega}, E) \tag{I.5}$$

Combined, these two terms form the left-hand side of the BTE, representing all the ways a neutron can leave a specific point in phase space:

$$\mathcal{L}\psi = \mathbf{\Omega} \cdot \nabla\psi(\mathbf{r}, \mathbf{\Omega}, E) + \Sigma_t(\mathbf{r}, E)\psi(\mathbf{r}, \mathbf{\Omega}, E) \tag{I.6}$$

Turning now to the gains, neutrons can enter the state $(\mathbf{r}, \mathbf{\Omega}, E)$ when neutrons that previously had other directions $\mathbf{\Omega}'$ and energies E' scatter into the outgoing direction and energy. This redistribution is described by a scattering kernel (a differential macroscopic scattering cross section):

$$\Sigma_s(\mathbf{r}; \mathbf{\Omega}' \to \mathbf{\Omega}, E' \to E) \tag{I.7}$$

which encodes the probability rate of scattering from $(\mathbf{\Omega}', E')$ into $(\mathbf{\Omega}, E)$. To get the total in-scattering rate into the current state, we integrate over all possible incoming directions and energies:

$$\mathcal{S}\psi = \int_0^\infty \int_{4\pi} \Sigma_s(\mathbf{r}, \mathbf{\Omega}' \to \mathbf{\Omega}, E' \to E)\psi(\mathbf{r}, \mathbf{\Omega}', E')\, d\mathbf{\Omega}' dE' \tag{I.8}$$

If fission is present, it provides an additional source of neutrons that is proportional to the flux. In its isotropic form, the fission production term can be written as [5, 6]:

$$\mathcal{F}\psi(\mathbf{r}, \mathbf{\Omega}, E) = \frac{1}{4\pi} \int_0^\infty \chi(\mathbf{r}, E' \to E)\, \nu(\mathbf{r}, E')\, \Sigma_f(\mathbf{r}, E')\, \phi(\mathbf{r}, E')\, dE' \tag{I.9}$$

where χ is the fission spectrum, ν is the average number of neutrons emitted per fission, Σ_f is the macroscopic fission cross section, and $\phi(\mathbf{r}, E) = \int_{4\pi} \psi(\mathbf{r}, \mathbf{\Omega}, E)\, d\mathbf{\Omega}$ is the scalar flux.

Finally, we include $Q_{ext}(\mathbf{r}, \mathbf{\Omega}, E)$, which represents any independent injection of neutrons, such as a startup source or a fixed source in a non-multiplying medium.

When we assemble these pieces, we arrive at the complete description of particle balance:

Steady-State Boltzmann Transport Equation

$$\underbrace{\boldsymbol{\Omega}\cdot\nabla\psi(\mathbf{r},\boldsymbol{\Omega},E)}_{\text{Leakage}}+\underbrace{\Sigma_t(\mathbf{r},E)\,\psi}_{\text{Collision}}=\underbrace{\int_0^\infty\int_{4\pi}\Sigma_s(\mathbf{r},\boldsymbol{\Omega}'\to\boldsymbol{\Omega},E'\to E)\,\psi\,d\boldsymbol{\Omega}'\,dE'}_{\text{In-Scattering}}$$
$$+\underbrace{\frac{1}{4\pi}\int_0^\infty\chi(\mathbf{r},E'\to E)\,\nu(\mathbf{r},E')\,\Sigma_f(\mathbf{r},E')\,\phi(\mathbf{r},E')\,dE'}_{\text{Fission Source}}$$
$$+\underbrace{Q_{ext}(\mathbf{r},\boldsymbol{\Omega},E)}_{\text{External Source}} \tag{I.10}$$

For theoretical completeness, the time dependence of the angular flux is briefly introduced. In a transient system, the neutron population at a given point in phase space may change with time, which is described by the time derivative of the angular flux. Because the angular flux is defined as the product of neutron density and speed, this temporal variation naturally appears scaled by the inverse of the neutron speed. This ensures that the balance is written in terms of neutron density rather than flux [5].

To keep the notation compact, a point in the steady-state phase space is represented as $z = (\mathbf{r}, \boldsymbol{\Omega}, E)$. By expressing the angular flux as $\psi(z,t)$, the time-dependent BTE can be written as:

$$\frac{1}{v(E)}\frac{\partial\psi}{\partial t}(z,t)+\mathcal{L}\psi(z,t)=\mathcal{S}\psi(z,t)+\mathcal{F}\psi(z,t)+Q_{ext}(z,t) \tag{I.11}$$

When the time derivative vanishes, the system is in steady state and the equation reduces to the form presented earlier in Eq. (I.10). While time dependence is essential for describing transients and kinetic phenomena, it is not required for the developments considered in this work.

This equation forms the fundamental description of neutron transport in nuclear systems. It is an integro-differential equation with high dimensionality (six independent variables in steady state and seven when time dependence is included), and its solutions are generally not smooth [5]. As a result, exact analytical solutions are possible only for the simplest cases, and virtually all neutron transport problems of practical interest must be addressed using computational methods, such as deterministic techniques and Monte Carlo simulations [5].

1.2. Computational Neutron Transport

Practical solutions to the Boltzmann transport equation rely on two fundamentally different computational philosophies: Stochastic (Monte Carlo) and Deterministic methods.

- **Deterministic methods:** solves the Boltzmann equation as a mathematical system. By discretizing the independent variables—energy, angle, and space—the integro-differential equation is converted into a large system of algebraic equations. This approach provides a global solution (the flux everywhere in the system) without statistical noise, but it introduces systematic errors due to discretization.
- **Stochastic method:** treats particle transport as a probabilistic process. It does not solve the equation directly; instead, it simulates the random histories of individual neutrons,

tracking them from birth to death based on interaction probabilities. Physical quantities, such as flux or reaction rates, are estimated by averaging these histories. This approach introduces no discretization of the phase space and handles complex geometries exactly, but it is computationally expensive, yields results with statistical variance, and remains limited by the resolution of the tabulated nuclear data it samples.

1.2.1. Deterministic Methods

Deterministic transport begins by replacing the continuous phase space $(\mathbf{r}, \mathbf{\Omega}, E)$ with discrete representations. The result is a coupled set of equations for unknowns such as $\psi_{g,m}(\mathbf{r})$: the angular flux in energy group g and direction m, defined over a spatial mesh. The central challenge is not writing the discretized equations, but choosing approximations that keep the problem computationally manageable while preserving the physics relevant to the application. Beyond discretization itself, energy, angle, and space are treated at different levels of refinement and coupled through outer iterations rather than solved simultaneously, which keeps the problem tractable at the cost of additional approximation error.

All deterministic discretizations share the same consequence: once energy, angle, and space are discretized, neutron transport becomes a large coupled algebraic system for unknowns such as $\psi_{g,m}(\mathbf{r})$. For a 3D problem with G energy groups, M discrete directions, and a spatial mesh of size $I \times J \times K$, the number of unknowns scales as

$$N_{tot} = G \times M \times I \times J \times K. \tag{I.12}$$

Even moderate resolutions can therefore produce systems with hundreds of millions of unknowns, which rules out direct factorization and motivates iterative solution strategies and acceleration schemes tailored to transport structure [5, 7, 9, 10]. The detailed treatment of these algorithms is beyond the scope of this discussion; instead, the remainder of this section focuses on the discretization choices that define the structure and properties of the resulting algebraic problem.

Energy Discretization: The Multigroup Approximation

Energy is typically handled with a multigroup model because nuclear cross sections vary sharply with energy (especially in resonance regions), making a fine continuous-energy grid impractical for deterministic solvers. The energy axis is divided into G intervals (groups), and the transport equation is integrated over each group g.

This transforms the continuous dependence into a set of coupled groupwise equations. All detailed energy dependence is condensed into multigroup cross sections (σ_g), which are defined to preserve reaction rates within each group:

$$\sigma_g(\mathbf{r}) = \frac{\int_g \sigma(\mathbf{r}, E)\phi(\mathbf{r}, E)\, dE}{\int_g \phi(\mathbf{r}, E)\, dE} \tag{I.13}$$

The underlying pointwise cross sections $\sigma(\mathbf{r}, E)$ are fixed physical quantities taken from evaluated nuclear data libraries and are not modified by the calculation. However, the group-averaged values σ_g depend on the weighting flux $\phi(\mathbf{r}, E)$, which is not known a priori and itself depends on the very group constants being defined. This circularity means that deterministic calculations must rely on representative spectra and approximate treatments to generate usable group constants.

In most deterministic workflows, these group constants are produced by nuclear data processing codes (e.g., NJOY [11]), which convert evaluated nuclear data libraries into multigroup libraries usable by lattice physics and core simulators, including temperature treatment and the generation of group-averaged cross sections and scattering matrices.

A major difficulty arises in resonance energy regions. There, large absorption cross sections strongly depress the neutron flux locally in energy—a phenomenon known as self-shielding [7]. Because the flux is no longer smooth, simple spectrum weighting can significantly misrepresent effective cross sections. One way to address this is to refine the energy group structure, but doing so quickly becomes computationally impractical. Instead, deterministic methods use specialized self-shielding approaches to capture resonance effects without explicitly resolving fine energy structure [12, 11].

Angle Discretization: The Discrete Ordinates Method

Once the energy is grouped, the angular variable $\mathbf{\Omega}$ must be addressed. In reality, neutrons travel in every possible direction on the unit sphere. To make this computable, the Discrete Ordinates (S_N) method is used. Instead of solving for a continuous function, we restrict particles to travel only along a specific set of discrete directions (ordinates), $\mathbf{\Omega}_m$, each assigned a specific weight, w_m.

The goal of this discretization is not just to track particles, but to accurately calculate physical quantities like the scalar flux (ϕ) and current ($\mathbf{J}$). These quantities are essentially integrals of the angular flux over the sphere. In the S_N method, these integrals are replaced by a weighted sum, a process known as numerical quadrature:

$$\int_{4\pi} \psi(\mathbf{\Omega})\, d\mathbf{\Omega} \approx \sum_{m=1}^{M} w_m \psi(\mathbf{\Omega}_m) \tag{I.14}$$

The choice of these discrete directions is crucial. We cannot simply pick random points; we need a set of directions that treats all axes symmetrically (so the solution doesn't change if we rotate the reactor) and accurately integrates the angular moments of the flux. This is why we typically use Level-Symmetric quadrature sets (like Gauss-Legendre). These sets are mathematically designed to integrate spherical harmonics (polynomials representing angular shapes) exactly up to a certain order N [5, 7].

The N in S_N refers to the angular resolution. An S_4 set typically uses 24 directions, while an S_8 set uses 80 [5]. Increasing N improves accuracy but drastically increases the number of equations to solve. While S_N is the industry standard, it has a distinct weakness known as the ray effect, which is particularly troublesome in shielding calculations. Because particles are forced to travel along specific rays, they cannot reach regions of space that lie between these fixed angles [5, 7]. These characteristics can be particularly limiting in shielding problems involving complex geometries, streaming paths, or localized sources, where accurate treatment of highly anisotropic radiation fields is required. For simpler configurations, such as bulk attenuation through relatively homogeneous media, deterministic methods can remain well suited. In a reactor core where scattering is frequent, collisions randomize directions and fill in the gaps, masking the ray effect.

This discrete limitation contrasts with the Spherical Harmonics (P_N) method, which represents the angular flux using a continuous expansion in spherical harmonics (the 3D generalization of

Legendre polynomials). This continuous representation avoids ray effects, but it struggles to accurately model tightly collimated radiation beams, often producing unphysical oscillations [9].

Spatial Discretization

Spatial discretization turns the streaming operator ($\mathbf{\Omega} \cdot \nabla\psi$) from Eq. (I.10) into algebraic couplings between neighboring spatial unknowns by partitioning the geometry into cells/regions and choosing how ψ varies inside each region. In practice, this ranges from structured finite-difference / finite-volume schemes to finite-element and discontinuous finite-element formulations that handle curved boundaries and unstructured meshes more naturally [9].

For lattice and assembly calculations, a widely used alternative is track-based (characteristic) discretization, where the transport equation is integrated along straight rays crossing the geometry; the geometry is decomposed into material regions, and the within-region source is typically approximated as piecewise constant (flat source) or piecewise linear (linear source) to better capture strong gradients without exploding the mesh size [10]. Because very fine spatial resolution can make transport iterations expensive, production solvers often combine the fine-mesh transport solve with a low-order coarse-mesh diffusion-like correction (e.g., CMFD) to stabilize and accelerate convergence while preserving the detailed heterogeneous representation in the high-order solution [5, 10].

1.2.2. Stochastic Methods

In the physical process of particle transport, a vast number of neutrons undergo random, independent histories. Each step of a particle's life is governed by specific probability distribution functions derived from nuclear data. The number of physical neutron histories is enormous: at nominal power a 900 MWe reactor produces on the order of 10^{20} fission neutrons per second [2]. Because so many independent histories contribute, the collective behavior is effectively deterministic, characterized by a mean with very small statistical fluctuations [5]. Monte Carlo methods allow us to model this process not by solving the equation, but by simulating the random histories of N_{MC} fictitious particles and averaging their outcomes.

A major distinction between the simulation and reality is the number of histories. A Monte Carlo calculation follows on the order of $N_{MC} \approx 10^7$ fictitious particles, far fewer than the physical population. Consequently, Monte Carlo simulations exhibit significantly higher statistical noise.

For a fixed physical model and set of input data, Monte Carlo results contain only statistical errors (unlike deterministic methods, which contain discretization errors). According to the Central Limit Theorem (CLT), for a sufficiently large N_{MC}, the statistical error in the estimation of a mean quantity is proportional to:

$$\text{Statistical Error} \propto \frac{\sigma}{\sqrt{N_{MC}}} \tag{I.15}$$

where σ is the standard deviation specific to the problem. This statistical nature of Monte Carlo simulations leads to several important advantages and limitations:

- **Accuracy through sampling.** Because Monte Carlo simulations operate in continuous phase space, they can naturally model complex geometries and rapid energy-dependent variations in cross sections, such as resonance behavior, without the averaging required

by deterministic methods [5]. For a fixed model, arbitrarily small statistical uncertainty can, in principle, be achieved by increasing N_{MC}.

- **Slow convergence.** The convergence rate is inherently slow, scaling as $1/\sqrt{N_{MC}}$. Reducing the statistical error by one order of magnitude therefore requires an increase in computational effort by approximately two orders of magnitude.
- **Rare-event inefficiency.** In deep-penetration or rare-event problems (such as deep shielding calculations) only a very small fraction of simulated particles may contribute to the quantity of interest. In an analog simulation, this fraction may be as low as one particle in 10^6 [13]. Consequently, most computational effort is spent on particle histories that do not contribute to the tally, resulting in large statistical uncertainties despite significant computational cost.

To overcome these statistical limitations, particularly in deep-penetration problems, Variance Reduction techniques are used. These methods modify the sampling procedure (essentially "loading the dice") to artificially encourage particles toward regions of interest while adjusting particle weights to ensure the final result remains unbiased. A detailed discussion of specific variance reduction techniques is provided in Sec. 2. Furthermore, the challenge of selecting optimal parameters for these techniques leads us to the intersection of stochastic and deterministic approaches, known as Hybrid Methods.

1.2.3. Hybrid Methods

As we have seen, deterministic and stochastic transport methods exhibit complementary strengths and limitations. Deterministic solvers provide fast, global estimates of the particle distribution across the entire system but are affected by spatial, angular, and energy discretization errors. Monte Carlo methods, by contrast, offer high-fidelity, continuous-energy solutions with exact geometry treatment, yet suffer from statistical noise, especially in large or highly attenuating systems where particles rarely reach regions of interest.

Hybrid methods address this trade-off by coupling both approaches within a single computational framework. A fast, approximate deterministic calculation is first used to obtain a global estimate of the particle importance throughout the system. Although approximate, this solution contains crucial information about the relative importance of different regions, which is then used to guide the Monte Carlo simulation. By systematically biasing particle transport toward important pathways while preserving unbiased results, hybrid methods dramatically reduce variance and computational cost, effectively automating variance-reduction strategies that traditionally required significant expert intervention [5].

Several widely used codes implement this hybrid philosophy. MAVRIC [12], part of the SCALE code system, tightly integrates deterministic importance calculations with Monte Carlo transport to efficiently solve deep-penetration shielding problems. Similarly, the coupling of ADVANTG [14] with MCNP [13] provides a hybrid workflow in which ADVANTG generates variance-reduction parameters from a deterministic adjoint solution that are then applied in MCNP simulations. Further details on MCNP and ADVANTG are provided in Sec. 3.

2. Variance Reduction

This section presents the variance-reduction techniques needed to understand the work of this thesis, following the formulations of MCNP and ADVANTG, the codes used in subsequent chapters.

In Monte Carlo radiation transport, variance is the measure of statistical dispersion of a random variable around its expected value. Because Monte Carlo methods are stochastic, every result is an estimate with an associated statistical uncertainty. Variance reduction techniques (VRTs) are strategies used to increase the precision of these estimates without increasing the computational cost, or conversely, to reduce the computational time required to reach a target precision [13].

It is important to distinguish between accuracy and precision.

- **Precision:** Describes the spread of the estimator due to statistical noise and is typically summarized by a standard error or relative error.
- **Accuracy:** Represents how close the estimated mean is to the true physical value. Accuracy is governed by systematic factors such as [15]:
  - (i) *Nuclear Data:* Errors in the evaluated cross sections, angular distributions, etc.
  - (ii) *Modeling:* Simplified geometries or incorrect material compositions.
  - (iii) *Physics:* Approximations in the transport physics (e.g., group discretization).
  - (iv) *User Error:* Incorrect source definitions or the abuse of variance reduction (e.g., biasing a problem so heavily that regions of phase space are never sampled).

As illustrated in Figure I.1, a calculation can be highly precise yet inaccurate if systematic errors are present. Conversely, a statistically noisy result may remain unbiased. While VRTs are powerful tools for reducing stochastic uncertainty, they do not correct for underlying modeling or data inadequacies; in fact, aggressive biasing schemes must be monitored to ensure they do not introduce new systematic biases by failing to sample the problem's full phase space [13].

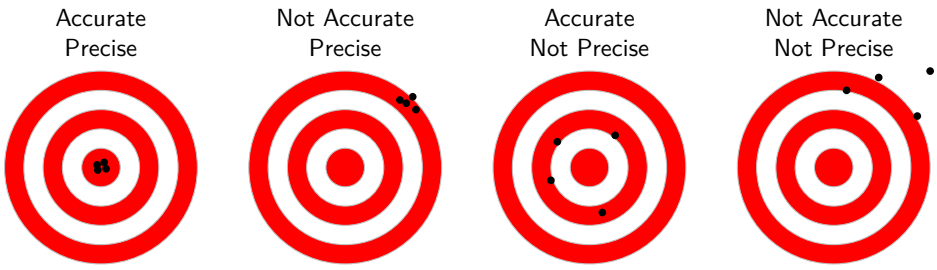


Figure I.1: Comparison between accuracy and precision.

Although MC methods are flexible, their primary limitation is computational cost: many simulated histories contribute negligibly to the tally of interest, while rare but important events may dominate the result. The objective of variance reduction is to concentrate sampling effort on the regions of phase space that contribute most to the tally—using techniques such as importance sampling, weight windows, or source biasing—to reduce statistical uncertainty at a fixed computational cost [16].

2.1. Statistical Reliability and Convergence Diagnostics

Before attempting to reduce variance, we must define what a "good" result looks like. In a Monte Carlo simulation, an estimator is said to be *converged* when three conditions hold simultaneously: the score distribution has finite variance, so that the Central Limit Theorem applies; the reported uncertainty is itself stable, in the sense that the estimate of the variance does not change appreciably with additional histories; and the rare but high-weight tail of the score distribution has been sampled enough that no further outliers are expected [13]. The diagnostics introduced below test these three conditions in turn.

2.1.1. The Relative Error (R)

The most fundamental indicator is the relative error, R, defined as the ratio of the standard deviation of the mean ($S_{\bar{x}}$) to the estimated mean ($\bar{x}$):

$$R = \frac{S_{\bar{x}}}{\bar{x}} \tag{I.16}$$

As established in Sec. 1.2.2, R follows a $1/\sqrt{N}$ proportionality in well-behaved simulations. Given that the computational time, T, for any MCNP run scales with N, we may represent the relative error in terms of computational time as follows:

$$R \sim \frac{C}{\sqrt{T}} \tag{I.17}$$

where C is a positive constant dependent on the tally choice. While R provides a quantitative measure of precision, specific thresholds are generally accepted in the field to judge the quality of a tally (see Table I.1). Notably, a low R does not guarantee a "correct" answer (accuracy), but rather signifies that the result is statistically reproducible.

Table I.1: Guidelines for interpreting the relative error (R) [13].

Range of R	**Quality of the tally**
0.5 to 1.0	Not meaningful
0.2 to 0.5	Factor of a few
0.1 to 0.2	Questionable
<0.10	Generally reliable

2.1.2. Figure of Merit (FOM)

Because R^2 is proportional to $1/T$, the product of the variance and the time is approximately constant for a given problem setup. This leads to the definition of the Figure of Merit (FOM):

$$FOM = \frac{1}{R^2 T} \tag{I.18}$$

The FOM is a powerful indicator for comparing different variance reduction strategies. A higher FOM indicates a more efficient calculation. In a well-converged run, the FOM should remain relatively constant as N increases. If the FOM fluctuates wildly, it suggests that the tally is being dominated by rare, high-weight events, indicating poor sampling [13, 15].

2.1.3. Variance of the Variance (VOV)

The Relative Error (R) is itself a random variable estimated during the simulation and, therefore, possesses its own statistical uncertainty. The Variance of the Variance (VOV) is a second-order diagnostic used to quantify the reliability of the estimated relative error. For a simulation with N histories, where x_i is the score of the ith history and w_i is its associated statistical weight (see Section 2.2), the VOV is defined as:

$$VOV = \frac{S^2(S_{\bar{x}}^2)}{[S_{\bar{x}}^2]^2} \tag{I.19}$$

Where $S^2(S_{\bar{x}}^2)$ is the estimated variance of the variance ($S_{\bar{x}}^2$). In practice, the VOV is calculated using the estimated central moments of the history scores. In terms of the history scores x_i, which already include the particle weights, it is given by:

$$VOV = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^4}{\left[\sum_{i=1}^{N}(x_i - \bar{x})^2\right]^2} - \frac{1}{N} \tag{I.20}$$

Because the VOV involves the fourth central moment, it is far more sensitive to spikes in history scores than the relative error is. A high VOV indicates that the sampling of the phase space is insufficient; specifically, it suggests that the tally is being dominated by rare, high-weight particles. For a result to be considered statistically converged, the following criteria are generally applied [13]:

- $VOV < 0.10$: This threshold ensures that the relative error estimate R is stable and unlikely to undergo large fluctuations with additional histories.
- Stability: The VOV should decrease monotonically as $1/N$. A sudden spike in VOV indicates that a high-weight particle has contributed a score significantly higher than the previous average, suggesting the tails of the distribution are not yet well-sampled.

2.1.4. Empirical Slope of the Score Distribution

To further assess the behavior of the large-score tail of the distribution, Monte Carlo codes monitor the Empirical Slope (β). This diagnostic ensures that the underlying Probability Density Function (PDF) of the scores, $f(x)$, decays rapidly enough for the CLT to remain valid [13]. For large scores, the PDF is assumed to follow a generalized Pareto distribution:

$$f(x) \approx \frac{1}{a}\left(1 + \frac{kx}{a}\right)^{-(1/k+1)} \quad \text{for } x > x_{threshold} \tag{I.21}$$

The Slope (β) is defined by the fit parameter k:

$$\beta = \frac{1}{k} + 1 \tag{I.22}$$

According to the MCNP methodology [13], the slope β is typically calculated using a Maximum Likelihood Estimation (MLE) fit. The code maintains a table of the largest scores (e.g., the top

201 scores) and fits them to the Pareto distribution to determine the rate of decay. The value of β provides the following insights into the simulation's convergence:

- $\beta > 3.0$: This is the theoretical minimum for the CLT to apply, as it indicates the second moment (variance) is finite.
- $\beta > 5.0$: At this level, the fourth moment is finite, indicating that the VOV estimate itself is statistically reliable.
- $\beta < 3.0$: If the slope is below 3.0, the distribution is "heavy-tailed." The variance is technically infinite, meaning the relative error R will not decrease predictably as $1/\sqrt{N}$, and the simulation results should be considered unreliable.

2.2. Particle Weights and Importance Sampling

As mentioned in the previous section, the statistical weight (w) is a fundamental parameter used in Monte Carlo estimators to calculate the mean, the relative error, and the VOV. Variance reduction aims to improve the Figure of Merit (FOM) by encouraging particles to move toward regions of interest (e.g., detectors) and preventing them from exploring irrelevant areas. This is achieved by systematically altering the weights and population in a way that preserves unbiasedness of the tally estimators.

While analog Monte Carlo simulations strictly emulate physical interactions, non-analog simulations modify the particle history to improve sampling efficiency. This modification is mathematically allowed provided that the fundamental conservation of the tally density is maintained:

$$\langle T \rangle = \sum_i w_i \cdot t_i \tag{I.23}$$

where w_i is the weight of the i-th particle and t_i is its score. The following subsections detail the mechanisms used to manipulate this weight and the theoretical frameworks used to optimize them.

2.2.1. Fundamental Principles: Weight and Importance

The central concept in variance reduction is the inverse relationship between particle weight (w) and regional importance (I). Importance is a scalar value representing the significance of a specific region in phase space (space, energy, angle) relative to the tally of interest.

To maintain a fair game (conserving the total "statistical mass" of the system) any increase in the sampling frequency (population) of a region must be accompanied by a proportional decrease in particle weight. This leads to the conservation law of variance reduction :

$$w \cdot I = \text{const} \tag{I.24}$$

Consequently:

- **Splitting:** When a particle moves to a region of higher importance, it is split into multiple particles with lower weights. This increases the sample size (N) in critical regions, reducing the variance ($\sigma \propto 1/\sqrt{N}$).

- **Russian Roulette:** When a particle moves to a region of lower importance, it undergoes a survival game (roulette). Surviving particles have their weights increased, while failing particles are terminated. This saves computational time (T) by ceasing to track irrelevant particles.

2.2.2. Common Weight Control Mechanisms

While the following mechanisms are discussed primarily within the context of the MCNP code framework [13], they represent fundamental non-analog techniques implemented across most modern Monte Carlo radiation transport codes. These methods serve to enforce the weight-importance conservation law through population control and interaction biasing.

Implicit Capture

Implicit capture, also known as survival biasing, modifies the sampling of absorption events to prevent particle histories from terminating prematurely. In an analog simulation, a particle colliding with a nucleus is absorbed with probability σ_a/σ_t. In implicit capture, the particle is forced to survive the collision. To compensate for this artificially increased survival rate, the particle's weight is reduced by the absorption probability [15]:

$$w_{new} = w_{old} \cdot \left(1 - \frac{\sigma_a}{\sigma_t}\right) \tag{I.25}$$

Implicit capture is enabled by default in MCNP [13]. This method is particularly effective in deep penetration problems where analog absorption would deplete the population before it reaches the tally region. However, it creates a "weight fluctuation" problem: particles may survive with infinitesimally small weights, consuming computational resources (T) without contributing significantly to the variance reduction [15].

Weight Cutoff and Russian Roulette

The use of implicit capture creates a secondary challenge: a particle could theoretically undergo an infinite number of collisions as its weight progressively approaches zero, wasting significant computer time on tracks with negligible statistical impact. To mitigate this, a Weight Cutoff mechanism is used as a population control method [15, 16].

When a particle's statistical weight falls below a user-defined threshold, the code triggers a game of Russian Roulette. The particle is either terminated or, if it survives, its weight is increased to a higher level (often referred to as the "survival weight") to ensure that the total weight in the system remains statistically conserved. This partnership between implicit absorption and weight cutoff allows the code to maintain a robust sample of particle tracks while automatically removing those that no longer hold significant value [16]. The weight cutoff is likewise active by default in MCNP.

Geometry Splitting

Geometry splitting occurs at the boundary between geometric cells. If a particle crosses from a cell of importance I_1 to a cell of importance I_2 (where $I_2 > I_1$), the particle is split into n independent particles, where $n = I_2/I_1$. Each daughter particle is assigned a weight of

w/n. Conversely, moving to a lower importance triggers Russian Roulette. While effective, this method is limited because it assigns a single importance value to an entire geometric cell, ignoring spatial variations within the cell or energy dependencies.

Source Biasing

Source biasing improves sampling efficiency by artificially starting more particles in regions of phase space (energy, position, or angle) that are most likely to contribute to the tally. To maintain unbiased results, the weight of the source particles is adjusted according to the ratio of the true PDF, $p(x)$, to the biased PDF, $p^*(x)$ [16]:

$$w_{source} = w_0 \cdot \frac{p(x)}{p^*(x)} \tag{I.26}$$

This is generally implemented through two primary domains:

- **Energy Biasing:** The physical source energy distribution is replaced by a biased distribution that oversamples energies expected to contribute most to the tally. For example, in a shielding problem, one might bias the source to sample the high-energy part of the fission spectrum more frequently, as these neutrons are more likely to penetrate the shield.
- **Spatial Biasing:** The source position distribution is modified to start more particles in spatial regions that are expected to contribute more to the tally. This is useful, for example, when only part of an extended source has a significant probability of streaming through a penetration or reaching a detector.
- **Angle Biasing:** This forces particles to start in a preferred direction (e.g., towards a detector). MCNP, for instance, offers two methods:
  1. *Cone Biasing:* A cone is defined around a preferred direction vector. A specified fraction of particles are forced to start within this cone.
  2. *Exponential Biasing:* The probability of starting in a direction Ω falls off exponentially as the angle deviates from the preferred direction Ω_{ref}. The density follows $e^{k(\Omega \cdot \Omega_{ref})}$, where k is a biasing parameter.

2.2.3. Weight Windows

The Weight Window is a more sophisticated population control method that addresses the limitations of geometry splitting. Unlike geometry splitting, which acts primarily at surfaces, the weight window acts as a global constraint on particle weight throughout the phase space (space and energy).

A weight window consists of defined lower (W_L) and upper (W_U) weight bounds for every region. It enforces that all particles in a given region maintain a weight within the interval ("width"):

$$W_L < w < W_U \tag{I.27}$$

Weight Window Operational Logic

The weight window enforces particle weights through two primary mechanisms at surfaces and/or collision sites:

- **Above the Window ($w > W_U$):** The particle is split into m particles (typically $m = w/W_{target}$) to bring weights within the interval.
- **Below the Window ($w < W_L$):** The particle undergoes Russian Roulette; survivors have their weight increased to a target value, others are terminated.
- **Inside the Window ($W_L < w < W_U$):** The particle tracks normally without adjustment.

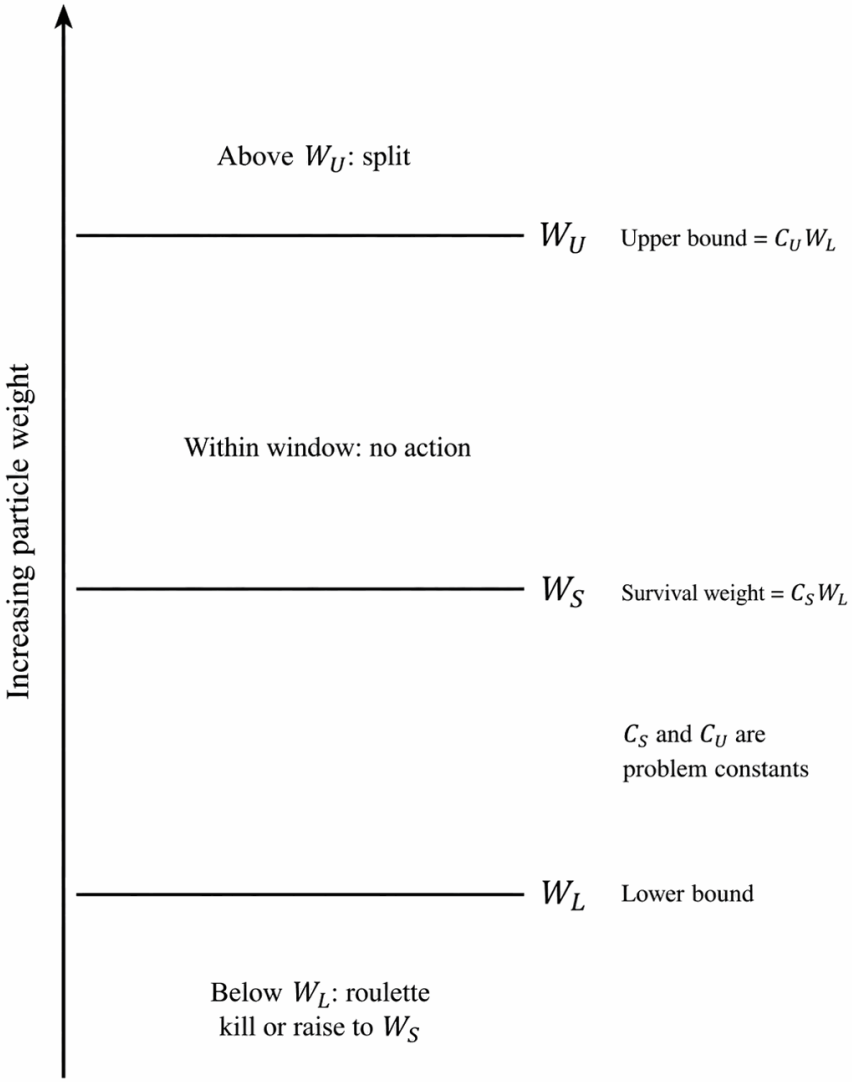


Figure I.2: Logic of the weight window technique showing the splitting and Russian Roulette regimes relative to the weight bounds [15]

To prevent instability, the window must be sufficiently wide. If the window is too narrow, a particle might be split (reducing its weight) only to immediately trigger the lower bound and play roulette. Therefore, the ratio is typically constrained such that:

$$\frac{W_U}{W_L} \geq C \quad (\text{typically } C \approx 5) \tag{I.28}$$

The effectiveness of this method relies on accurately defining the bounds for each cell, which are inversely proportional to the cell's importance. In principle, the importance of a region can be estimated from a forward simulation, by scoring how much the particles passing through it contribute to the tally [15]. For the deep-penetration shielding problems of interest, however, too few histories reach the regions far from the source for this forward estimate to be reliable, which motivates the deterministic approach developed below.

In modern MCNP shielding workflows, weight windows are typically implemented using a superimposed mesh. This mesh is independent of the problem's physical geometry, allowing for a highly detailed discretization of the phase space without complicating the geometry.

Each voxel (i) in this mesh contains the parameters necessary to define the window locally:

- **Lower Weight Bound ($W_{L,i}$):** The base threshold for the window. The upper bound and target weights are then derived from this value using a constant ratio.
- **Energy Dependence:** While the transport itself may use continuous energy physics, the weight window bounds are discretized into energy bins. A single voxel will contain different W_L values for different energy ranges to account for the varying importance of, for example, fast versus thermal neutrons.

In modern high-fidelity simulations, these bounds are stored in a superimposed mesh (such as an MCNP `wwinp` file [13]). Manually defining these values for complex geometries is practically impossible; instead, they are generated using deterministic adjoint calculations. This relationship between particle importance and optimal weight is defined by the Adjoint Transport Theory.

2.3. Adjoint Transport Theory

The effectiveness of the weight window technique depends heavily on the accurate selection of the lower (W_L) and upper (W_U) weight bounds. As previously discussed, these bounds should be inversely proportional to the particle importance (see Eq. (I.24)). While estimating importance manually is practically impossible for complex geometries, a mathematical solution to optimal importance selection exists in Adjoint Transport Theory.

In standard forward transport, we solve for the particle flux density $\psi(\mathbf{r}, \mathbf{\Omega}, E)$ using the transport equation as defined in Eq. (I.3):

$$\mathcal{H}\psi = q \tag{I.29}$$

where q (previously denoted as Q_{ext}) represents the external source distribution. The quantity of interest, such as a detector response R, is calculated as the inner product of this flux and a response function $\sigma_d(\mathbf{r}, \mathbf{\Omega}, E)$ [14]:

$$R = \langle \psi, \sigma_d \rangle = \int_0^\infty \int_{4\pi} \int_V \psi(\mathbf{r}, \mathbf{\Omega}, E)\sigma_d(\mathbf{r}, \mathbf{\Omega}, E)\ d\mathbf{r}d\mathbf{\Omega}dE \tag{I.30}$$

In adjoint transport, we solve the mathematically transposed equation for the adjoint function $\psi^\dagger$:

$$\mathcal{H}^\dagger \psi^\dagger = \sigma_d \tag{I.31}$$

As shown in the operator equation above, the source for the adjoint calculation is the physical response function σ_d. The fundamental identity of adjoint theory states that the system response can be calculated using either the forward flux or the adjoint solution [14, 17]:

$$R = \langle \psi, \sigma_d \rangle = \langle q, \psi^\dagger \rangle \tag{I.32}$$

This identity, visualized in Figure I.3, provides $\psi^\dagger$ with a clear physical meaning: the adjoint function represents the expected contribution of a particle at a specific position, energy, and direction to the final detector response. If $\psi^\dagger(\mathbf{r}, E, \mathbf{\Omega})$ is large, a particle at that point in phase space has a high probability of contributing to the tally. Consequently, the adjoint flux is the formal mathematical definition of particle importance.

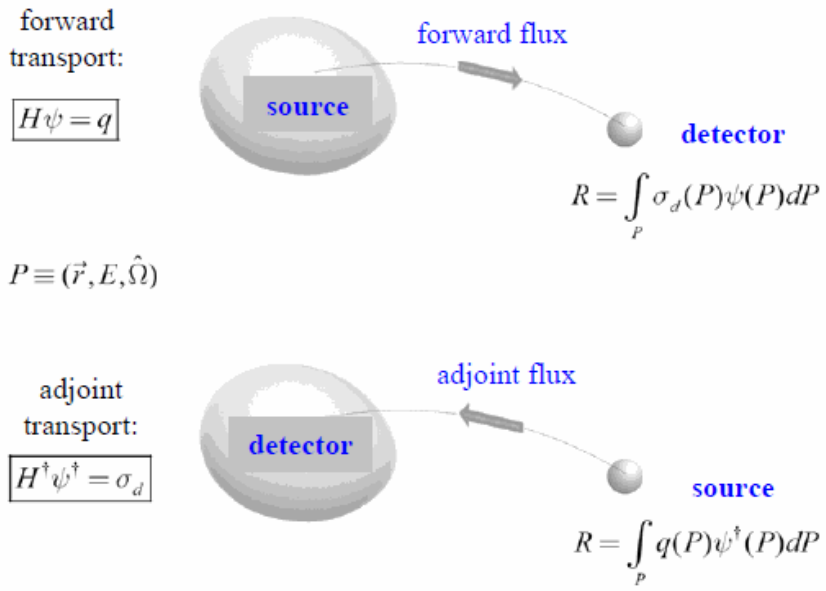


Figure I.3: Schematic illustrating the duality between forward and adjoint transport. In the adjoint mode, particles effectively travel "backwards" from the detector to the source [18].

A "Zero-Variance solution" represents a theoretical ideal where every history contributes the exact same score; to approximate this and minimize variance, the weight window is made inversely proportional to the importance [19]. By solving the adjoint equation (typically using deterministic methods), we can generate an optimal importance map to define the target weights for our weight windows:

$$w(\mathbf{r}, \mathbf{\Omega}, E) \propto \frac{1}{\psi^\dagger(\mathbf{r}, \mathbf{\Omega}, E)} \tag{I.33}$$

2.4. Adjoint-Driven Importance Sampling

As discussed in Sec. 2.3, the adjoint flux $\psi^\dagger(\mathbf{r}, \mathbf{\Omega}, E)$ provides a formal definition of particle importance for a chosen response, and the reciprocity relation (Eq. (I.32)) connects the forward

source to the adjoint solution. In practical shielding problems, however, obtaining a useful importance map by manual tuning is rarely feasible. Adjoint-driven hybrid workflows address this by using a deterministic solver to compute an approximate adjoint field at relatively low cost, and then converting that field into Monte Carlo variance reduction parameters (e.g., space- and energy-dependent weight windows and a consistent biased source) [14]. In this section, the focus is on how the adjoint solution is translated into Monte Carlo variance reduction via the CADIS and FW-CADIS methods.

2.4.1. Consistent Adjoint Driven Importance Sampling (CADIS)

CADIS is intended to reduce the variance of a single scalar response (for example, the response of one detector tally) [14]. The starting point is the same interpretation introduced in Sec. 2.3: regions of phase space with larger $\psi^\dagger$ are, in expectation, more influential for the selected response.

In CADIS, an adjoint deterministic calculation is performed with a source equal to the detector response function. This corresponds to solving Eq. (I.31) where σ_d characterizes the tally of interest. Once $\psi^\dagger$ is available, CADIS defines a target (or "ideal") Monte Carlo weight that is inversely proportional to importance, consistent with the discussion around the zero-variance (see Eq. (I.33)). A commonly used normalization is

$$w(\mathbf{r}, \mathbf{\Omega}, E) = \frac{R}{\psi^\dagger(\mathbf{r}, \mathbf{\Omega}, E)}, \tag{I.34}$$

where the scalar R is the same response that appears in the reciprocity identity (Eq. (I.32)). In practice, the weight-window bounds used by the Monte Carlo code are constructed from this target weight using the window width parameters introduced earlier (Eq. (I.27)).

A second key component of CADIS is consistent source biasing. If particles were sampled from the analog source distribution but the target weight at the source location differed substantially from the analog birth weight, the simulation would typically spend many events immediately splitting or rouletting near the source, which is inefficient. CADIS instead samples particles from a biased source distribution proportional to the product of the analog source and the adjoint importance:

$$\hat{q}(\mathbf{r}, \mathbf{\Omega}, E) = \frac{\psi^\dagger(\mathbf{r}, \mathbf{\Omega}, E)\, q(\mathbf{r}, \mathbf{\Omega}, E)}{R}. \tag{I.35}$$

To keep estimators unbiased, each history is born with the appropriate statistical weight associated with this change of sampling distribution (as in standard source-biasing theory), so that the Monte Carlo estimates remain correct while more histories are started in phase-space regions that are important for the chosen response [14].

Overall, CADIS is most effective when the problem objective can be expressed as a single dominant response. When multiple tallies are equally important, for example flux tallies at several azimuthal positions on the reactor pressure vessel and at the surveillance capsule inside it (as in the PWR fluence calculation of Chapter IV), a single-response importance map tends to over-emphasize some regions and under-emphasize others; FW-CADIS is the standard choice in that setting.

2.4.2. Forward-Weighted CADIS (FW-CADIS)

FW-CADIS extends the CADIS idea to objectives that are spatially and/or energetically distributed, where the goal is often to achieve a reasonably uniform relative uncertainty over many tally bins, for example, a flux map [14]. The main difference is how the adjoint source is defined.

In a typical FW-CADIS workflow, a deterministic forward calculation first estimates the scalar flux $\phi(\mathbf{r}, E)$. A target response density $f(\mathbf{r}, E)$ then sets which regions should be well resolved: $f(\mathbf{r}, E) = 1$ for a uniform flux map, or a chosen weighting to emphasize particular locations or energies. FW-CADIS then forms the adjoint source by weighting this target with the inverse of the forward flux:

$$q^{\dagger}(\mathbf{r}, E) = \frac{f(\mathbf{r}, E)}{\phi(\mathbf{r}, E)}, \tag{I.36}$$

The adjoint transport equation is then solved with this source, producing an adjoint field $\psi^{\dagger}$ that aims to distribute Monte Carlo effort more uniformly over the chosen objective [14].

After this point, the construction of weight windows and the consistent biased source follows the same pattern as CADIS (Eqs. (I.34)–(I.35)), but now with the FW-CADIS adjoint solution. In many applications this provides a more balanced variance reduction setup for problems where several spatially or energetically distinct detectors are of interest.

3. Codes

This thesis relies on a set of well-established computational tools to model neutron transport, generate variance-reduction parameters, and support sensitivity and reactor-core analyses. The codes used in this work are briefly introduced in the following subsections.

3.1. MCNP6.3

MCNP (Monte Carlo N-Particle) [13] is a general-purpose particle transport code developed at Los Alamos National Laboratory. It simulates the coupled transport of neutrons, photons, electrons, and other particle types across wide energy ranges by tracking individual particle histories through user-defined materials and three-dimensional geometries.

In MCNP, the geometry is typically described using constructive solid geometry (CSG), where cells are bounded by analytical surfaces and assigned material compositions and densities. Radiation transport is then solved stochastically by sampling interaction types and secondary-particle production from evaluated nuclear and atomic data. Quantities of interest (e.g., fluxes, reaction rates, or detector responses) are estimated with tallies, and the associated statistical uncertainty is reported through relative errors that decrease as more particle histories are simulated.

A practical consequence of the Monte Carlo approach is that deep-shielding problems can become computationally expensive: if only a very small fraction of particles contribute to the tally region, analog simulations may converge slowly and require very large numbers of histories. For this reason, MCNP serves in this thesis both as the baseline tool for reference calculations and as the transport engine to which variance-reduction parameters generated by ADVANTG are applied in challenging fixed-source configurations.

MCNP is well suited to shielding and deep-penetration transport: it represents detailed three-dimensional geometries exactly, treats neutron interactions in continuous energy without group averaging, and transports neutrons and photons in a single coupled calculation, so secondary-photon production and the associated energy deposition are captured consistently with the neutron field. It is widely used for reactor-pressure-vessel fluence and PWR ex-core response analyses [20, 21, 22], and serves in this thesis as the high-fidelity reference against which the deterministic importance maps and the approximations of the two-step vessel scheme are checked. The aspects of MCNP most central to the present work, namely its perturbation and sensitivity capabilities, are described in detail in Chapter II (Sec. 2), where they support the nuclear-data uncertainty propagation used across this work.

3.2. ADVANTG 3.2.0

ADVANTG (Automated Variance Reduction Generator) [14] is a companion tool designed to automate variance reduction for MCNP fixed-source calculations, most commonly through space- and energy-dependent weight windows and, optionally, biased source descriptions. The goal is to improve the efficiency of Monte Carlo simulations without changing the underlying physical model.

ADVANTG follows a hybrid deterministic–Monte Carlo workflow. It first builds a discretized representation of the problem and runs a deterministic discrete-ordinates calculation with Denovo [23] to obtain an approximate importance map, commonly derived from an adjoint solution. ADVANTG then converts this information into variance-reduction parameters (e.g., weight-window bounds) that are written to files compatible with MCNP. When MCNP reads these parameters, it preferentially splits particles that move toward important phase-space regions and applies Russian roulette to particles moving away from them, concentrating computational effort where it most improves tally precision.

To perform the deterministic transport step within Denovo, ADVANTG uses internal multigroup (MG) nuclear data libraries. It reads coupled neutron–photon MG libraries in ANISN format [24]. The code can interpret the library data, perform the material mixing required by the MCNP composition definitions, and write a working library in the format expected by Denovo. The set of MG libraries distributed with ADVANTG spans different group structures, isotope coverage, and nuclear data evaluations, which can be advantageous when the problem spectrum or application domain differs substantially (e.g., reactor shielding vs. fusion-oriented spectra). Table I.2 lists the libraries documented in the ADVANTG distribution and highlights their main characteristics.

Within the workflow used here, these multigroup libraries enter only the deterministic calculation that produces the variance-reduction parameters; the final Monte Carlo simulation uses continuous-energy libraries (such as ENDF/B-VIII.1 or JEFF-4.0). The choice of multigroup library therefore affects the quality of the weight-window guidance rather than the physical accuracy of the tally itself. The accuracy demand on the multigroup data is consequently modest, with one important caveat: the library must contain the reaction channels that drive the dominant contributors to the tally. If a relevant channel is absent, the deterministic calculation misses its contribution and the resulting weight windows can steer the simulation along an other path; provided the dominant contributor paths are represented, sub-percent accuracy in the multigroup data is not required.

In the context of this thesis, ADVANTG is used specifically to make otherwise inefficient shield-

Table I.2: Multigroup libraries available for Denovo calculations in ADVANTG [14].

Library	# of groups (N/G)	# of isotopes or elements	Evaluation
27n19g	27/19	393	ENDF/B-VII.0
200n47g	200/47	393	ENDF/B-VII.0
BUGLE-96	47/20	120	ENDF/B-VI.3
BPLUS	47/20	393	ENDF/B-VII.0
DABL69	46/23	80	ENDF/B-V
DPLUS	46/23	393	ENDF/B-VII.0
FENDL67	46/21	71	FENDL-2.1
HILO2K	83/42	32	ENDF/Bertini INC model

ing/attenuation fixed-source calculations computationally feasible, especially in configurations involving strong attenuation, long streaming paths, and/or small tally regions. As with any variance-reduction approach, the resulting simulations still require standard convergence checks (e.g., stability of mean values, reasonable relative errors, and consistency across runs) to ensure that improved efficiency does not come at the expense of poor phase-space coverage or nonphysical artifacts.

3.3. Serpent 2.2

Serpent 2.2 [25], developed at VTT Technical Research Center of Finland, is a Monte Carlo code originally optimized for reactor physics and burnup calculations. Serpent is particularly valuable in this work because both its sensitivity calculation methods and their implementation differ from those of MCNP. It was used to perform sensitivity analysis on angular distributions and to verify cross section sensitivities. Because Serpent's underlying algorithms differ from those in MCNP, comparing the two provides a robust layer of verification, ensuring that the sensitivity of the final tallies to nuclear data uncertainties is accurately characterized.

3.4. CASMO5/CMSLINK5/SIMULATE5

For full PWR simulations, the chain begins with the characterization of the neutron source within the reactor core, a task accomplished using the Studsvik Core Management System (CMS5). This process is structured as a hierarchical three-step sequence. It starts with CASMO5 [10], a multi-group, two-dimensional transport code that performs the lattice physics calculations for individual fuel segments. These results are then processed by CMSLINK5 [26], which serves as an essential data bridge, compiling the microscopic cross sections and assembly-level data into comprehensive libraries. The final step in the core modeling phase is performed by SIMULATE5 [27], which uses these libraries to execute three-dimensional nodal diffusion calculations, coupled with thermal-hydraulic feedback, for the entire reactor core. This allows for a high-fidelity determination of the 3D power distribution and burnup state of the fuel.

For the purposes of this thesis, the Studsvik suite is used to convert the SIMULATE5 output of a PWR into a fixed source term, which provides the spatial and spectral inputs for the subsequent neutron transport simulations beyond the core. The implementation of this coupling is described in Sec. 5.

Beyond its nominal calculation role, CASMO5 includes a built-in capability for nuclear data sampling directly from covariance matrices. For each sample i, a perturbed set of nuclear data parameters is generated as $\vec{x}_i = \vec{x}_0 + A\vec{z}_i$, where $\vec{x}_0$ is the nominal parameter vector, A is the sampling matrix derived from the eigendecomposition of the covariance matrix, and $\vec{z}_i$ is a vector of independent standard normal variates (see Sec. 3.2.2, Eq. (II.85) for the general formalism). The perturbable quantities span approximately 180 isotopes and cover the reaction types listed in Table I.3, all based on ENDF/B-VIII.0 covariance data discretized on 586 energy groups [10]. This capability is exploited in Chapter IV (Sec. 5) to propagate nuclear data uncertainties through the source term calculation.

Table I.3: Reaction types available for nuclear data sampling in CASMO5 [10]. Perturbations are generated from ENDF/B-VIII.0 covariance matrices via eigendecomposition.

MT	Symbol	Description
13	σ_s	Scattering cross section
16	σ_{2n}	$(n, 2n)$ cross section
18	σ_f	Fission cross section
101	σ_d	Disappearance (including (n, γ), (n, α), etc.)
452	ν	Average neutrons per fission
457	λ	Decay constants
458	κ	Energy release per fission
459	FY	Fission yields (without correlations)
461	DBR	Decay branching ratios
462	CBR	Capture branching ratios
1018	χ	Fission spectrum

4. Nuclear Data

Particle transport simulations rely on nuclear data to describe the interaction probabilities and kinematics of reactions between incident particles and target nuclei. For neutron transport, the most central quantities are microscopic reaction cross sections as a function of incident energy, $\sigma_r(E)$, and secondary-particle distributions that describe how particles are emitted after an interaction (e.g., scattering angle distributions and outgoing energy spectra). These quantities are not directly measured or computed in a complete and self-consistent manner for every isotope, reaction channel, and energy range of interest. Instead, they are assembled through nuclear data evaluation, where experimental measurements are combined with theoretical models and physical constraints to produce a coherent representation of the state of the art of the current knowledge of the interaction physics over broad energy ranges. This evaluated knowledge is not static: as new measurements and improved models become available, evaluated libraries are revised and reissued.

Evaluated nuclear data are typically distributed in standardized formats designed for archival and exchange, while transport solvers usually require code-optimized "application libraries". Consequently, nuclear data processing is required to transform evaluations into forms that are directly usable by Monte Carlo or deterministic transport codes, including temperature-dependent continuous-energy libraries and multigroup libraries with consistent self-shielding and uncertainty information.

4.1. Nuclear Data Cycle

The development of high-quality nuclear data is not a static one-time effort but a continuous, iterative process known as the *Nuclear Data Cycle*. This cycle ensures that data remains accurate, consistent, and relevant to modern applications. As illustrated in Figure I.4, the process is driven by the specific needs of end-users—ranging from commercial power reactors and medical isotope production to fusion technology and space exploration.

These application-driven needs define target accuracies and priorities, which in turn motivate new experimental measurements and theoretical developments. The cycle proceeds through several distinct but interconnected stages:

1. **Measurements & Theory:** Experimentalists perform differential measurements which are compiled into the EXFOR database [28]. Simultaneously, theoreticians develop nuclear reaction models to predict data where measurements are sparse or impossible.
2. **Evaluation:** Evaluators combine experimental data and theoretical models to produce "evaluated" files, which comprehensive datasets describing the interaction of particles with a specific nuclide over a wide energy range.
3. **Processing & Validation:** Evaluated files are processed into application-ready formats and tested against integral benchmarks: experiments whose measured response folds together contributions from many isotopes, energies, and reaction channels, providing a stringent global test of the underlying differential data.
4. **Release:** Validated evaluations are released in major libraries (such as JEFF, ENDF/B, or JENDL; Table I.4) for use by the wider community.

Feedback from the application side eventually flows back into the cycle, identifying discrepancies or new requirements, thus restarting the process.

4.1.1. EXFOR

The foundation of any nuclear data evaluation is experimental evidence. The **EXFOR** (EXchange FORmat) database is the global repository for experimental nuclear reaction data [30]. It contains an extensive compilation of reaction cross sections, angular distributions, and other differential quantities derived from experiments performed worldwide. EXFOR is maintained by the International Network of Nuclear Reaction Data Centres (NRDC), a cooperative effort coordinated by the International Atomic Energy Agency (IAEA).

The content of EXFOR is not random; it is often driven by specific community needs. To ensure that experimental efforts target the most critical gaps in knowledge, the OECD Nuclear Energy Agency (NEA) maintains the High Priority Request List (HPRL) [31]. The HPRL acts as a formal bridge between data users (who identify accuracy targets for applications like advanced reactors or medical isotopes) and data producers (experimentalists). Requests listed in the HPRL are rigorously reviewed and serve as a guide for planning new measurements.

The role of EXFOR in the nuclear data cycle is effectively to serve as the unified source of experimental evidence. When evaluators begin work on a specific isotope, they draw on the available EXFOR datasets to constrain their nuclear reaction models. The completeness and accuracy of the EXFOR entries are therefore critical for the quality of the final evaluated libraries.

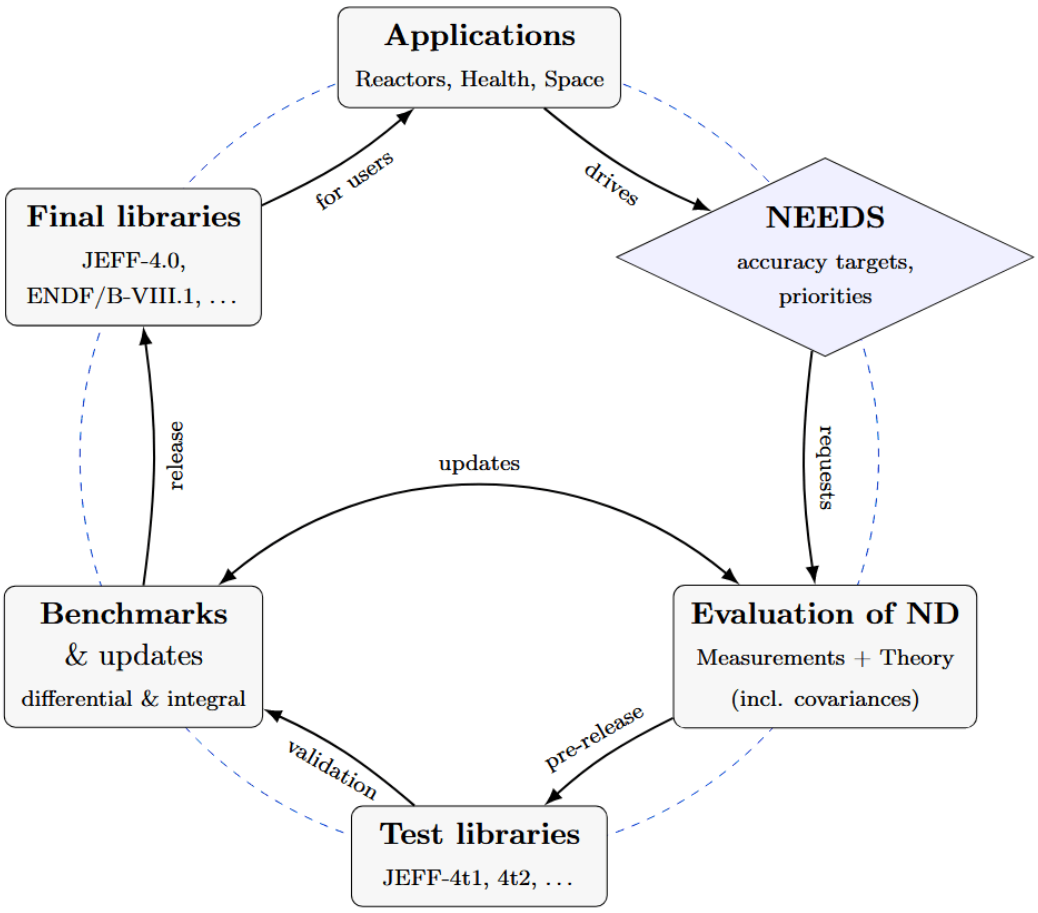


Figure I.4: The Nuclear Data Cycle. This schematic illustrates the iterative process of data refinement, starting from application-driven needs (e.g., reactors, health, space) to the measurement, evaluation, validation, and eventual release of finalized libraries such as JEFF-4.0 and ENDF/B-VIII.1 [29].

4.1.2. Evaluated Nuclear Data Libraries

Evaluated nuclear data libraries provide recommended ("best estimate") reaction and emission data for a large set of nuclides, usually accompanied—at least for a subset of reactions and materials—by quantified uncertainties and correlations. Several internationally maintained "general-purpose" libraries are widely used in reactor physics, shielding, criticality safety, and fusion applications. In this thesis, the analyses focus on libraries in the JEFF (Joint Evaluated Fission and Fusion), ENDF/B (Evaluated Nuclear Data File), JENDL (Japanese Evaluated Nuclear Data Library), and TENDL (TALYS Evaluated Nuclear Data Library) families, summarized in Table I.4; other significant regional libraries include BROND (Russia) and CENDL (China).

Table I.4: Overview of major international nuclear data libraries.

Library	**Origin / Coordinator**	**Details**	**Ref.**
JEFF	OECD / NEA community	European reference library Latest release JEFF-4.0 (2025)	[32]
ENDF/B	CSEWG / NNDC (BNL)	U.S. reference library Latest release ENDF/B-VIII.1 (2024)	[33]
JENDL	JAEA	Japanese reference library Latest release JENDL-5 (2021)	[34]
TENDL	TALYS-based	Broad coverage, strongly model-based Latest release TENDL-25 (2025)	[35]

Historically, these libraries have been distributed in the ENDF-6 format, a legacy punched-card-based format that has served as the international standard for decades (detailed further in Section 4.2). While robust, the rigid structure of ENDF-6 imposes limitations on numerical precision and the complexity of data that can be represented. To address these challenges, the community has developed the Generalized Nuclear Data Structure (GNDS) [36]. GNDS uses modern hierarchical data structures to allow for greater flexibility, higher precision, and the inclusion of more detailed physical information, aiming to replace ENDF-6 as the standard for data exchange.

Even when libraries target the same applications, differences may arise from (i) different selections or interpretations of experimental datasets, (ii) different resonance analyses and parameterizations, (iii) different nuclear model choices and parameter adjustments in higher-energy regions, and (iv) different treatments of correlated quantities, all of which lead to differences in the completeness and structure of the covariance information that each library provides. Historically, covariance data received little attention and were often absent from evaluations; today they are regarded as essential to any quantitative use of nuclear data, and the fraction of isotopes and reactions for which covariances are supplied has grown steadily across recent releases. These differences become especially relevant when simulation results are sensitive to particular reaction channels (for example, elastic scattering in shielding problems or fission and capture in criticality problems), and motivate the systematic comparison of evaluations and the development of the uncertainty-propagation methods presented in later chapters.

4.1.3. Benchmarking and Validation

To close the nuclear data cycle, the full simulation chain must be confronted with integral experiments. Integral benchmarks measure a global response that folds the cross sections of many nuclides over a neutron spectrum and the full transport process, including multiple scattering and self-shielding. They therefore test the data in combination and in the conditions in which they are used, and are sensitive to energy ranges and reaction channels that differential measurements constrain only sparsely. Because many contributions are folded together, however, opposing errors in different cross sections can compensate one another, so agreement at the integral level does not by itself guarantee correct differential data. Integral and differential testing are therefore complementary.

Several international benchmark programs provide curated, peer-reviewed datasets. For shielding and deep-penetration transport problems, the Shielding Integral Benchmark Archive and Database (SINBAD) compiles benchmark experiments relevant to reactor, accelerator, and fusion shielding applications [37]. For criticality safety, the International Criticality Safety Benchmark Evaluation Project (ICSBEP) provides a large set of evaluated criticality benchmark configurations [38]. Reactor-physics benchmarks are similarly curated through IRPhEP and related efforts [39]. Each program is assembled to validate simulation tools for one application class, shielding, criticality, or reactor physics, rather than to test nuclear data. The nuclear-data community can nonetheless reuse the best-characterized of these benchmarks to compare evaluated libraries and test nuclear data, but only when the geometry, materials, and uncertainties are specified precisely enough to attribute a discrepancy to the data rather than to the model or the experiment.

For this work, the ASPIS-88 benchmark [40] from the SINBAD database was used as a case study and as a way to test the developed methodology against experimental data. The results are presented in Chapter IV.

4.2. ENDF-6 Files

The Evaluated Nuclear Data File (ENDF-6) format [41] is the established standard for storing and exchanging nuclear data used in reactor physics, radiation shielding, and criticality safety. Maintained by the Cross Section Evaluation Working Group (CSEWG), the format is characterized by a strict 80-character, fixed-width structure. While this layout is a legacy of punched-card conventions, it ensures that data remains both human-readable and easily parsed by machine, providing a consistent framework for all represented data.

4.2.1. Hierarchical Organization

ENDF-6 data are organized hierarchically in three primary identification layers:

$$\texttt{TAPE} \rightarrow \texttt{MATERIAL (MAT)} \rightarrow \texttt{FILE (MF)} \rightarrow \texttt{SECTION (MT)}.$$

Each **MAT** identifies a target material (typically a nuclide evaluation), **MF** denotes a category of data (e.g., cross sections or angular distributions), and **MT** specifies a particular reaction channel (e.g., elastic scattering or radiative capture). Table I.5 summarizes these conceptual identifiers. The labels date from the era when ENDF libraries were distributed on magnetic tape: `MF` stands for “magnetic file” and `MT` for “magnetic tape”.

Table I.5: Conceptual identifiers in the ENDF-6 hierarchy.

Level	Identifier	Meaning
Material	`MAT`	Nuclide/element evaluation identifier
File	`MF`	Data category (e.g., cross sections, angular distributions)
Section	`MT`	Reaction channel (e.g., elastic, fission, capture)

The material number is described by:

$$\mathrm{MAT} = 100 \cdot Z + I, \tag{I.37}$$

where Z is the atomic number and I is an isotope/evaluation index relative to the element. For general purpose files, I is assigned according to the following conventions:

- For naturally occurring elemental evaluations, I=00.
- For isotopes, I is assigned based on increasing mass number in steps of 3. To accommodate neutron-excess nuclides, the lightest stable isotope is conventionally assigned I=25.
- The in-between numbers (e.g., Z26, Z27) are kept for isomeric (metastable) states.

For example, `MAT` =2631 is used for ^{56}Fe and `MAT` =9237 for ^{238}U.

The mapping of ENDF-6 physics sections to their corresponding nominal and covariance file numbers (`MF`) is summarized in Table I.6.

The `MT` numbers follow a standardized reaction numbering scheme defined in the ENDF-6 format manual [41]. The main `MT` groupings are illustrated in Figure I.5.

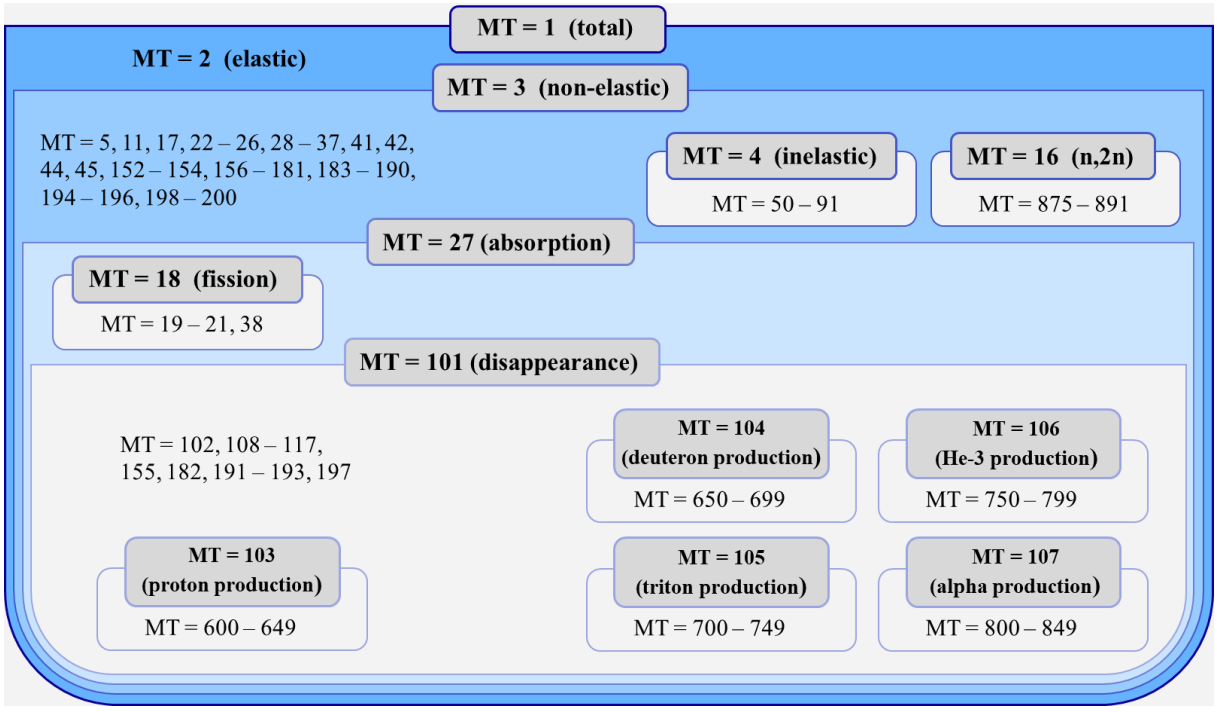


Figure I.5: Hierarchical structure of ENDF-6 `MT` reaction numbers.

Table I.6: Mapping of ENDF-6 physics sections to their corresponding nominal and covariance file numbers (`MF`).

Physics Description	Nominal	Covariance
Structure & Resonance		
General Information	1	31
Resonance Parameters	2	32
Cross Sections		
Reaction Cross Sections	3	33
Thermal Scattering	7	–
Secondary Distributions		
Angular Distributions	4	34
Energy Distributions	5	35
Energy-Angle Distributions	6	–
Production & Yields		
Radioactivity & Yields	8	8
Radioactive Multiplicities	9	39[a]
Radioactive CS	10	40
Photon Production		
Multiplicities	12	–
Cross Sections	13	–
Angular Distributions	14	–
Energy Distributions	15	–

[a] Reserved in the format manual, with no format defined; not used by current libraries.

4.2.2. File Formatting

Every ENDF-6 record is contained within a line of exactly 80 characters. To ensure consistent alignment and straightforward parsing, each line is divided into two primary segments:

- **Data fields (columns 1–66):** Six fixed-width fields of 11 characters each, used for numeric values.
- **Control fields (columns 67–80):** Identifiers and sequencing metadata used to navigate the file.

The control portion is further subdivided into the Material Identifier `MAT`, File identifier `MF`, Section/Reaction identifier `MT`, and an optional line sequence number `SEQ`. Table I.7 summarizes the standard column assignments for every line in an ENDF-6 file.

Table I.7: Fields in each line of the ENDF-6 file.

Columns	Width	Symbol	Description
1–66	66	–	Data
67–70	4	`MAT`	Material identifier
71–72	2	`MF`	File identifier
73–75	3	`MT`	Section (reaction) identifier
76–80	5	`SEQ`	Line sequence number (optional)

To preserve structural integrity and allow streaming parsers, ENDF-6 defines explicit terminators for these hierarchical boundaries. Section, file, material, and tape ends are indicated using dedicated records, commonly referred to as `SEND` (Section END), `FEND` (File END), `MEND` (Material END), and `TEND` (Tape END). These terminators allow a parser to know exactly when a specific block of data has concluded without needing to scan ahead.

To maximize precision within the 11-character data fields, numbers are stored in a compact scientific notation where the "E" is omitted. The exponent is written as a trailing signed integer; for example, 1.234567+5 represents 1.234567×10^{5}, and -3.141590-1 represents -3.141590×10^{-1}.

Example of the ENDF-6 layout

```
 2.605400+4 5.534700+1          0          1          0          02625 31  2   1
|<-- C1 -->|<-- C2 -->|<- C3 ->|<- C4 ->|<- C5 ->|<- C6 ->|MAT|MF|MT|SEQ|
  (11 chr)   (11 chr)  (11 chr) (11 chr) (11 chr) (11 chr) (4) (2)(3) (5)
```

While every line follows this 80-character format, the information itself is organized into standardized record layouts. These include header/control records (used for labeling sections and providing flags or counters), text records (for human-readable descriptions), and structured data records (for lists and tables) [41].

The most common use of these records is to store tabulated values, such as a physics quantity (y) as a function of incident energy (x). These tables store (x, y) point pairs along with specific metadata that defines the interpolation law for one or more regions [41]. By encoding these laws alongside the data, ENDF-6 ensures that the interpretation of values between tabulated points is entirely unambiguous.

4.2.3. Elastic-Scattering Angular Distributions

Angular distributions of secondary particles describe the direction in which reaction products are emitted. In the ENDF-6 format, this data is primarily stored in `MF4` for nominal values and `MF34` for covariances. They characterize the anisotropy of the scattering process. As the elastic-scattering distributions play a central role in the analysis presented in this thesis, the description below focuses on them.

Representations in ENDF-6

The angular distribution is typically described by a normalized probability density function (PDF), $f(\mu, E)$, where $\mu = \cos\theta$ is the cosine of the scattering angle and E is the incident energy (see Figure I.6). The ENDF-6 format allows for several representations to balance precision with storage compactness:

1. **Legendre Polynomial Coefficients:** The distribution is represented as a truncated series of Legendre polynomials. This is the historical standard due to its compactness.
2. **Tabulated Distributions:** The PDF $f(\mu, E)$ is given explicitly as a table of μ vs. probability. This is preferred when distributions are highly anisotropic (e.g., highly forward-peaked scattering at high energies).
3. **Isotropic:** A simplified case where emission is uniform in all directions.
4. **Mixed Representation:** An evaluation may switch between representations depending on the energy range (e.g., Legendre at low energies and Tabulated at high energies).

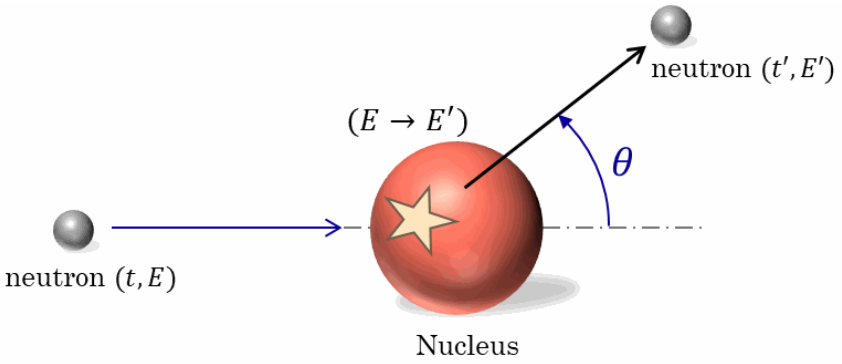


Figure I.6: Scattering geometry in the laboratory (LAB) frame. An incident neutron scatters from a target initially at rest, emerging at the laboratory angle θ relative to the beam direction.

While Legendre coefficients are efficient for storage, they suffer from limitations when representing sharp peaks. To accurately describe a highly forward-peaked distribution, a very high order of expansion ($l > 30$) may be required. Furthermore, truncating the Legendre series can introduce unphysical "wiggles" or even negative probability values in certain angular ranges (Gibbs phenomenon) [42]. Despite these limitations, the Legendre representation is the one adopted throughout this thesis: the ^{56}Fe elastic angular distribution evaluated in Chapter III, and its covariance, are expressed in this basis.

Mathematical Formalism: Legendre Polynomials

To understand the storage of these distributions, one must define the basis functions used. The angular dependence is expanded using Legendre polynomials, $P_l(\mu)$, which form a complete orthogonal set over the interval $\mu \in [-1, 1]$.

Legendre Expansion of the Angular Distribution

In ENDF-6, the normalized angular distribution $f(\mu, E)$ is expanded in Legendre polynomials as:

$$f(\mu, E) = \frac{1}{2} \sum_{l=0}^{L} (2l + 1)\, a_l(E)\, P_l(\mu) \tag{I.38}$$

where l is the order of the polynomial, $a_l(E)$ are the Legendre coefficients stored in the evaluation file and $P_l(\mu)$ are the Legendre polynomials shown in Figure I.7.

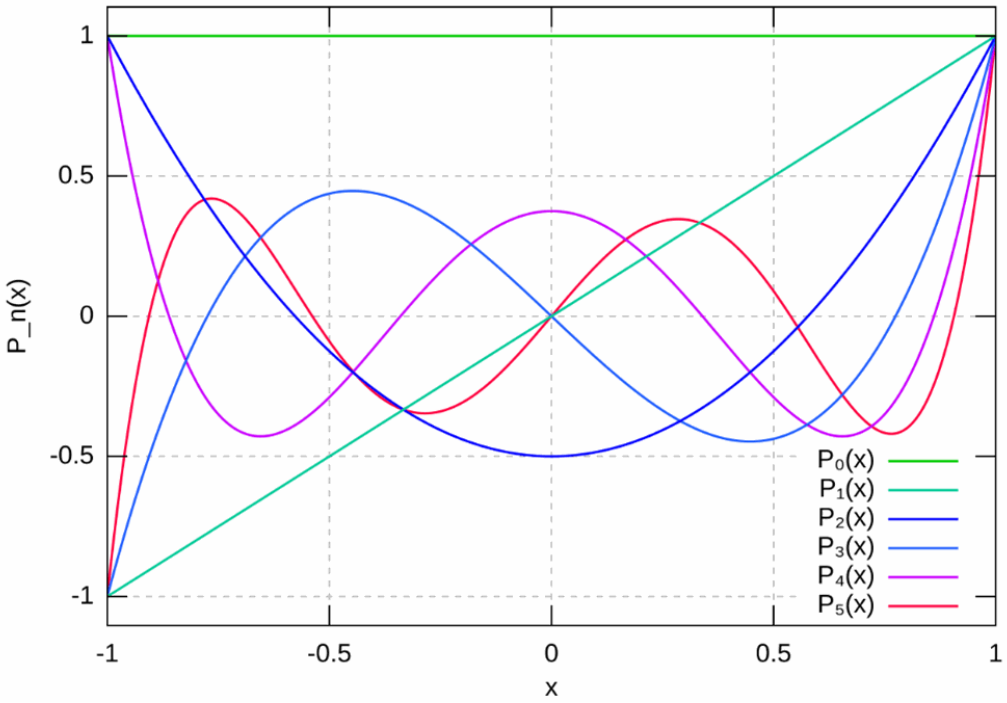


Figure I.7: The first six Legendre polynomials $P_n(x)$ in the interval $[-1, 1]$, where x here corresponds to the scattering cosine μ. All orders are normalized such that $P_n(1) = 1$. The zeroth order P_0 (green) represents an isotropic distribution, while higher orders introduce increasing angular complexity.

This formulation effectively treats the angular shape as a linear combination of different moments. The coefficient a_l determines the "weight" or contribution of the l-th order shape to the total distribution. By definition of the normalization ($\int f(\mu) d\mu = 1$), the zeroth coefficient is always unity ($a_0 = 1$), corresponding to the isotropic component.

Of particular physical significance is the first-order coefficient, a_1. Due to the orthogonality of the polynomials, a_1 corresponds to the mean scattering cosine, $\bar{\mu}$, which quantifies the average

forward momentum retention of the neutron:

$$\bar{\mu}(E) = \int_{-1}^{1} \mu f(\mu, E) d\mu = a_1(E) \tag{I.39}$$

In diffusion theory and transport approximations, $\bar{\mu}$ is the primary parameter used to correct for anisotropy.

Connection to Experimental Data

Evaluated files store the shape of the distribution, whereas experimental data (e.g., from EXFOR) typically provide the Differential Cross Section, denoted as $d\sigma/d\Omega$. To compare an evaluation with experimental data, one must combine the angular shape from `MF4` with the integrated cross section from `MF3`:

$$\frac{d\sigma}{d\Omega}(\mu, E) = \frac{\sigma(E)}{2\pi} f(\mu, E) = \frac{\sigma(E)}{4\pi} \sum_{l=0}^{L} (2l+1)\ a_l(E) P_l(\mu) \tag{I.40}$$

The factor of 4π ensures that integrating the differential cross section over the full solid angle returns the total cross section $\sigma(E)$ of the considered reaction.

Comparing these evaluated shapes against experimental constraints is a non-trivial task due to resolution effects, normalizations, and coordinate system differences. A deeper investigation into this subject is presented in Chapter III.

Frame of Reference

A critical distinction in nuclear data is the frame of reference. Angular distributions can be defined in either the Center of Mass (CM) System, which simplifies the interaction physics and is widely used in evaluations and Monte Carlo transport codes, or the Laboratory (LAB) System, where measurements are taken.

For elastic scattering, the relationship between the cosine of the scattering angle in the CM system (μ_{CM}) and the LAB system (μ_L) is governed by the target-to-projectile mass ratio $A = M/m$, where M is the target nucleus mass and m is the neutron mass:

$$\mu_L = \frac{1 + A\mu_{CM}}{\sqrt{A^2 + 2A\mu_{CM} + 1}} \tag{I.41}$$

While ENDF-6 allows evaluations to be stored in either frame (indicated by the `LCT` flag), experimental data in EXFOR is almost exclusively in the LAB system. Therefore, transformations using the equation above are needed when comparing evaluations with experimental benchmarks.

4.3. ACE Files

Transport codes typically do not read ENDF-6 evaluations directly. For continuous-energy Monte Carlo solvers, a widely used application format is ACE ("A Compact ENDF") [43], originally developed in the MCNP ecosystem and since adopted by several other Monte Carlo codes such as Serpent [44], PHITS [45] or MORET [46].

ACE files are produced from evaluated nuclear data libraries by processing codes like NJOY [11], which reconstruct resonance cross sections, Doppler-broaden to the desired temperature, compute heating numbers, convert the evaluated angular distributions into the pointwise sampling laws stored in the file, and generate probability tables for the unresolved resonance range. Because Doppler broadening is usually performed at processing time, a separate ACE file is typically produced for each nuclide–temperature combination. Some codes can instead account for the target temperature on the fly during transport [44].

The format is designed to be compact and efficient for runtime use: cross sections are stored on unionised energy grids with fast look-up, and secondary-particle emission is described through angle-energy sampling laws that the transport solver can sample directly. A companion directory file (xsdir in MCNP) maps each nuclide, temperature, and evaluation to its corresponding ACE file. A consequence of this pre-processing step is that ACE is a one-way format: covariance information, resonance parameters, and other evaluation metadata present in ENDF-6 are not carried into the ACE representation. Uncertainty and sensitivity studies therefore require either reprocessing from the evaluated files or dedicated perturbation techniques at the transport level.

4.4. Nuclear Data Processing

Raw evaluated nuclear data files are rarely suitable for direct implementation in reactor physics simulations. Nuclear data processing serves as the essential bridge between these evaluations and application-specific libraries. This intermediate step involves mathematically transforming the data: reconstructing continuous cross sections from resonance parameters, Doppler-broadening to reactor operating temperatures, generating thermal scattering laws, and formatting the output for specific transport solvers. These tasks are performed by specialized nuclear data processing codes.

The **NJOY** system [11], originally developed at Los Alamos National Laboratory (LANL), represents a global standard in this domain. NJOY processes evaluated files to produce derived datasets tailored to both stochastic and deterministic codes. For continuous-energy Monte Carlo solvers, NJOY generates pointwise data that preserves the detailed energy dependence of cross sections. Conversely, for deterministic transport codes, it collapses pointwise data into multigroup averages weighted by a representative flux spectrum.

A typical processing pipeline using NJOY to generate a continuous-energy library and corresponding multigroup data—as illustrated in Figure I.8—involves the following sequence of modules [11]:

- **MODER**: Converts between ASCII and binary ENDF/PENDF formats. Every pipeline begins with `MODER` to convert the input ENDF tape into binary form for efficient internal processing.
- **RECONR**: Reconstructs resonance cross sections from evaluated resonance parameters (MF=2) and linearizes them with tabulated background cross sections. This yields a pointwise representation, $\sigma(E)$, on a unified, fine energy grid typically containing tens of thousands of points.
- **BROADR**: Performs Doppler broadening of the pointwise cross sections to account for the thermal motion of target nuclei. This step effectively smooths resonance peaks based on the target material temperature.

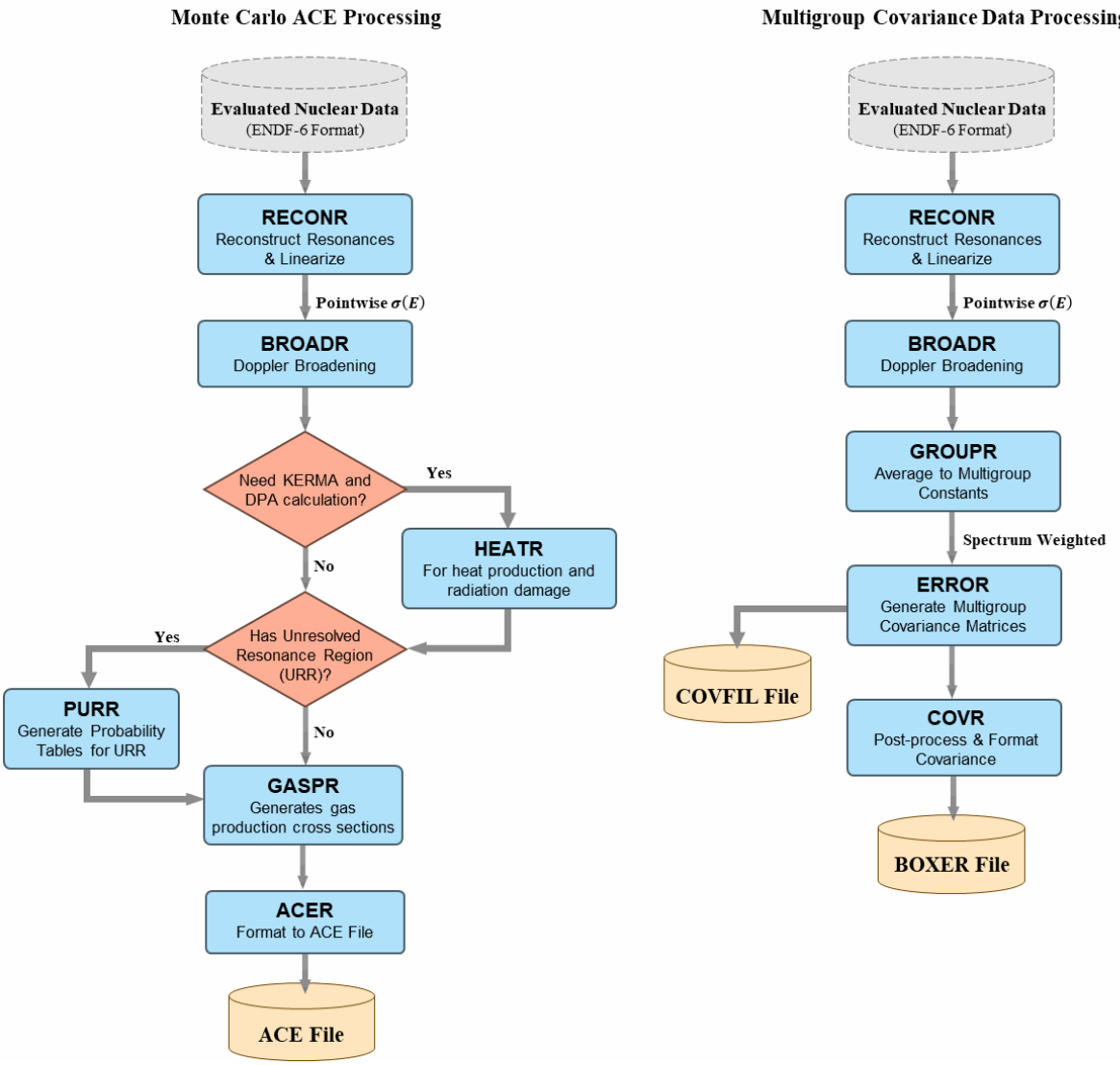


Figure I.8: Schematic overview of typical NJOY processing sequences: the generation of continuous-energy ACE libraries for Monte Carlo codes (left) and the production of multigroup covariance matrices (right).

- **HEATR**: Computes pointwise heating (KERMA) factors and radiation damage-energy production (dpa) cross sections from reaction kinematics.
- **PURR**: Generates probability tables for the unresolved resonance region (URR) that allow Monte Carlo codes to stochastically sample self-shielded cross sections, properly accounting for resonance self-shielding effects.
- **GASPR**: Adds gas production cross sections (hydrogen and helium isotopes) summed from the reactions that emit these light charged particles, necessary for material damage and activation analyses.
- **ACER**: Formats the processed data into ACE files. This module consolidates the outputs from prior steps—including Doppler-broadened cross sections, probability tables, and heating factors—into the standard format read by many continuous-energy Monte Carlo codes.
- **GROUPR**: Generates multigroup self-shielded cross sections. `GROUPR` averages the pointwise data over broad energy groups using a user-specified weighting flux. This produces group constants, $\bar{\sigma}_g$, essential for deterministic diffusion or transport calculations.
- **ERRORR**: Processes covariance data (MF=31–35) to quantify nuclear data uncertainties. It produces multigroup covariance matrices tailored to the same energy structure used in `GROUPR`, enabling sensitivity and uncertainty analysis in downstream codes.
- **COVR**: Post-processes the multigroup covariance data generated by `ERRORR`. Its primary function is to export these covariance matrices into standard user formats such as BOXER format, which are sometimes required for external codes or processing.

The workflow shown in Figure I.8 represents a common processing scheme, but NJOY provides additional specialized modules for other applications. For instance, `THERMR` generates thermal scattering cross sections and angular distributions using TSL laws, essential for deterministic transport codes and, together with `LEAPR`, for producing thermal scattering libraries for moderator materials (e.g. H in H_2O, graphite). Similarly, `UNRESR` calculates effective self-shielded cross sections via the Bondarenko method for deterministic multigroup applications.

Although NJOY is the most common tool internationally, other processing systems are also widely used. These alternatives are essential for independent verification of results and are often developed to support specific national data formats or reactor physics suites. Table I.8 highlights the most widely used nuclear data processing codes.

The continuous development of these codes ensures that the high-fidelity physics captured in evaluated nuclear data is accurately preserved when transmitted to the solvers used for reactor design and safety evaluation.

4.5. Covariance Data

In evaluated nuclear data, covariance data quantify the uncertainties of tabulated quantities, such as multigroup cross sections, and how these uncertainties co-vary across energy points, reactions, and isotopes. If a set of evaluated quantities is collected into a vector $\mathbf{x} = \{x_1, x_2, \ldots, x_N\}$, their covariance matrix $\mathbf{C}$ is defined element-wise as:

$$C_{ij} = \mathrm{Cov}(x_i, x_j) = \mathbb{E}[(x_i - \bar{x}_i)(x_j - \bar{x}_j)] \tag{I.42}$$

Table I.8: Overview of main nuclear data processing codes.

Code	Developer	Primary Use & Characteristics	Ref.
NJOY	USA (LANL)	The international reference code for processing ENDF-formatted data into continuous-energy and multigroup libraries.	[11]
AMPX	USA (ORNL)	Tightly integrated with the SCALE system; generates covariance data via `PUFF-IV` and creates MG/CE libraries.	[47]
FUDGE	USA (LLNL)	A Python-based tool designed to handle the hierarchical GNDS data structure.	[48]
GAIA	France (ASNR)	A nuclear data processing software developed at ASNR for nuclear-safety-related data processing and assessment needs.	[49]
GALILÉE-1	France (CEA)	New generation code developed in synergy with the CEA reactor-physics suite, producing libraries for the deterministic solver APOLLO3 and the Monte Carlo solver TRIPOLI-4.	[50]
FRENDY	Japan (JAEA)	Developed with modern software engineering practices; handles both ENDF-6 and GNDS formats.	[51]
GRUCON	Russia (KI)	Package used for preparing libraries for Russian reactor physics codes.	[52]
PREPRO	IAEA	A collection of standalone modules for ENDF data processing, including linearization, resonance reconstruction, Doppler broadening, and multigroup averaging.	[53]

The diagonal elements represent individual variances ($C_{ii} = \sigma_i^2$), while the off-diagonal terms encode the degree to which two quantities vary together. To remain physically meaningful for uncertainty propagation, $\mathbf{C}$ must be symmetric and positive semi-definite.

Representing uncertainty through a covariance matrix alone implicitly assumes that the underlying distribution is approximately Gaussian and that downstream propagation can be linearized about the mean; both assumptions are formalized and revisited in Chapter II (Sec. 3).

4.5.1. Data Representations

Covariance data is typically reported in one of two forms, depending on the application:

- **Absolute Covariance:** Defined as C_{ij}, this form retains the physical units of the product of the two variables (e.g., barns squared for cross sections).
- **Relative Covariance:** Commonly used in nuclear data evaluations, this expresses covariance relative to the mean values:

$$C_{ij}^{(\mathrm{rel})} \approx \frac{C_{ij}}{x_i\, x_j} \tag{I.43}$$

On the diagonal, the relative covariance simplifies to the square of the relative standard deviation: $C_{ii}^{(\mathrm{rel})} = (\sigma_i/x_i)^2$. One can easily toggle between these representations using the relationship:

$$C_{ij} = x_i\, x_j\, C_{ij}^{(\mathrm{rel})} \tag{I.44}$$

4.5.2. The Correlation Matrix

While covariance magnitude depends on scaling, correlation is dimensionless and isolates the strength of the linear dependence. The Pearson correlation coefficient is defined as:

$$\rho_{ij} = \frac{C_{ij}}{\sigma_i\, \sigma_j}, \qquad -1 \le \rho_{ij} \le 1 \tag{I.45}$$

where $\sigma_i = \sqrt{C_{ii}}$. An important property is that ρ_{ij} is invariant under positive rescaling; whether calculated from absolute or relative covariances, the coefficients remain identical.

This normalization makes correlation matrices convenient for visualization. In Figure I.9, the diagonal ($\rho_{ii} = 1$) represents perfect self-correlation, coherent regions close to ± 1 indicate strong linear coupling between energy bins, and values near zero indicate weak coupling.

For discretized data, the indices i and j run over energy bins, reactions, and isotopes, so the matrix acquires a natural block structure. The figure above shows the simplest case, the energy–energy block of a single reaction; concatenating several quantities produces the broader pattern:

- **Within-block (energy–energy):** correlations within a single quantity. Smooth positive correlations across neighboring groups often reflect shared model parameters or normalization constraints.
- **Off-diagonal blocks (reaction–reaction):** correlations driven by shared experimental data, common systematic uncertainties, or channel competition (e.g., elastic vs. inelastic).
- **Inter-isotope blocks:** correlations arising when evaluations share experimental datasets or modeling assumptions across different isotopes.

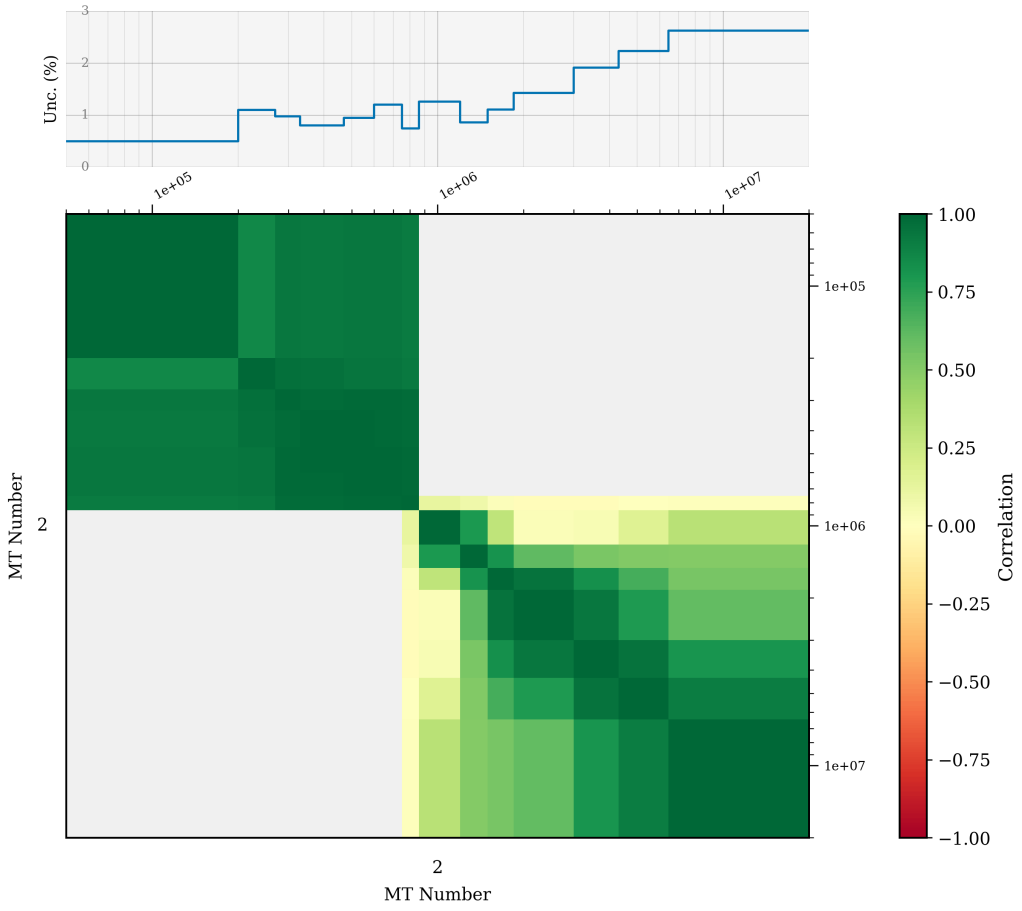


Figure I.9: Correlation matrix and associated relative uncertainties for the ^{56}Fe elastic scattering cross section (`MT2`) in JENDL-5. The top panel displays the percentage uncertainty as a function of incident energy, while the main heatmap illustrates the correlation coefficients between energy bins. Grey cells mark energy ranges where the evaluation provides no covariance data.

4.5.3. Angular-Distribution Covariances (MF34)

Angular-distribution covariances (`MF34`) extend the block structure above with an additional dimension: the Legendre order ℓ. In `MF33`, the matrix describes correlations between energy groups for a single reaction. For angular distributions, the uncertain quantity at each energy is not a scalar but a set of coefficients $a_\ell(E)$, and the parameter vector is a concatenation of energy-dependent vectors for each order:

$$\mathbf{x} = \Big[\underbrace{a_1(E_1)\dots a_1(E_N)}_{\ell=1},\quad \underbrace{a_2(E_1)\dots a_2(E_N)}_{\ell=2},\quad \dots\Big] \tag{I.46}$$

The full covariance matrix $\mathbf{C}_{\mathrm{MF34}}$ therefore has a block-symmetric structure, partitioned by Legendre order. For orders up to L, it is composed of $L \times L$ sub-matrices:

$$\mathbf{C}_{\mathrm{MF34}} = \begin{pmatrix} \mathbf{V}_{1,1} & \mathbf{V}_{1,2} & \cdots & \mathbf{V}_{1,L} \\ \mathbf{V}_{2,1} & \mathbf{V}_{2,2} & \cdots & \mathbf{V}_{2,L} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{V}_{L,1} & \mathbf{V}_{L,2} & \cdots & \mathbf{V}_{L,L} \end{pmatrix} \tag{I.47}$$

Each sub-block $\mathbf{V}_{\ell,k}$ is $G \times G$ (where G is the number of energy groups), with elements:

$$(\mathbf{V}_{\ell,k})_{i,j} = \mathrm{cov}\left(a_\ell(E_i), a_k(E_j)\right) \tag{I.48}$$

The physical interpretation of the blocks is:

- **Diagonal blocks** ($\mathbf{V}_{\ell,\ell}$): energy-energy correlations for a single Legendre order. For example, $\mathbf{V}_{1,1}$ describes how the uncertainty in a_1 is correlated across the energy range.
- **Off-diagonal blocks** ($\mathbf{V}_{\ell,k}$, $\ell \neq k$): correlations between different Legendre orders. Non-zero values indicate that a perturbation in a_ℓ is statistically linked to a change in a_k.

A special case is the zeroth-order coefficient a_0. Since $a_0(E) \equiv 1$ by normalization, its variance is zero. The ENDF-6 format reuses the $\ell = 0$ index to store a different quantity: the correlation between the absolute cross section and the angular shape parameters:

$$(\mathbf{V}_{0,\ell})_{i,j} = \mathrm{cov}\left(\sigma(E_i), a_\ell(E_j)\right) \tag{I.49}$$

Non-zero values in $\mathbf{V}_{0,\ell}$ capture the coupling between the cross-section magnitude and the angular shape. In practice, most evaluated libraries leave the $\ell = 0$ slot empty for a given reaction, so this σ–shape coupling is typically absent from propagation.

4.5.4. Covariance Data Formats

Multigroup covariance data is stored in several formats depending on the processing toolchain:

- **COVFIL:** An intermediate output generated by NJOY's `ERRORR` module [54].
- **BOXER:** An efficient library format used by NJOY's `COVR` module [54].
- **COVERX:** A format standard in the SCALE/AMPX ecosystem [12].

4.5.5. Uncertainty Propagation

A primary application of nuclear covariance data is propagating input uncertainties to a calculated response $y = f(\mathbf{x})$. For a single response, linearized propagation is expressed as:

$$\mathrm{Var}(y) \approx S\mathbf{C}S^{\mathsf{T}} \tag{I.50}$$

where S is the sensitivity vector. For multiple responses, S is generalized to the Jacobian matrix $\mathbf{J}$, yielding a response covariance matrix $\mathbf{C}_y \approx \mathbf{JCJ}^{\mathsf{T}}$. A comprehensive derivation of these concepts is provided in Chapter II.

5. VACS: Vessel Aging Calculation Scheme

The aging of a reactor pressure vessel (RPV) under fast neutron irradiation must be predicted over the entire operating life of the reactor, while the underlying physics changes on much shorter time scales. Within a single PWR fuel cycle the core composition evolves with burnup, control rods are repositioned, the soluble boron concentration is gradually reduced, and moderator density and temperature respond to the local power distribution. From cycle to cycle, the loading pattern is modified and the assemblies that face the vessel are not the same. No coupled neutronics–thermohydraulics–burnup Monte Carlo code can currently perform such a simulation across the full vessel lifetime within practical computing resources. The Vessel Aging Calculation Scheme (VACS) was developed at ASNR during the doctoral thesis of R. Vuiart [2] to address this problem by decoupling the calculation into two steps with very different fidelity requirements.

VACS combines a deterministic, low-cost evaluation of the fission neutron source term in the core with a high-fidelity Monte Carlo transport of those neutrons through the reactor internals to the vessel wall and the surveillance capsules. The source term is computed by the CASMO5/SIMULATE5 chain (Sec. 3.4); the attenuation is computed by MCNP6 (Sec. 3.1) with variance reduction parameters produced by ADVANTG (Sec. 3.2). The two steps exchange information through a per-pin, per-energy fission neutron source that is averaged over a fuel cycle.

VACS belongs to the same family of two-step deterministic–Monte Carlo schemes adopted internationally for vessel fluence calculations. The EDF reference methodology couples an `APOLLO2`/`COCCINELLE` deterministic core calculation with a TRIPOLI-4 Monte Carlo attenuation step [4]. A recent study at EPFL in Switzerland [3] applies the same two-step structure to a 3-loop reactor, coupling a SCALE/Polaris/PARCS core-follow with a Serpent Monte Carlo shielding step. The JSI (Jožef Stefan Institute) [22] adopts an MCNP-to-MCNP variant for the Krško NPP, coupling an MCNP core source term to an MCNP ex-core attenuation calculation accelerated by ADVANTG. Compared to a single-step Monte Carlo criticality calculation that scores the vessel response directly, VACS is faster and easier to deploy, at the cost of approximations on the source term. Compared to a fully deterministic decoupled approach, it retains the high-fidelity attenuation calculation needed in the deep-penetration region between the core and the vessel.

The methodology has been validated on the H.B. Robinson-2 vessel dosimetry benchmark [55, 2], with dosimeter activities and fast neutron flux predictions falling within the experimental uncertainty. The uncertainty budget for VACS and the comparison with other published studies are presented in Chapter IV.

5.1. Displacement Damage Cross Sections

The fast neutron fluence $\phi_{E>1\,\mathrm{MeV}}$ is the historical surrogate quantity used in vessel surveillance programs and is the input variable of the empirical embrittlement correlations adopted in France [2]. However, the underlying physical damage is more directly described by the number of atomic displacements per atom (dpa), which folds the entire neutron spectrum with a displacement cross section $\sigma_d(E)$. The VACS calculation reports both quantities, and Chapter IV presents results for the fast neutron fluence and for the displacement damage under both the NRT and ARC formulations.

Two formulations of σ_d are considered in this work. The first, the Norgett–Robinson–Torrens model (NRT-dpa) [56], is the long-standing international reference and gives the number of displaced atoms per primary knock-on atom (PKA) as:

$$N_d^{\mathrm{NRT}}(T_d) = \begin{cases} 0, & T_d < E_d \\ 1, & E_d \le T_d < 2E_d/0.8 \\ \dfrac{0.8\,T_d}{2E_d}, & T_d \ge 2E_d/0.8 \end{cases} \tag{I.51}$$

where E_d is the displacement threshold energy and T_d is the damage energy available to the PKA after subtracting the energy lost to electronic excitations. The factor 0.8 is the displacement efficiency, which corrects the hard-sphere Kinchin–Pease estimate for realistic scattering between the recoiling atoms [56]. NRT-dpa is known to overestimate the number of stable defects in metals such as iron by a factor of approximately three, because it ignores the athermal recombination of Frenkel pairs during the heat-spike phase of the cascade [57].

The athermal-recombination-corrected model (ARC-dpa) introduced by Nordlund *et al.* [57] multiplies the NRT count by a material-dependent efficiency function ξ_{arc}:

$$N_d^{\mathrm{ARC}}(T_d) = N_d^{\mathrm{NRT}}(T_d)\cdot\xi_{\mathrm{arc}}(T_d), \qquad \xi_{\mathrm{arc}}(T_d) = (1-c_{\mathrm{arc}})\left(\frac{T_d}{2E_d/0.8}\right)^{b_{\mathrm{arc}}} + c_{\mathrm{arc}} \tag{I.52}$$

The factor ξ_{arc} never exceeds one: it equals one for low-energy events and decreases toward $c_{\mathrm{arc}} \approx 0.286$ for the most energetic ones, so for iron the ARC model counts roughly a third of the displacements predicted by NRT. The constants E_d, b_{arc} and c_{arc} are fitted to atomistic simulations. For iron, the dominant constituent of the 16MND5 vessel steel, the recommended values are $E_d = 40$ eV, $b_{\mathrm{arc}} = -0.568 \pm 0.020$, $c_{\mathrm{arc}} = 0.286 \pm 0.005$ [57].

In practice, $\sigma_d(E)$ is generated by NJOY's HEATR module from an evaluated nuclear data library and then folded with the incident neutron flux to obtain the dpa rate R_{dpa}. Both formulations are available in NJOY2016. For iron under a fast neutron spectrum, ARC-dpa predictions are typically a factor of ~ 3 lower than NRT-dpa, reflecting the post-cascade athermal recombination ignored by the NRT model [57]. The estimates given in Chapter IV are reported with both NRT-dpa and ARC-dpa cross sections. Both NRT and ARC reduce the damage to a single displacement-per-atom value derived from an iron-based displacement cross section; more complete descriptors of primary radiation damage that go beyond this metric have been proposed [58, 59, 60].

5.2. Step 1: Fission Neutron Source Term

Returning to the two-step structure outlined above, the first step builds the fission neutron source distribution that drives the attenuation calculation. It determines the spatial and spectral

distribution of fission neutrons across the reactor core using the deterministic chain described in Sec. 3.4. CASMO5 computes homogenized macroscopic cross sections for each fuel assembly type. These are passed through CMSLINK5 to SIMULATE5, which solves the three-dimensional diffusion equation for the full core and produces the fission neutron production rate νR_f for each assembly and axial node.

The production rate at the pin level is then reconstructed using form factors X_p derived from the pin-power reconstruction in SIMULATE5. The fission neutron source at pin p in assembly m is written as:

$$S_p(E', t) = \sum_i R_{f,i,m}(t) \cdot \nu_i \cdot X_p(t) \cdot \chi_m(E', t) \tag{I.53}$$

where, for isotope i, $R_{f,i,m}$ is the space- and energy-integrated fission rate in assembly m, ν_i is the spectrum-averaged number of neutrons per fission, X_p is the pin form factor, normalized to unit average over the pins of the assembly so that it redistributes the assembly source among pins without changing its total, and $\chi_m(E')$ is the assembly-averaged fission spectrum, normalized to unity in energy ($\int \chi_m(E')\,\mathrm{d}E' = 1$). The spectrum χ_m is constructed as a production-weighted average over the dominant fissile isotopes (^{235}U, ^{238}U, ^{239}Pu, ^{241}Pu), which together account for more than 95% of fission neutrons in both UOX and MOX fuel [2]. The isotope spectra are analytic, a Watt form for ^{235}U, ^{238}U, and ^{239}Pu and a Maxwellian for ^{241}Pu, and VACS writes the mixed spectrum of each assembly on the source cards of the MCNP input. Mixing at the assembly level reproduces a reference eigenvalue calculation of the vessel flux within its statistical uncertainty [61]. SIMULATE5 evaluates the source on 20 axial nodes per assembly. The source can be passed to Step 2 either as a snapshot at a given burnup state (instantaneous source) or as a trapezoidal average over the burnup steps of the cycle (cycle-averaged source); Chapter IV uses the cycle-averaged form.

Three properties of this step are worth emphasizing because they carry most of the assumptions of VACS:

1. The deterministic chain includes a simplified thermohydraulic feedback that updates the moderator density and the fuel temperature self-consistently with the local power.
2. The burnup-dependent isotopic composition of every assembly is updated automatically at each step by solving the Bateman equations for 16 actinides and roughly 40 fission products and absorbers [2].
3. The critical boron concentration is adjusted by SIMULATE5 to maintain criticality at each burnup step.

As a consequence, the only state variable that is approximated in time is the power history of the reactor, which is replaced by constant operation at the cycle-averaged power.

5.3. Step 2: Neutron Transport to the Vessel

The cycle-averaged source term from Step 1 is used as a fixed source in MCNP6 to transport neutrons from the core through the baffle, the moderator bypass, the core shroud, the neutron pad, and the downcomer into the RPV wall. The MCNP model represents the major reactor internals, the surveillance capsules, and the vessel itself, with explicit moderator density gradients derived from the SIMULATE5 thermohydraulic solution. The simulation estimates the fast neutron flux ($E > 1$ MeV) and the dpa rate (using both NRT-dpa and ARC-dpa cross sections,

Sec. 5.1) at multiple azimuthal and axial positions on the vessel inner surface and inside the surveillance capsules. Variance reduction is applied through ADVANTG. VACS performs a single attenuation calculation per cycle, driven by the cycle-averaged source term; this is justified by the weak sensitivity of the attenuation kernel to intra-cycle variations of the reactor state.

6. Summary

This chapter has assembled the background on which the rest of the thesis builds. The Boltzmann transport equation was introduced as the governing balance for neutron transport, together with the deterministic, Monte Carlo, and hybrid methods used to solve it. Variance reduction was then developed in detail, from the diagnostics that define a converged Monte Carlo result to the adjoint-based importance maps behind the CADIS and FW-CADIS schemes. The computational tools used throughout the thesis were presented, namely MCNP, ADVANTG, Serpent, and the CASMO5/SIMULATE5 chain, followed by the nuclear data foundations: the evaluation cycle, the major evaluated libraries, the ENDF-6 format, and the covariance information that underpins uncertainty quantification. The VACS scheme then tied these elements together into the two-step vessel-fluence calculation used as the application case of this work.

The next chapter develops the sensitivity-analysis and uncertainty-quantification framework that carries these nuclear data uncertainties through to transport observables.

Chapter II

Sensitivity Analysis and Uncertainty Quantification

Contents

1 Introduction and Scope 50
1.1 Definitions 51
1.2 Classification of Uncertainties 51
1.3 Responses of Interest 52
1.4 Input Parameters 53
2 Sensitivity Analysis Framework 53
2.1 Local Sensitivity Analysis 53
2.1.1 Direct Perturbation 54
2.1.2 Perturbation Theory 55
2.1.3 Generalized Perturbation Theory (GPT) 58
2.1.4 Differential Operator Sampling (DOS) 65
2.1.5 Correlated Sampling (CS) 70
2.1.6 Scope of the local sensitivity framework 72
3 Uncertainty Quantification Framework 72
3.1 First-Order Second-Moment Propagation (FOSM) 72
3.1.1 Response Linearization and the Sandwich Rule 73
3.1.2 Relative Form and Connection to Sensitivity 73
3.1.3 Applicability and Limitations 74
3.2 Monte Carlo Sampling (MCS) for Uncertainty Quantification . 74
3.2.1 Sampling Techniques 74
3.2.2 Matrix Decomposition and Correlated Sampling . . . 75
3.2.3 Decoupling Parametric and Statistical Uncertainties . 77
4 The KIKA Computational Framework 81
4.1 Motivation 81
4.2 Architecture Overview 82
4.3 Supported Formats and External Codes 82

4.4 Multigroup Collapse of Angular Distribution Covariances . . . 83
4.5 Nuclear Data Perturbation Pipelines . . . 84
4.5.1 Cross Section Perturbation . . . 84
4.5.2 Angular Distribution Perturbation . . . 84
4.5.3 Combined Perturbation . . . 86
5 Methodology Verification: The PWR Sphere Toy Model . . . 87
5.1 Model Description . . . 87
5.2 Sensitivity Analysis and FOSM Propagation . . . 90
5.2.1 Sensitivity Profiles . . . 90
5.2.2 First-Order Applicability . . . 91
5.2.3 Energy-Integrated Sensitivity . . . 92
5.2.4 Uncertainty Propagation . . . 94
5.3 Monte Carlo Sampling Propagation . . . 94
5.3.1 Sampling and Decomposition . . . 94
5.3.2 Simulation Sets and Response Distribution . . . 95
5.3.3 Convergence and Efficiency . . . 97
5.4 Comparison of Methods and Discussion . . . 100
5.5 Angular Distribution Sensitivity and Uncertainty . . . 101
5.5.1 Sensitivity . . . 101
5.5.2 Uncertainty Quantification . . . 102
6 Conclusions . . . 103

This chapter develops the sensitivity analysis (SA) and uncertainty quantification (UQ) framework used throughout the thesis to propagate nuclear data uncertainties to neutron transport observables. The propagation methods themselves (sensitivity coefficients, the sandwich rule, and Monte Carlo sampling) are general and apply to any input–response pair; this thesis specializes them to nuclear data and reaction-rate responses. The chapter begins with definitions and scope (Sec. 1), then presents the local sensitivity analysis framework and its implementation in transport codes (Sec. 2). The uncertainty quantification methods, including the sandwich rule and Monte Carlo sampling, are developed in Sec. 3. Section 4 describes the `kika-nd` computational framework built during this thesis to implement the full SA&UQ chain. Finally, Sec. 5 applies the complete methodology to a simplified benchmark, comparing codes, methods, and propagation strategies for both cross section and angular distribution uncertainties.

1. Introduction and Scope

Any prediction from a computational model carries uncertainty, conventionally split into a statistical part that reflects random variability or finite sampling and a systematic part that reflects imperfect knowledge of the model and its inputs. This section introduces these terms in the setting of nuclear data and neutron transport and establishes the scope of the SA&UQ analysis developed in this chapter.

1.1. Definitions

Sensitivity Analysis (SA)

Sensitivity Analysis is the study of how the variation in a model response R can be attributed to variations in the input parameters $\mathbf{x}$ [5]. Within this work, a distinction is made between two primary approaches:

- **Local Sensitivity Analysis (LSA):** Evaluates the impact of small perturbations around a nominal point $\mathbf{x}_0$, typically represented by the gradient or first-order derivatives $\frac{\partial R}{\partial \mathbf{x}}$.
- **Global Sensitivity Analysis (GSA):** Examines the influence of inputs over their entire feasible range, accounting for interactions between parameters and non-linearities across the full parameter space.

Uncertainty Quantification (UQ)

Uncertainty Quantification (UQ) is the process of characterizing the reliability of a model response R by mapping input uncertainties (such as nuclear cross sections) to the final result [5]. In this work, this mapping is categorized into:

- **Linear Propagation:** An analytic approach using sensitivity coefficients and covariance matrices ("Sandwich Rule").
- **Sampling-based Methods:** Statistical approaches, such as Monte Carlo Sampling (MCS), that propagate uncertainties through repeated simulations.

While SA identifies which parameters most influence the system behavior, UQ determines the magnitude of the resulting uncertainty in the response. In this work, we emphasize LSA both as the bridge to Linear Uncertainty Propagation and because the sensitivity coefficients themselves yield a per-parameter decomposition of the total uncertainty, exposing the dominant contributors and their relative impact. This local treatment is well suited to the predominantly linear regime of the problems considered here.

1.2. Classification of Uncertainties

The international metrology guidance [62] provides the standard framework for expressing measurement uncertainty, classifying contributions by their method of evaluation into Type A, from the statistical analysis of repeated observations, and Type B, from other information such as reference data or expert judgment. The contributions below are instead grouped by their physical source. In a general computational analysis, the total uncertainty on a response combines three conceptually different contributions: a statistical component from finite sampling, a systematic component (also called bias or model error) introduced by the solver and the modeling assumptions, and a parametric component from imperfect knowledge of the input data. When these contributions are independent they combine in quadrature into the total uncertainty on the response. Monte Carlo neutron transport is one of the few cases in which the solver itself introduces no systematic error beyond statistical noise, having no discretization or model error to speak of. This is not the general situation: deterministic neutron transport and diffu-

sion solvers, as well as thermal-hydraulics and mechanics codes, carry a numerical model error that can dominate the budget. This work therefore retains only the statistical and parametric components:

- **Statistical Uncertainty:** Arises from the stochastic nature of the Monte Carlo random walk. This decreases as $1/\sqrt{N}$ with the number of simulated histories and can be reduced by using variance reduction techniques (see Chapter I Sec. 2).
- **Parametric Uncertainty:** Arises from imperfect but quantifiable knowledge of the input parameters $\mathbf{x}$ that describe the modeled system, such as its geometry, the material compositions, and the nuclear data. Each of these can be assigned a known uncertainty, for example fabrication tolerances on dimensions and densities, or the evaluated covariances on cross sections. This uncertainty is intrinsic to the physical inputs and does not decrease with more simulation histories. The nuclear data parameters are the component analyzed in this work.

These two categories must be carefully distinguished: the reported variance should reflect a genuine lack of physical knowledge, not a limitation of the simulation's convergence. Both are quantifiable, and both differ from epistemic uncertainty, which arises from what the model does not represent at all. Typical examples are components left out of the geometry, or design details that are simply not known, such as the internal layout of a surveillance capsule. Whereas parametric uncertainty describes inputs that are known well enough to be assigned a distribution, epistemic uncertainty reflects missing knowledge that cannot be assigned a covariance, and it is therefore not part of the parametric budget quantified here from evaluated nuclear data. In practice, the statistical and parametric components interact during sampling-based uncertainty propagation, where parametric variability must be estimated in the presence of per-run Monte Carlo noise [63].

1.3. Responses of Interest

In neutron transport simulations, the primary quantity of interest is the particle flux, obtained as the solution of the steady-state Boltzmann transport equation (see Chapter I, Sec. 1.1). However, most experimentally observable quantities are not the flux itself but response functionals derived from it. A general linear response $R(\mathbf{x})$ (e.g., a detector signal) is expressed as a functional of the flux:

$$R(\mathbf{x}) = \langle h(\mathbf{x}), \psi(\mathbf{x}) \rangle \equiv \int_V \int_{4\pi} \int_0^\infty h(\mathbf{r}, \mathbf{\Omega}, E; \mathbf{x})\, \psi(\mathbf{r}, \mathbf{\Omega}, E; \mathbf{x})\, dE\, d\mathbf{\Omega}\, d\mathbf{r} \tag{II.1}$$

where h is the response kernel characterizing the physical observable. Specifically, for a reaction of type r involving nuclide n in a detector region V_d, with the macroscopic cross section Σ, the reaction rate is:

$$R_{n,r}(\mathbf{x}) = \int_{V_d} \int_0^\infty \Sigma_{n,r}(\mathbf{r}, E; \mathbf{x})\, \phi(\mathbf{r}, E; \mathbf{x})\, dE\, d\mathbf{r} \tag{II.2}$$

This formulation reflects the operating principle of neutron detectors such as activation foils or fission chambers, which measure integrated reaction rates rather than the flux itself.

1.4. Input Parameters

The input vector $\mathbf{x} = (x_1, x_2, \ldots, x_N)^T$ represents the physical input data defining the system. In this thesis, we distinguish between:

1. **Nuclear Data Parameters:** These are the primary sources of parametric uncertainty analyzed here. They include parameters such as microscopic cross sections (capture, fission, scattering), angular distributions, fission spectra (χ), and average number of neutrons per fission ($\bar{\nu}$). These parameters are energy-dependent and isotope-specific.
2. **System Parameters:** These can be grouped into material uncertainties (density and isotopic composition), geometric uncertainties (dimensional tolerances and component positioning), and operating-condition uncertainties such as temperature. While the methodology presented is general, these parameters are treated as fixed nominal values in this work unless otherwise specified.

2. Sensitivity Analysis Framework

This section develops the local sensitivity analysis methods used to relate perturbations in nuclear data inputs to changes in transport observables. The mathematical foundations are presented first, followed by their implementation in the Monte Carlo transport codes used in this work: Serpent and MCNP.

2.1. Local Sensitivity Analysis

To quantify the relationship between input parameters and the system response, it is necessary to evaluate how perturbations in the input vector $\mathbf{x}$ (primarily nuclear data such as cross sections) propagate to change the calculated reaction rate R. Following the foundational treatments of perturbation theory and sensitivity analysis in reactor physics [64, 65, 66, 67], the most fundamental measure is the absolute sensitivity coefficient, defined as the partial derivative of the response with respect to an input parameter x_i, evaluated at the nominal parameter value $\mathbf{x}_0$:

$$S_{x_i}^{R,(abs)} = \left.\frac{\partial R}{\partial x_i}\right|_{\mathbf{x}_0} \tag{II.3}$$

However, absolute sensitivities can be difficult to compare across different parameters due to varying units and magnitudes. Therefore, it is standard practice in reactor physics to use the relative sensitivity coefficient, $S_{x_i}^R$, which is a dimensionless quantity defined as [64, 66]:

$$S_{x_i}^R = \frac{x_i}{R}\frac{\partial R}{\partial x_i} = \frac{\partial \ln R}{\partial \ln x_i} \tag{II.4}$$

This relative coefficient represents the percentage change in the response R induced by a 1% change in the input parameter x_i. The magnitude of S_i serves as a ranking tool for parameter importance: a high magnitude means the response is strongly driven by that specific input. The sign indicates the nature of the physical relationship: a positive sign implies a direct correlation, such as increasing a detector's own cross section to increase its reaction rate, while a negative sign

indicates an inverse relationship, often seen when an increase in a competing neutron absorber "steals" neutrons from the reaction of interest. For example, $S_\sigma = 1.0$ indicates that a 5% increase in the cross section σ leads to a 5% increase in the reaction rate, whereas $S_\sigma = -1.0$ would result in a 5% decrease.

The above definitions apply to any scalar response of interest. For instance, when the response is the fast-neutron flux ϕ at the reactor pressure vessel, a quantity that drives radiation damage to structural steels, the relative sensitivity:

$$S_{x_i}^{\phi} = \frac{x_i}{\phi}\frac{\partial \phi}{\partial x_i} \tag{II.5}$$

indicates the fractional change in vessel fluence per fractional change in nuclear data x_i. A positive S_i identifies cross sections whose increase raises the predicted fluence, for example forward elastic scattering on ^{56}Fe in the MeV range, which redirects neutrons toward the vessel; a negative S_i identifies absorbers or back-scatterers that reduce it. Similarly, if the response is a ratio of two quantities A and B (for example, a reaction rate ratio $R = A/B$), the sensitivity can be derived from the logarithmic definition: $\ln R = \ln A - \ln B$. A small change in input x_i will cause A and B to change, and to first order the sensitivity of a ratio is the difference between the sensitivities of the numerator and denominator [68]:

$$S_{x_i}^{R} = S_{x_i}^{A} - S_{x_i}^{B} \tag{II.6}$$

This property is useful for complex responses like detector ratios, as it allows building ratio sensitivities from constituent parts.

In the specific context of nuclear engineering, these parameters typically correspond to multi-group cross sections for specific reactions and isotopes. Because cross sections are continuous functions of energy, sensitivity is often presented as an energy-dependent sensitivity profile, $S(E)$. These profiles are useful for identifying the energy ranges most critical to a specific response. In a thermal system, for instance, the sensitivity of k_{eff} to ^{235}U fission will likely peak in the thermal energy region, while its sensitivity to ^{238}U capture may be concentrated at specific resonance energies.

These local sensitivities are first-order derivatives and are strictly valid for small perturbations where the system response remains approximately linear. If a parameter changes by a very large amount, higher-order effects (nonlinearities, feedbacks, threshold reactions, parameter interactions) may come into play and the true change in R will deviate from that predicted by S_{x_i} For nuclear data perturbations of typical magnitude, however, the linearity assumption is often valid, and sandwich-rule propagation gives uncertainties consistent with those obtained from sampling-based methods [69]. Despite this formal limitation, sensitivities remain the primary tool for prioritizing nuclear data improvements and provide the weighting factors for combining input covariance data into a final uncertainty estimate for the response.

2.1.1. Direct Perturbation

The sensitivity coefficients defined in Eq. (II.3) and (II.4) rely on the partial derivatives of the system response with respect to input parameters. In principle, these can be estimated through a direct calculation. For this process, one could increase a specific cross section σ by a fixed percentage (e.g., 10%), run a transport simulation to find the perturbed response R', and then compare it to the nominal response R.

$$S_{x_i}^R = \frac{x_i}{R}\frac{\partial R}{\partial x_i} \approx \frac{x_i}{R}\frac{R(x_i + \Delta x_i) - R(x_i)}{\Delta x_i} \tag{II.7}$$

To improve numerical robustness, a central difference approximation can be used. This requires three separate simulations: the nominal case (R_0) and one perturbation in each direction. Comparing the positive and negative perturbations also provides a direct test of the linearity assumption: a measurable asymmetry about R_0 signals the presence of higher-order effects. The relative sensitivity is then expressed as:

$$S_{x_i}^R \approx \frac{x_i}{R_0}\left(\frac{R(x_i + \Delta x_i) - R(x_i - \Delta x_i)}{2\Delta x_i}\right) \tag{II.8}$$

While this is manageable for energy-integrated sensitivities, it becomes computationally prohibitive for energy-dependent sensitivity profiles. To obtain an energy profile, each cross section must be perturbed independently within each energy bin g. For a library with G energy groups, the number of required simulations is:

- **Forward difference:** $G + 1$ simulations.
- **Central difference:** $2G + 1$ simulations.

Given that a typical reactor analysis involves several isotopes and reaction types, the direct perturbation method is not feasible for large-scale nuclear data studies.

2.1.2. Perturbation Theory

Perturbation theory provides the mathematical framework to bypass the computational limitation of direct perturbations. By making use of the properties of the adjoint function (introduced in Chapter I, Sec. 2.3), it is possible to determine the sensitivity of a response to every input parameter across all energy groups simultaneously, using only two calculations: one for the forward flux and one for the adjoint flux, which is the importance function of the response of interest [5, 6, 12].

Fixed-Source Problems

Consider the steady-state Boltzmann transport equation for a fixed source problem, as previously defined in Eq. (I.3):

$$\mathcal{H}\psi(z) = q(z), \tag{II.9}$$

where $\mathcal{H}$ is the forward transport operator acting on $\psi(z)$, $q(z)$ is the fixed external source and $z = (\mathbf{r}, \mathbf{\Omega}, E)$ is the phase space. We introduce a small perturbation δx_i to a system parameter, resulting in perturbed operators and sources:

$$\mathcal{H} = \mathcal{H}_0 + \delta\mathcal{H}, \qquad q = q_0 + \delta q, \qquad \psi = \psi_0 + \delta\psi, \qquad h = h_0 + \delta h \tag{II.10}$$

Neglecting second-order terms, substituting into Eq. (II.9) and subtracting the nominal state ($\mathcal{H}_0\psi_0 = q_0$), gives the first-order flux perturbation equation:

$$\mathcal{H}_0\delta\psi = \delta q - \delta\mathcal{H}\psi_0 \tag{II.11}$$

The response of interest is defined in Eq. (II.1) as a functional $R = \langle h, \psi \rangle$, where h is the response kernel and $\langle \cdot, \cdot \rangle$ denotes the inner product over the phase space. The first-order response variation δR is:

$$\delta R = \underbrace{\langle \delta h, \psi_0 \rangle}_{\text{Direct term}} + \underbrace{\langle h_0, \delta \psi \rangle}_{\text{Flux (indirect) term}} \tag{II.12}$$

A direct evaluation of $\langle h_0, \delta\psi \rangle$ for each parameter would require a separate transport solve for every perturbation. To avoid the explicit calculation of $\delta\psi$ for every parameter variation, we introduce the adjoint flux $\psi_0^\dagger$ associated with the response kernel h_0, defined as the solution of [64]:

$$\mathcal{H}_0^\dagger \psi_0^\dagger = h_0 \tag{II.13}$$

Here the response kernel h_0 acts as the adjoint source, so $\psi_0^\dagger$ depends on the response of interest and a different adjoint solution is required for each detector.

Using the adjoint identity (Eq. (I.32)), $\langle \mathcal{H}_0^\dagger \psi_0^\dagger, \delta\psi \rangle = \langle \psi_0^\dagger, \mathcal{H}_0 \delta\psi \rangle$, the flux term becomes:

$$\langle h_0, \delta\psi \rangle = \langle \psi_0^\dagger, \mathcal{H}_0 \delta\psi \rangle = \left\langle \psi_0^\dagger, \delta q - \delta\mathcal{H}\, \psi_0 \right\rangle \tag{II.14}$$

where Eq. (II.11) was used in the last step. Substituting into Eq. (II.12) yields the standard first-order adjoint-weighted perturbation expression [70]:

$$\delta R = \underbrace{\langle \delta h, \psi_0 \rangle}_{\text{Direct term}} + \underbrace{\langle \psi_0^\dagger, \delta q \rangle}_{\text{Source term}} - \underbrace{\langle \psi_0^\dagger, \delta\mathcal{H}\, \psi_0 \rangle}_{\text{Operator term}} \tag{II.15}$$

This form enables the evaluation of δR for many perturbations using only the nominal forward solution ψ_0 and one adjoint solution $\psi_0^\dagger$ per detector [12, 70]. For nuclear-data perturbations with an unchanged external source, $\delta q = 0$ and only the direct and operator terms contribute.

Fixed-Source Sensitivity Coefficient

The relative sensitivity $S_{x_i}^R$ for a parameter x_i in a fixed-source system is:

$$S_{x_i}^R = \frac{x_{i,0}}{R_0} \left[\left\langle \left. \frac{\partial h}{\partial x_i} \right|_0, \psi_0 \right\rangle + \left\langle \psi_0^\dagger, \left. \frac{\partial q}{\partial x_i} \right|_0 \right\rangle - \left\langle \psi_0^\dagger, \left. \frac{\partial \mathcal{H}}{\partial x_i} \right|_0 \psi_0 \right\rangle \right] \tag{II.16}$$

This expression is the starting point for adjoint-based sensitivity methods and for the more systematic developments grouped under Generalized Perturbation Theory (GPT) [70].

Criticality Problems

In the absence of an external source ($q = 0$), a steady-state neutron population can only be sustained if the system is exactly critical. To treat the general case, the effective multiplication factor k is introduced as an eigenvalue that scales the fission production term. This leads to the k-eigenvalue equation [6, 17]:

$$\mathcal{H}\psi = \frac{1}{k} \mathcal{F}\psi \tag{II.17}$$

where $k \equiv k_{\text{eff}}$ represents the ratio of neutrons produced in one generation to those lost in the previous one. The corresponding adjoint eigenvalue problem is [17]:

$$\mathcal{H}^{\dagger}\psi^{\dagger} = \frac{1}{k}\,\mathcal{F}^{\dagger}\psi^{\dagger} \tag{II.18}$$

Because these equations are homogeneous, the amplitudes of ψ and $\psi^{\dagger}$ are mathematically arbitrary. For perturbation expressions, one typically imposes a bilinear normalization [12]:

$$\langle \psi_0^{\dagger}, \mathcal{F}_0\psi_0 \rangle = 1 \tag{II.19}$$

Under a small perturbation of an input parameter x_i, the operators change as $\mathcal{H} \to \mathcal{H}_0 + \delta\mathcal{H}$ and $\mathcal{F} \to \mathcal{F}_0 + \delta\mathcal{F}$, producing a change δk in the eigenvalue. Neglecting second-order terms, first-order perturbation theory yields [6, 71]:

$$\frac{\delta k}{k} = \frac{\left\langle \psi_0^{\dagger}, \left(\frac{1}{k}\delta\mathcal{F} - \delta\mathcal{H}\right)\psi_0 \right\rangle}{\left\langle \psi_0^{\dagger}, \frac{1}{k}\mathcal{F}_0\psi_0 \right\rangle} \tag{II.20}$$

Using the normalization condition in Eq. (II.19), the denominator reduces to $1/k$, so that Eq. (II.20) becomes:

$$\frac{\delta k}{k} = \left\langle \psi_0^{\dagger}, \delta\mathcal{F}\,\psi_0 \right\rangle - k\left\langle \psi_0^{\dagger}, \delta\mathcal{H}\,\psi_0 \right\rangle. \tag{II.21}$$

Criticality (Eigenvalue) Sensitivity Coefficient

The relative sensitivity of the multiplication factor k to a parameter x_i is:

$$S_{x_i}^{k} \equiv \frac{x_{i,0}}{k}\left.\frac{\partial k}{\partial x_i}\right|_0 = x_{i,0}\left[\left\langle \psi_0^{\dagger}, \left.\frac{\partial\mathcal{F}}{\partial x_i}\right|_0 \psi_0 \right\rangle - k\left\langle \psi_0^{\dagger}, \left.\frac{\partial\mathcal{H}}{\partial x_i}\right|_0 \psi_0 \right\rangle\right] \tag{II.22}$$

This formulation shows that the sensitivity of k_{eff} to nuclear-data perturbations can be obtained from the nominal forward and adjoint eigenfunctions, without explicitly computing $\delta\psi$ for each parameter [12, 71].

Implementation in Transport Codes

Equations (II.15)–(II.16) provide a general sensitivity identity, but transport codes evaluate them through different numerical strategies. Broadly, two approaches are used in practice.

- **Adjoint-Based Evaluation:** This approach is the most direct application of first-order perturbation theory and serves as the foundation for Generalized Perturbation Theory (GPT). It requires two independent transport solutions: the nominal forward distribution and a corresponding adjoint distribution, where the adjoint source is defined by the response kernel [5]. Deterministic solvers typically implement this by performing numerical integration over the discretized spatial cells, angular moments, and energy groups using the stored forward and adjoint flux fields.
- **Sampling-Based Evaluation:** Monte Carlo codes often estimate sensitivities without solving an explicit adjoint equation. Instead, they use statistical estimators to correlate

perturbed and unperturbed particle histories. These methods effectively sample the sensitivity information during the forward random walk. As discussed in the following sections, this is typically achieved through Differential Operator Sampling (DOS) [13, 72], which evaluates the derivatives of the response, or Correlated Sampling (CS) [13], which reuses random walk histories to compare multiple system states with reduced statistical variance.

In summary, first-order perturbation theory defines the mathematical foundation of sensitivity analysis, while the transport solver governs how it is carried out in practice. The sections that follow focus on the methods most relevant to the present work, covering the GPT framework and the specialized sampling techniques of DOS and CS.

2.1.3. Generalized Perturbation Theory (GPT)

Standard first-order perturbation theory, as derived in Sec. 2.1.2, applies directly when the response R is a linear functional of the forward flux, such as a reaction rate in a fixed-source problem (Eq. (II.1)), and admits a standard extension to eigenvalue responses such as k_{eff} (Eq. (II.17)). However, many parameters of interest in reactor physics are nonlinear functionals of the flux, most commonly ratios of linear functionals such as spectral indices, or bilinear forms involving both forward and adjoint modes such as kinetic parameters [5, 66]. GPT extends the adjoint framework to these complex responses while maintaining the first-order consistency established in Eq. (II.15) [66].

Generalized Perturbation Theory Formulation

As in the fixed-source development, consider the nominal forward state ψ_0 satisfying:

$$\mathcal{H}_0\,\psi_0 = q_0 \tag{II.23}$$

and define a generic response as a functional:

$$\mathcal{R} \equiv \mathcal{G}(\psi; x) \tag{II.24}$$

The response $\mathcal{R}$ depends explicitly on the parameter x and implicitly through the state $\psi(x)$. Linearizing $\mathcal{R}$ about the nominal values (x_0, ψ_0) and applying the chain rule gives:

$$\delta\mathcal{R} = \underbrace{\left\langle \left(\frac{\partial \mathcal{G}}{\partial x}\right)_0, \delta x \right\rangle_E}_{\text{Direct effect}} + \underbrace{\left\langle \left(\frac{\partial \mathcal{G}}{\partial \psi}\right)_0, \delta\psi \right\rangle}_{\text{Indirect effect}} \tag{II.25}$$

where $\langle \cdot, \cdot \rangle$ denotes the phase-space inner product introduced in Eq. (II.1), and $\langle \cdot, \cdot \rangle_E$ denotes an integral (or sum if multigroup) over the energy-dependent parameter.

The indirect term involves $\delta\psi$, which is eliminated using the same reciprocity argument used in the derivation of Eq. (II.15). Specifically, define the generalized adjoint $\psi^\dagger_{\mathcal{R}}$ as the solution of:

$$\mathcal{H}_0^\dagger\,\psi^\dagger_{\mathcal{R}} = \left(\frac{\partial \mathcal{G}}{\partial \psi}\right)_0 \tag{II.26}$$

Then, using the adjoint identity in Eq. (I.32) together with the forward perturbation relation (Eq. (II.11)), the indirect term becomes:

$$\left\langle \left(\frac{\partial \mathcal{G}}{\partial \psi}\right)_0, \delta\psi \right\rangle = \left\langle \mathcal{H}_0^\dagger \psi^\dagger_{\mathcal{R}}, \delta\psi \right\rangle = \left\langle \psi^\dagger_{\mathcal{R}}, \mathcal{H}_0 \delta\psi \right\rangle = \left\langle \psi^\dagger_{\mathcal{R}}, \delta q - \delta\mathcal{H}\,\psi_0 \right\rangle \tag{II.27}$$

Substituting this into Eq. (II.25) yields the GPT analogue of the standard perturbation identity:

$$\delta\mathcal{R} = \left\langle \left(\frac{\partial \mathcal{G}}{\partial x}\right)_0, \delta x \right\rangle_E + \left\langle \psi^\dagger_{\mathcal{R}}, \delta q \right\rangle - \left\langle \psi^\dagger_{\mathcal{R}}, \delta\mathcal{H}\,\psi_0 \right\rangle \tag{II.28}$$

This identity serves as the foundation for GPT; it allows the response variation to be evaluated for any number of perturbations using only the nominal forward solution ψ_0 and a single, response-specific generalized-adjoint solution $\psi^\dagger_{\mathcal{R}}$ [5, 66].

Generalized Sensitivity Coefficient (GPT)

For a general functional $\mathcal{R}$, the first-order relative sensitivity is:

$$S^{\mathcal{R}}_{x_i} = \frac{x_{i,0}}{\mathcal{R}_0}\left[\left\langle \frac{\partial \mathcal{G}}{\partial x_i}, 1 \right\rangle_E + \left\langle \psi^\dagger_{\mathcal{R}}, \frac{\partial q}{\partial x_i} \right\rangle - \left\langle \psi^\dagger_{\mathcal{R}}, \frac{\partial \mathcal{H}}{\partial x_i}\psi_0 \right\rangle\right] \tag{II.29}$$

Eq. (II.29) gives the first-order relative sensitivity of a general response functional $\mathcal{R} = \mathcal{G}(\psi; x)$ in terms of the nominal forward solution ψ_0 and a single response-dependent generalized adjoint $\psi^\dagger_{\mathcal{R}}$.

Sensitivity to Scattering Angular Distributions

The perturbation framework described above applies to more than just reaction probabilities like cross sections or $\bar{\nu}$. It also covers scattering laws, which define the probability density for sampling outgoing directions and energy-angle pairs [71]. For elastic scattering, we represent this as an angular probability density function (PDF) for the scattering cosine μ, conditioned on the incident energy E:

$$f_x(\mu \mid E), \qquad \int_{-1}^{1} f_x(\mu \mid E)\, d\mu = 1 \tag{II.30}$$

where μ is typically defined in the center-of-mass frame and x identifies the specific nuclide–reaction–material (and, if relevant, a spatial bin) whose scattering distribution is being perturbed.

As described in Chapter I, Sec. 4.2.3, angular scattering distributions are commonly parameterized using a Legendre expansion. Adopting that representation here for the perturbed PDF,

$$f_x(\mu \mid E) = \sum_{\ell=0}^{L} \frac{2\ell+1}{2} P_\ell(\mu)\; a_{x,\ell}(E) \tag{II.31}$$

the expansion coefficients are the Legendre moments of the PDF,

$$a_{x,\ell}(E) = \int_{-1}^{1} P_\ell(\mu)\, f_x(\mu \mid E)\, d\mu, \tag{II.32}$$

with $a_{x,0}(E) = 1$, $a_{x,1}(E) = \bar{\mu}(E)$, and higher moments encoding progressively finer angular-shape information.

To quantify how a response $\mathcal{R}$ depends on the Legendre coefficients, we define the energy-dependent relative sensitivity of $\mathcal{R}$ to $a_{x,\ell}(E)$ as:

$$S^{\mathcal{R}}_{a_{x,\ell}}(E) \;\equiv\; \frac{d\mathcal{R}/\mathcal{R}}{da_{x,\ell}(E)/a_{x,\ell}(E)} \;=\; \frac{a_{x,\ell}(E)}{\mathcal{R}}\,\frac{d\mathcal{R}}{da_{x,\ell}(E)} \tag{II.33}$$

Since $a_{x,\ell}(E)$ enters $\mathcal{R}$ only through the angular PDF $f_x(\mu \mid E)$, the chain rule for functionals expresses the total response variation as an integral over angle:

$$\frac{\delta \mathcal{R}}{\mathcal{R}} = \int_{-1}^{1} S^{\mathcal{R}}_{f_x}(\mu, E)\, \frac{\delta f_x(\mu \mid E)}{f_x(\mu \mid E)}\, d\mu \tag{II.34}$$

where $S^{\mathcal{R}}_{f_x}(\mu, E)$ is the relative sensitivity of $\mathcal{R}$ to a localized perturbation of the angular PDF at (μ, E). It weighs how strongly trajectories that sample the cosine μ at incident energy E contribute to the response, and plays the role of an angular adjoint function. Unlike the Legendre coefficients themselves, $S^{\mathcal{R}}_{f_x}$ is a transport-dependent quantity and is not available a priori from the nuclear data alone [71].

For a perturbation that affects only one coefficient $a_{x,\ell}(E)$ at a fixed energy, the induced first-order change in the PDF follows directly from Eq. (II.31):

$$\delta f_x(\mu \mid E) = \frac{\partial f_x(\mu \mid E)}{\partial a_{x,\ell}(E)}\, \delta a_{x,\ell}(E) = \frac{2\ell + 1}{2}\, P_\ell(\mu)\, \delta a_{x,\ell}(E) \tag{II.35}$$

Substituting into Eq. (II.34) and normalizing by the fractional change $\delta a_{x,\ell}/a_{x,\ell}$ yields the moment-level sensitivity:

Legendre Moment Sensitivity Coefficient

The relative sensitivity of a response $\mathcal{R}$ to the Legendre moment $a_{x,\ell}(E)$ is given by:

$$S^{\mathcal{R}}_{a_{x,\ell}}(E) = \int_{-1}^{1} S^{\mathcal{R}}_{f_x}(\mu, E) \left[\frac{2\ell + 1}{2}\, P_\ell(\mu)\, \frac{a_{x,\ell}(E)}{f_x(\mu \mid E)} \right] d\mu \tag{II.36}$$

Two different quantities appear under the integral. The bracketed factor depends only on the nuclear data (P_ℓ, $a_{x,\ell}$, f_x) and can be evaluated directly from nuclear data at any sampled cosine μ and incident energy E. The angular adjoint $S^{\mathcal{R}}_{f_x}$, by contrast, contains how trajectories at (μ, E) contribute to the response and depends on the full transport solution. Rather than constructing $S^{\mathcal{R}}_{f_x}$ explicitly, Serpent estimates the entire integral by accumulating the bracketed weight along the sampled scattering events of each particle history, as described in the next section [71].

Serpent Collision-History GPT-Equivalent Estimators

Equation (II.28) evaluates GPT sensitivities by explicitly solving the response-dependent generalized adjoint problem in Eq. (II.26). In continuous-energy Monte Carlo, however, storing and solving for $\psi^\dagger_{\mathcal{R}}$ over the full phase space is generally impractical. Serpent therefore implements a first-order GPT-equivalent approach based on collision histories: instead of constructing $\psi^\dagger_{\mathcal{R}}$, it estimates the same first-order sensitivity coefficients through correlations between detector scores and a bookkeeping variable that summarizes the occurrence of perturbed events along particle genealogies [71, 73].

To simplify the presentation, consider a linear response $R_j = \langle h_j, \psi \rangle$ and assume an unchanged source ($\delta q = 0$). Under these assumptions, the general first-order perturbation result (Eq. (II.15)) collapses to the sum of a kernel (direct) term and a flux (indirect) term; dividing

by R_j yields the usual direct–indirect relative sensitivity:

$$S_x^{R_j} = \underbrace{\frac{\left\langle \frac{\partial h_j}{\partial x}, \psi \right\rangle}{\langle h_j, \psi \rangle}}_{\text{Direct term}} + \underbrace{\frac{\left\langle h_j, \frac{\partial \psi}{\partial x} \right\rangle}{\langle h_j, \psi \rangle}}_{\text{Indirect term}} \tag{II.37}$$

The direct term is non-zero only when the response kernel h_j explicitly depends on the perturbed parameter x. However, it is not generally a binary contribution. For a relative sensitivity coefficient, it is more generally written as:

$$S_{x,\text{dir}}^{R_j} = \frac{\left\langle x \frac{\partial h_j}{\partial x}, \psi \right\rangle}{\langle h_j, \psi \rangle}. \tag{II.38}$$

This term is equal to unity only in the special case where the complete response kernel is proportional to the perturbed parameter over the whole scoring domain, for example a ^{56}Fe elastic reaction-rate tally with respect to a uniform perturbation of the ^{56}Fe elastic cross section. In more general cases, such as group-wise perturbations or composite responses like dpa, the direct term represents the fractional contribution of the perturbed component to the total response.

The indirect term is where Serpent's collision-history approach becomes essential. The method is based on an accept/reject oversampling scheme. For each event type associated with a perturbed parameter x (e.g. a specific reaction–nuclide–energy–region bin), the sampling frequency of that event is artificially increased by a factor f (often $f = 2$) [71]. When an oversampled event is selected during tracking, it is accepted with probability $1/f$ (a physical, analog collision) and rejected with probability $1 - 1/f$ (a virtual event that leaves the particle state unchanged). This construction preserves unbiased transport while avoiding any modification of the particle weight used for tracking and tallying.

For each tracked particle n, Serpent maintains a net collision-history score:

$$C_x^{(n)} \equiv \sum_{g=a-k}^{a} \left(N_{\text{ACC},x}^{(n,g)} - N_{\text{REJ},x}^{(n,g)} \right) \tag{II.39}$$

where $N_{\text{ACC},x}^{(n,g)}$ and $N_{\text{REJ},x}^{(n,g)}$ are the numbers of accepted and rejected x-events found in the ancestry of particle n over the stored collision history [73]. Here, a is the current generation index, while k controls how many ancestor generations are retained in the collision-history bookkeeping. In criticality calculations, this retained history is used to approximate the asymptotic importance associated with the converged fission source. In fixed-source calculations, the relevant history is instead the particle genealogy generated from the prescribed external source, in principle back to the initial source particle. Thus, in fixed-source mode, k should be interpreted primarily as a truncation parameter for the stored collision genealogy rather than as a source-convergence parameter.

The indirect contribution is then estimated by correlating each detector score with the instantaneous collision-history score at the time of scoring. If T_j denotes an individual score contribution to detector j, Serpent accumulates the buffers:

$$B_j \equiv \sum_{\text{scores}} T_j, \qquad B_{j,x} \equiv \sum_{\text{scores}} T_j \, C_x^{(n)} \tag{II.40}$$

and forms the collision-history estimate:

$$S^{R_j}_{x,ind} = \frac{\left\langle h_j,\, \frac{\partial \psi}{\partial x} \right\rangle}{\langle h_j, \psi \rangle} \approx \frac{B_{j,x}}{B_j} \tag{II.41}$$

In practice, for detector-based responses (linear or bilinear), Serpent's GPT interface is formulated in terms of ratios, as is the GPT capability of TRIPOLI-4 [74]:

$$\mathcal{R} = \frac{\langle h_1, \psi \rangle}{\langle h_2, \psi \rangle} \equiv \frac{R_1}{R_2} \tag{II.42}$$

In this case, Eq. (II.37)–(II.41) apply separately to the numerator and denominator responses ($j \in \{1, 2\}$). Using the ratio property in Eq. (II.6), the sensitivity of $\mathcal{R}$ can be written as the difference of two response sensitivities, which in Serpent corresponds to maintaining two sets of buffers:

$$S^{\mathcal{R}}_x = \underbrace{\frac{\left\langle \frac{\partial h_1}{\partial x}, \psi \right\rangle}{\langle h_1, \psi \rangle} - \frac{\left\langle \frac{\partial h_2}{\partial x}, \psi \right\rangle}{\langle h_2, \psi \rangle}}_{\text{Direct}} + \underbrace{\frac{\left\langle h_1, \frac{\partial \psi}{\partial x} \right\rangle}{\langle h_1, \psi \rangle} - \frac{\left\langle h_2, \frac{\partial \psi}{\partial x} \right\rangle}{\langle h_2, \psi \rangle}}_{\text{Indirect}} \tag{II.43}$$

If one wishes to obtain the sensitivity of a single reaction-rate response (i.e. without forming a ratio), this can be done within the ratio interface by placing the denominator detector in the outside cell of the geometry, where it never scores. Both its direct and indirect terms then vanish, and $S^{\mathcal{R}}_x$ reduces to the sensitivity of the numerator response.

For the eigenvalue response, Serpent does not rely on the detector-ratio interface: k_{eff} sensitivities are enabled as a basic response, and the associated adjoint weighting is obtained through Iterated Fission Probability together with the latent-generation collision history [75]. In other words, Serpent treats k_{eff} as its own response class (rather than a detector ratio), while using the same collision-history principle to avoid an explicit deterministic adjoint solve.

Legendre Moment Estimators in Serpent

Applying the collision-history estimator to the Legendre sensitivity identity (Eq. (II.36)) requires determining the value added to the particle's history at each collision. Inspection of the term in brackets in Eq. (II.36) reveals the quantity that must be accumulated whenever a scattering cosine μ is sampled at incident energy E:

$$G_{x,\ell}(\mu, E) \;\equiv\; \frac{2\ell + 1}{2}\, P_\ell(\mu)\, \frac{a_{x,\ell}(E)}{f_x(\mu \mid E)} \tag{II.44}$$

Intuitively:

- $P_\ell(\mu)$ selects the angular shape associated with the ℓ-th mode;
- $a_{x,\ell}(E)$ sets the magnitude of that mode in the current nuclear data;
- dividing by $f_x(\mu \mid E)$ converts an absolute PDF change into a relative (log-like) change, which is what appears in Eq. (II.34).

A key practical point is that $f_x(\mu \mid E)$ and $a_{x,\ell}(E)$ must be evaluated consistently with the sampling law. For tabulated angular distributions, Serpent evaluates $f_x(\mu \mid E)$ using the same

interpolation rules used during sampling. To reduce computational cost, the Legendre moments are precomputed at each incident-energy grid point E_i by numerically evaluating Eq. (II.32), and $a_{x,\ell}(E)$ at an arbitrary E is then obtained by interpolating between the surrounding grid points with the same scheme used for the tabular law [71]. The per-event score $G_{x,\ell}(\mu_g, E_g)$ itself is then evaluated on the fly at each scattering event from the sampled cosine μ_g and the interpolated nuclear-data quantities; only $a_{x,\ell}$ is tabulated in advance.

The collision-history GPT estimator for Legendre moments uses the same set of recorded events introduced for cross section perturbations (Eq. (II.39)). The only modification is in the history variable: instead of accept/reject counts, $C^{(n)}_{x,\ell}$ accumulates the continuous per-event score $G_{x,\ell}$ evaluated at the cosine and energy sampled at each scattering event:

$$C^{(n)}_{x,\ell} = \sum_{g=a-k}^{a} G_{x,\ell}(\mu_g, E_g) \tag{II.45}$$

Reusing the detector buffers introduced earlier, we define the perturbation buffer:

$$B_{j,x,\ell} \equiv \sum_{\text{scores}} T_j \, C^{(n)}_{x,\ell} \tag{II.46}$$

The corresponding relative sensitivity estimate for a linear detector response is then:

$$S^{R_j}_{a_{x,\ell}} \approx \frac{B_{j,x,\ell}}{B_j} \tag{II.47}$$

Finally, for a ratio response $\mathcal{R} = R_1/R_2$, the difference relation applies:

$$S^{\mathcal{R}}_{a_{x,\ell}} \approx \left(\frac{B_{1,x,\ell}}{B_1}\right) - \left(\frac{B_{2,x,\ell}}{B_2}\right) \tag{II.48}$$

Overall, the practical message is that Legendre-moment sensitivities are obtained with the same collision-history correlation used for cross sections, but replacing "event counts" by an event-wise score $G_{x,\ell}$ computed from the sampled cosine and the underlying nuclear-data PDF. This yields a fully continuous estimator in angle, without introducing explicit angular binning.

Example with Spectral Index Sensitivity in Serpent

To demonstrate the application of the collision-history method, consider the sensitivity analysis of a spectral index, defined as the ratio of two reaction rates: $\mathcal{R} = R_1/R_2$. In this example, the response of interest is the ratio of ^{238}U(n,f) (R_1) to ^{235}U(n,f) (R_2) in a specific detector region. The sensitivity is calculated with respect to the ^{10}B elastic scattering cross section (x) in an adjacent control cell.

Assume three source histories contribute to the detectors as pictured in Figure II.1: Tracks 1 and 2 score in the ^{235}U(n,f) detector (R_2), while Track 3 scores in the ^{238}U(n,f) detector (R_1). Track 1 sees a single rejected ^{10}B trial, Track 2 sees one rejected trial followed by two accepted scatters, and Track 3 passes through the control cell without interacting. Table II.1 lists each history's score contributions (T_1, T_2), the accept/reject counts for x, the resulting $C^{(n)}_x$, and the products needed for $B_{j,x}$.

Compute the buffers for R_1 (^{238}U fission):

$$B_1 = 0.000 + 0.000 + 0.040 = 0.040$$
$$B_{1,x} = 0.000 + 0.000 + 0.000 = 0.000$$

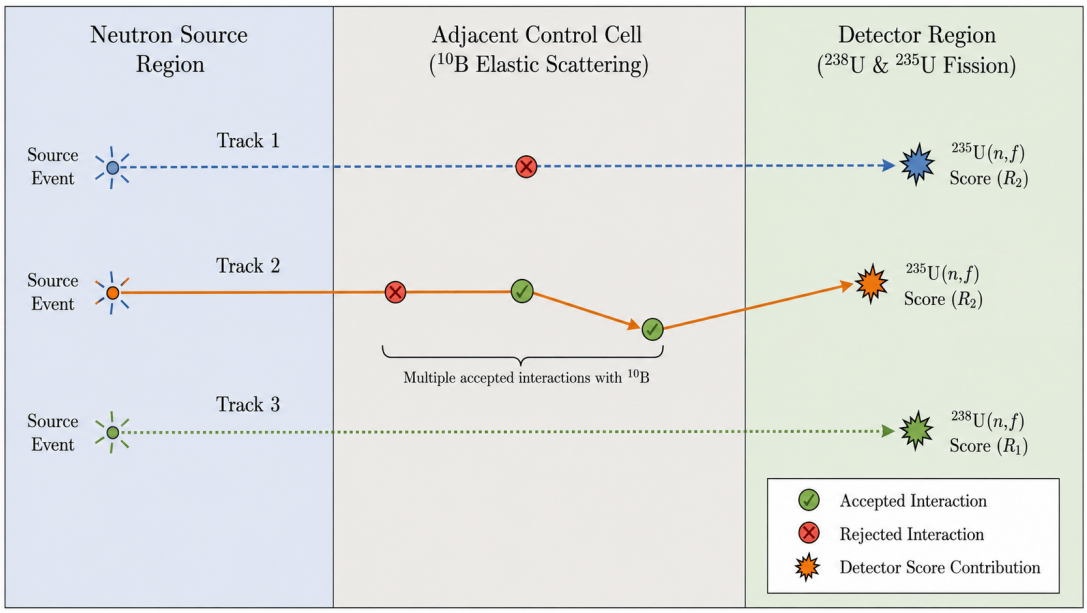


Figure II.1: Illustration of representative particle tracks used in collision-history sensitivity analysis, highlighting accepted versus rejected interactions in the perturbed zone.

Table II.1: Collision-history example for a perturbed event type x.

History n	T_1	T_2	$N_{\mathrm{ACC},x}$	$N_{\mathrm{REJ},x}$	$C_x^{(n)}$	$T_1 C_x$	$T_2 C_x$
1	0.000	0.050	0	1	-1	0.000	-0.050
2	0.000	0.120	2	1	$+1$	0.000	0.120
3	0.040	0.000	0	0	0	0.000	0.000

The indirect sensitivity for R_1 is:

$$S^{R_1}_{x,\text{ind}} \approx \frac{B_{1,x}}{B_1} = \frac{0.000}{0.040} = 0.000 \tag{II.49}$$

Compute the buffers for R_2 (^{235}U fission):

$$B_2 = 0.050 + 0.120 + 0.000 = 0.170$$
$$B_{2,x} = -0.050 + 0.120 + 0.000 = 0.070$$

The indirect sensitivity for R_2 is:

$$S^{R_2}_{x,\text{ind}} \approx \frac{B_{2,x}}{B_2} = \frac{0.070}{0.170} \approx 0.412 \tag{II.50}$$

Using the ratio property for sensitivity, where $\mathcal{R} = R_1/R_2$:

$$S^{\mathcal{R}}_{x,\text{ind}} \approx S^{R_1}_{x,\text{ind}} - S^{R_2}_{x,\text{ind}} = 0.000 - 0.412 = -0.412 \tag{II.51}$$

Since the detector reactions (fission) differ from the perturbed reaction (elastic scattering), the direct terms are zero ($d \neq x$). This result implies that a 1% increase in the ^{10}B elastic scattering cross section produces an approximate 0.412% decrease in the spectral index. The numerator response R_1 receives no indirect contribution from the perturbed reaction in these histories, while the denominator R_2 has a positive indirect sensitivity. Because $\mathcal{R} = R_1/R_2$, the ratio sensitivity is negative.

For a terminating reaction such as capture, an accepted event ends the history at the perturbed site, so the corresponding particle never scores at the downstream detectors and contributes zero to $B_{j,x}$; the sensitivity to that reaction is then carried entirely by the surviving particles, whose histories accumulate negative contributions to $C^{(n)}_x$ from rejected (virtual) capture events.

2.1.4. Differential Operator Sampling (DOS)

Differential Operator Sampling (DOS) is a sampling-based local sensitivity technique in which response derivatives are obtained by differentiating the probability measure of the forward Monte Carlo random walk. In the neutron transport literature this approach is also referred to as the differential operator method or the Taylor-series perturbation method [65, 76].

The Differential-Operator identity

Let $\omega \in \Omega_\omega$ denote a complete particle history (random walk), and let $\mathbf{x} = (x_1, \ldots, x_d)$ be the vector of physical input parameters. A scalar response $R(\mathbf{x})$ is the theoretical mean of the tally score,

$$R(\mathbf{x}) = \mathbb{E}_{\mathbf{x}}\left[T(\omega;\mathbf{x})\right] = \int_{\Omega_\omega} T(\omega;\mathbf{x})\; p(\omega;\mathbf{x})\; d\omega \tag{II.52}$$

where $p(\omega;\mathbf{x})$ is the path probability density induced by the transport sampling rules and $T(\omega;\mathbf{x})$ is the tally score[1]. This integral represents a probability-weighted average of the scores across

[1] The semicolon in $T(\omega;\mathbf{x})$ is used to distinguish the random variable ω from the physical parameters $\mathbf{x}$. This notation emphasizes that the score is calculated for a specific random walk ω that is parameterized by the inputs $\mathbf{x}$, rather than implying T is a function of two independent coordinate variables as a comma might suggest.

all possible random walks.

To derive the DOS identity, differentiate Eq. (II.52) with respect to x_i and use the log-derivative identity $\frac{\partial p}{\partial x_i} = p\,\frac{\partial \ln p}{\partial x_i}$:

$$\begin{aligned}\frac{\partial R}{\partial x_i} &= \int_{\Omega_\omega} \left[\frac{\partial T}{\partial x_i}\,p + T\,\frac{\partial p}{\partial x_i}\right] \mathrm{d}\omega \\ &= \int_{\Omega_\omega} \left[\frac{\partial T}{\partial x_i} + T\,\frac{\partial \ln p}{\partial x_i}\right] p(\omega;\mathbf{x})\,\mathrm{d}\omega \\ &= \mathbb{E}_{\mathbf{x}}\left[\frac{\partial T(\omega;\mathbf{x})}{\partial x_i} + T(\omega;\mathbf{x})\,\mathcal{D}_i(\omega;\mathbf{x})\right],\end{aligned} \tag{II.53}$$

where the differential-operator (score-function) weight is:

$$\mathcal{D}_i(\omega;\mathbf{x}) \equiv \frac{\partial \ln p(\omega;\mathbf{x})}{\partial x_i} \tag{II.54}$$

This yields the usual direct/indirect decomposition:

$$\frac{\partial R}{\partial x_i} = \underbrace{\mathbb{E}_{\mathbf{x}}\left[\frac{\partial T(\omega;\mathbf{x})}{\partial x_i}\right]}_{\text{Direct}} + \underbrace{\mathbb{E}_{\mathbf{x}}\left[T(\omega;\mathbf{x})\cdot\mathcal{D}_i\right]}_{\text{Indirect}} \tag{II.55}$$

which is the analogue expression of the direct/indirect split introduced in Eq. (II.12) for the perturbation theory [65, 76].

Differential-Operator Sensitivity Coefficient (DOS)

Evaluated at nominal state $\mathbf{x}_0$, the first-order relative sensitivity is:

$$S^R_{x_i} = \frac{x_{i,0}}{R_0}\,\mathbb{E}_0\left[\frac{\partial T(\omega;\mathbf{x})}{\partial x_i}\bigg|_{\mathbf{x}_0} + T(\omega;\mathbf{x}_0)\,\mathcal{D}_i(\omega;\mathbf{x}_0)\right] \tag{II.56}$$

with $R_0 \equiv R(\mathbf{x}_0)$ and $\mathbb{E}_0[\cdot] \equiv \mathbb{E}_{\mathbf{x}_0}[\cdot]$.

Score-function centering (zero-mean baseline)

A useful identity follows from normalization of the path probability. Since $\int_{\Omega_\omega} p(\omega;\mathbf{x})\,\mathrm{d}\omega = 1$, differentiating with respect to x_i gives

$$\mathbb{E}_{\mathbf{x}}[\mathcal{D}_i(\omega;\mathbf{x})] = \int_{\Omega_\omega} \frac{\partial \ln p}{\partial x_i}\,p\,\mathrm{d}\omega = \int_{\Omega_\omega} \frac{\partial p}{\partial x_i}\,\mathrm{d}\omega = \frac{\partial}{\partial x_i}\int_{\Omega_\omega} p\,\mathrm{d}\omega = 0 \tag{II.57}$$

Because $\mathbb{E}_{\mathbf{x}}[\mathcal{D}_i] = 0$, the indirect term can be written in a centered form,

$$\mathbb{E}_{\mathbf{x}}[T\,\mathcal{D}_i] = \mathbb{E}_{\mathbf{x}}[(T - C)\,\mathcal{D}_i] \tag{II.58}$$

for any deterministic constant C (i.e., not depending on ω). This is exactly a control-variate (baseline) construction for score-function estimators [77, 78]. In the scalar case, the variance-minimizing constant is:

$$C^\star = \frac{\mathbb{E}_{\mathbf{x}}[T\,\mathcal{D}_i^2]}{\mathbb{E}_{\mathbf{x}}[\mathcal{D}_i^2]} \tag{II.59}$$

Choosing C close to $C^\star$ can reduce the variance of the Monte Carlo estimator substantially [77]. A common, simple heuristic is $C \approx R(\mathbf{x}) = \mathbb{E}_{\mathbf{x}}[T]$ (the same $R(\mathbf{x})$ defined in Eq. (II.52)). With $C = R(\mathbf{x})$, the indirect term becomes a covariance:

$$\mathbb{E}_{\mathbf{x}}\left[(T - R)\,\mathcal{D}_i\right] = \mathrm{Cov}_{\mathbf{x}}\left(T, \mathcal{D}_i\right) \tag{II.60}$$

highlighting that DOS sensitivities arise from correlation between the tally score and the weight of the differential operator.

In practice, an exact $R(\mathbf{x})$ (or $C^\star$) is not available, so implementations either (i) use no explicit baseline, (ii) use a pilot estimate of R (separate run or early batches), or (iii) use a running/batched estimate of C that is effectively constant within a batch [77]. Regardless of centering, sensitivity estimates often converge more slowly than the base tally because the estimator involves the product $(T - C)\mathcal{D}_i$, whose variability depends on both T and $\mathcal{D}_i$ and on their correlation [65, 79].

Event-wise form for macroscopic cross section perturbations

To connect the abstract score $\mathcal{D}_i = \partial \ln p / \partial x_i$ to practical transport tallies, we now specialize to fixed-source analog random walks with perturbations to macroscopic cross sections. In that setting, only two sampling steps contribute x_i-dependence to the path probability [80]:

1. Free-flight distance sampling.
2. Reaction channel selection at collision.

For a history with K collisions, the cross section-dependent part of the path probability can be written as a product of per-collision factors:

$$p(\omega; \mathbf{x}) \;\propto\; \prod_{k=1}^{K} \left[\Sigma_t(E_{k-1})\, e^{-\Sigma_t(E_{k-1})\,\ell_k}\right] \left[\frac{\Sigma_{r_k}(E_{k-1})}{\Sigma_t(E_{k-1})}\right], \tag{II.61}$$

where ℓ_k is the k-th flight distance, Σ_t is the total macroscopic cross section, and r_k is the reaction realized at collision k [80]. (Dependence on geometry and scattering kinematics is omitted here because only cross section perturbations are considered.)

Taking a log-derivative converts the product in Eq. (II.61) into an additive operator score:

$$\mathcal{D}_i(\omega) = \sum_{k=1}^{K} \left[\frac{\partial \ln \Sigma_{r_k}(E_{k-1})}{\partial x_i} - \ell_k \frac{\partial \Sigma_t(E_{k-1})}{\partial x_i}\right] \tag{II.62}$$

Equation (II.62) shows the two canonical DOS contributions for cross section perturbations [65, 79]:

- A collision term, present only when the realized reaction is the perturbed reaction.
- A track-length term, present on every flight segment through the perturbed domain.

Let the perturbation be a relative change ε applied to a targeted macroscopic reaction cross section Σ_x in a specified region and energy interval, so that $\Sigma_x \to \Sigma_x^{(0)}(1 + \varepsilon)$ within that

domain. Since only Σ_x is perturbed, both Σ_x and the total cross section Σ_t shift by the same absolute amount, giving $\partial\Sigma_x/\partial\varepsilon = \partial\Sigma_t/\partial\varepsilon = \Sigma_x^{(0)}$ at $\varepsilon = 0$. Inserting these relations into Eq. (II.62) yields the practical eventwise DOS score:

$$\mathcal{D}_\varepsilon(\omega) = \sum_{k=1}^{K} \left[\mathbb{I}_{x,k} - \ell_k^{(\mathrm{dom})}\Sigma_x(E_{k-1})\right], \tag{II.63}$$

where $\mathbb{I}_{x,k} = 1$ if collision k is of type x and occurs in the selected region/energy domain (and 0 otherwise), and $\ell_k^{(\mathrm{dom})}$ is the portion of the k-th flight length that lies inside the same domain. This "realized minus expected" structure is consistent with $\mathbb{E}[\mathcal{D}_\varepsilon] = 0$ in Eq. (II.57) and underlies practical differential-operator Monte Carlo sensitivity estimators [65, 79].

Implementation in MCNP: `PERT` cards (METHOD=2/3) and Taylor coefficients

In MCNP, the differential-operator (Taylor-series) capability is invoked with the `PERT` card and provides Monte Carlo estimates of the first- and optionally second-order tally variations induced by changes to material density, composition, or nuclide-reaction cross sections [13, 81, 82]. Within the differential-operator framework, the first-order coefficient is the DOS score derived in Eq. (II.63) and is returned by `METHOD`=2. The second-order coefficient, returned by `METHOD`=3, follows from differentiating the log path probability a second time with respect to the perturbation; this produces additional cross terms over pairs of collisions and flight segments, accumulated within the same nominal random walk [76, 79]. A clear practical guide to configuring `PERT` cards efficiently is the paper by Favorite [81]; we follow here the workflow recommended there and focus on `METHOD`=2 (linear/first-order) and `METHOD`=3 (second-order curvature).

The `PERT` capability discussed here is intended for fixed-source applications. For criticality sensitivities, MCNP provides dedicated options (e.g., `KPERT` and `KSEN`) based on adjoint/IFP methods; these are outside the present scope.

Let $c(\varepsilon)$ denote a tally result as a function of a relative perturbation magnitude ε applied to a parameter (e.g., a reaction cross section σ_x). Expanding about the nominal case $\varepsilon = 0$,

$$c(\varepsilon) \;\approx\; c_0 + c_1\,\varepsilon + c_2\,\varepsilon^2 \tag{II.64}$$

where $c_0 \equiv c(0)$ is the nominal tally and c_1, c_2 are the first- and second-order Taylor coefficients.

`PERT` does not directly report sensitivity coefficients. Instead, for a user-chosen reference perturbation ε_r, MCNP reports tally variations such as:

$$\Delta c_1(\varepsilon_r) \;=\; c_1\,\varepsilon_r \qquad (\texttt{METHOD} = 2), \qquad \Delta c_2(\varepsilon_r) \;=\; c_2\,\varepsilon_r^2 \qquad (\texttt{METHOD} = 3) \tag{II.65}$$

Therefore, the coefficients are obtained by simple rescaling:

$$c_1 = \frac{\Delta c_1(\varepsilon_r)}{\varepsilon_r}, \qquad c_2 = \frac{\Delta c_2(\varepsilon_r)}{\varepsilon_r^2} \tag{II.66}$$

A common choice recommended for convenience is $\varepsilon_r = 1$ (a 100% relative change), which removes the rescaling and simplifies post-processing.

The first-order relative sensitivity of the tally to the perturbed parameter is:

$$S \;\equiv\; \left.\frac{\partial \ln c}{\partial \ln x}\right|_0 \;=\; \left.\frac{1}{c_0}\frac{\partial c}{\partial \varepsilon}\right|_0 \;\approx\; \frac{c_1}{c_0} \;\approx\; \frac{\Delta c_1(\varepsilon_r)}{\varepsilon_r\, c_0}, \tag{II.67}$$

which is the standard post-processing step: MCNP gives c_0 and Δc_1, and the user computes S.

Internally, `PERT` uses the differential-operator method: the first-order coefficient corresponds to the DOS derivative evaluated in the nominal run. For tallies whose definition does not contain the perturbed quantity (e.g., an `F4` flux tally under a cross section perturbation), the direct term is zero and the first-order variation is an indirect correlation of the form:

$$c_1 \equiv \left.\frac{\partial c}{\partial \varepsilon}\right|_0 = \mathbb{E}[T(\omega)\,\mathcal{D}_\varepsilon(\omega)]\,, \tag{II.68}$$

The parameter to be perturbed is a cross section, but MCNP applies the perturbation through a change in the cell density of a fictitious material. To target the cross section of a specific isotope j within a material, the user defines a fictitious material in which j appears with a perturbed composition fraction and rescales the cell density so that all other isotopes retain their nominal absolute number densities [82]. Let the original (unnormalized) material fractions be $\{\omega_i^{(0)}\}$ and choose a target isotope j:

$$\omega_j' = (1+\varepsilon)\,\omega_j^{(0)}, \qquad \omega_{i\neq j}' = \omega_i^{(0)}. \tag{II.69}$$

The compensating cell density is

$$\rho' = \rho^{(0)}\,\frac{\sum_i \omega_i'}{\sum_i \omega_i^{(0)}}. \tag{II.70}$$

After MCNP's internal normalization of the material fractions, the absolute number density of isotope j has been scaled by $(1+\varepsilon)$ while all other isotopes retain their nominal densities, so any macroscopic cross section attributable to j shifts by the intended relative amount ε. The reaction channel (`MT`) and the energy interval to which the perturbation applies are then selected on the `PERT` card itself, through the `RXN` and `ERG` keywords [13, 81].

When `METHOD`=3 is included, $\Delta c_2(\varepsilon_r)$ provides c_2 through Eq. (II.66), and the quadratic estimate of Eq. (II.64) becomes available. The ratio $|c_2\,\varepsilon/c_1|$ then quantifies the size of the second-order correction relative to the linear term at perturbation magnitude ε. When this ratio is small, the linear estimate is enough; when it approaches unity, the quadratic term must be retained to capture the response curvature [81].

Regarding uncertainties for these values, MCNP reports statistical uncertainties for c_0 and for Δc_1 (and Δc_2 if `METHOD`=3 is used). A common approximate propagation for the sensitivity uncertainty treats c_0 and Δc_1 as uncorrelated [81]:

$$\sigma_S \approx \sqrt{\left(\frac{\sigma_{\Delta c_1}}{\varepsilon_r\,c_0}\right)^2 + \left(\frac{\Delta c_1\,\sigma_{c_0}}{\varepsilon_r\,c_0^2}\right)^2} \tag{II.71}$$

Using the methodology described in [81], one can compute exact uncertainties by incorporating correlations via additional `PERT` cards. In practice, the impact of these corrections are often negligible.

Example with `F4` flux tally in MCNP

Consider a fixed-source calculation with an `F4` flux tally in a detector cell. A `PERT` card applies a relative perturbation ε to a capture cross section in a nearby region. Because `F4` does not explicitly contain the perturbed cross section, the direct term is zero and the first-order effect is the DOS correlation in Eq. (II.68).

Let T_n be the per-history `F4` contribution and $\mathcal{D}_{\varepsilon,n}$ the per-history differential-operator score accumulated for that `PERT` specification as in Eq. (II.63). Over N histories,

$$c_0 = \frac{1}{N}\sum_{n=1}^{N} T_n, \qquad c_1 = \frac{1}{N}\sum_{n=1}^{N} T_n\,\mathcal{D}_{\varepsilon,n}, \qquad \Delta c_1(\varepsilon) = \varepsilon\, c_1. \tag{II.72}$$

Table II.2: DOS/`PERT` illustration for an `F4` flux tally with three particle histories.

History (n)	`F4` score (T_n)	Operator score ($\mathcal{D}_{\varepsilon,n}$)	Product ($T_n\mathcal{D}_{\varepsilon,n}$)
1	$1.0 \cdot 10^{-4}$	-0.5	$-5.0 \cdot 10^{-5}$
2	$2.0 \cdot 10^{-4}$	$+0.1$	$+2.0 \cdot 10^{-5}$
3	$2.5 \cdot 10^{-4}$	-0.2	$-5.0 \cdot 10^{-5}$

From Eq. (II.72), $c_0 \approx (5.5\times10^{-4})/3 = 1.83\times10^{-4}$ and $c_1 \approx (-8.0\times10^{-5})/3 = -2.67\times10^{-5}$. If we follow the common `PERT` convention $\varepsilon_r = 1$ (a 100% reference perturbation), then `METHOD=2` reports $\Delta c_1(\varepsilon_r) \approx c_1$. The first-order sensitivity is then

$$S \approx \frac{\Delta c_1(\varepsilon_r)}{\varepsilon_r\, c_0} = \frac{c_1}{c_0} \approx -0.146.$$

This means that a 1% increase in the perturbed parameter produces a 0.146% decrease in the `F4` tally.

2.1.5. Correlated Sampling (CS)

Correlated sampling (CS) is another well-known Monte Carlo strategy for estimating small response changes due to small input perturbations. Although CS was not used in this work, it is included here for completeness because it is implemented in several Monte Carlo codes, including MORET [83], TRIPOLI-4 [84], and MCNP [13], and a recent review surveys these and related developments [85]. CS is also conceptually simple: instead of estimating a derivative inside one run, it estimates a difference between a nominal and a perturbed calculation while reducing the statistical uncertainty of that difference by enforcing strong correlation between the two simulations [86].

Mathematical Formulation

Let the nominal model be defined by an input vector $\mathbf{x}_0$ and consider a perturbation of one parameter x_i by a small relative amount ε:

$$x_i(\varepsilon) = x_i(0)\,(1+\varepsilon), \qquad |\varepsilon| \ll 1 \tag{II.73}$$

where $R_0 \equiv R(\mathbf{x}_0)$ is the nominal response and $R_\varepsilon \equiv R(x_i(\varepsilon))$ is the response for the perturbed system (all other inputs unchanged).

In CS, the nominal and perturbed responses are estimated from paired particle histories that start from the same pseudo-random-number state. If $T_n^{(0)}$ and $T_n^{(\varepsilon)}$ are the per-history contributions to the response from the n-th history in the nominal and perturbed runs, respectively, then the correlated-sampling estimator of the response change is:

$$\Delta\widehat{R}_{\mathrm{CS}} \;\equiv\; \widehat{R}_\varepsilon - \widehat{R}_0 \;\approx\; \frac{1}{N}\sum_{n=1}^{N}\left(T_n^{(\varepsilon)} - T_n^{(0)}\right) \tag{II.74}$$

The variance of $\Delta\widehat{R}_{\mathrm{CS}}$ depends on the variance of each tally and on their covariance:

$$\mathrm{Var}\left(\widehat{R}_{\varepsilon} - \widehat{R}_0\right) = \mathrm{Var}(\widehat{R}_{\varepsilon}) + \mathrm{Var}(\widehat{R}_0) - 2\,\mathrm{Cov}(\widehat{R}_{\varepsilon}, \widehat{R}_0) \tag{II.75}$$

so that a positive correlation between the two runs reduces the uncertainty of the difference. In practice, this covariance can be estimated from paired batch means from the nominal and perturbed runs, provided that corresponding batches use the same pseudo-random-number sequence; applying the standard sample-covariance formula to these paired batch means gives the batch-statistics estimate [87].

To convert the response change into a relative sensitivity coefficient as in Eq. (II.4), CS uses a finite-difference approximation. While a simple forward difference can be used, it introduces a truncation error of $\mathcal{O}(\varepsilon)$. To reduce this error to $\mathcal{O}(\varepsilon^2)$, the central-difference approach is generally preferred in sensitivity studies [87] as discussed in Sec. 2.1.1.

The general formulation for the CS sensitivity coefficient using the central-difference method is summarized below. For a more detailed derivation of these estimators and their associated uncertainties using batch statistics, the reader is referred to Favorite [87].

Correlated-Sampling Sensitivity Coefficient (CS)

For a parameter x_i perturbed by a relative amount ε such that $x_{i,\pm} = x_{i,0}(1 \pm \varepsilon)$, the central-difference sensitivity estimator is:

$$S^R_{x_i} \approx \frac{1}{R_0}\,\frac{R_{+\varepsilon} - R_{-\varepsilon}}{2\varepsilon} \tag{II.76}$$

where $R_{+\varepsilon}$ and $R_{-\varepsilon}$ are the correlated responses calculated at the positive and negative perturbations, respectively.

Note that while Eq. (II.76) is formally identical to the direct perturbation central-difference in Eq. (II.8), CS uses correlated histories to maximize the covariance in Eq. (II.75). This allows it to resolve small differences that would otherwise be hidden by Monte Carlo noise.

Implementation Strategy in MCNP

In MCNP, correlated sampling is not exposed through a dedicated card for fixed-source tallies. The user instead runs paired MCNP calculations, one nominal and one perturbed, configured so that history n in both runs starts from the same pseudo-random-number state via the `RAND` card [13]. Each perturbed parameter then requires its own paired run, so the workflow scales poorly when many parameters must be studied. The covariance term in Eq. (II.75) is not reported directly and must be reconstructed from the batched outputs, for which dedicated tools such as COSUBS are available [87].

Other Monte Carlo codes implement CS as a built-in capability. In MORET, for example, every neutron carries an array of weight corrections along its random walk, so that a single calculation simultaneously estimates the response change for many perturbed systems [83].

CS can be applied to essentially any small model change (not only cross sections), but its efficiency deteriorates as the perturbation grows because the nominal and perturbed random

walks decorrelate (reducing the magnitude of the covariance term in Eq. (II.75)). For very small perturbations and a small number of parameters, CS can be highly effective; for large sensitivity studies involving many parameters, the differential-operator approach is often more practical [86].

2.1.6. Scope of the local sensitivity framework

The methods presented so far are local: they characterize the response behavior in the neighborhood of a nominal parameter set. Global sensitivity analysis (GSA), by contrast, explores the full range of input variability and can capture interactions and nonlinearities that local derivatives miss. While GSA was investigated during this thesis, the high dimensionality of the nuclear data input space and the computational cost of the transport simulations limited the practical applicability of these methods. A brief overview of the main GSA concepts and the lessons learned is provided in Appendix A.

3. Uncertainty Quantification Framework

Uncertainty quantification (UQ) aims to map uncertainties in the input parameters $\mathbf{x}$ to uncertainties in responses R computed by particle-transport simulations. As defined in Sec. 1.1, two complementary strategies are considered:

(i) Linear uncertainty propagation based on local sensitivity coefficients and covariance matrices.

(ii) Sampling-based methods in which uncertain nuclear data are drawn from a prescribed joint distribution and propagated by repeated transport calculations.

These two strategies differ mainly in (1) the assumptions made on the response function and the input distributions, and (2) the computational cost required to obtain a stable estimate of the output uncertainty. Linear methods are traditionally less expensive but require the ability to calculate sensitivity coefficients to the parameters of interest, and second-moment information from them. Sampling-based methods are more general (nonlinearity and non-Gaussian effects can be captured) but are computationally demanding, requiring explicit treatment of correlations and of the separation between nuclear-data and Monte Carlo statistical uncertainties. Beyond these differences, both strategies are ultimately limited by the quality of the input uncertainty information itself: how well the correlations between parameters are known and which distributions can justifiably be assumed for them, that is, how well the inputs are known in the first place.

3.1. First-Order Second-Moment Propagation (FOSM)

The term First-Order Second-Moment (FOSM) originates from classical reliability and uncertainty analysis [88]. “First-order” refers to the linearization of the model response around the nominal input vector; “second-moment” refers to the use of only variances and covariances to characterize the input and propagated output uncertainties. FOSM therefore produces estimates of the response mean and variance but does not capture higher-order features such as skewness or heavy tails.

3.1.1. Response Linearization and the Sandwich Rule

Let the uncertain input vector be denoted by $\mathbf{x} = (x_1, \ldots, x_N)^T$, with mean $\mathbf{x}_0$ and covariance matrix $\mathbf{C}_x = \mathrm{Cov}(\mathbf{x})$. We linearize the scalar response $R(\mathbf{x})$ about the nominal parameter set $\mathbf{x}_0$. A first-order Taylor expansion gives:

$$R(\mathbf{x}) \approx R(\mathbf{x}_0) + \sum_{i=1}^{N} \frac{\partial R}{\partial x_i}\Big|_{\mathbf{x}_0} (x_i - x_{0,i}) \tag{II.77}$$

Using the absolute sensitivity coefficients defined in Eq. (II.3), define the absolute sensitivity row vector:

$$\mathbf{S}^{(abs)} \equiv \left(S_{x_1}^{(abs)}, \ldots, S_{x_N}^{(abs)}\right) \tag{II.78}$$

Then Eq. (II.77) can be written compactly as:

$$R(\mathbf{x}) \approx R(\mathbf{x}_0) + \mathbf{S}^{(abs)}(\mathbf{x} - \mathbf{x}_0) \tag{II.79}$$

Thus, the propagated variance is defined as:

$$\mathrm{Var}(R) \approx \mathbf{S}^{(abs)}\, \mathbf{C}_x \left(\mathbf{S}^{(abs)}\right)^T \tag{II.80}$$

Equation (II.80) is commonly called the "sandwich rule" (or "sandwich formula") because the input covariance matrix is sandwiched between two sensitivity vectors [89]. Accordingly, the standard deviation of R is:

$$\sigma_R \approx \sqrt{\mathbf{S}^{(abs)}\, \mathbf{C}_x \left(\mathbf{S}^{(abs)}\right)^T} \tag{II.81}$$

For multiple responses collected in a vector $\mathbf{R}(\mathbf{x}) \in \mathbb{R}^m$, the same derivation yields a full response covariance matrix:

$$\mathbf{C}_R \approx \mathbf{J}\, \mathbf{C}_x\, \mathbf{J}^T, \tag{II.82}$$

where $\mathbf{J} = \partial \mathbf{R}/\partial \mathbf{x}$ is the Jacobian. This is useful when response correlations are required.

3.1.2. Relative Form and Connection to Sensitivity

In nuclear data applications, it is often convenient to work with relative variations and relative sensitivity coefficients. Recalling the definition of the relative sensitivity coefficient from Eq. (II.4), $S_{x_i}^R \equiv \frac{\partial \ln R}{\partial \ln x_i}$, and collecting these into a row vector $\mathbf{S}^R = [S_1^R, \ldots, S_d^R]$, and introducing the covariance matrix of relative variations:

$$\mathbf{C}_{\mathrm{rel}} \equiv \mathrm{Cov}\left(\frac{\delta x_i}{x_i}, \frac{\delta x_j}{x_j}\right) \tag{II.83}$$

the relative response variance becomes:

$$\left(\frac{\sigma_R}{R}\right)^2 \approx \mathbf{S}\, \mathbf{C}_{\mathrm{rel}}\, \mathbf{S}^T. \tag{II.84}$$

Equation (II.84) is the standard form used in sensitivity/uncertainty (S/U) analyses, where $\mathbf{S}$ is obtained from the local SA techniques described in Sec. 2.1, and $\mathbf{C}_{\mathrm{rel}}$ is taken from evaluated nuclear data covariance information processed into a consistent format [89].

A practical advantage of FOSM is that it provides an efficient route to variance decomposition: once sensitivities are available, the effect of selecting subsets of $\mathbf{x}$ (e.g. a single isotope or a specific energy range) can be assessed by restricting $\mathbf{S}$ and the corresponding block of $\mathbf{C}_{\mathrm{rel}}$ [89].

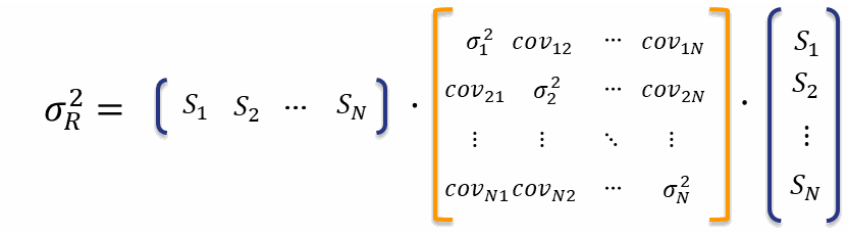


Figure II.2: Matrix representation of the Sandwich Rule for the propagation of uncertainties to the response variance.

3.1.3. Applicability and Limitations

FOSM is exact if (i) the response is truly linear in the perturbed parameters within the uncertainty range and (ii) the input uncertainty is fully captured by the covariance matrix. In reactor physics applications, these conditions are often only approximately met, but FOSM remains widely used because it is computationally efficient, transparent, and directly linked to the physical interpretation of sensitivity profiles [85]. However, FOSM can be biased when the response exhibits strong nonlinearity, threshold effects, or feedbacks, or when the output distribution deviates significantly from Gaussian behavior (in which case higher moments matter) [88]. These limitations motivate the complementary use of sampling-based propagation methods.

3.2. Monte Carlo Sampling (MCS) for Uncertainty Quantification

Sampling-based propagation methods treat the uncertain inputs $\mathbf{x}$ as random variables and approximate the distribution of the response $R(\mathbf{x})$ by repeated code executions [90]. In the nuclear data context, the sampling step must represent not only the marginal uncertainties of individual parameters (e.g. cross sections in a given energy bin) but also their often strong correlations across energy, reactions, and isotopes [90]. Each sample of nuclear data produces one transport estimate of the response, which carries its own statistical noise when the transport solver is a Monte Carlo code. The set of runs is then used to estimate the response mean and standard deviation.

3.2.1. Sampling Techniques

Consider a d-dimensional uncertain input vector $\mathbf{x}$. We generate a set of N samples $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$ from the prescribed joint distribution. Common sampling strategies include:

- **Simple random sampling (SRS)**: samples are drawn independently from the target distribution using pseudo-random numbers.
- **Latin Hypercube Sampling (LHS)**: each marginal distribution is stratified into N equiprobable intervals, and one value is drawn from each interval, which improves coverage of each variable for a fixed sample size [91]. Intuitively, for a one-dimensional Gaussian, you would place N vertical cut lines so that each strip contains probability $1/N$, then draw one sample from each strip.
- **Low-discrepancy (quasi-random) sequences**: deterministic point sets, such as the Sobol sequence, are constructed to cover the unit hypercube as evenly as possible, reducing gaps and clusters compared with pseudo-random samples. The name refers to

the *discrepancy*, a measure of how far a finite point set departs from a uniform spread; these sequences keep this measure low for every prefix of the first N points. In practice, this often yields more accurate estimates than SRS for the same N, especially when the quantity of interest varies smoothly with the inputs [92].

A practical difference among these strategies is the nested property. With SRS, any prefix of a sample set is still a valid random sample; similarly, Sobol sequences are commonly constructed in sizes that are powers of two and maintain a nested structure across sample-set refinements. By contrast, standard LHS is designed for one fixed sample size and is not naturally nested; increasing the sample size requires augmentation strategies [93]. These properties are relevant when assessing convergence as a function of the number of samples.

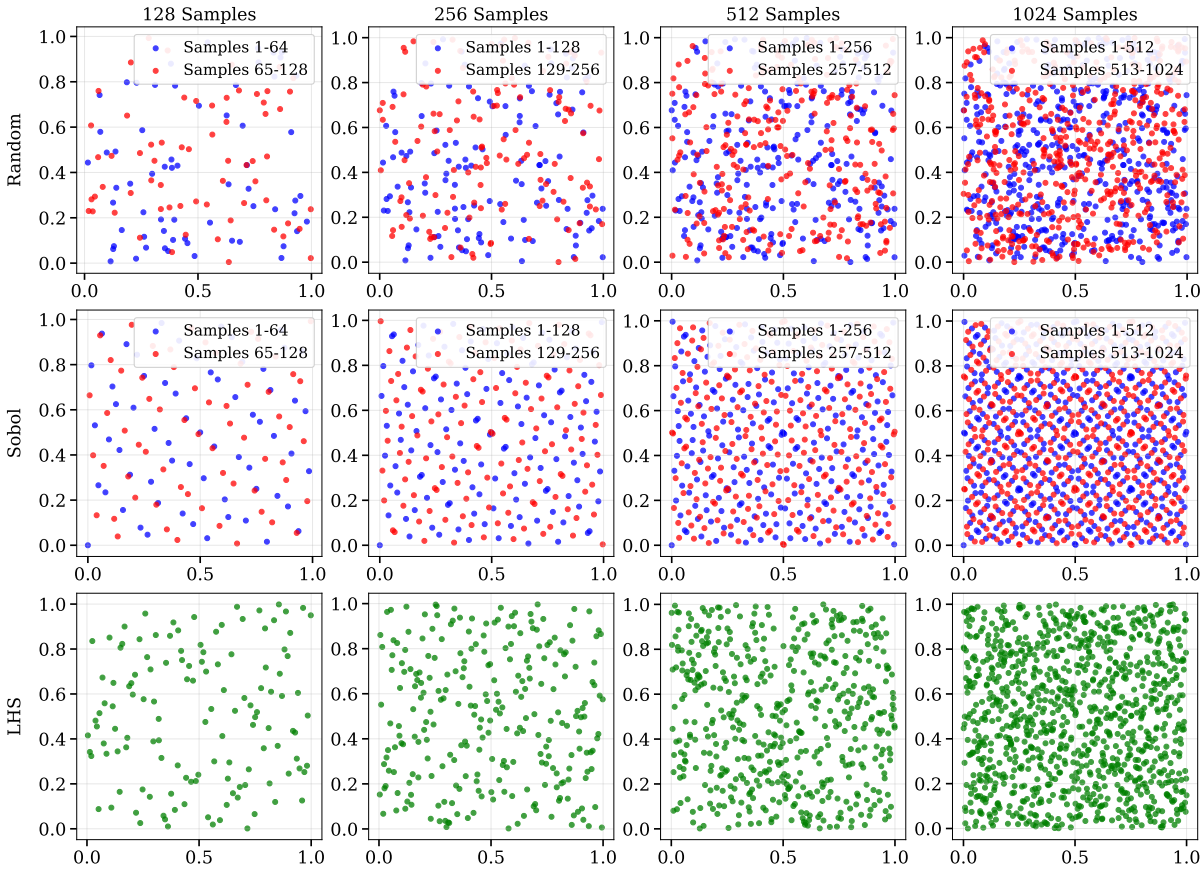


Figure II.3: Comparison of sampling techniques in 2D Space: Random, Sobol, LHS. A well-distributed sampling approach minimizes gaps and ensures better domain coverage.

Both LHS and Sobol sequences naturally provide points $\mathbf{u}^{(i)} \in [0,1]^d$ that are uniform. We convert them into standard normal samples by applying the inverse standard normal cumulative distribution function (CDF) to each component, $z_j^{(i)} = \Phi^{-1}\left(u_j^{(i)}\right)$. In the next section, we impose the desired correlations by transforming $\mathbf{z}^{(i)}$ with a matrix decomposition of the covariance matrix, and then obtain samples in the final target distribution.

3.2.2. Matrix Decomposition and Correlated Sampling

To represent correlated nuclear data, sampling is commonly performed in two steps. First, independent standard normal vectors $\mathbf{z}^{(k)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ are generated. Second, these are mapped

to correlated perturbations through a factorization of the covariance matrix. Let $\mathbf{C}$ denote the desired covariance matrix. If a matrix $\mathbf{L}$ satisfies $\mathbf{C} = \mathbf{L}\mathbf{L}^T$, then:

$$\boldsymbol{\delta}^{(k)} = \mathbf{L}\,\mathbf{z}^{(k)} \tag{II.85}$$

has covariance $\mathrm{Cov}(\boldsymbol{\delta}) = \mathbf{C}$. The perturbed nuclear data are then constructed by applying $\boldsymbol{\delta}^{(k)}$ to the nominal data as relative perturbation factors.

By definition, a covariance matrix $\mathbf{C}$ is positive semidefinite (PSD), so all of its eigenvalues are non-negative [94]. In practice, covariance matrices obtained from evaluated nuclear data can exhibit small negative eigenvalues due to processing, rounding, or consistency issues [95]. These negative directions have no physical interpretation as variances and prevent the use of certain sampling factorizations without additional conditioning.

Cholesky Decomposition

The Cholesky factorization goes for a direct square root of the covariance:

$$\mathbf{C} = \mathbf{L}\mathbf{L}^T, \tag{II.86}$$

with $\mathbf{L}$ being the lower triangular matrix [96]. Algorithmically, the Cholesky factorization builds the matrix $\mathbf{L}$ one row/column at a time. At each step it computes a diagonal element of $\mathbf{L}$ as the square root of a remaining "unexplained" diagonal term of $\mathbf{C}$ (after subtracting contributions from previously computed columns). For a covariance matrix this remaining term corresponds to a variance-like quantity and it is non-negative when $\mathbf{C}$ is PSD. If $\mathbf{C}$ is not PSD, this remaining quantity can become negative. In that case the next diagonal entry would require taking the square root of a negative number, so the factorization cannot continue with real numbers. This is why Cholesky is sensitive: even very small negative eigenvalues can propagate into a negative value at some step and cause the algorithm to fail.

Eigenvalue Decomposition (EVD)

For a symmetric matrix, the eigenvalue decomposition always exists [94]:

$$\mathbf{C} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T, \tag{II.87}$$

and it reveals PSD violations directly through negative eigenvalues in $\boldsymbol{\Lambda}$. Sampling requires a real square root factor, commonly written as:

$$\mathbf{L} = \mathbf{Q}\boldsymbol{\Lambda}^{1/2}, \tag{II.88}$$

which is only real-valued if $\boldsymbol{\Lambda} \succeq 0$. This notation ($\succeq$) indicates that the matrix is positive semi-definite, which for a diagonal matrix simply means that all its entries are greater than or equal to zero. Therefore, when small negative eigenvalues are present, a common remedy is to replace $\boldsymbol{\Lambda}$ by a conditioned diagonal matrix $\tilde{\boldsymbol{\Lambda}}$ (e.g. clipping eigenvalues below a threshold to 0 or to a small $\epsilon > 0$), and then construct:

$$\tilde{\mathbf{C}} = \mathbf{Q}\tilde{\boldsymbol{\Lambda}}\mathbf{Q}^T, \qquad \tilde{\mathbf{L}} = \mathbf{Q}\tilde{\boldsymbol{\Lambda}}^{1/2}. \tag{II.89}$$

In this way, EVD-based approaches typically do not fail abruptly; instead they make the non-PSD directions explicit and provide a controlled route to a nearby PSD approximation.

Singular Value Decomposition (SVD)

The singular value decomposition exists for any matrix [97],

$$\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \qquad \mathbf{\Sigma} \succeq 0, \tag{II.90}$$

and the non-negativity of the singular values makes SVD numerically robust in the presence of rank deficiency or near-singularity. However, for covariance sampling one needs a factor $\mathbf{L}$ such that $\mathbf{C} = \mathbf{L}\mathbf{L}^T$. A common SVD-based construction is

$$\mathbf{L}_{\text{svd}} = \mathbf{U}\mathbf{\Sigma}^{1/2}, \tag{II.91}$$

which yields a matrix that is always PSD by construction:

$$\mathbf{L}_{\text{svd}}\mathbf{L}_{\text{svd}}^T = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^T \equiv \tilde{\mathbf{C}}_{\text{svd}} \tag{II.92}$$

Importantly, $\tilde{\mathbf{C}}_{\text{svd}}$ equals the original $\mathbf{C}$ only in the special case where $\mathbf{U} = \mathbf{V}$ (which holds for symmetric PSD matrices) [97]. If $\mathbf{C}$ is not PSD, then $\mathbf{L}_{\text{svd}}$ does not provide an exact square root of $\mathbf{C}$; instead, it implicitly replaces $\mathbf{C}$ by the PSD surrogate $\tilde{\mathbf{C}}_{\text{svd}}$.

Summary of Decompositions

In summary, the choice of decomposition involves a trade-off between mathematical rigor and numerical stability. Cholesky is computationally the most efficient but requires a strictly PSD matrix [96]. EVD provides the most control, allowing the user to physically interpret and "clip" non-physical negative eigenvalues [94]. SVD offers maximum robustness for automated pipelines, though it implicitly modifies the covariance structure without user intervention [97].

Verification of Sample Convergence

Regardless of the decomposition chosen, the stochastic nature of the sampling process introduces sampling noise. To assess how well a finite sample of size N reproduces the target covariance $\mathbf{C}$, the sample covariance $\hat{\mathbf{C}}$ is calculated. The convergence is quantified using the relative Frobenius norm [98, 99]:

$$\varepsilon = \frac{\|\mathbf{C} - \hat{\mathbf{C}}\|_F}{\|\mathbf{C}\|_F}, \qquad \text{where} \quad \|\mathbf{A}\|_F = \sqrt{\sum_{i,j} a_{ij}^2} \tag{II.93}$$

This metric provides a global measure of the discrepancy between the intended and the realized correlations. For simple random sampling, the error is expected to scale as $\varepsilon = \mathcal{O}(N^{-1/2})$, consistent with standard Monte Carlo convergence rates [98]. In our application, each sample corresponds to one independent simulation, so N is directly limited by computational cost. This makes the choice of sampling strategy important. LHS or low-discrepancy sequences, as discussed in the previous section, can reduce the error in comparison to random sampling for a fixed N, thereby lowering the number of expensive runs required to achieve a prescribed tolerance in ε.

3.2.3. Decoupling Parametric and Statistical Uncertainties

The convergence study above addresses how accurately the input sampling reproduces the prescribed covariance structure. In addition, each transport calculation introduces its own MC

statistical noise. As a result, the run-to-run variability observed over a set of N calculations reflects both (i) variability induced by the sampled nuclear data and (ii) MC statistical fluctuations. A dedicated separation procedure is therefore required to estimate the nuclear-data uncertainty σ_{ND} alone.

Observed spread and variance decomposition

The run-to-run variability observed over a set of N calculations reflects two different components:

1. **Parametric uncertainty:** Variability induced by the sampled nuclear data (σ_{ND}).
2. **Statistical uncertainty:** Monte Carlo statistical fluctuations inherent to the transport algorithm (σ_{MC}).

To isolate the nuclear data component, we first define the observed variance (σ^2_{obs}) and the average MC statistical variance ($\overline{\sigma^2_{\mathrm{MC}}}$) across the N samples. Here $q_i = R(\mathbf{x}^{(i)})$ is the response of the i-th run and $\overline{q}$ its sample mean:

$$\sigma^2_{\mathrm{obs}} = \frac{1}{N-1}\sum_{i=1}^{N}(q_i - \overline{q})^2\,, \qquad \overline{q} = \frac{1}{N}\sum_{i=1}^{N} q_i \tag{II.94}$$

$$\overline{\sigma^2_{\mathrm{MC}}} = \frac{1}{N}\sum_{i=1}^{N}\sigma^2_{\mathrm{MC},i} \tag{II.95}$$

The total observed variance is the sum of these independent contributors:

$$\sigma^2_{\mathrm{obs}} = \sigma^2_{\mathrm{ND}} + \overline{\sigma^2_{\mathrm{MC}}} \tag{II.96}$$

Depending on the specific configurations used for simulations, different strategies can be used to estimate σ^2_{ND} from σ^2_{obs} and $\overline{\sigma^2_{\mathrm{MC}}}$. We briefly summarize the three approaches used in the literature: (i) classic Total Monte Carlo (TMC), (ii) Fast TMC, and (iii) Fast GRS.

Total Monte Carlo (TMC)

The classic Total Monte Carlo approach [90] keeps the statistical term in Eq. (II.96) small enough to be neglected. The number of particle histories per run is chosen so that the average statistical uncertainty stays well below the observed spread, $\overline{\sigma_{\mathrm{MC}}} \leq 0.05\,\sigma_{\mathrm{obs}}$ [100]. The nuclear-data uncertainty is then read directly from the observed spread,

$$\sigma_{\mathrm{ND}} \approx \sigma_{\mathrm{obs}} \tag{II.97}$$

All runs share the same random seed, so the residual statistical fluctuations are correlated from one run to the next and add little to the spread. The reliability of the method rests on the small statistical uncertainty, which is also its cost: pushing $\overline{\sigma_{\mathrm{MC}}}$ below the threshold requires a large number of histories per run.

In practice, σ_{obs} is typically unknown a priori. One must either perform a preliminary pilot study to estimate the required number of histories or iteratively increase the particle count until the ratio $\overline{\sigma_{\mathrm{MC}}}/\sigma_{\mathrm{obs}}$ falls below the 0.05 threshold.

Fast TMC

Fast TMC [100] improves computational efficiency by allowing for larger per-run statistical uncertainties. By using a different seed for each run, the MC statistical contribution becomes an independent, approximately zero-mean random term. The decomposition in Eq. (II.96) can then be rearranged to recover the nuclear-data variance by subtracting the average statistical variance from the observed variance,

$$\sigma^2_{\mathrm{ND}} = \sigma^2_{\mathrm{obs}} - \overline{\sigma^2_{\mathrm{MC}}} \tag{II.98}$$

Because the seeds are independent, this subtraction remains accurate even when the statistical uncertainty is substantial.

In practice, this allows the target statistical uncertainty to range between one-third and one-half of the total observed uncertainty. This significantly relaxes the 5% threshold required by classic TMC, enabling a substantial reduction in the required particle histories per run without compromising the accuracy of the nuclear-data uncertainty estimate.

Fast GRS

A third strategy, Fast GRS [101], involves running two independent transport realizations for each nuclear data sample using two different seeds. For each sample, let the outputs be denoted as $q_i^{(1)}$ and $q_i^{(2)}$ for $i = 1, \ldots, N$. The nuclear-data variance is then estimated by the covariance between these two series:

$$\sigma^2_{\mathrm{ND}} = \mathrm{Cov}\left(q^{(1)}, q^{(2)}\right) = \frac{1}{N-1} \sum_{i=1}^{N} \left(q_i^{(1)} - \overline{q}^{(1)}\right)\left(q_i^{(2)} - \overline{q}^{(2)}\right) \tag{II.99}$$

where $\overline{q}^{(1)}$ and $\overline{q}^{(2)}$ are the means of the respective series. Under the assumption that $q^{(1)}$ and $q^{(2)}$ are identically distributed and conditionally independent given the sampled nuclear data, the covariance cancels the MC statistical component and isolates the variability due to nuclear data. Unlike TMC or Fast TMC, this approach does not rely on the explicit MC uncertainty provided by the transport code; it only requires that individual realizations have sufficient convergence for the covariance to remain stable.

Discussion and practical implications

The sampling technique (SRS/LHS/Sobol), covariance factorization (Cholesky/EVD/SVD), and separation strategy (TMC/Fast TMC/Fast GRS) jointly determine the computational efficiency and the robustness of σ_{ND}. The sampling method affects space-filling and convergence with N; the decomposition affects how accurately correlations are represented; and the separation strategy fixes the trade-off between histories per run and number of runs [69]. Section 5 applies these trade-offs to a case study.

Advanced methods

High-dimensional nuclear data spaces motivate methods that reduce the number of expensive high-fidelity transport runs. Lasso Monte Carlo (LMC) [102] combines a sparse linear surrogate model with a multifidelity Monte Carlo correction to compute unbiased estimates of output mean

and variance at reduced cost. The approach has been demonstrated on nuclear engineering UQ problems and represents a potential direction for future extensions of the present work.

4. The KIKA Computational Framework

The SA&UQ methods presented in Secs. 2–3.2 require a chain of operations that spans several data file formats (ENDF-6, ACE, MCTAL, SDF, COVERX and COVFIL covariance files) and external codes (NJOY, MCNP, Serpent). In practice, each step (covariance parsing, matrix decomposition, correlated sampling, file perturbation, transport execution, and output post-processing) must be performed consistently and reproducibly. We developed the `kika-nd` Python library (v0.2.4) during this thesis to unify these operations within a single framework [103]. The library is open-source under the BSD-3-Clause license. This section first motivates the decision to build a new tool, then describes the framework capabilities and the nuclear data perturbation pipelines that it implements. Sec. 5 applies the framework to a verification case, and Chapter III uses it for an ^{56}Fe angular distribution evaluation.

4.1. Motivation

Several existing tools address parts of the SA&UQ chain described in the preceding sections, but none covers it entirely for both cross section and angular distribution uncertainties.

The ACE perturbation tool of FRENDY [104] was initially used for cross section sampling. However, extending the study to angular distributions, `MF34` covariance handling, and custom sampling strategies (e.g. LHS, Sobol sequences) required modifications at the source-code level. A self-contained implementation ultimately proved more practical. FRENDY later introduced an angular distribution perturbation method based on the maximum entropy method [105], but this feature became available after the `kika-nd` pipeline was already operational. SANDY [106] documents angular distribution perturbation as a supported capability. However, as of 2026 a code restructure has left this functionality non-operational. The C++ toolkits ACEtk [107] and ENDFtk [108] provide low-level nuclear data I/O, but at the time of development they were still in development, with incomplete documentation and missing features for `MF34` and `MF4` manipulation. More broadly, limited or absent documentation is a recurrent difficulty across nuclear data tools: adapting an unfamiliar codebase to a specific research need can require as much effort as building a focused solution from scratch.

Implementing the full chain (parsing, processing, decomposition, sampling, perturbation, and writing) also provided the in-depth understanding of nuclear data file structures and numerical methods that a thesis centered on this topic demands. The development was guided by two goals: making the API as intuitive and well-documented as possible, and providing comprehensive diagnostic logging so that every numerical decision (eigenvalue clipping, variance capping, positivity corrections) is transparent and traceable.

A further motivation was accessibility. Researchers who need to perform nuclear data perturbation studies but did not develop the underlying code often face a steep learning curve when working directly with a library API. To lower this barrier, a companion desktop application, `KIKA` [109], wraps the library in an interactive graphical interface for nuclear data visualization and pipeline configuration. Although developed alongside the research presented here, it lies outside the main focus of the thesis. A brief description is provided in Appendix B.

4.2. Architecture Overview

Figure II.4 shows the library organized into three functional categories: *Data* parsers, *Tools* implementing the methods of Secs. 2–3.2, and *Codes* interfaces to external processing and transport codes. A central design choice is *round-trip fidelity*: nuclear data files can be parsed, selectively modified (e.g. a single `MF/MT` section or a single energy-dependent cross section), and re-written to disk without disturbing the remainder of the file. Further details on the programming interfaces and data model are given in Appendix C.

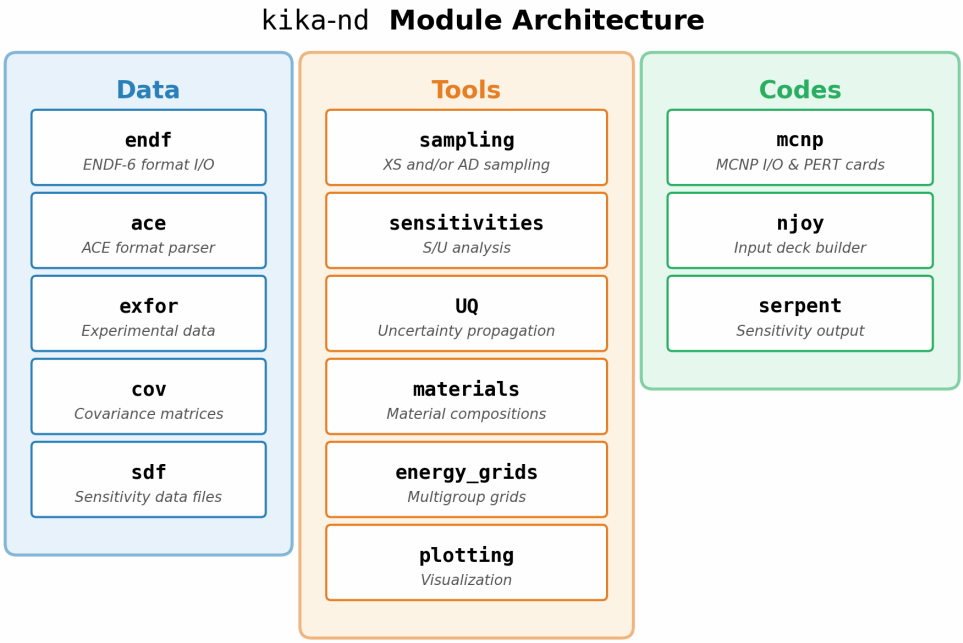


Figure II.4: Module architecture of the `kika-nd` library. Fourteen modules are grouped into three categories: *Data* parsers for the nuclear data formats handled by the SA&UQ chain, *Tools* implementing the sensitivity, sampling, and UQ methods of this chapter, and *Codes* interfaces to the external processing and transport codes. The perturbation pipelines (Sec. 4.5) chain modules across all three columns.

4.3. Supported Formats and External Codes

The library reads and writes the formats encountered along the SA&UQ chain: ENDF-6 (every file type except `MF26` and `MF30`, covariance files included), ACE, MCTAL tally output, SDF sensitivity profiles, and several multigroup covariance formats (COVERX, COVFIL, BOXER, and SCALE/NJOY direct outputs). Round-trip parsing supports selective replacement of individual `MF/MT` sections within an evaluation, which is the mechanism behind the perturbation pipelines of Sec. 4.5. The library also drives NJOY through a fixed input template covering the standard MODER→RECONR→BROADR→HEATR→PURR→GASPR→ACER sequence (full listing in Appendix D); arbitrary input decks can be assembled for non-standard processing. For MCNP, input files and MCTAL tallies are parsed, `PERT` card generation is automated for the DOS sensitivity extraction of Sec. 2, and fictitious materials with adjusted densities are constructed following Eq. (II.70). Serpent's native sensitivity output is parsed directly, and an

EXFOR interface provides access to experimental data for the workflow of Chapter III. Decomposition methods (Cholesky, EVD, SVD) are exposed through a common interface for all covariance types and feed the sampling routines of Sec. 3.2.2. The full programming interface, supported sections within each format, and the data model are described in Appendix C.

4.4. Multigroup Collapse of Angular Distribution Covariances

The FOSM sandwich rule (Eq. (II.84)) requires multigroup covariance matrices aligned with the energy structure of the sensitivity profiles. Angular distribution covariances are read directly from ENDF `MF34` sections, where each sub-matrix corresponds to a Legendre order pair (ℓ, ℓ') and is defined on the evaluator's own energy-group grid. This evaluator grid is usually not the energy structure on which the sensitivity profiles are computed, which typically follows a standard multigroup grid such as VITAMIN-J. The covariances must therefore be re-binned onto that target structure. NJOY's ERRORR module can collapse `MF34` angular distribution covariances into multigroup form, but only for the first Legendre order ($\ell = 1$). A multigroup collapse was therefore implemented in KIKA, outside NJOY, following the same methodology used by NJOY's GROUPR and ERRORR modules [11], extended to higher Legendre orders.

Both the Legendre coefficients from `MF4` and the covariance matrices from `MF34` must be re-binned onto the target multigroup structure. The weighting function is a cross-section-weighted spectrum $\sigma(E)\,\phi(E)$, where $\sigma(E)$ is the reconstructed pointwise cross section and $\phi(E)$ is a user-selected base spectrum. The multigroup-averaged Legendre coefficient for order ℓ and target group g is:

$$\bar{a}_{\ell,g} \;=\; \frac{\displaystyle\int_g \sigma(E)\,\phi(E)\,a_\ell(E)\,\mathrm{d}E}{\displaystyle\int_g \sigma(E)\,\phi(E)\,\mathrm{d}E} \tag{II.100}$$

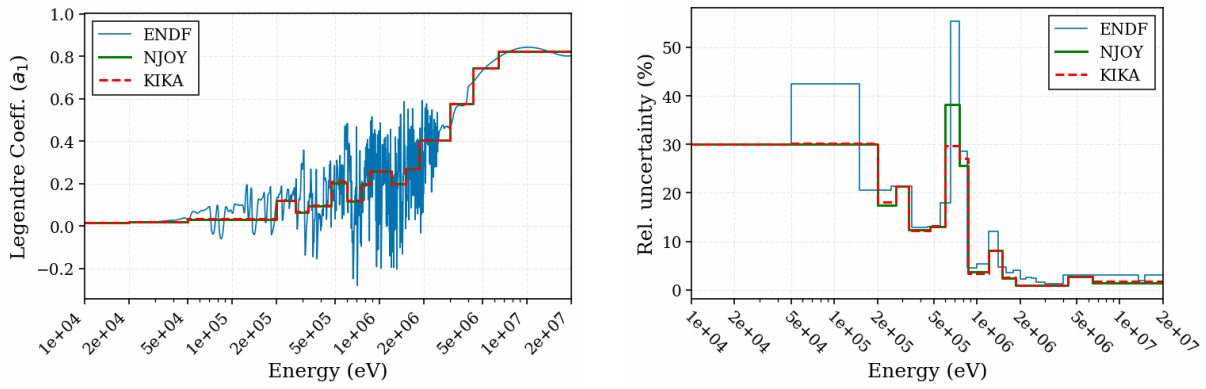
For `MF34` the original energy grid is intersected with the target multigroup grid to form a union grid. For each union cell i and target group g, the overlap weight is the $\sigma\phi$ integral over their intersection:

$$w_{ig} \;=\; \int_{E_i \cap G_g} \sigma(E)\,\phi(E)\,\mathrm{d}E \tag{II.101}$$

The flux fraction $f_i^g = w_{ig}/W_g$, with $W_g = \sum_i w_{ig}$, gives the share of group g's total weight that falls in union cell i. The multigroup relative covariance is then collapsed directly using these fractions, following the same approach as NJOY ERRORR:

$$R^{\mathrm{MG}}_{gg'} \;=\; \sum_i \sum_j f_i^g \; R_{ij} \; f_j^{g'} \tag{II.102}$$

After collapse, the matrices are symmetrized and projected to the nearest positive semi-definite matrix via eigenvalue clipping. Figure II.5 compares the KIKA collapse against NJOY for the first Legendre order of ^{56}Fe elastic scattering, using the SCALE 56-group structure with $\sigma(E)/E$ lethargy weighting. The coefficient and uncertainty agree across the energy range. The small residual differences in the relative uncertainty occur in groups that span a `MF34` energy boundary where a_1 varies steeply. There the two codes assign slightly different coefficients within the group: NJOY collapses its group-averaged values onto the integration grid, whereas KIKA integrates the pointwise coefficient directly. The group-averaged coefficients themselves match to better than 0.01%.



(a) (b)

Figure II.5: Multigroup collapse of ^{56}Fe elastic scattering: (a) first-order Legendre coefficient a_1 and (b) first-order relative uncertainty.

4.5. Nuclear Data Perturbation Pipelines

The MCS method (Sec. 3.2) requires N sets of perturbed nuclear data files, each reflecting a correlated draw from the covariance matrix. Two pipelines produce these files depending on the data type being perturbed: the ACE pipeline for cross sections and the ENDF pipeline for angular distributions.

4.5.1. Cross Section Perturbation

The ACE pipeline perturbs one or several ACE files, each corresponding to a single isotope, using multigroup covariance data. For each isotope, the pipeline generates N correlated sample vectors from the corresponding covariance block and creates a perturbed ACE file for each sample. The orange path in Figure II.6 illustrates this workflow.

For each sample, the continuous-energy cross section data points in the nominal ACE file are mapped to multigroup bins. Each cross section value is then multiplied by the perturbation factor of its energy group. Samples that produce non-positive factors for any group are rejected for that reaction (leaving it unperturbed), since a negative cross section is unphysical. After applying factors to all requested reactions, composite cross sections (MT=1 total, MT=101 total absorption, MT=3 non-elastic) are recalculated from their components to maintain sum-rule consistency.

Covariance data are typically provided for lumped reaction channels (e.g. MT=4 for total inelastic scattering, MT=103 for total (n,p)). The pipeline automatically expands these to their constituent partial reactions: MT=4 maps to MTs 51–91, MT=103 to MTs 600–649, and so forth. The same factors are applied to all members of a group, ensuring consistency between the lumped covariance representation and the detailed reaction channels stored in the ACE file. Figure II.7 shows the relative differences obtained for a set of perturbed ^{56}Fe elastic-scattering cross sections.

4.5.2. Angular Distribution Perturbation

For Legendre coefficient uncertainties stored in `MF34`, an analogous pipeline operates at the ENDF level. Scaling factors are generated from the `MF34` covariance, organized by Legendre

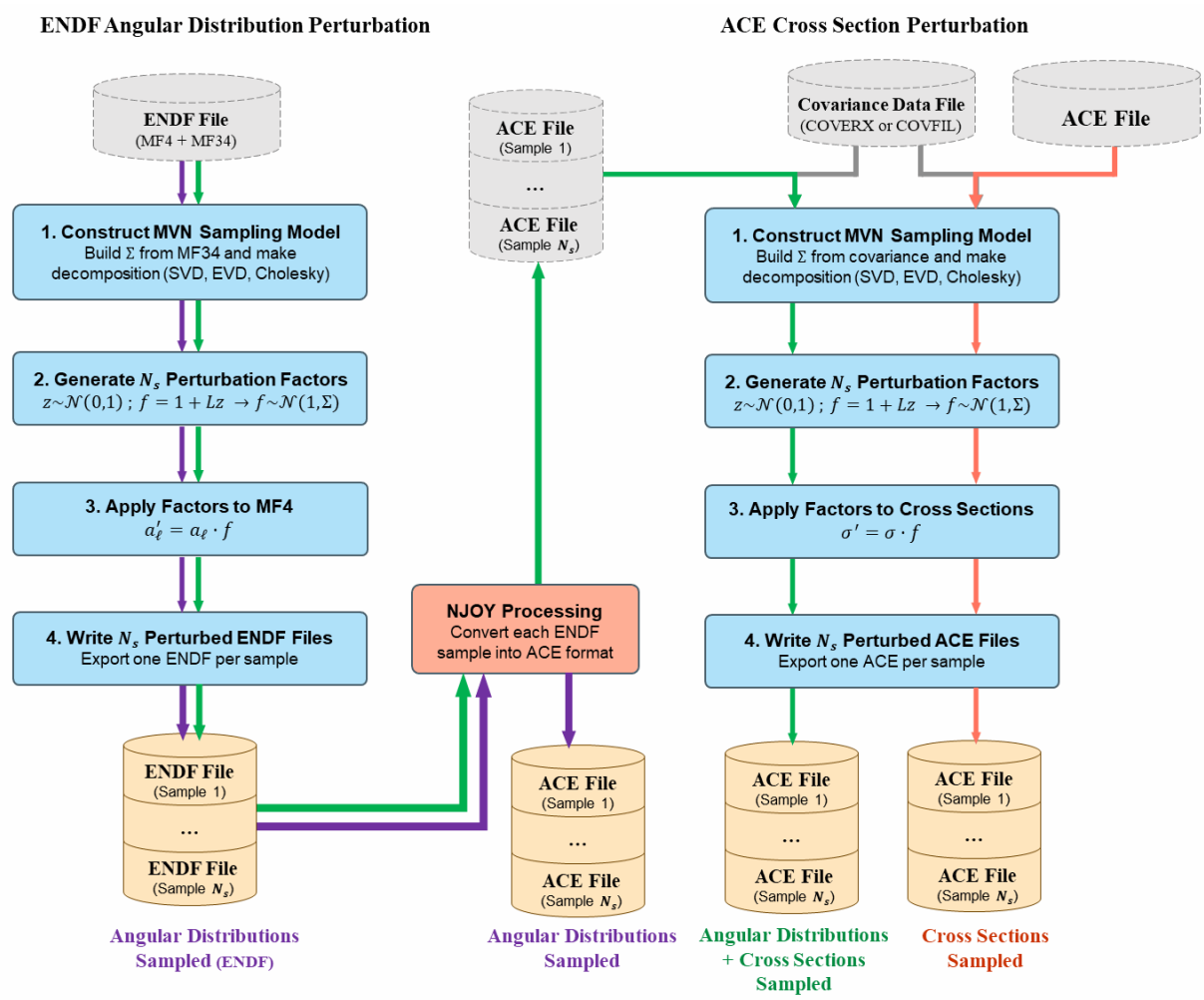


Figure II.6: Overview of the nuclear data perturbation pipelines implemented in `kika-nd`. The ENDF pipeline (left) perturbs angular distributions at the Legendre coefficient level; the ACE pipeline (right) perturbs cross sections directly. Three paths are color-coded: purple for angular distributions only, orange for cross sections only, and green for both combined.

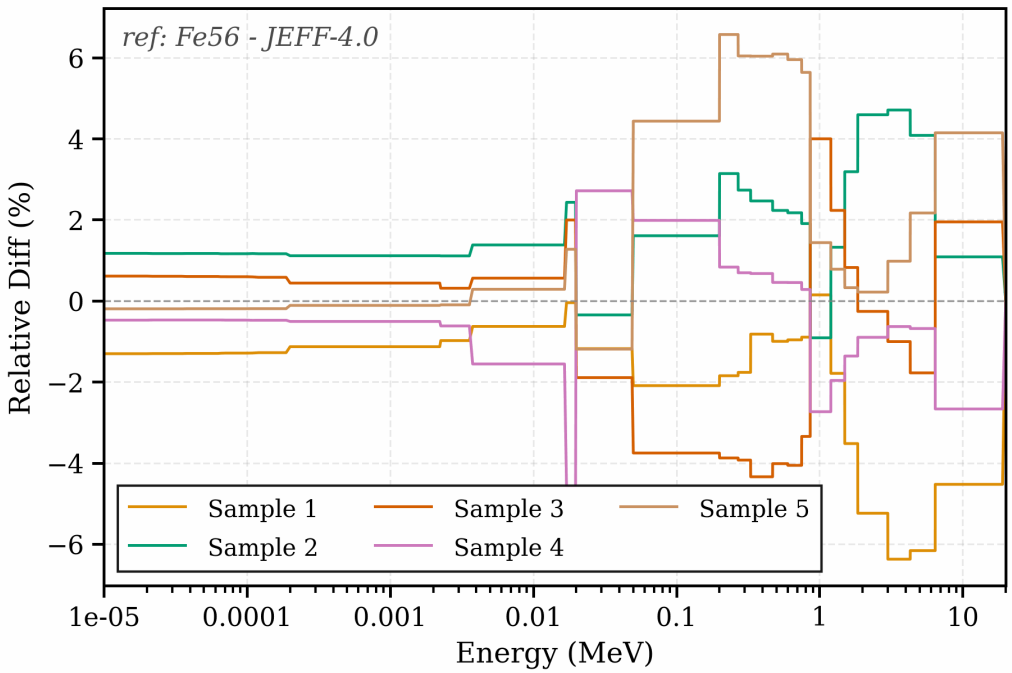


Figure II.7: Relative difference in the ACE ^{56}Fe elastic-scattering cross section (MT=2) for perturbed realizations obtained using multigroup scaling factors sampled from the covariance data.

order and energy node, and applied multiplicatively to the coefficients $a_\ell(E)$ at every energy point in the nominal evaluation. This produces N perturbed ENDF files, as illustrated by the purple path in Figure II.6. Figure II.8 shows the perturbed $a_1(E)$ realizations against the nominal evaluation and its $\pm 1\sigma$ band. Chapter III applies this pipeline to the ^{56}Fe evaluation.

Each perturbed ENDF file can then be processed through the NJOY automation (Sec. 4.3) to produce a perturbed ACE library, enabling angular distribution uncertainties to be propagated through Monte Carlo transport codes.

4.5.3. Combined Perturbation

The two pipelines can be chained to produce nuclear data files that carry correlated perturbations in both cross sections and angular distributions, as shown by the green path in Figure II.6. The procedure is: (1) perturb the ENDF file (angular distributions), (2) process each perturbed ENDF through NJOY to obtain a perturbed ACE library, and (3) perturb the resulting ACE file (cross sections). The two perturbation sets are drawn independently; cross-covariances between `MF33` and `MF34` can be derived from nuclear-model parameter covariances [110] but are rarely provided in current evaluations and are not considered here.

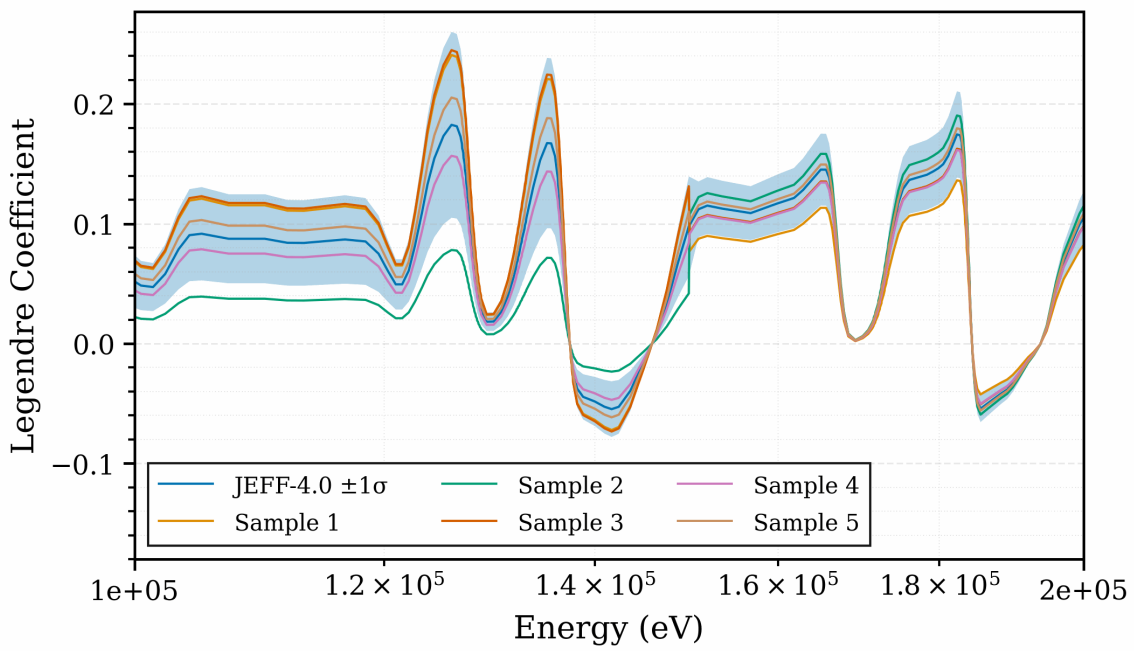


Figure II.8: ENDF angular distribution perturbation for ^{56}Fe elastic scattering. The nominal $a_1(E)$ (blue) with $\pm 1\sigma$ bands from `MF34` covariance is shown together with perturbed realizations sampled from the covariance.

5. Methodology Verification: The PWR Sphere Toy Model

This section applies the SA&UQ framework developed in Secs. 2–4 to a simplified spherical benchmark designed to provide an idealized representation of neutron transport from the core through the reactor vessel in a PWR. The model is intentionally simple so that (i) multiple codes and methods can be compared on identical geometry, (ii) first-order linearity assumptions can be tested explicitly, and (iii) both FOSM and MCS uncertainty propagation can be carried out at moderate computational cost. Results for cross section uncertainties were published in [69]; the angular distribution analysis presented here is new.

5.1. Model Description

Geometry

Figure II.9 illustrates the spherical geometry. A neutron point source at the center emits particles with energies uniformly distributed between 1 and 3 MeV, which provides a simple approximation to the dominant energy range of the fission spectrum commonly represented by a Watt distribution. Successive concentric shells of water (10 ppm boron) and 16MND5 steel surround the source, all enclosed by air. A 1 cm spherical shell positioned in the surrounding air serves as a neutron flux detector. Table II.3 lists the radial dimensions of each layer.

Table II.4 details the isotopic compositions of the three materials. The isotopes retained for uncertainty propagation are highlighted in bold; for steel, these 10 isotopes account for more than 98% of the total weight fraction.

All transport calculations use the ENDF/B-VIII.0 nuclear data library with identical ACE files

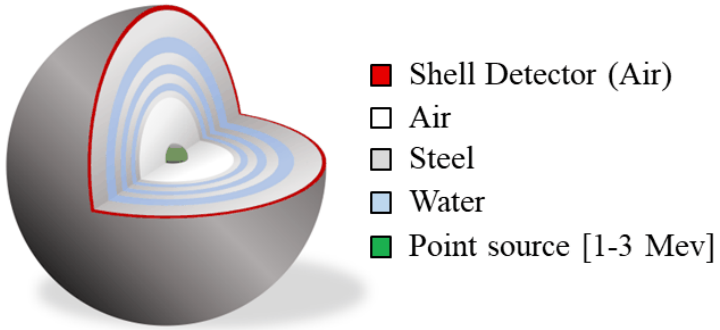


Figure II.9: Schematic representation of the PWR sphere geometry (not to scale). Alternating layers of steel and borated water surround a central point source, with a thin spherical-shell detector located in the outermost air region.

Table II.3: Layer specifications of the spherical geometry. The detector shell is a 1 cm air region scoring the neutron flux in three energy groups: 0–0.1 MeV, 0.1–1.0 MeV, and 1.0–3.0 MeV.

Material	**Inner radius (cm)**	**Thickness (cm)**
Air	0	162
Steel	162	3
Water	165	5
Steel	170	5
Water	175	6.15
Steel	181.15	6.5
Water	187.65	11.75
Steel	199.4	20
Air / Detector	219.4	1
Air	220.4	79.6

Table II.4: Normalized isotopic compositions used in the model: (a) water, (b) air, and (c) steel. Isotopes used for uncertainty propagation are shown in bold.

(a)

Nuclide	Atomic Fraction
^{1}H	**6.6666×10^{-1}**
^{16}O	**3.3333×10^{-1}**
^{10}B	**1.1054×10^{-6}**
^{11}B	**4.4495×10^{-6}**

(b)

Nuclide	Atomic Fraction
^{14}N	7.8848×10^{-1}
^{16}O	2.1151×10^{-1}

(c)

Nuclide	Atomic Fraction	Nuclide	Atomic Fraction
^{12}C	9.1261×10^{-3}	**^{56}Fe**	**8.8106×10^{-1}**
^{13}C	1.0223×10^{-4}	**^{57}Fe**	**2.0634×10^{-2}**
^{28}Si	**3.6398×10^{-3}**	**^{58}Fe**	**2.7833×10^{-3}**
^{29}Si	1.8430×10^{-4}	^{59}Co	2.8212×10^{-4}
^{30}Si	1.2234×10^{-4}	**^{58}Ni**	**4.1907×10^{-3}**
^{31}P	1.4314×10^{-4}	**^{60}Ni**	**1.6014×10^{-3}**
^{32}S	1.3140×10^{-4}	^{61}Ni	6.9335×10^{-5}
^{33}S	1.0372×10^{-6}	^{62}Ni	2.2027×10^{-4}
^{34}S	5.8221×10^{-6}	^{64}Ni	5.5836×10^{-5}
^{36}S	2.3509×10^{-8}	^{63}Cu	4.8281×10^{-4}
^{50}V	2.7198×10^{-7}	^{65}Cu	2.1489×10^{-4}
^{51}V	1.0852×10^{-4}	^{92}Mo	4.2747×10^{-4}
^{50}Cr	1.1591×10^{-4}	^{94}Mo	2.6861×10^{-4}
^{52}Cr	**2.2327×10^{-3}**	^{95}Mo	4.5924×10^{-4}
^{53}Cr	2.5314×10^{-4}	^{96}Mo	4.8234×10^{-4}
^{54}Cr	6.2886×10^{-5}	^{97}Mo	2.7727×10^{-4}
^{55}Mn	**1.3618×10^{-2}**	^{98}Mo	6.9608×10^{-4}
^{54}Fe	**5.5666×10^{-2}**	**^{100}Mo**	**2.7727×10^{-4}**

across simulations. Sensitivity coefficients are computed on the SCALE 6.1 44-group energy grid [89]. The associated variance–covariance matrices, constructed from various ENDF/B evaluations and JENDL-3.3 [89], are used to propagate nuclear data uncertainties. The following reaction channels are included: elastic scattering (MT 2), inelastic scattering (MT 4), radiative capture (n, γ) (MT 102), proton production (n, p) (MT 103), and alpha production (n, α) (MT 107). Since this section focuses on methodology verification, the consistency between the transport library and the covariance library is not examined further.

5.2. Sensitivity Analysis and FOSM Propagation

This subsection presents the sensitivity analysis and first-order uncertainty propagation for cross sections.

5.2.1. Sensitivity Profiles

Sensitivity coefficients as defined in Eq. (II.4) are computed for all isotope–reaction pairs on the 44-group energy grid, using the MCNP differential-operator method (Sec. 2.1.4) and the Serpent collision-history estimator (Sec. 2.1.3). Results are presented for the 1.0–3.0 MeV detector response, the most relevant energy range for irradiation damage studies.

Figure II.10 presents the energy-dependent sensitivity profiles for ^{56}Fe total, elastic, and total inelastic cross sections, comparing four configurations: MCNP, Serpent, MCNP with weight windows, and MCNP with corrected weight windows. MCNP and Serpent are in good agreement across all channels.

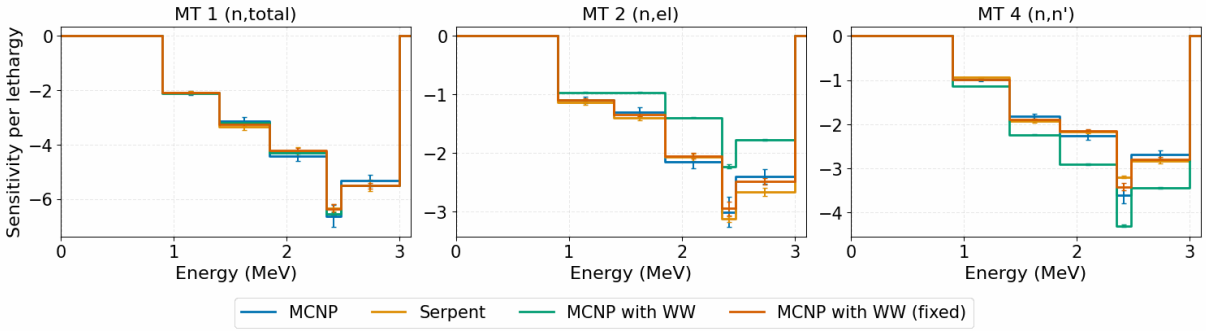


Figure II.10: Sensitivity profiles for ^{56}Fe in the 1–3 MeV detector response: total (MT 1), elastic (MT 2), and total inelastic (MT 4). Four configurations are compared: MCNP, Serpent, MCNP with weight windows, and MCNP with corrected weight windows, each with $\pm 1\sigma$ statistical uncertainty bands.

For ^{56}Fe inelastic scattering, the MCNP `PERT` card returned MT 4 values approximately seven orders of magnitude below the expected sum of individual level contributions. The total inelastic sensitivity is therefore obtained throughout this work by summing MT 51 and MT 52. Sensitivities for radiative capture (MT 102) are 2–3 orders of magnitude lower than elastic and inelastic channels, and Serpent does not resolve any sensitivity for this reaction. Proton production (MT 103) and alpha production (MT 107) are negligible in all configurations.

Weight window parameters were generated with ADVANTG 3.2.0 [14]. As shown in Figure II.10, combining these mesh-based weight windows with the `PERT` card biases the sensitivity coefficients: elastic sensitivities are underestimated and inelastic sensitivities overestimated, although

the errors partially compensate in the total cross section. The weight windows are standard; the bias originates in MCNP's transport when weight windows and `PERT` cards are used together, not in ADVANTG. After this issue was posted on the MCNP Forum, the development team suggested a workaround: setting the `nmfp` parameter (10th entry on the `wwp` card) to a large value (e.g., 9×10^9), which disables the mean-free-path weight check and removes the bias. The developers identified this check, which mesh-based weight windows apply every mean free path by default and cell-based windows do not, as the probable cause, pending confirmation in the source code. This workaround recovers agreement with the reference without degrading variance reduction efficiency.

Table II.5 compares the statistical quality of the `PERT` tally for ^{56}Fe elastic scattering (2.479–3.0 MeV bin) between MCNP and MCNP with weight windows. The figure of merit improves by roughly an order of magnitude when weight windows are applied.

Table II.5: Statistical checks for the first-order Taylor coefficient from the `PERT` card, for ^{56}Fe elastic scattering in the 2.479–3.0 MeV energy bin (10^7 histories).

Configuration	Error	VOV	Slope	FOM
MCNP	0.0219	0.0014	4.7	1.6
With weight windows	0.0060	<0.0001*	10.0	17

*Below the reporting precision of the MCNP output.

Figure II.11 shows the corresponding profiles for ^{1}H. Both codes yield consistent results across all reaction channels. For isotopes that do not undergo inelastic scattering, the default ADVANTG weight windows parameters do not introduce any bias in the sensitivity coefficients.

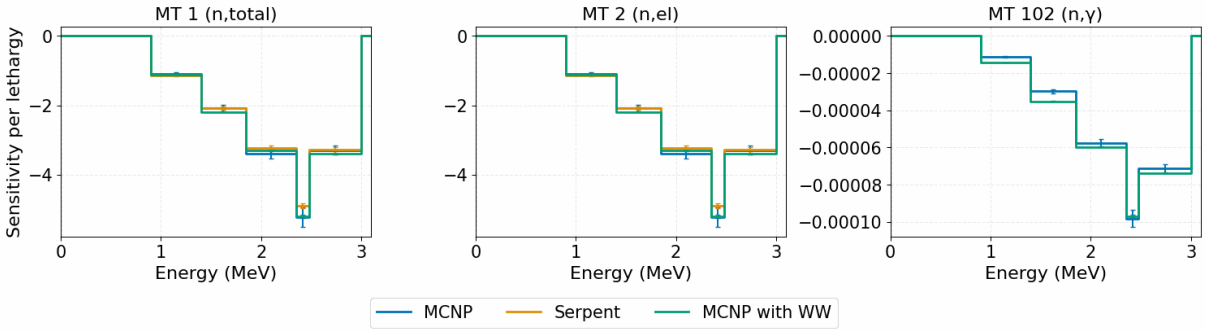


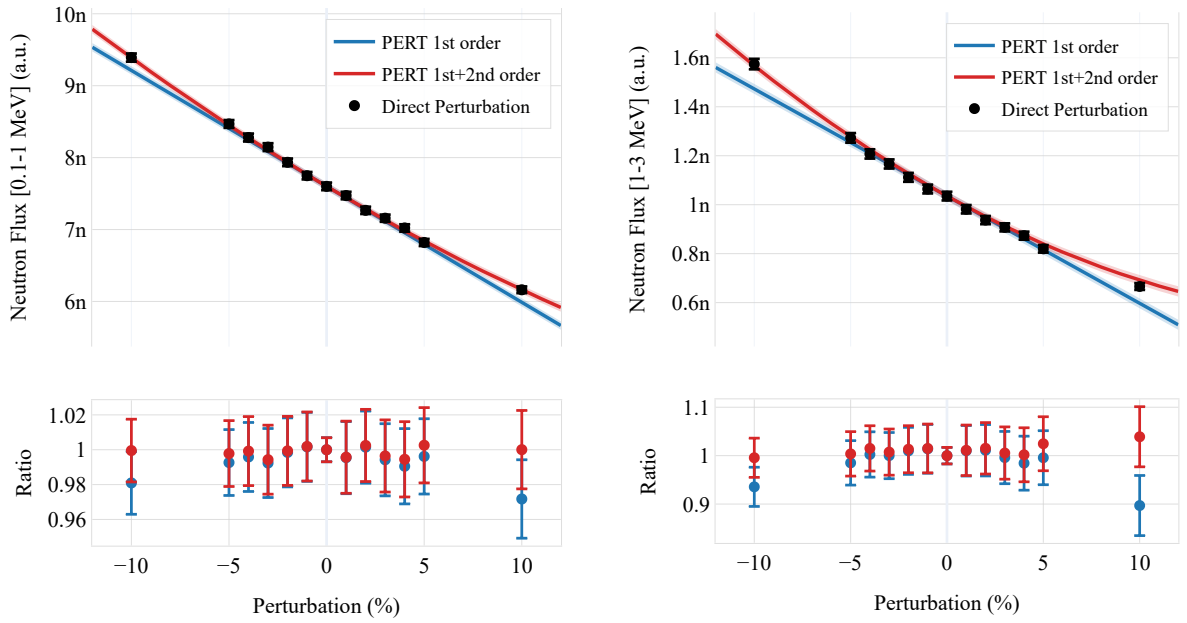
Figure II.11: Sensitivity profiles for ^{1}H reactions in the 1–3 MeV detector response: total (MT 1), elastic (MT 2), and radiative capture (MT 102). MCNP and Serpent results agree within statistical uncertainties.

Computationally, MCNP required roughly twice the CPU time of Serpent for the same statistical quality, both without weight windows, although Serpent's memory consumption was considerably higher, limiting the achievable number of particle histories. In the remainder of this work, all FOSM propagations use sensitivity coefficients from MCNP without weight windows.

5.2.2. First-Order Applicability

The range of validity of the first-order linearization must be verified for the perturbation magnitudes implied by the covariance data. MCNP provides both first- and second-order Taylor coefficients through the `PERT` card, enabling a direct assessment [81].

Figure II.12 compares the perturbed neutron flux as a function of the ^{56}Fe total cross section perturbation. The first-order and first-plus-second-order approximations are compared against direct perturbation results. The linear approximation remains consistent with direct perturbations up to approximately 5%; beyond that threshold, deviations become significant, particularly at higher energies.



(a) (b)

Figure II.12: Perturbed neutron flux as a function of the ^{56}Fe total cross section perturbation, computed with first-order only (blue), first-plus-second-order (red), and direct perturbation (black points). Bottom panels show the ratio of the approximate to the direct result with $\pm 3\sigma$ error bars. (a) 0.1–1.0 MeV; (b) 1.0–3.0 MeV.

A more systematic assessment uses the ratio of second-order to first-order contributions from Eq. (II.64):

$$R_{2/1} = \frac{c_2\, p^2}{c_1\, p}, \tag{II.103}$$

where c_1 and c_2 are the first- and second-order Taylor coefficients and p is the perturbation fraction. For ^{56}Fe, the SCALE 44-group covariance matrix contains maximum uncertainties of approximately 7% for elastic scattering, less than 5% for inelastic scattering, and up to 31% for radiative capture. Figure II.13 shows $R_{2/1}$ for the elastic channel. The steepest slope is approximately 1, meaning that at a 7% perturbation the second-order correction amounts to at most 7% of the first-order effect. For radiative capture, the slope is negligible (3.4×10^{-3}). The first-order approximation therefore remains valid for this system.

5.2.3. Energy-Integrated Sensitivity

As a final cross-check, energy-integrated total cross section sensitivities for ^{56}Fe are compared across all configurations. Figure II.14 shows the integral sensitivity coefficients obtained with a 5% perturbation, including 3σ Monte Carlo uncertainty bars. All methods yield consistent values. The sensitivity in the 1–3 MeV range is approximately twice as large as at lower energies, reflecting the greater importance of ^{56}Fe cross sections in this energy range.

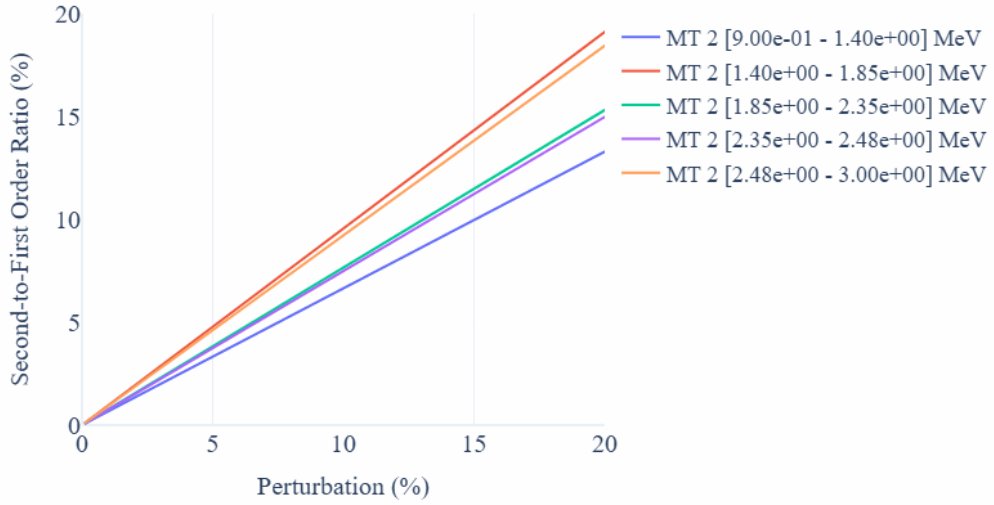


Figure II.13: Second-to-first-order ratio $R_{2/1}$ for ^{56}Fe elastic scattering (MT 2) in the 1–3 MeV detector response, plotted as a function of perturbation magnitude for different energy bins. The steepest slope is approximately 1, confirming that first-order effects dominate for the uncertainty magnitudes present in the covariance data.

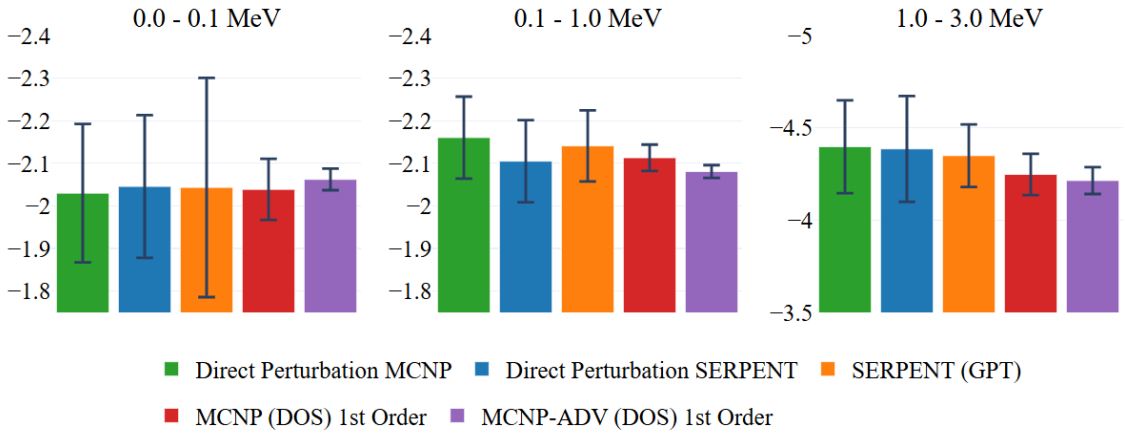


Figure II.14: Energy-integrated sensitivity coefficients for the ^{56}Fe total cross section with 5% perturbation. Results from MCNP, MCNP with weight windows, and Serpent are compared with $\pm 3\sigma$ error bars.

5.2.4. Uncertainty Propagation

Uncertainty propagation is carried out by applying the sandwich rule (Eq. (II.84)), accounting for energy-group and inter-reaction correlations within each isotope when available; cross-isotope correlations are not considered.

Table II.6 presents the propagated nuclear data uncertainty of the neutron flux for each detector energy group. It increases with energy, reaching 11.61% in the 1.0–3.0 MeV range. The accompanying 95% confidence interval is obtained by bootstrap resampling: the sensitivity coefficients are drawn from their Monte Carlo statistical distribution, each propagated through the sandwich rule, and the interval is taken as the 2.5th–97.5th percentile range of the resulting uncertainty. This interval reflects the Monte Carlo uncertainty of the sensitivity coefficients; it does not include the uncertainty of the covariance evaluation itself.

Table II.6: FOSM-propagated nuclear data uncertainty for the neutron flux in each detector energy group. The 95% confidence interval is obtained by bootstrap resampling the Monte Carlo statistical uncertainty of the sensitivity coefficients through the sandwich rule.

Detector energy (MeV)	**Flux per source neutron**	**Propagated ND uncertainty (%)**	**95% CI (%)**
0–0.1	2.64e-09	5.13	4.99–5.27
0.1–1.0	7.65e-09	6.49	6.38–6.59
1.0–3.0	1.07e-09	11.61	11.24–11.96

Table II.7 identifies the dominant variance contributors in the 1.0–3.0 MeV range: ^{56}Fe elastic scattering (55.1%), ^{56}Fe inelastic scattering (27.7%), and ^{1}H elastic scattering (13.8%) together account for over 96% of the total variance.

Table II.7: Major contributors to the propagated variance of the neutron flux in the 1.0–3.0 MeV detector energy group.

Nuclide	**Reaction**	**Integral sensitivity**	**Variance contribution (%)**
^{56}Fe	Elastic	−1.98	55.14
^{56}Fe	Inelastic	−2.19	27.70
^{1}H	Elastic	−2.79	13.83
^{54}Fe	Elastic	−0.22	1.82
^{16}O	Elastic	−0.61	1.11

5.3. Monte Carlo Sampling Propagation

This subsection presents the MCS uncertainty propagation, covering covariance reconstruction, transport simulations, convergence analysis, and computational efficiency.

5.3.1. Sampling and Decomposition

Table II.8 reports the Frobenius-norm relative error (Eq. (II.93)) between the reconstructed and original covariance matrices for ^{56}Fe and ^{1}H, for all reactions with available covariance data, as a function of sample size and sampling–decomposition combination. Sobol sampling with

SVD decomposition consistently yields the lowest relative errors. Among sampling techniques, Sobol outperforms the alternatives, followed by LHS. Aside from the Sobol–SVD combination, no significant differences are observed between decomposition methods.

Table II.8: Frobenius-norm relative error (%) between reconstructed and original covariance matrices for different sampling methods (RS, Sobol, LHS) and decomposition techniques (SVD, Cholesky, EVD), for ^{56}Fe and ^{1}H across varying sample sizes. Best values for each column are shown in bold.

		^{56}Fe				^{1}H			
Sampling	**Decomposition**	**1024**	**512**	**256**	**128**	**1024**	**512**	**256**	**128**
RS	SVD	8.09	15.55	21.55	27.65	0.80	1.68	5.88	4.71
RS	Cholesky	12.00	16.08	20.80	31.05	1.53	5.61	1.07	6.76
RS	EVD	10.11	13.41	20.62	37.12	5.96	0.73	4.56	14.70
Sobol	SVD	**1.36**	**2.33**	**5.20**	**10.28**	**0.05**	0.34	**0.39**	1.16
Sobol	Cholesky	6.67	13.89	18.60	31.14	0.30	0.19	1.96	4.80
Sobol	EVD	5.27	14.72	27.00	39.04	0.68	0.25	0.79	3.00
LHS	SVD	9.25	13.11	17.74	24.90	0.26	**0.10**	0.92	1.87
LHS	Cholesky	9.13	15.03	17.97	29.32	1.34	4.24	1.93	1.78
LHS	EVD	9.37	13.30	20.69	22.53	0.23	0.22	0.80	**0.99**

Based on these results, SVD is adopted as the default decomposition method. All three sampling techniques (RS, Sobol, LHS) are carried forward into the transport stage to enable a convergence comparison.

5.3.2. Simulation Sets and Response Distribution

Perturbed ACE files were generated with SVD decomposition for all three sampling techniques using the `kika-nd` ACE pipeline (Sec. 4.5.1). Transport calculations used MCNP without variance reduction.

Different numbers of particle histories (NPS) were used to achieve different ratios of Monte Carlo statistical uncertainty to observed uncertainty (Eq. (II.96)): 10^8 histories yield approximately 5%, 10^7 approximately 15%, 10^6 approximately 45%, and 10^5 exceeds 100%. Table II.9 summarizes all 34 simulation sets and the fraction of MCNP statistical checks passed (see Chapter I, Sec. 2.1).

For simulations with 10^8 histories, the slope test failed despite satisfactory convergence in relative error, VOV, and FOM. After additional investigation, these results were still considered reliable.

The neutron flux distribution obtained from 1024 Sobol–SVD samples (ID 1) was examined for normality. Figure II.15 shows the resulting histogram. The nominal (unperturbed) flux aligns with the distribution mean. Skewness and kurtosis are small, indicating approximately normal behavior. A slight upper tail produces a low Shapiro–Wilk p-value, but the distribution is considered sufficiently close to normal for the present analysis.

Table II.9: Summary of MCS simulation sets. For each configuration, the table reports the sampling technique, strategy, number of particle histories per run, and the percentage of runs passing all ten MCNP statistical checks for the single spherical-shell flux detector. No variance reduction was applied.

ID	**Sampling**	**Strategy**	**NPS**	**Stat. checks passed (%)**
1	Sobol	TMC	10^8	15.63
2	Sobol	TMC	10^8	15.72
3	Sobol	TMC	10^8	12.60
4	Sobol	TMC	10^8	6.25
5	Sobol	TMC	10^8	31.38
6	RS	TMC	10^8	15.72
7	LHS	TMC	10^8	16.08
8	Sobol	TMC	10^7	88.09
9	RS	TMC	10^7	88.38
10	LHS	TMC	10^7	87.35
11	Sobol	TMC	10^6	70.02
12	RS	TMC	10^6	71.39
13	LHS	TMC	10^6	72.18
14	Sobol	Fast TMC	10^7	88.48
15	RS	Fast TMC	10^7	88.77
16	LHS	Fast TMC	10^7	88.46
17	Sobol	Fast TMC	10^6	84.77
18	RS	Fast TMC	10^6	81.84
19	LHS	Fast TMC	10^6	81.96
20	Sobol	Fast TMC	10^5	2.93
21	RS	Fast TMC	10^5	3.22
22	LHS	Fast TMC	10^5	3.23
23	Sobol	Fast GRS	10^7	89.75
24	RS	Fast GRS	10^7	88.09
25	LHS	Fast GRS	10^7	88.81
26	Sobol	Fast GRS	10^7	92.87
27	RS	Fast GRS	10^7	91.31
28	LHS	Fast GRS	10^7	91.68
29	Sobol	Fast GRS	10^6	70.21
30	RS	Fast GRS	10^6	72.95
31	LHS	Fast GRS	10^6	74.19
32	Sobol	Fast GRS	10^6	92.97
33	RS	Fast GRS	10^6	92.68
34	LHS	Fast GRS	10^6	93.65

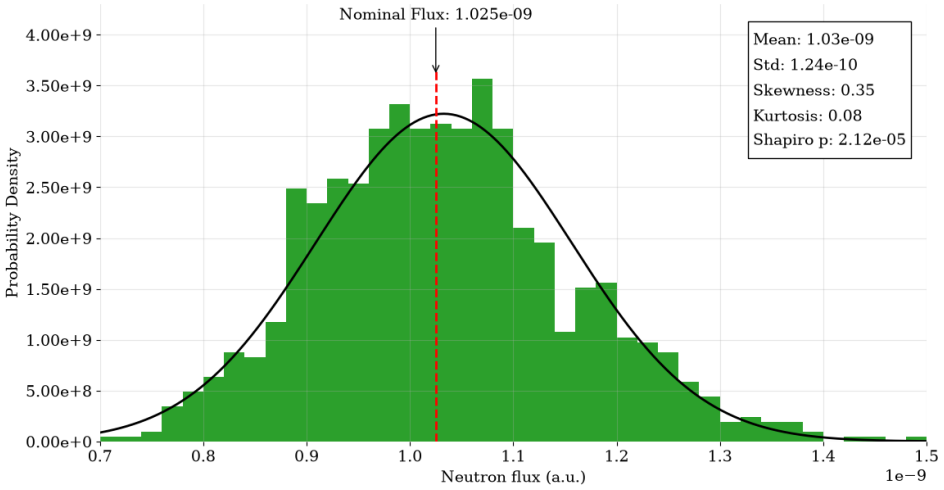


Figure II.15: Distribution of neutron flux values in the 1–3 MeV detector group from 1024 Sobol–SVD samples (TMC, 10^8 histories per run). The fitted normal distribution (solid curve) and statistical metrics (mean, standard deviation, skewness, kurtosis, Shapiro–Wilk p-value) are overlaid.

5.3.3. Convergence and Efficiency

The convergence behavior of the MCS uncertainty estimate as a function of sample size is examined for each combination of sampling technique and simulation strategy. A reference uncertainty value is established by averaging the results from five independent 1024-sample sets (IDs 1–5, Table II.9). The mean of the response converged faster than the standard deviation in all cases, remaining within ±1% of its reference across all relevant configurations; therefore, only the standard deviation convergence is discussed below.

Figure II.16 presents the evolution of the estimated nuclear data standard deviation for the 1–3 MeV range as a function of sample count. Because Sobol sequences possess a nested property (every power-of-two subset of N samples is itself a valid space-filling set), the 1024 samples can be partitioned into 8 sets of 128, 4 of 256, 2 of 512, and 1 of 1024. RS samples are all independent and can be partitioned into arbitrary subsets. The shaded region spans the range of estimates obtained from all such subsets at each sample size. LHS does not share this nested property: each sample size requires a completely new design, so independent LHS runs are shown as individual points without a shaded band. Across all strategies, Sobol converges faster than RS, while LHS yields robust results from small sample sizes.

To quantify the confidence in these estimates, bootstrapping was used to compute 95% confidence intervals at each sample size. Figure II.17 shows the results for Sobol and LHS sampling across three NPS levels. The error bar at each sample size represents the union of the bootstrapped confidence intervals from all nested subsets. For TMC, significant deviations occur when 10^6 histories are used, as Monte Carlo noise dominates. Fast TMC remains close to the reference even with 10^5 histories, although the confidence interval widens to approximately ±12%.

An efficiency criterion was defined to identify the most practical configurations: (1) the reference value must lie within the 95% confidence interval, and (2) the confidence interval width must not exceed ±8%. Table II.10 summarizes the most efficient combinations meeting both criteria. RS is omitted because its performance was consistently inferior to Sobol under the same conditions.

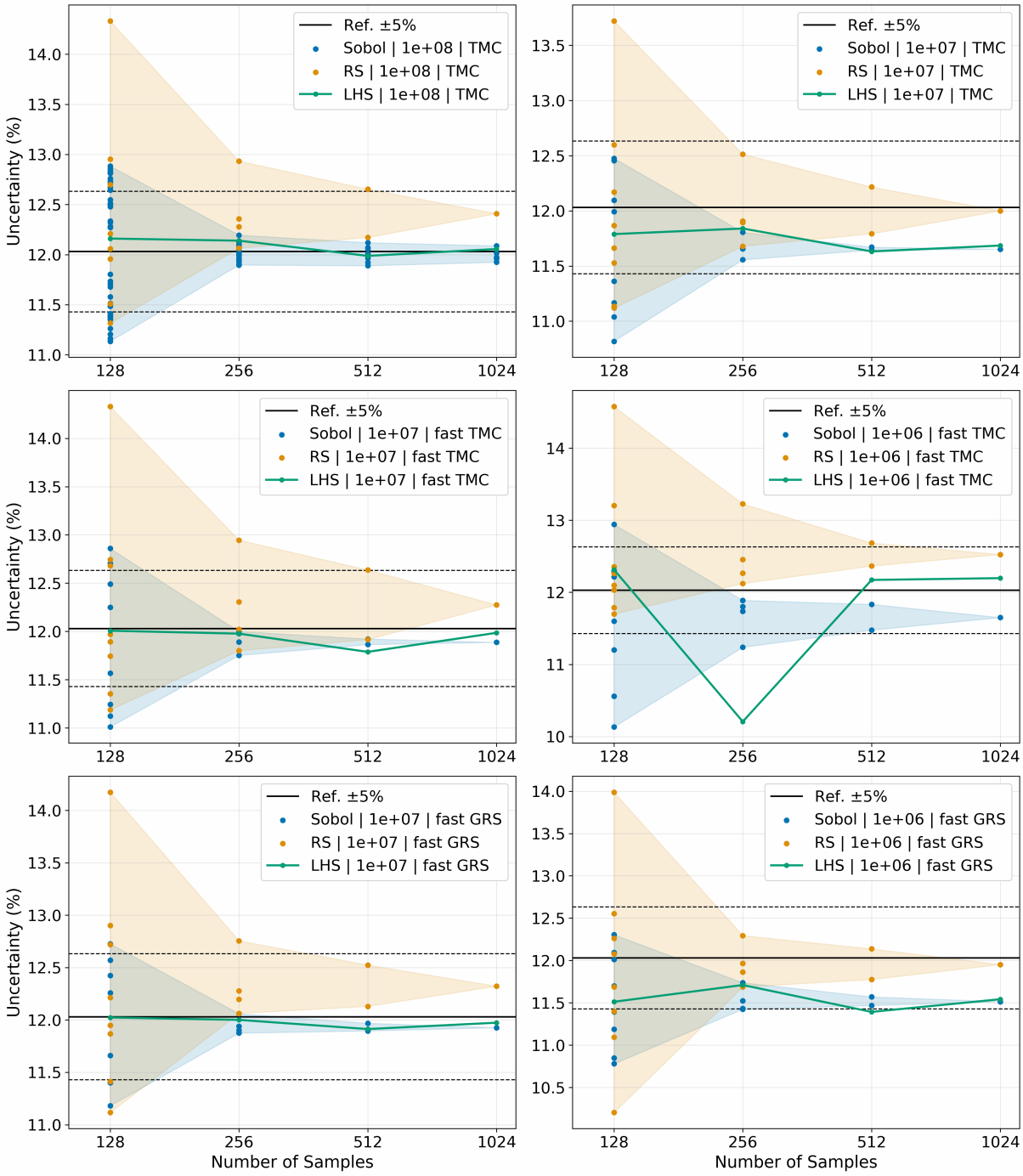


Figure II.16: Convergence of the nuclear data uncertainty estimate (1–3 MeV) as a function of sample size for three simulation strategies (TMC, fast TMC, fast GRS), each shown at two NPS levels. The shaded band spans the range of estimates from nested subsets of the 1024 samples for Sobol and RS; LHS points are independent runs at each sample size. The dashed band indicates ±5% of the reference value.

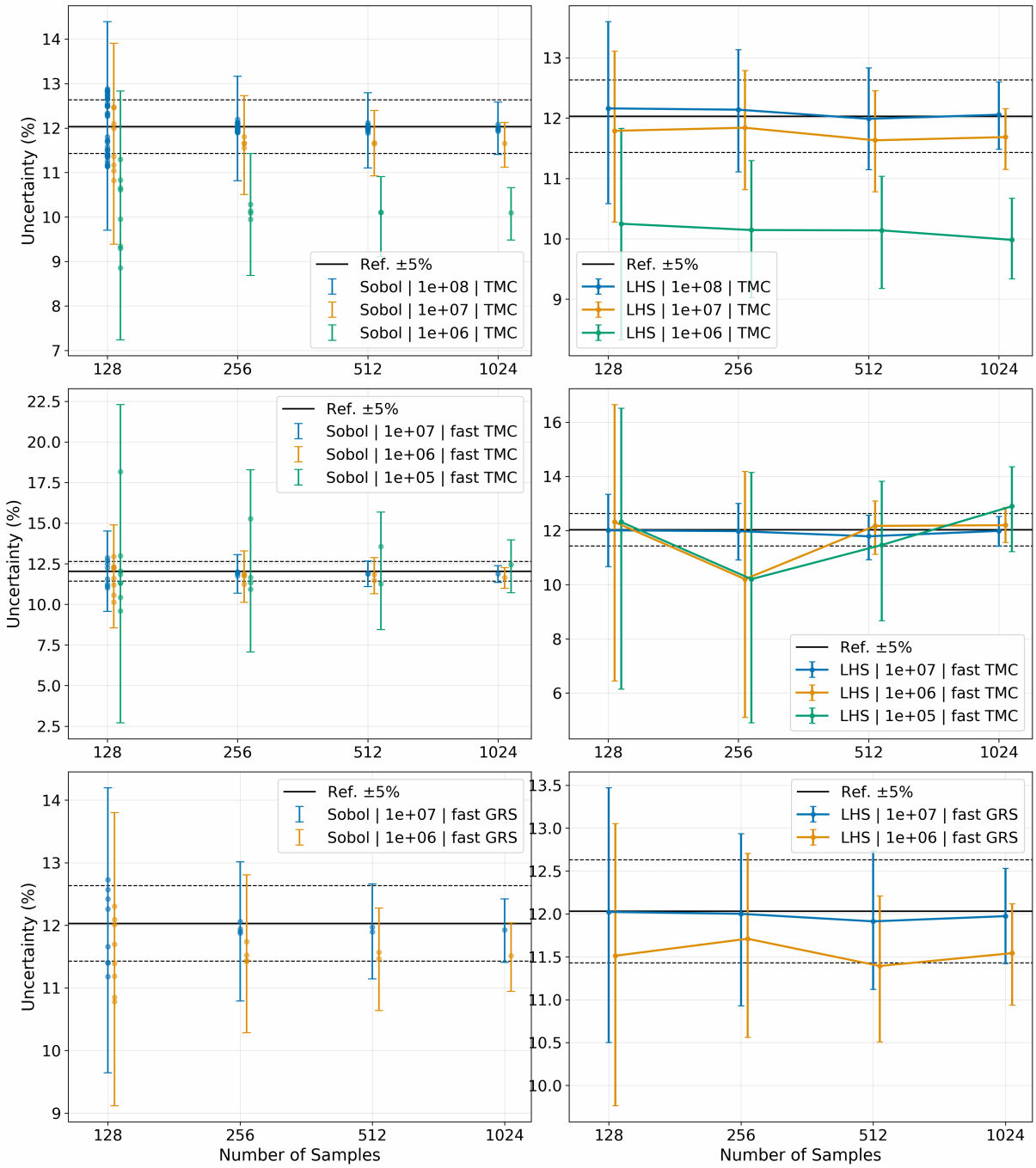


Figure II.17: Convergence of the nuclear data uncertainty estimate with 95% confidence intervals (error bars), grouped by sampling technique and simulation strategy: (a, b) TMC; (c, d) fast TMC; (e, f) fast GRS. Results for Sobol and LHS sampling are shown at three different NPS levels.

Table II.10: Most efficient MCS configurations meeting the convergence criteria (reference within 95% CI, CI width $\leq \pm 8\%$) for the 1–3 MeV detector response.

Strategy	**Sampling**	**NPS**	**Samples**	**ND σ (%)**	**95% CI**	**CI width (%)**
TMC	Sobol	10^7	512	11.67	10.88–12.30	6.12
TMC	LHS	10^7	512	11.63	10.77–12.44	7.19
Fast TMC	Sobol	10^6	1024	11.65	11.01–12.28	5.45
Fast TMC	LHS	10^6	1024	12.20	11.57–12.83	5.16
Fast GRS	Sobol	2×10^6	512	11.52	10.73–12.28	6.74
Fast GRS	LHS	2×10^6	512	11.39	10.54–12.21	7.34

5.4. Comparison of Methods and Discussion

Table II.11 compares the final FOSM and MCS uncertainty estimates with a reference value computed by averaging five independent 1024-sample sets (IDs 1–5). For MCS, the efficient configuration from Table II.10 is used (fast TMC, Sobol, 10^6 histories, 1024 samples). Both methods agree well with the reference across all energy groups, with confidence intervals that overlap the reference interval in every group.

Table II.11: Comparison of propagated nuclear data uncertainty from FOSM and MCS (fast TMC, Sobol, 10^6 histories) with the MCS reference value (average of IDs 1–5). All intervals are 95% confidence intervals obtained by bootstrap resampling: MCS resamples the nuclear-data sample ensemble; FOSM resamples the Monte Carlo statistical uncertainty of the sensitivity coefficients.

Detector energy (MeV)	**Reference (%) [95% CI]**	**MCS (%) [95% CI]**	**FOSM (%) [95% CI]**
0–0.1	5.26 (5.16–5.36)	5.06 (4.75–5.34)	5.13 (4.99–5.27)
0.1–1.0	6.56 (6.44–6.69)	6.59 (6.26–6.88)	6.49 (6.38–6.59)
1.0–3.0	12.00 (11.77–12.22)	11.65 (11.01–12.28)	11.61 (11.24–11.96)

The close agreement between the two methods confirms that both provide consistent uncertainty estimates for this system. Differences are expected to become more pronounced for problems exhibiting stronger nonlinear behavior.

For MCS, the systematic exploration of sampling techniques, decomposition methods, and simulation strategies identifies Sobol–SVD as the most efficient combination for reproducing the target covariance structure with a limited number of samples. Both Sobol and LHS consistently outperform RS under identical conditions, with no notable differences among decomposition methods aside from the Sobol–SVD pairing.

In terms of strategies, fast TMC and fast GRS achieve comparable accuracy to TMC with an order of magnitude fewer particle histories per run; fast TMC and fast GRS are comparable in total computational cost, while TMC requires approximately five times more resources. The efficient fast TMC configuration (10^6 histories, 1024 samples) achieves accurate results ($\sim$5% CI width). A direct comparison of computational cost with FOSM is not straightforward, since the FOSM runs in this work were not optimized for efficiency: each isotope was computed in a separate run, and additional perturbation cards were included for the linearity study, both of which

increased the total CPU time beyond what is strictly necessary for uncertainty propagation.

These findings are consistent with Rochman et al. [100], with one notable difference: the present work demonstrates that the Monte Carlo noise threshold for TMC can be relaxed to approximately 15% of the observed uncertainty (compared to the recommended 5%) while still producing satisfactory results. For fast TMC, acceptable performance is obtained at approximately 45% Monte Carlo uncertainty relative to the observed value, consistent with the reported range of one-third to one-half. As discussed in Sec. 3.2.3, when the MC statistical uncertainty exceeds these thresholds the propagated nuclear data uncertainty can only be approximated, as the subtraction in Eq. (II.96) becomes numerically unstable.

FOSM can effectively propagate nuclear data uncertainties to the neutron flux provided that: (1) weight-window parameters do not alter sensitivity coefficient calculations; (2) the problem exhibits sufficient linearity to justify a first-order approximation; and (3) uncertainties associated with angular distributions and other nuclear data quantities whose sensitivities cannot be computed by MCNP `PERT` cards are negligible. When any of these conditions is not met, MCS is the preferable option. Nevertheless, the FOSM sensitivity analysis remains useful for identifying the most influential input parameters regardless of the propagation method chosen.

5.5. Angular Distribution Sensitivity and Uncertainty

The preceding subsections address only cross section uncertainties (`MF3` data) and used the SCALE 44-group covariance library, which does not provide angular distribution covariances for ^{56}Fe. Angular distribution uncertainties (`MF4`/`MF34` data) are an independent source, examined here using nuclear data libraries that do evaluate them. The main goal of this subsection is the angular distribution covariance study; the cross section covariances of the same libraries are also compared as a secondary check. Since the MCNP `PERT` card cannot perturb Legendre coefficients, angular distribution sensitivities are computed with Serpent.

Two nuclear data libraries are compared: JEFF-4.0 and JENDL-5. JEFF-4.0 and ENDF/B-VIII.1 share the same ^{56}Fe evaluation but do not include angular distribution covariances; for JEFF-4.0, these are taken from JEFF-3.3. JENDL-5 provides its own independent evaluation and covariances.

Covariance data are collapsed to the SCALE 56-group energy grid, the cross section covariances with NJOY and the angular-distribution covariances with the KIKA collapse of Sec. 4.4. Sensitivity coefficients for cross sections are calculated with MCNP and for angular distributions with Serpent, all on the same 56-group structure. FOSM uncertainties are propagated with the sandwich rule; MCS uncertainties are obtained by perturbing the Legendre coefficients with the `kika-nd` ENDF pipeline (Sec. 4.5.2) and processing the perturbed files with NJOY to generate ACE libraries for transport.

5.5.1. Sensitivity

Figure II.18 presents the sensitivity of the 1–3 MeV detector response to the Legendre moments of ^{56}Fe elastic scattering angular distributions. For JEFF-4.0, sensitivities to the first six Legendre orders are shown; for JENDL-5, only the first-order moment is shown, because JENDL-5 evaluated angular distribution covariances for a single Legendre order. Sensitivities to higher JENDL-5 orders could be computed but carry no covariance to propagate. Both libraries yield similar sensitivities for the first-order moment (P_1). The first three orders (P_1–P_3) are the most

sensitive; orders above 3 have sensitivities close to zero.

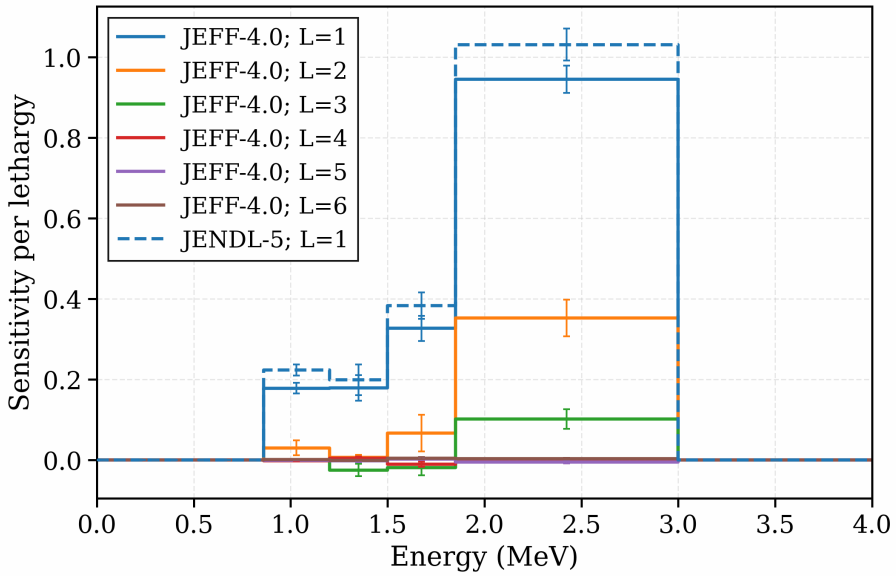


Figure II.18: Sensitivity of the 1–3 MeV detector response to the Legendre moments of ^{56}Fe elastic scattering angular distributions. JEFF-4.0 results (solid lines) are shown for orders 1–6; JENDL-5 (dashed) for order 1.

5.5.2. Uncertainty Quantification

Table II.12 summarizes the propagated uncertainties for the 1–3 MeV detector response, decomposed into angular distribution (AD) and cross section (XS) contributions. FOSM and MCS results agree within their statistical uncertainties for both libraries. For JEFF-4.0, the angular distribution contribution is negligible, accounting for less than 1% of the total variance. For JENDL-5, angular distributions are significant: the propagated AD uncertainty is approximately 11.5%, contributing about 20% of the total variance. The cross section contributions show the same library dependence. The JEFF-4.0 XS uncertainty (about 11%) is close to the SCALE 44-group result of 11.61% reported earlier (Table II.6), whereas the JENDL-5 XS uncertainty (about 23%) is roughly twice as large.

Table II.12: Propagated nuclear data uncertainty for the 1–3 MeV detector response from ^{56}Fe, decomposed into angular distribution (AD) and cross section (XS) contributions, for the FOSM and MCS methods. The last column shows the relative variance decomposition between the two sources.

Library	Method	AD (%)	XS (%)	Total (%)	Var. AD / XS (%)
JEFF-4.0	FOSM	0.82	11.19	11.22	0.5 / 99.5
JEFF-4.0	MCS	0.75	10.94	10.97	0.5 / 99.5
JENDL-5	FOSM	11.53	23.00	25.73	20.1 / 79.9
JENDL-5	MCS	11.47	22.18	24.97	21.1 / 78.9

Table II.13 presents the FOSM variance decomposition by Legendre order for each library. For JEFF-4.0, P_1 dominates (66.6%), followed by P_6 (16.2%) and P_2 (14.7%). The relatively large contribution of P_6, despite its near-zero sensitivity, arises because the covariance assigns a_6

an absolute uncertainty comparable to that of a_1: the stored value is a large fraction of a small coefficient. Negative variance contributions for P_4 and P_5 arise because these orders have sensitivities of opposite sign to the dominant contributors; the positive correlations between orders then produce negative cross terms in the sandwich rule. For JENDL-5, the entire angular distribution variance comes from P_1, since the library provides covariance data for only one Legendre order.

Table II.13: FOSM variance decomposition of the angular distribution uncertainty by Legendre order for ^{56}Fe elastic scattering (1–3 MeV detector response). Negative contributions arise from opposite-sign sensitivities between Legendre orders.

		JEFF-4.0		**JENDL-5**	
Order	**MT**	**Variance**	**% Total**	**Variance**	**% Total**
P_1	4001	4.52×10^{-5}	66.6	1.33×10^{-2}	100.0
P_2	4002	1.00×10^{-5}	14.7	—	—
P_3	4003	3.82×10^{-6}	5.6	—	—
P_4	4004	-5.95×10^{-7}	-0.9	—	—
P_5	4005	-1.54×10^{-6}	-2.3	—	—
P_6	4006	1.10×10^{-5}	16.2	—	—

Cross sections and angular distributions were also perturbed both separately and simultaneously via MCS. Since no cross-correlations between these two data types are considered, the perturbations are independent. In both libraries, the combined propagation yielded a total variance equal to the sum of the individual contributions, confirming that the two sources can be propagated either jointly or separately without introducing bias. A joint propagation reduces the total number of transport simulations, while separate propagations provide a direct decomposition of the uncertainty budget.

These results show that the contribution of angular distribution uncertainties to the total response depends strongly on the nuclear data library. The reasons for the large differences between JEFF-4.0 and JENDL-5 covariances, and whether these covariance data adequately represent the state of knowledge, are examined in detail in Chapter III.

6. Conclusions

This chapter has developed the SA&UQ framework used throughout the thesis and verified it on a simplified benchmark. The main outcomes are summarized below.

Sensitivity analysis

First-order perturbation theory, its generalized extension (GPT), and the differential-operator sampling (DOS) method provide the mathematical basis for computing sensitivity coefficients from forward and adjoint transport solutions. The collision-history estimator implemented in Serpent and the differential-operator method available in MCNP were compared on the PWR sphere model. Both codes yield consistent sensitivity profiles for cross sections; for angular distributions (Legendre coefficients), only Serpent provides the necessary estimator. A practical contribution of this work is the identification of a bug in MCNP whereby mesh-based weight

windows bias the sensitivity coefficients. Disabling the mean-free-path weight check removes the bias on this benchmark, but the workaround does not always succeed: it did not work for the full PWR model of Chapter IV, where weight windows could not be used for the sensitivity calculation.

Uncertainty propagation

Two complementary propagation methods were applied: FOSM (sandwich rule) and Monte Carlo sampling. FOSM is computationally efficient and provides a direct variance decomposition by isotope, reaction, and energy group. MCS is more general and captures nonlinear effects, but requires careful treatment of the separation between nuclear data and Monte Carlo statistical uncertainties. Three separation strategies (TMC, fast TMC, fast GRS) were compared. Fast TMC with Sobol–SVD sampling achieves accurate results with an order of magnitude fewer histories per run than classic TMC.

The `kika-nd` framework

The `kika-nd` library was developed to implement the full chain from covariance parsing through nuclear data perturbation to transport post-processing. Two pipelines were presented and verified on the PWR sphere model: ACE-level perturbation for cross sections and ENDF-level perturbation for angular distributions. The two can be combined to propagate cross section and angular distribution uncertainties simultaneously.

Angular distribution uncertainties

Application of the framework to angular distribution covariances revealed a strong library dependence. With JEFF-4.0 covariances (inherited from JEFF-3.3), angular distributions contribute less than 1% of the total variance in the 1–3 MeV detector response. With JENDL-5 covariances, the contribution rises to about 20%. This discrepancy motivates the data-driven evaluation of ^{56}Fe angular distribution covariances presented in Chapter III.

Chapter III

Data-Driven Evaluation of ^{56}Fe Elastic-Scattering Angular Distributions and Covariances

Contents

1 Introduction . . . 106
1.1 Role of Angular Distributions in RPV Fluence . . . 106
1.2 Status of ^{56}Fe Angular Distributions in Major Libraries . . . 107
1.3 Scope of the Evaluation . . . 110
2 Mathematical Formalism for Uncertainty Quantification . . . 111
2.1 Propagation of MF34 Uncertainties to the Differential Cross Section . . . 111
2.1.1 General first-order propagation . . . 111
2.1.2 Simplification adopted in this work (shape-only bands) 112
2.2 Physical Constraints and Cross-Section Coupling . . . 112
3 Experimental Database and Comparison Methodology . . . 113
3.1 Experimental Configurations and Measurement Extractions . . 113
3.2 Energy Resolution Folding and Normalization Bias . . . 115
3.3 Treatment of Experimental Uncertainties . . . 117
3.4 Goodness-of-Fit Metric . . . 119
3.5 Preliminary Library Comparison . . . 121
4 Evaluation Pipeline: Fitting and Sampling Methodology . . . 122
4.1 Pipeline Overview . . . 122
4.2 Data Selection and Energy Grid . . . 125
4.3 Nominal Fit . . . 125
4.3.1 Block-correlated generalized least squares . . . 125
4.3.2 Per-point weights for experimental heterogeneity . . . 126
4.3.3 Automatic degree selection via AIC . . . 126
4.3.4 Angular-band discrepancy model . . . 127
4.3.5 Model averaging and degree sampling . . . 128

4.4 Per-Bin Variance Pass — Monte Carlo Pass 1 129
4.4.1 Perturbation model 130
4.4.2 Refit with frozen c_0 and per-sample degree 130
4.4.3 Positivity enforcement 131
4.5 Kernel-Weight Correlation Pass — Monte Carlo Pass 2 131
4.5.1 Overlap weights from TOF resolution 132
4.5.2 Perturbation model 132
4.6 Covariance Assembly 133
4.6.1 Sample covariance and block structure 134
4.6.2 Near-zero regularization 134
4.6.3 Adaptive multigroup collapse 135
4.7 Pipeline Output: Use and Limitations 135
5 Results and Comparison 136
5.1 Pipeline Output: Use and Limitations 137
5.2 Legendre Coefficient Comparison Across Evaluations 137
5.3 Uncertainty Band Comparison at Selected Energies 143
5.4 Goodness-of-Fit Analysis 145
5.5 Integral Cross-Check: Criticality Benchmarks 149
6 Conclusions 152

This chapter presents a data-driven evaluation of elastic-scattering angular distributions and their covariances for ^{56}Fe in the 847 keV–4 MeV neutron energy range. The evaluation is built directly from experimental data in the EXFOR database and compared against the JEFF-4.0 and JENDL-5 nuclear data libraries. The chapter starts with the physics motivation, the status of the existing libraries in this energy range, and a quantitative measure of their agreement with experimental data (Sec. 1). It then introduces the mathematical formalism for propagating Legendre-coefficient uncertainties to the differential cross section (Sec. 2), the experimental database and the comparison metric (Sec. 3), and the evaluation pipeline itself (Sec. 4). Results and comparison against experiment are presented in Sec. 5, and the main findings are summarized in Sec. 6. Supporting material that complements the main thread of the chapter is collected in Appendix E.

1. Introduction

1.1. Role of Angular Distributions in RPV Fluence

Accurate prediction of neutron fluence at the RPV requires reliable nuclear data for the materials located between the core and the vessel [3]. In light-water reactors, stainless steels constitute a significant fraction of structural components and shielding regions. ^{56}Fe, the dominant isotope of natural iron, plays a major role in the attenuation and angular redistribution of neutrons traveling through these regions.

Quantifying nuclear-data-driven uncertainties in RPV fluence requires not only nominal evaluated data but also covariance information that enables uncertainty propagation through transport calculations [100]. While angular distributions are sometimes approximated as isotropic in core calculations, this approximation is inadequate for deep-penetration problems such as RPV fluence [111]. The scattering angle after each interaction determines whether neutrons continue outward (forward scattering) or are redirected back toward the core (backward scattering). Uncertainties in anisotropic elastic scattering on ^{56}Fe can therefore contribute non-negligibly to the predicted vessel fluence, particularly where scattering is strongly anisotropic and cross sections exhibit resonance-driven fluctuations [112]. Without consistent and credible covariance data, propagated uncertainty estimates may be incomplete or misleading [100].

Two routes exist for carrying nuclear-data uncertainty into a vessel-fluence prediction: validating the calculation against representative integral measurements, or propagating the evaluated covariances through the transport calculation. The first is the more direct and reliable when a benchmark closely matches the application, but representative integral data for the vessel configuration are scarce, so the covariance route carries much of the burden. That route is only as sound as the covariances it rests on, which for ^{56}Fe elastic angular distributions are sparse and inconsistent across libraries, and this is what motivates their close examination in this chapter.

1.2. Status of ^{56}Fe Angular Distributions in Major Libraries

Modern evaluations provide updated nominal elastic-scattering angular distributions for ^{56}Fe (ENDF-6 `MF4`), but the corresponding covariance information (`MF34`) is uneven across libraries. JEFF-4.0 and ENDF/B-VIII.1 share the same `MF4` evaluation, and neither library carries its own angular-distribution covariance for this reaction. The `MF34` block associated with JEFF-4.0 in this work is therefore sourced from JEFF-3.3, where it had itself been inherited from JEFF-3.0 (released in 2005); it covers the first six Legendre orders and predates the current `MF4`. Since `MF34` stores relative covariances, each entry is a fraction of the coefficients of the evaluation it was derived from, so the absolute uncertainty it implies for the current `MF4` is not the one it originally described. JENDL-5 (2021) supplies covariance information only for the first Legendre coefficient. TENDL-2023 reports model-based covariance matrices in which the entire `MF34` content for this reaction is condensed into a single very broad energy group extending up to hundreds of MeV, and the relative uncertainties assigned to some Legendre coefficients are very large.

The nominal elastic cross section $\sigma_{\mathrm{el}}(E)$ and the first Legendre coefficient $a_1(E)$ in this window are shown in Figures III.1 and III.2. The cross section fluctuates on the keV scale, following the underlying resonance structure. The evaluated $a_1(E)$ shows analogous resonance-like structure below about 2.5 MeV, where the experimental angular-distribution grid is fine enough to resolve it, though it fluctuates at a much lower rate in energy than the cross section; above 2.5 MeV the experimental energies are spaced more widely, the evaluated coefficients are correspondingly smoother, and the resonant fluctuations of the underlying cross section are no longer reflected in a_1.

Beyond the central values, the available `MF34` sections show different patterns in how uncertainty is distributed across orders. In JEFF-4.0 the low-order coefficients ($\ell = 1, 2, 3$) carry relative uncertainties of a few percent, while the higher-order coefficients ($\ell = 4, 5, 6$) carry values one to two orders of magnitude larger (Figure III.3). A large fractional value is not on its own the sign of an implausible uncertainty: a_4 to a_6 cross zero repeatedly in this range, so the ratio $\sigma(a_\ell)/|a_\ell|$ diverges wherever a coefficient passes near zero while the absolute uncertainty stays comparable

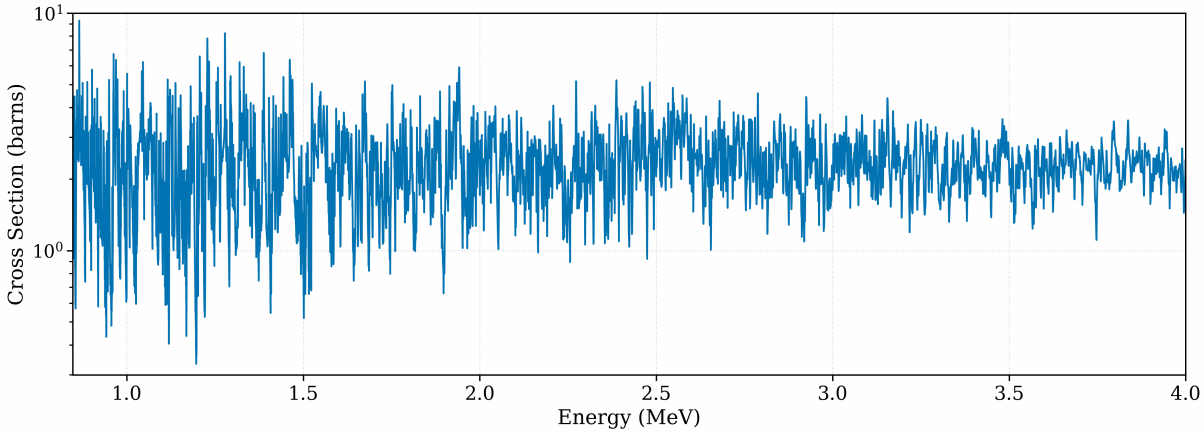


Figure III.1: ^{56}Fe elastic-scattering cross section from JEFF-4.0 in the 847 keV–4 MeV range.

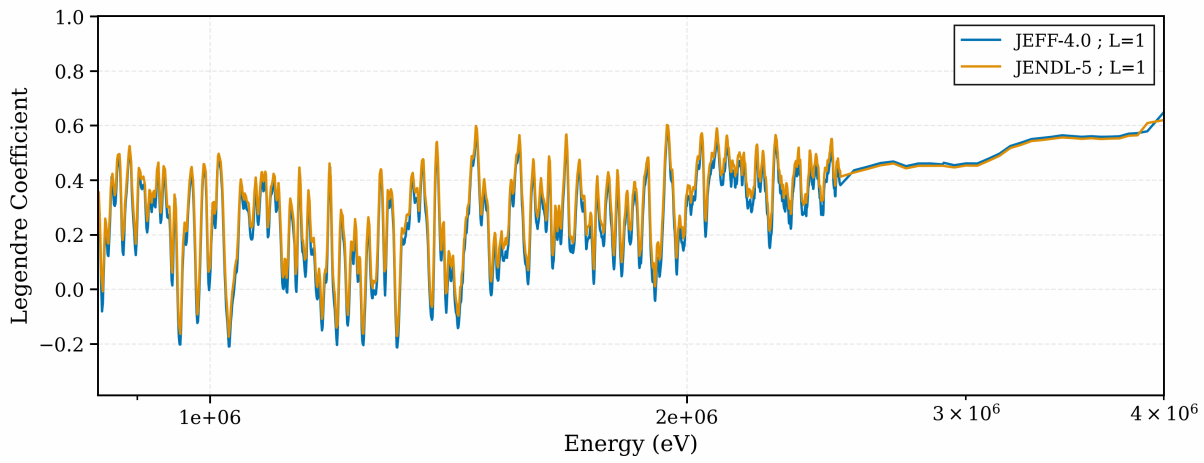
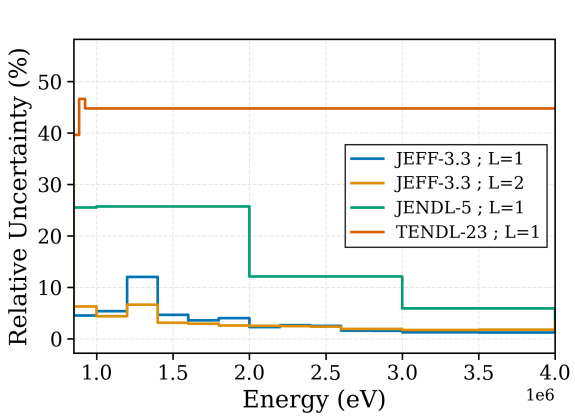
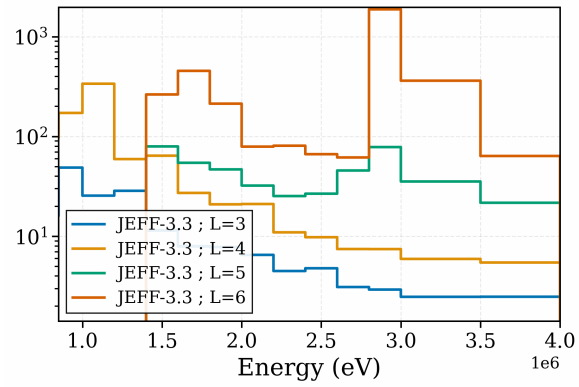


Figure III.2: First Legendre coefficient $a_1(E)$ for ^{56}Fe elastic scattering in the 847 keV–4 MeV range, shown for JEFF-4.0 and JENDL-5.

to the one carried by the low orders. What makes these values consequential is their combination with a covariance energy grid much coarser than the structure of the coefficients, so that one value covers a range across which $|a_\ell|$ moves by orders of magnitude. When propagated to the differential cross section, the result is oscillatory uncertainty bands (Figure III.4a). JENDL-5's single-order representation has the opposite limitation: with only $\ell = 1$ uncertain, the band is dipolar in μ and collapses to zero at $\cos\theta = 0$, an artefact of carrying covariance only for $\ell = 1$ and not a physical property of the differential cross section at 90° (Figure III.4b).

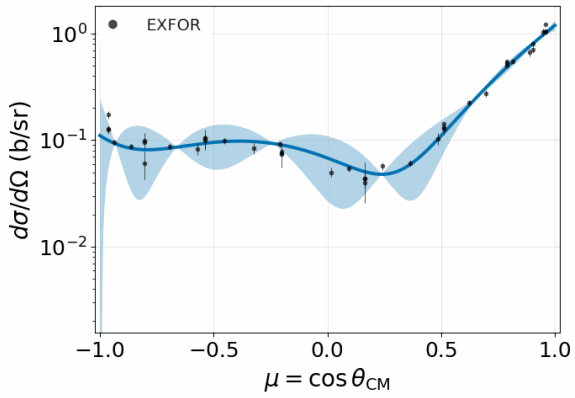


(a) Low-order coefficients ($\ell = 1$, $\ell = 2$).

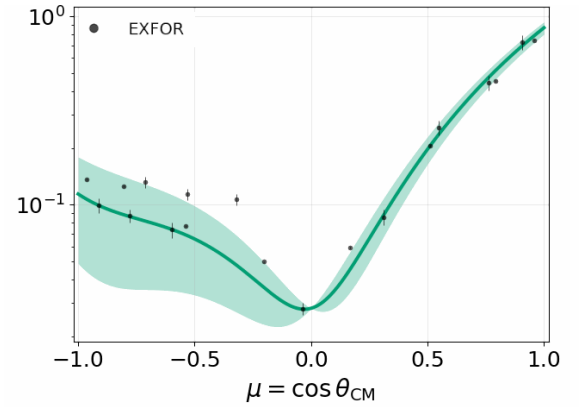


(b) High-order coefficients ($\ell = 3$ to $\ell = 6$), on a logarithmic scale.

Figure III.3: Relative uncertainties of the Legendre coefficients a_ℓ for ^{56}Fe elastic scattering from the available angular-distribution covariances.



(a) $\ell = 6$ contribution in JEFF-4.0 at $E_n = 2.800$ MeV.

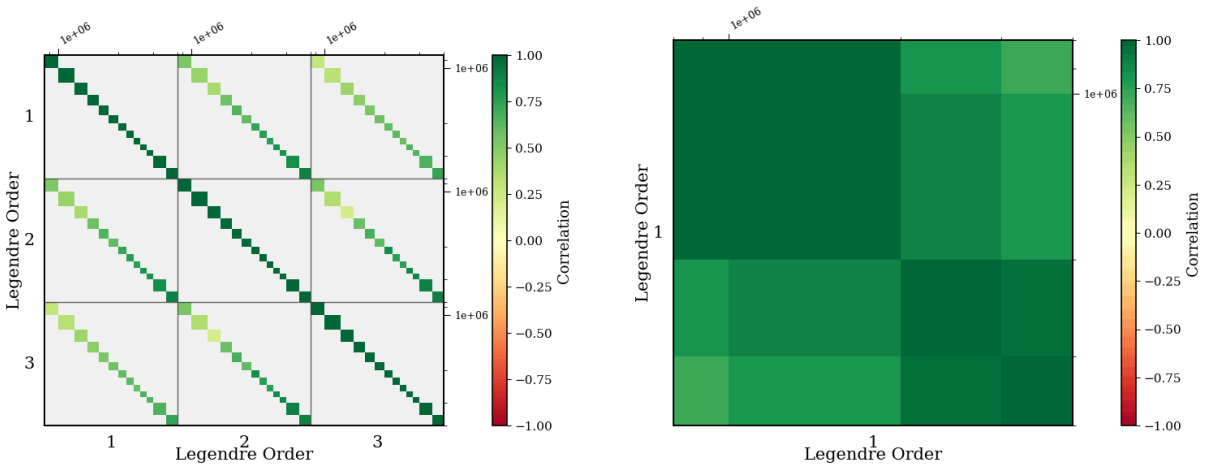


(b) $\ell = 1$ contribution in JENDL-5 at $E_n = 1.571$ MeV.

Figure III.4: Contribution of a single Legendre order to the propagated $\pm 1\sigma$ uncertainty band on the ^{56}Fe elastic differential cross section. Black markers are EXFOR points within the corresponding energy bin.

The energy structure of the covariances is also coarse, and in JEFF-4.0 it differs from one Legendre order to the next: orders $\ell = 1$ to 4 are partitioned into thirteen energy groups across the 847 keV–4 MeV interval, while $\ell = 5$ and $\ell = 6$ carry ten groups above 1.4 MeV and no declared uncertainty below it. No energy correlation is assigned between groups. JENDL-5 covers the same range with only four broad groups and assigns strong positive energy correlations between them. Both choices coexist with a nominal `MF4` grid that contains more than a thousand pointwise energies in the same interval (Figure III.1), so the covariance resolution is much

coarser than the resolution of the nominal data on which it acts. The contrasting bin structures and correlation assumptions of the two libraries are visible in their `MF34` correlation matrices (Figure III.5); the block structure of an `MF34` covariance is defined in Chapter I, Sec. 4.5.3.



(a) JEFF-4.0 (orders $\ell = 1$–3). (b) JENDL-5 (order $\ell = 1$ only).

Figure III.5: Correlation matrices from the `MF34` sections of (a) JEFF-4.0 (orders 1–3) and (b) JENDL-5 (order 1) for ^{56}Fe elastic scattering in the 847 keV–4 MeV range.

As documented in the CIELO evaluation of ^{56}Fe [113], the central values of the Legendre coefficients between 847 keV and 2.5 MeV were obtained by refitting the Kinney measurement [114]; between 2.5 MeV and 4.0 MeV the values were taken from Smith [115]. The angular shape was adjusted on a shape-only basis, with the absolute normalization left to the integrated elastic cross section. This choice fixes the central values to two campaigns and leaves the other EXFOR measurements unused at this stage. The integrated elastic cross section itself was obtained by subtraction, as the total minus the other partial cross sections, independently of the differential data. The refit of the Kinney data also raised its backward-angle scattering slightly, smoothing a forward-backward discontinuity toward the Perey measurement.

1.3. Scope of the Evaluation

The new evaluation is restricted in energy to 847 keV–4 MeV. Fast neutrons in the MeV range dominate radiation damage in RPVs, and this interval is the range over which the latest JEFF and JENDL evaluations build their elastic angular distributions primarily from experimental data and evaluator adjustments. Above 4 MeV the same evaluations transition to model-based results obtained with EMPIRE [116] for JEFF-4.0 and CCONE [117] for JENDL-5; restricting the present work to the data-driven window keeps the new evaluation and the comparison libraries on the same footing.

The output is a new `MF4` together with an associated `MF34` covariance, both built directly from the full EXFOR set introduced in Sec. 3 rather than from a single calibration campaign. The evaluation modifies the angular shape only: `MF3`, `MF33`, and the ENDF-6 $\ell = 0$ slot from `MF34` that would couple σ_{el} to the Legendre coefficients are not modified. The rationale for this magnitude–shape decoupling, and how it propagates through both the fit and the goodness-of-fit metric, is given in Sec. 2.1.

2. Mathematical Formalism for Uncertainty Quantification

Chapter I (Sec. 4.2.3 and 4.5.3) introduces the Legendre representation of elastic angular distributions in `MF4` and the block structure of the associated covariance `MF34`. This section sets out how the angular shape is reconstructed and how its covariances propagate to the differential cross section; the data-driven fit that produces the new central values is the subject of Sec. 4.

For the energy range of interest, the probability density $f(\mu, E)$ is reconstructed from the coefficients $a_\ell(E)$ stored in `MF4`. To compare these evaluations with experimental differential cross section (DCS) data, the angular shape is combined with the integrated elastic cross section σ_{el} from `MF3`. Experimental DCS points are not strictly differential: each is an average over the detector's finite angular aperture and over the time-of-flight energy-resolution window, not a value at a single angle and energy. The energy-resolution averaging, which is significant in this resonance-rich range, is treated in Sec. 3.2.

Recalling Eq. (I.40), the reconstructed differential cross section is:

$$\frac{d\sigma}{d\Omega}(E,\mu) = \frac{\sigma_{\mathrm{el}}(E)}{4\pi}\sum_{\ell=0}^{L}(2\ell+1)a_\ell(E)P_\ell(\mu) \tag{III.1}$$

where the normalization condition requires $a_0(E) \equiv 1$ for all energies. This means that the variation in the angular shape is entirely contained within the higher-order coefficients $(a_1, a_2, \ldots, a_L)$.

2.1. Propagation of MF34 Uncertainties to the Differential Cross Section

For each incident energy E, the differential cross section is reconstructed from $\sigma_{\mathrm{el}}(E)$ (`MF3`) and $a_\ell(E)$ (`MF4`) via Eq. (III.1). In what follows, the dependence on E is omitted for readability.

2.1.1. General first-order propagation

Let the uncertain parameters at fixed energy be $\{\sigma_{\mathrm{el}}, a_1, \ldots, a_L\}$, with a covariance structure given by `MF33` (for σ_{el}) and `MF34` (for the a_ℓ and, when available, their coupling to σ_{el}). To first order, the variance of the reconstructed DCS at a given μ can be written as:

$$\begin{aligned}\mathrm{Var}\left[\frac{d\sigma}{d\Omega}(\mu)\right] \approx \left(\frac{\partial}{\partial\sigma_{\mathrm{el}}}\frac{d\sigma}{d\Omega}\right)^2 \mathrm{Var}[\sigma_{\mathrm{el}}] + \sum_{\ell=1}^{L}\sum_{k=1}^{L}\left(\frac{\partial}{\partial a_\ell}\frac{d\sigma}{d\Omega}\right)\left(\frac{\partial}{\partial a_k}\frac{d\sigma}{d\Omega}\right)\mathrm{Cov}[a_\ell, a_k] \\ + 2\sum_{\ell=1}^{L}\left(\frac{\partial}{\partial\sigma_{\mathrm{el}}}\frac{d\sigma}{d\Omega}\right)\left(\frac{\partial}{\partial a_\ell}\frac{d\sigma}{d\Omega}\right)\mathrm{Cov}[\sigma_{\mathrm{el}}, a_\ell].\end{aligned} \tag{III.2}$$

From Eq. (III.1), the required derivatives are:

$$\frac{\partial}{\partial\sigma_{\mathrm{el}}}\frac{d\sigma}{d\Omega}(\mu) = \frac{1}{4\pi}\sum_{\ell=0}^{L}(2\ell+1)a_\ell P_\ell(\mu), \qquad \frac{\partial}{\partial a_\ell}\frac{d\sigma}{d\Omega}(\mu) = \frac{\sigma_{\mathrm{el}}}{4\pi}(2\ell+1)P_\ell(\mu) \quad (\ell \geq 1) \tag{III.3}$$

Variance Decomposition of the Differential Cross Section

Substituting (III.3) derivatives into (III.2) yields a decomposition into three different physical contributions:

$$\begin{aligned}\mathrm{Var}\left[\frac{d\sigma}{d\Omega}(\mu)\right] \approx & \underbrace{\left[\frac{1}{4\pi}\sum_{\ell=0}^{L}(2\ell+1)a_\ell P_\ell(\mu)\right]^2 \mathrm{Var}[\sigma_{\mathrm{el}}]}_{\text{(i) MF33: cross-section magnitude}} \\ & + \underbrace{\left(\frac{\sigma_{\mathrm{el}}}{4\pi}\right)^2 \sum_{\ell=1}^{L}\sum_{k=1}^{L}(2\ell+1)(2k+1)\,P_\ell(\mu)P_k(\mu)\,\mathrm{Cov}[a_\ell, a_k]}_{\text{(ii) MF34: angular-shape term}} \\ & + \underbrace{\frac{2\sigma_{\mathrm{el}}}{(4\pi)^2}\left[\sum_{\ell=0}^{L}(2\ell+1)a_\ell P_\ell(\mu)\right]\sum_{k=1}^{L}(2k+1)P_k(\mu)\,\mathrm{Cov}[\sigma_{\mathrm{el}}, a_k]}_{\text{(iii) MF34 } \ell=0\text{: } \sigma\text{–shape coupling}} \end{aligned} \tag{III.4}$$

The `MF34` covariances $\mathrm{Cov}[a_\ell, a_k]$ at a given energy are obtained from the same-energy elements of the `MF34` blocks $\mathbf{V}_{\ell,k}$ introduced in Sec. 4.5.3. When present, the coupling terms $\mathrm{Cov}[\sigma_{\mathrm{el}}, a_k]$ are obtained from the `MF34` blocks with index $\ell = 0$.

2.1.2. Simplification adopted in this work (shape-only bands)

The objective of the present analysis is to quantify the uncertainty on the angular shape of the elastic channel. Cross section and angular distribution are physically coupled, but in this evaluation they are treated independently: the angular shape $a_\ell(E)$ is fitted to EXFOR data, while the integrated cross section $\sigma_{\mathrm{el}}(E)$ enters the reconstructed DCS only as a normalization factor and is not modified. The same decoupling is applied in the goodness-of-fit metric of Sec. 3.4: the σ_{el} used to reconstruct each library's predicted DCS is the one derived from a fit to the experiments that drove the corresponding evaluation (Kinney and Smith for JEFF-4.0 and JENDL-5; the full EXFOR database for this work), instead of the value tabulated in any library's `MF3`, which is generally not coherent with the differential data in this range and would leave the normalization shifted relative to the measured points (Sec. 3.2). The a_ℓ here are fitted to the differential data, whereas σ_{el} comes from the host evaluation, built by a separate method that does not use those measurements; in this resonance region a magnitude integrated from the differential data does not even match it at the same energy. With no common source constraining both, the coupling cannot be estimated and is left at zero (Appendix E.8). Model-based evaluations of ^{56}Fe find this coupling strong [110]. Term (iii) then raises or lowers the variance at a given angle, since $P_k(\mu)$ changes sign, but it leaves the variance of the angle-integrated cross section unchanged.

Setting $\mathrm{Var}[\sigma_{\mathrm{el}}] = 0$ and $\mathrm{Cov}[\sigma_{\mathrm{el}}, a_\ell] = 0$ in Eq. (III.4) leaves only term (ii), which is the working expression for the shape-only bands reported throughout this chapter.

2.2. Physical Constraints and Cross-Section Coupling

The simplifications above deliberately isolate the angular-shape uncertainty. In a complete evaluation, however, the cross-section magnitude and the elastic angular distribution are not

independent. They are coupled by the optical theorem, which relates the total cross section to the imaginary part of the forward elastic scattering amplitude [118]. This implies Wick's inequality [119]:

$$\frac{d\sigma_{\mathrm{el}}}{d\Omega}(0^\circ) = |f_{\mathrm{el}}(0)|^2 \geq \left(\frac{k\sigma_{\mathrm{tot}}}{4\pi}\right)^2. \tag{III.5}$$

Here, k is the wave number and σ_{tot} is the total cross section. Thus, for a given total cross section, the forward elastic differential cross section cannot be arbitrarily small. Evaluations that treat the total cross-section magnitude and the Legendre coefficients of the elastic angular distribution as independent may therefore violate this physical constraint [119].

3. Experimental Database and Comparison Methodology

Differential elastic scattering data for ^{56}Fe and natural iron were collected from the EXFOR database for neutron energies between $E_n = 0.847$ and 4 MeV. Throughout this chapter, the term *experiment* refers to a single EXFOR subentry (e.g., `10571.002`), corresponding to one specific measurement series; an *entry* groups one or more related experiments from the same publication or campaign; and a *dataset* denotes the angular distribution points contributed by one experiment at a single nominal energy. From here, we filtered out datasets characterized by systematic deviations from the global trend *and* unrealistically small uncertainties that would disproportionately weight a fit. Only one entry met these criteria for removal: Tostkii–1957 (`32246.002`), which showed significant systematic offsets despite its narrow reported uncertainties. This affected only eight points, all measured at $E_n = 2.8$ MeV. The resulting collection of datasets used for the analysis is summarized in Table III.1.

The campaigns listed in Table III.1 differ substantially in measurement structure. Kinney covers the densest energy grid at 8 angles per energy; Cierjacks spans a comparable range at finer energy spacing with only 2–7 angular points per energy; several smaller campaigns contribute broad angular coverage at one or a few energies; and a number of single-energy measurements add as few as one point. Treating every angular point as an independent measurement misrepresents these data because the angles within each experiment share normalization-like systematics (monitor calibration, sample mass, detector solid angle): a calibration shift moves every angle of one experiment together, not each angle independently. Section 3.3 therefore decomposes each EXFOR point into an uncorrelated statistical component and a correlated systematic component, and this decomposition is what later allows the fit to weight each experiment by its effective number of independent measurements rather than by its raw point count.

3.1. Experimental Configurations and Measurement Extractions

Most fast-neutron elastic angular-distribution measurements are performed in time-of-flight (TOF) facilities. In these setups, the incident neutron energy is not measured directly but is inferred from the measured flight time t over a known flight path L. Because the neutron kinetic energy in the nonrelativistic regime depends on the square of the velocity, the relative energy resolution is directly tied to the timing resolution and the flight path uncertainty:

$$E = \frac{1}{2}m_n\left(\frac{L}{t}\right)^2, \qquad \frac{\Delta E}{E} \approx 2\sqrt{\left(\frac{\Delta t}{t}\right)^2 + \left(\frac{\Delta L}{L}\right)^2} \tag{III.6}$$

Table III.1: EXFOR experimental datasets contributing data points in the neutron-energy interval E_n = 0.847–4.000 MeV. For datasets spanning a wider interval, the number of points listed corresponds only to this interval. The listed counts are those entering the fit after the per-bin closest-energy selection of Sec. 4.2.

Author	Year	Energy (MeV)	Points	Subentry	Ref.
Kinney	1976	0.8470–2.5000	13232	10571.002	[114]
Cierjacks	1978	0.8470–3.0485	4900	20743.002	[120]
Pirovano	2019	1.9978–3.9928	1720	23365.004 to 23365.005	[121]
Gkatis	2025	1.0098–3.9338	824	27673.002	[122]
Smith	1980	1.6840–3.9050	490	10886.002	[115]
Cox	1966	0.8480–1.2320	376	11519.016	[123]
Barnard	1968	0.9400–1.1800	247	30076.004	[124]
Tomita	1973	1.4260–2.1520	225	20008.002 to 20008.018	[125]
Jacquot	1966	1.0000–2.2800	198	20379.003	[126]
Malmskog	1972	0.8500–1.4500	175	20197.006	[127]
Wells	1963	0.8480–0.9820	156	11121.006 to 11121.011	[128]
Ramirez	2017	1.3000–4.0000	134	14462.002	[129]
Holmqvist	1970	1.7700–2.7600	92	20020.012	[130]
Tsukada	1969	1.3700–3.2600	59	20304.002	[131]
Muehlhause	1956	1.4800–1.6600	55	11316.003	[132]
Holmqvist	1969	2.9600–3.9500	40	20019.099	[130]
Perey	1991	0.8477–0.8515	30	13511.004	[133]
Korzh	1977	1.5000–3.0000	36	40532.004	[134]
Cranberg	1956	2.2500–2.4500	25	11396.011	[135]
Machwe	1959	3.6600	24	11572.011	[136]
Morozov	1972	1.8000	17	40075.002	[137]
Beyster	1956	2.5000	17	11220.013	[138]
Landon	1958	2.2000	14	11717.002	[139]
Becker	1966	3.2000	14	11511.009	[140]
Vanhoy	2018	1.7500	12	14451.003	[141]
Schweitzer	1978	3.4000	12	30463.020	[142]
Cabe	1967	0.9700	11	20482.005	[143]
Begum	1981	2.9000	11	30633.002	[144]
Bostrom	1959	3.6700	11	11341.005	[145]
Bredin	1964	1.9500	10	11496.004	[146]
Walt	1954	1.0000	9	11637.007	[147]
Galloway	1979	2.9000	9	20761.002	[148]
Cox	1972	0.9021	8	10332.004	[149]
Korzh	1968	1.5000	8	40042.016	[150]
Kazakova	1965	2.0000	8	40706.046	[151]
Pasechnik	1958	2.8000	7	40336.101	[152]
Darden	1955	1.0000–1.5500	6	11638.003	[153]
Popov	1957	2.9000	6	40367.002	[154]
Lovchikova	1957	0.9000	5	40168.003	[155]
Hopkins	1964	2.0000–4.0000	5	11706.005	[156]
Rodgers	1967	2.3300	5	11276.009	[157]
Salnikov	1957	2.3400	5	40372.004	[158]
Tsukada	1961	3.4400–4.0700	3	22531.007	[159]
Stelson	1952	1.8700	1	11711.002	[160]
Poole	1953	2.5000	1	21377.005	[161]
Total Points			**23253**		

In many TOF setups, the flight-path uncertainty is negligible compared to the timing resolution, meaning the relative energy uncertainty is approximately twice the relative timing uncertainty. For example, in high-resolution measurements performed at GELINA using the ELISA spectrometer, a timing resolution of roughly 10 ns translates to an energy resolution of about 10 keV at 1 MeV, which broadens to the order of 100 keV at several MeV due to the decreasing flight times at higher energies [122].

In several dedicated experiments, detectors are positioned at angles whose cosines match the roots of a Legendre polynomial, enabling Gauss–Legendre quadrature to compute the angle-integrated elastic cross section directly from the discrete differential measurements. Some datasets therefore provide both differential cross sections and a consistent, experimentally derived σ_{el} from the same campaign. For instance, ELISA uses eight angles matching the zeros of P_8:

$$\sigma_{\mathrm{el}}(E) = 2\pi \sum_{i=1}^{8} w_i \left.\frac{d\sigma_{\mathrm{el}}}{d\Omega}(E,\theta)\right|_{\theta=\theta_i} \tag{III.7}$$

with quadrature weights w_i [122].

Experimentalists often summarize their angular distributions by fitting a truncated Legendre series in the center-of-mass frame, yielding unnormalized coefficients $\alpha_\ell(E)$ in units of b/sr. The ENDF representation uses normalized coefficients a_ℓ with $a_0 = 1$; the two are related by:

$$\alpha_0(E) = \frac{\sigma_{\mathrm{el}}(E)}{4\pi}, \qquad a_\ell(E) = \frac{\alpha_\ell(E)}{(2\ell+1)\,\alpha_0(E)} \quad (\ell \geq 1). \tag{III.8}$$

3.2. Energy Resolution Folding and Normalization Bias

A major complication in this energy range is the fast fluctuation of the ^{56}Fe elastic cross section. Because TOF measurements have a finite energy resolution, a cross section reported at nominal energy E_0 represents a resolution-broadened average over an energy window, not a monochromatic measurement. This resolution is characterized by the Full Width at Half Maximum (FWHM) of a Gaussian distribution representing the energy response. Throughout this chapter, the energy resolution is parameterized by the standard deviation σ_E of this Gaussian kernel, related to the FWHM by $\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma_E \approx 2.355\,\sigma_E$. For a representative time-of-flight setup, σ_E ranges from a few keV near 850 keV to a few tens of keV near 4 MeV. Each experiment is folded with its own σ_E, derived from its flight path and timing resolution (Sec. 4.5); the values quoted in the figures of this section are illustrative. The evaluated elastic cross section carries structure finer than this window, so a time-of-flight experiment sees only its window-average and must be folded before comparison. The evaluated angular distribution comes from these same experiments and shares their energy resolution; its tabulated points can sit closer together than σ_E because the measured data were unfolded, but that spacing is not finer resolution. In this window the JEFF-4.0 `MF4` grid is tabulated at 1 keV intervals below 2.5 MeV and about 50 keV above, so at 1.4 MeV some 34 tabulated angular distributions fall within $\pm\sigma_E$ of the nominal energy. Figure III.6 shows both effects together at 1.4 MeV: across the resolution window the magnitude σ_{el} and the angular shape both change, and their resolution-kernel average follows the measured points more closely than the distribution at the nominal energy. The magnitude-only and shape-only contributions are separated in Appendix E.1.

This creates a challenge when comparing multiple datasets. One standard approach is to fold the evaluated cross section with a Gaussian resolution kernel based on each experiment's TOF parameters. However, TOF parameters are not always available in EXFOR, and the folding can

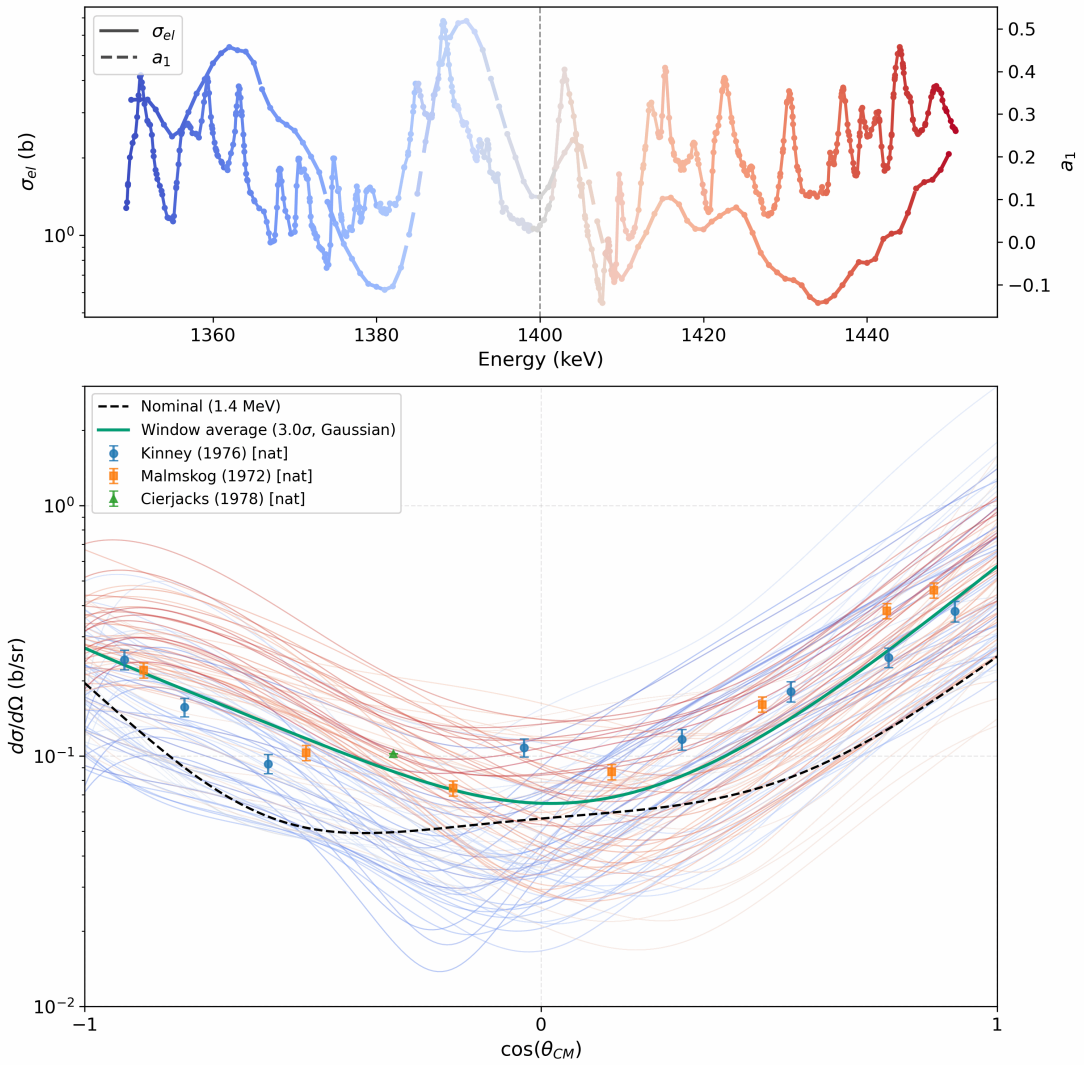


Figure III.6: Resolution folding for ^{56}Fe(n, el) at a nominal 1.4 MeV (JEFF-4.0, center-of-mass frame; $\sigma_E \approx 17$ keV for a 27 m, 10 ns setup). Top: σ_{el} (solid) and a_1 (dashed) across the window, colored by energy. Bottom: $d\sigma/d\Omega$ at each energy in the window (faint), the nominal 1.4 MeV distribution (black dashed), the Gaussian window average (green), and EXFOR data within the window.

only match one experimental configuration at a time, making simultaneous multi-experiment comparison impossible.

An alternative is to normalize the experimental data directly to the evaluated angular shape (MF4), stripping away the absolute scale. However, the "true" cross-section value at each experimental energy is unknown, and discrepancies in absolute magnitude can mask underlying shape deficiencies. Figure III.7 compares both alternatives at 1.4 MeV.

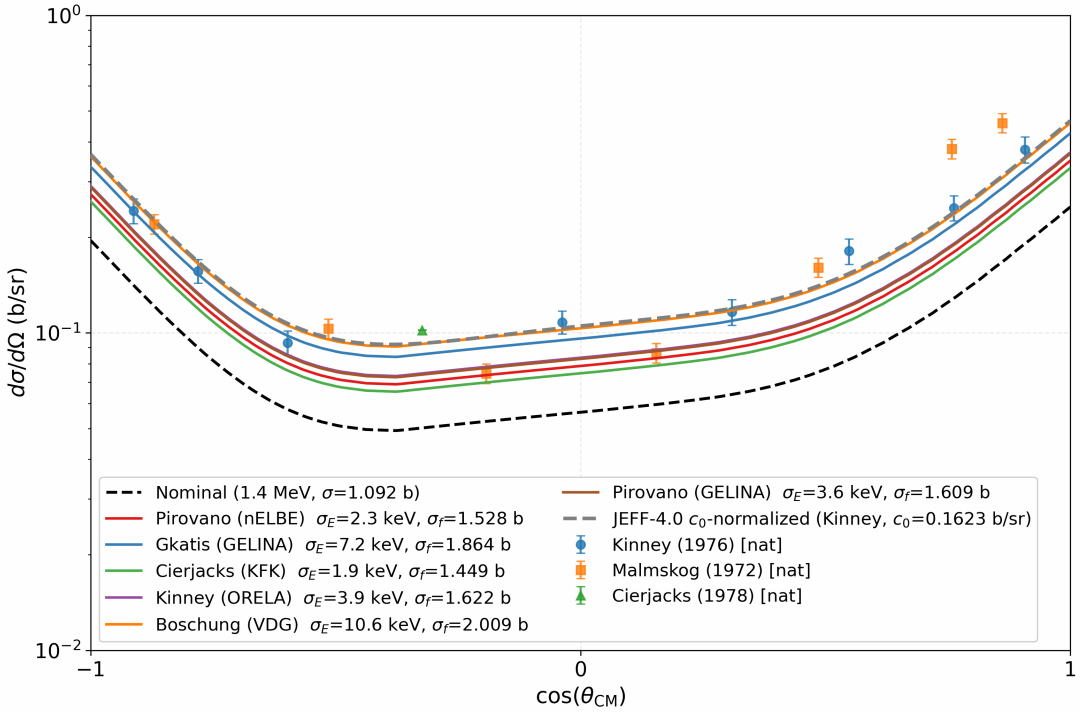


Figure III.7: Elastic differential cross section of ^{56}Fe at $E_n = 1.4$ MeV (center-of-mass frame). Black dashed: nominal JEFF-4.0. Colored lines: same shape scaled by the folded total cross section (σ_f) for six TOF resolutions σ_E. Grey dashed: shape-normalized JEFF-4.0 with the isotropic component re-fitted to Kinney data [114].

The approach adopted here builds the predicted differential cross section by decoupling the absolute normalization from the evaluated angular shape. The Legendre shape coefficients a_ℓ ($\ell \geq 1$) are retained from the library, but the isotropic component is replaced with a scaling factor $c_0(E)$ extracted as the zeroth-order coefficient of a weighted least-squares Legendre fit to the reference experimental data that drove the original evaluation. The resulting shape-normalized differential cross section is:

$$\frac{d\sigma}{d\Omega}(\mu) = c_0(E)\left[1 + \sum_{\ell=1}^{L}(2\ell+1)\,a_\ell\,P_\ell(\mu)\right] \tag{III.9}$$

This construction applies the same magnitude–shape decoupling as the shape-only variance term (ii) of Eq. (III.4).

3.3. Treatment of Experimental Uncertainties

EXFOR reports per-point uncertainties for each angular measurement but not always a decomposition into uncorrelated and correlated components. The pipeline organizes each point along this

axis: a statistical part $\sigma_{\mathrm{stat},i}$ that is uncorrelated point to point, and a systematic part that is correlated across all points of one experiment, both in angle and in energy (monitor calibration, sample mass, detector solid angle). Because EXFOR rarely reports this split, the pipeline has to assume it. The assumption matters: a shared calibration offset and independent noise of the same size affect the chi-squared and the fit very differently.

The ERR-ANALYS prose of each EXFOR entry is reviewed once, and the resulting decomposition is recorded in a per-subentry uncertainty manifest.

Each point i is one measurement at a single angle and energy. The systematic part scales every point of experiment g by the same relative factor σ_g^{sys}. With y_i the measured cross section at point i (in b/sr), the systematic standard deviation is:

$$\sigma_{\mathrm{sys},i} \equiv \sigma_g^{\mathrm{sys}} |y_i|, \tag{III.10}$$

on the same footing as the already-absolute statistical component $\sigma_{\mathrm{stat},i}$.

When the prose does not provide a decomposition, the systematic is assigned a default value of 5%. This default is the median of σ_g^{sys} across the experiments whose prose does specify a decomposition, whose reported values lie between 3% and 10%. It is taken by 31 of the 67 experiments. Thirty of them report no decomposition at all, but together they hold only 5% of the points. The weight of the default comes from the remaining one, Cierjacks 1978, which alone supplies 61% of the points and whose reported total names its sources without separating them. Kinney 1976, by contrast, does resolve a decomposition, at 8.1% in total. The declared systematics of the largest campaigns are therefore the most valuable target for improvement in this database. The reported total is taken as the per-point $\sigma_{\mathrm{tot},i}$, and the residual statistical component is recovered by quadrature subtraction:

$$\sigma_{\mathrm{stat},i} = \sqrt{\sigma_{\mathrm{tot},i}^2 - \sigma_{\mathrm{sys},i}^2}. \tag{III.11}$$

The same subtraction is used for curated entries whose prose names σ_g^{sys} together with a reported total. A small number of experiments require a treatment beyond this template; their per-experiment routings and the resolved values for the database are documented in Appendix E.2.

For experiment g contributing n_g points, this decomposition assembles into an $n_g \times n_g$ data covariance with a rank-one-plus-diagonal block structure. Written out element by element, the block reads:

$$\Sigma^{(g)} = \begin{pmatrix} \sigma_{\mathrm{stat},1}^2 + \sigma_{\mathrm{sys},1}^2 & \sigma_{\mathrm{sys},1}\,\sigma_{\mathrm{sys},2} & \cdots & \sigma_{\mathrm{sys},1}\,\sigma_{\mathrm{sys},n_g} \\ \sigma_{\mathrm{sys},2}\,\sigma_{\mathrm{sys},1} & \sigma_{\mathrm{stat},2}^2 + \sigma_{\mathrm{sys},2}^2 & \cdots & \sigma_{\mathrm{sys},2}\,\sigma_{\mathrm{sys},n_g} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{\mathrm{sys},n_g}\,\sigma_{\mathrm{sys},1} & \sigma_{\mathrm{sys},n_g}\,\sigma_{\mathrm{sys},2} & \cdots & \sigma_{\mathrm{stat},n_g}^2 + \sigma_{\mathrm{sys},n_g}^2 \end{pmatrix} \tag{III.12}$$

Every off-diagonal entry is the rank-one outer product $\sigma_{\mathrm{sys},i}\,\sigma_{\mathrm{sys},j}$, reflecting the single calibration draw that perturbs every point of experiment g together. The diagonal entries gather the uncorrelated noise $\sigma_{\mathrm{stat},i}^2$ and the same-point contribution $\sigma_{\mathrm{sys},i}^2$ of the rank-one mode.

Splitting the diagonal contributions from the rank-one outer product gives the compact representation:

$$\Sigma^{(g)} = D^{(g)} + u^{(g)}\,u^{(g)\,T}, \qquad D_{ii}^{(g)} = \sigma_{\mathrm{stat},i}^2, \qquad u_i^{(g)} = \sigma_{\mathrm{sys},i}, \tag{III.13}$$

Points from different experiments are assumed uncorrelated, so the full data covariance for the experimental database, written Σ_{data}, is block-diagonal in g with the blocks $\Sigma^{(g)}$ of Eq. (III.13).

This neglects genuine cross-experiment correlations that can arise from shared measurement standards, detectors, or reference cross sections; EXFOR does not document these in a form that could be propagated here, and assuming independence is the standard practice. This is the covariance used both in the goodness-of-fit metric below and in the generalized least-squares fit of Sec. 4.3; building it once at the data-preparation stage ensures that every downstream stage operates on a single consistent representation of the experimental uncertainties, with the systematic uncertainty multiplied by the prediction wherever a normalization is fitted (Sec. 3.4).

3.4. Goodness-of-Fit Metric

The agreement between an evaluated angular distribution and the experimental database is measured separately for each experiment g. The residual vector is defined as:

$$\mathbf{r}_g = \mathbf{y}_g^{\mathrm{exp}} - c_0(E_g)\,\mathbf{s}_g \tag{III.14}$$

where $c_0(E_g)\,\mathbf{s}_g$ is the shape-normalized evaluated prediction defined in Sec. 3.2. The covariance used in the comparison is the sum of the experimental data covariance and, when included, the propagated covariance of the evaluated library:

$$C_g = \Sigma_{\mathrm{data}}^{(g)} + \Sigma_{\mathrm{eval}}^{(g)} \tag{III.15}$$

Here $\Sigma_{\mathrm{data}}^{(g)}$ is the experimental covariance of Eq. (III.13), with one change. The relative systematic uncertainty σ_g^{sys} is converted to b/sr by multiplying it by the prediction, $u_i^{(g)} = \sigma_g^{\mathrm{sys}}\, c_0(E_g)\, s_{g,i}$, and not by the measured value y_i. Only the systematic part $u^{(g)}u^{(g)\,T}$ changes, both its diagonal terms u_i^2 and its off-diagonal products $u_i u_j$. The statistical variances $D^{(g)}$, the relative uncertainty σ_g^{sys} and the measured values stay as they are.

The reason is a bias known as Peelle's Pertinent Puzzle [162]. When the systematic uncertainty is proportional to the measured value, a point that happens to fall low receives a smaller uncertainty in b/sr, and therefore more weight in a least-squares fit. A level fitted to such points is pulled below the data. The size of this pull depends on how far the points scatter about the fitted curve. If every point lay exactly on the curve, multiplying by the measured value or by the prediction would give the same uncertainties, and there would be no bias. The wider the scatter, the more the low points dominate and the lower the fitted level falls. Multiplying by the prediction, as in the Chiba–Smith prescription [163], removes the bias [162].

In the χ^2 comparison, the bias matters most for the normalization $c_0(E_g)$ assigned to JEFF-4.0 and JENDL-5. So that each library is judged on its angular shape alone, the metric does not take the magnitude from the library's `MF3`. It keeps the library shape and fits only its magnitude at each energy, to the Kinney data below 2.5 MeV and to the Smith data above (Sec. 3.2). This fit is a step of the χ^2 calculation and is separate from the evaluation fit of Sec. 4.3. This fit also multiplies the systematic uncertainty by the prediction. Since the prediction depends on c_0, the quantity being fitted, the uncertainty is recomputed from each new estimate and the fit is repeated until c_0 no longer changes. With the uncertainty proportional to the measured values, the fitted c_0 would be lower by a median of 18% for JEFF-4.0 at the Kinney energies, and by 1.4% for JENDL-5. The difference comes from the scatter. JENDL-5 reproduces the Kinney angular distributions closely, so its curve passes near every point. The backward-angle shape of JEFF-4.0 does not match those data, so the points scatter widely about its curve and the low ones pull its normalization down.

Within the χ^2 itself, the same scaling acts on the residuals. A prediction lying above the data is compared with the smaller uncertainties of the low points, so it is penalized more than one lying equally far below. Table III.2 is computed with the correction applied to both the normalization fit and the χ^2. Without it, JEFF-4.0 would score $\chi^2/N = 19.2$ on the full database instead of 21.8, helped by its lower normalization, and JENDL-5 would score 24.3 instead of 22.4, penalized where it lies above the data.

The evaluation fit of this work (Sec. 4.3), which uses every experiment available in each energy bin, keeps the uncertainty proportional to the measured values, so its fitted c_0 can carry the bias. The bias lowers all coefficients c_ℓ by the same factor, which cancels, to first order, in the normalized shape $a_\ell = c_\ell/c_0$ stored in `MF4`. The fitted c_0 itself is not used in the file, whose magnitude comes from the host `MF3`. In the χ^2, this work is treated like the two libraries. Its normalization is refitted with the shape of each energy bin held fixed, on the same points as the evaluation fit, and with the systematic uncertainty multiplied by the prediction. The refitted level differs from that of the evaluation fit by a median of 0.2%. The difference exceeds 20% in 30 of the 1 738 bins, each constrained by two or three experiments that disagree, where the low points had pulled the level of the evaluation fit down. The comparison of Sec. 5.4 uses this refitted level.

When included, $\Sigma^{(g)}_{\text{eval}}$ is obtained by sandwich-propagating the library's `MF34` blocks $\mathbf{V}_{\ell,k}$ of Eq. (I.48) to the (E_i, μ_i) of each experimental point:

$$\left(\Sigma^{(g)}_{\text{eval}}\right)_{ij} = \sum_{\ell,k=1}^{L} \left(\frac{\partial}{\partial a_\ell}\frac{d\sigma}{d\Omega}\right)_i \left(\frac{\partial}{\partial a_k}\frac{d\sigma}{d\Omega}\right)_j \left(\mathbf{V}_{\ell,k}\right)_{E_i,E_j} \tag{III.16}$$

using the Legendre sensitivities of Eq. (III.3). The block inherits the library's cross-order (ℓ, k) and cross-energy (E_i, E_j) correlations; on the diagonal it reduces to the variance term (ii) of Eq. (III.4). The chi-squared contribution of experiment g is then computed as the Mahalanobis quadratic form:

$$\chi^2_g = \mathbf{r}_g^T C_g^{-1} \mathbf{r}_g \tag{III.17}$$

The total statistic is obtained by summing over experiments:

$$\chi^2 = \sum_g \chi^2_g, \qquad N = \sum_g n_g \tag{III.18}$$

The quantity reported in this work is χ^2/N, where N is the total number of experimental angular points. This should be interpreted as a normalized measure of the average discrepancy between the evaluation and the database under the adopted covariance model.

A formal reduced chi-squared would use the degrees of freedom in the denominator:

$$\nu = N - p \tag{III.19}$$

with p the number of fitted parameters. The angular fit selects a Legendre order independently at each energy bin, so p varies across bins and no single value characterizes the procedure; the reported χ^2/N uses N in the denominator, with the same definition applied to every evaluation in the comparison.

When only the experimental covariance is used (the first term of Eq. (III.15)), χ^2/N tests the agreement of the evaluated central values with the database. When the evaluated-library

covariance is also included (the second term of Eq. (III.15)), it tests whether the residuals are consistent with the combined experimental and evaluated uncertainties. Values much larger than unity indicate residuals larger than expected from the assigned covariances, while values much smaller than unity may indicate overestimated uncertainties or overly strong correlations.

A value of $\chi^2/N \approx 1$ is nevertheless not the target of this metric. Two contributions that would be needed to reach it are left out on purpose, so that the comparison isolates the angular distribution and its covariance. The first is the uncertainty of the evaluated magnitude. The cross-section covariance (`MF33`) differs from one library to the next, and adding it to C_g would lower χ^2/N by a different amount for each evaluation, for reasons unrelated to the angular distribution; the effect of the different `MF34` covariances, which is what the comparison is meant to expose, would be hidden behind it. The second is the energy-resolution folding of Sec. 3.2. Folding the prediction with the resolution of each experiment requires the evaluated cross section across the resolution window, whereas the metric takes the magnitude from the reference level $c_0(E_g)$ fitted to the data and does not use the evaluated $\sigma_{\mathrm{el}}(E)$. The prediction is therefore the angular distribution at the nominal energy, and the difference between it and the resolution-averaged measurement remains in the residuals. A third contribution comes from the database itself: its campaigns are offset from one another by more than their declared uncertainties, so no single set of central values can reach unity against it. All three affect every evaluation in the same way. The absolute χ^2/N is therefore expected to stay above unity, and the quantity that carries meaning is the comparison between evaluations built under the same construction.

3.5. Preliminary Library Comparison

Applying the metric defined above to the EXFOR database (67 experiments, 46 742 raw angular (E, μ) points) gives an approximate first quantitative measure of how well JEFF-4.0 and JENDL-5 reproduce the data.

Two point counts appear in this chapter and refer to different stages. The fit uses one dataset per experiment in each energy bin, the one closest to the bin center, which gives the 23 253 points of Table III.1. The goodness-of-fit compares every measured point once and so uses the full database between 0.85 and 4 MeV that carries a declared uncertainty, 46 742 points. The two counts are not nested. Most of the difference comes from Cierjacks, whose measurements are densely spaced in energy, so its 28 631 points in the goodness-of-fit are reduced to 4 900 by the per-bin selection. For the other experiments the fit count is larger, by 242 points in total, because the fit starts at 0.847 MeV and its last bin extends slightly above 4 MeV, keeps the points without a declared uncertainty, and counts a distribution once in every bin that uses it.

The data covariance follows the rank-one-plus-diagonal per-experiment construction of Eq. (III.13): statistical contributions on the diagonal, and the per-experiment systematic as a single rank-one correlated mode. Two variance budgets are used throughout this comparison and the goodness-of-fit analysis of Sec. 5.4. The first, written σ_{exp}, uses only the per-experiment EXFOR uncertainties. The second, written $\sigma_{\mathrm{exp}} \oplus \sigma_{\mathrm{eval}}$, additionally folds in each library's propagated `MF34` dense covariance block of Eq. (III.16). Reporting both separates the agreement of each evaluation's central values, tested by σ_{exp}, from the consistency of the residuals with the library's own uncertainty assignment, tested by $\sigma_{\mathrm{exp}} \oplus \sigma_{\mathrm{eval}}$. Table III.2 reports the reduced χ^2/N under three data splits and these two budgets.

The normalization $c_0(E_g)$ at each energy is fitted to the Kinney and Smith data, so each library is judged only on its angular shape. Both libraries stay well above $\chi^2/N = 1$ on the full database,

Table III.2: Reduced χ^2/N of JEFF-4.0 and JENDL-5 against the EXFOR experimental database for ^{56}Fe elastic scattering in the 847 keV–4 MeV range, under the two variance budgets defined in the text. Bold marks the lower χ^2/N between the two libraries in each column.

	All 67 experiments		**Excluding Kinney**		**Kinney only**	
Library	σ_{exp}	$\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$	σ_{exp}	$\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$	σ_{exp}	$\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$
JEFF-4.0	**21.8**	**9.6**	**27.9**	**12.2**	6.4	3.0
JENDL-5	22.4	15.1	31.0	20.9	**0.53**	**0.53**

a level partly set by the construction of the metric (Sec. 3.4), so the informative reading is the comparison between subsets.

Read this way, both libraries fit Kinney much more closely than the rest of the data: dropping that single campaign worsens JEFF-4.0 by about a factor of 1.3 and JENDL-5 by about 1.4, and on Kinney alone JENDL-5 falls below unity. Whether the remaining level is a defect of the libraries or a feature of the database is answered in Sec. 5.4, where the same comparison is applied to an evaluation fitted to the whole EXFOR set.

These observations, together with the structural limitations of Sec. 1.2, motivate an independent evaluation that draws on the full EXFOR database and produces both the nominal angular distributions and a covariance whose energy and order structure is consistent with the data it was built from.

4. Evaluation Pipeline: Fitting and Sampling Methodology

This section describes the pipeline, developed for this work, that produces the new evaluation: from the selection of experimental data, through the determination of the Legendre coefficients at each energy, to the construction of an `MF34` covariance matrix in ENDF-6 format.

4.1. Pipeline Overview

The primary goal of the pipeline is to produce, for each energy bin on a data-driven energy grid, a set of Legendre coefficients $a_\ell(E)$ that best represents the available experimental angular distributions for ^{56}Fe and natural iron elastic scattering in the 847 keV to 4 MeV range. The pipeline also constructs a complete `MF34` covariance matrix characterizing the uncertainties and correlations of these coefficients.

The database is uneven in angular and energy coverage, and many bins rest on only a few experiments. The fitting, weighting, and discrepancy choices described below were shaped by these constraints.

The workflow proceeds through seven stages, illustrated in Figure III.8:

1. **Experimental database.** Angular-distribution measurements are loaded from EXFOR, the per-point uncertainty decomposition of Sec. 3.3 is applied, and the data are organized into a per-experiment block-correlated covariance.

2. **Energy grid.** The fitting grid is built from the union of energies reported in two experimental subentries (Sec. 4.2).

3. **Nominal fit.** In each bin, the pooled data are fitted with a Legendre expansion using a ridge-regularized, block-correlated generalized least-squares method. The method uses the data covariance from Sec. 3.3 and per-point weights to account for experimental heterogeneity. The polynomial order is chosen using the Akaike Information Criterion (AIC), and per-angular-band scale factors s_b are estimated from the fit residuals (Sec. 4.3).

4. **Per-energy bin variances (MC pass 1).** A per-bin Monte Carlo pass resamples the experimental data and refits, yielding the variance of the Legendre coefficients at each energy (Sec. 4.4).

5. **Correlation matrix (MC pass 2).** A kernel-weight Monte Carlo pass applies shared perturbations across all bins to which each experiment contributes, producing the full parameter correlation matrix — both cross-order within a bin and cross-energy between bins — from the natural data overlap (Sec. 4.5).

6. **Covariance assembly.** The per-bin variances from Pass 1 are combined with the correlation matrix from Pass 2 through a diagonal congruence transform to form the full Legendre-coefficient covariance (Sec. 4.6).

7. **Output.** The nominal Legendre coefficients are written to `MF4` and the covariance matrix, collapsed onto an adaptive multigroup grid, is written to `MF34` (Sec. 4.7).

The pipeline requires four inputs: (a) the EXFOR experimental database summarized in Table III.1, (b) the per-subentry uncertainty manifest documented in Appendix E.2, which records how the `ERR-ANALYS` prose of each entry is decomposed into the (σ_{stat}, σ_g^{sys}) components, including any energy-dependent contributions, (c) experiment-specific time-of-flight parameters for computing energy-resolution overlap weights, with defaults applied where no values are reported, and (d) a reference ENDF file. The reference file provides the base file structure for writing the output and supplies Legendre coefficients outside the pipeline energy range for optional merging. The fitting and covariance construction within the 847 keV–4 MeV range are independent of the reference evaluation.

The primary output is an ENDF file containing both the best-estimate `MF4` Legendre coefficients and their `MF34` covariance on an adaptively collapsed energy grid, with the Legendre order selected per energy up to a maximum of $\ell = 6$. Pre-existing `MF34` data outside the pipeline energy range are preserved, and correlations between the pipeline range and the outer regions are set to zero because the two analyses are independent. Once the nominal file is produced, perturbed ENDF samples for uncertainty propagation can be generated by sampling the `MF34` covariance using the perturbation pipeline described in Chapter II.

Each of the following subsections describes one pipeline step in order. No nuclear-model calculations or prior evaluations enter the fitting or covariance construction within the 847 keV–4 MeV range. The optical-model treatment that nuclear-reaction codes such as TALYS apply to elastic angular distributions describes the smooth, energy-averaged behavior and does not reproduce the strong resonance fluctuations that characterize this region. A resonance-based analysis with an R-matrix code such as SAMMY [164] or REFIT [165] could in principle provide a model description here, but building one is a separate evaluation effort; and starting from an existing evaluation in a Bayesian framework would compromise the independence of the result. The evaluation instead relies on the experimental data directly so that over 23 000 deduplicated points from more than 35 campaigns (Table III.1) drive the fit, without an informative model prior.

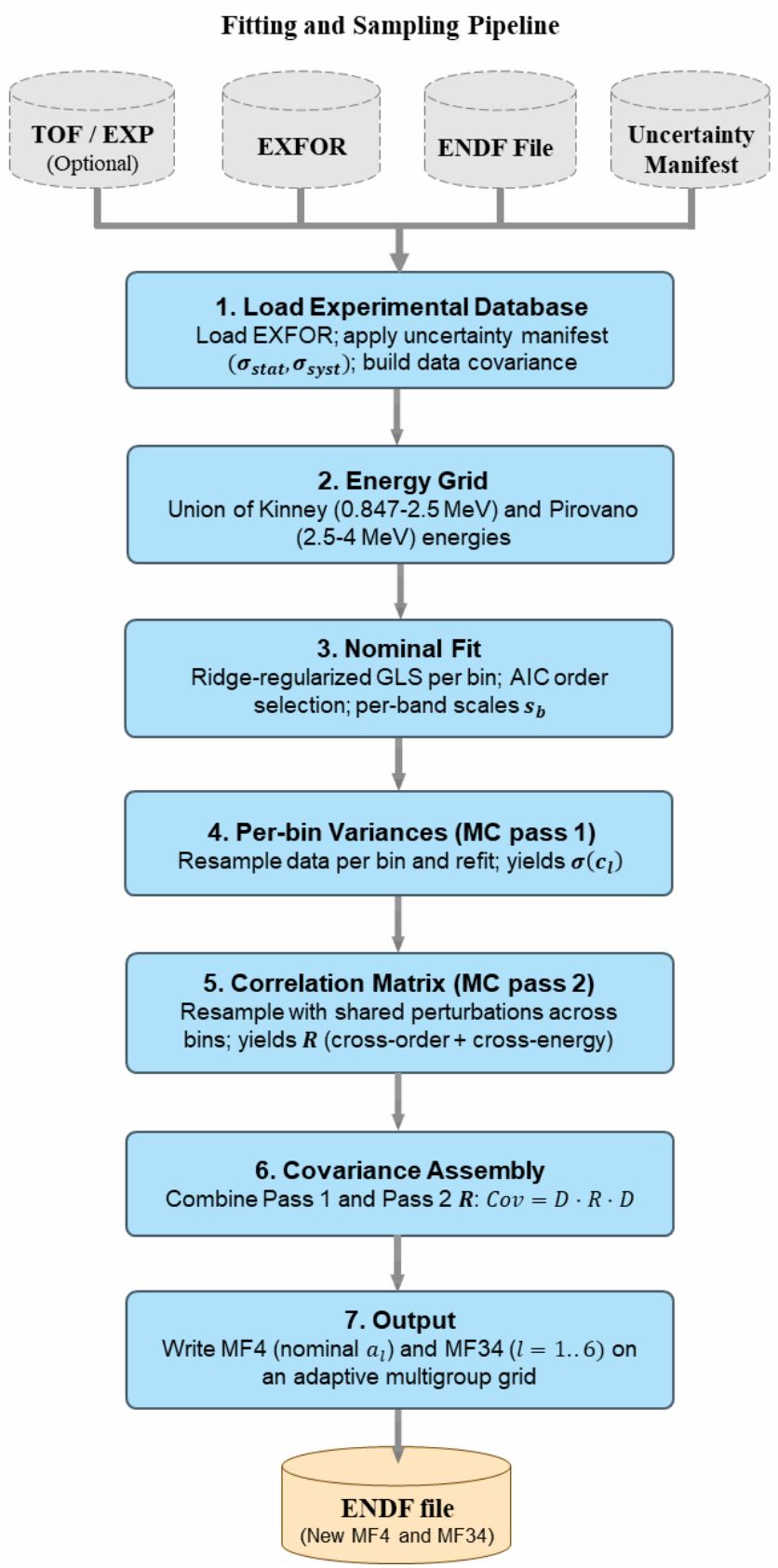


Figure III.8: Overview of the evaluation pipeline. EXFOR angular distributions are fitted to Legendre coefficients per energy bin; two Monte Carlo passes estimate the per-bin variances (Pass 1) and the full parameter correlation matrix (Pass 2), assembled into the covariance written to `MF4` and `MF34`.

4.2. Data Selection and Energy Grid

The fitting procedure uses a data-driven energy grid constructed from the union of energies reported in two subentries: Kinney [114] (847 keV to 2.5 MeV) and Pirovano [121] subentry 23365.005 (2.5 to 4 MeV). Previous evaluations such as JEFF-4.0 and JENDL-5 used Smith [115] to define the grid above 2.5 MeV; Pirovano is adopted here because it is a more recent experiment (2019 vs. 1980) and provides a considerably denser energy mesh in this range (83 energies vs. 29). The resulting grid contains 1738 points in the 847 keV–4 MeV range.

To pool the experimental data for each grid point E_i, an energy bin is defined by the midpoints to the adjacent grid energies:

$$E_i^{\text{low}} = \frac{E_{i-1} + E_i}{2}, \qquad E_i^{\text{up}} = \frac{E_i + E_{i+1}}{2}. \tag{III.20}$$

This partitioning provides continuous coverage with no gaps or overlaps. The bin widths are naturally denser where the experimental campaigns measured at finer energy intervals.

For each bin, all EXFOR experiments with angular-distribution data at a nominal energy within the bin boundaries are identified. If a single experiment provides distributions at multiple energies within the same bin, only the one closest to the bin center is retained. This keeps each experiment to one representative point per bin and prevents a high-resolution campaign from dominating it with many closely spaced measurements. When the data retained in a bin have fewer than four points or do not cover the forward, middle, and backward angular bands, the pooling window is widened by up to three neighboring bins on each side, so a distribution can then enter more than one bin.

The per-point statistical uncertainty assembled by the procedure of Sec. 3.3 is used as-is in the fit. A small numerical guard ($10^{-4}\,|y_i|$) prevents literal-zero statistical uncertainties from yielding infinite least-squares weights; this threshold sits well below every credible reported value in the database and does not bind in practice. The sensitivity of the results to this guard and to the floor protecting the quadrature subtraction of Eq. (III.11) is examined in Appendix E.3.

4.3. Nominal Fit

At each energy bin, the pooled experimental angular-distribution data are fitted with a Legendre polynomial expansion to extract the coefficients c_ℓ that best describe the differential cross section as a function of the cosine of the scattering angle $\mu = \cos\theta$:

$$y(\mu) = \sum_{\ell=0}^{L} c_\ell \, P_\ell(\mu) \tag{III.21}$$

This subsection describes how the nominal (best-estimate) coefficients are determined. The Monte Carlo passes that follow perturb the inputs to this fit to estimate uncertainties and correlations.

4.3.1. Block-correlated generalized least squares

For the pooled data $\{(\mu_i, y_i)\}_{i=1}^{n}$ in a bin, the nominal coefficients $\hat{\mathbf{c}}$ minimize a ridge-regularized generalized least-squares objective [166]:

$$\mathcal{L}(\mathbf{c}) = (\mathbf{y} - \mathbf{X}\mathbf{c})^T \, \widetilde{\Sigma}^{-1} \, (\mathbf{y} - \mathbf{X}\mathbf{c}) \; + \; \sum_{\ell=0}^{L} \lambda_\ell \, c_\ell^2 \tag{III.22}$$

where $\mathbf{X}$ is the matrix of Legendre polynomials evaluated at the measurement angles ($X_{i\ell} = P_\ell(\mu_i)$), $\mathbf{y}$ the vector of measured cross sections in the bin, and $\widetilde{\Sigma}$ the block-diagonal data covariance Σ_{data} of Sec. 3.3 after rescaling by the per-point weights $w_{p,i}$ defined in the next subsection.

Minimization of Eq. (III.22) is a deterministic ridge-regularized generalized least-squares problem with the closed-form normal-equation solution:

$$\hat{\mathbf{c}} = \left(\mathbf{X}^T \widetilde{\Sigma}^{-1} \mathbf{X} + \Lambda\right)^{-1} \mathbf{X}^T \widetilde{\Sigma}^{-1} \mathbf{y}, \qquad \Lambda = \text{diag}(\lambda_0, \lambda_1, \ldots, \lambda_L) \tag{III.23}$$

which the pipeline solves once per bin to obtain the nominal coefficients c_ℓ.

The ridge penalty $\lambda_\ell = \lambda\, \ell^p$ uses a base strength $\lambda = 10^{-4}$ and a power $p = 4$. The quartic scaling makes the penalty negligible for low orders ($\lambda_2 = 1.6 \times 10^{-3}$) and substantial for higher orders ($\lambda_6 = 0.13$), which suppresses high-order oscillatory artefacts when the angular coverage is sparse without biasing the low-order terms.

The upper limit $L_{\max} = 6$ is adopted for three reasons. First, it matches the maximum order retained by JEFF-4.0 and JENDL-5 in this energy range. Second, fits at $L > 6$ on this dataset produce spurious oscillations not supported by the angular coverage of the available experiments. Third, an expansion of order six is enough to capture the forward-scattering peak up to 4 MeV; the strongly forward-peaked distributions that genuinely require higher orders only appear at incident energies above this work's range.

4.3.2. Per-point weights for experimental heterogeneity

As discussed in Sec. 3, the experimental campaigns differ markedly in point count and intra-experiment correlation, so an unweighted fit would over-represent dense-coverage campaigns whose angles share the same systematic.

A Kish effective-sample-size collapse [167] addresses this by reweighting each (experiment, nominal-energy) cell according to its internal redundancy. For a cell with n angles, an equicorrelation approximation gives the mean intra-cell correlation through the shared systematic and the corresponding shrinkage factor:

$$\rho = \frac{(\sigma^{\text{sys}}_{g,E})^2}{(\sigma^{\text{sys}}_{g,E})^2 + \bar{\sigma}^2_{\text{stat}}}; \qquad w_{p,i} = g_{\text{Kish}} = \frac{1}{1 + (n-1)\,\rho} \tag{III.24}$$

where $\bar{\sigma}_{\text{stat}}$ is the mean relative statistical uncertainty of the cell's points. When $\bar{\sigma}_{\text{stat}} \gg \sigma^{\text{sys}}_{g,E}$, the points are nearly independent and $g_{\text{Kish}} \to 1$; when $\bar{\sigma}_{\text{stat}} \ll \sigma^{\text{sys}}_{g,E}$, the points are strongly correlated and the cell contributes approximately as a single observation. The factor is computed once per cell from the assembled $\sigma^{\text{sys}}_{g,E}$ and the mean statistical uncertainty and is frozen for the Monte Carlo passes.

The combined weights of any single experiment in a bin are additionally capped at 80% of the bin's total weight to prevent a single dense campaign from dominating a bin in which other contributors are individually informative but jointly outweighed.

4.3.3. Automatic degree selection via AIC

Rather than fitting at a fixed Legendre order, the pipeline evaluates all orders from $L = 1$ to $L_{\max}$ and selects the one that best balances goodness of fit against model complexity. The

selection criterion is the Akaike Information Criterion [168]:

$$\mathrm{AIC}(L) = \chi^2(L) + 2k \tag{III.25}$$

where $\chi^2(L)$ is the residual sum of squares of the GLS fit at order L and k is the effective number of fitted parameters, computed as the trace of the hat matrix to account for the ridge regularization. The degree L^* minimizing AIC is selected as the nominal order for the bin.

The small-sample-corrected variant (AICc) and the Bayesian Information Criterion (BIC) were also tested on this dataset; both perform comparably to or worse than AIC, and the detailed comparison with the corresponding per-bin order distributions is reported in Appendix E.4.

4.3.4. Angular-band discrepancy model

The measurement uncertainties reported in EXFOR generally reflect statistical and systematic contributions within a single experiment. They do not account for discrepancies that arise when data from multiple experiments are combined in a single fit. In such fits, the scatter around the best-fit curve may exceed what the reported uncertainties predict. A classical approach is the Birge ratio method [169], which inflates all uncertainties uniformly by $\sqrt{\chi^2_{\mathrm{red}}}$. In angular distribution measurements, however, the excess scatter is often not uniform across the angular range. Different angular regions may show different levels of inter-experiment consistency depending on the incident energy and the experimental technique. A uniform Birge inflation, tested on this dataset, left the resulting uncertainties underestimated under the adopted goodness-of-fit metric.

To address this, an angular-band discrepancy model is introduced that estimates a band-specific multiplicative scale factor s_b. The angular range is divided into three bands:

- Forward band: $\mu > 0.5$ (scattering angles $\theta < 60°$).
- Mid band: $-0.5 \leq \mu \leq 0.5$ ($60° \leq \theta \leq 120°$).
- Backward band: $\mu < -0.5$ ($\theta > 120°$).

For each band b at a given energy, the estimation proceeds as follows:

1. Compute the normalized residuals $r_i = (y_i - y_{\mathrm{fit},i})/\sigma_i$ for all data points in the band.
2. Compute the median absolute deviation $\hat{\sigma}_b = 1.4826 \cdot \mathrm{median}_i\,|r_i - \mathrm{median}(r)|$ [170], a robust analogue of the standard deviation that is insensitive to outliers.
3. Compute the band scale factor $s_b = \min\!\big(\max(1,\ \hat{\sigma}_b),\ s_{\max}\big)$, where $s_{\max} = 5$ is a safety cap. If $\hat{\sigma}_b > 1$, the data scatter exceeds the reported uncertainties and s_b inflates them proportionally; otherwise no inflation is applied. The cap prevents bands with very few data points from producing unstable, excessively large inflation factors.

The effective uncertainty for each data point is obtained by multiplicative scaling of the per-point statistical uncertainty:

$$\sigma_{\mathrm{eff},i} = s_b\, \sigma_{\mathrm{stat},i} \tag{III.26}$$

where b is the band to which point i belongs. This scaling preserves the relative weights within a band: a well-measured point retains more influence on the fit than a poorly measured one.

The band-scale estimation is iterated jointly with the nominal fit through an iteratively reweighted least-squares (IRLS) loop. After the initial s_b values are computed, the GLS fit is repeated with the inflated σ_{eff} entering the diagonal of $D^{(g)}$ in Eq. (III.13), the residuals are reread, and the band scales are recomputed. The loop finishes when the maximum change in any s_b falls below a tolerance of 10^{-2}.

Estimating s_b from a band's own residuals is only reliable when the band has enough points, so bands with fewer than four are not self-estimated and instead inherit a floor equal to the median s_b of the bins where the band can be estimated, taken band by band. This partial pooling propagates the inter-experiment scatter measured in data-rich bins to those where it cannot be measured locally. Band-specific biases peculiar to a single dataset remain undetectable, but the floor prevents data-poor bins from claiming sharper agreement with the data than the rest of the dataset supports.

4.3.5. Model averaging and degree sampling

In many energy bins, two or three Legendre orders yield comparable AIC scores, which reflects a genuine ambiguity in the optimal polynomial order. To account for this model-form uncertainty, Akaike weights [171] are computed for each candidate order:

$$w_L = \frac{\exp\!\left(-\tfrac{1}{2}\,\Delta\mathrm{AIC}(L)\right)}{\sum_{L'} \exp\!\left(-\tfrac{1}{2}\,\Delta\mathrm{AIC}(L')\right)}; \qquad \Delta\mathrm{AIC}(L) = \mathrm{AIC}(L) - \mathrm{AIC}(L^*) \tag{III.27}$$

where the sum runs over all orders with $w_L > 0.01$. The nominal fit uses the single best order L^*. During the variance-estimation pass (Sec. 4.4), each Monte Carlo sample draws its Legendre order from the full w_L distribution, which feeds the model-form component into the coefficient variance.

Figure III.9 and Table III.3 illustrate this selection for ^{56}Fe elastic scattering at $E = 1.200$ MeV, where 21 data points from three experiments are pooled. Orders 0 and 1 clearly underfit the data, while orders 2 through 6 produce visually similar curves of comparable quality. The AIC scores spread the weight across the lowest well-fitting orders rather than committing to a single one, so the model averaging carries this residual ambiguity into the variance estimate.

Table III.3: Fit statistics and Legendre coefficients c_ℓ for the order comparison at $E = 1.200$ MeV illustrated in Figure III.9. The information-criterion column uses the AIC form $\chi^2 + 2k$ of Eq. (III.25). The ridge penalty uses $\lambda = 10^{-4}$ with power $p = 4$. Dashes indicate coefficients not present at a given order.

L	χ^2_ν	AIC	w_L (%)	c_0	c_1	c_2	c_3	c_4	c_5	c_6
0	8.11	164.25	0.0	0.1051	—	—	—	—	—	—
1	8.50	165.48	0.0	0.1052	0.0110	—	—	—	—	—
2	1.08	25.51	26.6	0.1653	0.0367	0.2009	—	—	—	—
3	1.04	25.75	23.6	0.1662	0.0510	0.2020	0.0267	—	—	—
4	0.98	25.67	24.6	0.1695	0.0534	0.2243	0.0330	0.0347	—	—
5	0.95	26.26	18.3	0.1691	0.0439	0.2179	0.0035	0.0292	−0.0386	—
6	1.02	28.23	6.8	0.1683	0.0434	0.2137	0.0031	0.0206	−0.0382	−0.0081

Table III.3 also shows that the value of a coefficient depends on the order at which it is fitted. At $E = 1.200$ MeV, c_2 moves from 0.201 at $L = 2$ to 0.224 at $L = 4$ before settling near 0.214

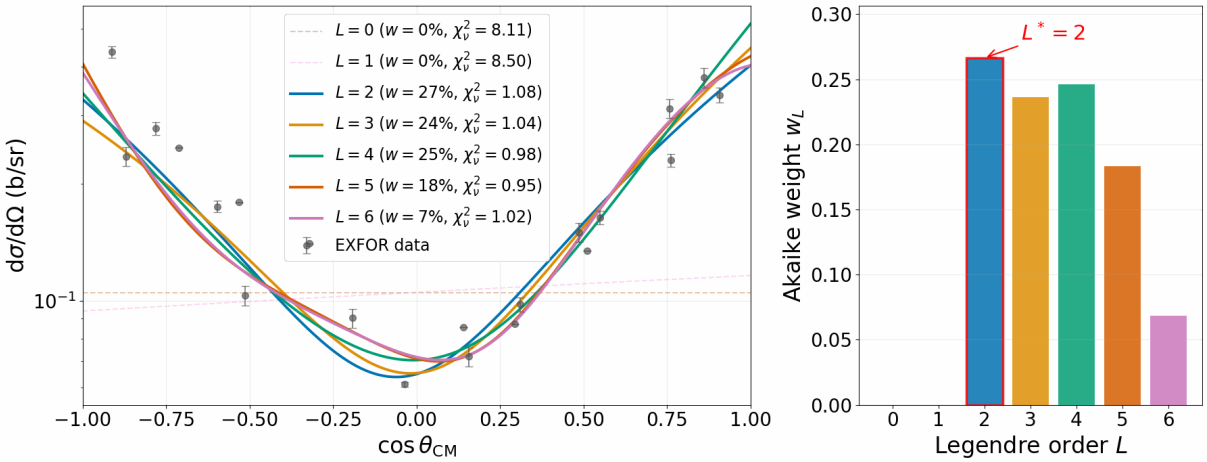


Figure III.9: Legendre fits and AIC-based model selection for ^{56}Fe(n,n) at $E = 1.200$ MeV. Left: ridge-regularized fits at orders $L = 0$ through 6 overlaid on the pooled experimental data (21 points from three experiments). Solid lines denote orders with Akaike weight $w_L > 0.5\%$; dashed lines denote orders with negligible weight. Right: Akaike weights w_L for each candidate order, with the selected degree L^* highlighted.

at higher orders, a shift of about 10% driven by the polynomial structure of the fit alone. In the variance pass (Sec. 4.4), each Monte Carlo sample is fitted at an order drawn from the Akaike weights of Eq. (III.27), so the sampled variance of c_2 combines two effects: the scatter of the refits performed at the same order, and the extra spread between samples fitted at different orders, which assign c_2 different central values. This second effect is the model-form component of the published uncertainty.

The nominal fit produces, for each energy bin: the nominal coefficients c_ℓ at the AIC-selected degree L^*, the Akaike weights w_L, and the converged band scale factors s_b. These quantities are fixed inputs to the subsequent Monte Carlo stages.

4.4. Per-Bin Variance Pass — Monte Carlo Pass 1

In a standard linear least-squares fit, the parameter covariance follows directly from the matrix of Legendre basis functions and the data weights. Let $\mathbf{X}$ denote the $n \times (L+1)$ matrix whose element $X_{i\ell} = P_\ell(\mu_i)$ evaluates the ℓ-th Legendre polynomial at the i-th measurement angle, so that the product $\mathbf{Xc}$ gives the fitted distribution at the measured angles. Let $\mathbf{W} = \mathrm{diag}(w_{p,i}/\sigma_i^2)$ be the diagonal weight matrix, with the per-point weights $w_{p,i}$ of Eq. (III.24). The least-squares solution is $\hat{\mathbf{c}} = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{y}$, and its covariance is:

$$\mathrm{Cov}(\hat{\mathbf{c}}) = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1} \tag{III.28}$$

This closed-form expression assumes a fixed model structure and purely linear error propagation. In the present pipeline, two steps break these assumptions. The Legendre order may differ between samples through degree sampling (Eq. (III.27)), which changes the model structure, and the positivity constraint (Sec. 4.4.3) is non-linear. Propagating uncertainties through these operations analytically would require approximations that are difficult to validate. The pipeline therefore uses a two-pass Monte Carlo approach: Pass 1 estimates the per-bin variances of the Legendre coefficients, and Pass 2 estimates the parameter correlation matrix. Their outputs combine at assembly to form the parameter covariance.

4.4.1. Perturbation model

For each energy bin independently, $N_s = 10\,000$ Monte Carlo realizations are generated by applying the following perturbations to the experimental data. The first two perturbations are drawn from the uncertainty decomposition assembled in Sec. 3.3:

1. **Per-experiment systematic.** Each experiment g is assigned a single multiplicative factor drawn from the per-experiment systematic of Eq. (III.10),

$$\mathcal{N}_g^{\mathrm{sys}} \sim \mathrm{LogNormal}\big(-\tfrac{1}{2}(\sigma_g^{\mathrm{sys}})^2,\, \sigma_g^{\mathrm{sys}}\big), \qquad y_{g,i}^{(s)} = \mathcal{N}_g^{\mathrm{sys}\,(s)} \cdot y_{g,i}, \tag{III.29}$$

   applied to every angular point of experiment g in the current bin. The amplitude σ_g^{sys} is the systematic resolved for that experiment by the manifest of Sec. 3.3. The mean parameter $-\frac{1}{2}(\sigma_g^{\mathrm{sys}})^2$ is a bias correction that fixes $\mathbb{E}[\mathcal{N}_g^{\mathrm{sys}}] = 1$. Within Pass 1 the factor is drawn independently for each energy bin.

2. **Per-point noise.** Independent Gaussian noise is added to each data point, $y_i^{(s)} \to y_i^{(s)} + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0,\, \sigma_{\mathrm{eff},i})$, where $\sigma_{\mathrm{eff},i} = s_b\, \sigma_{\mathrm{stat},i}$ uses the frozen band scale factors s_b from the nominal fit (Eq. (III.26)).

3. **Degree sampling.** The Legendre order L_{draw} for the refit is drawn from the Akaike weight distribution w_L (Eq. (III.27)), which lets different samples be fitted with different polynomial orders.

Drawing one per-experiment systematic factor that acts multiplicatively on all points of an experiment, together with independent per-point noise, is the sampling mechanism that reproduces the rank-one-plus-diagonal block structure of Eq. (III.13) ensemble by ensemble: averaging over realizations recovers the data covariance $\Sigma^{(g)}$. Sampling the normalization and refitting each realization is also free of the downward bias that a multiplicative covariance scaled by the measured values introduces into a least-squares fit (Sec. 3.4), so the sampled shape covariance carries no such bias [162].

4.4.2. Refit with frozen c_0 and per-sample degree

The perturbed data in each bin are refitted with the same ridge-regularized least-squares procedure of Sec. 4.3. The refit weights each point by the diagonal matrix $\mathbf{W} = \mathrm{diag}(w_{p,i}/\sigma_i^2)$ introduced above, which combines the per-point weight $w_{p,i}$ of Eq. (III.24) with the point's own variance. The block-correlated part of the data covariance is left out of this fit, because the rank-one systematic correlation is already injected through the shared per-experiment factor of Eq. (III.29), and reapplying it in the fit covariance would double-count it. Two further changes, absent from the nominal fit, are applied to each refit.

First, the zeroth-order coefficient c_0 is frozen at its nominal value, so the sampled variance is carried by the shape coefficients $c_1, \ldots, c_L$. This follows the ENDF convention of storing the cross-section magnitude (`MF3`) and the angular distribution (`MF4`) separately, and the variance-propagation formula of Sec. 2.1 treats the magnitude as fixed in the same way. Freezing c_0 does not renormalize each experiment to a common angle-integrated cross section. The disagreement in overall normalization among the experiments is retained in the pooled data and broadens the variance of the shape coefficients. The fit is therefore not a strict shape-only fit performed on renormalized data. This broadening only enlarges the shape uncertainty. It does not reintroduce

a coupling between σ_{el} and a_ℓ, because the evaluated magnitude is taken from the host `MF3` independently of the fitted shape (Sec. 2.2).

Second, each Monte Carlo sample is fitted at its drawn degree L_{draw}. Coefficients $c_1, \ldots, c_{L_{\mathrm{draw}}}$ are estimated from the perturbed data; coefficients of orders $\ell > L_{\mathrm{draw}}$ are absent from the sample's coefficient vector. When the per-bin samples are stacked to form the covariance over the full set of published orders $\ell = 1, \ldots, L_{\mathrm{cov}} = 6$, the higher-order slots of samples with $L_{\mathrm{draw}} < \ell$ are filled with zero. The resulting per-order variance therefore has two contributions: the within-subset variance from the samples that were actually fitted at order ℓ or above, and a model-form term that comes from mixing the zero-padded slots with the fitted ones across samples. When the Akaike weight w_ℓ is small, the second term dominates and the published variance reflects the data's lack of support for order ℓ; when w_ℓ is close to one, the first term dominates and the published variance is essentially the fitted-subset variance. Orders above the nominal degree L^* are not sampled. The Monte Carlo order cap is L^*, or lower where the angular coverage of the bin supports fewer orders, and above it every sample keeps the nominal coefficient, so these orders carry zero variance. The model-form term therefore acts only on orders up to L^*, through the samples that draw a lower degree.

For example, if the AIC weights in a bin are $w_3 = 60\%$, $w_4 = 25\%$, $w_5 = 12\%$, and $w_6 = 3\%$, the nominal degree is $L^* = 3$. Then c_1 to c_3 are refitted in every sample, while c_4 to c_6 keep their nominal value and carry zero variance, because the Monte Carlo order cap does not exceed L^*.

4.4.3. Positivity enforcement

A truncated Legendre expansion can produce small negative values at certain angles, which is unphysical since the differential cross section is a non-negative quantity [172]. After each Monte Carlo refit, the reconstructed angular distribution is checked on a fine angular grid; if it violates non-negativity, the coefficient vector is projected onto the nearest non-negative one by a constrained least-squares step. The adjustment is small and preserves the overall angular shape. The projection formulation and an illustration are given in Appendix E.5.

This per-bin pass yields the marginal variance σ_ℓ^2 of each Legendre coefficient at each energy bin, with contributions from normalization uncertainty, measurement noise, and model-order ambiguity. Because every bin is fitted independently with its own set of perturbations, the samples carry no information about correlations between energy bins. They do constrain the correlations between Legendre orders within a bin, which the assembled covariance takes from Pass 2 as well (Appendix E.6).

4.5. Kernel-Weight Correlation Pass — Monte Carlo Pass 2

Neighboring energy bins share experimental datasets because the finite TOF energy resolution of each experiment spreads its measurements across multiple bins. This data overlap creates natural statistical correlations between bins. The kernel-weight Monte Carlo pass exploits this overlap to estimate the full parameter correlation matrix in a single self-consistent realization.

4.5.1. Overlap weights from TOF resolution

For each experiment g and energy bin i, an overlap weight $\omega_{g,i}$ is computed as the fraction of the experiment's energy-resolution kernel that falls within the bin boundaries:

$$\omega_{g,i} = \Phi\left(\frac{E_i^{\mathrm{up}} - E_g}{\sigma_{E,g}}\right) - \Phi\left(\frac{E_i^{\mathrm{low}} - E_g}{\sigma_{E,g}}\right), \tag{III.30}$$

where Φ is the standard normal CDF, E_g is the experiment's nominal energy, and $\sigma_{E,g}$ is its TOF energy resolution derived from the flight path and timing resolution (Sec. 3.1). When experiment-specific TOF parameters are available from the EXFOR entry or associated bibliography, they are used directly. Otherwise, default values of $\Delta t = 5$ ns and $L = 27.037$ m are adopted, corresponding to the Pirovano measurement [121] at the GELINA facility. Experiments with poorer resolution (larger $\sigma_{E,g}$) have broader kernels and therefore contribute to more bins, creating stronger correlations.

4.5.2. Perturbation model

The perturbation model differs from Pass 1 in four respects:

1. **Per-experiment systematic.** A single multiplicative factor is drawn once per experiment per sample,

$$\mathcal{N}_g^{\mathrm{sys}} \sim \mathrm{LogNormal}\left(-\tfrac{1}{2}(\sigma_g^{\mathrm{sys}})^2, \sigma_g^{\mathrm{sys}}\right), \qquad y_{g,i}^{(s)} = \mathcal{N}_g^{\mathrm{sys}\,(s)} \cdot y_{g,i}, \tag{III.31}$$

   and applied across every bin to which experiment g contributes. This is the mechanism that generates the long-range cross-bin correlations: an experiment that covers a wide energy range, such as Kinney (0.847–2.5 MeV) or Cierjacks (0.847–3.0 MeV), links every bin in its range through the same realization of $\mathcal{N}_g^{\mathrm{sys}\,(s)}$.

2. **Per-point noise.** A dataset here is one experiment at one nominal EXFOR energy, that is, a single angular distribution. Gaussian noise is drawn independently for each of its angular points, once per dataset, and the perturbed dataset is then reused in every bin it contributes to. The amplitude uses $\sigma_{\mathrm{eff},i} = s_b\,\sigma_{\mathrm{stat},i}$ from Eq. (III.26), with the frozen band scale factors s_b from the nominal fit.

3. **Degree.** The Legendre order is fixed at the AIC-selected nominal degree L^*. The purpose of Pass 2 is to capture the cross-bin correlation structure; the variance is taken from Pass 1.

4. **Weights.** Each data point in the bin receives a per-point weight $w_{\mathrm{pp},i} = \omega_{g,i}\, g_{\mathrm{Kish}}$ that combines the energy-resolution overlap weight from Eq. (III.30) with the Kish factor of Eq. (III.24). Every bin performs its weighted least-squares fit using all experiments whose overlap weight exceeds a minimum threshold ($\omega_{g,i} > 10^{-3}$). Because Pass 2 does not deduplicate energies, an experiment may enter a bin at several nearby nominal energies through its resolution. The overlap weight $\omega_{g,i}$ spreads its contribution across those energies, the Kish factor accounts for the angular redundancy within each measurement, and the per-experiment weight cap of Sec. 4.3 keeps any single campaign from dominating the bin.

Because the same perturbed datasets contribute to multiple bins through the overlap weights, the resulting coefficient samples are naturally correlated across energy. Two mechanisms drive

these correlations. The energy-resolution overlap creates short-range correlations spanning a few bins: an experiment with broader resolution contributes to more bins, which links neighboring bins through the shared per-point noise. The per-experiment systematic creates long-range correlations spanning the full energy range of each experiment: a single calibration offset affects all energies of one experiment together, which gives these correlations a direct physical interpretation.

The refit uses the same ridge-regularized least-squares procedure as Pass 1 (Sec. 4.4), again with diagonal per-point weights, c_0 frozen at the nominal value, and positivity enforced.

Figure III.10 illustrates this mechanism at a representative energy bin. The data points whose nominal energy lies at the bin center carry the highest overlap weight; points contributed by the same experiment from neighboring nominal energies enter with diminishing weight as their distance from the bin center grows.

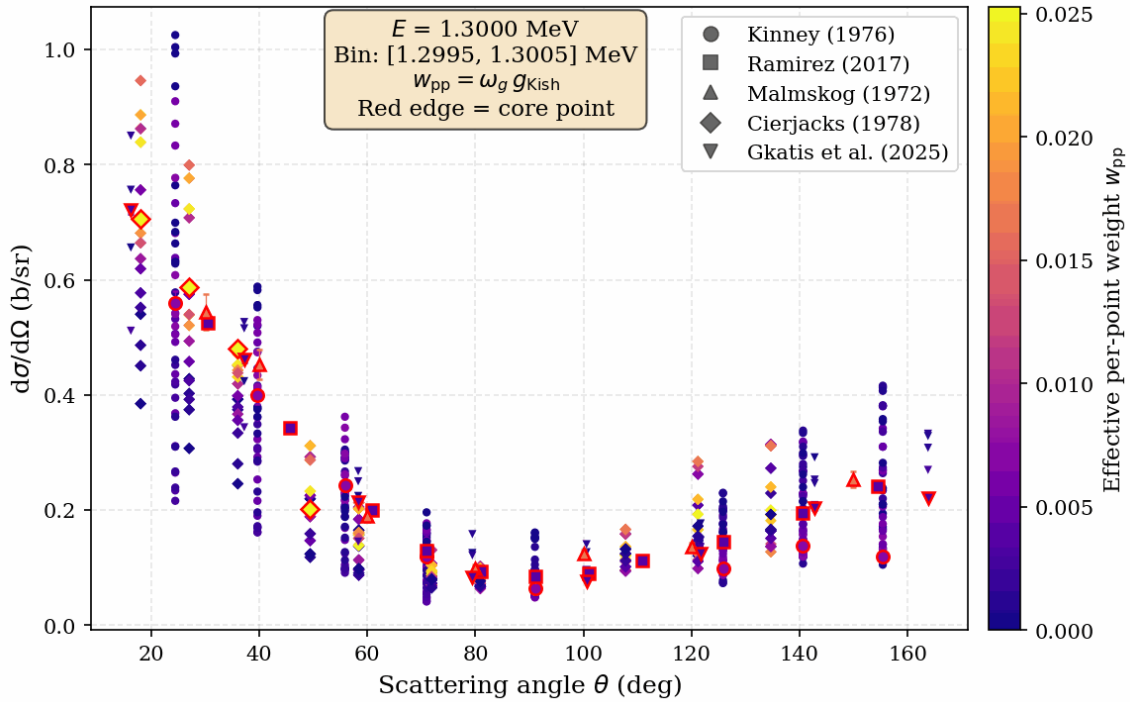


Figure III.10: Effective per-point weights $w_{\mathrm{pp},i} = \omega_{g,i}\, g_{\mathrm{Kish}}$ for the kernel-weight Monte Carlo pass at $E = 1.30$ MeV. Each marker shape represents a different experiment; the color encodes the combined weight from the CDF-based overlap $\omega_{g,i}$ (Eq. (III.30)) and the Kish factor g_{Kish} (Eq. (III.24)). Red-edged markers denote the core data point of each experiment (highest overlap weight). Error bars are shown for core points only.

The output of this pass is the full parameter correlation matrix $\mathbf{R}_{\mathrm{KW}}$. Both the within-bin (cross-order) blocks on the diagonal and the cross-bin (cross-energy) blocks off the diagonal are estimated jointly from the same Monte Carlo ensemble.

4.6. Covariance Assembly

During assembly, the parameter vector is ordered by energy bin, with L_{cov} Legendre orders per bin:

$$\mathbf{x}_{\mathrm{asm}} = \Big[\underbrace{a_1(E_1)\ \ldots\ a_L(E_1)}_{E_1},\quad \underbrace{a_1(E_2)\ \ldots\ a_L(E_2)}_{E_2},\quad \ldots\Big] \tag{III.32}$$

The correlation matrix $\mathbf{R}_{\mathrm{KW}}$ from Pass 2 is rescaled by the vector of per-bin standard deviations $\boldsymbol{\sigma}$ from Pass 1 through a diagonal congruence transform:

$$\mathbf{C} = \mathrm{diag}(\boldsymbol{\sigma})\, \mathbf{R}_{\mathrm{KW}}\, \mathrm{diag}(\boldsymbol{\sigma}) \tag{III.33}$$

This per-bin ordering differs from the per-order layout used in the `MF34` format (Sec. 4.5.3), where the parameter vector groups all energies for order $\ell = 1$, then all energies for $\ell = 2$, and so on. The two representations contain the same information; the matrix is reordered to the `MF34` $\mathbf{V}_{\ell,k}$ block structure before writing the output file.

The kernel-weight correlation matrix captures the actual structure of the experimental database, including asymmetric energy coverage across facilities, varying energy resolution, and the specific energy ranges of each dataset. In energy regions with sparse experimental overlap, the correlation estimates are less reliable because fewer shared data points are available to constrain them. Sourcing the full $\mathbf{R}_{\mathrm{KW}}$ from Pass 2 alone is what guarantees positive semi-definiteness of $\mathbf{C}$; the argument is given in Appendix E.6.

4.6.1. Sample covariance and block structure

This final step only changes the basis of the covariance; it does not add any new perturbation. The assembled covariance $\mathbf{C}$ describes the fitted coefficients c_ℓ, whereas `MF34` stores the normalized coefficients $a_\ell = (c_\ell/c_0)/(2\ell+1)$. Because this normalization divides by c_0, which in general differs from one Monte Carlo sample to the next, the map from c_ℓ to a_ℓ is nonlinear and cannot be applied to $\mathbf{C}$ directly. The pipeline therefore draws a synthetic Gaussian sample set from $\mathbf{C}$ by Cholesky decomposition,

$$\mathbf{C} = \mathbf{L}\mathbf{L}^T, \qquad \mathbf{c}^{(s)} = \bar{\mathbf{c}} + \mathbf{L}\,\mathbf{z}^{(s)}, \qquad \mathbf{z}^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{III.34}$$

applies the normalization $a_\ell^{(s)} = (c_\ell^{(s)}/c_0^{(s)})/(2\ell+1)$ to each sample, and recomputes the covariance in the a_ℓ basis. These samples are an intermediate numerical device for the basis change, not perturbed ENDF files for transport calculations (those are generated separately in Chapter II). In the current configuration c_0 is frozen at the nominal value (Sec. 4.4), so the transformation reduces to a linear scaling; the sample-based route is retained for generality.

Let $\mathbf{a}^{(s)}$ denote the normalized parameter vector for sample s, ordered as:

$$\mathbf{a}^{(s)} = [a_1(E_1), \ldots, a_1(E_N),\ a_2(E_1), \ldots, a_2(E_N),\ \ldots] \tag{III.35}$$

The sample covariance is defined as:

$$\hat{\mathbf{C}} = \frac{1}{N_s - 1} \sum_{s=1}^{N_s} \left(\mathbf{a}^{(s)} - \bar{\mathbf{a}}\right)\left(\mathbf{a}^{(s)} - \bar{\mathbf{a}}\right)^T, \tag{III.36}$$

where $\bar{\mathbf{a}} = N_s^{-1} \sum_s \mathbf{a}^{(s)}$ is the sample mean. Because the parameter vector is ordered by Legendre order and then by energy, $\hat{\mathbf{C}}$ inherits the $\mathbf{V}_{\ell,k}$ block structure of Sec. 4.5.3. With $N_s = 10\,000$ samples, the statistical noise on the estimated covariance entries is small enough for the resulting `MF34` to be used directly in uncertainty propagation.

4.6.2. Near-zero regularization

The ENDF-6 `MF34` format stores covariance data in relative form. When a Legendre coefficient passes through zero as a function of energy, the relative standard deviation diverges even though

the absolute variance is well-behaved, which would create artificial spikes in the stored MF34 data. Coefficients whose signal-to-noise ratio falls below unity are flagged and have their relative variance replaced by the median of the nearest well-behaved neighbors of the same Legendre order, applied through a diagonal congruence transform that preserves correlations and positive semi-definiteness. The two-stage check and its rationale are detailed in Appendix E.7.

4.6.3. Adaptive multigroup collapse

An adaptive multigroup collapse reduces the size of the MF34 section by merging adjacent energy bins that show highly correlated behavior. Grouping decisions are based on the first-order coefficient ($\ell = 1$), with the resulting energy grid applied uniformly to all Legendre orders so that the published MF34 carries a single grid across the block.

The grouping algorithm proceeds as follows:

1. For adjacent energy bins i and $i+1$, compute the $\ell = 1$ correlation $\rho(a_1(E_i),\, a_1(E_{i+1}))$.
2. Extend the current group by one bin if $\rho \geq 0.85$ and the running ratio $\max(\sigma_{a_1})/\min(\sigma_{a_1})$ within the candidate group stays below 5. Otherwise, finalize the group and start a new one at the next bin.
3. Collapse each group via a linear aggregation: $\mathbf{C}_{\text{group}} = \mathbf{A}\,\hat{\mathbf{C}}\,\mathbf{A}^T$, where $\mathbf{A}$ is a bin-width-weighted averaging matrix.
4. Compensate the variance lost through averaging. For each group, the target variance is set at an adaptive percentile of the fine-bin variances. The percentile scales with the intra-group variance heterogeneity: groups where all bins have similar σ_{a_1} receive a base percentile (67th), while groups that merge bins with very different variances receive a higher percentile (up to the 85th). A no-deflation guard ensures that scale factors are always ≥ 1, so the compensation can only restore variance, never reduce it.

The correlation threshold is the primary gate: it keeps the merged bins coherent in their angular-shape behavior. The variance-ratio gate is intentionally loose because differences in σ_{a_1} between adjacent bins often reflect uneven experimental coverage rather than genuinely different uncertainty levels, and merging a sparsely sampled bin with a better-sampled neighbor lets the grouped variance reflect the richer information available nearby. The adaptive percentile compensation in step 4 is the downstream safeguard: when bins with moderately different variances are merged, the averaging collapse pulls the grouped variance below the larger fine-bin values, and raising the percentile target in proportion to the intra-group variance ratio restores it.

The collapsed covariance is written in the standard ENDF-6 MF34 format using the $\mathbf{V}_{\ell,k}$ block structure, making it compatible with processing codes.

4.7. Pipeline Output: Use and Limitations

The evaluation pipeline produces a nominal ENDF file containing the best-estimate MF4 Legendre coefficients and their MF34 covariance matrix, together with auxiliary outputs (fitted coefficient tables, covariance matrices, and a detailed log file). This subsection describes how the MF34 output can be used for uncertainty propagation and its limitations.

Uncertainty propagation strategies

From the nominal ENDF file, uncertainty can be propagated to application quantities through two standard approaches introduced in Chapter II. The first is the first-order sensitivity method (the "sandwich rule," Eq. (II.80)), provided sensitivities to the individual Legendre moments are available. The second is random sampling from the multivariate normal distribution defined by the `MF34` covariance, using the general ENDF perturbation pipeline described in Chapter II. The `MF34` covariance is also well suited for direct comparison with covariance data from other evaluations (e.g., JEFF, JENDL, TENDL).

Limitations of the MF34 representation

The `MF34` representation has two limitations. First, it operates on the grouped energy grid produced by the adaptive collapse rather than the original fine grid. Second, samples drawn from `MF34` use a fixed Legendre-order structure (the published orders $\ell = 1$–6), whereas the internal Monte Carlo procedure draws a degree per sample from the full Akaike weight distribution (Sec. 4.3). This adds a model-form component that the covariance reflects only through the marginal variance of each order; the joint distribution of orders that the Monte Carlo ensemble carries is richer than a second-moment summary can capture. The published `MF34` also treats the integrated cross section and the angular coefficients as independent, carrying no $\ell = 0$ cross-covariance row; this is the correct value under the EXFOR-only methodology, as explained in Appendix E.8.

Validity of the Gaussian covariance representation

Representing the uncertainty as a covariance matrix assumes the joint distribution of the Legendre coefficients is approximately Gaussian, so that a second-moment summary captures the dependence between coefficients and downstream propagation can be linearized about the mean. This assumption was checked against the Monte Carlo ensemble. For the great majority of coefficient pairs the dependence is linear enough for the Pearson correlation to capture it, so the covariance matrix is an adequate representation; the departures are confined to the adjacent-bin pairs of the resonance band that produce the strong negative correlations of Appendix E.9, where the ensemble carries model-form structure that a second-moment summary only partially reflects. This is consistent with the systematic residual chi-squared excess seen in the goodness-of-fit analysis, and is examined further in Appendix E.10.

5. Results and Comparison

This section presents the results of the ^{56}Fe elastic-scattering angular distribution evaluation. It begins with a summary of the pipeline execution and the sampled angular distributions (Sec. 5.1), then compares the fitted Legendre coefficients and their uncertainties against JEFF-4.0 and JENDL-5 (Sec. 5.2), inspects the propagated uncertainty bands (Sec. 5.3), and concludes with a goodness-of-fit analysis (Sec. 5.4).

5.1. Pipeline Output: Use and Limitations

The pipeline processed 67 experiments comprising 23 253 angular-distribution data points across 1 738 energy bins in the 847 keV–4 MeV range. The energy grid is the union of the Kinney and Pirovano experimental energies described in Sec. 4.2. For reference, the JEFF-4.0 `MF4` uses 1 693 energy points and JENDL-5 uses 1 129 points in this range.

Each Monte Carlo pass draws its own 10 000 samples. From these (Sec. 4.6), the pipeline assembles a pointwise covariance matrix with 10 428 parameters (1 738 energy bins × 6 Legendre orders). The adaptive collapse merges these 1 738 fine energy bins into 526 multigroups, a single grid shared across all six orders. For comparison, the JEFF-4.0 `MF34` uses 13 energy groups for its first four orders and 10 for $\ell = 5$–6, and JENDL-5 uses 4. The resulting ENDF file is 94.3 MB, and the full pipeline runs in approximately 5.0 hours. The runtime breakdown, parallelization scheme, and additional execution diagnostics are reported in Appendix E.11.

A first qualitative assessment of the evaluation comes from the Monte Carlo samples themselves. Each sample corresponds to a complete, self-consistent angular distribution that can be reconstructed into a differential cross section via Eq. (III.1). Figure III.11 displays the reconstructed differential cross sections at four representative incident energies (1.00, 1.30, 2.15 and 3.10 MeV). Alongside the three evaluated libraries, the figure includes two TALYS [173] calculations obtained with different optical-model prescriptions: the default Koning–Delaroche phenomenological potential [174] and the microscopic Jeukenne–Lejeune–Mahaux (JLM) model. Because the present evaluation is purely data-driven, these model-based curves serve as an independent physics cross-check on the fitted distributions.

Each panel overlays the experimental data in the bin of the nominal energy. The 25 Monte Carlo samples give a visual approximation of the spread covered by the full 10 000-sample set: narrower where experimental coverage is denser, wider where fewer measurements are available. The three evaluations show broadly similar behavior, and the two TALYS calculations are in reasonable agreement with the data-driven evaluations, although they differ from each other more than the three evaluations do.

5.2. Legendre Coefficient Comparison Across Evaluations

A more complete picture than the discrete-energy view of Figure III.11 is obtained by examining the Legendre coefficients $a_\ell(E)$ as continuous functions of energy. Figure III.12 compares $a_1(E)$ for JEFF-4.0, JENDL-5, and this work across the full 847 keV–4 MeV range and in two zoomed windows.

The full-range view (Figure III.12a) shows a trend that is close for all three evaluations, with the differences becoming more visible above 2.5 MeV. Below 2.5 MeV all three draw on Kinney's data (Sec. 1.3) and follow the same trend, although this work shows somewhat more fluctuations. This is driven less by the energy grid, whose density is comparable across the three evaluations, than by the fitting methodology: JEFF and JENDL always fit six Legendre orders, which acts as a smoothing constraint, whereas this work selects the order adaptively via AIC. The effect is most visible in the zoomed view of Figure III.12b. Above 2.5 MeV the data sources also diverge: JEFF and JENDL rely on Smith, while this work uses Pirovano, whose finer energy mesh resolves more structure. These additional fluctuations are not unphysical. The elastic cross section itself (Figure III.1) varies on a similar scale across the whole range, so there is no reason to expect the angular distribution to be smoother above 2.5 MeV than below. The relative smoothness of

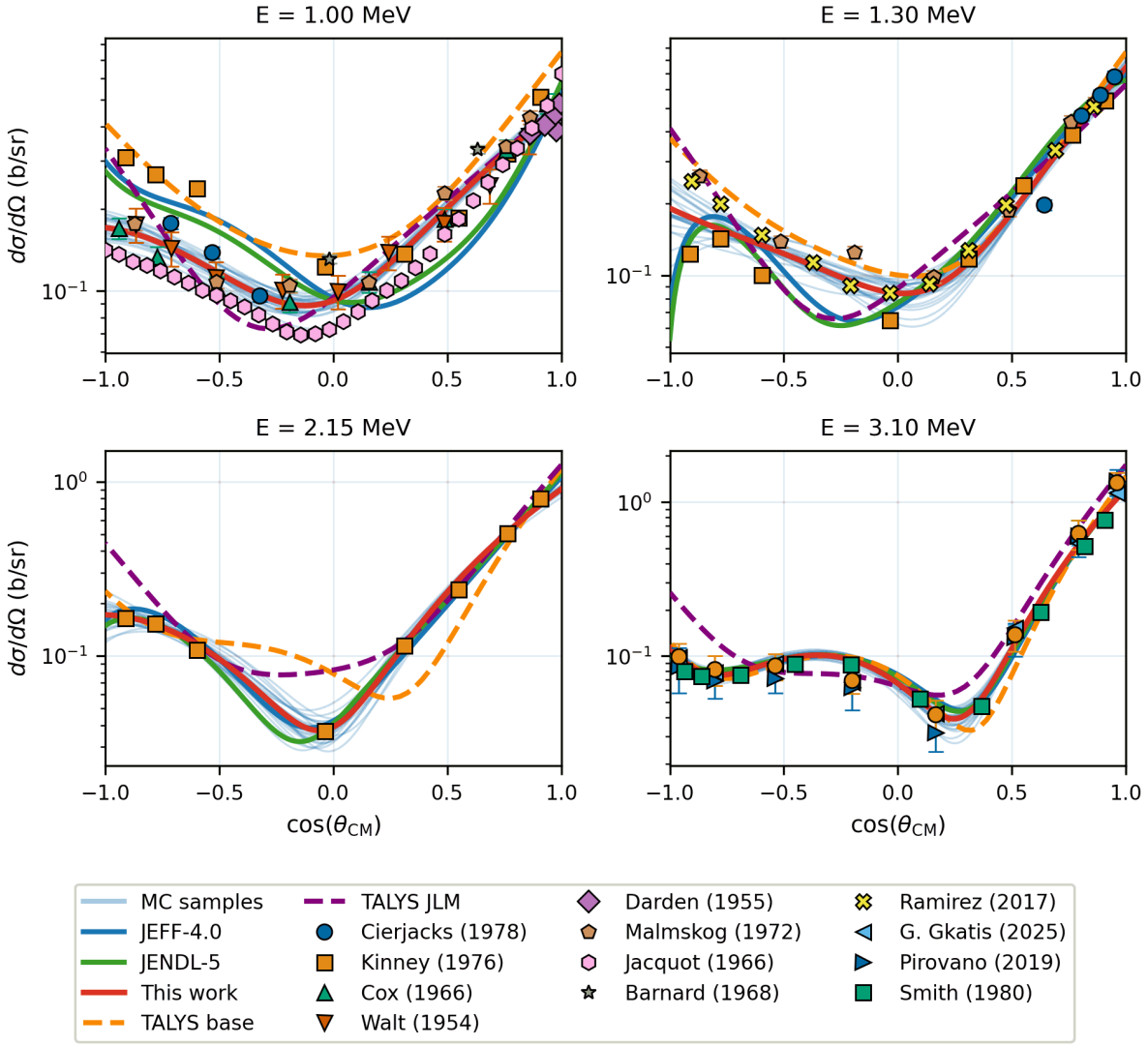
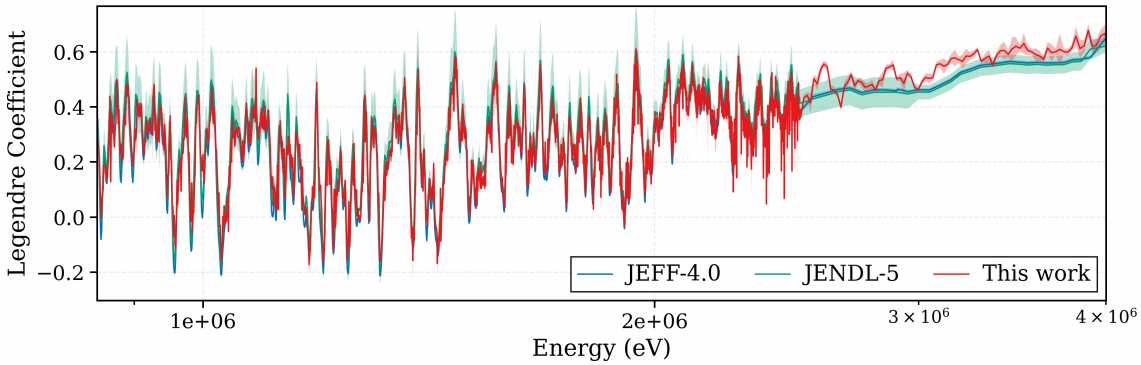
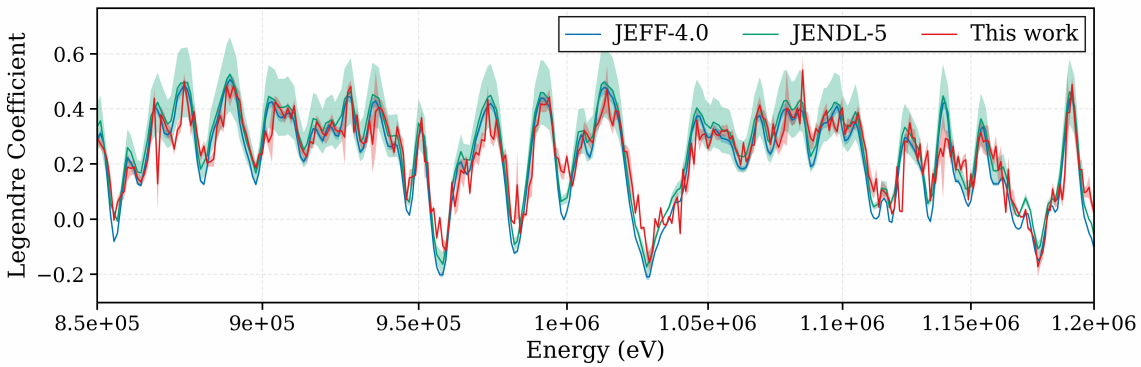


Figure III.11: Differential elastic cross sections of ^{56}Fe at four representative energies. Each panel shows the evaluation of this work (red), JEFF-4.0 (blue), JENDL-5 (green), and two TALYS calculations using the default Koning–Delaroche OMP and the JLM model. Colored markers denote experimental data points whose nominal energy falls inside the energy bin being evaluated. Twenty-five Monte Carlo samples are drawn as thin curves to illustrate the spread of the sampled distributions.

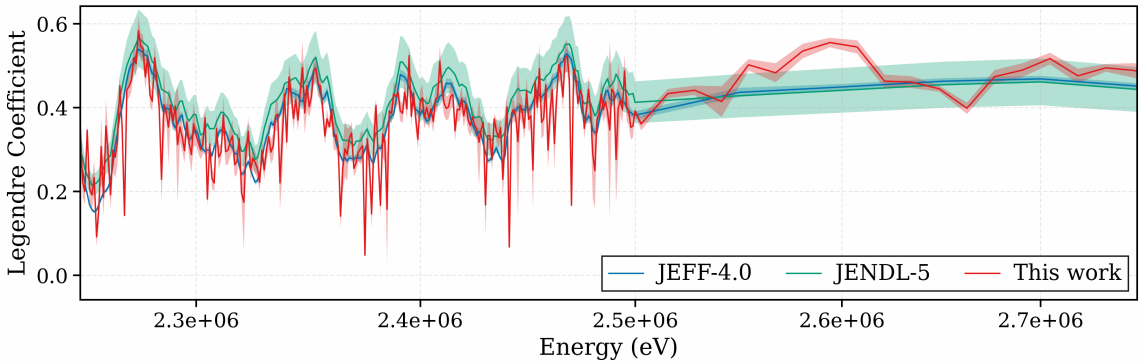
JEFF and JENDL, and to some extent this work, reflects the grid resolution and fitting choices rather than the underlying physics, and even the finer Pirovano grid is likely still too coarse to resolve all of the structure.



(a) Full range.



(b) Zoomed: 0.85–1.2 MeV.



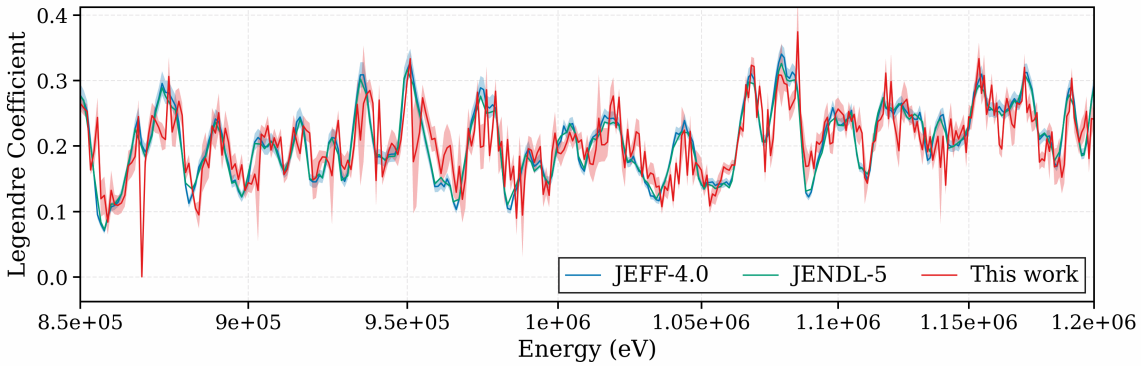
(c) Zoomed: 2.25–2.75 MeV.

Figure III.12: First Legendre coefficient a_1 for ^{56}Fe elastic scattering comparing JEFF-4.0, JENDL-5, and this work. (a) Full energy range 847 keV–4 MeV. (b) Zoomed view of the resonance region 0.85–1.2 MeV. (c) Zoomed view of the 2.25–2.75 MeV region.

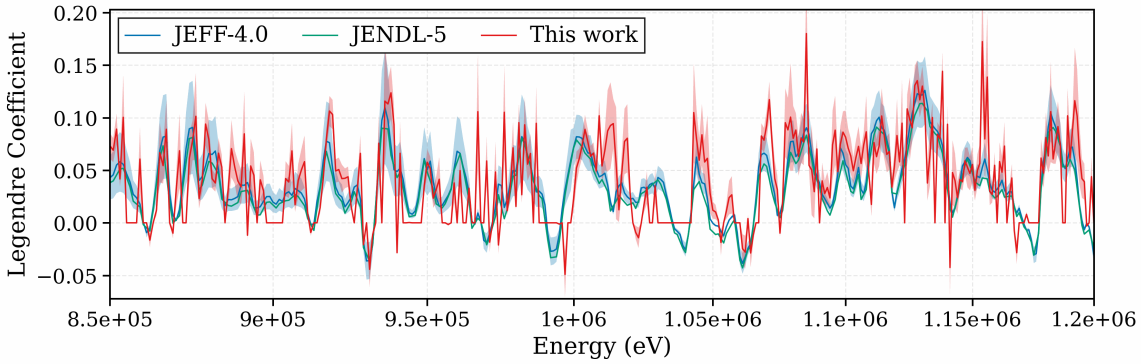
Below 2.5 MeV the agreement is generally within 1–2σ of the uncertainties from this work. Above 2.5 MeV the a_1 of this work lies systematically above both libraries, by up to about 0.1, which is more than its own uncertainty. In the 0.85–1.2 MeV region (Figure III.12b), the three curves agree closely because Kinney's data dominate the fit for all evaluations. In the 2.25–2.75 MeV window (Figure III.12c), the general energy dependence is shared but this work shows more fluctuations, consistent with its finer grid.

Figure III.13 presents a zoomed comparison of a_2 and a_3 in the 0.85–1.2 MeV region. The

same grid and fitting effects apply, with one further consequence for these orders: fitting six Legendre orders even when fewer would suffice redistributes variance across orders and can shift the lower-order coefficients. For a_3, this work shows flat stretches at zero in the bins where AIC selected only two orders, so a_3 vanishes identically; the fixed-order fits of JEFF and JENDL instead carry small, poorly determined a_3 values there.



(a) Second-order coefficient a_2.



(b) Third-order coefficient a_3.

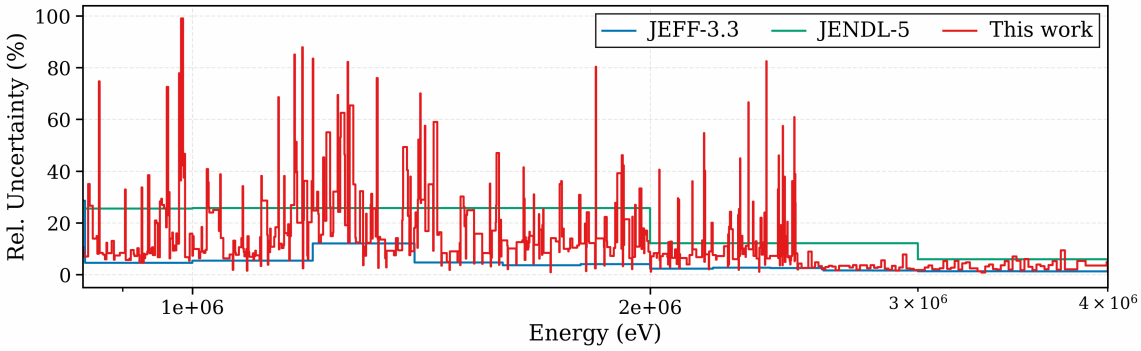
Figure III.13: Zoomed comparison of a_2 and a_3 in the 0.85–1.2 MeV resonance region for JEFF-4.0, JENDL-5, and this work.

Figure III.14 compares the relative uncertainties $\sigma(a_\ell)/|a_\ell|$ for the first three Legendre orders from the `MF34` covariance data of each evaluation. The energy grid of this work is finer than that of JEFF or JENDL for the covariance representation (Sec. 5.1), with 526 multigroups compared to 13 (JEFF) and 4 (JENDL).

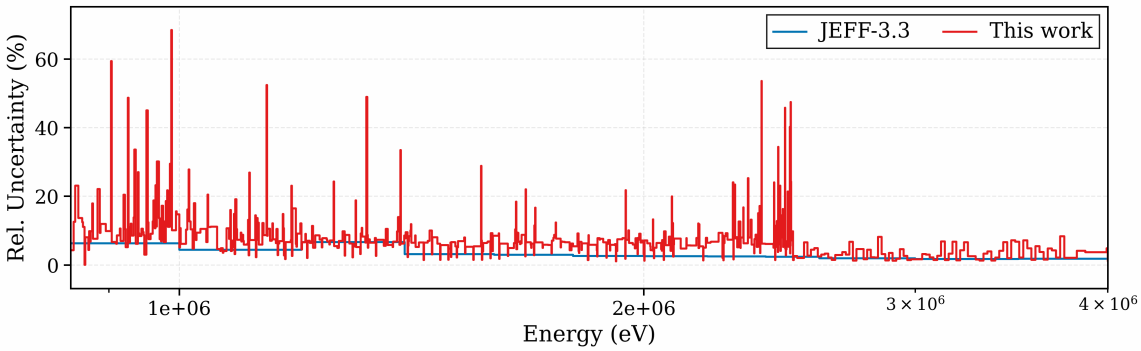
For a_1 (Figure III.14a), the uncertainties from this work generally fall between those of JEFF and JENDL, with several spikes at energies where the experimental datasets disagree more severely. JENDL-5 provides covariance information only for $\ell = 1$, leaving higher-order uncertainties undefined.

For a_2 (Figure III.14b), the uncertainties from this work are roughly similar to those of JEFF, slightly larger overall, and again show spikes at energies where experimental datasets disagree. For a_3 (Figure III.14c), this work shows flat stretches at zero for the same reason as the central values: at energy bins where the AIC selected only two Legendre orders, the a_3 uncertainty is identically zero.

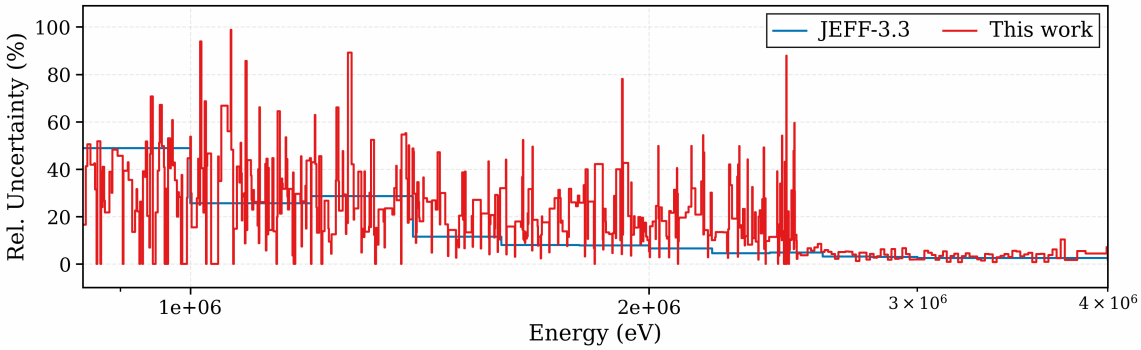
These spikes reflect how the covariance is built. Each coefficient uncertainty is estimated independently in every energy bin, so the value can change sharply from one bin to the next. A bin constrained by a single campaign, most often Kinney, carries a small variance, close to



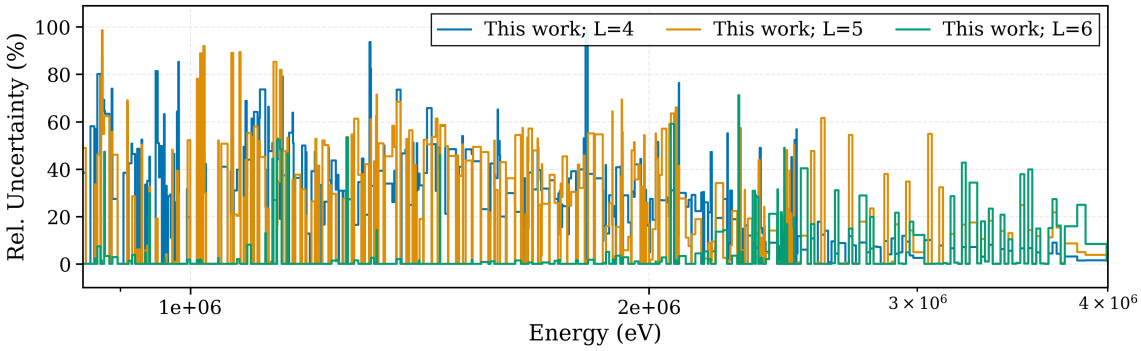
(a) a_1 relative uncertainty.



(b) a_2 relative uncertainty.



(c) a_3 relative uncertainty.



(d) Higher orders a_4, a_5, a_6 relative uncertainty.

Figure III.14: Relative uncertainties of the Legendre coefficients from the `MF34` covariance data of JEFF-4.0, JENDL-5, and this work: (a)–(c) the first three orders a_1, a_2, a_3, and (d) the higher orders a_4, a_5, a_6.

the uncertainty that experiment reports for its own points. A bin where several campaigns fall together and disagree on the differential cross section carries a variance inflated by their spread. The peaks therefore mark the bins with the strongest between-experiment disagreement.

The higher orders a_4–a_6 (Figure III.14d) show the same contrast. The large relative uncertainties of JEFF-4.0 on these orders reflect the near-zero coefficients in their denominator (Sec. 1.2) as much as the covariance itself. This work keeps the coefficient uncertainties bounded where the orders are fitted and zero in the bins where AIC does not select them, and resolves them on a grid fine enough to follow the coefficients.

Figure III.15 shows the energy correlation matrix of the first Legendre coefficient a_1 produced by the pipeline. The full $\ell = 1$–6 correlation matrix is reported in Appendix E.11, Figure E.4.

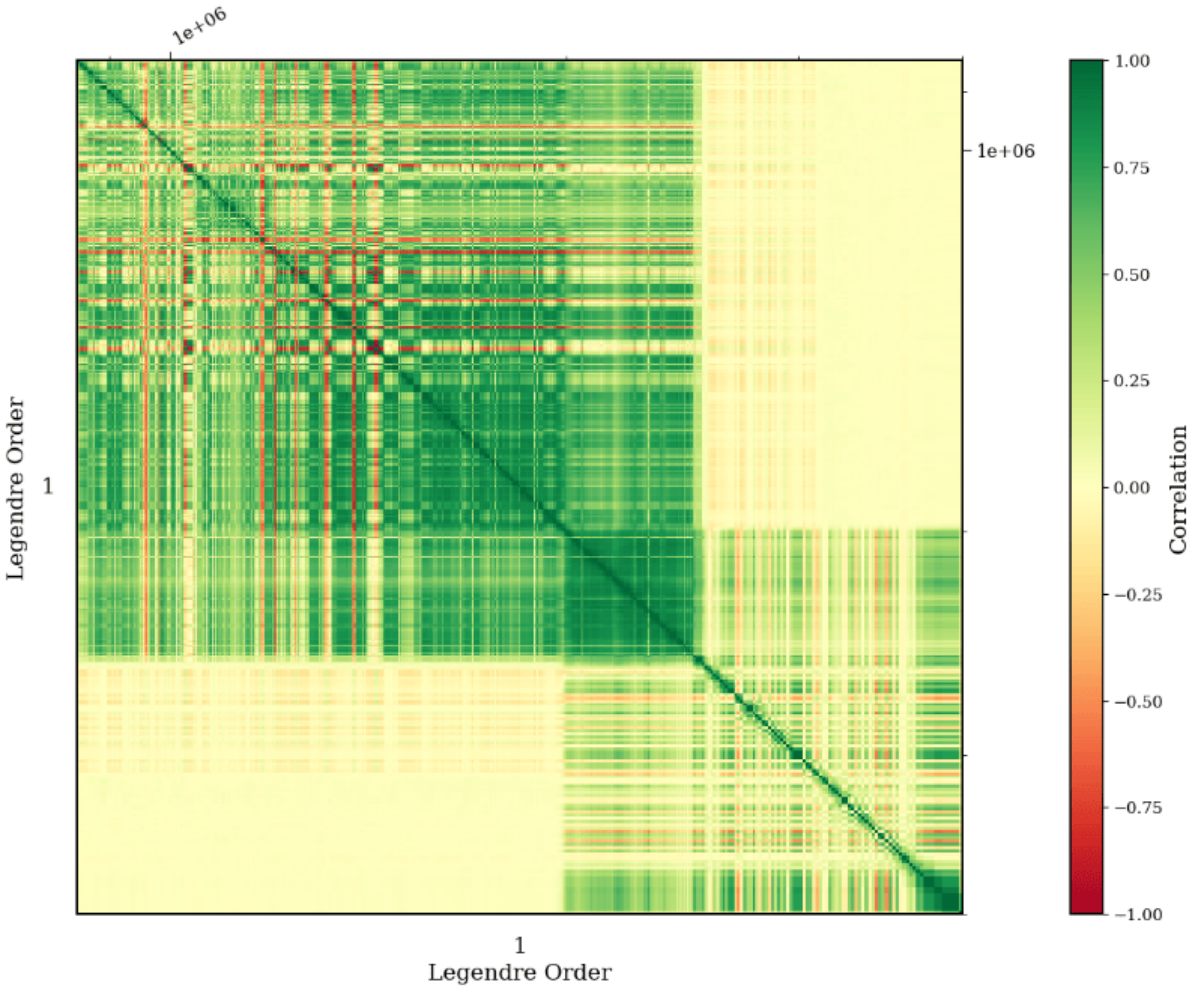


Figure III.15: Energy correlation matrix of the first Legendre coefficient a_1 for ^{56}Fe elastic scattering from this work, across the 847 keV–4 MeV range. The full $\ell = 1$–6 matrix is shown in Figure E.4.

Two broad positively correlated blocks stand out. The block below about 2.5 MeV is set mainly by the experiments that dominate that range, Kinney, Cierjacks, and Gkatis, while the block above 2.5 MeV reflects the different measurements available at higher energies, with the largest contributions from Pirovano, Smith, and Ramirez, and Gkatis spanning both. These blocks come from the per-experiment normalization uncertainty (Sec. 4.5): a single calibration offset shifts all energies of one experiment together, so each campaign imprints a positive-correlation

band over the energy range it constrains.

Superimposed on this background is a small population of strongly negative correlations between some adjacent energy bins, most visible for $\ell = 1$ in the 1.0–1.8 MeV resonance band. These occur where neighboring bins have different AIC-selected Legendre orders L^*: with c_0 frozen, changing the fitted order redistributes leverage across the coefficients, so a perturbation that raises a coefficient in one bin lowers it in the next. They reflect a genuine model-averaging ambiguity, and are discussed in detail in Appendix E.9.

5.3. Uncertainty Band Comparison at Selected Energies

While the Legendre coefficient comparison (Sec. 5.2) describes uncertainties in the parametric domain, the quantity of direct physical interest is the uncertainty on the differential cross section itself. To assess how the different `MF34` covariance structures translate into angular-distribution uncertainty bands, we propagate the Legendre covariances to the differential cross section using the sandwich formula (Sec. 2.1) and compare the resulting bands at four representative energies.

Figures III.16–III.19 present this comparison in a three-panel side-by-side layout (JEFF-4.0, JENDL-5, and this work) at four representative energies.

Two general features of the band shapes should be noted. Because JENDL-5's `MF34` is restricted to $\ell = 1$, its uncertainty band captures only the forward–backward asymmetry, producing a characteristic dipolar shape without higher-order modulations. JEFF's bands, by contrast, can become notably wide in energy bins where the absolute uncertainty carried by the higher-order coefficients dominates the propagated variance.

At 1.0 MeV (Figure III.16), numerous experimental datasets are available. This work incorporates all of them and finds a balanced fit. At backward angles, the Kinney data points used to fit JEFF and JENDL deviate noticeably from the other experiments available at those angles, which may introduce a bias in the existing evaluations at this energy. The JEFF band shows a pronounced lobed angular pattern driven by high-order coefficient uncertainties. The JENDL band is narrow relative to the spread of the experimental data.

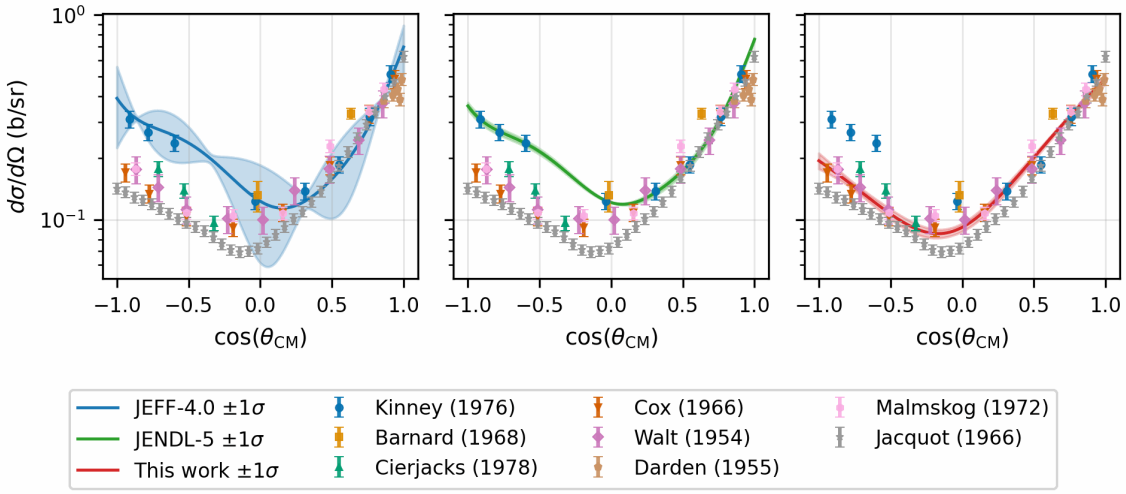


Figure III.16: Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 1.0$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data.

At 1.4 MeV (Figure III.17), this work achieves a good fit to the available experimental data using only two Legendre orders, while JEFF and JENDL always fit six. The resulting uncertainty band is well calibrated to the scatter of the data points.

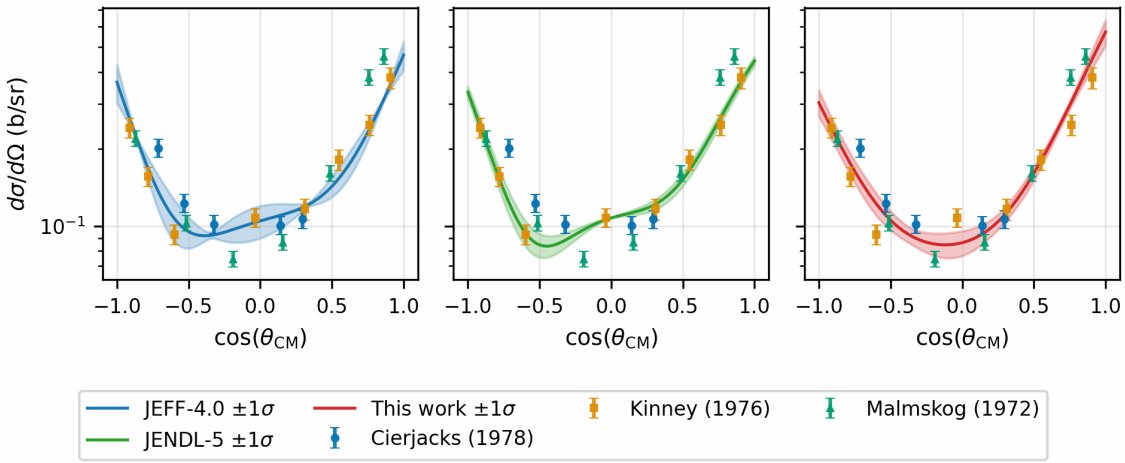


Figure III.17: Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 1.4$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data.

At 2.38 MeV (Figure III.18), only two experiments constrain the fit. The three evaluations produce similar central values. This work shows somewhat wider uncertainty bands, reflecting the reduced experimental constraint at this energy. The conservative band is appropriate given the limited data coverage.

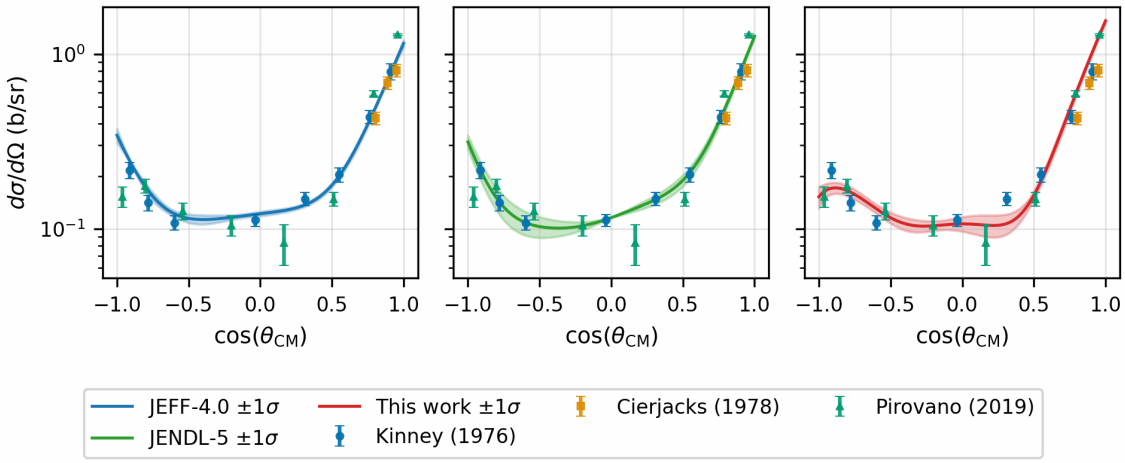


Figure III.18: Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 2.38$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data.

At 2.85 MeV (Figure III.19), several experimental datasets are available and show good mutual consistency. The fits are similar across all three evaluations, and the uncertainty bands from JEFF and this work are comparable in width. The consistent low uncertainties for this bin reflect the well-constrained experimental situation.

Across these four energies, the evaluation adapts to the available data: wider bands where experimental coverage is sparse (2.38 MeV), narrower bands where multiple datasets agree (2.85 MeV),

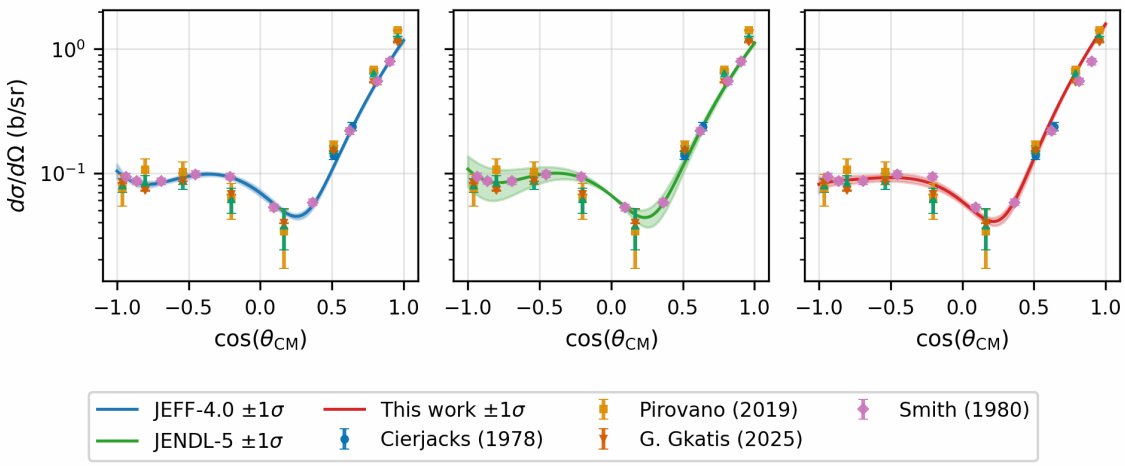


Figure III.19: Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 2.85$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data.

and physically smooth angular profiles throughout.

5.4. Goodness-of-Fit Analysis

This section applies the chi-squared framework of Sec. 3.4 to the three evaluations. The normalization $c_0(E_g)$ is obtained for each evaluation from the experiments that drove it (Sec. 3.2). The covariance combines the rank-one EXFOR structure of Eq. (III.13) with each library's propagated `MF34` block of Eq. (III.16).

The database is unevenly distributed, and two campaigns carry most of it. Cierjacks 1978 [120] alone contributes 28 631 of the 46 742 points, about 61%. Kinney 1976 [175] contributes 13 208 and is the campaign on which both libraries build their angular distributions below 2.5 MeV (Sec. 1.2); Smith 1980 [115], their anchor above it, stays with the other experiments. The aggregate χ^2/N is therefore shaped by these two experiments, and results are reported on five subsets that isolate them, in the order in which they are discussed below: all experiments; Kinney 1976 only; all experiments except Kinney; Cierjacks 1978 only; and all experiments except Kinney and Cierjacks, which leaves the other 65 experiments. Most campaigns use natural iron, which is over 91% ^{56}Fe, and the enriched datasets show no systematic offset with respect to the natural-iron ones.

Two variance budgets are shown. The first uses the EXFOR-declared uncertainties alone, σ_{exp} (Figure III.20), and measures how well each evaluation's central values agree with the data. The second adds each library's propagated `MF34` block, $\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$ (Figure III.21). Neither budget is expected to bring χ^2/N to unity (Sec. 3.4), so the analysis below compares evaluations on the same subset.

Under the EXFOR-only budget (Figure III.20), this work is the lowest of the three on every subset except Kinney taken alone, and adding each library's `MF34` block (Figure III.21) lowers every value without changing this ordering. The subsets are read below in the order of the figures.

On the whole database this work is the lowest of the three, but its margin over JEFF-4.0 is modest. This aggregate is strongly shaped by Kinney, whose points, more than a quarter of the

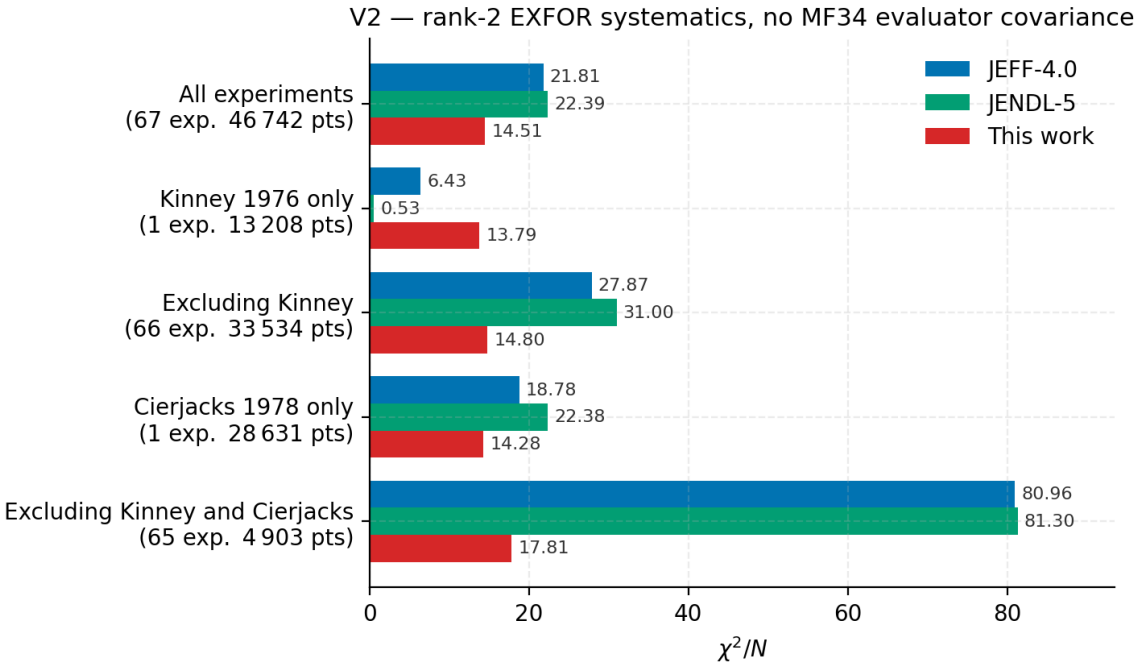


Figure III.20: Reduced χ^2/N for JEFF-4.0, JENDL-5, and this work across the five experiment subsets under the EXFOR-only budget σ_{exp}. The statistic responds only to how well each evaluation's central values agree with the experimental data, without the cushion of the library's own covariance.

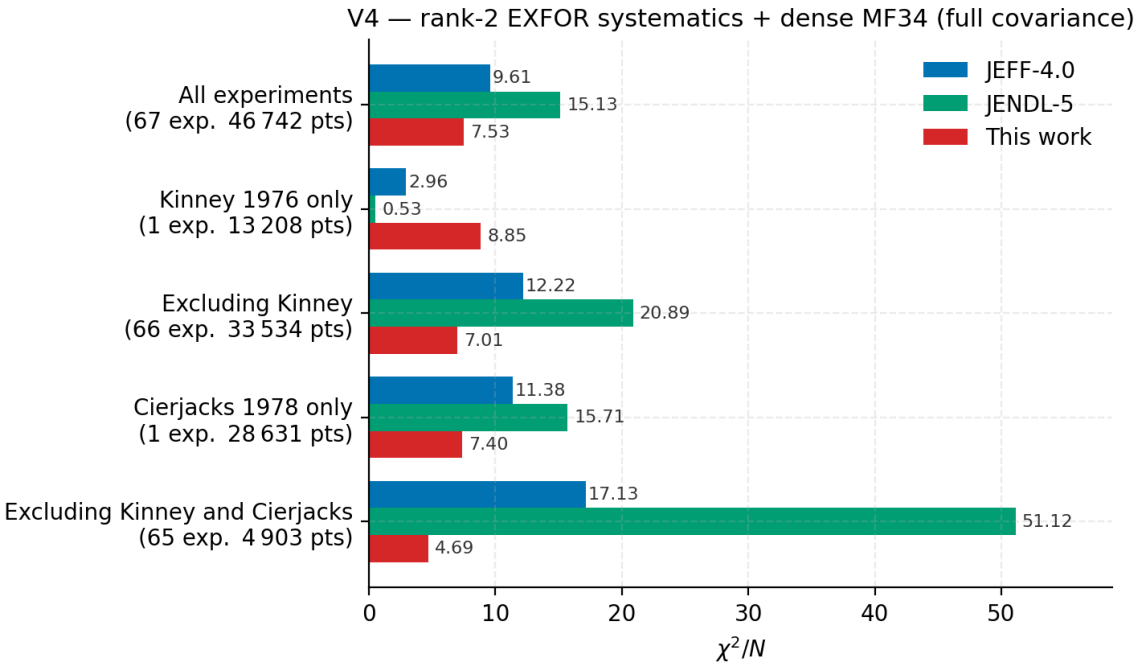


Figure III.21: Same comparison as Figure III.20 under the full-covariance budget $\sigma_{\mathrm{exp}} \oplus \sigma_{\mathrm{eval}}$, which adds each library's propagated `MF34` block.

database, are the ones both libraries were fitted to.

On Kinney alone the ordering reverses. JENDL-5 falls below unity, meaning that its residuals against Kinney are smaller than the uncertainties the campaign reports for itself, the signature of an evaluation fitted too closely to those same points. JEFF-4.0 is also low but stays above unity, although both libraries build on the same campaign. The difference lies in how the campaign was used. The CIELO evaluation of ^{56}Fe [113], from which JEFF-4.0 inherits its angular distributions, corrected the normalization of the Kinney data at backward angles, raising the backward-angle scattering slightly toward the Perey measurement (Sec. 1.2), so JEFF-4.0 differs from the published Kinney points there. JENDL-5, a separate JAEA evaluation, fits the same points as reported. This work weights Kinney no more heavily than any other experiment and therefore does not reproduce it as tightly.

Removing Kinney widens the gap between this work and the libraries: both libraries rise by roughly 25–40%, while this work changes by less than 7% under either budget. The remaining points are nevertheless still dominated by a single experiment, Cierjacks, which supplies about 85% of them.

On Cierjacks alone none of the three evaluations reproduces the angular shape within the declared uncertainties, but this work does clearly better than both libraries. The three evaluations match the overall level of Cierjacks closely, with a median residual within about 2% for each of them, so the differences between them come from the angular shape. Any uniform shift that remains is absorbed by the experiment's normalization uncertainty, which the metric treats as one shared mismatch for the whole experiment and not as 28 631 separate disagreements.

Removing both Kinney and Cierjacks leaves about 4 900 points spread across the 65 other experiments. With neither dominant campaign present, this subset is the clearest measure of how each evaluation agrees with independent measurements, and it gives the largest separation: JEFF-4.0 sits more than three times and JENDL-5 more than ten times above this work. Read in this order, the subsets show that the agreement of the libraries on the full database rests on the Kinney campaign they were fitted to and on a relatively favorable match to the Cierjacks block, not on uniform agreement across the data.

The same comparison can be made experiment by experiment. Across the 67 experiments, this work has the lowest χ^2/N on 60, JEFF-4.0 on 4, and JENDL-5 on 3. The per-experiment values, ranked by ascending value for this work, are shown in Figure III.22. The distribution for this work sits below those of JEFF-4.0 and JENDL-5 across the full ranking, with the two libraries extending one to two orders of magnitude higher on the experiments where their central values disagree most with the data.

The seven experiments on which a library is at least as good as this work are listed in Table III.4, ranked by this work's value. They fall into three kinds. The first is the calibration anchor Kinney 1976, the largest dataset in the table at 13 208 points: JENDL-5 reproduces it almost exactly ($\chi^2/N = 0.53$) and JEFF-4.0 closely (2.96), while this work sits at 8.85, the same anchoring effect already discussed for Kinney taken alone. The second is a small campaign on which this work sits close to $\chi^2/N \approx 1$ alongside both libraries (Schweitzer); the nominal ranking places a library first, but the difference is not meaningful. The third is a handful of mostly single-energy measurements where this work's central values genuinely differ from the closest library (Machwe, Pasechnik, Begum, Tomita, and Holmqvist).

A closer look at the third group shows that the better-fitting library is mostly JEFF-4.0, which is best on four of its five experiments, Machwe being the exception. On several of them, JENDL-5

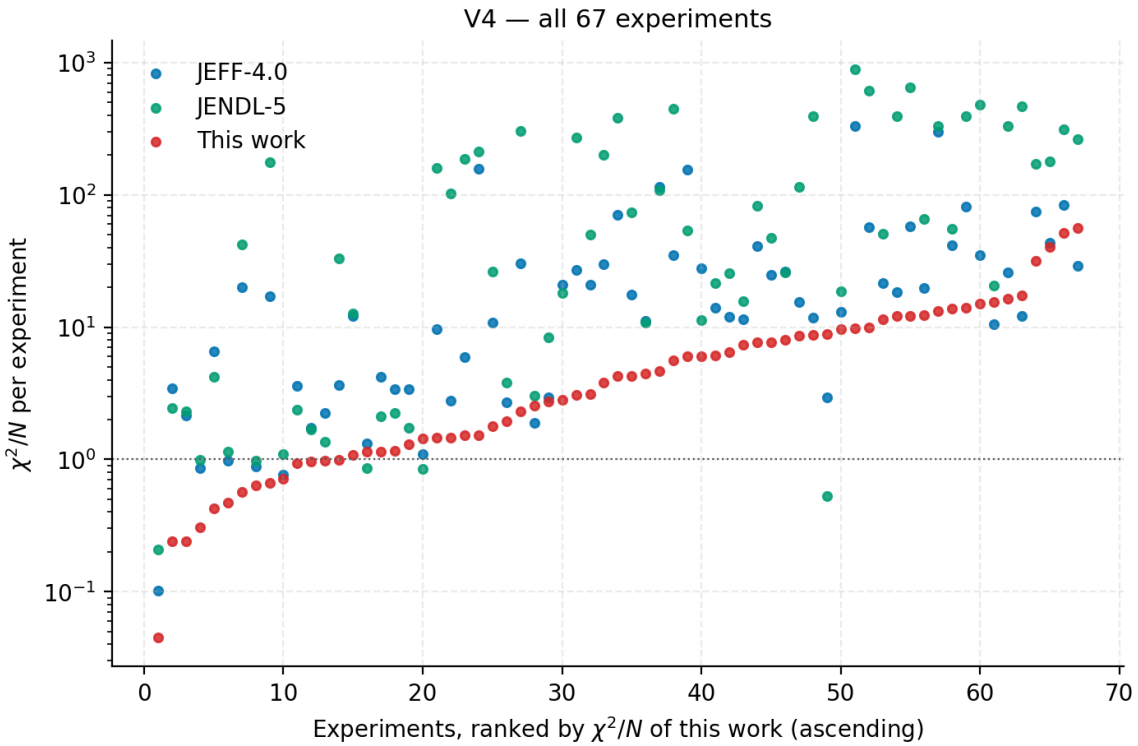


Figure III.22: Per-experiment χ^2/N on all 67 experiments, ranked by ascending value for this work. The vertical axis is logarithmic; the dotted horizontal line marks $\chi^2/N = 1$.

is in fact far worse than this work: on Holmqvist it reaches 264.0 against 56.2 for this work, and on Tomita it exceeds 400. Outside Kinney and the near-ties, then, a closer library fit means JEFF-4.0 specifically, on an isolated older dataset whose angular shape its Kinney- and Smith-based evaluation happens to match. On these experiments this work trades a slightly worse fit at one energy for the lower χ^2/N it reaches across the rest of the database, consistent with a fit that does not privilege any single campaign.

Table III.4: The seven experiments (of 67) on which JEFF-4.0 or JENDL-5 reaches a χ^2/N at least as low as this work, under the full-covariance budget $\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$, ranked by this work's value. N is the number of angular points entering the metric; bold marks the lowest χ^2/N in each row. EXFOR identifiers and references are given in Table III.1.

Experiment		N	E (MeV)	JEFF-4.0	JENDL-5	This work
Schweitzer	1978	12	3.40	1.32	**0.86**	1.14
Machwe	1959	8	3.66	1.10	**0.84**	1.43
Pasechnik	1958	7	2.80	**1.89**	3.02	2.56
Kinney	1976	13 208	0.85–2.50	2.96	**0.53**	8.85
Begum	1981	11	2.90	**10.52**	20.53	15.58
Tomita	1973	13	1.47	**12.08**	467.91	17.39
Holmqvist	1969	40	2.96–3.95	**28.91**	264.04	56.20

Measurements published after the libraries

Three campaigns measured ^{56}Fe elastic angular distributions after the evaluated distributions of JEFF-4.0 and JENDL-5 were built: Ramirez [129], Pirovano [121], and Gkatis [122][1], together 2 622 points. Both libraries take their angular distributions from Kinney and Smith (Sec. 5.2), so none of the three entered their fit. All three are part of the input data of this work. The comparison is therefore asymmetric: for the libraries it measures how well they extrapolate to recent data, and for this work it is an in-sample residual.

Table III.5: Reduced χ^2/N on the three campaigns published after the evaluated angular distributions of JEFF-4.0 and JENDL-5, under the full-covariance budget $\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$. Bold marks the lowest value in each row.

Campaign		N	E (MeV)	JEFF-4.0	JENDL-5	This work
Gkatis	2025	784	1.01–3.93	20.80	18.17	**2.81**
Ramirez	2017	134	1.30–4.00	81.27	394.05	**14.06**
Pirovano	2019	664	1.99–3.99	0.86	0.99	**0.31**
Pirovano	2019	1 040	2.00–3.99	0.98	1.14	**0.47**

Table III.5 reports the result. This work is the lowest of the three on every campaign, and the ordering holds under the EXFOR-only budget as well. The libraries separate by energy. On the two Pirovano subentries, which start at 2 MeV, JEFF-4.0 and JENDL-5 both sit near unity and reproduce the data within its uncertainties. On Gkatis and Ramirez, which reach down to 1.01 and 1.30 MeV, they rise to between 18 and 394. The modern data therefore place the disagreement of the libraries below 2 MeV, the part of the window where the cross section fluctuates most strongly. Because all three campaigns are part of the fit of this work, its own values are in-sample residuals and should be read as a consistency check.

5.5. Integral Cross-Check: Criticality Benchmarks

The goodness-of-fit analysis measures the evaluation against the differential data it was built from. The fitted central values stay close to JEFF-4.0 and JENDL-5, within 1–2σ below 2.5 MeV and with a higher a_1 above it (Sec. 5.2), yet they do move, and a change in the angular distribution carries into any application that depends on how neutrons scatter off iron. This section is a check on the `MF4` central values: it confirms that the changed values stay consistent with integral criticality measurements before the covariance is propagated in Chapter IV. The ^{56}Fe evaluation in JEFF-4.0 is replaced with the present one, every other nuclide left in place, and k_{eff} is recomputed with the MORET 6 Monte Carlo code [46], converged to a statistical uncertainty of 20 pcm (1σ) on every case. The benchmarks were selected by their sensitivity to ^{56}Fe elastic scattering: of the 5 459 configurations of the ICSBEP handbook [38], 1 931 carry a non-zero sensitivity to the ^{56}Fe elastic cross section, and 88 of them were run. The indicator is that sensitivity integrated over 850 keV–4 MeV, computed by MORET 6 and ranked with CALINS [176], restricted to the evaluation window so that energies the new evaluation leaves untouched stay out of the ranking.

Across the full suite the agreement is preserved. The reduced χ^2/N against the benchmark values moves from 4.50 to 4.55 and the mean bias from −311 to −342 pcm, both small against the

[1] At the time of writing, the Gkatis measurement has been submitted and assigned EXFOR entry 27673002, but the entry is not yet publicly released.

789 pcm root-mean-square of the suite (Table III.6). The swap lowers k_{eff} by a median of 25 pcm and at most 192 pcm, which stays below the experimental uncertainty in 87 of the 88 benchmarks. On the 56 benchmarks that combine a high ^{56}Fe sensitivity with an experimental k_{eff} known to better than 300 pcm, the mean bias improves from +50 to +23 pcm. The shift grows with the hardness of the spectrum (Figure III.24), measured by the energy of average lethargy causing fission (EALF). Systems whose EALF exceeds 100 keV shift by a median of −78 pcm, while those below 1 keV shift by a median of only −19 pcm, comparable to the Monte Carlo uncertainty on the difference. The per-benchmark values are given in Appendix E.12 (Figure E.5).

The benchmarks that move most are fast plutonium- and highly enriched uranium-metal assemblies surrounded by a steel reflector (Figure III.23). The largest shift falls on a steel-reflected δ-phase plutonium sphere (pu-met-fast-028). The k_{eff} of these systems depends on the fast neutrons reflected back into the core, a fraction set by the ^{56}Fe elastic angular distribution at backward angles, which is the quantity re-evaluated here. The three most sensitive plutonium systems all move toward their measured value. The central values produced here therefore leave integral criticality performance intact. Criticality is nevertheless a weak test of an angular distribution, because k_{eff} integrates over the whole spectrum and over all directions. The shielding benchmarks of Chapter IV are far more sensitive to the fast angular shape, and there the evaluation is propagated with its covariance as well.

Table III.6: Agreement with the ICSBEP criticality benchmarks before and after replacing the ^{56}Fe evaluation in JEFF-4.0 with the present one, for the full suite, for the 56 benchmarks that combine the highest ^{56}Fe elastic sensitivities with an experimental k_{eff} known to better than 300 pcm, and for the remainder. $\overline{C-E}$ is the mean calculated-minus-experimental bias and RMS $(C-E)$ its root-mean-square, both in pcm.

		χ^2/N		$\overline{C-E}$ (pcm)		RMS $(C-E)$ (pcm)	
Benchmark set	N	JEFF-4.0	This work	JEFF-4.0	This work	JEFF-4.0	This work
All benchmarks	88	4.50	4.55	−311	−342	789	797
Sensitive and well measured	56	5.43	5.45	+50	+23	318	306
Remainder of the suite	32	2.87	2.96	−941	−979	1238	1259

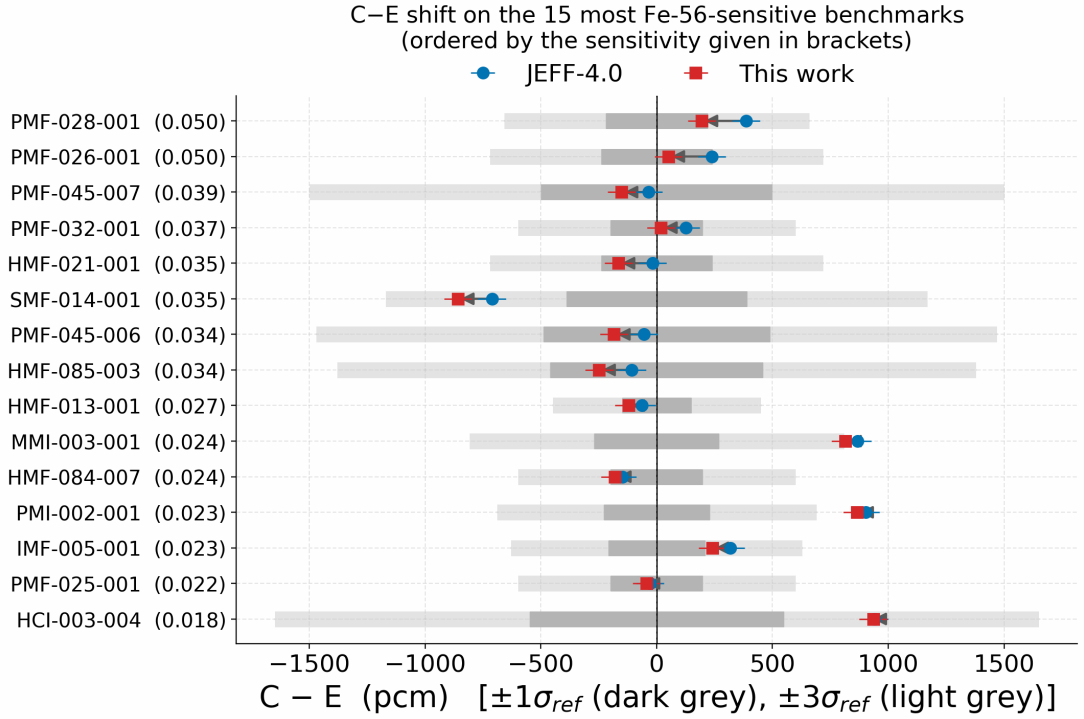


Figure III.23: Calculated-minus-experimental bias $C - E$ on the fifteen benchmarks most sensitive to the ^{56}Fe elastic cross section, for JEFF-4.0 and after the ^{56}Fe swap, ordered by that sensitivity, whose value is given in brackets after each benchmark name. The bands mark each benchmark's experimental 1σ and 3σ.

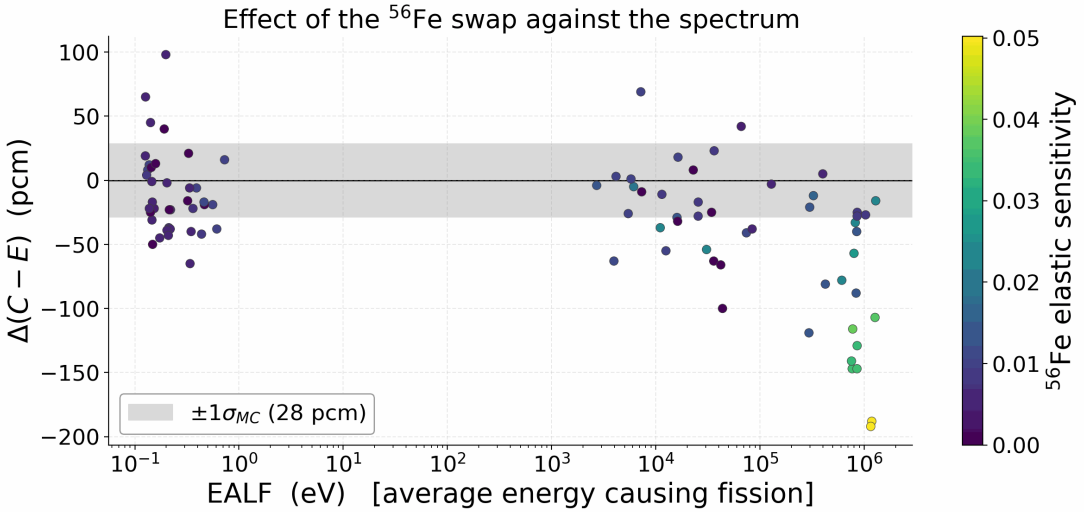


Figure III.24: Shift $\Delta(C - E)$ produced by the ^{56}Fe swap against the EALF of each benchmark, colored by its ^{56}Fe elastic sensitivity over 850 keV–4 MeV. The band marks the Monte Carlo uncertainty on the difference.

6. Conclusions

This chapter has presented a complete, data-driven evaluation of ^{56}Fe elastic-scattering angular distributions and their covariances in the 847 keV–4 MeV energy range, within the shape-only `MF4`/`MF34` framework. Starting from the EXFOR experimental database, the evaluation pipeline produces a nominal ENDF file containing both the best-estimate `MF4` Legendre coefficients and their `MF34` covariance matrix on an adaptively collapsed energy grid. The main findings are summarized below.

Evaluation pipeline

The methodological contributions of this work include: (i) a data-driven energy grid constructed from the Kinney and Pirovano experimental subentries; (ii) ridge-regularized Legendre fitting with automatic order selection via AIC; (iii) model-form uncertainty captured through Akaike-weight degree sampling across the sample ensemble; (iv) systematic treatment of normalization bias, which acts as a correlation between all angular points of the same experiment, together with angular-band discrepancies; (v) a two-pass Monte Carlo scheme that pairs a per-bin variance pass with a kernel-weight pass providing the full parameter correlation matrix, combined through a diagonal congruence transform that preserves positive semi-definiteness; and (vi) adaptive multigroup collapse for compact `MF34` construction covering orders $\ell = 1$–6. Together, these components provide a reproducible, end-to-end pipeline from experimental data to a complete ENDF evaluation. All codes and inputs used in this evaluation will be made available once the work is published.

Central values and covariance

The fitted Legendre coefficients follow JEFF-4.0 and JENDL-5 across the energy range (Sec. 5.2). Below 2.5 MeV the differences between the three evaluations are generally within 1–2σ; above it, the a_1 of this work lies systematically higher than both libraries. A criticality cross-check confirms that these small central-value changes leave integral performance intact. Swapping the present ^{56}Fe into 88 ICSBEP benchmarks holds the reduced χ^2/N against experiment essentially fixed and modestly improves the fast metal systems most sensitive to iron reflection (Sec. 5.5). The covariance differs more between the evaluations than the central values do. This work spreads the uncertainty across the fitted Legendre orders and produces smooth, physically interpretable bands on the differential cross section (Sec. 5.3).

Goodness of fit

The goodness-of-fit analysis (Sec. 5.4) compares the three evaluations on five subsets of the experimental database, under two variance budgets, the EXFOR-declared uncertainties alone and those combined with each library's propagated `MF34` block. This work reaches the lowest reduced chi-squared on the large majority of the individual experiments and on every aggregate subset once Kinney 1976 is set aside. On that campaign alone JENDL-5 reaches $\chi^2/N = 0.53$, below the uncertainties Kinney reports for itself, and the same close tracking lowers both libraries on every aggregate that contains its 13 208 points. The separation is largest on the subset that excludes Kinney and the Cierjacks block, which shows that the libraries' agreement on the full database rests on those two campaigns and not on uniform agreement across the data.

Limitations

Several limitations should be noted. First, the assessment is restricted to the 847 keV–4 MeV interval; outside it, evaluated files transition to model-guided representations, and the present conclusions should not be extrapolated without an extended mixed data–model evaluation. Second, the correlation between the angular points of one experiment is modeled as the single rank-one mode implied by its declared systematic uncertainty (Eq. (III.13)), since EXFOR reports the uncertainty components but not the correlation matrix itself. Third, the evaluation treats the cross-section magnitude and the angular shape as independent (Sec. 2.2). A more complete evaluation would also produce a covariance for the elastic scattering cross section and, where the two are coupled, the cross-correlation terms between magnitude and shape. This is left as a line of future work and is described among the extensions below. Finally, the reduced chi-squared on the database stays well above unity for all three evaluations. Part of this level is set by the construction of the metric (Sec. 3.4), but the uncertainties produced here, although larger and more evenly distributed than those of the existing libraries, may still underestimate the spread the experimental data imply. The robust statement is the relative one, that this evaluation sits closer to the data across the database than JEFF-4.0 or JENDL-5 under an identical metric.

A more basic limitation underlies the covariance itself. Most energy bins are constrained by only two or three independent campaigns, which often disagree by more than their stated uncertainties. With so few independent measurements, the between-experiment spread that the covariance is meant to capture is itself poorly determined. A Student-t treatment of this spread, which widens the band when only a few campaigns constrain it, would account for this explicitly. The reported uncertainties should therefore be read as the best estimate the present data allow, and a denser, more independent database could shift them.

A second basic limitation is set by the energy resolution of the measurements themselves. Every campaign observes the same underlying angular distribution through its own resolution window (Sec. 3.2), so a fine campaign such as Kinney and a coarse one do not report the same quantity at the same nominal energy. Combining them without losing information is the central difficulty: folding the fine data to a common kernel discards the structure that only they resolve, while treating every point as monochromatic discards the averaging that the coarse data carry. Several stages of the pipeline exist to limit that loss, among them the resolution-derived energy binning and the overlap weights of Sec. 4.5. What they cannot reach is the ideal behavior, an evaluation that follows the fine measurements over a narrow energy interval, with rapid variation wherever those data support it, and that reproduces the coarse measurements once averaged over a wider one. That behavior calls for an estimator in which the resolution of each experiment enters the model itself, and fitting the energy dependence with B-splines is one candidate worth studying.

Recommendations for future experiments

The evaluation points to a few measurements that would most improve future angular-distribution covariances, all within the 847 keV–4 MeV range where the present data and the intended applications concentrate. The clearest need is for more independent measurements. Many energy bins now rest on one or two campaigns, often Kinney alone or Kinney together with Cierjacks, and a variance and correlation estimated from so few experiments is weak. Because so many bins rest on Kinney alone, and because the present methodology fits each bin on its own, the single most useful addition would be a campaign matching the density of

Kinney in both energy and angle, which would let the variance in those bins be estimated from the spread between experiments instead of from a single reported uncertainty. This part of the recommendation follows from the method: under an estimator that carries each experiment's resolution inside the model, a campaign of any resolution contributes, as long as that resolution is known. A finer energy grid would not help, since the binning is already limited by the experimental energy resolution and by the number of points available per bin. Angular coverage matters for the intended applications, since forward angles drive deep-penetration transport and backward angles drive reflection, so both hemispheres should be well characterized. In practice the measured angles are largely fixed by the Gaussian-quadrature nodes of the setup, which makes broad and consistent sampling across campaigns more valuable than fine control of individual angles. Finally, experiments should report their full uncertainty decomposition, including normalization and angular point-to-point correlations, together with the flight path and timing parameters that fix the energy resolution. EXFOR entries often omit both, which forces the evaluator to assume values such as σ_{norm} and to fall back on default resolution parameters.

Extensions of the pipeline

The pipeline is not specific to ^{56}Fe. It can produce an angular-distribution evaluation and an `MF34` covariance for any isotope whose differential elastic data over the range of interest come mainly from EXFOR, as long as enough independent campaigns exist to constrain both the fit and its covariance. One caveat carries over from the iron case. Most measurements here use natural iron, which is a mild approximation because ^{56}Fe makes up more than 91% of it, and the enriched and natural datasets are statistically consistent. For an isotope that forms a smaller fraction of its natural element, measurements on the natural element would mix in a larger contribution from the other isotopes, which would have to be modeled or removed before the data could feed the pipeline.

A second extension, and the natural next step of this work, concerns the cross section magnitude. The present evaluation fixes the angular shape and leaves `MF3` and `MF33` untouched (Sec. 2.2). The same differential measurements also constrain the elastic cross section, and although their energy resolution is too coarse to reconstruct $\sigma_{\mathrm{el}}(E)$ at the resolution of the evaluated files, they can constrain an energy-averaged cross section and yield a covariance for it. Adding this would extend the pipeline from the angular shape to the magnitude. Consistency argues for going further. The energy-averaged constraint these data carry is most valuable as input to the evaluation of the cross section itself, so that the magnitude and its covariance rest on the same information. That is a joint evaluation of magnitude and shape, a considerably larger undertaking, and it is where the cross-correlation between the two would be quantified and carried in the covariance.

Outlook

The evaluation and its `MF34` covariance provide the basis for propagating angular-distribution uncertainties to application-level quantities. Perturbed ENDF samples are generated from the `MF34` using the general perturbation pipeline of Chapter II. This propagation is carried through Chapter IV to the shielding benchmarks and to the fast neutron fluence at the pressure vessel of the PWR900 model.

Chapter IV

Integral Validation and Application to PWR Vessel Fluence

Contents

1 Introduction . . . 156
2 The ASPIS-88 Iron Benchmark . . . 157
2.1 Benchmark Description . . . 157
2.2 MCNP Model and Variance Reduction . . . 157
2.3 C/E Comparison with Nuclear-Data Uncertainties . . . 159
2.3.1 C/E Comparison . . . 159
2.3.2 Propagated Nuclear-Data Uncertainties . . . 161
2.3.3 Integral Effect of the Chapter-3 Angular Evaluation . 162
2.4 Sensitivity Analysis and the Origin of Library Differences . . . 163
2.4.1 Sensitivity Profiles . . . 163
2.4.2 Isolating ^{56}Fe by a Single-Isotope Library Swap . . . 163
3 The VENUS-3 Benchmark . . . 166
3.1 Benchmark Description . . . 166
3.2 MCNP Model and Uncertainty Propagation . . . 166
3.3 C/E Comparison . . . 167
3.4 Propagated Uncertainties and the Processing of the ^{16}O Covariances . . . 168
4 Reactor Model, Tallies, and Variance Reduction . . . 171
5 Nuclear Data Uncertainty in the Source Term . . . 175
5.1 Calculation Setup . . . 176
5.2 Outputs of the Core Calculation . . . 177
5.3 Perturbation of the Emitted Fission Spectrum . . . 178
5.4 Propagation to the Vessel . . . 178
5.5 Summary . . . 180
6 Nuclear Data Uncertainty in the Vessel Fluence and Capsule Responses . . . 182
6.1 Calculation Setup . . . 182

6.2 Fast Neutron Flux at the Vessel 182
6.3 Vessel Atomic-Displacement Damage 184
6.4 Capsule Dosimetry 184
6.5 Cross Section and Angular-Distribution Contributions 185
6.6 Comparison of ^{56}Fe Angular Evaluations 186
6.7 Per-Isotope Decomposition of the Cross Section Channel 187
6.8 Cross-Check: MCS Against FOSM for Cross-Section Propagation 189
7 Comparison with Published Studies 191
8 Conclusions 194

This chapter takes the sensitivity and uncertainty framework of Chapter II and the ^{56}Fe angular evaluation of Chapter III and follows them from measured experiments to the reactor vessel that motivates the thesis. Section 1 sets out the aging problem of the reactor pressure vessel and the reason for validating the method before applying it. Section 2 and Section 3 test the evaluated libraries against the ASPIS-88 and VENUS-3 shielding experiments, with ASPIS-88 also testing the ^{56}Fe angular evaluation of Chapter III. Section 4 then builds the PWR900 reactor model, and the sections that follow propagate the source-term and transport uncertainties to the vessel fluence, the displacement damage, and the surveillance dosimeters, decompose the result by isotope and by reaction channel, and place the budget against the published literature.

1. Introduction

The aging of reactor pressure vessels (RPVs) under neutron irradiation is governed by the accumulated fast neutron fluence and the associated atomic displacement damage. Surveillance capsules placed inside the vessel measure this field at a few positions, but its coverage over the whole vessel and over the full plant lifetime rests on simulation. The VACS methodology described in Sec. 5 of Chapter I performs that simulation in two steps, a deterministic source term followed by a Monte Carlo transport calculation, so it draws on nuclear data at every stage of the chain and is exposed to their uncertainty. Published estimates place the total uncertainty on the vessel fast fluence at roughly 10% at 1σ, with nuclear data among the dominant contributors and the modeling and manufacturing effects usually smaller [4]. Quantifying the nuclear-data part of that budget, and finding where it comes from, is the goal of this chapter.

The difficulty is that the vessel fluence cannot be measured directly. The prediction there rests entirely on the transport calculation and on the nuclear data it uses, with no experiment to confirm it. Before the method is trusted on the vessel, it is therefore checked where measurements do exist, on shielding benchmarks that isolate the same physics. This is the thread the chapter follows, first validating the method and the data where they can be compared with experiment, then applying the validated method where they cannot.

Two benchmarks are used, chosen to climb toward the reactor in steps. The ASPIS-88 experiment is a deep-penetration measurement through a thick mild-steel wall. Its clean one-dimensional geometry isolates ^{56}Fe, the isotope that governs attenuation through the vessel, and its activation foils span the fast neutron spectrum along the full shield depth. The VENUS-3 experiment adds a fission source, a three-dimensional layout, and water along the path, which brings the test close to a power reactor and extends it to the oxygen of the surrounding water.

The PWR900 vessel is the final step, a full reactor geometry where the same ^{56}Fe and ^{16}O data now set the predicted fluence.

The ^{56}Fe elastic-scattering angular evaluation built in Chapter III is carried through the ASPIS-88 benchmark and the PWR900 vessel, so its effect on measured reaction rates and on the vessel budget can be followed on the same footing as the published library data. On VENUS-3, where the steel paths are shorter and the angular effect is small, this channel is not propagated. Throughout the chapter the uncertainty is propagated with the sampling scheme verified in Chapter II. The libraries compared change with the setting. ASPIS-88 and the vessel use the three modern evaluations ENDF/B-VIII.1, JEFF-4.0, and JENDL-5, while VENUS-3 follows successive releases from ENDF/B-VIII.0 to VIII.1 and from JEFF-3.3 to 4.0.

2. The ASPIS-88 Iron Benchmark

The Iron-88 benchmark is a single-material deep-penetration shielding experiment performed in 1988 on the ASPIS facility at Winfrith [177], and compiled in the SINBAD database [37]. It measures the attenuation of a fast neutron field through a thick mild-steel assembly, which makes it a direct integral test of the iron cross sections and angular distributions.

2.1. Benchmark Description

ASPIS was installed on the NESTOR reactor, a light-water-cooled, graphite-and-light-water-moderated reactor operating at up to 30 kW and used as a neutron source. The shield array is driven by a fission plate, a thin slab of 93%-enriched uranium-aluminum alloy placed at the upstream face of the assembly. Thermal neutrons leaking from the NESTOR graphite reflector induce fissions in the plate, which then acts as a broad fission-spectrum surface source feeding the shield. The detailed plate loading and source distribution are given in the benchmark specification [177].

The shield is a stack of 13 mild-steel slabs, each approximately 5.1 cm thick, for a total of about 67 cm of steel along the penetration axis. Mild steel is iron-dominated, so ^{56}Fe (about 91.7% of natural iron) sets the attenuation. Reaction rates are measured with activation foils inserted between the slabs along the central axis. The five reactions span a wide range of effective energy thresholds, so that together they cover the neutron spectrum from the thermal range up to several MeV (Table IV.1); their evaluated activation cross sections are taken from the IRDFF-II dosimetry library [178] and are shown in Figure IV.2. The geometry and the foil positions are sketched in Figure IV.1.

2.2. MCNP Model and Variance Reduction

The MCNP [13] model of the benchmark comes from previous ASNR studies, with the fission plate represented as a surface source carrying the specified spatial and energy distribution. Because the deepest foils sit many mean free paths from the source, an analog simulation cannot converge the reaction rates at acceptable cost. Each foil is therefore run from its own MCNP input, with weight windows generated in this work for that detector position and energy response by ADVANTG [14] through the FW-CADIS method. The resulting statistical error stays below about 3% for the threshold reactions at every measured position, and below 7% in the worst case, the ^{197}Au foil at the deepest positions.

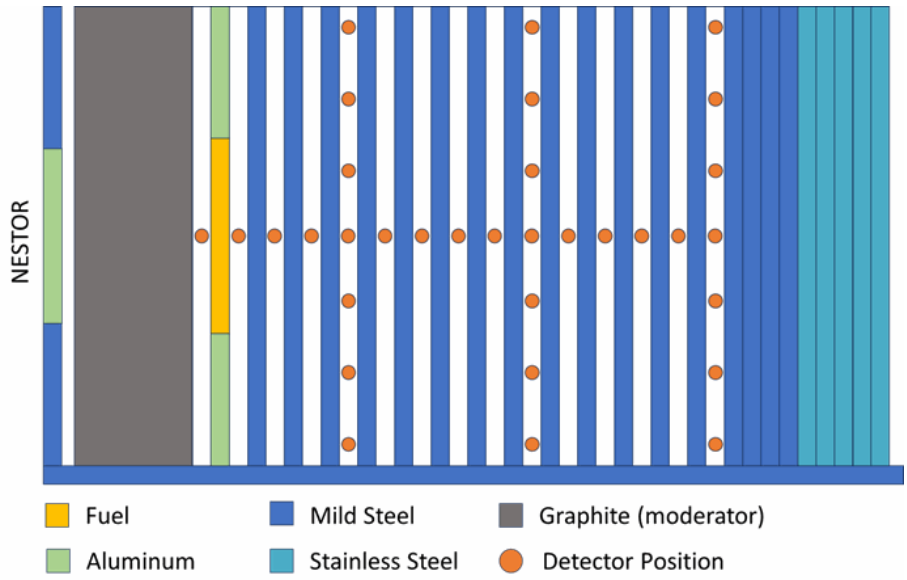


Figure IV.1: Side elevation of the ASPIS-88 single-material iron benchmark: fission-plate source, mild-steel slab stack, and the foil measurement positions along the central axis (not to scale).

Table IV.1: Activation foils of the ASPIS-88 benchmark, ordered by effective energy threshold. The threshold is the approximate neutron energy above which each reaction responds; the number of measured axial positions and the deepest position are taken from the benchmark dataset.

Reaction	Product	Effective threshold	Positions	Max. depth
^{197}Au(n,γ)	^{198}Au	epithermal ($>$ 0.4 eV)	14	67.0 cm
^{103}Rh(n,n$'$)	^{103m}Rh	$\sim$ 0.1 MeV	13	61.8 cm
^{115}In(n,n$'$)	^{115m}In	$\sim$ 1.0 MeV	10	46.4 cm
^{32}S(n,p)	^{32}P	$\sim$ 2.0 MeV	14	67.0 cm
^{27}Al(n,α)	^{24}Na	$\sim$ 7.0 MeV	4	25.6 cm

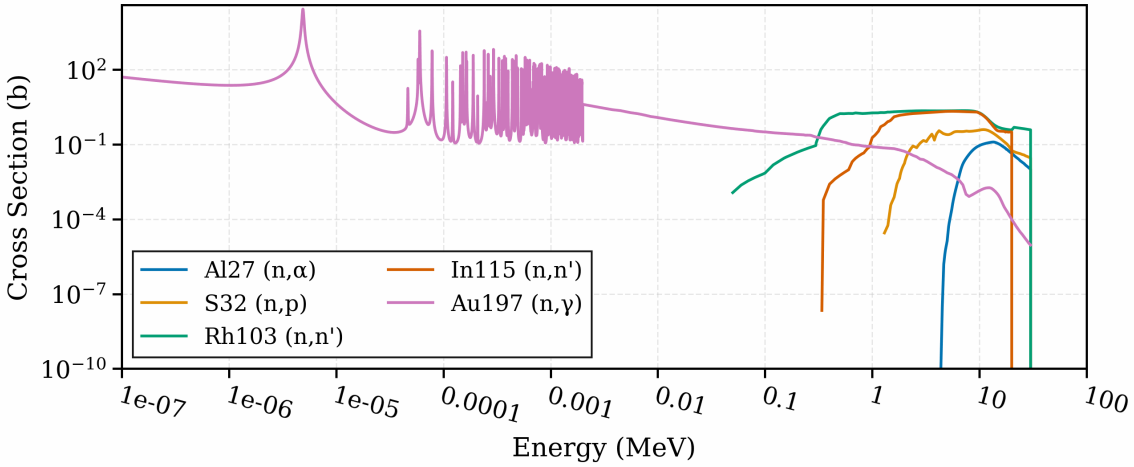


Figure IV.2: Activation cross sections of the five ASPIS-88 foil reactions.

2.3. C/E Comparison with Nuclear-Data Uncertainties

This section presents the C/E ratios for the JEFF-4.0 [32], JENDL-5 [34], and ENDF/B-VIII.1 [33] libraries together with the propagated ^{56}Fe nuclear-data uncertainties. The reaction rates are normalized to the fission-plate source emission rate, so the comparison is absolute. The uncertainties are propagated with the Monte Carlo sampling scheme verified on the PWR sphere model in Chapter II, here perturbing only the ^{56}Fe data: the ^{56}Fe covariances are sampled, each sample is run through the MCNP model of the corresponding foil, and the propagated relative uncertainty is taken from the resulting population, with 95% confidence intervals from a bootstrap over the samples. The propagation covers the three libraries; since JEFF-4.0 and ENDF/B-VIII.1 share the same ^{56}Fe evaluation, their propagated uncertainties are identical.

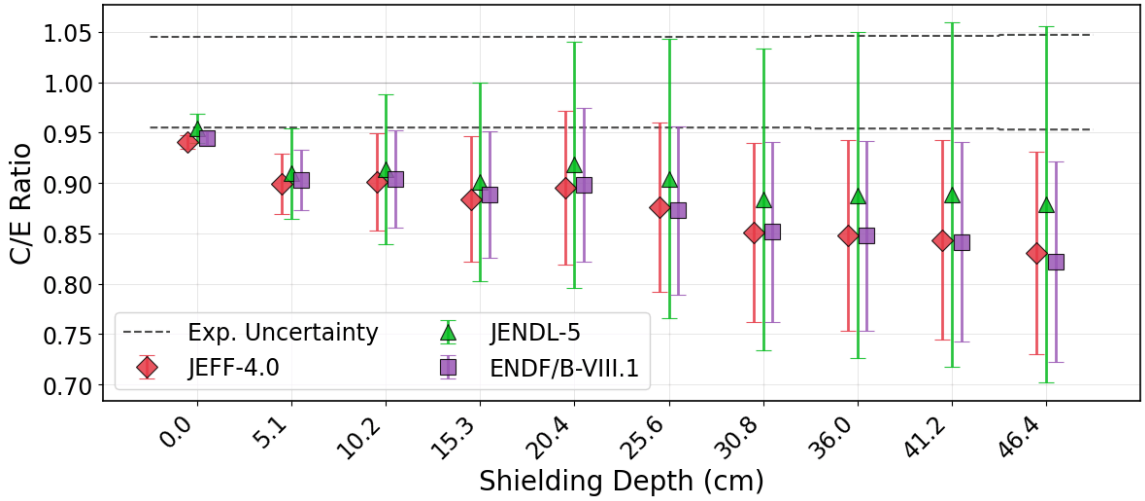
The uncertainty is split into a cross section channel and an angular-distribution (`MF34`) channel, propagated separately and combined in quadrature. For the angular channel, three ^{56}Fe `MF34` evaluations are propagated: the JEFF-4.0 covariance (the JEFF-3.3 `MF34`), the JENDL-5 covariance, and the EXFOR-based evaluation developed in Chapter III; the latter is propagated for the three inelastic-threshold foils. The body shows ^{115}In and ^{103}Rh, the two foils whose activation response is closest to the neutron spectrum found in a reactor pressure vessel; the corresponding figures for the remaining foils, together with the per-position tables of propagated uncertainties and their confidence intervals, are collected in Appendix F.

2.3.1. C/E Comparison

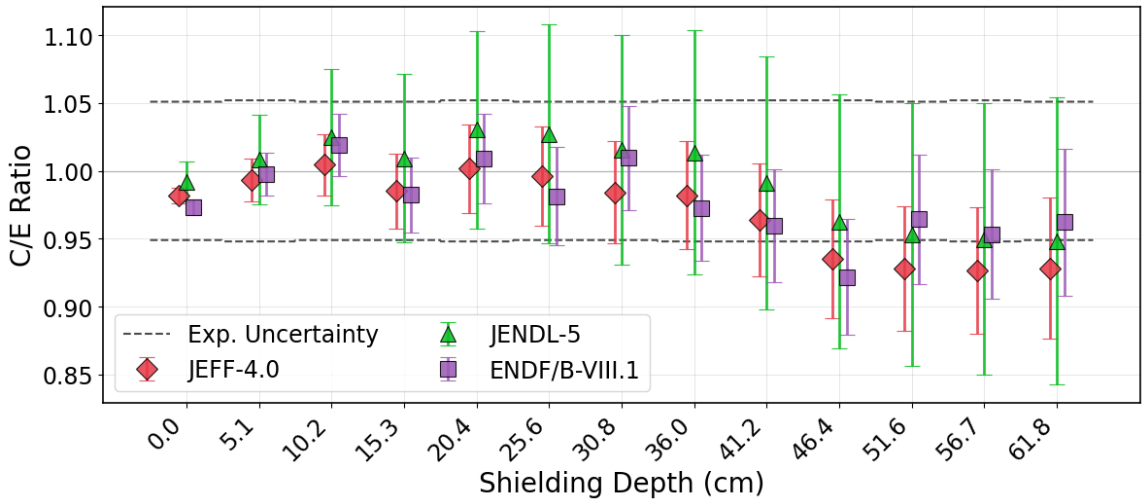
Figure IV.3 shows the C/E ratios of ^{115}In and ^{103}Rh at every measured position; the figures for the other three foils follow the same layout and are given in Appendix F. Foil by foil, the comparison shows the following.

- ^{115}In(n,n′) is systematically underpredicted already at the first measured position, which points to the model or the source definition rather than to the iron data. On top of this bias, the C/E shows a negative trend with depth in all three libraries, slightly flatter for JENDL-5 (Figure IV.3a).
- ^{103}Rh(n,n′) agrees well with the measurement, with some negative trend at the larger depths in all libraries (Figure IV.3b).
- ^{197}Au(n,γ) agrees well at all depths.
- ^{27}Al(n,α) mirrors ^{115}In: a positive systematic bias from the first position, followed by a rising trend that is stronger for JEFF-4.0 and ENDF/B-VIII.1 and essentially flat for JENDL-5.
- ^{32}S(n,p) carries a systematic negative bias like ^{115}In, with a positive trend toward the largest depths in all libraries.

The same C/E values were compared against independent calculations of the benchmark: MCBEND Monte Carlo results provided by Tim Ware and Graham Lee (Amentum, United Kingdom) and the SUSD3D deterministic results of Kodeli [112]. The codes agree well for all foils except ^{27}Al(n,α), where SUSD3D overestimates the positive systematic bias even further than the MCNP and MCBEND results.



(a) ^{115}In(n,n′).



(b) ^{103}Rh(n,n′).

Figure IV.3: C/E ratios along the shield depth for the three libraries. Error bars show the propagated ^{56}Fe nuclear-data uncertainty, with the cross section and angular-distribution channels combined in quadrature; dashed lines mark the experimental uncertainty.

2.3.2. Propagated Nuclear-Data Uncertainties

Table IV.2 reports the propagated uncertainty of each channel at the deepest measured position of each foil; the per-position values, with their 95% bootstrap confidence intervals, are tabulated in Appendix F. The cross section uncertainty grows with depth and with the energy threshold of the foil, from a few percent on the low-threshold foils to about 30% (JEFF-4.0) and 41% (JENDL-5) on the deepest ^{32}S position. JENDL-5 propagates the larger cross section uncertainty for every foil except ^{27}Al, where the two libraries reach the same 18%.

Table IV.2: Propagated ^{56}Fe nuclear-data uncertainty at the deepest measured position of each foil: cross section (XS) channel for JEFF-4.0 and JENDL-5, and angular-distribution (AD) channel for the three `MF34` evaluations. Values in percent at 1σ.

		XS unc. (%)		AD unc. (%)		
Reaction	Depth	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5	This work
^{197}Au(n,γ)	67.0 cm	3.2	10.6	0.3	0.5	—
^{103}Rh(n,n$'$)	61.8 cm	4.6	9.0	3.2	6.6	3.1
^{115}In(n,n$'$)	46.4 cm	8.4	16.8	8.7	11.1	3.3
^{32}S(n,p)	67.0 cm	29.6	40.9	3.3	19.1	7.1
^{27}Al(n,α)	25.6 cm	18.0	18.1	3.5	3.9	—

The angular-distribution channel depends on the foil:

- For ^{197}Au and ^{27}Al it is negligible. The ^{197}Au response comes from low-energy neutrons and the ^{27}Al response from fast neutrons above 7 MeV. In both energy ranges the cross sections set the attenuation, and the scattering anisotropy has little effect.
- For ^{115}In the library uncertainties at the deepest position, 8.7% for JEFF-4.0 and 11.1% for JENDL-5, are as large as the corresponding cross section ones. The Chapter III evaluation gives a much smaller 3.3%.
- For ^{103}Rh the Chapter III evaluation (3.1%) matches JEFF-4.0 (3.2%), and JENDL-5 is twice as large (6.6%).
- For ^{32}S the Chapter III evaluation (7.1%) falls between JEFF-4.0 (3.3%) and JENDL-5 (19.1%). All three are small compared with the 30–41% cross section uncertainty of this foil.

The angular channel can therefore not be neglected for ^{115}In and ^{103}Rh under the library covariances, while it remains a secondary contribution for the other foils and under the Chapter III evaluation.

The differences between the three evaluations reflect their covariances in the energy window that drives each foil: the JEFF-4.0 `MF34` carries an absolute uncertainty on its high-order Legendre coefficients comparable to the one it carries on a_1, the JENDL-5 covariance concentrates a strongly correlated uncertainty on the first coefficient alone, and the Chapter III evaluation distributes a smaller, experimentally constrained uncertainty across the orders. A finer attribution to individual Legendre coefficients would require the corresponding sensitivities, which could not be computed for this model because of the memory demands of the perturbation scoring noted in Chapter II.

The propagated uncertainties are now compared with the C/E results. The depth-dependent trends are covered: for ^{115}In and ^{103}Rh, the combined nuclear-data band at depth is comparable to both the slope of the C/E and the spread between JEFF-4.0 and JENDL-5. The width of that coverage depends strongly on the angular evaluation: the very wide JENDL-5 angular covariance inflates the band well beyond the C/E trend, whereas the tighter Chapter III evaluation suggests that a more realistic angular uncertainty still covers the ^{115}In trend without overstating it. The standing biases are a different matter: the offsets of ^{115}In, ^{27}Al, and ^{32}S are present already at the first measured position, where the propagated iron uncertainty is still small, so they cannot be explained by the ^{56}Fe data and point to the model, the source definition, or the dosimetry data. The large JEFF–JENDL difference in propagated uncertainty, and in particular the very wide JENDL-5 angular band, is the integral counterpart of the covariance differences found in Chapter III, and motivates the data-driven evaluation developed there.

2.3.3. Integral Effect of the Chapter-3 Angular Evaluation

The final test substitutes the elastic angular distribution developed in Chapter III into the JEFF-4.0 transport, holding every ^{56}Fe cross section fixed, so that any change in the calculated reaction rates comes from the angular distribution alone. Table IV.3 reports the agreement with the measured reaction rates before and after the substitution, for each foil.

Table IV.3: Agreement with the ASPIS-88 foil measurements before and after replacing the JEFF-4.0 elastic angular distribution with the Chapter III evaluation, with the ^{56}Fe cross sections held fixed. χ^2/N is the reduced chi-square of the C/E against unity, weighted by the experimental uncertainty, and $\overline{C/E}$ the mean ratio, each over the N measured positions of the foil. The foils are ordered by increasing effective threshold.

		χ^2/N		$\overline{C/E}$	
Foil	N	JEFF-4.0	This work	JEFF-4.0	This work
^{197}Au(n,γ)	14	0.38	0.32	1.015	1.011
^{103}Rh(n,n$'$)	13	0.66	0.74	0.969	0.968
^{115}In(n,n$'$)	10	8.26	7.39	0.873	0.880
^{32}S(n,p)	14	2.30	1.29	0.903	0.945
^{27}Al(n,α)	4	11.5	11.1	1.158	1.154

The change is concentrated in the fast-threshold foils and grows with depth, as expected of a scattering change that builds up along the shield. It is largest for ^{32}S(n,p), the highest-threshold reaction measured along the full 67 cm. Its reduced χ^2/N against the measurement drops from 2.3 to 1.3 and its mean C/E rises from 0.90 to 0.95. Of the fourteen ^{32}S positions, eleven resolve beyond the Monte Carlo statistics, ten of them moving toward the measured value and one away. The improvement is strongest at depth. JEFF-4.0 underpredicts the deep reaction rates by about ten percent, and the Chapter III evaluation reaches agreement beyond 30 cm, while the offset at the front is unchanged. ^{115}In(n,n$'$) moves the same way but only slightly, its χ^2/N easing from 8.3 to 7.4. For ^{197}Au, ^{103}Rh, and ^{27}Al the agreement is essentially unchanged, their standing offsets carrying no sensitivity to the angular distribution. The overall C/E trends are the same as with JEFF-4.0. The per-position C/E curves for ^{32}S and ^{115}In are given in Appendix F.3.

2.4. Sensitivity Analysis and the Origin of Library Differences

This section presents two complementary analyses. The first computes sensitivity profiles for the elastic channel and the first two inelastic levels, expected to be the dominant contributors, and shows the energy ranges over which each foil responds to them. The second is a single-isotope library swap, which exchanges the complete ^{56}Fe evaluation between JEFF-4.0 and JENDL-5 to test whether the C/E differences between the libraries are a ^{56}Fe effect.

2.4.1. Sensitivity Profiles

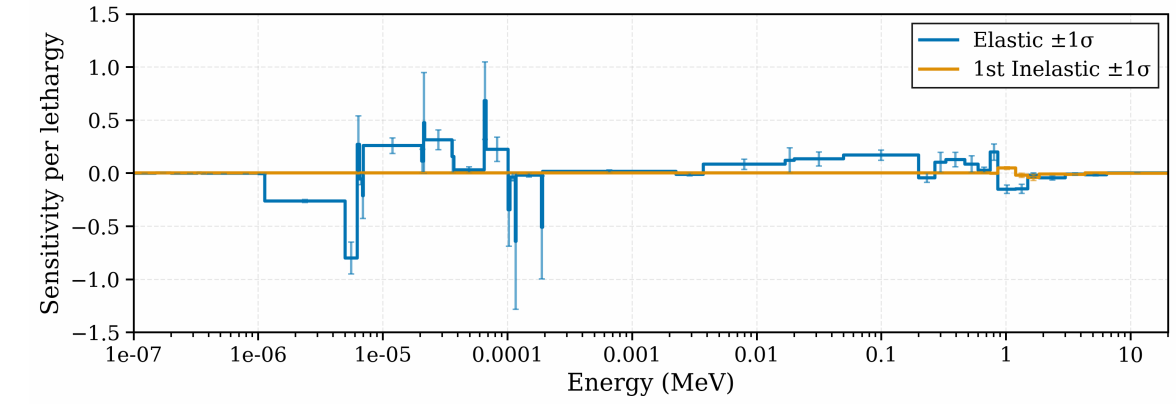
The sensitivity of each foil response to the ^{56}Fe cross sections is obtained with the MCNP `PERT` capability described in Chapter II, Sec. 2.1.4. The profiles separate the elastic channel (`MT` 2) from the first two inelastic levels (`MT` 51 and 52, summed). Figure IV.4 shows the profiles of the five foils at their deepest measured position.

Across the five foils the sensitivity sorts by effective threshold. The ^{197}Au response is dominated by elastic scattering, with a negligible inelastic contribution. For the four threshold foils the elastic and inelastic channels are of similar size, and the sensitive energy window shifts upward, from ^{103}Rh through ^{115}In and ^{32}S to ^{27}Al, as the threshold of each reaction increases.

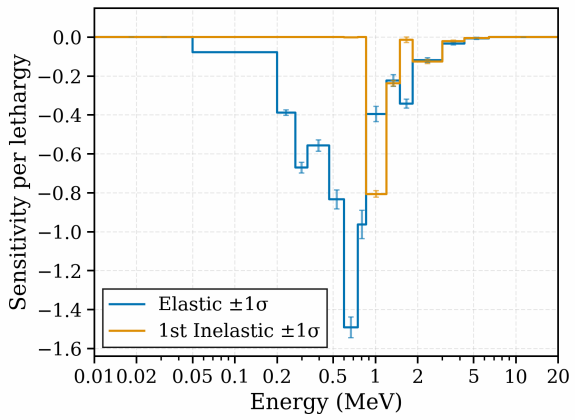
Overlaying these profiles on the relative difference between the JEFF-4.0 and JENDL-5 ^{56}Fe cross sections, shown in Figure IV.5, explains where each library performs better. In the few-hundred-keV-to-MeV window that drives ^{103}Rh and ^{115}In, the elastic and inelastic cross sections of the two libraries differ, and the JENDL-5 values give the better C/E at depth. Above the ^{27}Al threshold near 7 MeV the same comparison again favors JENDL-5. The ^{32}S foil is the one case where JEFF-4.0 is closer, although its standing bias makes that comparison less conclusive.

2.4.2. Isolating ^{56}Fe by a Single-Isotope Library Swap

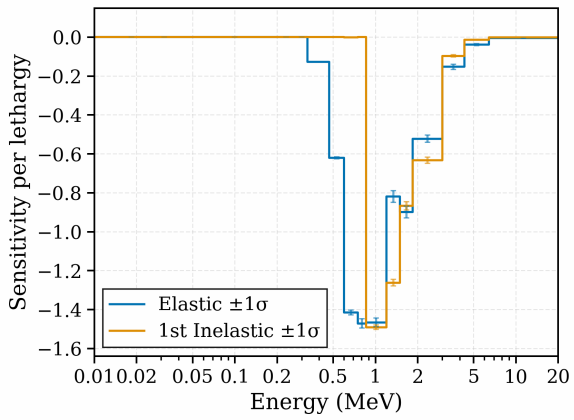
That the cross section difference is specifically the ^{56}Fe difference is confirmed by a library-swap calculation: the transport is repeated with one library throughout except for ^{56}Fe, which is taken from the other. Swapping only ^{56}Fe reproduces essentially the entire JEFF–JENDL C/E difference for the threshold foils; the per-foil C/E curves are collected in Appendix F.2. For ^{27}Al(n,α) (Figure F.8), replacing the JEFF-4.0 ^{56}Fe with the JENDL-5 one moves the C/E from about 1.13–1.20 down to about 1.10, matching the native JENDL-5 result, and the reverse swap raises the JENDL-5 C/E back to the JEFF-4.0 values. The ^{32}S foil behaves the same way, its inter-library C/E gap closing once ^{56}Fe is exchanged. The remaining isotopes of the steel leave only a percent-level residual. The inter-library C/E differences are therefore explained almost entirely by the ^{56}Fe evaluation, although small compensating contributions from the other iron isotopes cannot be fully ruled out. This is what makes the ^{56}Fe uncertainty the relevant quantity to propagate in Sec. 2.3.



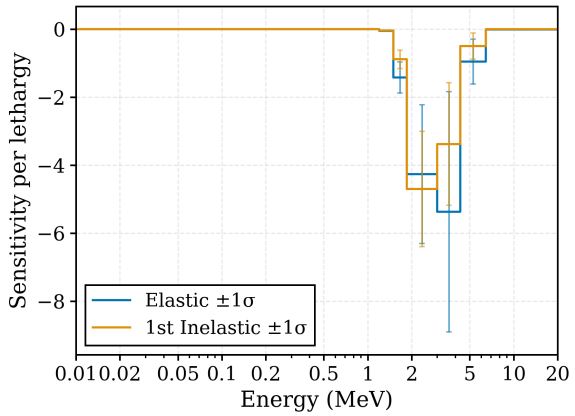
(a) ^{197}Au(n,γ).



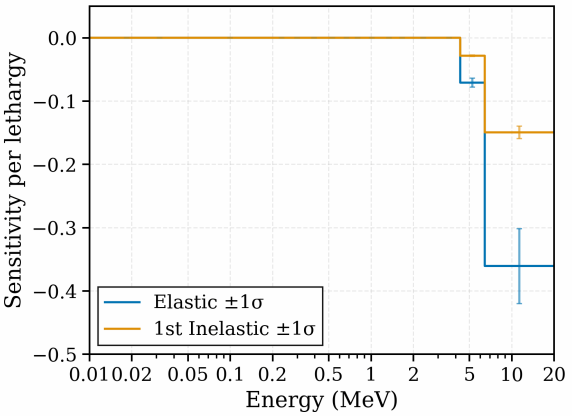
(b) ^{103}Rh(n,n$'$).



(c) ^{115}In(n,n$'$).

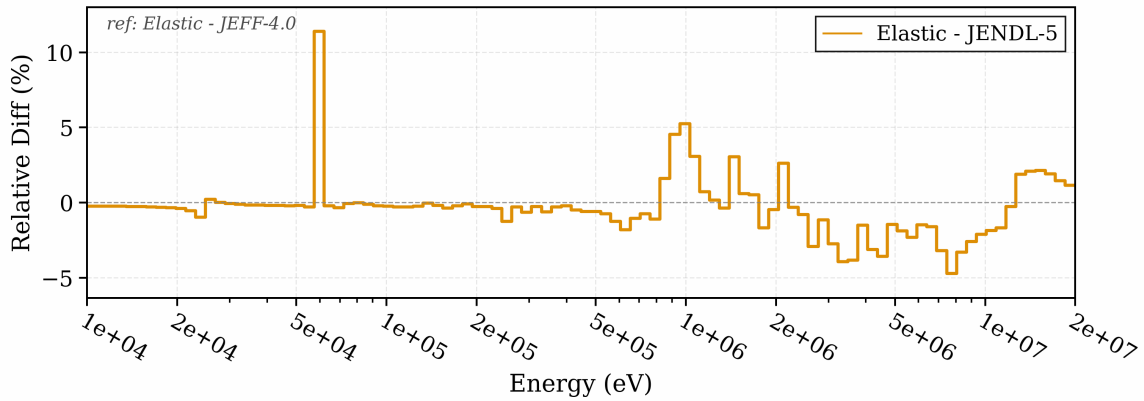


(d) ^{32}S(n,p).

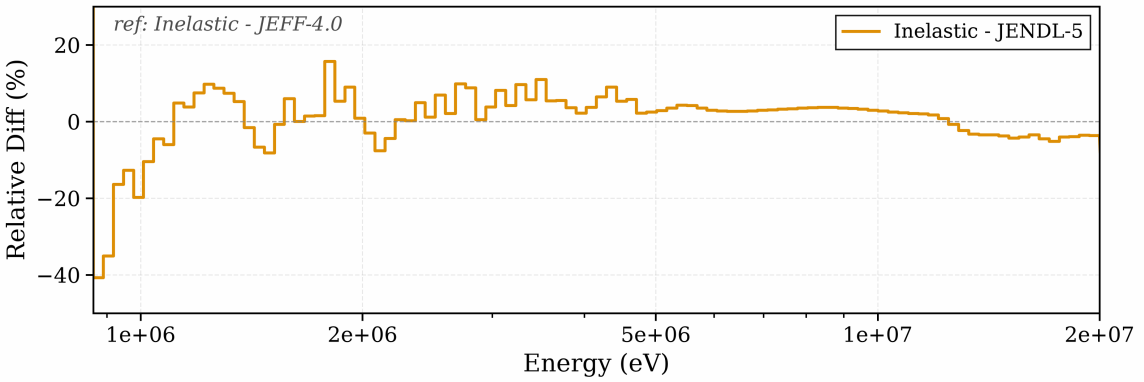


(e) ^{27}Al(n,α).

Figure IV.4: Sensitivity of each foil response at its deepest measured position to the ^{56}Fe elastic (MT 2) and first two inelastic levels (MT 51+52) cross sections, as a function of incident neutron energy.



(a) Elastic scattering (MT 2).



(b) First-level inelastic scattering (MT 51).

Figure IV.5: Relative difference between the JEFF-4.0 and JENDL-5 ^{56}Fe cross sections as a function of incident neutron energy, for the elastic and first-level inelastic channels. The difference is averaged over 100 equal-lethargy groups.

3. The VENUS-3 Benchmark

VENUS-3 is a three-dimensional fission-source shielding benchmark that complements the one-dimensional iron attenuation of ASPIS-88 with a more reactor-like geometry. ASPIS-88 isolates ^{56}Fe along a single deep-penetration path; VENUS-3 has a fission source, several structural materials, and detectors both inside and outside the core. The benchmark is used here to follow how successive releases of the evaluated libraries, from ENDF/B-VIII.0 to VIII.1 and from JEFF-3.3 to 4.0, change the agreement with the measured reaction rates, and to quantify the propagated nuclear-data uncertainty across the full set of structural isotopes. The analysis also reports a problem found in the processing of the ^{16}O covariance data.

3.1. Benchmark Description

VENUS-3 was a reactor assembly built to reproduce the geometry and neutron field of a PWR core surrounded by a steel pressure vessel [179]. The core used uranium dioxide fuel enriched to 3.3% in ^{235}U and included four peripheral assemblies fitted with partial-length shielding assemblies (PLSA), in which the upper half of the fuel rods was replaced by solid stainless steel to reduce the fast neutron flux reaching the vessel. Reaction rates were measured at 386 positions, inside the core and in the ex-core region near the vessel, with ^{58}Ni(n,p), ^{115}In(n,n$'$), and ^{27}Al(n,α) activation foils (Figure IV.6). The benchmark geometry, materials, and measured reaction rates are documented in the NEA benchmark report [180], and the experiment is distributed through the SINBAD database [37]. The experimental uncertainty is about 6% for ^{115}In and ^{58}Ni and about 7% for ^{27}Al [180]. These values are upper bounds, quoted for the detectors at the reactor barrel, and decrease for positions closer to the core; the C/E comparison applies a single band per reaction set to these upper bounds.

3.2. MCNP Model and Uncertainty Propagation

The MCNP [13] model of the benchmark comes from previous ASNR studies and runs in criticality (`KCODE`) mode. A preliminary run with 5×10^5 particles per cycle over 500 cycles produces a converged fission source, which is stored and used as the starting source for all subsequent calculations. The reaction rates are computed with cross sections from the IRDFF-II dosimetry library [178], for consistency with the measured activation foils.

The nuclear-data uncertainties are propagated with the same sampling methodology developed and verified on the PWR sphere model in Chapter II, applied here to the VENUS-3 model. The covariances are sampled with the `kika-nd` framework, drawing 1024 Sobol samples per library from the multigroup matrices processed with the NJOY ERRORR module [11] in a 56-group structure; each sample is run through the MCNP model with 2×10^6 particles per cycle, the first 20 cycles discarded as inactive to settle the source and the tallies accumulated over the following 250 active cycles. The propagation covers every isotope of the attenuation path with available covariance data, excluding the fuel and cladding: 30 isotopes in JEFF-3.3, 44 in JEFF-4.0, 26 in ENDF/B-VIII.0, and 29 in ENDF/B-VIII.1, listed per library in Table F.6 of Appendix F.5. The nuclear-data and statistical contributions to the spread are separated with the Fast TMC scheme of Chapter II, which is reliable where the Monte Carlo uncertainty stays below about half the observed uncertainty; the detectors that exceed this condition are flagged in the results.

Unlike the ASPIS-88 analysis, the angular-distribution channel is not propagated for VENUS-3. Preliminary checks indicated that its contribution to the reaction rates was generally small in

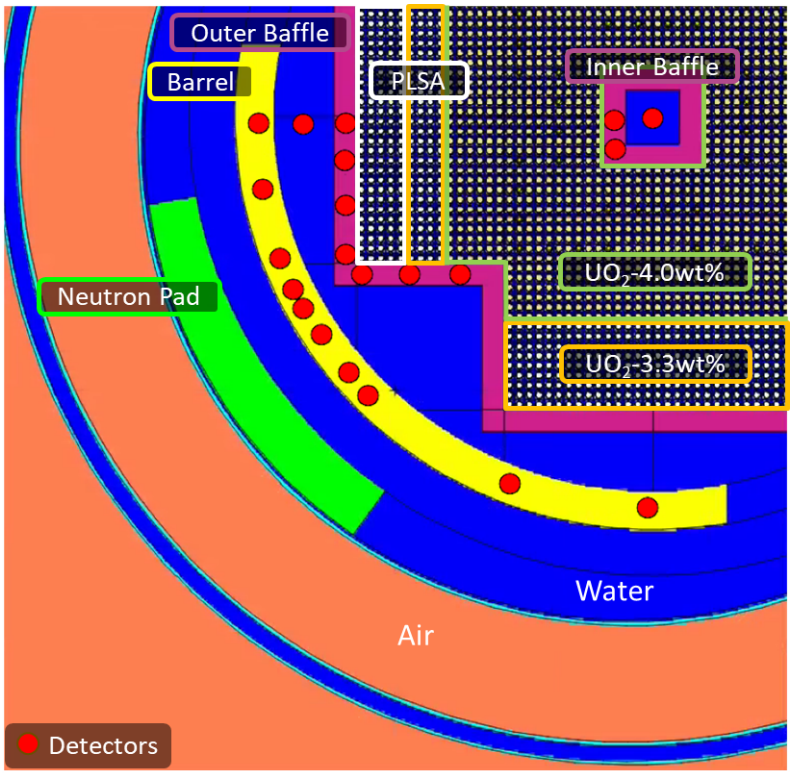


Figure IV.6: Radial cross section of the VENUS-3 configuration showing the main components and detector locations. Only some detector positions are shown for clarity.

this geometry where the steel paths are shorter, so the propagation here covers only the cross section covariances of the full isotope set.

3.3. C/E Comparison

Table IV.4 summarizes the C/E ratios for the four libraries: the mean C/E (an unweighted average over the 386 detectors), the root-mean-square error (RMSE), and the percentage of detectors within the 1σ experimental band. The detector-level C/E distribution is close to symmetric about unity, so the median agrees with the mean closely and the summary is not distorted by the few outlying detectors. ENDF/B-VIII.1 gives the best agreement, with a mean C/E of 0.999 and more than 93% of the detectors within the experimental uncertainty. Both the ENDF/B and the JEFF series improve over their previous releases.

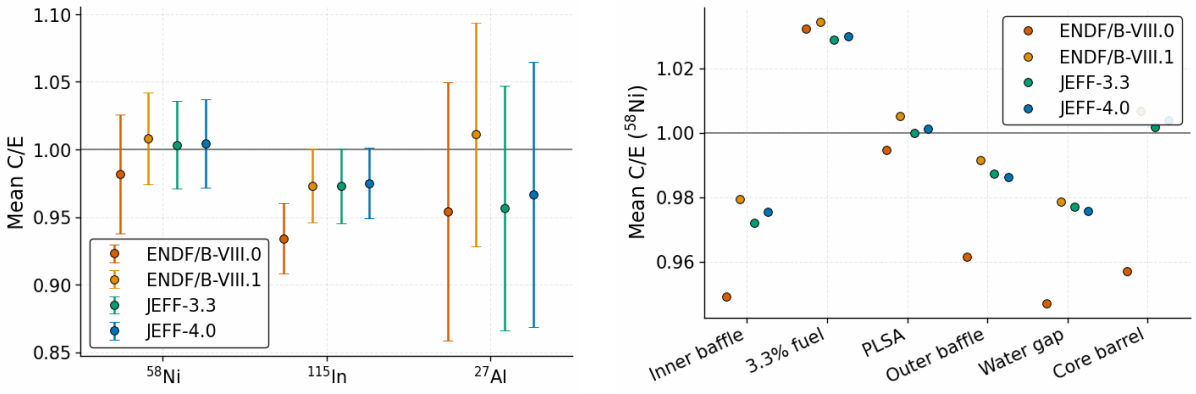
Table IV.4: Mean C/E values, root-mean-square error (RMSE), and percentage of detectors within 1σ experimental uncertainty for each evaluated library.

Library	Mean C/E	RMSE	Within 1σ (%)
ENDF/B-VIII.0	0.9662	0.0620	70.47
ENDF/B-VIII.1	0.9991	0.0428	93.26
JEFF-3.3	0.9906	0.0450	91.19
JEFF-4.0	0.9928	0.0456	92.75

The largest gain, from ENDF/B-VIII.0 to ENDF/B-VIII.1, follows the revised ^{56}Fe cross sections that dominate transport through the steel. ENDF/B-VIII.1 and JEFF-4.0 share this recent ^{56}Fe

evaluation and both reach a mean C/E near unity, whereas ENDF/B-VIII.0, which keeps the earlier data, stays about 3% low; JEFF-3.3 also lies close to unity. That two independently maintained evaluations agree once they adopt the same ^{56}Fe data supports the attribution. The spatial pattern of the C/E change, examined next, reinforces it.

Grouping the detectors by reaction and by position helps interpret the spread (Figure IV.7). On average, the ^{115}In detectors read about 2–3% below unity in the three recent libraries, and lower still in ENDF/B-VIII.0, whose whole set is shifted down. The ^{58}Ni detectors in the 3.3% enriched fuel inside the core (detectors 189–244 in Figure IV.9) read about 3% high in every library; this offset is nearly identical across the four libraries, which points to the dosimetry cross sections or the experimental normalization rather than the transport data. The ^{27}Al reaction shows by far the largest scatter, mainly from the higher statistical uncertainty of its tally: its high energy threshold scores few reactions and inflates the variance.



(a) Mean C/E by reaction.

(b) ^{58}Ni C/E by detector position, core to vessel.

Figure IV.7: Nominal C/E for the four libraries, grouped by reaction and, for ^{58}Ni, by detector position a. Error bars in are the standard deviation across detectors.

The C/E change between ENDF/B-VIII.0 and ENDF/B-VIII.1 follows the amount of steel along the neutron path (Figure IV.8). It is smallest for the in-core fuel detectors, about 0.4%, and grows to about 5% for the outermost detectors at the core barrel; the detectors behind the PLSA show an intermediate change, and no clear axial or azimuthal dependence is found. This growth with steel thickness is what the revised ^{56}Fe cross sections would produce. The two releases differ in more than ^{56}Fe, so a single-isotope calculation would be needed to make the attribution exact, but the spatial pattern already points to ^{56}Fe as the main driver. Across the ^{58}Ni detectors along the steel path (roughly detectors 1–100), JEFF-4.0 shows more detector-to-detector structure in the C/E than ENDF/B-VIII.1 (Figure IV.9); identifying the isotope responsible would require a dedicated isotope-by-isotope study.

3.4. Propagated Uncertainties and the Processing of the ^{16}O Covariances

Figure IV.9 shows the C/E ratios with their propagated 1σ nuclear-data uncertainties for the four libraries. The points in pink mark the detectors that fail the separation condition, where the Monte Carlo uncertainty exceeds about half the observed uncertainty. At these detectors the nuclear-data uncertainty is smaller than the residual statistical noise and cannot be separated reliably, so the band shown is the total 1σ value, an upper bound that still contains both the statistical and the nuclear-data contributions. The statistical uncertainty at these pink-marked points is already low, and the nuclear-data part is smaller still. Resolving it would

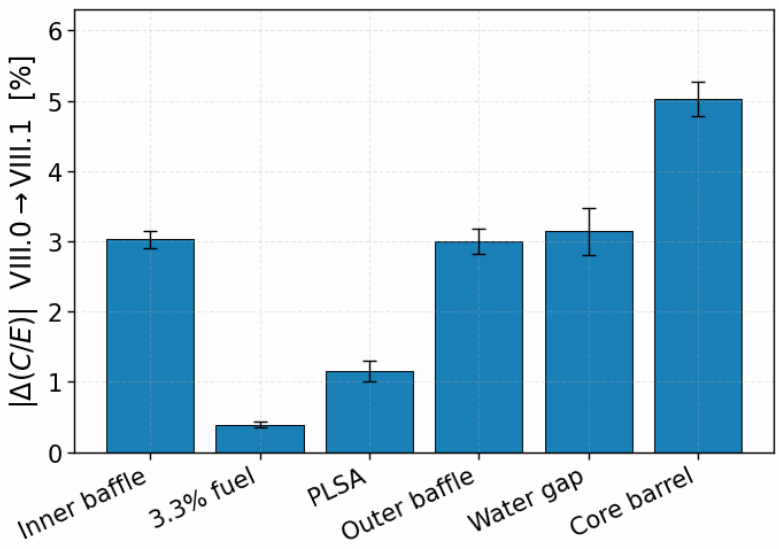


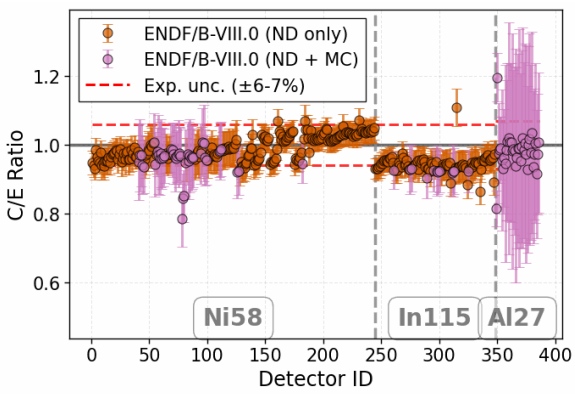
Figure IV.8: Change in the ^{58}Ni C/E between ENDF/B-VIII.0 and ENDF/B-VIII.1, by detector region ordered from the core to the vessel. The change grows with the steel thickness along the neutron path. Error bars are the standard error of the mean.

require much longer runs to further reduce the statistical noise, at a high computational cost and with little practical value, since the nuclear-data uncertainty is small. For ENDF/B-VIII.0, ENDF/B-VIII.1, and JEFF-3.3, the propagated nuclear-data uncertainties are consistently low, with average values between about 1% and 4.5%, and the overall mean uncertainty stays below 7% once the statistical contribution is included. JEFF-4.0, in contrast, shows much larger propagated uncertainties, averaging around 10%.

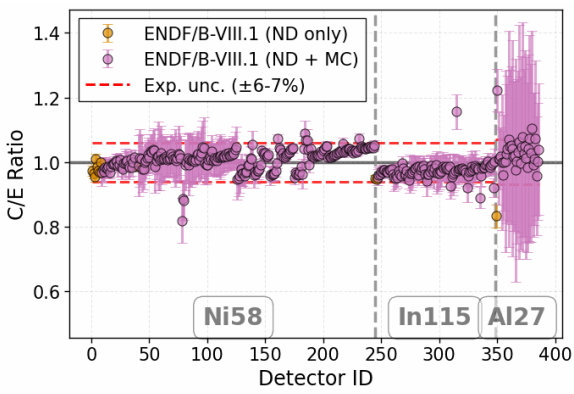
The larger JEFF-4.0 uncertainty was traced to its ^{16}O covariance data. During NJOY processing, the multigroup covariance matrices of ^{16}O showed unrealistically large variances on the elastic scattering and the radiative capture, exceeding 1000% for elastic scattering above 1 MeV. The effect was concentrated in the detectors located where water dominates the neutron transport, consistent with an ^{16}O origin. Two checks confirmed it: removing ^{16}O from the sampling, and reprocessing its covariance data with the PUFF module of AMPX [47]. In both cases the propagated uncertainties returned to the level of the other libraries, and in particular to JEFF-3.3, whose covariance data are largely shared with JEFF-4.0. The inflated band came not from the evaluation but from how it was processed, so the covariance processing must be checked before the propagated uncertainties can be trusted.

Once the ^{16}O processing is corrected, the propagated nuclear-data uncertainties of VENUS-3 are small, between 2% and 4% for all libraries and below the 6–7% experimental band. The calculations stay consistent with the measurements for both the nominal data and their covariances. The small uncertainties also mean that VENUS-3 offers little room for reducing nuclear-data covariances through data assimilation in PWR aging studies.

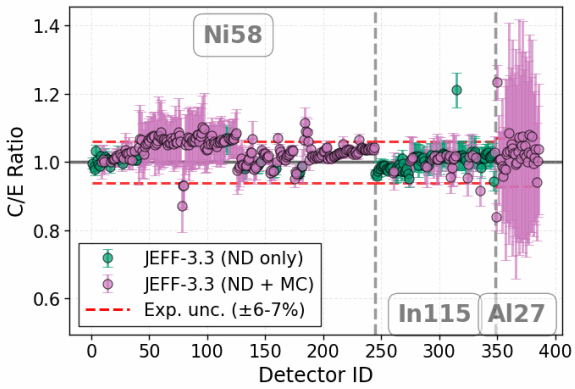
The two benchmarks confirm that the modern libraries reproduce the measured reaction rates, that the differences between the libraries are carried mainly by the ^{56}Fe cross sections, and that the propagated nuclear-data uncertainty covers the observed spread. On ASPIS-88 the Chapter III angular evaluation reproduces the measured rates on the same footing as the published data. The same propagation is now taken to the reactor vessel itself, where no direct measurement of the fluence exists and the prediction rests on the transport calculation alone.



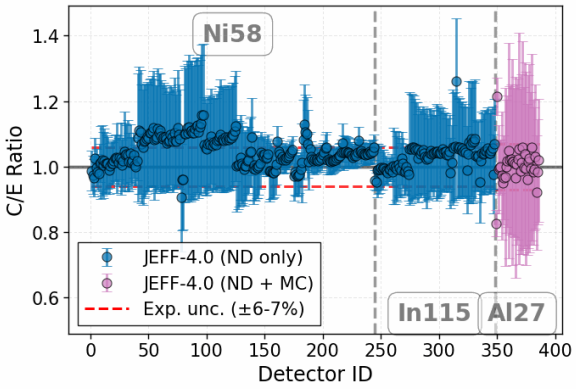
(a) ENDF/B-VIII.0.



(b) ENDF/B-VIII.1.



(c) JEFF-3.3.



(d) JEFF-4.0.

Figure IV.9: C/E ratios with the propagated 1σ nuclear-data uncertainties for the four libraries. Detectors are grouped by reaction; dashed lines mark the experimental uncertainty. Pink points are detectors that fail the separation condition, for which the total 1σ band is shown.

4. Reactor Model, Tallies, and Variance Reduction

The analysis uses a single reactor model throughout, the French 3-loop 900 MWe pressurized water reactor of the reference VACS scheme [4]. The CASMO5/SIMULATE5 core model and the MCNP6 transport model are those of that ASNR study. The core operates at equilibrium cycle conditions with a cycle length of 11 GWd/t and a mixed UOX/MOX loading. From the core outward, the radial structure follows the standard PWR architecture shown in Figure IV.10, with the baffle (2), the moderator bypass (3), the core barrel (4), the asymmetric neutron pad (5), the downcomer annular zone (6), and the vessel inner wall (7).

The vessel response is scored at three azimuths, 0°, −20°, and −28°, spanning the hotspot region of the vessel fluence identified by Vuiart [4]; each detector is 4° wide and 1 cm deep into the vessel wall. The active core height is divided into nine axial segments of 40 cm, and the analysis is restricted to the three central segments, spanning the ±60 cm window about the core midplane where the fast flux of the spectrum is largest. The surveillance capsule, modeled as a vessel-steel cylinder one meter tall and centered on the active core, is placed at −17° and carries three axial detectors of 33.33 cm.

The capsule contains six threshold dosimeters: ^{237}Np(n,f), ^{238}U(n,f), ^{58}Ni(n,p), ^{54}Fe(n,p), ^{46}Ti(n,p), and ^{63}Cu(n,α). Their effective threshold energies span from ~ 0.7 MeV (the ^{237}Np fission threshold) to ~ 5 MeV (the ^{63}Cu(n,α) threshold), so the six reactions collectively probe the fast tail of the neutron spectrum at the surveillance position. The corresponding activation cross sections, taken from the IRDFF-II dosimetry library [178], are shown in Figure IV.11.

Variance reduction

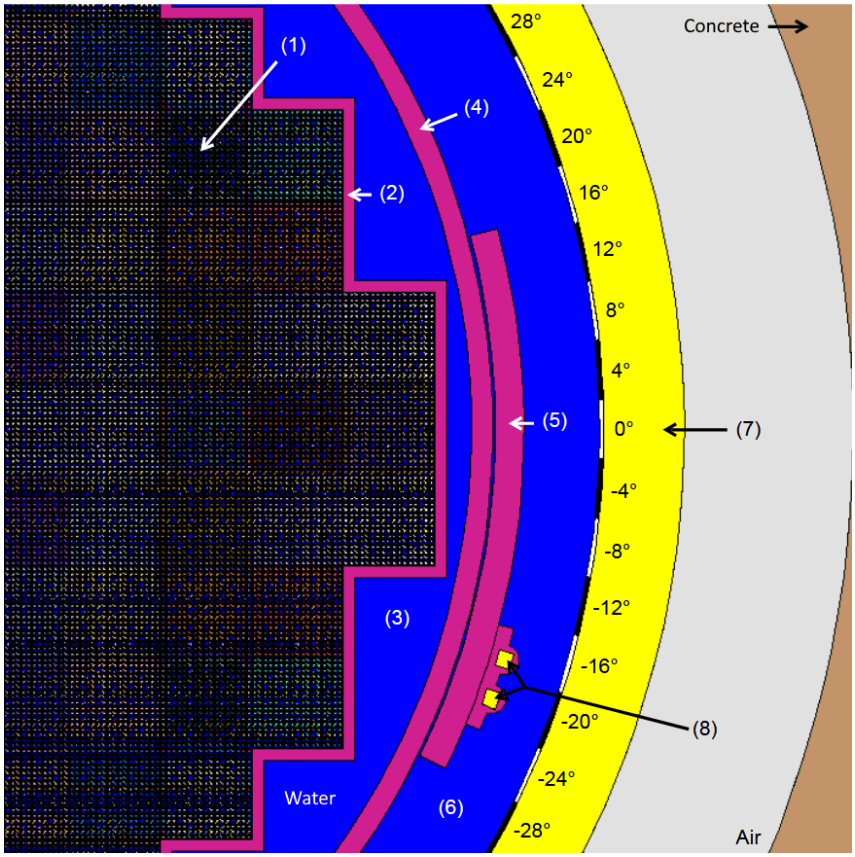
The vessel and capsule tallies sit several mean free paths from the source. Convergence at acceptable statistical error is achieved with weight windows generated in this work with the FW-CADIS scheme of ADVANTG, which generates source and weight-window biasing from a forward-adjoint deterministic pre-calculation. The adjoint solver uses the BPLUS multigroup library; the spatial mesh is 1×1 cm^2 in the x–y plane in the detector regions and their immediate surroundings, and 4×4 cm^2 elsewhere, with a 40 cm grid along the axial direction. A single weight-window map, shown in Figure IV.12, covers the vessel and capsule tally regions, and one MCNP run delivers all reported responses in approximately 9 CPU-hours, with statistical errors below 1% on the vessel and below 2% on the capsule dosimeters.

Reference tally values

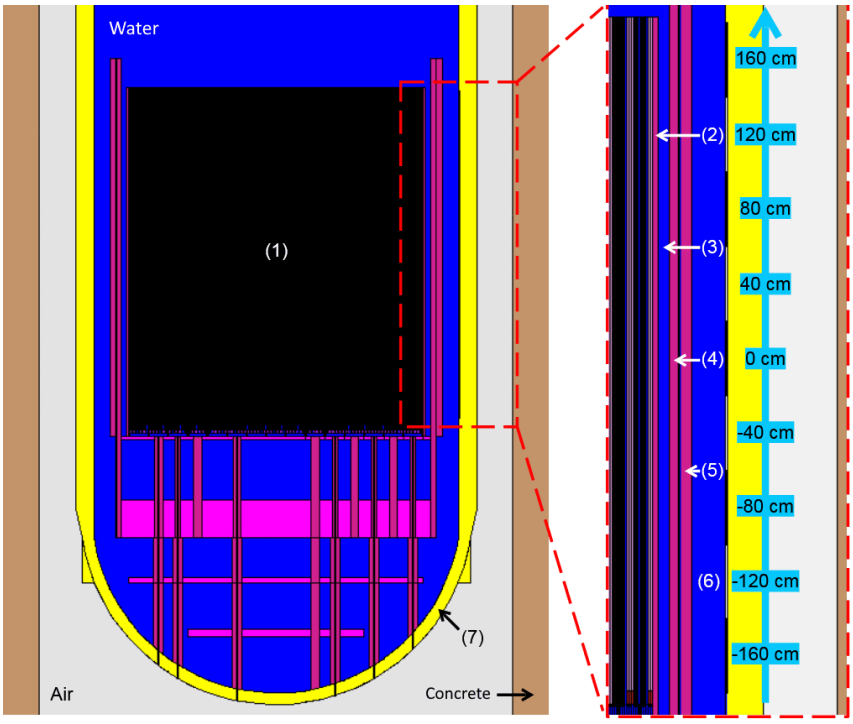
Table IV.5 reports the central values of the vessel and capsule tallies for JEFF-4.0, in physical units. Each value comes from the nominal run, a single MCNP calculation with the unperturbed JEFF-4.0 library. The last column is the statistical error of that run; it fixes the precision of the central value and stays well below the propagated uncertainties of Sec. 6. The same quantities for ENDF/B-VIII.1 and JENDL-5 are reported in Appendix F.6. ENDF/B-VIII.1 agrees with JEFF-4.0 to within 0.3% on the vessel, and JENDL-5 is 3–5% lower at the 0° and −20° azimuths and at the capsule (Sec. 6.2).

The physical values in Table IV.5 follow from the native per-source-neutron MCNP tallies by scaling with the total fission neutron source rate of the core:

$$S = \frac{P_{\mathrm{th}}\,\bar{\nu}}{E_R} = \frac{(2.785\times10^{9}\ \mathrm{W})\,(2.5)}{3.2\times10^{-11}\ \mathrm{J}} \approx 2.2\times10^{20}\ \mathrm{n\,s^{-1}} \tag{IV.1}$$



(a) Horizontal cross-sectional view.



(b) Vertical cross-sectional view.

Figure IV.10: Cross-sectional views of the MCNP6 PWR900 model showing the tally positions. (1) fuel assemblies; (2) core baffle; (3) moderator bypass region; (4) core barrel; (5) asymmetric neutron pad; (6) annular moderator downcomer; (7) reactor pressure vessel; (8) surveillance capsule [4].

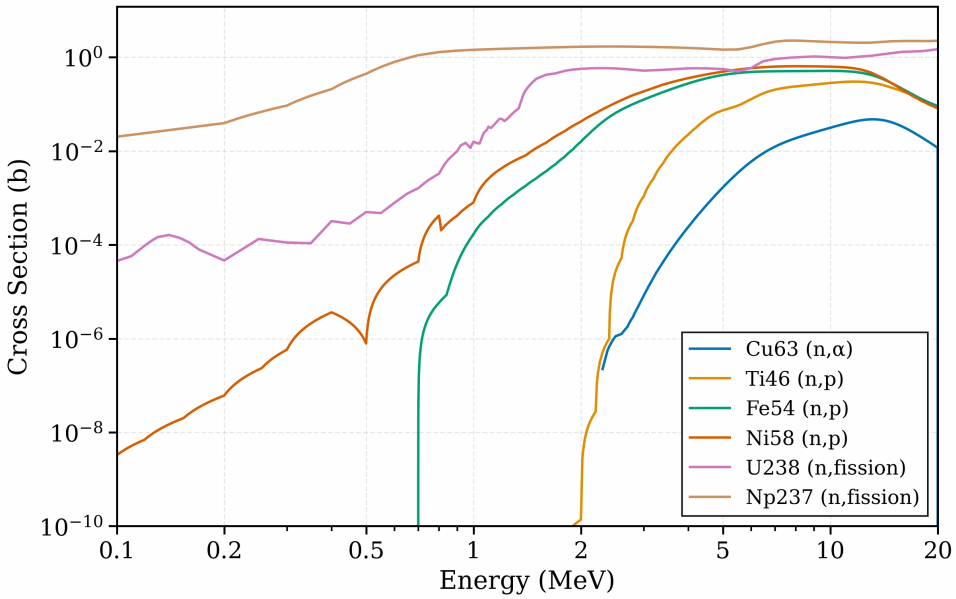


Figure IV.11: Evaluated cross sections of the six capsule dosimetry reactions as a function of incident neutron energy above 0.1 MeV.

Table IV.5: Reference central values of the vessel and capsule tallies for JEFF-4.0, in physical units, from the nominal run; the last column is its MCNP statistical error. Vessel quantities are volume-averaged over the ±60 cm central window, capsule quantities over the three axial detectors at −17°. Reaction rates are per target atom (s^{-1} $atom^{-1}$), the dosimeters being scored as fictitious materials.

Quantity	Location	Value	Stat. (%)
Flux $\Phi_{E>1\,\mathrm{MeV}}$ ($n\,cm^{-2}\,s^{-1}$)	Vessel, 0°	3.24×10^{10}	0.36
	Vessel, −20°	1.66×10^{10}	0.39
	Vessel, −28°	2.29×10^{10}	0.57
	Capsule, −17°	1.68×10^{11}	0.33
NRT-dpa rate ($dpa\,s^{-1}$)	Vessel, 0°	4.51×10^{-11}	0.36
	Vessel, −20°	2.42×10^{-11}	0.36
	Vessel, −28°	3.11×10^{-11}	0.56
ARC-dpa rate ($dpa\,s^{-1}$)	Vessel, 0°	1.46×10^{-11}	0.38
	Vessel, −20°	7.93×10^{-12}	0.36
	Vessel, −28°	9.92×10^{-12}	0.56
^{237}Np(n,f) rate	Capsule	5.69×10^{-13}	0.29
^{238}U(n,f) rate	Capsule	5.33×10^{-14}	0.37
^{58}Ni(n,p) rate	Capsule	1.31×10^{-14}	0.53
^{54}Fe(n,p) rate	Capsule	9.14×10^{-15}	0.59
^{46}Ti(n,p) rate	Capsule	1.49×10^{-15}	1.05
^{63}Cu(n,α) rate	Capsule	7.99×10^{-17}	1.61

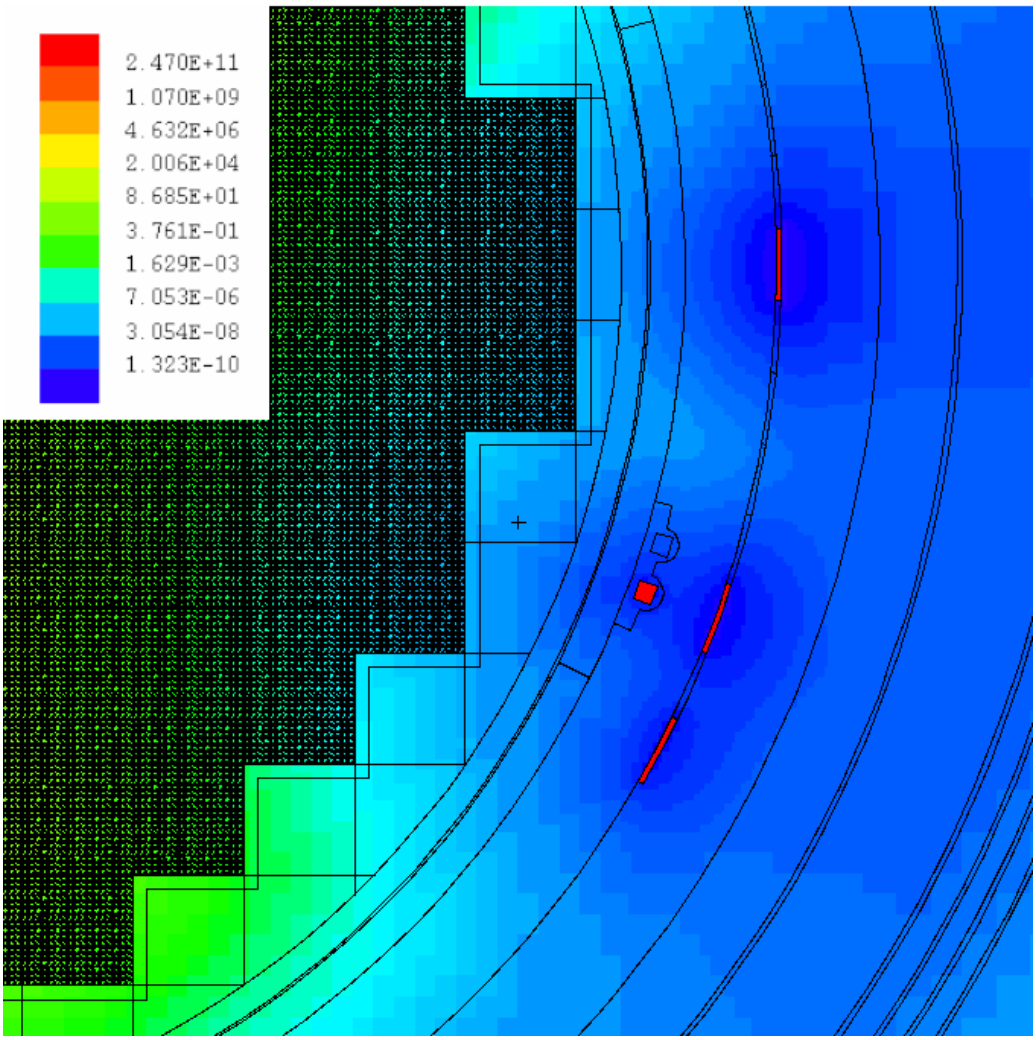


Figure IV.12: Horizontal cross-sectional view showing the ADVANTG-generated weight-window map used for detector-response variance reduction. The color scale represents the generated weight-window values, with the optimized detector regions in red (not to be confused with the highest WW values, also in red).

where $P_{\mathrm{th}} = 2785$ MWth is the core thermal power of the 900 MWe reactor, $\bar{\nu} \approx 2.5$ the average number of neutrons per fission, and $E_R \approx 200$ MeV ($= 3.2 \times 10^{-11}$ J) the recoverable energy per fission [4]. Scaling by S places the fast neutron flux ($E > 1$ MeV) at the inner wall at the 0° hotspot at 3.2×10^{10} n cm^{-2} s^{-1}. Accumulated over several decades of operation, this is of the same order as the French regulatory fast-fluence limit of 6.5×10^{19} n cm^{-2} that drives the embrittlement correlations [4].

The azimuthal pattern of the central values is consistent with the geometry of the model: the highest fast flux is at the 0° hotspot, the lowest at the −20° position behind the surveillance capsule, and an intermediate value at −28° where the spectrum no longer crosses the asymmetric neutron pad. The spectrum at each location is shown in Figure IV.13: the capsule sits closest to the core and is the least attenuated of the four positions.

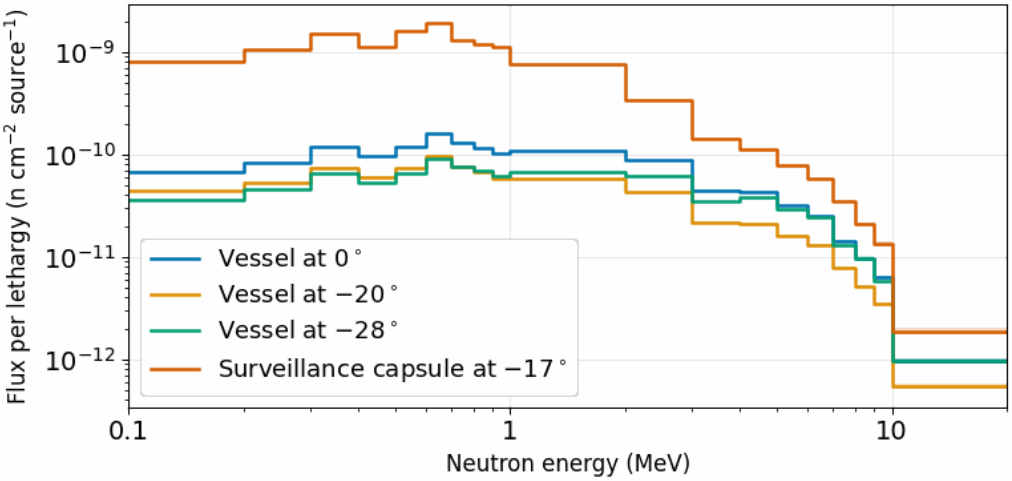


Figure IV.13: Neutron energy spectrum at the four detector positions of the PWR900 model, height-integrated over the central ±60 cm window. The spectra are scored from the JEFF-4.0 nominal run and lethargy-normalized.

5. Nuclear Data Uncertainty in the Source Term

The fission neutron source term is the first step of the VACS calculation chain. As introduced in Chapter I, the source at pin p in assembly m is:

$$S_p(E', t) = \sum_i R_{f,i,m}(t) \cdot \nu_i \cdot X_p(t) \cdot \chi_m(E', t) \tag{IV.2}$$

the product of the assembly fission rate, the neutron multiplicity, the pin form factor, and the assembly-averaged fission spectrum. The pin-wise source amplitude is $X_p(t) \sum_i \nu_i R_{f,i,m}(t)$, while $\chi_m(E', t)$ defines the emitted-neutron energy distribution. The source uncertainty is carried by both the fission neutron production term, including its pin-wise redistribution, and the fission spectrum. Two independent propagations treat them. The first samples all the nuclear data of the core calculation and follows the response of the production term (Secs. 5.1 and 5.2). The second samples the fundamental fission spectra of the source isotopes and follows the response of the emitted spectrum of the source term (Sec. 5.3). Both are carried to the vessel in Sec. 5.4.

5.1. Calculation Setup

The loading pattern of the PWR900 core (Sec. 4) at equilibrium cycle is shown in Figure IV.14, with UOX assemblies at various burnup stages and MOX assemblies in three plutonium enrichment zones.

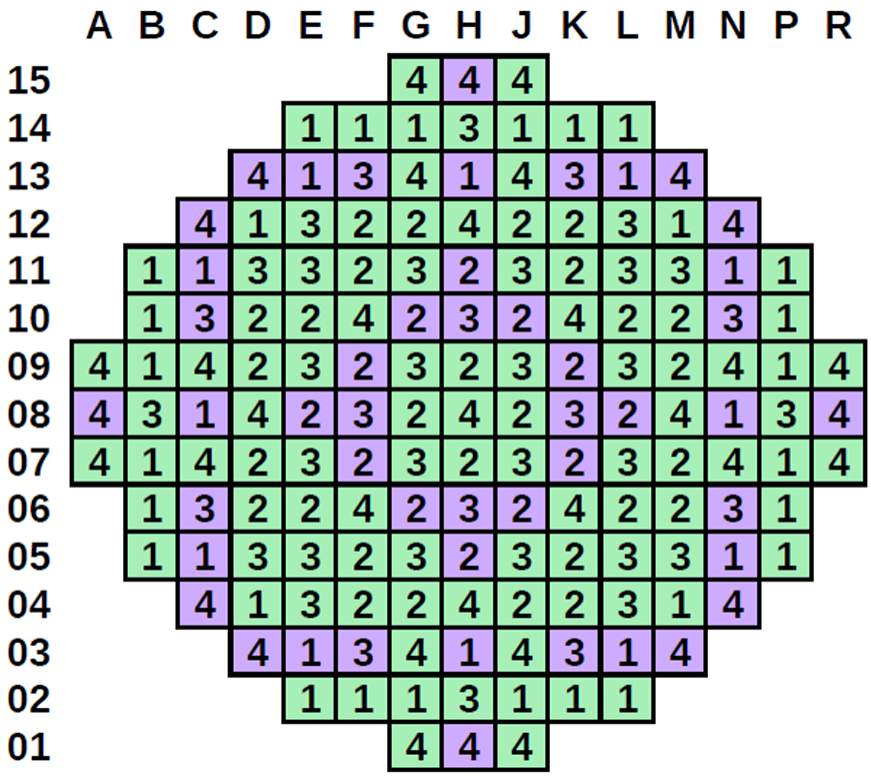


Figure IV.14: Fuel loading pattern for the PWR900 core at equilibrium cycle. UOX assemblies are shown in green and MOX assemblies in purple. Numbers indicate the irradiation cycle of each assembly, so that an assembly labeled "4" has already completed three irradiation cycles in the core [4].

The sampling of the core calculation was carried out by Flavien Bazon during a 2025 internship at ASNR, supervised as part of this thesis. The nominal CASMO5 lattice calculations use the ENDF/B-VIII.1 library [33]. Nuclear data uncertainties are propagated by drawing 1000 perturbed samples from the ENDF/B-VIII.0 covariance matrices. On input, all available isotopes and reaction types (Table I.3) are perturbed simultaneously, the fission spectrum (`MT` 1018) included. The covariance library at this step is fixed to ENDF/B-VIII.0 by CASMO5 and is not a user choice. This global strategy captures the combined effect of all nuclear data uncertainties and their correlations, at the cost of preventing a breakdown into individual contributors. On output, each sample gives the SIMULATE5 core state, namely the fission neutron production rate of each isotope in each assembly, the pin form factors, and the moderator temperature. For each sample, the burnup-dependent results from SIMULATE5 are averaged over the 28 steps of the equilibrium cycle using a trapezoidal weighting, and subsequently over the 20 axial nodes, yielding cycle-averaged quantities per assembly. This 28-step temporal mesh follows the burnup-step convergence study of Vuiart [4].

5.2. Outputs of the Core Calculation

The primary observable for the source term is the fission neutron production rate, νR_f, defined as the product of the average fission neutron multiplicity and the macroscopic fission reaction rate per unit volume. This quantity determines the spatial intensity of the neutron source that drives the fluence at the vessel wall. The per-assembly distributions obtained from the 1000 samples are approximately Gaussian and centered on the nominal value (Appendix F.7), so the spread is reported as a relative uncertainty, the sample standard deviation divided by the mean.

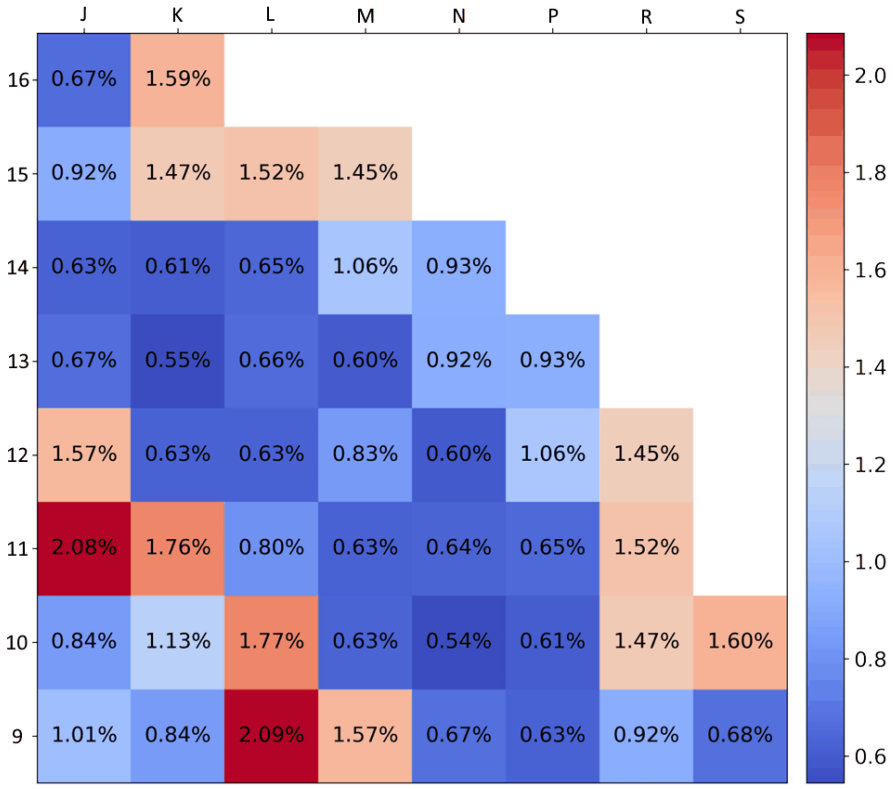


Figure IV.15: Relative uncertainty (standard deviation over the mean) of the assembly-averaged fission neutron production rate νR_f on the PWR900 core in quarter-symmetry representation, propagated from 1000 ENDF/B-VIII.0 covariance samples through CASMO5/SIMULATE5.

Figure IV.15 shows the spatial distribution of the relative uncertainty. The highest values, reaching 2.1%, are found in the MOX assemblies near the center of the core, whose isotopic composition includes several plutonium isotopes with larger relative nuclear data uncertainties than ^{235}U. The peripheral assemblies closest to the vessel wall (J16, K16, K15, L15, M15) carry uncertainties of $\sim 1.5\%$. These assemblies are of particular interest because, despite their lower absolute production rates, they face the vessel directly and dominate the fast neutron flux at the RPV inner surface (Appendix F.8).

Two other outputs of the same samples respond much less. The first is the assembly spectrum $\chi_m(E')$ that VACS builds from each sample. VACS mixes the analytic Watt and Maxwell spectra of the four source isotopes (Sec. 5.2), so χ_m changes from one sample to the next only through the production weights $\nu_i R_{f,i,m}$ of Eq. (IV.2). Its relative uncertainty stays below 0.1% up to 10 MeV and rises to $\sim 0.5\%$ at 20 MeV. This value is the uncertainty of the isotope mixture. The perturbation of the fission spectrum on input (`MT` 1018) acts inside the core calculation and does not reach the spectrum of the VACS source. The second output is the moderator temperature

computed by the SIMULATE5 thermal-hydraulic feedback, whose relative uncertainty stays below 0.08%.

5.3. Perturbation of the Emitted Fission Spectrum

This second propagation measures the uncertainty of the fundamental fission spectrum $\chi_i(E \to E')$ carried into the emitted spectrum of the VACS source term. The assembly spectrum $\chi_m(E')$ of Eq. (IV.2) is not computed by the core calculation. CASMO5/SIMULATE5 supply, for each assembly, the cycle-averaged share of fission neutrons produced by each of ^{235}U, ^{238}U, ^{239}Pu, and ^{241}Pu, and VACS mixes the isotope spectra with these shares (Sec. 5.2). Because the plutonium spectra are harder, MOX assemblies emit a harder spectrum than UOX ones. The MCNP source samples birth sites pin by pin, and each site takes the spectrum of its assembly, which is written directly on the source cards of the transport input.

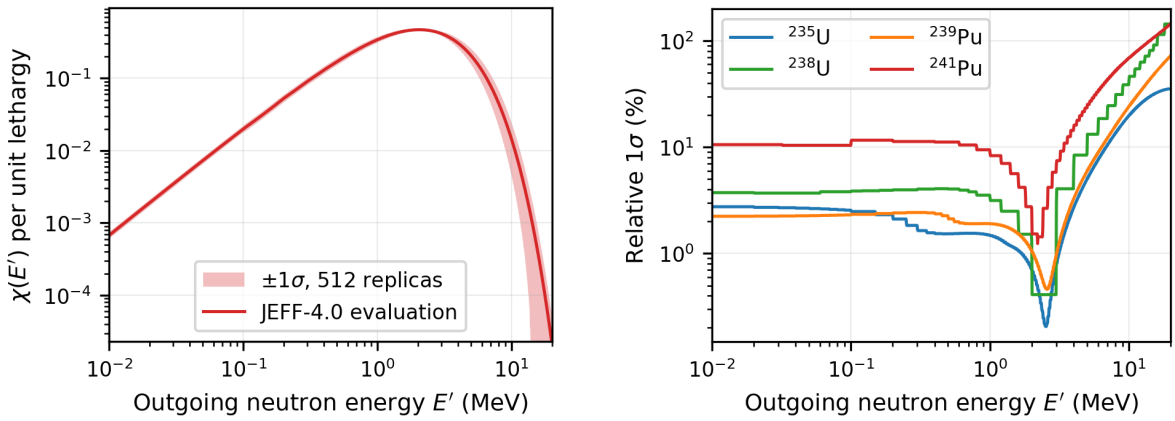
The analytic forms are an approximation of the measured spectra [181]. This work replaces them with the tabulated prompt fission neutron spectra (`MF5`) of JEFF-4.0 and samples them with KIKA from their evaluated covariance (`MF35`), as is done for the transport cross sections. JEFF-4.0 is used because it supplies this covariance for all four isotopes, while ENDF/B-VIII.0 and ENDF/B-VIII.1 give none for ^{241}Pu. Where the libraries overlap, the JEFF-4.0 covariance is the larger one. On the mean energy of the thermal ^{235}U spectrum it states a 2.0% uncertainty, against 0.5% in ENDF/B-VIII.0 and ENDF/B-VIII.1 [182, 33], and for thermal ^{239}Pu about seven times the ENDF/B-VIII.1 value. The fission-spectrum contribution obtained here is therefore specific to the JEFF-4.0 covariance. That covariance states the uncertainty of the emission probability in each outgoing-energy group, for one incident-energy band at a time.

VACS assigns a single spectrum to each isotope, while the evaluated spectrum and its covariance depend on the incident energy. Each isotope is therefore taken at one representative incident energy, with the covariance band that contains it. The thermal point (0.0253 eV) is used for ^{235}U, ^{239}Pu, and ^{241}Pu, and 2 MeV for ^{238}U, which fissions only above its threshold. No weighting by the fission rate over incident energy is applied. The fissile spectra change little below ~ 1 MeV, and this choice moves the mean energy of the source spectrum by about 0.1%. The isotopes are mixed in each assembly with the cycle-averaged weights of the nominal VACS source.

A population of 512 samples is drawn with the same Sobol scheme as the transport campaign of Sec. 6.1, independently for each isotope, and each sample is kept positive and normalized to unity. The samples were checked against the evaluation. With a zero perturbation they reproduce the evaluated spectra exactly, and their standard deviation matches the evaluated covariance in every group that carries significant probability. Figure IV.16 shows the evaluated spectra and the spread of the samples. The largest relative uncertainty sits in the high-energy tail, and ^{241}Pu carries the largest one.

5.4. Propagation to the Vessel

The two propagations reach the vessel through the MCNP6 model of Sec. 4. The source spatial distribution is carried by three representative cases, and the emitted spectrum by the full population of 512 samples.



(a) ^{241}Pu emission spectrum with the $\pm1\sigma$ envelope of the samples.

(b) Relative 1σ of the four source isotopes.

Figure IV.16: Prompt fission neutron spectra of the source isotopes from JEFF-4.0 and the spread of the 512 perturbed samples, per outgoing-energy bin. Each curve carries the `MF35` incident-energy band its source spectrum is taken from.

Source spatial distribution

Propagating the full set of 1000 source term samples through the MCNP6 transport calculation is computationally costly, since each simulation requires billions of neutron histories. Vuiart's analysis on the same model [4] did not propagate the full source-term covariance through MCNP; it relied on the prompt fission neutron spectrum (PFNS) result of Berge [183] and found the source term to be one of the largest contributors to its fluence uncertainty budget. The assembly-level production rates are already available from the source term sampling and respond roughly linearly, so a few representative cases are enough to bound their effect on the vessel flux. The present work propagates the source spatial distribution directly on the MCNP geometry using three such cases from the source term sample distributions: the nominal case, computed with unperturbed nuclear data; sample 330, which produces source strengths above $+2\sigma$ on the peripheral assemblies closest to the vessel; and sample 921, which produces source strengths below -2σ on the same peripheral assemblies. The peripheral assemblies (K16, K15, L15, M15, J16) were used as the selection criterion because they dominate the fast neutron flux incident on the RPV.

Table IV.6: Fast neutron flux (E = 1–20 MeV) at the RPV inner surface, 0° vessel azimuth (the hot spot), for three representative source term cases, in MCNP normalization (per source neutron). The relative deviation is computed with respect to the nominal case.

Case	Flux (n/cm^2)	Statistical error	Relative deviation
Nominal	1.476×10^{-10}	0.20%	—
Sample 330	1.525×10^{-10}	0.19%	+3.3%
Sample 921	1.434×10^{-10}	0.20%	−2.8%

The results for the fast neutron flux tally at the vessel inner surface are summarized in Table IV.6. The spread of [−2.8%, +3.3%] corresponds to source term perturbations at roughly $\pm2\sigma$ on the peripheral assemblies. Under a linear propagation assumption, this implies a 1σ contribution of the source spatial distribution to the RPV fast neutron fluence of $\sim$ 1.5%, consistent with

the relative uncertainty observed on the peripheral assembly production rates (Figure IV.15). The three cases confirm that this nuclear-data-driven variation of the source spatial distribution sits well below the transport-side cross section uncertainty quantified in Sec. 6, so the full 1000-sample propagation is not warranted and the $\sim 1.5\%$ estimate is kept as an approximation. The tallies are per source neutron and are scaled by the source rate S of Eq. (IV.1), computed at fixed thermal power. The variation of $\bar{\nu}$ and of the energy release per fission between samples therefore stays inside this normalization and is not propagated. The uncertainty of the moderator density and temperature is a separate contribution, still to be quantified.

Emitted spectrum

Each of the 512 spectrum samples of Sec. 5.3 is run through the full MCNP model. Only the source cards change from one sample to the next, since the spatial source distribution and the transport cross sections are common to the whole population, so the propagated spread measures the emitted spectrum alone. It gives 6.1% on the fast flux at the vessel hot spot and ranges from 2.5% to 11.6% across the responses of the campaign (Table IV.7). At the vessel, the contribution is largest at 0° and smallest at −20° for every response. At the capsule, it grows with the threshold of each dosimeter, from 2.8% on ^{237}Np(n,f) to 11.6% on ^{63}Cu(n,α), because the largest evaluated uncertainty sits in the high-energy tail of the emitted spectrum (Figure IV.16).

5.5. Summary

This section propagated the nuclear data uncertainty of the source term through two independent studies, one on the core calculation and one on the emitted spectrum. The main findings are:

- With all ENDF/B-VIII.0 data sampled in the core calculation, the relative uncertainty on the fission neutron production rate reaches up to 2.1% for central MOX assemblies and $\sim 1.5\%$ for the peripheral assemblies that dominate the RPV fluence.
- The other outputs of the core calculation respond much less, with less than 0.1% on the assembly spectrum rebuilt from the sampled isotope weights and less than 0.08% on the moderator temperature.
- A three-case MCNP propagation of the source spatial distribution only gives a spread from −2.8% to +3.3% on the RPV fast neutron flux at $\pm 2\sigma$ source perturbations, corresponding to a 1σ contribution of $\sim 1.5\%$.
- The JEFF-4.0 fission spectra, perturbed from their `MF35` covariance and propagated through 512 MCNP calculations, contribute 6.1% at the vessel hot spot and from 2.5% to 11.6% across the responses.

The source spatial distribution, at $\sim 1.5\%$, is a modest but not negligible contribution. It is consistent with the sensitivity of the source rate to the fission cross sections and $\bar{\nu}$ of the in-core actinides, although the simultaneous sampling of all in-core data does not allow a clean reaction-by-reaction decomposition. The emitted spectrum carries the larger term and enters the vessel-flux budget in Sec. 7. The transport-side contribution, propagated through Monte Carlo from the same MCNP model, is quantified in Sec. 6.

Table IV.7: Emitted-spectrum contribution to the vessel and capsule responses, propagated from the JEFF-4.0 `MF35` covariance through 512 MCNP calculations. Values are relative 1σ uncertainties in percent, with their 95% bootstrap confidence interval (CI). Vessel responses are height-integrated over the ±60 cm central window, capsule responses averaged over the three capsule positions.

Response	Location	PFNS unc. (%)	95% CI
Flux $\Phi_{E>1\,\mathrm{MeV}}$	Vessel, 0°	6.08	[5.72, 6.42]
	Vessel, −20°	4.90	[4.62, 5.17]
	Vessel, −28°	5.55	[5.23, 5.84]
	Capsule	3.20	[3.02, 3.38]
Flux $\Phi_{E>0.1\,\mathrm{MeV}}$	Vessel, 0°	4.97	[4.68, 5.25]
	Vessel, −20°	4.02	[3.79, 4.24]
	Vessel, −28°	4.96	[4.67, 5.22]
	Capsule	2.47	[2.32, 2.60]
NRT-dpa	Vessel, 0°	6.12	[5.76, 6.46]
	Vessel, −20°	4.85	[4.57, 5.12]
	Vessel, −28°	5.77	[5.44, 6.08]
ARC-dpa	Vessel, 0°	5.98	[5.63, 6.31]
	Vessel, −20°	4.74	[4.47, 5.00]
	Vessel, −28°	5.69	[5.37, 5.99]
^{237}Np(n,f)	Capsule	2.76	[2.60, 2.92]
^{238}U(n,f)	Capsule	3.89	[3.66, 4.10]
^{58}Ni(n,p)	Capsule	5.73	[5.41, 6.05]
^{54}Fe(n,p)	Capsule	6.17	[5.82, 6.51]
^{46}Ti(n,p)	Capsule	9.23	[8.70, 9.74]
^{63}Cu(n,α)	Capsule	11.58	[10.90, 12.23]

6. Nuclear Data Uncertainty in the Vessel Fluence and Capsule Responses

The transport step of the VACS chain carries the source term through the steel and water surrounding the core to the vessel inner surface. Three response functions are evaluated on the model of Sec. 4: the fast neutron flux above 1 MeV, the atomic-displacement damage, and a set of capsule reaction-rate dosimeters. All three depend on the nuclear data of the structural isotopes along the transport path. This section propagates those nuclear data uncertainties to the three responses for three modern evaluated libraries (ENDF/B-VIII.1, JEFF-4.0, JENDL-5), using the KIKA nuclear-data sampling framework (Sec. 4 of Chapter II).

6.1. Calculation Setup

A set of perturbed cross sections and Legendre coefficients is propagated for each of three modern evaluations: JEFF-4.0, ENDF/B-VIII.1, and JENDL-5. For every library and every channel, 512 samples are drawn with a Sobol low-discrepancy sequence and propagated through the PWR900 model with Fast TMC (Sec. 3.2.3 of Chapter II). The cross section and angular-distribution channels are propagated independently and combined in quadrature; the angular channel covers elastic scattering, the only reaction with evaluated `MF34` covariance in the libraries. Since neither JEFF-4.0 nor ENDF/B-VIII.1 provides an `MF34` file for ^{56}Fe, the JEFF-3.3 evaluation is used in its place, the most recent available (Chapter III, Sec. 1.2). JEFF-4.0 carries no ^{16}O `MF34` file, so its combined uncertainty omits the ^{16}O angular channel. All uncertainties are relative and given in percent, and a difference between two of them is the difference of the two percentages: a rise from 7.0% to 7.5% is a rise of 0.5%.

Table IV.8 lists the isotopes carrying covariance data in each evaluation and the Legendre orders taken from the `MF34` data. For JEFF-4.0, the uncertainties were propagated both with all isotopes perturbed together and with each isotope perturbed individually. The two agree, showing no interaction between isotopes, and the per-isotope runs additionally give a decomposition of the nuclear-data variance by isotope, which is adopted as the JEFF-4.0 reference throughout this chapter.

6.2. Fast Neutron Flux at the Vessel

The fast neutron flux above 1 MeV at the inner surface of the vessel is the historical fluence metric used by surveillance programs (Sec. 5.1 of Chapter I) and is the input variable of the empirical embrittlement correlations adopted in France. It is evaluated at the three vessel azimuths introduced in Sec. 4: the 0° hotspot, the −20° position behind the surveillance capsule, and the −28° position off the asymmetric neutron pad, where the path crosses proportionally less steel and more water, so the spectrum reaching the vessel is the softest of the three. Figure IV.17 shows the combined nuclear-data uncertainty obtained for each library. Spatial maps of the flux above 0.1 MeV and azimuth-by-height hotspot maps for all responses are given in Appendix F.9 and Appendix F.10.

The largest uncertainty is obtained at the −20° azimuth, where the path between the core and the vessel passes through the additional steel of the surveillance capsule, amplifying the dominant ^{56}Fe contribution to the variance. ENDF/B-VIII.1 and JEFF-4.0 agree to within $\sim 0.3\%$ at every azimuth, with height-integrated values of $\sim 5.9\%$ at 0°, 7.1–7.2% at −20°, and 5.5–5.8% at −28°. JENDL-5 departs from both in a way that follows the amount of steel along

Table IV.8: Isotopes carrying covariance data in the three evaluations propagated in this chapter. A check mark indicates an isotope is perturbed; `MF33` covers cross sections, `MF34` covers elastic-scattering Legendre coefficients.

	ENDF/B-VIII.1		JEFF-4.0		JENDL-5	
Isotope	MF33	MF34	MF33	MF34	MF33	MF34
^{1}H	✓	—	✓	—	✓	—
^{10}B	✓	—	✓	—	—	—
^{16}O	✓	$L = 1$	✓	—	✓	$L = 1$
^{28}Si	✓	—	✓	—	—	—
^{52}Cr	✓	—	✓	—	✓	—
^{55}Mn	✓	—	✓	—	✓	—
^{54}Fe	✓	—	✓	—	—	—
^{56}Fe	✓	$L = 1$–$6^{\dagger}$	✓	$L = 1$–$6^{\dagger}$	✓	$L = 1$
^{58}Ni	✓	—	✓	—	✓	—
^{238}U	—	—	✓	—	—	—

† Neither JEFF-4.0 nor ENDF/B-VIII.1 provides an `MF34` file for ^{56}Fe; the JEFF-3.3 evaluation is used in its place in both cases (Chapter III, Sec. 1.2).

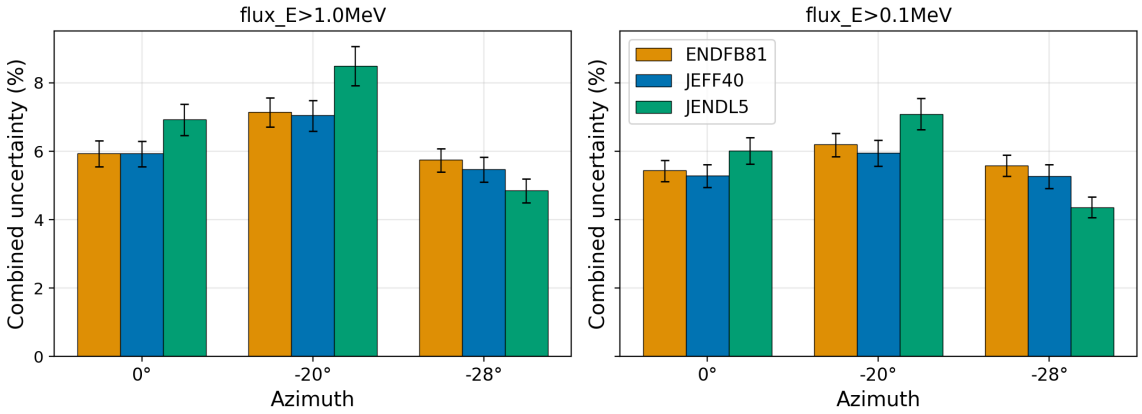


Figure IV.17: Combined cross section and angular-distribution uncertainty on the fast neutron flux at the vessel inner surface, height-integrated over the ±60 cm central window, at each of the three vessel azimuths and for the three evaluated libraries. The left panel shows the > 1 MeV threshold and the right panel the > 0.1 MeV threshold. Error bars are 95% bootstrap confidence intervals.

the path. At 0° and −20° it returns the highest uncertainty, 6.9% and 8.5% height-integrated and up to 8.9% in the middle height segment of the −20° azimuth, about 1% above the other two libraries. At −28°, where the path crosses proportionally less steel and more water, it returns the lowest, 4.9%. The JENDL-5 ^{56}Fe cross section covariance therefore weighs more than those of the other two evaluations where iron governs the attenuation. The highest-uncertainty tallies and the largest library spreads across the campaign are tabulated in Appendix F.11.

Table IV.9 sets the central flux values of the three libraries against the propagated uncertainty. ENDF/B-VIII.1 and JEFF-4.0 agree to within 0.2% at every azimuth. JENDL-5 is 3.3–3.4% lower at 0° and −20°, and at −28° the three libraries agree to within 0.1%. The JENDL-5 difference therefore follows the amount of steel along the path, as its uncertainty does, and vanishes at the most water-attenuated azimuth. The cross-library spread of the central values stays below the propagated 1σ uncertainty at every azimuth, so the libraries are consistent within nuclear-data uncertainty. The vessel displacement rates follow the same pattern (Appendix F.6).

Table IV.9: Central values of the height-integrated fast neutron flux ($E > 1$ MeV), in $\mathrm{n\,cm^{-2}\,s^{-1}}$, at the three vessel azimuths for the three libraries, with the cross-library spread (maximum minus minimum over the mean) and the range of the propagated nuclear-data uncertainty across libraries. Central values from the nominal run of each library.

Azimuth	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	Spread (%)	Propagated σ (%)
0°	3.24×10^{10}	3.24×10^{10}	3.13×10^{10}	3.5	5.9–6.9
−20°	1.66×10^{10}	1.66×10^{10}	1.61×10^{10}	3.3	7.1–8.5
−28°	2.29×10^{10}	2.29×10^{10}	2.29×10^{10}	0.1	4.9–5.7

6.3. Vessel Atomic-Displacement Damage

The atomic-displacement damage at the vessel inner surface is computed under both the NRT and the ARC damage models (Sec. 5.1 of Chapter I). Figure IV.18 shows that the two damage models carry essentially identical nuclear-data uncertainties, within 0.1% of each other at every azimuth and for every library. The choice of damage model therefore affects only the central displacement rate, not its uncertainty. The azimuthal and library pattern follows the one observed on the fast flux: the highest uncertainty is at −20° for every library, reaching $\sim$ 7.3% height-integrated for JENDL-5 and 6.5–6.7% for ENDF/B-VIII.1 and JEFF-4.0, and the largest cross-library spread is at −28°, where JENDL-5 returns the lowest value (4.4%).

6.4. Capsule Dosimetry

The activation cross sections of the six dosimeters are taken from IRDFF-II [178] and held fixed across the sample population, so the propagated spread on the capsule reaction rates reflects the uncertainty on the spectrum delivered to the capsule, weighted by each dosimeter's response function (Figure IV.11. Figure IV.19 reports the combined uncertainty on the capsule-averaged reaction rates. For ENDF/B-VIII.1 and JEFF-4.0 the threshold dosimeters ^{63}Cu(n,α)^{60}Co (9.6–9.9%) and ^{46}Ti(n,p)^{46}Sc (9.0–9.2%) carry the largest values because they respond only to the highest-energy neutrons reaching the capsule, and at the opposite end of the threshold range ^{237}Np(n,f) is the most precise dosimeter at 3.7–3.9%. JENDL-5 reverses this ordering. Its uncertainty is largest on the fission dosimeters, 9.1% on ^{238}U(n,f) and 6.0% on ^{237}Np(n,f), and smallest on the highest-threshold ones, 7.0% on ^{46}Ti(n,p) and 8.1% on ^{63}Cu(n,α). The JENDL-5

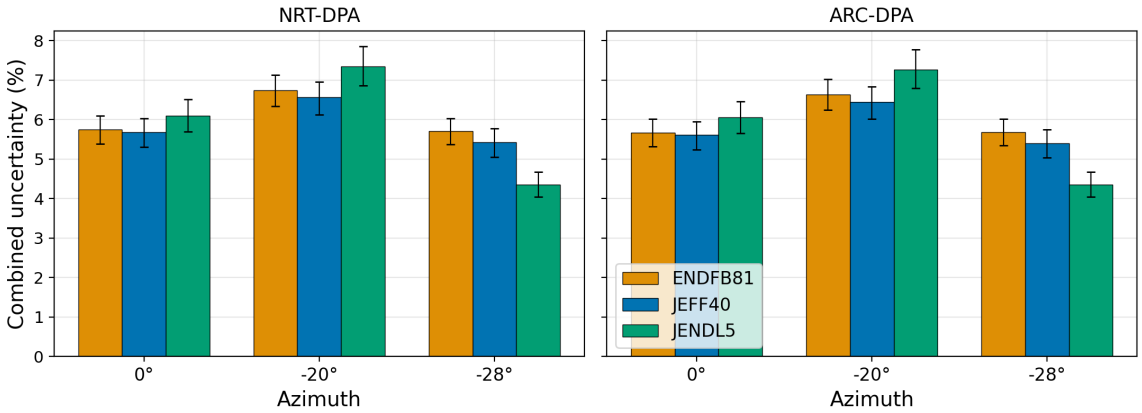


Figure IV.18: Combined nuclear-data uncertainty on the NRT-dpa and ARC-dpa at the vessel inner surface, height-integrated, for the three azimuths and the three libraries. Error bars are 95% bootstrap confidence intervals.

covariance thus places more uncertainty than the other two evaluations on the 1–3 MeV range that drives the fission dosimeters, the same range that drives the vessel fast flux, and less on the highest-energy tail of the spectrum.

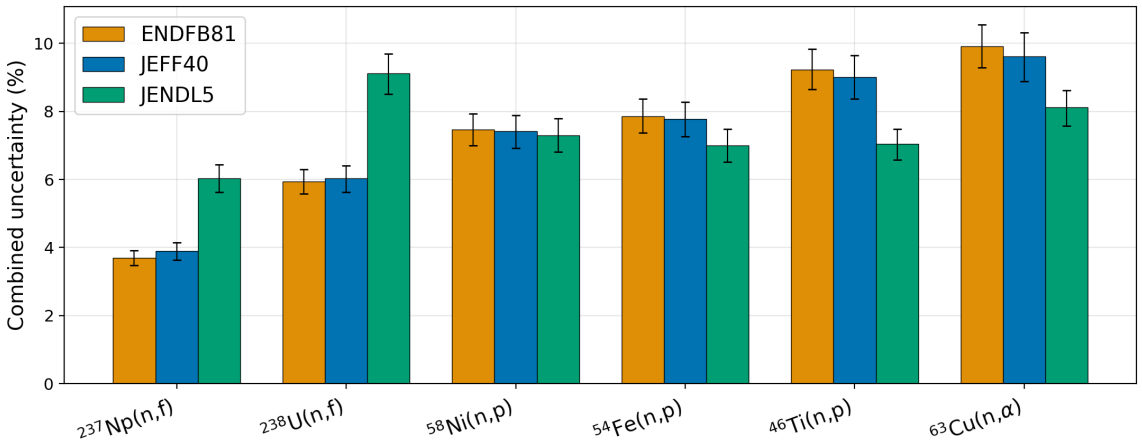


Figure IV.19: Nuclear-data uncertainty on the six capsule dosimetry reaction rates, with the cross section and angular-distribution channels combined in quadrature, averaged over the three capsule positions, for the three libraries. Error bars are 95% bootstrap confidence intervals.

6.5. Cross Section and Angular-Distribution Contributions

Throughout this chapter, the combined uncertainty is the quadrature sum of the three nuclear-data perturbation channels: the cross sections (Table IV.8), the ^{56}Fe angular distributions, and the ^{16}O angular distributions. Their relative weight, evaluated tally-by-tally across the simulations, supports the following findings:

- **Cross sections dominate the vessel budget.** For the ^{56}Fe angular evaluation shared by JEFF-4.0 and ENDF/B-VIII.1, the ^{56}Fe and ^{16}O angular channels together raise the combined uncertainty by at most 0.1% at any vessel tally, and by $\sim 0.06\%$ at the peak. JENDL-5 is the exception: its native ^{56}Fe `MF34` ($L = 1$ only) carries a 1.1%–2.7% angular uncertainty, which raises its combined vessel uncertainty by up to $\sim 0.5\%$.

- **Per-channel contributions at the vessel.** For the same shared evaluation, the ^{56}Fe angular channel adds less than 0.2% everywhere in the vessel structure and only 0.04% at the peak tally; the ^{16}O angular channel adds at most 0.06% and never exceeds 0.9% on its own.
- **Capsule exception.** The most pronounced angular contribution is the JENDL-5 propagation of the ^{238}U(n,f) capsule reaction rate, where the ^{56}Fe angular component reaches $\sim 3.9\%$ and raises the combined uncertainty by $\sim 0.9\%$ above the cross section value.
- **Library coverage caveat.** ^{16}O `MF34` is only carried by ENDF/B-VIII.1 and JENDL-5; JEFF-4.0 has no ^{16}O angular covariance, so its combined uncertainty does not include this channel. Leaving it out lowers the combined uncertainty by at most 0.06% anywhere on the vessel.

The small angular contribution found here holds for the covariance the evaluations report, which is sparse for ^{56}Fe elastic scattering; JENDL-5 is the only library whose native ^{56}Fe `MF34` contributes appreciably at the vessel.

6.6. Comparison of ^{56}Fe Angular Evaluations

The angular weights of the preceding subsection rest on the published `MF34` files, which are uneven for ^{56}Fe elastic scattering. The JEFF-3.3 covariance adopted here for JEFF-4.0 and ENDF/B-VIII.1 implies an absolute uncertainty of comparable size on every Legendre order it covers, so the relative values it carries run from a few percent on a_1 to order unity on a_6; and because it is stored on an energy mesh of a few hundred keV, across which the high-order coefficients change sign, the absolute uncertainty it implies swings by orders of magnitude inside a single group. JENDL-5 supplies covariance for the first Legendre coefficient only (Chapter III, Sec. 1.2). The ^{56}Fe `MF34` evaluation developed in Chapter III distributes its uncertainty more evenly across Legendre orders. Propagating each evaluation's ^{56}Fe `MF34` covariance as a standalone angular channel through the PWR900 model isolates its effect on the vessel and capsule budgets. For JEFF-4.0 and the Chapter III evaluation this is a direct swap on the same transport base, so the two differ only in the ^{56}Fe angular covariance. Table IV.10 reports the result.

On the vessel fast flux the new Chapter III evaluation falls between the published JEFF-4.0 and JENDL-5 values at every azimuth. The same ordering holds on the low-threshold ^{237}Np(n,f) and ^{238}U(n,f) dosimeters, where JENDL-5's single-coefficient covariance produces by far the widest angular spread, reaching $\sim 3.9\%$, and the Chapter III evaluation stays near 1%. The three evaluations converge on the two highest-threshold dosimeters, ^{46}Ti(n,p) and ^{63}Cu(n,α), whose response is set by the highest-energy neutrons where the three covariances assign comparable uncertainty.

Added in quadrature to the JEFF-4.0 cross section budget, the Chapter III angular channel raises the combined uncertainty by at most 0.08% on the vessel and 0.16% at the ^{63}Cu(n,α) capsule rate. On the vessel this increment is almost indistinguishable from the one obtained with the official JEFF-3.3 covariance (Table IV.10). The angular channel therefore stays a minor contributor whether the official or the independent EXFOR-based evaluation is used, and the conclusion of Sec. 6.5 does not depend on which ^{56}Fe angular covariance is adopted.

Table IV.10: Propagated ^{56}Fe angular-distribution (`MF34`) relative uncertainty at the three vessel azimuths (height-integrated > 1 MeV flux) and the six capsule reaction rates, for the published JEFF-4.0 and JENDL-5 evaluations and the EXFOR-based evaluation of Chapter III. Each value is the isolated ^{56}Fe angular channel (cross section and ^{16}O channels excluded), reported as a relative 1σ uncertainty in percent with its 95% bootstrap confidence interval in brackets.

Response	JEFF-4.0	JENDL-5	Chapter III
Vessel flux, 0°	0.64 [0.59, 0.68]	2.10 [1.97, 2.21]	0.81 [0.75, 0.87]
Vessel flux, −20°	0.76 [0.70, 0.82]	2.74 [2.58, 2.90]	1.05 [0.98, 1.12]
Vessel flux, −28°	0.52 [0.44, 0.58]	1.05 [0.97, 1.13]	0.61 [0.54, 0.68]
^{237}Np(n,f)	0.46 [0.42, 0.50]	3.03 [2.85, 3.21]	0.89 [0.83, 0.95]
^{238}U(n,f)	0.58 [0.53, 0.63]	3.85 [3.62, 4.08]	1.25 [1.16, 1.33]
^{58}Ni(n,p)	1.16 [1.07, 1.24]	2.77 [2.60, 2.92]	1.26 [1.17, 1.36]
^{54}Fe(n,p)	1.31 [1.22, 1.40]	2.54 [2.39, 2.68]	1.37 [1.27, 1.48]
^{46}Ti(n,p)	1.76 [1.62, 1.89]	1.60 [1.48, 1.71]	1.86 [1.69, 2.01]
^{63}Cu(n,α)	1.64 [1.45, 1.81]	1.23 [1.04, 1.40]	1.78 [1.55, 1.97]

6.7. Per-Isotope Decomposition of the Cross Section Channel

To identify which isotope drives the budget, the per-isotope runs are reorganized as a variance decomposition. Figure IV.20 reports the resulting decomposition at five representative locations.

The decomposition varies with location:

- **Vessel, 0° and −20°:** ^{56}Fe carries $\sim 65\%$ of the cross section variance, followed by ^{16}O ($\sim 18\%$) and ^{52}Cr ($\sim 9\%$). The path between core and vessel stays within steel (and, at −20°, within the surveillance capsule), so the iron elastic-scattering chain dominates the response.
- **Vessel, −28°:** The path misses the neutron pad and crosses proportionally less steel and more water. The ^{56}Fe share falls to 47% while ^{16}O rises sharply to 35% and ^{1}H joins ^{52}Cr at $\sim 6\%$ each, as the lower steel-to-water ratio shifts the response toward the oxygen and hydrogen of the moderator. The total cross section uncertainty changes little, however (5.5% versus 5.9% at 0°), so the variance is redistributed differently across isotopes rather than removed.
- **Capsule activation rates:** ^{56}Fe again leads, carrying 55% to 86% of the variance, with ^{52}Cr at 7%–13% as the second contributor.

Every other contributor is a structural material in the path between core and vessel. ^{238}U is the exception: it sits only in the core, yet its cross section uncertainty still affects these tallies, because the neutrons that reach them are produced in the core and their energies are shaped by the core materials. It contributes 3%–6% at the vessel and up to $\sim 11\%$ at the ^{237}Np(n,f) capsule rate. Summed in quadrature, the per-isotope variances reproduce the combined cross section budget, so the decomposition is complete and can be re-summed to match the isotope set of any comparison library.

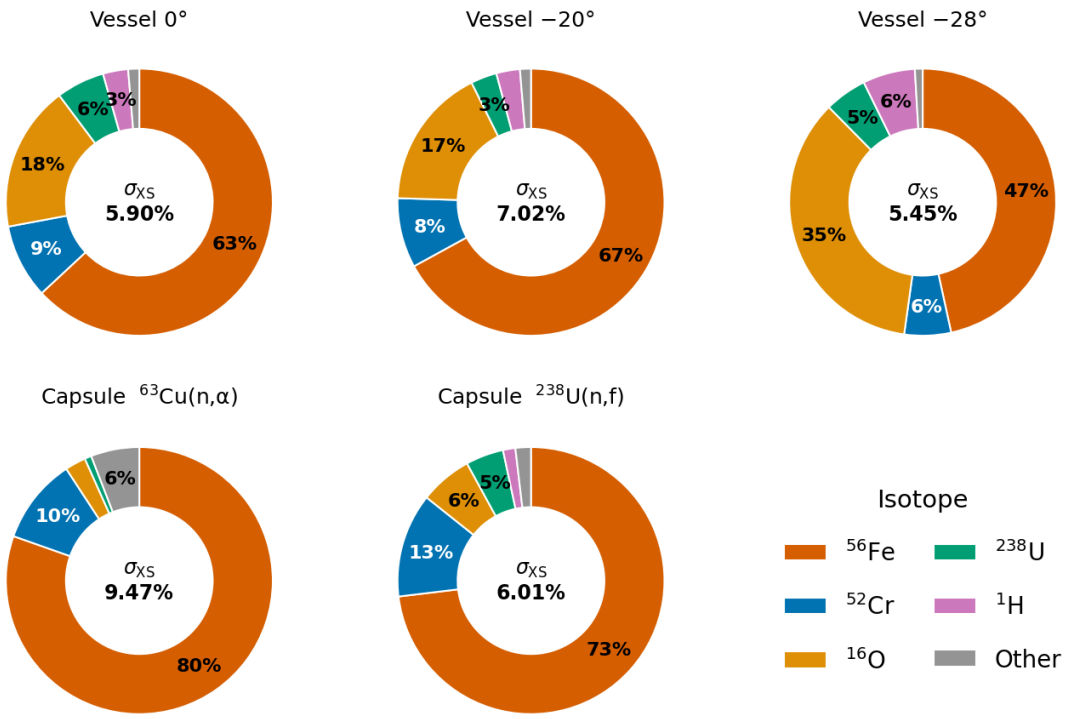


Figure IV.20: Per-isotope decomposition of the JEFF-4.0 cross section uncertainty at five representative locations: the three vessel azimuths above 1 MeV and two capsule reaction rates (^{63}Cu(n,α), ^{238}U(n,f)). Each wedge has area $\sigma^2_{\mathrm{iso}}/\sigma^2_{\mathrm{XS}}$, so the wedges add to 100%; the combined σ_{XS} for each location is printed at the center of the donut. Sub-1% isotopes (^{58}Ni, ^{54}Fe, ^{55}Mn, ^{28}Si, ^{10}B) and the bootstrap residual are pooled as "Other".

6.8. Cross-Check: MCS Against FOSM for Cross-Section Propagation

The two propagation methods used in this work carry very different costs. The sandwich rule (FOSM, Sec. 3.1 of Chapter II) returns the response variance from a single sensitivity calculation, while Monte Carlo sampling (MCS, Sec. 3.2) requires a full population of perturbed transport runs. FOSM reproduces the MCS variance only when the response is linear in the perturbed parameters over their uncertainty range, a condition established for these reactions and uncertainty magnitudes in Chapter II (Sec. 5.2.2). This subsection tests that equivalence directly on the full PWR900 model, at the height-integrated 0° vessel flux for ^{56}Fe / JEFF-4.0. The sandwich pipeline verified on the PWR sphere model (Sec. 5.4 of Chapter II) is applied here to the complete geometry for this single tally. Extending it further is impractical here because the `PERT` sensitivities cannot use the FW-CADIS weight windows: the weight-window/`PERT` bias identified in Chapter II (Sec. 5.2.1) did not respond to the `nmfp` weight-window workaround on this model, leaving only costly analog sensitivity runs.

Both estimators draw from the same 56-group GENDF covariance at 600 K, and both are restricted to the same four reactions: elastic scattering (`MT` 2), the first two inelastic levels (`MT` 51 and `MT` 52), and radiative capture (`MT` 102). FOSM combines these MCNP `PERT`-card sensitivities with the covariance through the sandwich rule (Eq. (II.84)). The matched MCS perturbs the same reactions and propagates them with the scheme used for every other propagation in this chapter: 512 samples drawn with a Sobol low-discrepancy sequence and run through Fast TMC.

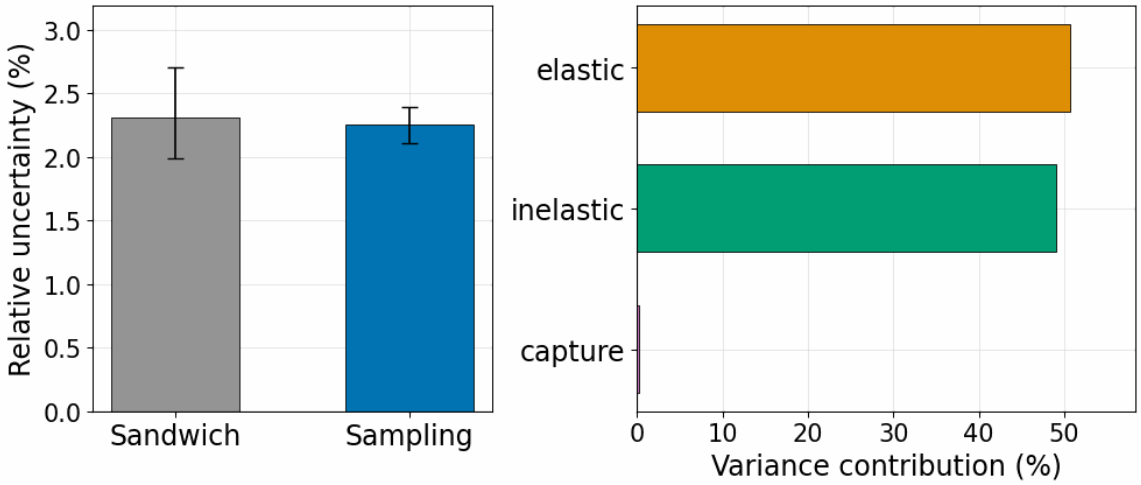
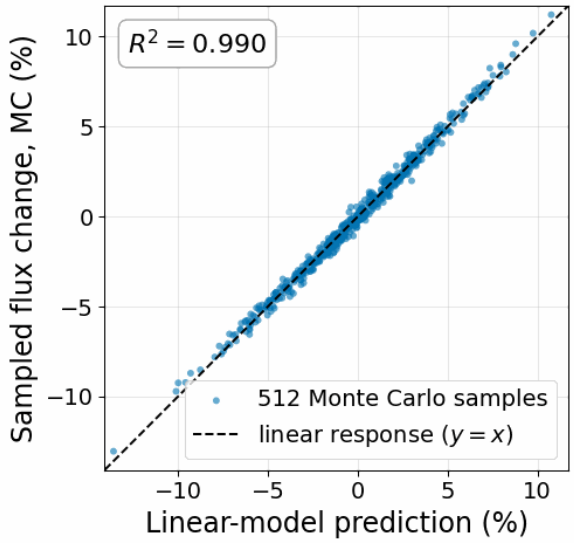


Figure IV.21: Matched-scope cross-check and sandwich variance decomposition for the height-integrated 0° vessel fast flux from ^{56}Fe / JEFF-4.0. *Left*: FOSM versus MCS over the same reactions, with 95% confidence intervals. *Right*: the matched-scope sandwich variance split by reaction channel.

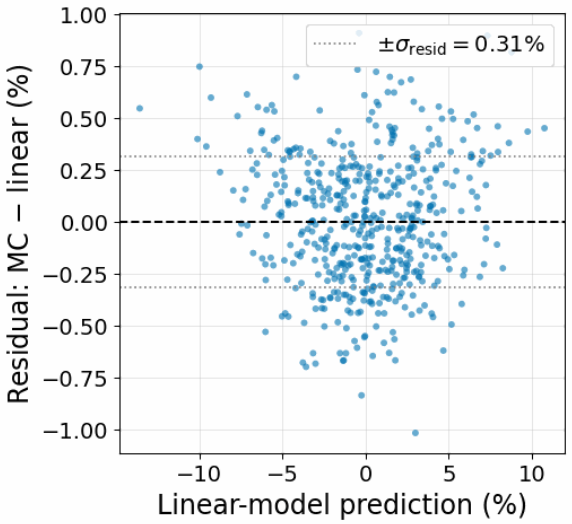
The two estimators agree (Table IV.11 and Figure IV.21). Their central values differ by 0.06% and their 95% confidence intervals overlap. Each interval is the 2.5th–97.5th percentile of a non-parametric bootstrap, resampling the `PERT` sensitivity groups for FOSM and the 512-sample population for MCS, following the procedure of Chapter II (Sec. 5.2.4). This agreement holds because the vessel flux responds linearly to the iron cross sections over their uncertainty range, which is the condition under which the sandwich rule reproduces the sampled variance.

That linearity is verified directly on the 512-sample run. Within each sample the ^{56}Fe cross sections are perturbed simultaneously across all 56 energy groups and the three reactions (elastic, inelastic, and capture), so a sample is described by a high-dimensional perturbation vector and

not by a single magnitude. To test whether the flux follows these perturbations linearly, a linear model is fitted that maps each sample's full perturbation vector to its flux response, collapsing the many perturbations into one predicted flux change per sample. Figure IV.22a plots this predicted change on the horizontal axis against the change the transport calculation actually produced on the vertical axis, both as relative departures from the sample mean. A linear response places the points on the diagonal, where they fall: the fitted model reproduces 99% of the sampled variance (coefficient of determination $R^2 \approx 0.99$, confirmed by a 5-fold cross-validation that guards against overfitting). The residuals of the same fit (Figure IV.22b) scatter around zero with no trend, and their spread of $\sim 0.3\%$ is set by the Poisson statistics of the MCNP tallies, not by response curvature. The sampled flux distribution is itself Gaussian (Shapiro–Wilk $p \approx 0.95$, skewness ≈ 0.01, excess kurtosis ≈ 0.09), as expected when Gaussian-like input perturbations propagate linearly, and the largest second-order term in a Taylor expansion of the response stays below 0.5% of the first-order term. First-order propagation is therefore adequate for the iron cross section uncertainty at the vessel flux.



(a) Sampled flux change against the linear-fit prediction; points on the diagonal ($y = x$) indicate a linear response.



(b) Residuals of the same fit, scattered around zero with no trend.

Figure IV.22: Linearity check for the 512-sample ^{56}Fe / JEFF-4.0 run at the height-integrated 0° vessel flux. Each point is one Monte Carlo sample.

A second MCS extends the perturbation from the two resolved inelastic levels to all inelastic levels (the discrete levels `MT` 51–90 plus the continuum `MT` 91), which the lumped `MT` 4 covariance also describes. Extending the scope this way raises the propagated uncertainty from 2.25% to 3.64% (Table IV.11); the levels above `MT` 52 add about 2.9% in quadrature. Reaching all of these levels with FOSM would require a separate `PERT` card for every discrete inelastic level, because the lumped `MT` 4 `PERT` does not work in MCNP and must be replaced by the sum over individual levels (Chapter II, Sec. 5.2.1). Combined with the weight-window/`PERT` incompatibility noted above, this makes full-scope sensitivity runs cumbersome and costly. The right panel of Figure IV.21 splits the matched-scope variance across the three reaction channels FOSM resolves: elastic scattering and the two inelastic levels carry it in nearly equal share, and capture is negligible.

Table IV.11: Propagated relative uncertainty σ_R at the height-integrated 0° vessel flux for ^{56}Fe / JEFF-4.0: FOSM against MCS over the matched reaction scope (first two rows), and MCS extended to all inelastic levels (third row).

Method	Reactions	Basis	σ_R (%)	95% CI
FOSM	`MT` 2, 51, 52, 102	`PERT` sensitivities	2.31	[1.99, 2.70]
MCS	`MT` 2, 51, 52, 102	512 Sobol samples	2.25	[2.10, 2.39]
MCS	`MT` 2, 51–91, 102	512 Sobol samples	3.64	[3.40, 3.84]

7. Comparison with Published Studies

The nuclear-data uncertainty budget derived in Sec. 6 is now placed against the published literature on the same problem. Four works are relevant: Berge evaluates the fission-spectrum contribution; Clouvel propagates the transport cross sections through a deterministic scheme; Vuiart assembles a full VACS budget on the same reactor used here, reusing Clouvel's transport result; and Timpano reports an independent full budget on a similar Westinghouse reactor.

- **Berge (CEA) [183].** Re-evaluation of the prompt fission neutron spectrum (PFNS) and, above all, of its covariance, which the libraries available at the time provided incompletely, and JEFF-3 not at all. The Madland–Nix model parameters are fitted with CONRAD, and their covariance is propagated to the > 1 MeV vessel flux of a French PWR through an importance-function method. The result is strongly evaluation- and correlation-dependent, ranging from $\sim 1\%$ (ENDF/B-VII.1) to 7% (JENDL-4.0), with 6.5% for the fitted Madland–Nix spectrum.

- **Clouvel (CEA) [1].** Stochastic sampling of the JEFF-3.2 cross section covariances of 25 isotopes, propagated through the deterministic APOLLO3/MINARET transport solver on an eighth-of-core PWR900 model. Surveillance capsules at 17° and 20° and a vessel-hotspot tally cover essentially the same positions used here. Clouvel reports a cross section contribution of 8.3% at 1σ at the hotspot, with ^{1}H, ^{56}Fe, and ^{16}O scattering as the dominant contributors; the full budget, which adds the source-term and geometry contributions, is published only in normalized form. The angular channel is evaluated only for ^{56}Fe, from its JEFF-3.2 `MF34` elastic covariance, and found to carry 3.6% of the flux variance against 96.4% for the cross sections, so it is excluded from the final budget.

- **Vuiart (ASNR, former IRSN) [4].** Two-step VACS scheme on the same French 900 MWe model used here, coupling a deterministic CASMO5/SIMULATE5 source term with MCNP6 neutron attenuation. The combined fast-fluence uncertainty is 8.8% at 1σ, obtained as a quadratic sum of component uncertainties. The transport cross section contribution is reused directly from Clouvel, and the angular channel is not separately quantified. The source term combines a Berge [183] PFNS uncertainty with a fuel-composition estimate; the fuel-composition term is a first-approximation estimate that rests on an assumed correlation structure, and it is among the largest contributors to Vuiart's budget. Modeling, geometry, and manufacturing terms are listed but not quantified.

- **Timpano (EPFL) [3].** Full vessel-fluence budget for Turkey Point 3, a Westinghouse three-loop PWR of 693 MWe (smaller than the PWR900 considered here) with the surveillance capsule mounted on the neutron pad, computed with a Polaris/PARCS/Serpent

chain. The cross section uncertainties are propagated by covariance sampling of the ENDF/B-VII.1 covariances, and the ^{56}Fe and ^{16}O angular distributions by TALYS Total Monte Carlo, which samples the underlying nuclear-model parameters. Timpano reports a total of 9.9% at the RPV, with a ^{56}Fe+^{16}O angular contribution of 4.1% ($\sigma_{\mathrm{AD}-^{56}\mathrm{Fe}} = 3.58\%$, $\sigma_{\mathrm{AD}-^{16}\mathrm{O}} = 2.05\%$) and a transport cross section contribution of 7.3%.

No single study propagates the whole vessel-fluence budget with equal depth; each invests its effort in a different piece. The budget reported here is assembled from the work that characterizes each piece best, all referring to the same French 900 MWe model and source chain. The transport channel, the spatial source term, and the fission-spectrum shape are developed in this work; Berge provides the published point of comparison for the last of them. These pieces rely on different evaluated libraries, so the budget is not built from a single consistent one. They act on separate stages of the chain (the source rate, the source spectrum, and the transport path), are treated as independent, and combine in quadrature into a nuclear-data total. Integral experiments such as criticality benchmarks constrain the fission cross sections, $\bar{\nu}$, and the fission spectrum together, and adjusting the data to them correlates these quantities [184, 185]. The evaluated files carry no such correlation, so the source rate and source spectrum terms may not be independent, and their combined contribution can differ from the quadrature sum. This work does not model manufacturing and geometry; a manufacturing term borrowed from Timpano is added to form a total fluence uncertainty comparable to the published full budgets. Table IV.12 sets the result against Timpano, whose single scheme covers most of these pieces on a comparable reactor and provides an independent total. The budget is reported first as a nuclear-data total and then including the borrowed manufacturing term, at the vessel hot spot and at the surveillance capsule. The same nuclear-data budget is given for every response, at the three vessel azimuths and at the capsule, in Appendix F.12.

The transport cross section channel is resolved most fully here. It is propagated across three modern evaluated libraries rather than the single evaluation used in each reference, decomposed per isotope, and cross-checked against the sandwich rule at the 0° vessel tally (Sec. 6.8). Clouvel [1] provides an independent check from a fully deterministic scheme (APOLLO3/MINARET): she finds the same dominant scatterers (^{1}H, ^{56}Fe, ^{16}O) and the cross section channel dominant, consistent with the results here. Her absolute budget is published only in normalized form, so it corroborates the methodology rather than the numbers.

The secondary angular-distribution channel is where the studies diverge most. Propagated here from the evaluated `MF34` covariance, it carries 0.6% on its own for JEFF-4.0 (Table IV.10), which raises the combined uncertainty by less than 0.1% once added in quadrature to the cross section channel; Clouvel likewise finds the ^{56}Fe angular term negligible and excludes it. Timpano instead reports a large angular contribution, 4.1%, of which $\sigma = 3.58\%$ from ^{56}Fe alone. The difference is methodological. Timpano samples the TALYS nuclear-model parameters (Total Monte Carlo) on a separate TENDL base, not the ENDF/B-VII.1 library used for the rest of that budget, whereas this work and Clouvel propagate the evaluated `MF34` covariance, which is sparse and, for ^{56}Fe, limited to low Legendre orders. Timpano notes that covariance-sampling tools cannot perturb the angular distributions at all, flags the known limitations of TALYS for this quantity, and acknowledges the inconsistency between the two pipelines, presenting the value as a first estimate. The two results therefore reflect what each method can see: the evaluated covariance captures only what the libraries report, and the model-parameter sampling explores a wider range. The gap between 0.6% and 4.1% marks the current spread between the two approaches.

Table IV.12: Fast neutron flux uncertainty budget $\Phi_{E>1\,\mathrm{MeV}}$ for Timpano and the present work, separated into nuclear-data contributions and an additional manufacturing-and-geometry term. The per-contributor rows are quoted at the vessel hot spot (0° for the present work, the RPV tally for Timpano). Totals are given at the vessel and at the surveillance capsule, first for the nuclear-data contributions alone and then including the manufacturing term.

	Timpano [3]	This work
Configuration		
Reactor model	Turkey Point 3	French 900 MWe
Source calculation	Polaris/PARCS	CASMO5/SIMULATE5
Transport calculation	Serpent	MCNP6
Transport library	ENDF/B-VII.1	JEFF-4.0
Source library	ENDF/B-VII.1	ENDF/B-VIII.1
Source term	cycle-averaged	cycle-averaged
Nuclear-data contributions (%)		
Transport cross sections (`MF33`)	7.3	5.9
Secondary angular distributions (`MF34`)	4.1[†]	0.6[§]
Spatial source term	0.3	1.5[‡]
Fission spectrum (PFNS shape)	—	6.1
Nuclear-data total, vessel (1σ)	**8.4**	**8.6**
Nuclear-data total, capsule (1σ)	**7.1**	**6.4**
Additional contribution (%)		
Manufacturing and geometry	5.3	5.3[⋆]
Total fluence uncertainty, vessel (1σ)	**9.9**	**10.1**
Total fluence uncertainty, capsule (1σ)	**8.6**	**8.1**

[†] From a TALYS/TENDL angular-distribution ensemble (Total Monte Carlo), not the ENDF/B-VII.1 library listed above.

[§] With the JEFF-4.0 `MF34` covariance; the ^{56}Fe evaluation of Chapter III gives 0.8 at the same tally (Table IV.10).

[‡] Estimated from a three-case MCNP propagation at $\pm 2\sigma$ under a linear assumption, not the full 1000-sample set (Sec. 5).

[⋆] Borrowed from Timpano as an approximation, since this work does not model manufacturing tolerances. These are plant-specific and propagated to first order; the borrowed value is 5.3 at the vessel and 4.9 at the capsule.

The fission-spectrum contribution is propagated here, and Berge provides the independent comparison. Berge [183] fits the Madland–Nix model parameters with CONRAD and propagates their covariance to the > 1 MeV vessel flux of a French PWR of the same type through an importance-function method, where the propagation of Sec. 5.3 samples the evaluated `MF35` covariance and runs each sample through the full MCNP model. The value obtained here, 6.1%, sits inside the range Berge reports across evaluations, from $\sim 1\%$ for ENDF/B-VII.1 to $\sim 7\%$ for JENDL-4.0, and close to the 6.5% of his fitted Madland–Nix spectrum. The 4.6% carried into Vuiart's budget on the same reactor sits below both values.

Timpano provides the independent comparison. On a similar Westinghouse three-loop PWR (Turkey Point 3, 693 MWe), a single Polaris/PARCS/Serpent scheme covers the transport, angular, source, and manufacturing contributions in one consistent propagation, with less depth on each than the specialized works above. On the nuclear-data contributions alone, the pieces assembled here give 8.6% at the vessel and 6.4% at the capsule, against Timpano's 8.4% and 7.1%. The two budgets carry their weight differently: this work has the lower transport value together with the spatial source term and a fission-spectrum term that Timpano does not carry, while Timpano has higher transport and angular values. Adding the manufacturing term, borrowed from Timpano here and propagated directly there, brings both to a total fluence uncertainty of 10.1% and 9.9% at the vessel and 8.1% and 8.6% at the capsule. The nuclear-data totals at the vessel, 8.4% and 8.6%, are close, which indicates some consistency.

8. Conclusions

This chapter followed the sensitivity and uncertainty method of Chapter II and the ^{56}Fe evaluation of Chapter III from measured benchmarks to the reactor vessel. The benchmarks validated the method against experiment, and the PWR900 analysis then used it to assemble a nuclear-data uncertainty budget for the vessel fluence and the surveillance dosimeters. The main findings are gathered below.

Validation against the benchmarks

On ASPIS-88 the C/E comparison separated standing biases from depth-dependent trends. The ^{115}In and ^{32}S foils are underpredicted and ^{27}Al overpredicted already at the first position, before any significant iron attenuation, so these offsets point to the model, the source, or the dosimetry data and not to the iron evaluation. On top of them the C/E develops trends with depth that the propagated ^{56}Fe uncertainty covers for ^{115}In and ^{103}Rh. The sensitivity profiles and a single-isotope library swap showed that the differences between JEFF-4.0 and JENDL-5 are carried almost entirely by their ^{56}Fe cross sections in the few-hundred-keV-to-MeV range, with JENDL-5 the closer of the two at depth. VENUS-3 carried the comparison into a three-dimensional geometry of core, steel, and water, following successive library releases. The C/E ratios sit close to unity for the modern libraries, and the improvement from ENDF/B-VIII.0 to ENDF/B-VIII.1 grows with the steel along the path, which again points to ^{56}Fe. The propagated cross section uncertainties are small, between 2% and 4% and below the experimental band, once a processing error in the ^{16}O covariances was found and corrected. Together the two benchmarks show that the libraries reproduce the measurements and that the propagation gives a realistic uncertainty, which is what justifies applying it to the vessel.

The need for more integral experiments

A common thread of the two benchmarks is that the propagated nuclear-data uncertainty is often smaller than the experimental uncertainty of the benchmark itself. On VENUS-3 the transport uncertainty stays between 2% and 4%, below the experimental band, and on several ASPIS foils the measured spread is wide enough that more than one library reproduces the data within it. When the experimental uncertainty is larger than the effect being tested, the benchmark can neither confirm nor reject the evaluation, and it cannot support data assimilation, the adjustment of nuclear data to integral measurements. This points to a need for new integral experiments with smaller and better characterized uncertainties, aimed at the structural-material data that drive deep-penetration transport. PETALE, carried out in the CROCUS reactor [186], is a recent experiment of this kind. It measures neutron transmission through stacks of metallic plates in a well-characterized reactor environment and is designed to test structural data such as iron. Propagating the present evaluation through the PETALE configuration would be a direct way to extend the validation of this chapter, and is left as future work.

Nuclear-data uncertainty at the vessel

On the PWR900 vessel the transport cross sections dominate the transport budget, and ^{56}Fe carries the largest share of the variance, followed by ^{16}O and ^{52}Cr. The combined transport uncertainty on the height-integrated fast flux ranges from 4.9% to 8.5% across the three azimuths for the three libraries. It is largest at the $-20°$ position, where the surveillance capsule adds steel between the core and the vessel. ENDF/B-VIII.1 and JEFF-4.0 agree to within 0.3%, while JENDL-5, whose ^{56}Fe covariance weighs more on the iron-dominated paths, is the highest at $0°$ and $-20°$ and the lowest at $-28°$, where the path crosses more water and the ^{16}O and ^{1}H shares rise. The displacement-damage uncertainty follows the flux closely, and the NRT and ARC damage models carry the same uncertainty to within 0.1%, so the choice of damage model changes the central rate and not its uncertainty.

The angular channel and the Chapter-3 evaluation

The elastic angular distributions were followed through the ASPIS-88 benchmark and the PWR900 vessel. Under the evaluated `MF34` covariances the angular channel is a minor contributor at the vessel, raising the combined uncertainty by less than 0.1% for JEFF-4.0 and ENDF/B-VIII.1, and it matters only for JENDL-5, whose single-coefficient covariance grows mainly on the capsule fission rates. The data-driven evaluation of Chapter III was propagated as a standalone channel in both the ASPIS-88 benchmark and the vessel calculations. On the ASPIS foils it gives a tighter angular uncertainty band than the single-coefficient JENDL-5 covariance. Substituted into the JEFF-4.0 transport as central values, it matches or improves the deep-penetration C/E of the higher-threshold foils, most clearly for ^{32}S, although it leaves the overall C/E trends unchanged. On the vessel it adds an increment almost identical to the official covariance. The angular contribution therefore stays small under the libraries, and the Chapter III evaluation does not change that conclusion while giving a more balanced spread across the Legendre orders.

The source term and the full budget

On the source side, the fission neutron production rate carries a nuclear-data uncertainty of about

1.5% at the vessel, dominated by the actinide fission cross sections and $\bar{\nu}$. The fission-spectrum shape is the larger source-term piece. It is perturbed from the evaluated `MF35` covariance on the source cards of the transport input and gives about 6.1% at the vessel hot spot, ranging from 2.5% to 11.6% across the responses according to the threshold of each. The largest absolute uncertainties of the whole study are not at the vessel but at the high-threshold capsule dosimeters, where ^{63}Cu(n,α), ^{46}Ti(n,p), and ^{54}Fe(n,p) reach $\sim$ 9.9%, $\sim$ 9.2%, and $\sim$ 7.9% in the transport channel, again dominated by ^{56}Fe, and where the fission spectrum adds as much again on the two highest thresholds. Assembling the contributions gives a nuclear-data uncertainty on the fast fluence of about 8.6% at the vessel hot spot and 6.4% at the capsule at 1σ. Adding a manufacturing term borrowed from Timpano, which this work does not model, brings the total to about 10.1% at the vessel and 8.1% at the capsule, close to the 9.9% and 8.6% of Timpano's independent budget on a comparable reactor and consistent with the VACS total of Vuiart on the same model. No single published study propagates the whole chain, so the budget here is built from the piece that each work characterizes best, with the transport channel, the spatial source term, and the fission-spectrum shape developed in this work.

Conclusion

The fast neutron fluence at the vessel wall sets the limit on how long a reactor may operate, and it cannot be measured where it matters most, inside the steel. It is calculated, and every uncertainty in that calculation is carried forward as a safety margin. This thesis set out to quantify that uncertainty for the vessel of a French 900 MWe pressurized water reactor, the PWR900, with particular attention to the nuclear data that dominate it and to one channel within them that the evaluated libraries leave poorly constrained: the angular distribution of neutrons elastically scattered by iron.

The work began by building the tools to propagate nuclear data covariances through a Monte Carlo transport calculation and to attribute the resulting variance to its sources. These were verified on a simplified spherical model of the vessel materials, where the sensitivity profiles and the propagated uncertainties from first-order perturbation theory and from Monte Carlo sampling agreed. That verification also exposed the problem that shaped the rest of the thesis. In the megaelectronvolt range that drives the damage, the elastic angular distribution of ^{56}Fe contributed less than one percent of the variance when its covariance was taken from JEFF-4.0, and close to a fifth when it was taken from JENDL-5. The two libraries disagreed by more than an order of magnitude on a quantity that had never been weighed against data, and neither could be trusted as it stood.

Resolving that disagreement meant going back to the differential measurements over 847 keV–4 MeV, the range that drives the damage. The libraries describe almost the same angular distribution but treat its uncertainty in different ways. Several provide no covariance for ^{56}Fe elastic scattering at all; one places the whole uncertainty on a single Legendre order; those that report more spread it over different orders and on energy grids far coarser than the distribution itself. We therefore re-analyzed the elastic scattering data collected in EXFOR and built an independent evaluation of the ^{56}Fe angular distribution and its covariance over this range. Existing evaluations calibrate to a few selected campaigns; the present one weighs the full set of measurements available between 847 keV and 4 MeV. Its central values stay close to JEFF-4.0 and JENDL-5, within their stated uncertainties below 2.5 MeV and slightly more forward-peaked above it, so the cross section stays in place and what changes is mainly the covariance. The uncertainty is spread more smoothly across the fitted Legendre orders, which gives a physically continuous band on the differential cross section in place of the dipolar shape that a single-order covariance imposes. That covariance carries the spread between independent experiments and the correlation between neighboring energies, on a grid of 526 energy groups against the thirteen of JEFF-4.0 and the four of JENDL-5. Across the full set of experiments the evaluation also reaches a lower reduced χ^2 than either library. This is in part a difference of method. The libraries are calibrated on a single campaign and follow it closely, JENDL-5 so closely that its residuals there fall below the uncertainties the campaign reports, while the evaluation here agrees with the whole database more evenly and tracks no single campaign as tightly. A cross-check against a suite of criticality benchmarks confirmed that the changed cen-

tral values leave integral performance intact and slightly improve the fast metal systems most sensitive to iron reflection.

With an evaluation in hand, the question was whether it behaved correctly where iron governs the transport. The ASPIS-88 benchmark sends neutrons through nearly seventy centimeters of steel, a geometry in which ^{56}Fe is almost the only actor. The agreement with measurement there separates into two effects. Some activation foils show a constant offset already at the first measurement position, present in every library; these standing biases point to the benchmarks specifications and not to the iron evaluation. On top of them sits a trend with depth, and it is this trend that the iron data control. Swapping only the iron evaluation between JEFF-4.0 and JENDL-5 reproduced almost the entire difference between their predictions for the threshold foils, which places the inter-library spread in the few-hundred-keV-to-MeV window squarely on ^{56}Fe. The propagated cross section uncertainty grows with depth and with the foil threshold, reaching several tens of percent on the deepest, highest-threshold positions. The angular-distribution channel is smaller and depends on the foil. For the foils most sensitive to it, the new evaluation improves the deep-penetration agreement slightly compared with JEFF-4.0 when used as central values, and its uncertainty band is tighter than the wider one the single-order JENDL-5 covariance produces.

The reactor-like VENUS-3 benchmark told a complementary story. Its geometry places fission, steel, and water on the neutron path, closer to the application, and the modern libraries reproduce the measurements to within a few percent. The propagated cross section uncertainties there are small enough to sit below the experimental error bars, which leaves little room to sharpen the evaluated covariances by assimilating these measurements; this is why the data assimilation once planned for the vessel was not pursued. Two software faults surfaced along the way and were reported: a bias in MCNP sensitivity estimates when perturbation cards are combined with weight-window files, and an unphysical ^{16}O elastic covariance produced by NJOY processing in JEFF-4.0.

The final step propagated the transport covariances through the vessel aging scheme to the PWR900 and combined them with the source-term contributions into a single nuclear-data budget. The nuclear data raise the fast neutron fluence uncertainty to about 8.6% at the vessel and 6.4% at the surveillance capsule, carried in comparable parts by the iron cross sections and the fission spectrum; adding a manufacturing and geometry term brings the total to about 10% at the vessel and 8% at the capsule. These figures sit within the range of the independent estimates available for comparable reactors. They also answer the question the thesis was built around. Propagated to the vessel wall, the elastic angular distribution of iron is a minor contributor: it raises the fluence uncertainty by at most 0.1% in absolute terms with the shared JEFF-4.0 and ENDF/B-VIII.1 covariance and with the new evaluation, and by up to about 0.5% with the single-coefficient covariance of JENDL-5. Against a nuclear-data total near 8.6% that the iron cross sections and the fission spectrum carry between them, the angular channel stays small for every library. This conclusion is specific to the vessel wall: the ASPIS results show the angular contribution growing as neutrons penetrate deeper into steel, so observables that sit behind the vessel would be expected to carry more of it.

Principal contributions

- The first data-driven evaluation of the ^{56}Fe elastic angular distribution and its covariance over 847 keV–4 MeV, built from the EXFOR record after documented filtering and energy-

bin selection and delivered as `MF4` and `MF34`.

- A quantitative answer to a question left open in vessel dosimetry. The elastic angular distribution of iron is a minor contributor to the vessel fluence uncertainty for every library. Even JENDL-5, which carries the largest angular-distribution uncertainties, raises the fluence uncertainty by at most about 0.5%, well below the cross section and fission-spectrum channels that bring the nuclear-data total to about 8.6%.
- A nuclear data uncertainty budget for the PWR900 fast neutron fluence and surveillance dosimetry, decomposed by isotope and reaction, with a total consistent with the independent uncertainty budgets published for comparable reactors.
- The `kika-nd` framework and the KIKA application: validated nuclear data covariance propagation and analysis, delivered as a reproducible chain from covariance parsing through nuclear data perturbation to transport post-processing, with separate pipelines for cross sections and for angular distributions.
- Two reported software faults: MCNP perturbation cards combined with weight windows, and NJOY processing of the JEFF-4.0 ^{16}O covariance.

Limitations and robustness

The evaluation rests on a finite and uneven experimental record, and this bounds what can be claimed for it. Its central values are stable: they already agree with two independent modern libraries, so a future re-evaluation is unlikely to move the angular distribution itself by much. The angular-distribution covariance is the more fragile product. Even with each library's covariance included, the reduced χ^2 stays above unity for all three evaluations, so the uncertainties reported here, though larger and more evenly distributed than those of the libraries, may still understate the spread the data imply. Many energy bins rest on one or two campaigns. A handful of new independent measurements would therefore act mostly on the covariance, tightening it where the between-experiment spread is currently weak and shifting the bins that today rest on a single campaign. The pipeline was built to ingest such data without modification.

The evaluation covers 847 keV–4 MeV and treats the angular shape independently of the cross section magnitude. Within each experiment, the systematic uncertainty is carried as a component correlated across all angular points, approximated by a normalization term where EXFOR does not report its breakdown; only the point-to-point statistical scatter is treated as independent. The campaigns also differ in energy resolution, so each constrains a different average of the same underlying distribution, and combining them costs either the structure that the finest campaigns resolve or the averaging that the coarsest ones carry. A second-moment covariance cannot hold the full diversity of the fitted ensemble.

At the application level, one term enters from outside the present calculation: the manufacturing and geometry contribution, borrowed from a comparable study. The gap between the angular contribution found here, at the tenth-of-a-percent level from evaluated covariances, and the few-percent value reported elsewhere from sampling nuclear-model parameters with TALYS, reflects two ways of expressing the same uncertainty. The evaluation here ties the angular uncertainty to the measured spread of the differential data, while the model-parameter sampling explores a wider range that those data do not require.

Perspectives

Several directions follow naturally. Of the isotopes carrying cross section covariance in the vessel budget, only ^{56}Fe and ^{16}O carry angular-distribution covariance in any library, and hydrogen, chromium, and nickel carry none. The same data-driven approach can be applied to the other isotopes that shape the vessel spectrum, beginning with ^{16}O, which carries the next-largest share of the cross section variance after iron. A joint evaluation of the cross section magnitude and the angular shape, carrying their cross-correlations, would remove the independence assumption made here. A comparison with model-based covariances for ^{56}Fe [110] would separate the information carried by the measurements from that imposed by the nuclear model. An estimator that carries the energy resolution of each experiment inside the model, such as a spline fit of the energy dependence, would use the high-resolution and the coarse measurements together without degrading either. The budget can be carried to observables deeper than the vessel wall, where the angular contribution is expected to grow. A target accuracy requirement (TAR) exercise [187] would turn the variance decomposition into the accuracy each nuclear datum must reach for a chosen uncertainty on the vessel fluence. Richer differential measurements, with independent campaigns, forward and backward angular coverage, and complete reporting of uncertainties and of the parameters that fix the energy resolution, would let the covariance be sharpened where it is now weakest.

These results help move part of the vessel-fluence margin from caution toward measurement. The dominant sources of uncertainty in the vessel fluence, the iron cross sections and the fission spectrum, are now identified and quantified, and the angular channel that motivated much of the work is shown to be small where it is applied and well characterized where it is not. For a fleet whose continued operation depends on knowing how long its vessels can last, an uncertainty grounded in measurement is the firmer basis for that judgment.

Appendices

Appendix A

Global Sensitivity Analysis

This appendix provides a brief overview of global sensitivity analysis (GSA) methods and their applicability to nuclear data uncertainty studies. The main body of the thesis relies on local sensitivity analysis (Chapter II, Sec. 2.1), which is appropriate for the predominantly linear regime of the problems considered. In this framework, FOSM methods provide a computationally efficient decomposition of output variance based on first-order sensitivities. However, this approach relies on a linear approximation of the model response and therefore cannot capture nonlinear effects or interactions between uncertain inputs. Global sensitivity analysis offers a more general framework in which variance-based decompositions can be obtained without relying on linearization. Through Monte Carlo sampling, GSA methods can in principle account for nonlinear behavior and higher-order interaction effects, providing a generalized extension of the FOSM variance decomposition to nonlinear systems. During this work, GSA techniques were explored to assess the potential importance of such effects; the conclusions from that exploration are summarized at the end of this appendix.

A.1. Variance-Based Sensitivity Analysis

Variance-based methods decompose the total variance of a model output into contributions from individual input parameters and their interactions. The framework originates from the functional ANOVA decomposition introduced by Sobol [188] and developed into a practical sensitivity analysis tool by Saltelli et al. [189].

Consider a model $Y = f(\mathbf{X})$ where $\mathbf{X} = (X_1, X_2, \ldots, X_d)$ is a vector of d independent input parameters. Under mild regularity conditions, f can be decomposed as:

$$f(\mathbf{X}) = f_0 + \sum_{i=1}^{d} f_i(X_i) + \sum_{i<j} f_{ij}(X_i, X_j) + \cdots + f_{1,2,\ldots,d}(X_1, \ldots, X_d), \tag{A.1}$$

where $f_0 = \mathbb{E}[Y]$ is the overall mean, $f_i(X_i) = \mathbb{E}[Y \mid X_i] - f_0$ captures the main effect of parameter X_i, f_{ij} captures the pairwise interaction between X_i and X_j, and so on for higher-order terms. Taking the variance of both sides yields:

$$\mathrm{Var}(Y) = \sum_{i=1}^{d} V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,2,\ldots,d}, \tag{A.2}$$

where $V_i = \mathrm{Var}(f_i)$, $V_{ij} = \mathrm{Var}(f_{ij})$, and so on.

Two indices are of primary interest. The **first-order Sobol index** of parameter X_i:

$$S_i = \frac{V_i}{\mathrm{Var}(Y)} = \frac{\mathrm{Var}(\mathbb{E}[Y \mid X_i])}{\mathrm{Var}(Y)}, \tag{A.3}$$

measures the fraction of total output variance explained by X_i alone, without interactions. The **total-effect index**:

$$S_{T_i} = 1 - \frac{\mathrm{Var}(\mathbb{E}[Y \mid \mathbf{X}_{\sim i}])}{\mathrm{Var}(Y)} = \frac{\mathbb{E}[\mathrm{Var}(Y \mid \mathbf{X}_{\sim i})]}{\mathrm{Var}(Y)}, \tag{A.4}$$

where $\mathbf{X}_{\sim i}$ denotes all inputs except X_i, captures the total contribution of X_i including all interaction terms in which it participates [190].

These indices satisfy $0 \leq S_i \leq S_{T_i} \leq 1$ and $\sum_i S_i \leq 1$ (with equality only when the model is purely additive). A large gap between S_{T_i} and S_i signals that parameter X_i is involved in significant interactions with other inputs. Conversely, when $\sum_i S_i \approx 1$, interaction effects are negligible and the model behaves additively, which is the regime where local sensitivity analysis provides equivalent information.

Interpretation in the nuclear data context

In nuclear data applications, the inputs $\mathbf{X}$ are the multigroup cross sections (or other nuclear parameters) for all isotope–reaction pairs, and Y is a transport observable such as a reaction rate or k_{eff}. First-order indices S_i rank the importance of individual energy groups or isotope–reaction channels, while total-effect indices S_{T_i} reveal whether the effect of a given channel depends on the values of other channels. If $S_i \approx S_{T_i}$ for all inputs, the response is effectively linear in the nuclear data and the first-order sensitivity coefficients from Sec. 2.1 capture the full picture.

A.2. Estimation Methods

Computing Sobol indices requires evaluating conditional expectations, which in practice is done through Monte Carlo estimation. The most widely used approach is the Saltelli estimator [191, 192], which estimates all first-order and total-effect indices from a design of $N(2d + 2)$ model evaluations, where N is the base sample size and d is the number of input parameters. The estimator works by generating two independent sample matrices $\mathbf{A}$ and $\mathbf{B}$ of size $N \times d$, then constructing d hybrid matrices $\mathbf{A}_B^{(i)}$ in which column i of $\mathbf{A}$ is replaced by column i of $\mathbf{B}$. The first-order and total-effect indices are then estimated from the model outputs evaluated at these matrices [188].

For stable estimates, N must typically be at least 500–1000 [189]. In a nuclear data problem covering d energy groups across multiple isotope–reaction pairs, d can easily reach several hundred. With $d = 200$ and $N = 1000$, the Saltelli estimator requires $1000 \times (2 \times 200 + 2) = 402{,}000$ model evaluations. When each evaluation is a full Monte Carlo transport calculation, this cost is prohibitive for routine application. This computational barrier is the primary reason GSA has seen limited adoption in nuclear data uncertainty studies, despite its theoretical appeal.

Morris screening

The Morris method [193] offers a much cheaper alternative that both ranks input importance

and diagnoses where the response is nonlinear or driven by interactions, without the cost of a full variance decomposition. This second role is the more useful one here, since it indicates directly whether first-order propagation captures the response. Also known as the Elementary Effects (EE) method, it evaluates finite differences along one-at-a-time (OAT) perturbation trajectories. For each input X_i, the elementary effect is:

$$\mathrm{EE}_i = \frac{f(X_1, \ldots, X_i + \Delta, \ldots, X_d) - f(\mathbf{X})}{\Delta}, \tag{A.5}$$

where Δ is a prescribed step size. By averaging over r trajectories (typically $r = 10$–50), two summary statistics are computed: $\mu_i^* = \frac{1}{r}\sum_{k=1}^{r} |\mathrm{EE}_i^{(k)}|$, which measures the overall influence of X_i, and σ_i, the standard deviation of the elementary effects, which indicates either nonlinearity or interaction effects.

The total cost of a Morris analysis is $r(d+1)$ model evaluations, which for the same $d = 200$ case with $r = 20$ gives $20 \times 201 = 4{,}020$ runs. This is two orders of magnitude cheaper than the Saltelli estimator and makes Morris screening feasible as a pre-filter: inputs identified as unimportant can be fixed at their nominal values, reducing the effective dimensionality before a full variance-based analysis. Screening is only one application; the same trajectories also verify that a response is linear before first-order propagation is applied to it, among other uses of the method in nuclear data uncertainty analysis.

A.3. Practical Limitations for Nuclear Data Applications

Several features of nuclear data uncertainty problems make GSA particularly challenging in practice.

- **High dimensionality.** A multigroup covariance library defines uncertainties over hundreds of energy groups for each isotope–reaction pair. Even a modest problem involving a few key isotopes easily results in $d > 200$ correlated parameters. The $N(2d+2)$ scaling of variance-based estimators becomes impractical at this scale. This count reflects the multigroup parameterization. Expressed instead through a few nuclear-model parameters, such as the optical-model potential coefficients or the resonance parameters of the two or three dominant resonances, the input space could be small enough for variance-based GSA to be applied directly.
- **Correlated inputs.** The standard Sobol decomposition assumes independent inputs. When inputs are correlated, as nuclear data inherently are through the multigroup covariance matrix, the variance decomposition is no longer unique and the interpretation of individual indices becomes ambiguous. Extensions to correlated inputs exist [194, 195] but they add computational cost and interpretive complexity.
- **Expensive forward model.** Each model evaluation requires a full Monte Carlo transport simulation. Depending on the tally, convergence requirements, and geometry complexity, a single run may take minutes to hours. Combined with the large number of evaluations needed, this makes direct GSA impractical for most production calculations. Surrogate models can reduce this per-run cost, with the limitations discussed in Sec. A.4.
- **Statistical noise in the response.** The Monte Carlo transport output itself carries statistical uncertainty, which contaminates the Sobol index estimates. Separating parametric variance from simulation noise requires either very well-converged individual runs

or dedicated variance decomposition techniques such as the fast TMC approach discussed in Sec. 5.3.2.

These limitations explain why GSA remains uncommon in nuclear data uncertainty studies. The few published applications have typically focused on low-dimensional problems (a handful of integral parameters rather than full multigroup representations) [196] or have used surrogate models to reduce the per-evaluation cost [197]. For the reactor-vessel fluence problem in particular, uncertainty studies have relied on direct sampling of the cross section covariances with per-isotope importance ranking [1].

A.4. Relevance to This Work

For the problems considered in this thesis, local sensitivity analysis combined with the sandwich rule (Sec. 2.1 and Sec. 3) provides an efficient and adequate framework. The near-linear behavior of neutron transport observables with respect to nuclear data perturbations, confirmed by the second-order Taylor analysis in Sec. 5.2.2 and by the close agreement between FOSM and MCS results throughout Chapter II, implies that first-order Sobol indices would account for nearly all the total variance. In this regime, the local sensitivity coefficients already encode the same ranking information as S_i, and the sandwich rule reproduces the total uncertainty without the computational overhead of a full GSA.

GSA would become valuable where the linearity assumption breaks down. The clearest driver is the uncertainty itself: when an important parameter carries a large covariance, its perturbations span a wide range, and over that range the response can depart from a linear trend. Threshold reactions are a typical case, since near the onset energy the cross section rises steeply and the response depends sensitively on where the threshold falls within the neutron spectrum. Coupled multiphysics problems, where feedback loops introduce nonlinear dependencies, are another. In such cases, the first-order sensitivity coefficients may not capture the dominant variance contributions, and interactions between input parameters may become significant.

One way to make GSA affordable is to replace the transport code with a surrogate model, a fast approximation also called a metamodel or emulator. Methods such as polynomial chaos expansion (PCE) [192], which has been applied to nuclear data uncertainty propagation in reactor physics [198], Gaussian process regression (kriging) [199], and neural network emulators [200] learn how the output responds to the inputs from a limited set of training runs, after which the Sobol indices can be computed on the surrogate at almost no added cost. A surrogate is harder to fit as the number of inputs grows, so it is practical only once that number has been cut down: a cheap screening step removes the parameters that do not matter, and the surrogate is built over the few that remain. It then takes over the costly part of the calculation, putting the full analysis within reach. This helps only when screening removes most of the inputs. When many parameters matter at once, the surrogate has too few training runs to stay accurate and little is gained.

Appendix B

The KIKA Desktop Application

B.1. Overview

The sensitivity analysis and uncertainty quantification workflows described in Chapter II involve multiple file formats, processing steps, and analysis tools. Working with the `kika-nd` library through its Python API provides full control over every step, but it also presents a steep learning curve for researchers who did not develop the underlying code. Inspecting a covariance matrix, configuring a sampling run, or comparing perturbed cross sections against experimental data all require writing scripts, managing file paths, and interpreting textual output.

To lower this barrier we developed `KIKA` [109], a cross-platform desktop application that wraps the `kika-nd` library in an interactive graphical interface. The application is available for Windows, macOS, and Linux, and provides point-and-click access to the same operations that the library exposes programmatically. All computations run locally on the user's machine and no files are uploaded to external servers, so confidentiality of proprietary data is preserved at all times. An optional account system allows users to register and log in so that their files, settings, and workspace persist across sessions. A guest mode is also available for immediate offline use without registration.

Files are loaded into the workspace through a standard file dialog. Supported formats include ACE, ENDF-6, covariance (COVERX, COVFIL, and BOXER), SDF, MCNP input, MCTAL, and NJOY input files. All plots are interactive, with zoom, pan, and hover tooltips for data inspection. The sidebar is divided into three sections: *Data* (file management and nuclear data visualization), *Tools* (analysis utilities and SA&UQ workflows), and *Codes* (transport and processing code integration). The following sections describe each group from the user's perspective.

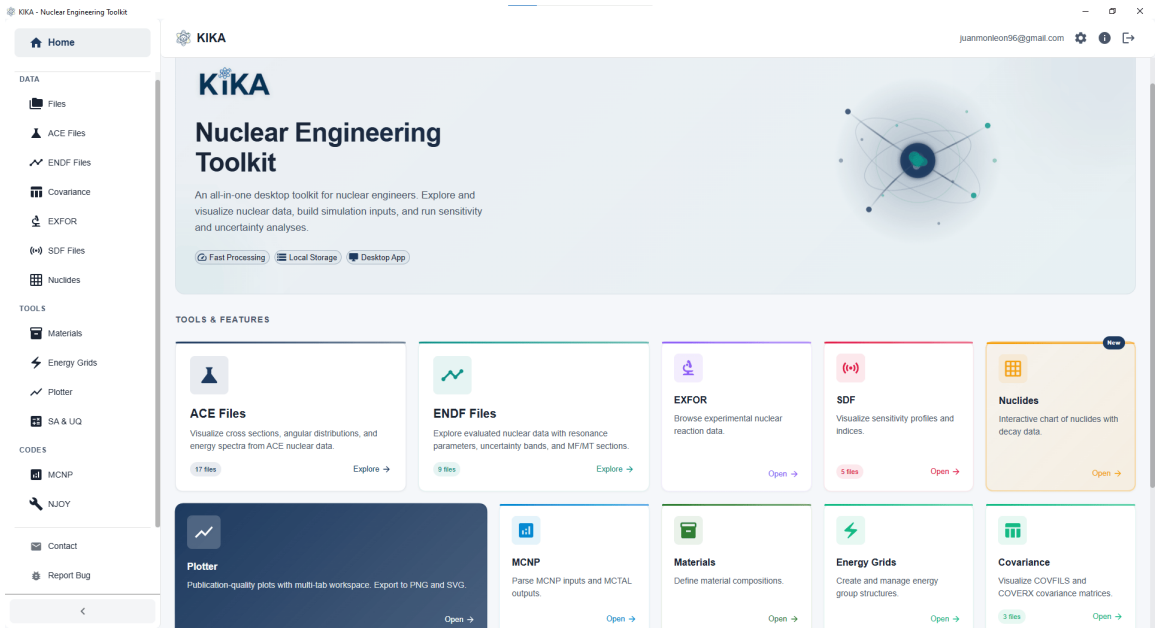


Figure B.1: The KIKA home dashboard. Each tile links to one of the three sidebar sections (Data, Tools, Codes) and displays the number of files currently loaded in the workspace.

B.2. Data Management and Visualization

KIKA provides dedicated viewers for each nuclear data format. In every case the workflow follows the same pattern: files are loaded into the workspace, their content is displayed through interactive plots and tables, and the results can be downloaded as figures or exported to CSV for further analysis.

ACE Files

The ACE viewer displays cross sections and angular distributions stored in ACE-formatted files. Reactions are indexed by `MT` number and can be selected for overlay on the same axes, making it straightforward to compare reaction channels across energy. Both plots and tabulated data can be exported.

ENDF Files

The ENDF viewer reads ENDF-6 formatted files and adapts its display to the sections present in each file. Dedicated panels cover general information and cross sections (`MF1`, `MF2`, `MF3`), angular distributions (`MF4`), and covariance data (`MF31`, `MF33`, `MF34`), each of which is described in more detail in Appendix C. Cross sections, angular distributions, and covariance matrices are rendered as interactive plots that can be explored, downloaded, and exported. A built-in download dialog allows users to fetch evaluated files directly from the IAEA online library without leaving the application.

Covariance Files

KIKA reads multigroup covariance data in the COVERX, COVFIL, and BOXER formats. The detail view displays diagonal standard deviations per reaction and energy group as line plots



Figure B.2: ACE file detail view showing interactive cross-section plots for ^{56}Fe. Reactions are selected by `MT` number from a side panel and overlaid on the same axes.

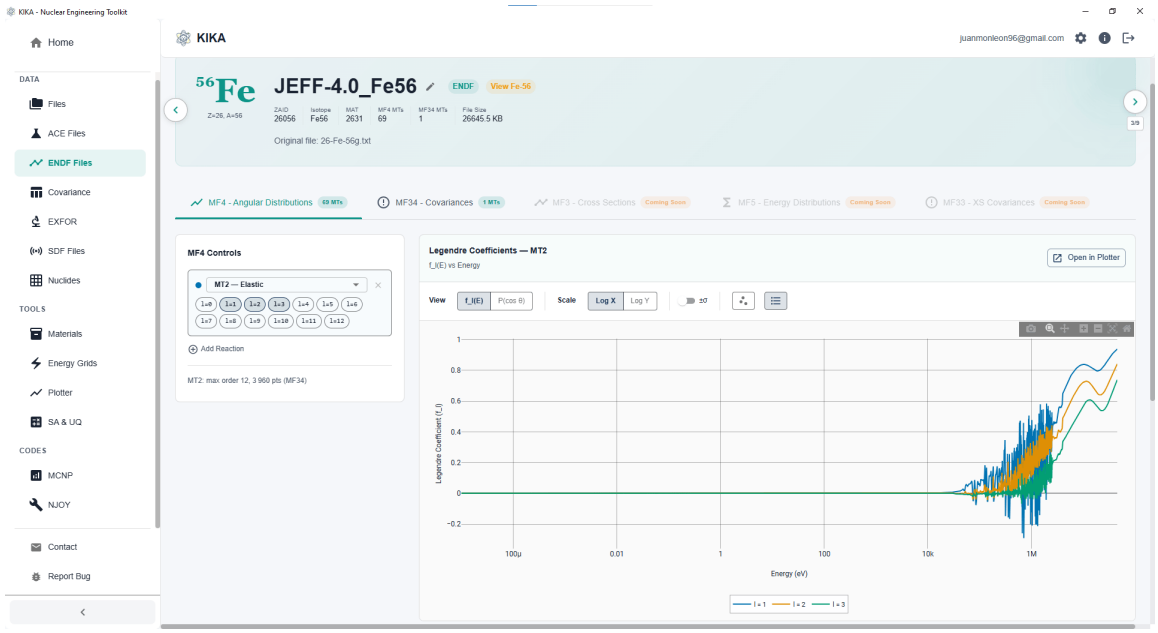


Figure B.3: ENDF file detail view showing the `MF34` covariance tab. The color map displays the correlation matrix for a selected Legendre order pair, with energy on both axes.

and renders the full correlation structure as two-dimensional color-map heatmaps. A dedicated tab converts a loaded covariance file from one of these formats to another, so the same data can be written out in whichever format a downstream code expects. Both the uncertainty and correlation representations can be downloaded as figures or exported as tabulated data.

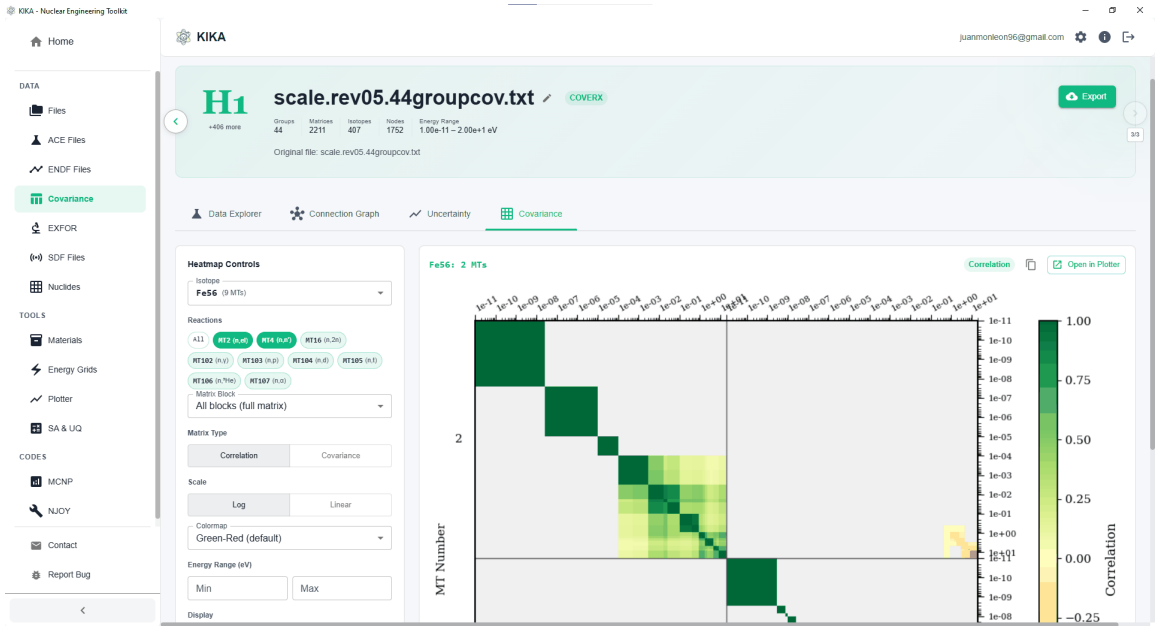


Figure B.4: Covariance file detail view showing a correlation heatmap for a selected reaction pair. The color scale represents the correlation coefficient between energy groups.

EXFOR Experimental Data

The EXFOR module provides access to the full EXFOR nuclear reaction database through a user-friendly search interface. To use this feature, the X4Pro database must first be downloaded from the IAEA website[1] and its path set in the application settings. Once configured, experiments can be queried by target element, reaction type, author, and other criteria. Search results include direct links to the corresponding EXFOR entries on the IAEA website. Experiments of interest can be bookmarked for later reference, and measured data points with their uncertainties can be plotted and exported.

Beyond standalone exploration, EXFOR experimental data can be overlaid on cross-section and angular-distribution plots produced from ACE or ENDF files. This overlay capability makes it possible to compare evaluated curves against measurements directly within the application, producing publication-quality comparison figures.

SDF Files

Sensitivity Data Files (SDF) are central to the FOSM uncertainty quantification workflow described in Chapter II. KIKA reads SDF files and displays the energy-dependent sensitivity profiles per nuclide and reaction. Multiple SDF files can be merged into a single object for combined analysis. SDF files can also be created directly within KIKA from MCNP fixed-source sensitivity calculations.

[1] https://www-nds.iaea.org/cdroms/#x4pro1trial

The UQ tab implements the sandwich formula interactively: the user selects a loaded covariance file, and the application computes the total relative uncertainty together with a per-reaction and per-isotope breakdown of variance contributions.

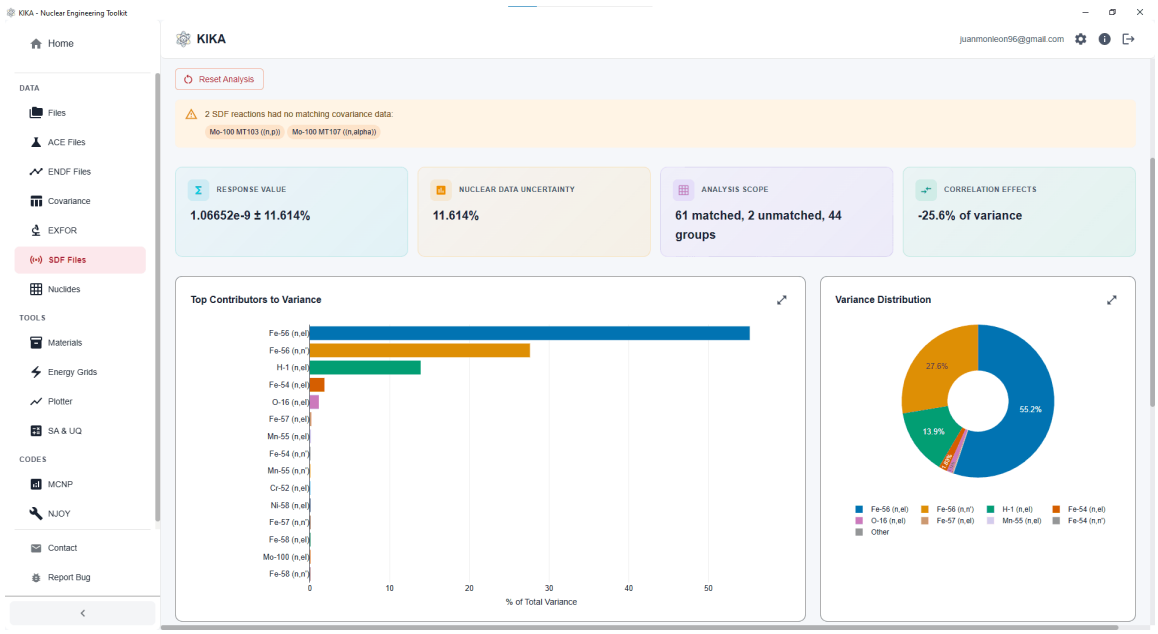


Figure B.5: SDF file detail view showing the sandwich-formula UQ tab. A covariance file is selected and the application computes the per-reaction variance contributions and total relative uncertainty.

Chart of Nuclides

An interactive chart of nuclides covers approximately 3 700 isotopes. The underlying nuclear data is obtained from the IAEA nuclear data services. Each nuclide is represented as a tile on a proton-number versus neutron-number grid, and a search bar locates specific nuclides and zooms the view to their position. Clicking a nuclide opens a detail page that includes an interactive decay chain diagram, radiation emission tables (alpha, beta, gamma, X-ray), and an energy-level scheme showing nuclear excited states and gamma transitions.



Figure B.6: The interactive chart of nuclides in KIKA. Each tile represents a nuclide colored by decay mode. The chart supports zoom, pan, and click-to-detail navigation.

B.3. Analysis Tools

Materials Manager

The materials manager allows users to create, edit, and store custom material compositions for reuse across analyses. Materials can be imported from existing MCNP input decks or defined manually by specifying isotopic compositions and densities. Stored materials can be exported as MCNP `M`-card syntax for direct use in transport input files. Additional utilities include composition normalization and conversion between weight fractions and atomic fractions.

Energy Grids

A collection of standard multigroup energy structures is available for visualization and export, including VITAMIN-B6 (199-group), SCALE-252, WIMS-69, and others commonly used in deterministic transport and sensitivity studies. Users can also define custom energy grids and store them for reuse. These grids serve as input to the multigroup covariance collapse (Sec. 4.4) and to SA&UQ workflows.

Plotter

The plotter is one of the most versatile tools in KIKA. Data from any combination of loaded files (ACE, ENDF, covariance, SDF, or EXFOR) can be overlaid on the same axes, and multiple independent tabs allow side-by-side comparisons of different datasets. Plots are highly customizable: axis labels, title, fonts, colors, line styles, markers, legend position, figure dimensions, and resolution can all be adjusted to produce publication-quality figures. Export targets include PNG, SVG, and PDF.

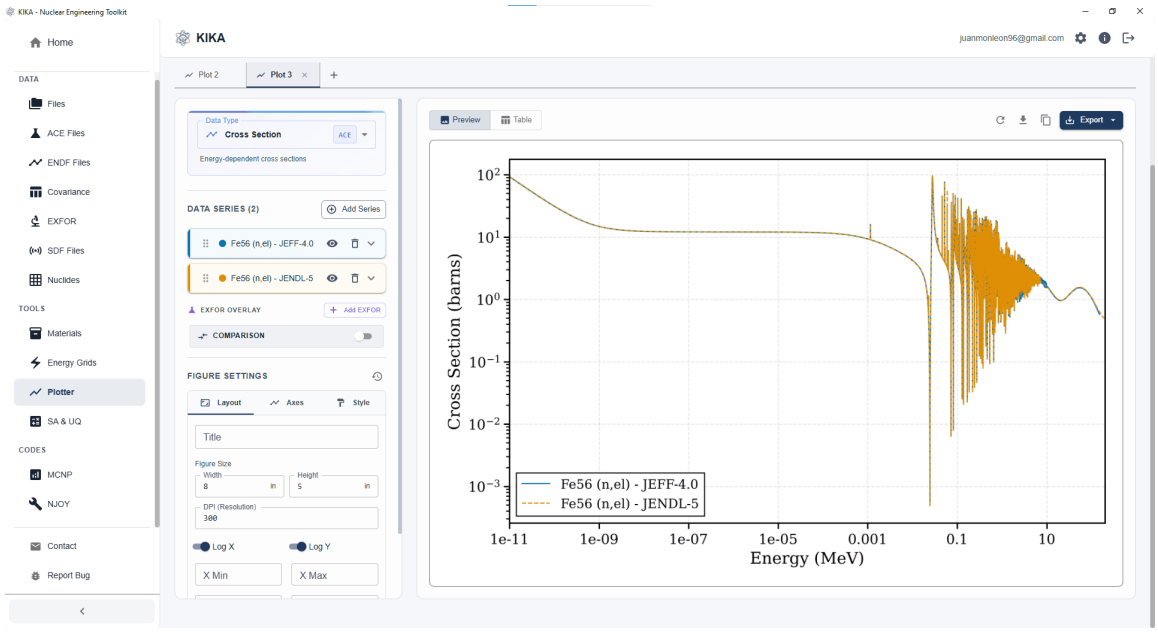


Figure B.7: The multi-tab plotter with an active cross-section comparison. Data series from different files are selected from a side panel and overlaid on the same axes. Each tab maintains an independent, fully customizable plot.

SA and UQ Hub

The SA&UQ hub consolidates the two main uncertainty quantification methods described in Chapter II: First-Order Second-Moment (FOSM) analysis and Monte Carlo Sampling (MCS).

The FOSM workflow begins with a local sensitivity analysis via PERT cards. A step-by-step wizard reads a loaded MCNP input file, extracts material definitions, and generates `PERT`-card blocks that implement the first-order calculation of sensitivity coefficients. These sensitivity coefficients are then combined with covariance data through the sandwich formula to obtain a full uncertainty quantification, including variance decomposition per reaction and per isotope.

The MCS workflow provides a guided interface for setting up Monte Carlo sampling runs. Users can sample perturbed nuclear data from ACE files (for cross sections), from ENDF files (for angular distributions), or from a combination of both. The wizard guides the configuration of each step and displays progress during execution.

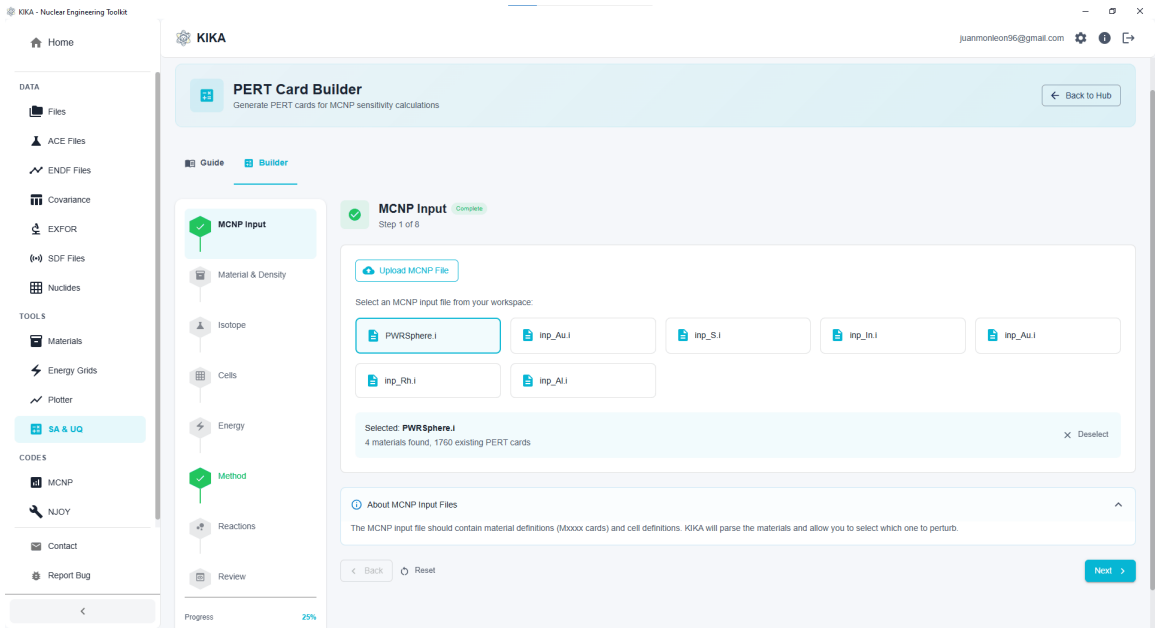


Figure B.8: The SA&UQ hub showing the available analysis workflows: FOSM-based sensitivity and uncertainty analysis via PERT cards and sandwich formula, and Monte Carlo sampling for cross-section and angular distribution perturbation.

B.4. Transport Code Integration

MCNP Hub

The MCNP hub manages MCNP input and MCTAL output files. Input files can be paired with their corresponding MCTAL output to create named runs that provide unified access to both. From a run, the application can extract material definitions (importing them into the Materials Manager), display tally results with figures of merit, and show `PERT`-card sensitivities per reaction channel. The parsed sensitivity data can be used to create an SDF file directly from within the application, which then feeds into the sandwich-formula UQ workflow for a full uncertainty analysis with variance decomposition.

NJOY

The NJOY section provides a visual pipeline builder for constructing NJOY input decks. Processing module blocks (RECONR, BROADR, HEATR, THERMR, ACER, ERRORR, among others) are added to a vertical pipeline and reordered by dragging, and each module's parameters are configured through a dedicated side panel. A live preview pane shows the generated input text, updating in real time as parameters change. Preset templates for common processing sequences are available, and an integrated terminal executes NJOY directly from the interface.

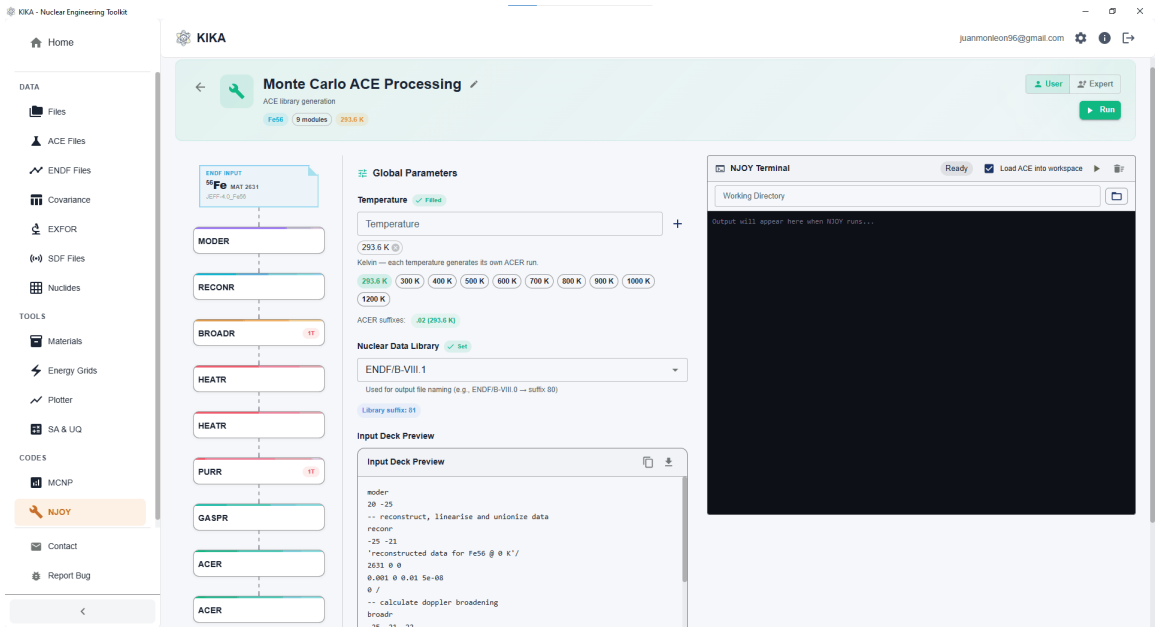


Figure B.9: The NJOY pipeline builder. A processing chain is assembled from module blocks arranged in a vertical pipeline (left). Parameters are configured in the side panel (right), and the generated input deck is shown in the live preview pane (bottom).

B.5. Distribution and Availability

KIKA is distributed as standalone installers for Windows (`.exe` setup wizard), Linux (AppImage), and macOS (`.dmg` disk image for Apple Silicon). A built-in auto-updater checks for new releases on startup and applies updates without manual reinstallation. The application is released under the BSD-3-Clause license and is publicly available at the repository cited in the text [109].

Appendix C

KIKA-ND Implementation Details

This appendix supplements the description of the `kika-nd` library given in Sec. 4. Where that section focuses on the physics and algorithms behind the perturbation pipelines, the material here describes the library's design choices, its current scope of supported formats, and practical usage patterns. Users who wish to extend the library or adapt its workflows to new applications will find here the context needed to do so.

The library is open-source under the BSD-3-Clause license and installable via `pip install kika-nd`. It targets Python 3.12 with core dependencies on NumPy, SciPy, Matplotlib, and pandas.

C.1. Design Principles

The library is organized around four principles:

(i) **Round-trip fidelity.** Nuclear data files can be parsed into structured Python objects and re-written to disk while preserving format-level details (record layout, significant figures, line sequencing). Individual sections of a file can be replaced without disturbing the remainder, which is the mechanism behind the perturbation pipelines of Sec. 4.5.

(ii) **Typed data model.** Every nuclear data entity (an ENDF material, an angular distribution, a covariance matrix, an ACE file) is represented as a typed Python dataclass with named attributes. This makes objects easy to inspect, to pass between pipeline stages, and to modify selectively.

(iii) **Shared interfaces.** Operations that apply identically to different covariance representations (Cholesky, EVD, SVD decomposition, sampling) are implemented once against a common interface. Both multigroup cross-section covariance objects and angular distribution covariance objects expose the same decomposition and sampling API, directly implementing the methods of Sec. 3.2.2.

(iv) **Pipeline automation.** High-level functions orchestrate multi-step workflows (covariance reading, sampling, file perturbation, NJOY processing) with `multiprocessing`-based parallelization. A logging system records every numerical decision (eigenvalue corrections, variance caps, positivity projections) to provide a complete audit trail for each run.

C.2. Supported Formats and Capabilities

The sections below describe each major module of the library and illustrate its usage through short code examples. The examples assume that the library has been imported as `import kika`.

C.2.1. ENDF-6

The library provides a full parser for the ENDF-6 format. The supported file types are:

- `MF1`: general information and directory, $\bar{\nu}$, fission energy release, and delayed photons.
- `MF2/MT151`: resolved and unresolved resonance parameters.
- `MF3`: reaction cross sections (any `MT` number).
- `MF4`–`MF6`: angular, energy, and energy–angle distributions of secondary particles.
- `MF7`: thermal neutron scattering law.
- `MF8`: radioactive decay, fission product yields, and radionuclide production.
- `MF9`, `MF10`: multiplicities and cross sections for radionuclide production.
- `MF12`–`MF15`: photon production.
- `MF23`, `MF27`, `MF28`: photo-atomic cross sections, form factors, and atomic relaxation.
- `MF31`–`MF35`, `MF40`: covariances of $\bar{\nu}$, resonance parameters, cross sections, angular distributions, energy distributions, and radionuclide production.

Individual `MF/MT` sections can be selectively replaced within an existing evaluation, which is the mechanism used by the ENDF perturbation pipeline (Sec. 4.5.2). New `MF34` covariance blocks can also be constructed from numerical arrays and merged into an evaluation, as done in the ^{56}Fe workflow of Chapter III. Evaluated files can be downloaded directly from the IAEA online library with local caching.

Reading and inspecting an ENDF file

```
import kika

endf = kika.read_endf("n-026_Fe_056.endf")
# Access angular distribution for elastic scattering
mf4_mt2 = endf.files[4][2]
# Access cross sections for a specific reaction
mf3_mt2 = endf.files[3][2]
```

C.2.2. ACE

The ACE module reads and writes the “A Compact ENDF” format used by MCNP, with support for continuous-energy cross sections, angular distributions (isotropic, equiprobable, tabulated,

Kalbach–Mann), energy distributions, delayed-neutron data, and photon production. A round-trip workflow (read, modify, write) produces perturbed ACE files with automatically generated MCNP directory entries for each file.

Reading an ACE file

```
import kika

ace = kika.read_ace("Fe56.ace")
# Inspect available reactions
print(ace.mt_numbers)
```

C.2.3. Covariance Data

Two covariance types are supported, each represented by a dedicated class. Multigroup cross-section covariances can be read from the COVERX, COVFIL, and BOXER formats produced by the SCALE and NJOY systems, with built-in support for common energy structures (SCALE 56-group, VITAMIN-J 175-group). The same data can be written back out in any of these formats, so a covariance file can be converted from one format to another. Angular distribution covariances are read directly from ENDF `MF34` sections, where each sub-matrix corresponds to a Legendre order pair (ℓ, ℓ') and is defined on the evaluator's own energy-group grid, with its own relative/absolute flag and reference frame (LAB/CM). Both types expose the same decomposition (Cholesky, EVD, SVD) and sampling interface described in Sec. 3.2.2. A multigroup counterpart for angular distribution covariances is obtained through the collapse algorithm of Sec. 4.4.

Reading covariance data and computing a decomposition

```
import kika

# Multigroup cross-section covariance from COVERX
covmat = kika.read_coverx("56Fe_coverx.dat")
L = covmat.cholesky_decomposition(space="log")

# Convert the same covariance to BOXER format
from kika.cov import write_boxer
write_boxer(covmat, "56Fe.boxer")

# Angular distribution covariance from ENDF MF34
from kika.cov import MF34CovMat
mf34_cov = MF34CovMat.from_endf("n-026_Fe_056.endf")
```

C.2.4. NJOY Processing

For workflows that require non-standard module sequences beyond the fixed perturbation template (Appendix D), the library provides a builder API for constructing arbitrary NJOY input decks. Module calls can be chained with automatic tape numbering. A companion parser reads and analyzes existing NJOY input files.

C.2.5. Transport Code Integration

The library provides interfaces to MCNP and Serpent, primarily for sensitivity-related workflows. For MCNP, input files and tally outputs (MCTAL) can be parsed, `PERT` card blocks can be generated automatically, and sensitivity profiles can be extracted from perturbation results and packaged into SDF objects compatible with the SCALE/TSUNAMI format. For Serpent, the native sensitivity output format is parsed directly and can also be converted to SDF format for use with the sandwich-formula UQ workflow.

C.2.6. EXFOR

The EXFOR module provides programmatic access to the EXFOR experimental reaction database [201]. Entries can be loaded directly from the X4Pro SQLite database [202], with built-in filters by target nuclide, reaction (`MT`), and incident-energy range. Angular distributions and cross sections are exposed as dedicated objects with helpers for laboratory-to-center-of-mass frame conversion, plotting, and overlay against ACE-evaluated data. This module is used in particular for the ^{56}Fe angular distribution evaluation workflow of Chapter III.

C.2.7. Materials

A dedicated materials module handles isotopic compositions and densities. Materials can be imported from MCNP input decks or defined programmatically, with conversion between weight and atomic fractions. The module also supports export to MCNP material card format for direct use in transport input files.

C.3. Sampling and Perturbation

The sampling module produces correlated perturbation vectors from a covariance matrix following the methodology of Sec. 3.2. It accepts any covariance object, a decomposition method (Cholesky, EVD, SVD, PCA), a sampling strategy (Sobol sequences, Latin Hypercube, or pseudo-random), and a perturbation space (linear or log-normal). The perturbation pipelines described in Sec. 4.5 are built on top of this sampling layer.

The following example illustrates a complete cross-section perturbation workflow:

ACE cross-section perturbation pipeline

```
from kika.sampling.ace_perturbation import perturb_ACE_files

perturb_ACE_files(
    ace_files="Fe56.ace",
    cov_files="56Fe_coverx.dat",
    mt_list=[2, 4, 102],
    num_samples=500,
    decomposition_method="svd",
    sampling_method="sobol",
    output_dir="perturbed_xs/",
    nprocs=4,
```

```
)
```

For angular distribution perturbations:

ENDF angular distribution perturbation pipeline

```python
from kika.sampling.endf_perturbation import perturb_ENDF_files

perturb_ENDF_files(
    endf_files="n-026_Fe_056.endf",
    mt_list=[2],
    legendre_coeffs=[1, 2, 3, 4],
    num_samples=500,
    output_dir="perturbed_ad/",
    generate_ace=True,        # Process through NJOY
    njoy_exe="/path/to/njoy",
    temperatures=293.6,
    nprocs=4,
)
```

C.4. Covariance Repair

Multigroup covariance matrices obtained from evaluated nuclear data or from processing codes are not always positive semi-definite (PSD) in practice. Sources of failure include numerical rounding during format conversions, extreme uncertainty values in isolated energy bins near reaction thresholds, and approximations in processing codes. Even small negative eigenvalues prevent Cholesky decomposition from succeeding and can produce unphysical perturbation samples.

The library repairs such matrices at two stages. During decomposition, an eigenvalue-based projection restores positive semi-definiteness before the matrix is factorized, and the strength of the correction adapts to the severity of the violation. A separate stage, disabled by default, can act on the covariance before decomposition and apply an additional structural repair that clamps extreme variances and, where necessary, removes the correlations or reactions responsible for the indefiniteness. Every correction, at either stage, is recorded in the run log so that the affected reactions and energy regions can be identified before committing to a production run.

Appendix D

NJOY Perturbation Processing Template

The perturbation pipelines described in Sec. 4.5 process every perturbed ENDF file through an identical NJOY sequence to produce a perturbed ACE library. Uniformity across all N samples is essential: if even one sample used a different processing chain, the resulting ACE files would not be statistically comparable. The template defines the full module sequence once, with four string placeholders substituted at runtime for each sample. Table D.1 documents these placeholders.

Table D.1: Placeholders in the NJOY perturbation template. Each placeholder is substituted at runtime with sample-specific values.

Placeholder	Type	Example	Description
`{title}`	string	`56Fe sample 042`	Run title embedded in the ACE file header.
`{mat}`	integer	`2631`	ENDF MAT number identifying the nuclide.
`{T}`	float	`293.6`	Target temperature in Kelvin.
`{suff}`	float	`0.03`	ACE ZAID suffix (e.g. the `.03c` in `26056.03c` for 300 K).

Listing D.1 shows the complete NJOY input template. The processing chain is:

$$\text{MODER} \rightarrow \text{RECONR} \rightarrow \text{BROADR} \rightarrow \text{HEATR} \rightarrow \text{PURR} \rightarrow \text{GASPR} \rightarrow \text{ACER}$$

NJOY communicates between modules through numbered “tapes”: positive numbers are formatted (ASCII) files and negative numbers are binary (PENDF) files held internally.

Listing D.1: NJOY input template used for all perturbation pipeline runs. Curly-brace tokens are Python placeholders substituted at runtime. Comment lines (prefixed with $) are added for readability.

```
$ Convert input ENDF from ASCII (tape 20) to binary (tape -25)
moder
 20 -25
$ Reconstruct resonance cross sections; tolerance 0.1%
reconr
 -25 -21
'{title}'/
 {mat} 0 0
 0.001 0 0.01 5e-08
 0 /
$ Doppler-broaden pointwise cross sections to temperature {T} K
broadr
 -25 -21 -22
 {mat} 1 0 0 0
 0.001 1e+06 0.01 5e-08
 {T}
 0 /
$ Compute heating (KERMA) factors -- first pass
heatr
 -25 -22 -21 /
 {mat} 5 0 0 0 0 /
 302 318 402 442 444 /
$ Compute heating (KERMA) factors -- second pass with local
$ photon energy deposition
heatr
 -25 -22 -23 /
 {mat} 6 0 1 0 2 /
 302 303 402 442 443 444 /
$ Compute probability tables for the unresolved resonance range
purr
 -25 -21 -24
 {mat} 1 9 20 64 1 0 /
 {T} /
 1e+10 1e+8 1e+6 1e+4 1e+3 3e+2 1e+2 3e+1 1e+1 /
 0 /
$ Add gas production recoil cross sections
gaspr
 -25 -24 -21  /
$ Write primary ACE file (tape 40) and xsdir line (tape 41)
acer
 -25 -21 0 40 41
 1 0 1 {suff} /
'{title}'/
 {mat} {T}
 1 1 1
 /
$ Check ACE file for internal consistency (iopt = 7)
acer
 0 40 0 43 44
 7 1 1 -1 /
'{title}'/
stop
```

Appendix E

Supplementary Material for Chapter III

This appendix collects supporting material for Chapter III that is not strictly needed to follow the main thread of the chapter. It separates the magnitude and shape contributions to the energy-resolution folding of Sec. 3.2 (Sec. E.1); it documents how the per-point statistical and systematic uncertainties are extracted from EXFOR for each experiment (Sec. E.2) and tests the sensitivity of the result to the σ_{stat} floor (Sec. E.3); compares the AIC, AICc, and BIC variants of the per-bin degree selection on this dataset (Sec. E.4); gives the positivity projection applied to each Monte Carlo refit (Sec. E.5); explains why the full coefficient correlation matrix is sourced from a single Monte Carlo pass (Sec. E.6); documents the near-zero regularization of near-vanishing relative variances (Sec. E.7); discusses why the published `MF34` carries no $\ell = 0$ row under the present methodology (Sec. E.8); explains the strongly negative adjacent-bin correlations observed in some bands of the multigroup covariance (Sec. E.9); checks the adequacy of the Gaussian covariance representation (Sec. E.10); summarizes pipeline-execution diagnostics (Sec. E.11); and gives the full-suite k_{eff} comparison for the criticality cross-check (Sec. E.12).

E.1. Energy-Resolution Folding: Magnitude and Shape Contributions

Figure III.6 of Sec. 3.2 folds the elastic cross section and the angular shape over the time-of-flight resolution window together. The two contributions are separated here. Figure E.1 varies only the magnitude σ_{el} across the window while holding the angular shape at its nominal value; Figure E.2 varies only the shape, reconstructed from all Legendre orders, while holding the magnitude fixed. The magnitude changes more strongly across the resonance structure of the window than the shape does, so most of the spread in the combined figure comes from the cross section, with the shape adding a smaller angle-dependent adjustment. Both enter the window average that a finite-resolution measurement records.

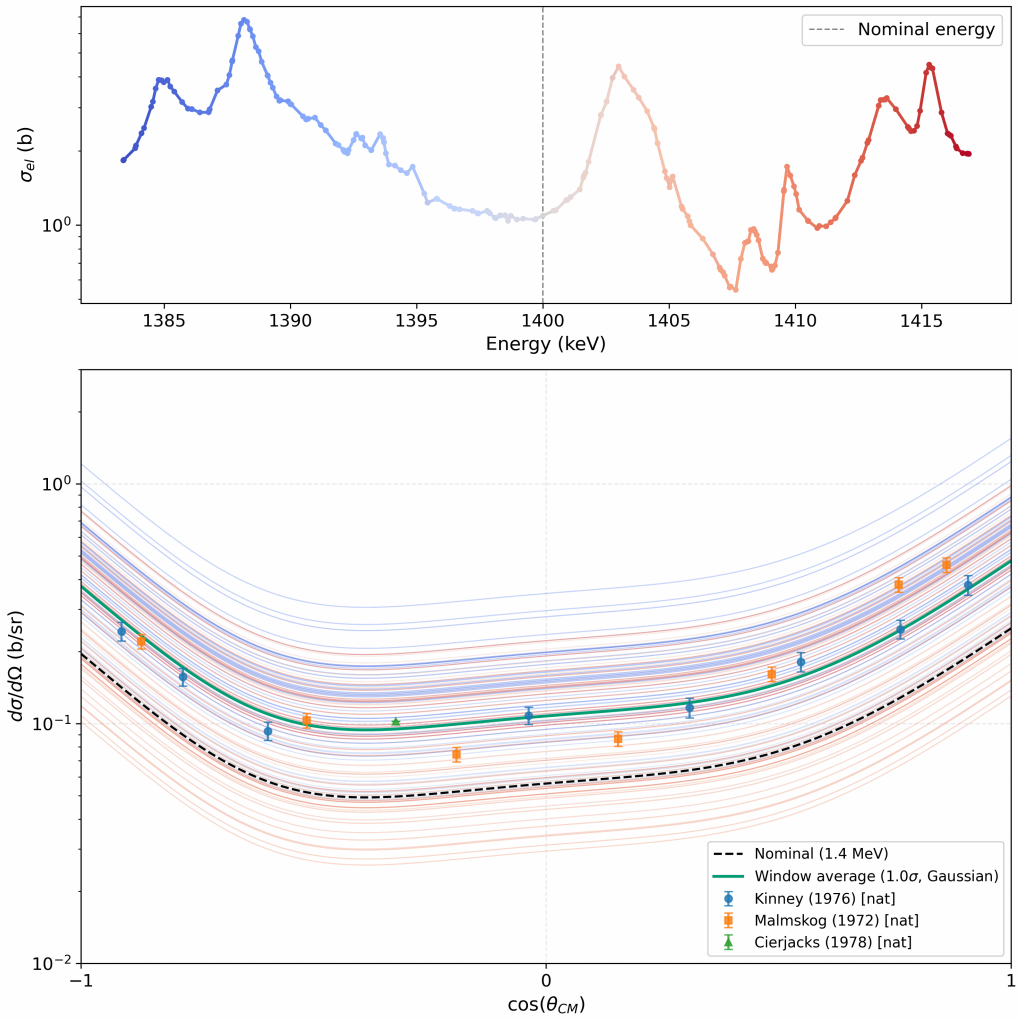


Figure E.1: Magnitude-only folding for ^{56}Fe(n, el) at a nominal 1.4 MeV (JEFF-4.0, center-of-mass frame): σ_{el} varies across the window while the angular shape is fixed at its nominal value. Panels and curves as in Figure III.6.

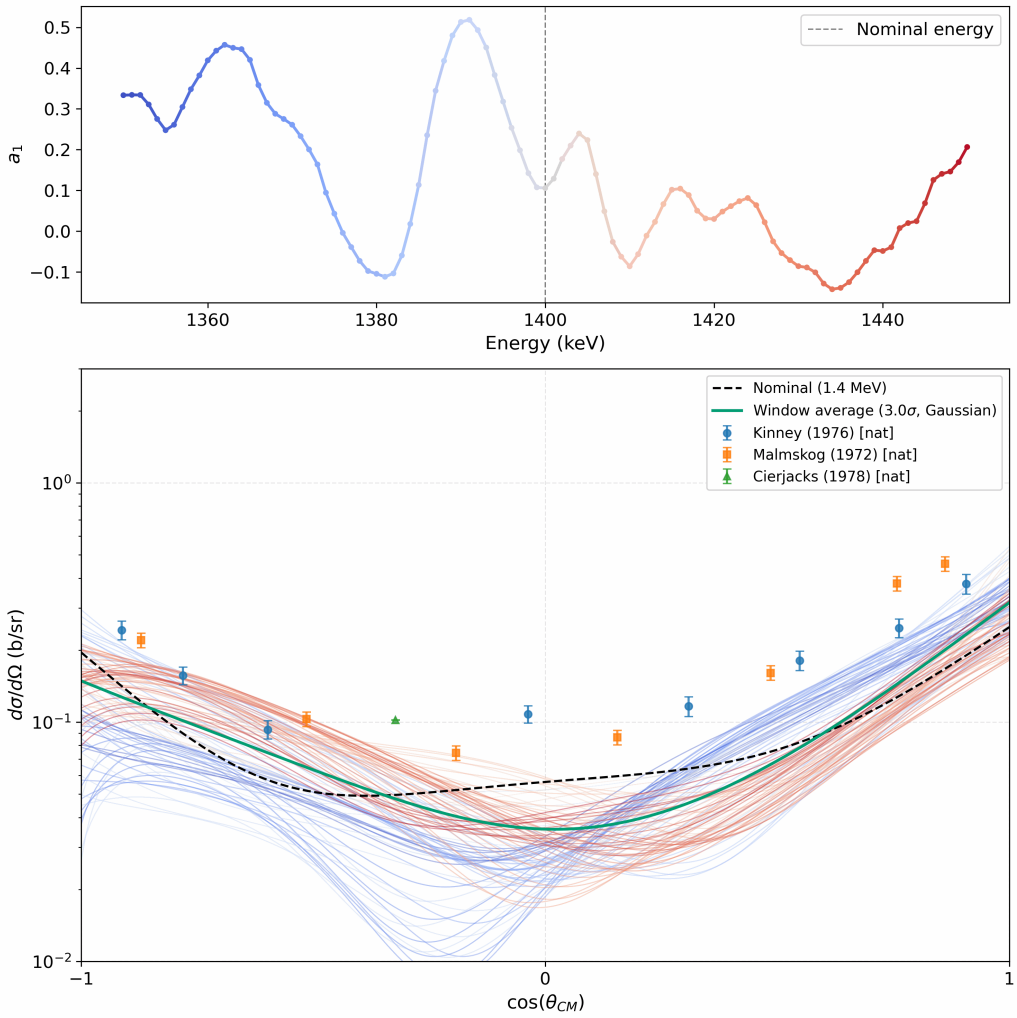


Figure E.2: Shape-only folding, plotted as in Figure E.1 with the roles reversed: the angular shape (all Legendre orders) varies across the window while σ_{el} is fixed. The top panel shows a_1 as a proxy for the full shape variation.

E.2. Per-Experiment Uncertainty Cases

Section 3.3 describes the per-point decomposition of the experimental uncertainties into a statistical component $\sigma_{\mathrm{stat},i}$ (uncorrelated point to point, entering only the diagonal of the per-experiment covariance block) and a per-experiment systematic σ_g^{sys} (one scalar per experiment, the rank-one normalization mode that contributes to both the diagonal and the full off-diagonal block). The construction follows directly from the ERR-ANALYS prose of each EXFOR entry, when available, and falls back to a fixed 5% default systematic otherwise (Sec. E.2.3). Most experiments fall under one of a small number of generic prose patterns documented in the next subsection; one requires a treatment beyond those patterns and is documented individually in Sec. E.2.2. This appendix reports the resolved values for the largest experiments in the database.

The EXFOR format does not standardize the meaning of per-point error columns. The same label (`DATA-ERR`, `ERR-T`, `ERR-S`, `ERR-1`, `ERR-DIG`, ...) carries different content across entries, and the ERR-ANALYS prose is the only definitive description of which physical contributions enter each column. The cases below describe how that prose maps to a stat/sys decomposition; entries without prose (Case F) are handled by a fixed-precedence inspection of whichever columns the EXFOR table exposes.

E.2.1. Generic Handling Cases

The handling of an experiment depends on what its ERR-ANALYS entry provides.

Case A: stat-only column with named systematic

The prose identifies one of the data columns (typically ERR-S) as a statistical-only error and reports the systematic components in the body of the entry. The named column is used directly as $\sigma_{\mathrm{stat},i}$ per point. The systematic components are summed in quadrature and routed to σ_g^{sys} when the prose describes them as normalization-like (monitor, sample mass, detector solid angle); experiments whose prose documents an additional contribution beyond simple normalization are routed individually in Sec. E.2.2.

Case B: total column with named decomposition

The prose says that one of the data columns (DATA-ERR, ERR-T, or similar) represents the total uncertainty per point and that the systematic components already enter that column. The column is taken as $\sigma_{\mathrm{tot},i}$, the systematic components are summed in quadrature to σ_g^{sys}, and the residual statistical contribution is obtained by subtraction at each point:

$$\sigma_{\mathrm{stat},i} = \sqrt{\max\!\left(\sigma_{\mathrm{tot},i}^2 - (\sigma_g^{\mathrm{sys}}\, y_i)^2,\ (0.01\, y_i)^2\right)}\,. \tag{E.1}$$

A floor at $1\%\,|y_i|$ protects the subtraction in the few points where the prose-derived systematic exceeds the reported total. When the floor is reached, the per-point $\sigma_g^{\mathrm{sys}}\, y_i$ is correspondingly capped so that the lab's reported total is preserved exactly. The Case-B treatment applies to Kinney 1976 (`10571.002`), Smith 1980 (`10886.002`), Barnard 1968 (`30076.004`), and Gkatis 2025 (`27673.002`); the floor is reached in practice only in Kinney, where the manifest's ∼8% correlated systematic exceeds `DATA-ERR` at the lowest-uncertainty points.

Case C: all-scalar specification

The prose gives a single relative statistical and a single relative systematic value, with no per-point column. Both quantities are applied as constant fractions to every point. Examples: Wells 1963 (11121), Tomita 1973 (20008).

Case D: multi-component combine

The prose identifies several per-point components in named columns, each with a different physical origin. The resolver evaluates each component independently and sums in quadrature, partitioning them between σ_{stat} and σ_g^{sys} according to the prose description. Example: Korzh 1968 (40042.016, combining ERR-S and ERR-DIG).

Case E: piecewise-energy systematic

The prose names a systematic whose magnitude varies with incident energy, beyond the energy-flat per-experiment normalization. This contribution is carried as an additional rank-one mode $v_i^{(g)} = \sigma_{g,E_i}^{\mathrm{sys},E}\, y_i$ in the per-experiment block, alongside the normalization mode $u^{(g)}$, so the block reads

$$\Sigma^{(g)} \;=\; D^{(g)} + u^{(g)} u^{(g)\,T} + v^{(g)} v^{(g)\,T}, \tag{E.2}$$

with $v^{(g)} = 0$ for every experiment outside this case. The only instance in the present database is Tsukada 1969 (20304.002), whose ^{3}H(p,n) neutron-source calibration contributes 10% above 2 MeV, rising to 20% below 1 MeV.

Case F: no decomposition in the prose

Most uncurated entries (or those whose ERR-ANALYS does not name a stat/sys split) fall here. The resolver consults the data columns in a fixed precedence order, DATA-ERR1 (combined with DATA-ERR2 in quadrature if both are present), then DATA-ERR, then ERR-T, then ERR-S, and uses the first column found. When the recovered column is a labeled total (DATA-ERR, ERR-T, DATA-ERR1), the Case B decomposition is applied with a default $\sigma_g^{\mathrm{sys}} = 5\%$. When the column is labeled statistical (ERR-S), it is used as-is for σ_{stat} and the same 5% default is added in quadrature as σ_g^{sys}. When no usable column is found, σ_{stat} is set to zero (the pipeline's $10^{-4}\,|y|$ guard then prevents infinite weights) and only the 5% systematic remains. The per-branch counts are reported in the coverage summary below.

E.2.2. Per-Experiment Special Cases

One experiment does not fit the generic patterns above and is routed individually.

Cierjacks 1978 (20743.002)

The prose provides a statistical-only column (used directly for $\sigma_{\mathrm{stat},i}$) and a per-point total `ERR-T` that lumps flux measurement, detector efficiency, sample thickness, background, multiple scattering, and flux depression into one 7–12% envelope, without per-source magnitudes. Because the envelope does not resolve the individual sources, the normalization-like part is carried by the default systematic of Sec. E.2.3, $\sigma_g^{\mathrm{sys}} = 5.00\%$. The residual between the reported total and the quadrature sum of $\sigma_{\mathrm{stat},i}$ and $\sigma_g^{\mathrm{sys}}\, y_i$ is therefore unattributed, and its correlation structure

cannot be determined from the prose alone: it could be a single fully correlated draw (a rank-one mode), uncorrelated point to point, or somewhere in between. Promoting it to a rank-one mode would claim a correlation structure the prose does not support, so the residual is added to $\sigma_{\mathrm{stat},i}$ as additional uncorrelated diagonal noise.

E.2.3. The 5% Default Systematic

The default value of $\sigma_g^{\mathrm{sys}} = 5\%$ is taken by 31 subentries: the 24 of Case F, the four Case A subentries without a named systematic, two of the three Case D subentries, and Cierjacks 1978. It is the median of the per-experiment systematic across the experiments whose prose does provide a decomposition. The reported values span 3% to 10%; the upper-end values come from a few precision-normalization-limited campaigns (Wells 1963, where the absolute normalization carries 18–24% on its own, is treated as a documented outlier and routed to its own value). The 5% choice is therefore a conservative central estimate consistent with the typical correlated calibration uncertainty in fast-neutron differential measurements.

E.2.4. Coverage Summary

Across the 67 EXFOR subentries that contribute to ^{56}Fe and natural-iron elastic scattering in the 847 keV–4 MeV range:

- Case A (curated, stat-only column with named sys): 19 subentries;
- Case A (curated, stat-only column with default sys): 4 subentries;
- Case B (curated, total-column decomposition): 5 subentries;
- Case C (curated, all-scalar): 10 subentries;
- Case D (curated, multi-component combine): 3 subentries;
- Case E (curated, piecewise-energy systematic): 1 subentry (Tsukada 1969);
- Case F (uncurated, default fallback): 24 subentries;
- Per-experiment special case (curated, individual routing): 1 subentry (Cierjacks 1978).

Within Case F, the breakdown by recovered column is 21 subentries with a labeled total (decomposed via Case B), 1 with ERR-S (treated as stat-only), and 2 with no usable column (treated as 5% systematic only).

E.2.5. Largest Contributors to the Fit

Table E.1 documents the resolved values for the experiments that contribute the most to the fit. Kinney and Smith are the two calibration anchors of JEFF and JENDL; Cierjacks is the largest contributor outside that pair; Pirovano and Gkatis are the most recent high-resolution campaigns.

Table E.1: Per-experiment uncertainty resolution for the largest contributors. "Case" refers to the classification in the text. σ_g^{sys} is the per-experiment relative systematic (rank-one normalization). The "$\sigma_{g,E}^{\text{sys},E}$ range" summarizes any documented additional contribution before routing; for Cierjacks this is folded into σ_{stat} as diagonal residual (see Sec. E.2.2). "Median $\sigma_{\text{stat}}/|y|$" is the resolved per-point statistical fraction.

Experiment	EXFOR	**Case**	σ_g^{sys}	$\sigma_{g,E}^{\text{sys},E}$ **range**	**Median** $\sigma_{\text{stat}}/\lvert y\rvert$
Kinney 1976	10571002	B	8.06%	–	4.3%
Smith 1980	10886002	B	5.00%	–	3.3%
Cierjacks 1978	20743002	A + special	5.00%	0.1–5.1%	4.9%
Pirovano 2019	23365004,005	F	5.0%	–	21.2%
Gkatis 2025	27673002	B	1.6%	–	4.3%

E.3. Statistical-Uncertainty Floor Sensitivity

The statistical uncertainty entering the diagonal $D_{ii}^{(g)} = \sigma_{\text{stat},i}^2$ of Sec. 3.3 is subject to two distinct floors that protect different pathologies.

The first is a $1\%\,|y|$ floor enforced once at the uncertainty-assembly stage, applied only in the Case B decomposition of Eq. (E.1). It prevents σ_{stat} from going to zero (or formally negative) where the prose-derived systematic saturates the lab's reported total, with the compensating cap on $\sigma_g^{\text{sys}}\, y_i$ that preserves the total described in the Case B definition above.

The second is a $10^{-4}\,|y|$ numerical guard applied at the GLS-fitting stage to prevent infinite weights when a loader returns $\sigma_{\text{stat}} = 0$ silently. This value is well below every credible reported statistical uncertainty in the database (Korzh 1972 ERR-S = 0.7%, Cierjacks per-point ERR-S median $\approx$ 0.05%, Tomita DATA-ERR $\approx$ 0.5%), so prose-reported small values pass through unchanged.

The two floors do not double-act. The Case B floor is conceptually a constraint on a decomposition derived from a prose statement and does not apply to entries whose prose does not require a decomposition. The numerical guard is a purely defensive value applied uniformly. Sensitivity tests in the range 0.5%–2% for the Case B floor produce χ^2/N changes below a few percent on every reported configuration, confirming that the headline numbers are not driven by the choice of this single tuning value.

E.4. Information-Criterion Comparison: AIC, AICc, BIC

This appendix documents the comparison of the three penalized-likelihood criteria that were tested as candidates for the per-bin degree selection of Sec. 4.3. The three production runs differ only in the selection criterion (AIC, AICC, or BIC); the data, the uncertainty assembly, the angular-band IRLS, the model averaging, and the multigroup collapse settings are identical.

E.4.1. Criteria

All three criteria have the same form $\text{score}(L) = \chi^2(L) + \text{pen}(k, n)$, with the penalty for one extra effective parameter being:

- AIC: $+2$ (independent of n);
- AICc: $+2 + 2k(k+1)/(n-k-1)$, with the second term dominating when $n \lesssim 12$;
- BIC: $+\ln(\max(1, n))$, which exceeds the AIC penalty for $n \geq 8$ and grows logarithmically with n.

In the present database the typical bin support is $n_{\text{eff}} \in [4, 30]$. AIC's penalty is constant at 2 throughout this range; BIC's $\ln n$ falls between 2 and 3; AICc's correction adds 5–10 to the per-extra-parameter penalty when n is at the low end. AICc is therefore strongly penalizing in the regime where most of the database's bins live.

E.4.2. Selected-Order Distribution

Table E.2 reports the distribution of the selected nominal degree across the 1 735 fitted bins in the production energy range.

Table E.2: Number of bins selecting each Legendre degree, per criterion, across the 1 735 fitted bins of the 0.847–4 MeV range.

L^*	AIC	AICc	BIC
1	2	21	2
2	200	561	230
3	614	764	667
4	453	262	460
5	328	66	271
6	138	61	105
mean	3.76	2.99	3.62

AICc concentrates around 76% of the mass at $L = 2$ or $L = 3$; AIC pushes nearly half the mass into $L \geq 4$. BIC sits between the two but closer to AIC, consistent with its $\ln n \approx 2$–3 penalty being only slightly stronger than AIC's fixed $+2$.

E.4.3. Per-Bin Reduced Chi-Squared

How closely each criterion's nominal fit follows the EXFOR angular data is measured by the per-bin reduced chi-squared $\tilde{\chi}^2_{\text{red}}$, the fit residual divided by the per-bin degrees of freedom. Table E.3 summarizes it over the fitted bins of the 0.847–4 MeV range. AIC and BIC give an almost identical, close fit: median $\tilde{\chi}^2_{\text{red}} = 0.39$ and 0.40, with about 88% of bins fitted within the data uncertainties. AICc is noticeably worse, with median 0.55 and only 76% of bins below unity. The gap is a direct consequence of the order selection of Table E.2: AIC and BIC use the orders the data support, whereas AICc's small-sample penalty forces lower orders even where the fit is still constrained, leaving structure in the residuals.

E.4.4. Choice Rationale

The empirical ranking AIC $\gtrsim$ BIC $\gg$ AICc on this dataset reflects the small-sample regime in which most bins live. AICc's correction is exact only under i.i.d. Gaussian residuals with known

Table E.3: Per-bin reduced chi-squared $\tilde{\chi}^2_{\text{red}}$ of the nominal fit against the EXFOR angular data, summarized over the fitted bins of the 0.847–4 MeV range, for the three model-selection criteria. Lower values indicates a closer fit; the last column is the fraction of bins fitted within the data uncertainties ($\tilde{\chi}^2_{\text{red}} < 1$).

Criterion	median $\tilde{\chi}^2_{\text{red}}$	mean $\tilde{\chi}^2_{\text{red}}$	bins with $\tilde{\chi}^2_{\text{red}} < 1$
AIC	0.39	0.56	88%
AICc	0.55	0.80	76%
BIC	0.40	0.56	87%

variance, an assumption that the angular-band IRLS of Sec. 4.3 violates: the variance entering each iteration of the criterion is the s_b-inflated σ_{eff}, so the underlying χ^2 surface is flatter than AICc assumes. The pipeline therefore uses AIC for the production runs; BIC and AICc remain available as configuration switches for sensitivity tests.

E.5. Positivity Enforcement

A truncated Legendre expansion can produce small negative values at certain angles, which is unphysical since the differential cross section is a non-negative quantity. This is a well-known artifact of series truncation [172]. Section 4.4.3 summarizes how the pipeline corrects it after each Monte Carlo refit; this appendix gives the projection in full.

The reconstructed angular distribution is evaluated on a grid of $M = 101$ uniformly spaced points $\{\mu_j\}_{j=1}^{M} \subset [-1, 1]$. If any negative values are detected, the coefficient vector is projected onto the nearest point (in the Euclidean norm) that satisfies non-negativity at all grid points:

$$\min_{\mathbf{c}^*} \ \|\mathbf{c}^* - \mathbf{c}\|_2^2 \quad \text{subject to} \quad \sum_{\ell=0}^{L} c_\ell^* P_\ell(\mu_j) \geq 0 \quad (j = 1, \ldots, M) \tag{E.3}$$

The constraints are linear in the coefficients: defining $\mathbf{P}_{j\ell} = P_\ell(\mu_j)$, the problem becomes $\min \|\mathbf{c}^* - \mathbf{c}\|^2$ subject to $\mathbf{P}\mathbf{c}^* \geq \mathbf{0}$ [203]. In practice, the adjustment is small: only the coefficients responsible for the negative excursion are modified, and the overall angular shape is preserved. The optimization is solved using the SLSQP algorithm [204] as implemented in SciPy [205]. A conceptual summary of this projection is illustrated in Figure E.3.

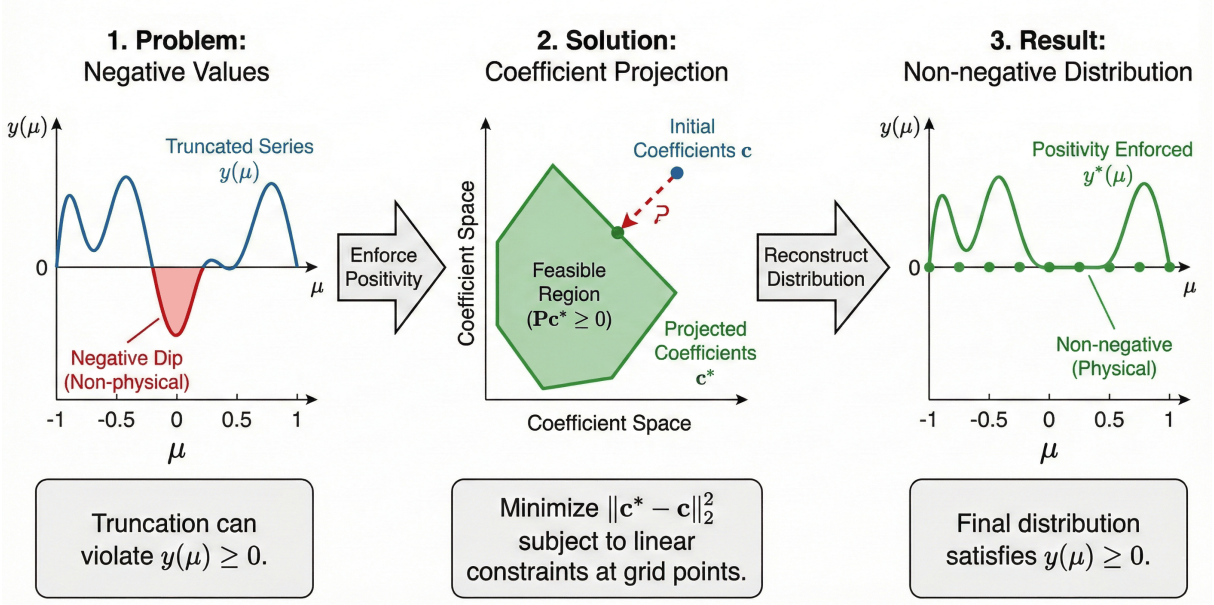


Figure E.3: Conceptual overview of the positivity enforcement procedure. An initial Legendre expansion exhibiting non-physical negative values (left) is corrected by projecting its coefficient vector $\mathbf{c}$ to the nearest feasible point $\mathbf{c}^*$ satisfying the linear constraints (middle), yielding a strictly non-negative angular distribution (right).

E.6. Single-Source Correlation Matrix and PSD

Sourcing the entire correlation matrix from Pass 2 is essential for positive semi-definiteness of the assembled covariance $\mathbf{C}$ (Eq. (III.33)). Pass 1 perturbs each bin independently, so it can supply the within-bin (cross-order) blocks but not the cross-bin blocks. Pass 2 shares perturbations across bins through the overlap weights and yields the full $\mathbf{R}_{\mathrm{KW}}$, including both block families, from a single self-consistent ensemble.

Combining the two sources, for example by taking the within-bin blocks from Pass 1 and the cross-bin blocks from Pass 2, mixes two different sampling models in a single matrix. The two ensembles are not jointly consistent, so the resulting matrix has no guaranteed positive semi-definiteness, and a Cholesky decomposition can fail without a clear physical reason. Drawing the full $\mathbf{R}_{\mathrm{KW}}$ from Pass 2 alone avoids this issue: the diagonal congruence transform of Eq. (III.33) cannot change the sign of any eigenvalue, so $\mathbf{C}$ inherits the positive semi-definiteness of $\mathbf{R}_{\mathrm{KW}}$.

E.7. Near-Zero Regularization

The ENDF-6 `MF34` format stores covariance data in relative form. When a Legendre coefficient passes through zero (sign change as a function of energy), the relative standard deviation diverges even though the absolute variance is well-behaved. Left uncorrected, these zero-crossings create artificial spikes in the relative covariance that corrupt the stored `MF34` data. Section 4.6 states the correction; this appendix gives the two-stage procedure.

First, during Monte Carlo sampling, each covariance matrix expressed relative to the sample

means is scanned for coefficients whose signal-to-noise ratio,

$$\mathrm{SNR}_\ell = \frac{|a_\ell|}{\sqrt{\hat{C}_{\ell\ell}}}, \tag{E.4}$$

falls below unity. Second, when the absolute covariance is converted to the nominal-relative form required by `MF34`, the same check is applied using the nominal Legendre coefficients as the reference. This second pass is necessary because the `MF34` format divides by the nominal coefficients, not by the Monte Carlo means. Near a zero-crossing the two references can differ significantly, so a coefficient that is well-behaved relative to the sample mean may still produce a divergent relative variance when divided by a smaller nominal value.

At both stages, flagged coefficients have their relative variance replaced by the median of the nearest well-behaved neighbors of the same Legendre order. The replacement is applied via a congruence transform $\hat{\mathbf{C}}' = \mathbf{D}\,\hat{\mathbf{C}}\,\mathbf{D}$, where $\mathbf{D}$ is a diagonal scaling matrix. As with the assembly transform of Eq. (III.33), this preserves correlations and positive semi-definiteness.

E.8. MF3–MF4 Cross-Correlations Under EXFOR-Only Methodology

ENDF `MF34` reserves an optional $L = 0$ row to carry the cross-covariance $\mathrm{Cov}(\sigma_{\mathrm{el}}, a_\ell)$ between the integrated cross section and the Legendre angular coefficients. A non-zero $L = 0$ row asserts that the two quantities share a source of uncertainty, so a fluctuation of σ_{el} implies a coherent fluctuation of the angular shape.

Under the methodology of this chapter, this assertion would not be meaningful. The integrated cross section is taken from the host evaluation's `MF3` (JEFF-4.0 in the present demonstration on ^{56}Fe), with uncertainties drawn from the corresponding `MF33`. The Legendre coefficients are fitted from the EXFOR angular distributions; their uncertainties come from the per-point statistical and per-experiment systematic decomposition of Sec. 3.3. These two sources are independent by construction: they share no experiments, no fitting code, and no model assumptions. The correct value of $\mathrm{Cov}(\sigma_{\mathrm{el}}, a_\ell)$ under this methodology is therefore zero, and the published `MF34` does not carry an $L = 0$ row.

Forcing a non-zero $L = 0$ row would require either (a) perturbing the EXFOR angular data by samples drawn from `MF33` and refitting, which makes the cross-block a function of the fit constraint rather than a physical channel, or (b) conditioning the angular fit on the host `MF3`, which couples the angular evaluation to a host-specific choice of integrated cross section. The first route returns zero in the natural normalization-floating fit and a spurious non-zero when c_0 is pinned; the second route makes the published a_ℓ depend on the host, and on the resonance-structured grid of this energy range the bin-by-bin agreement between $4\pi c_0^{\mathrm{EXFOR}}$ and the trapezoidally averaged σ_{MF3} is poor (only one in twenty bins agrees to within 3%, and one in four shows a relative discrepancy above 50%), so the constraint is either inactive or distorting.

A future joint evaluation in which σ_{el} and a_ℓ are constrained by the same set of measurements (for example through an R-matrix or optical-model fit that ingests both integrated and differential data) would produce a physically meaningful $\mathrm{Cov}(\sigma_{\mathrm{el}}, a_\ell)$ from the joint posterior. That is the regime in which the $L = 0$ row of `MF34` is appropriate.

E.9. Strong Negative Adjacent-Bin Correlations

The published multigroup `MF34` carries a small but persistent population of strongly negative correlations between adjacent energy bins, most visibly for $\ell = 1$ in the 1.0–1.8 MeV resonance band and to a lesser extent at $\ell = 3$. This appendix explains the mechanism and confirms that the structure is a feature of the EXFOR-plus-model-averaging methodology, not an artifact of the post-processing.

In the kernel-weight pass each fine bin is fitted at its own nominal degree L^* (Sec. 4.5). Adjacent fine bins share most of their experimental data through the TOF kernel; if the two bins had the same L^*, the resulting coefficient samples would be near-perfectly positively correlated. Negative correlations arise where adjacent bins have *different* nominal degrees. With c_0 frozen, the truncated basis $\{P_1, \ldots, P_L\}$ is not orthogonal at the realised set of μ_p values, so adding (or removing) a higher-order term changes the leverage of every lower-order coefficient on every data point. A perturbation that pushes c_3 up under the $L = 3$ basis can be partly absorbed by c_4 under the $L = 4$ basis, leaving c_3 to decrease there. The same data perturbation therefore drives c_ℓ up in one bin and down in the adjacent bin in every sample.

The strongest negatives concentrate in the 1.0–1.8 MeV resonance band, where the angular distribution changes rapidly with energy and adjacent bins genuinely prefer different polynomial orders. Each such pair carries three signatures: an AIC-mode change between the two bins, only two or three experiments contributing to both bins (so the basis-rotation effect is not diluted), and noticeably different per-bin band-scale factors. The negatives are not an artifact of the multigroup correlation threshold (the strong-negative tail is bit-identical with the threshold on or off), of Higham nearest-PSD repair (the assembled covariance is PSD by construction through the diagonal congruence of Sec. 4.6), or of stochastic kernel weights (the weights are deterministic functions of the TOF parameters).

A downstream user that sees $\rho \approx -1$ at adjacent bins should read it correctly: the two bins' c_ℓ values cannot be moved together, because moving one up forces the other down through the change of polynomial basis that the data and the AIC weights both prefer. This is an real expression of the model-form uncertainty exposed by the change of nominal degree between adjacent bins and not a defect of the construction.

E.10. Validity of the Gaussian Covariance Representation

The published `MF34` is a Gaussian, second-moment summary: it records the variance of each Legendre coefficient and the Pearson correlation between coefficients, and downstream sampling draws from the multivariate normal it defines. The adequacy of this representation was checked against the Monte Carlo ensemble from which it is built, asking whether the dependence between coefficients is linear enough for a Pearson correlation to capture it. The check compares, for each pair of coefficients, the mutual information estimated from the samples against the value a Gaussian of the same correlation would imply.

For most coefficient pairs the measured mutual information matches the value implied by a Gaussian of the same correlation, confirming that the second-moment summary captures the dependence and that the covariance reproduces the bulk of the sample structure. The departures are confined to the adjacent-bin pairs of the 1.0–1.8 MeV resonance band that also produce the strong negative correlations of Sec. E.9, where the change of polynomial basis between adjacent

bins makes the joint distribution genuinely non-Gaussian. There the ensemble carries model-form structure that a second-moment summary can only partially reflect, consistent with the systematic residual chi-squared excess of Sec. 5.4; a user that needs that structure should sample the pipeline ensemble directly rather than regenerate it from `MF34`.

E.11. Pipeline Diagnostics

This section reports detailed diagnostics from the pipeline execution summarized in Sec. 5.1.

Legendre order selection

The AIC criterion selects $L = 3$ or $L = 4$ for the majority of the fitted bins; the full per-order distribution, alongside the AIC vs. AICc vs. BIC comparison, is reported in Sec. E.4 (Table E.2).

Covariance assembly

The Monte Carlo sampling step generates 10 000 correlated Legendre coefficient samples from the combined covariance matrix of 10 428 parameters (Sec. 5.1). The two-pass kernel-weight scheme combines its correlation and per-bin variance passes through a congruence transform, which is positive semi-definite (PSD) by construction. The assembled covariance has a minimum eigenvalue of -1.4×10^{-14}, at the level of numerical round-off, so the 10 000 samples are drawn directly by Cholesky decomposition, with a small ridge nugget of 4.4×10^{-12} added for stability. The sample-to-target standard-deviation ratio averages 1.00, confirming that the draws reproduce the target variance structure. The near-zero regularization step identified 82 parameters with signal-to-noise ratio below 1.0 and replaced their variances with smoothed neighbor values. The full correlation matrix of the assembled covariance is shown in Figure E.4; the $\ell = 1$ sub-block appears alone in Chapter III, Figure III.15.

Multigroup collapse

The adaptive collapse reduces 1 738 fine bins to 526 multigroups (265 single-bin groups and 261 multi-bin groups), using a minimum adjacent-bin correlation threshold of $\rho_{\min} = 0.85$ on the $\ell = 1$ correlation. Among the multi-bin groups, the minimum within-group correlation ranges from 0.85 to near unity (mean 0.93), and the within-group standard-deviation ratio reaches its configured cap of 5.0 (mean 2.8). The multigroup covariance (3 156 parameters $= 526 \times 6$) passed PSD checks with a minimum eigenvalue of -6.9×10^{-15}, within numerical precision.

Parallelization and runtime

The kernel-weight Monte Carlo sampling accounts for approximately 4.8 of the 5.0 hours of total runtime. This step is parallelized across energy bins using `multiprocessing.Pool` with 40 workers; each worker processes independent energy bins without inter-process communication. The remaining steps (data loading, fitting, covariance assembly, multigroup collapse, ENDF writing) run sequentially and complete in under one minute combined.

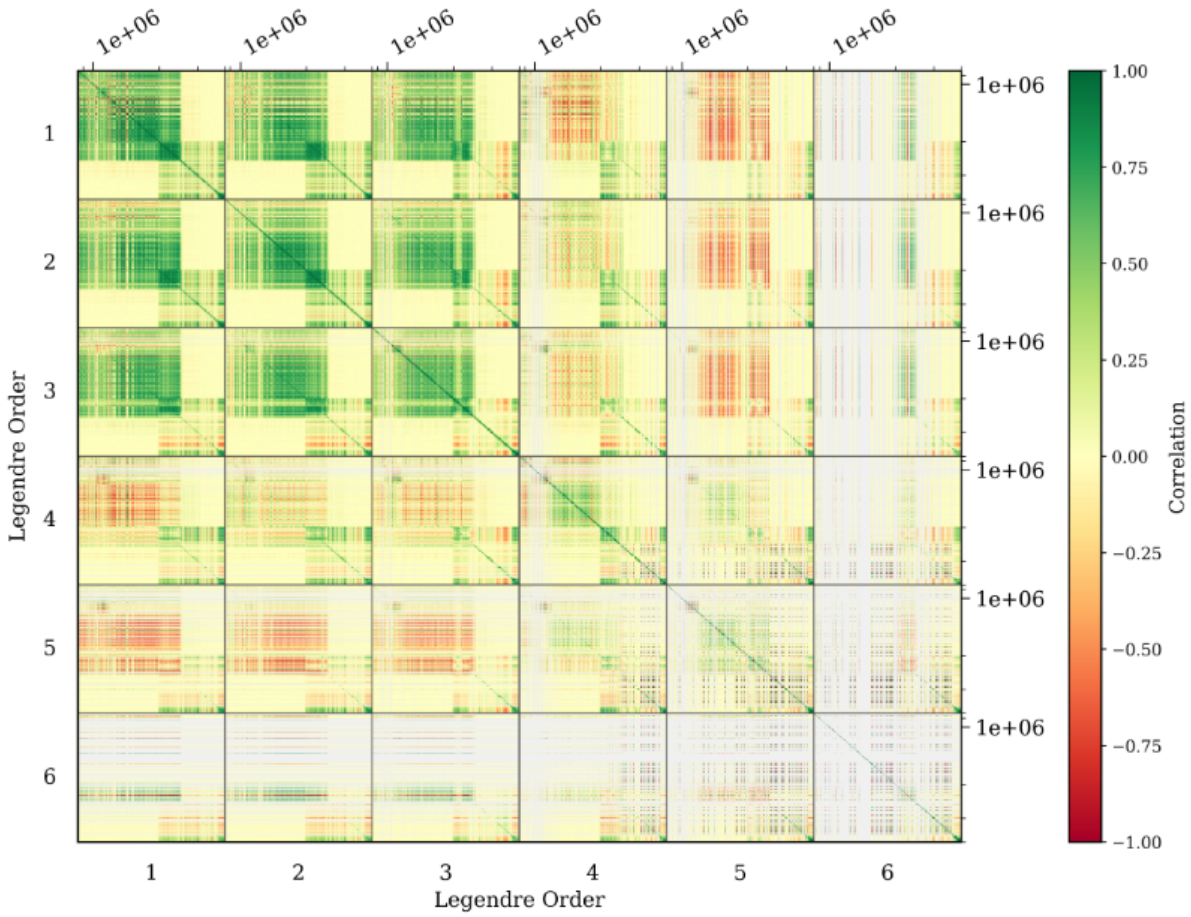


Figure E.4: Full correlation matrix of the Legendre coefficients ($\ell = 1$–6) of this work's evaluation for ^{56}Fe elastic scattering across the 847 keV–4 MeV range. The block structure groups parameters by Legendre order; within each block, rows and columns run over the 526 adaptive multigroups produced by the collapse step. The $\ell = 1$ sub-block on the diagonal is reproduced in Chapter III, Figure III.15.

E.12. Criticality Benchmark Cross-Check: Full-Suite Figure

Section 5.5 summarizes the criticality cross-check of the central values on two subsets and the fifteen most sensitive benchmarks. Figure E.5 extends that comparison to the full suite: it shows the calculated k_{eff} for both evaluations against the experimental values, and their per-benchmark difference $C-C$, over all 88 benchmarks.

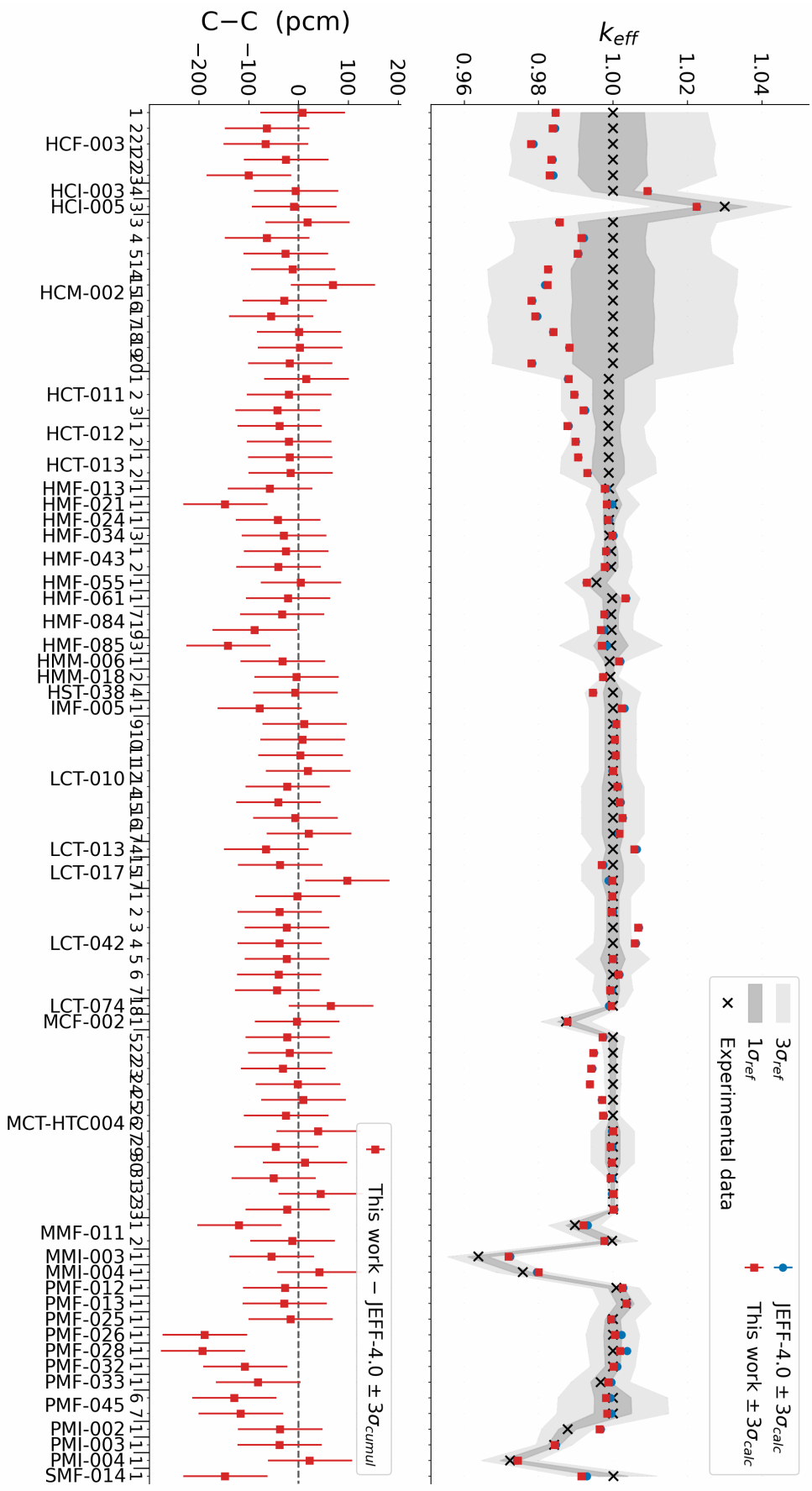


Figure E.5: Calculated k_{eff} for JEFF-4.0 and after the ^{56}Fe swap across all 88 ICSBEP benchmarks (top), with the experimental values and their 1σ and 3σ bands, and the per-benchmark difference $C-C$ between the two evaluations (bottom). Benchmarks are in alphabetical order.

Appendix F

Supplementary Material for Chapter IV

This appendix collects the supporting material for Chapter IV, ordered as the sections are first referenced in the main text. The first group supports the benchmark analysis. Section F.1 gives the C/E results with propagated nuclear-data uncertainties for the ASPIS-88 foils not shown in the body, Section F.2 the single-isotope library swap, Section F.3 the effect of the Chapter III angular evaluation, Section F.4 the per-foil uncertainty tables, and Section F.5 the isotopes perturbed in the VENUS-3 propagation. The second group supports the PWR900 analysis. Section F.6 extends the reference tally values to the three libraries, Sections F.7 and F.8 give the per-assembly distributions of the fission neutron production rate and their spatial contribution, and Sections F.9 to F.11 give the fast flux above 0.1 MeV, the azimuth-by-height uncertainty maps, and the highest-uncertainty tallies with the largest cross-library spreads. Section F.12 gives the nuclear-data budget of every response at the three vessel azimuths and at the capsule.

F.1. C/E with Nuclear-Data Uncertainties for the Remaining Foils

Figures F.1 to F.3 show the C/E ratios of the ^{197}Au(n,γ), ^{32}S(n,p), and ^{27}Al(n,α) foils at every measured position, with the same layout as Figure IV.3. The foil-by-foil behavior is discussed in Sec. 2.3.

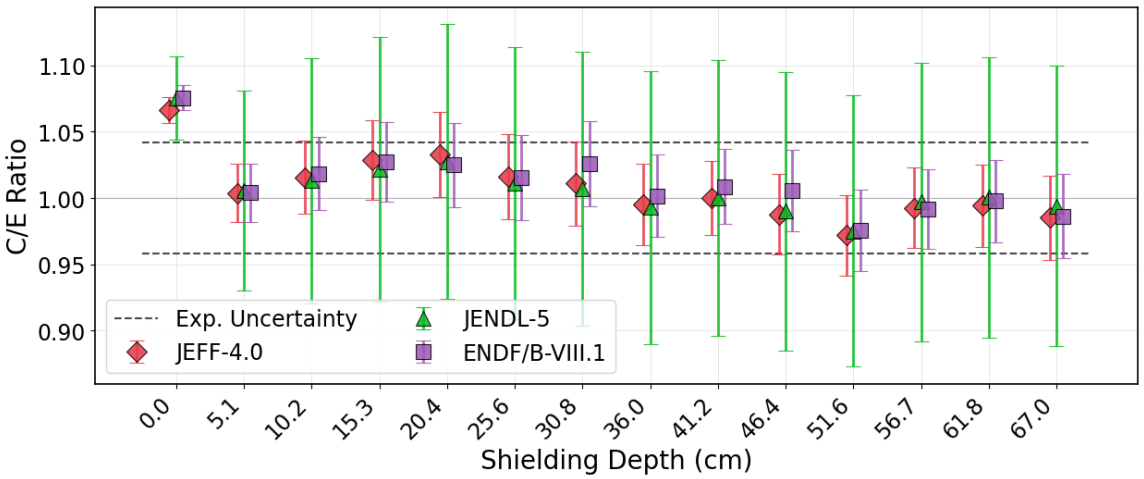


Figure F.1: C/E ratio for the ^{197}Au(n,γ) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3.

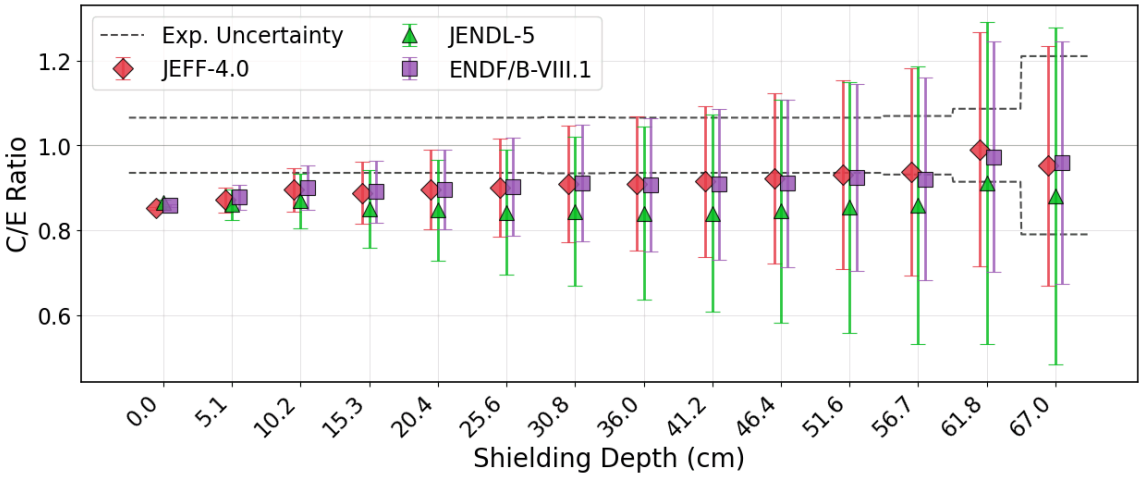


Figure F.2: C/E ratio for the ^{32}S(n,p) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3.

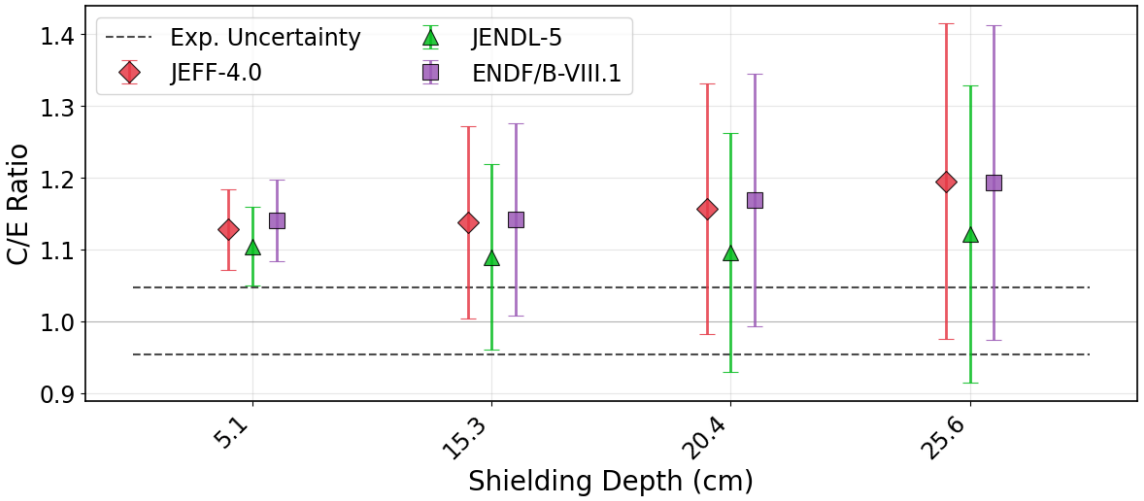


Figure F.3: C/E ratio for the ^{27}Al(n,α) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3.

F.2. Single-Isotope Library Swap

Figures F.4 to F.8 show, for each foil, the C/E of the nominal JEFF-4.0 and JENDL-5 calculations together with the two cross-library runs in which only the ^{56}Fe evaluation is exchanged between the libraries. Each swapped curve moves onto the other library's nominal result, confirming that the inter-library C/E difference is carried by the ^{56}Fe evaluation, as discussed in Sec. 2.4. The error bars are the statistical (Monte Carlo) uncertainty of the MCNP runs; the nuclear-data uncertainty is not propagated in this comparison.

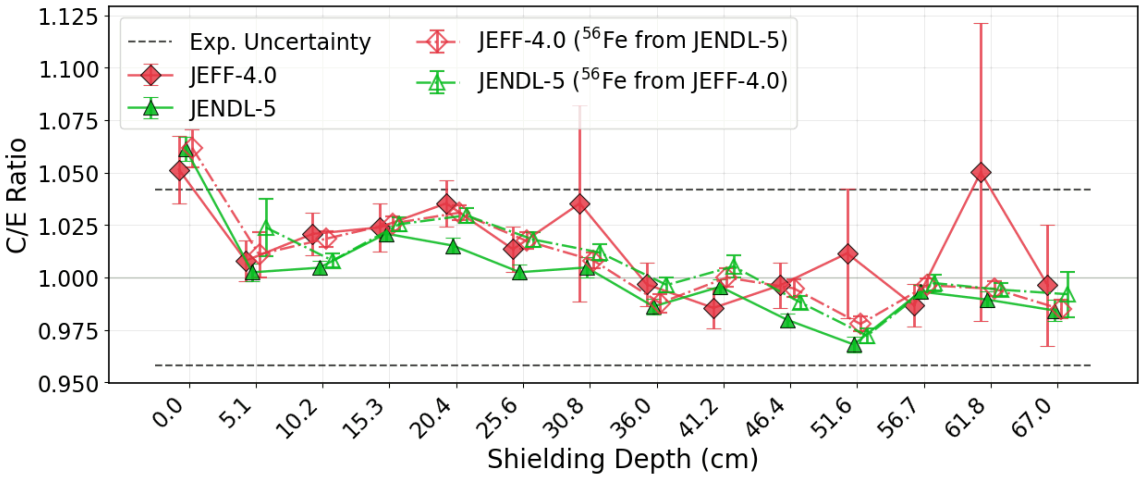


Figure F.4: C/E ratio for the ^{197}Au(n,γ) reaction rate, nominal libraries and ^{56}Fe swap.

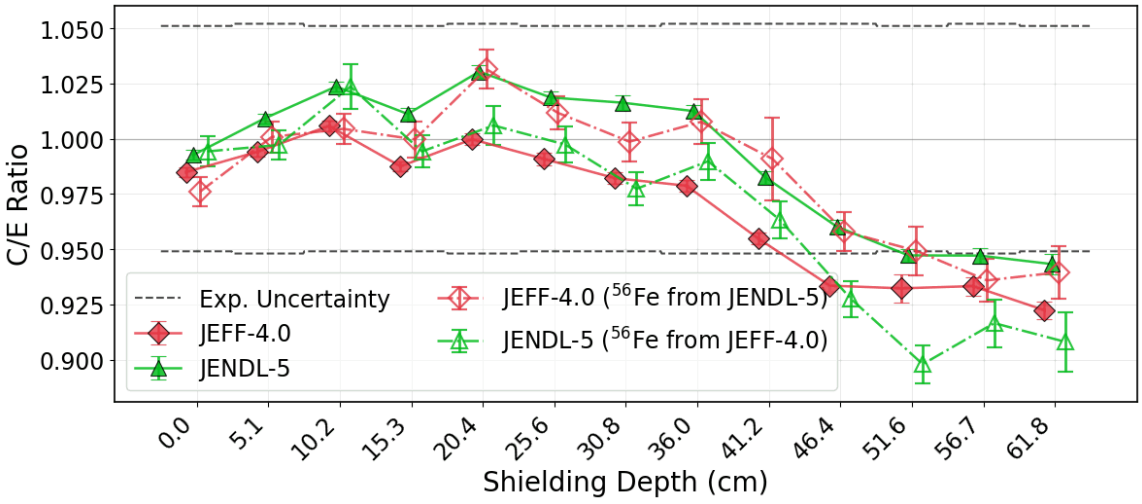


Figure F.5: C/E ratio for the ^{103}Rh(n,n$'$) reaction rate, nominal libraries and ^{56}Fe swap.

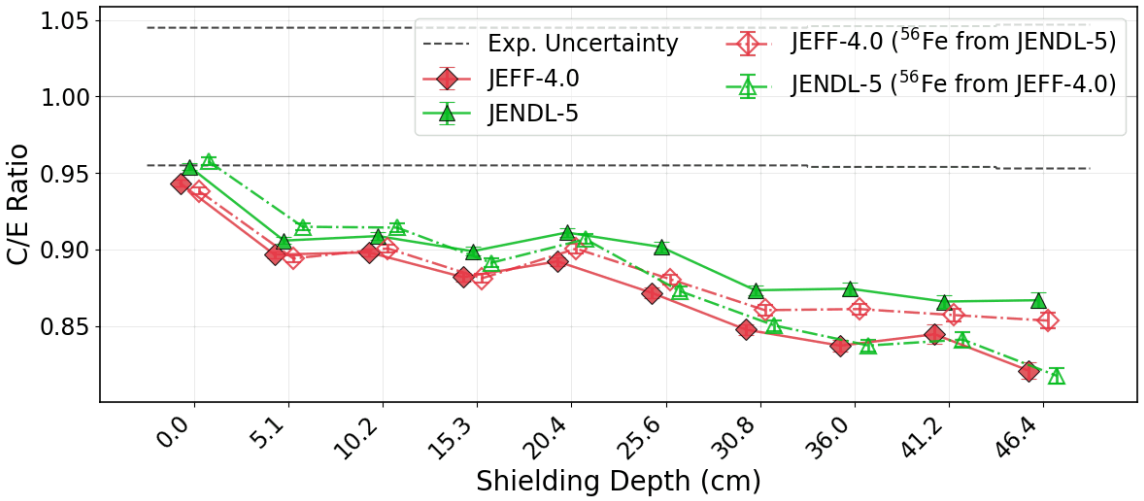


Figure F.6: C/E ratio for the ^{115}In(n,n′) reaction rate, nominal libraries and ^{56}Fe swap.

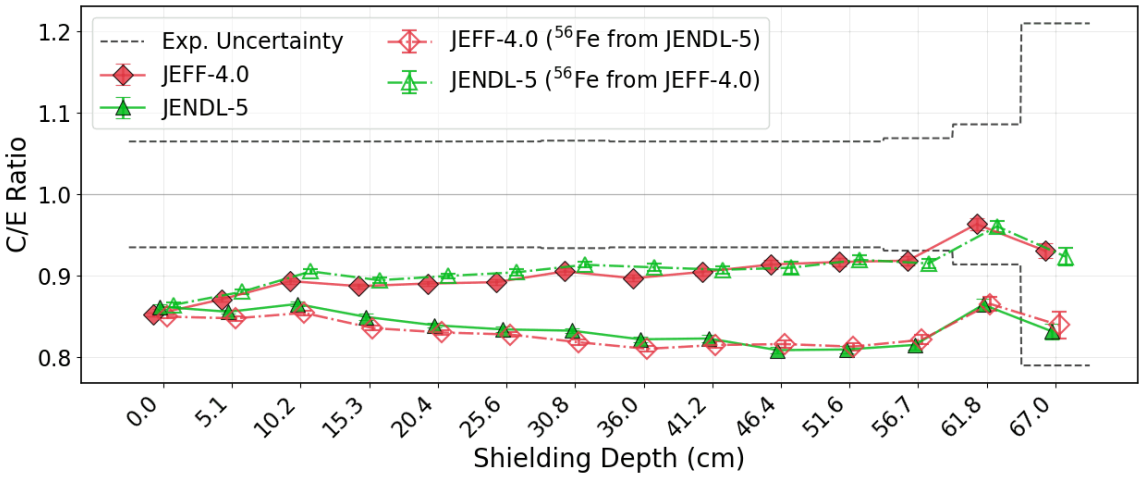


Figure F.7: C/E ratio for the ^{32}S(n,p) reaction rate, nominal libraries and ^{56}Fe swap.

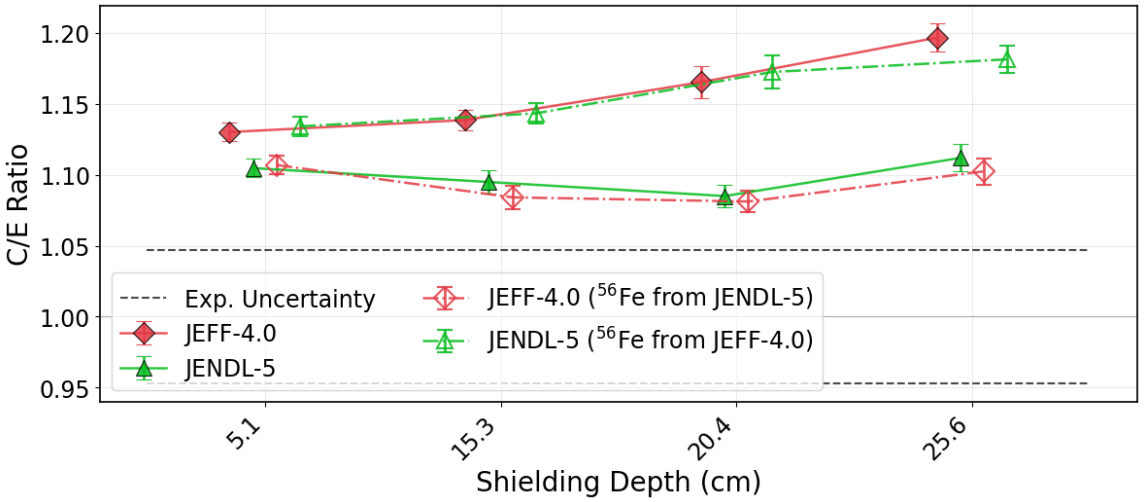


Figure F.8: C/E ratio for the ^{27}Al(n,α) reaction rate, nominal libraries and ^{56}Fe swap.

F.3. Chapter-3 Angular Evaluation Swap

Figures F.9 and F.10 show the per-position C/E of ^{32}S(n,p) and ^{115}In(n,n′) for the JEFF-4.0 angular distribution and for the Chapter III evaluation, with the ^{56}Fe cross sections held fixed. These are the two foils whose agreement changes beyond the Monte Carlo statistics, as summarized in Table IV.3. The error bars are the statistical uncertainty of the MCNP runs.

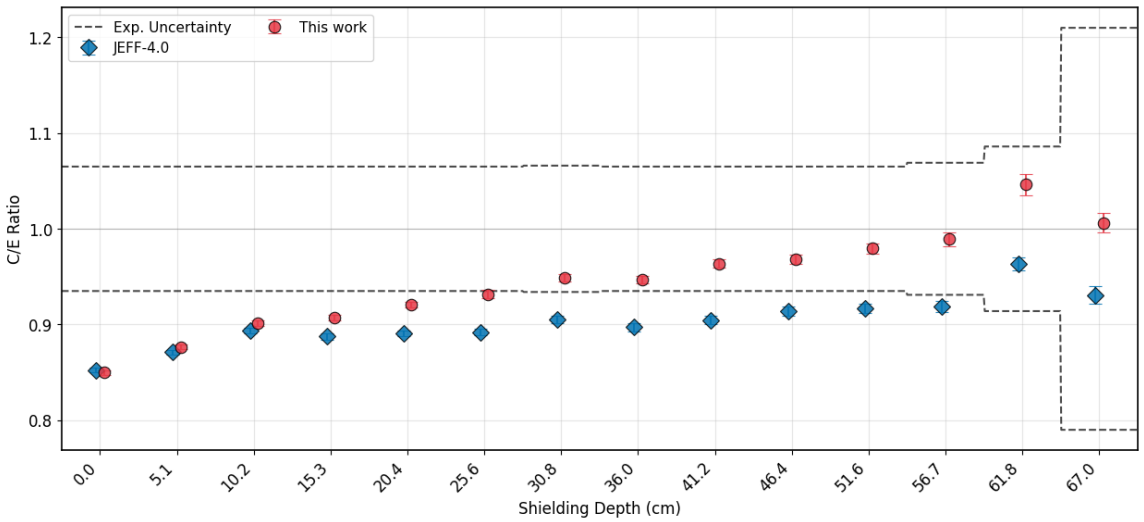


Figure F.9: C/E ratio for the ^{32}S(n,p) reaction rate, JEFF-4.0 angular distribution and Chapter III evaluation.

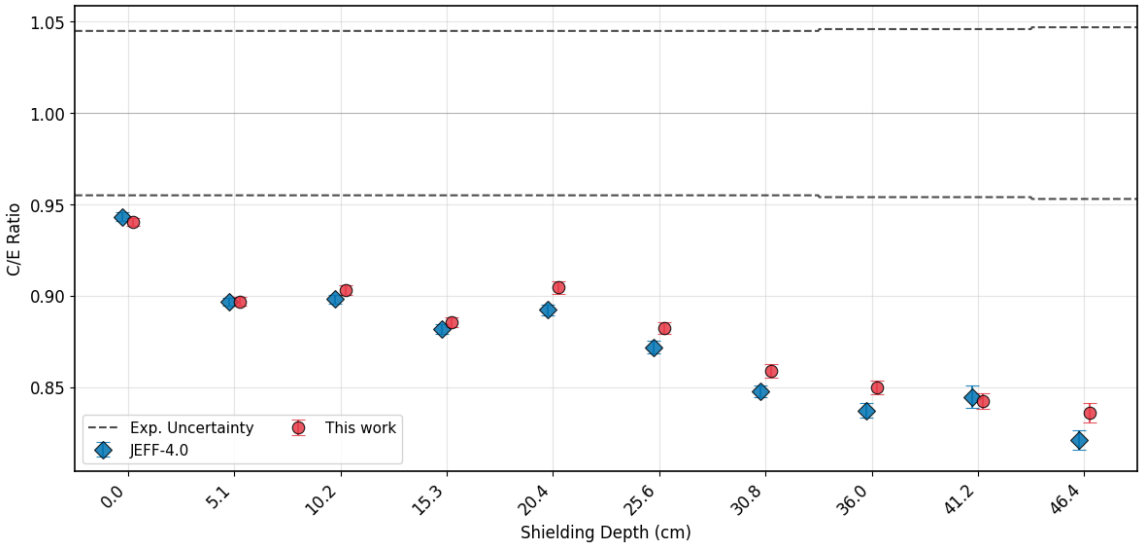


Figure F.10: C/E ratio for the ^{115}In(n,n′) reaction rate, JEFF-4.0 angular distribution and Chapter III evaluation.

F.4. Per-Foil Uncertainty Tables

Tables F.1 to F.5 report, for each foil and measured position of the ASPIS-88 iron benchmark (Sec. 2), the C/E ratio of the three libraries and the propagated ^{56}Fe nuclear-data uncertainties of the cross section (XS) and angular-distribution (AD) channels. For JEFF-4.0 and JENDL-5

the C/E is obtained from the mean of the perturbed populations, and for ENDF/B-VIII.1 from its nominal calculation. The uncertainties are given in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. Where the propagated component is small, the population spread is dominated by the statistical error of the MCNP runs and the decomposition is not resolved; the confidence interval then extends to zero. The combined uncertainty shown in the C/E figures is the quadrature sum of the XS and AD columns of the corresponding library.

Table F.1: Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{197}Au(n,γ) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses.

	C/E			XS unc. (%)		AD unc. (%)	
Depth (cm)	JEFF-4.0	JENDL-5	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5
0.0	1.066	1.075	1.076	0.9 (0.6–1.1)	2.9 (2.7–3.1)	0.2 (0.0–0.5)	0.0 (0.0–0.6)
5.1	1.004	1.005	1.004	2.2 (2.1–2.3)	7.5 (7.0–7.9)	0.0 (0.0–0.3)	0.4 (0.1–0.6)
10.2	1.016	1.013	1.018	2.7 (2.5–2.8)	9.1 (8.6–9.6)	0.5 (0.2–0.7)	0.3 (0.2–0.4)
15.3	1.029	1.022	1.027	2.9 (2.4–3.3)	9.8 (9.2–10.3)	0.0 (0.0–0.5)	0.1 (0.0–0.3)
20.4	1.033	1.028	1.025	3.1 (2.8–3.3)	10.1 (9.5–10.6)	0.4 (0.2–0.5)	0.2 (0.0–0.3)
25.6	1.016	1.011	1.015	3.1 (2.9–3.3)	10.1 (9.6–10.7)	0.4 (0.3–0.5)	0.1 (0.0–0.2)
30.8	1.011	1.007	1.026	3.1 (2.9–3.3)	10.3 (9.7–10.8)	0.5 (0.3–0.6)	0.2 (0.0–0.3)
36.0	0.995	0.993	1.002	3.1 (2.8–3.3)	10.3 (9.8–10.9)	0.4 (0.3–0.6)	0.3 (0.2–0.4)
41.2	1.000	1.000	1.009	2.8 (2.3–3.1)	10.4 (9.8–11.0)	0.4 (0.3–0.5)	0.5 (0.3–0.6)
46.4	0.988	0.990	1.005	3.0 (2.8–3.2)	10.6 (9.9–11.2)	0.1 (0.0–0.4)	0.4 (0.3–0.5)
51.6	0.972	0.975	0.976	3.1 (2.9–3.3)	10.4 (9.9–11.0)	0.4 (0.0–0.6)	0.6 (0.5–0.8)
56.7	0.993	0.997	0.992	3.0 (2.8–3.2)	10.5 (9.9–11.1)	0.2 (0.0–0.3)	0.6 (0.5–0.7)
61.8	0.994	1.000	0.998	3.1 (2.9–3.3)	10.6 (10.0–11.2)	0.4 (0.1–0.6)	0.7 (0.6–0.7)
67.0	0.985	0.994	0.986	3.2 (3.0–3.4)	10.6 (10.0–11.3)	0.3 (0.0–0.4)	0.5 (0.0–1.0)

Table F.2: Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{103}Rh(n,n′) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses.

	C/E			XS unc. (%)		AD unc. (%)		
Depth (cm)	JEFF-4.0	JENDL-5	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5	This work
0.0	0.982	0.992	0.973	0.6 (0.4–0.7)	1.3 (1.2–1.4)	0.1 (0.0–0.4)	0.7 (0.6–0.9)	0.3 (0.0–0.5)
5.1	0.993	1.008	0.997	1.5 (1.4–1.7)	3.2 (3.0–3.4)	0.3 (0.0–0.5)	0.7 (0.6–0.8)	0.0 (0.0–0.3)
10.2	1.005	1.025	1.019	2.2 (2.1–2.4)	4.6 (4.2–4.9)	0.3 (0.0–0.5)	1.7 (1.6–1.8)	0.5 (0.4–0.7)
15.3	0.985	1.009	0.982	2.7 (2.5–3.0)	5.4 (5.1–5.8)	0.5 (0.2–0.6)	2.8 (2.6–3.0)	0.9 (0.8–1.0)
20.4	1.002	1.030	1.009	3.1 (2.9–3.3)	6.1 (5.6–6.5)	0.9 (0.8–1.0)	3.6 (3.4–3.9)	1.2 (1.1–1.4)
25.6	0.996	1.027	0.981	3.5 (3.3–3.7)	6.6 (6.1–7.0)	1.2 (1.0–1.3)	4.3 (4.0–4.5)	1.3 (1.2–1.4)
30.8	0.984	1.015	1.010	3.6 (3.3–3.8)	6.8 (6.3–7.3)	1.4 (1.2–1.5)	4.8 (4.5–5.1)	1.7 (1.6–1.9)
36.0	0.982	1.014	0.973	3.7 (3.4–3.9)	7.2 (6.6–7.7)	1.7 (1.5–1.8)	5.2 (4.9–5.5)	1.7 (1.5–1.9)
41.2	0.964	0.991	0.960	3.9 (3.7–4.2)	7.6 (7.0–8.2)	1.8 (1.6–1.9)	5.6 (5.2–5.9)	2.1 (1.9–2.4)
46.4	0.935	0.963	0.922	4.2 (3.8–4.4)	7.8 (7.2–8.5)	2.1 (1.9–2.3)	5.8 (5.4–6.2)	2.3 (2.1–2.5)
51.6	0.928	0.953	0.964	4.3 (4.0–4.7)	8.2 (7.5–8.8)	2.4 (2.2–2.6)	6.1 (5.7–6.5)	2.7 (2.4–3.1)
56.7	0.926	0.950	0.953	4.3 (4.0–4.6)	8.5 (7.8–9.1)	2.6 (2.4–2.8)	6.3 (5.8–6.7)	2.7 (2.4–2.9)
61.8	0.928	0.948	0.962	4.6 (4.2–5.0)	9.0 (8.2–9.9)	3.2 (2.7–3.7)	6.6 (6.2–7.0)	3.1 (2.7–3.4)

Table F.3: Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{115}In(n,n′) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses.

	C/E			XS unc. (%)		AD unc. (%)		
Depth (cm)	JEFF-4.0	JENDL-5	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5	This work
0.0	0.941	0.954	0.944	0.5 (0.5–0.6)	1.4 (1.3–1.4)	0.5 (0.5–0.6)	0.6 (0.6–0.7)	0.2 (0.1–0.2)
5.1	0.899	0.909	0.903	2.3 (2.2–2.5)	4.8 (4.5–5.1)	2.4 (2.2–2.5)	1.2 (1.1–1.2)	0.4 (0.3–0.4)
10.2	0.901	0.914	0.904	3.8 (3.6–4.0)	7.6 (7.1–8.2)	3.8 (3.5–4.0)	3.0 (2.8–3.1)	0.9 (0.8–0.9)
15.3	0.884	0.901	0.888	5.0 (4.7–5.3)	9.9 (9.3–10.6)	5.0 (4.7–5.3)	4.6 (4.3–4.8)	1.3 (1.2–1.4)
20.4	0.895	0.918	0.898	6.0 (5.6–6.3)	11.9 (11.1–12.7)	6.1 (5.7–6.4)	6.0 (5.6–6.3)	1.8 (1.7–1.9)
25.6	0.876	0.904	0.873	6.7 (6.3–7.1)	13.5 (12.6–14.4)	6.8 (6.4–7.2)	7.3 (6.9–7.7)	2.1 (2.0–2.3)
30.8	0.851	0.884	0.852	7.4 (6.9–7.8)	14.7 (13.7–15.7)	7.4 (7.0–7.9)	8.4 (7.9–8.9)	2.5 (2.3–2.6)
36.0	0.848	0.888	0.848	7.8 (7.4–8.3)	15.6 (14.5–16.6)	7.9 (7.4–8.4)	9.4 (8.9–10.0)	2.8 (2.6–3.0)
41.2	0.843	0.888	0.841	8.2 (7.7–8.6)	16.2 (15.0–17.3)	8.4 (7.9–8.9)	10.3 (9.7–10.9)	3.1 (2.9–3.3)
46.4	0.831	0.879	0.822	8.4 (7.9–8.8)	16.8 (15.5–18.0)	8.7 (8.2–9.2)	11.1 (10.4–11.7)	3.3 (3.0–3.5)

Table F.4: Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{32}S(n,p) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses.

	C/E			XS unc. (%)		AD unc. (%)		
Depth (cm)	JEFF-4.0	JENDL-5	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5	This work
0.0	0.852	0.864	0.858	0.4 (0.4–0.5)	0.8 (0.8–0.9)	0.1 (0.0–0.1)	0.4 (0.3–0.4)	0.1 (0.1–0.2)
5.1	0.871	0.860	0.878	3.4 (3.2–3.6)	3.9 (3.7–4.2)	0.3 (0.2–0.3)	1.3 (1.2–1.4)	0.4 (0.3–0.4)
10.2	0.895	0.869	0.900	5.8 (5.4–6.1)	6.9 (6.4–7.4)	0.6 (0.5–0.6)	3.0 (2.8–3.1)	0.9 (0.8–0.9)
15.3	0.888	0.850	0.891	8.1 (7.6–8.6)	9.8 (9.1–10.5)	0.9 (0.8–1.0)	4.6 (4.3–4.9)	1.4 (1.3–1.5)
20.4	0.895	0.847	0.895	10.4 (9.7–11.0)	12.7 (11.9–13.7)	1.2 (1.1–1.3)	6.1 (5.7–6.5)	1.9 (1.8–2.0)
25.6	0.900	0.842	0.902	12.7 (11.9–13.4)	15.7 (14.7–16.9)	1.5 (1.4–1.6)	7.6 (7.1–8.1)	2.3 (2.2–2.5)
30.8	0.909	0.844	0.911	15.0 (14.0–15.8)	18.7 (17.4–20.1)	1.8 (1.7–1.9)	9.1 (8.5–9.7)	2.9 (2.7–3.0)
36.0	0.910	0.839	0.906	17.2 (16.1–18.1)	21.9 (20.4–23.4)	2.0 (1.9–2.2)	10.6 (9.9–11.3)	3.4 (3.2–3.6)
41.2	0.915	0.840	0.908	19.4 (18.2–20.5)	24.9 (23.1–26.7)	2.3 (2.2–2.5)	12.1 (11.3–12.8)	3.8 (3.6–4.1)
46.4	0.922	0.845	0.910	21.6 (20.2–22.8)	28.1 (26.1–30.1)	2.5 (2.3–2.6)	13.5 (12.6–14.3)	4.4 (4.1–4.6)
51.6	0.931	0.853	0.924	23.7 (22.2–25.1)	31.3 (28.9–33.6)	2.7 (2.5–2.9)	14.9 (14.0–15.9)	4.9 (4.6–5.2)
56.7	0.937	0.859	0.920	25.8 (24.1–27.3)	34.4 (31.7–37.0)	2.9 (2.7–3.2)	16.3 (15.3–17.4)	5.4 (5.0–5.7)
61.8	0.990	0.911	0.973	27.7 (25.8–29.3)	37.7 (34.6–40.6)	3.2 (3.0–3.6)	17.7 (16.5–18.8)	5.9 (5.5–6.2)
67.0	0.952	0.881	0.959	29.6 (27.6–31.3)	40.9 (37.5–44.2)	3.3 (3.1–3.6)	19.1 (17.8–20.3)	7.1 (6.0–8.4)

Table F.5: Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{27}Al(n,α) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses.

	C/E			XS unc. (%)		AD unc. (%)	
Depth (cm)	JEFF-4.0	JENDL-5	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	JEFF-4.0	JENDL-5
5.1	1.128	1.105	1.140	4.9 (4.6–5.2)	4.9 (4.6–5.2)	0.6 (0.6–0.7)	0.7 (0.5–0.7)
15.3	1.138	1.090	1.142	11.6 (11.0–12.3)	11.6 (10.8–12.3)	2.0 (1.9–2.2)	2.4 (2.2–2.5)
20.4	1.157	1.096	1.169	14.8 (14.0–15.7)	14.9 (13.9–15.7)	2.8 (2.6–3.0)	3.1 (3.0–3.3)
25.6	1.195	1.121	1.194	18.0 (17.0–19.1)	18.1 (16.9–19.1)	3.5 (3.3–3.7)	3.9 (3.6–4.1)

F.5. Perturbed Isotopes in the VENUS-3 Propagation

Table F.6 lists, for each of the four evaluated libraries, the isotopes of the VENUS-3 attenuation path that carry covariance data and are perturbed in the uncertainty propagation of Sec. 3.4. The coverage varies between libraries, which sets the per-library isotope counts quoted in the main text.

Table F.6: Isotopes perturbed in the VENUS-3 nuclear-data uncertainty propagation for each evaluated library. A check mark indicates the isotope carries covariance data in that library and is perturbed; a dash indicates it is absent.

Isotope	JEFF-3.3	JEFF-4.0	ENDF/B-VIII.0	ENDF/B-VIII.1
^{1}H	✓	✓	✓	✓
^{10}B	✓	✓	✓	✓
^{11}B	✓	✓	✓	✓
^{12}C	—	✓	✓	✓
^{13}C	✓	✓	✓	✓
^{16}O	✓	✓	✓	✓
^{23}Na	✓	✓	✓	✓
^{28}Si	✓	✓	✓	✓
^{29}Si	✓	✓	✓	✓
^{30}Si	✓	✓	✓	✓
^{31}P	✓	✓	—	✓
^{32}S	✓	✓	—	✓
^{33}S	✓	✓	—	—
^{34}S	✓	✓	—	—
^{36}S	✓	✓	—	—
^{39}K	✓	✓	—	—
^{41}K	✓	✓	✓	✓
^{50}Cr	✓	✓	✓	✓
^{52}Cr	✓	✓	✓	✓
^{53}Cr	✓	✓	✓	✓
^{54}Cr	✓	—	—	—
^{55}Mn	✓	✓	✓	✓
^{54}Fe	✓	✓	✓	✓
^{56}Fe	✓	✓	✓	✓
^{57}Fe	✓	—	—	—
^{58}Fe	✓	✓	—	—
^{59}Co	✓	—	—	✓
^{58}Ni	✓	✓	✓	✓
^{60}Ni	✓	✓	✓	✓
^{61}Ni	—	✓	—	—
^{62}Ni	✓	✓	—	—
^{64}Ni	—	✓	—	—
^{92}Mo	—	✓	✓	✓
^{94}Mo	—	✓	✓	✓
^{95}Mo	✓	✓	✓	✓
^{96}Mo	—	✓	✓	✓
^{97}Mo	—	✓	✓	✓
^{98}Mo	—	✓	✓	✓
^{100}Mo	—	✓	✓	✓
^{106}Cd	—	✓	—	—
^{108}Cd	—	✓	—	—
^{110}Cd	—	✓	—	—
^{111}Cd	—	✓	—	—
^{112}Cd	—	✓	—	—
^{113}Cd	—	✓	—	—
^{114}Cd	—	✓	—	—
^{116}Cd	—	✓	—	—
Total	30/47	44/47	26/47	29/47

F.6. Reference Tally Values Across Libraries

Table F.7 extends the reference values of Table IV.5 of Chapter IV to the three libraries propagated in this work. All values come from the nominal run of each library, with unperturbed nuclear data, in MCNP normalization (per source neutron). Vessel quantities are height-integrated over the ±60 cm central window; capsule reaction rates are averaged over the three capsule axial detectors at the −17° position. ENDF/B-VIII.1 and JEFF-4.0 agree to within 0.3% on the vessel and to within 1% on the capsule flux and the fission and (n,p) dosimeters of ^{58}Ni and ^{54}Fe; on the two highest-threshold dosimeters, ^{46}Ti(n,p) and ^{63}Cu(n,α), ENDF/B-VIII.1 is 2.6% and 5.5% higher. JENDL-5 is 3–5% lower than the other two libraries at the 0° and −20° azimuths and on the capsule flux, and agrees with them to within 0.5% at −28°. The flux spread is set against the propagated uncertainty in Sec. 6.2.

Table F.7: Reference central values of the vessel and capsule tallies, by library. Vessel quantities are height-integrated over the ±60 cm central window; capsule reaction rates are averaged over the three capsule positions.

Quantity	Location	ENDF/B-VIII.1	JEFF-4.0	JENDL-5
Flux $\Phi_{E>1\,\mathrm{MeV}}$ (n/cm^2/s)	Vessel, 0°	1.47×10^{-10}	1.47×10^{-10}	1.42×10^{-10}
	Vessel, −20°	7.55×10^{-11}	7.56×10^{-11}	7.31×10^{-11}
	Vessel, −28°	1.04×10^{-10}	1.04×10^{-10}	1.04×10^{-10}
	Capsule, −17°	7.56×10^{-10}	7.63×10^{-10}	7.25×10^{-10}
NRT-dpa (dpa/s)	Vessel, 0°	2.05×10^{-7}	2.05×10^{-7}	1.99×10^{-7}
	Vessel, −20°	1.10×10^{-7}	1.10×10^{-7}	1.07×10^{-7}
	Vessel, −28°	1.41×10^{-7}	1.41×10^{-7}	1.42×10^{-7}
ARC-dpa (dpa/s)	Vessel, 0°	6.66×10^{-8}	6.64×10^{-8}	6.45×10^{-8}
	Vessel, −20°	3.61×10^{-8}	3.61×10^{-8}	3.50×10^{-8}
	Vessel, −28°	4.50×10^{-8}	4.51×10^{-8}	4.52×10^{-8}
^{237}Np(n,f) rate (s^{-1})	Capsule	2.57×10^{-9}	2.59×10^{-9}	2.50×10^{-9}
^{238}U(n,f) rate (s^{-1})	Capsule	2.41×10^{-10}	2.42×10^{-10}	2.31×10^{-10}
^{58}Ni(n,p) rate (s^{-1})	Capsule	5.94×10^{-11}	5.96×10^{-11}	5.66×10^{-11}
^{54}Fe(n,p) rate (s^{-1})	Capsule	4.15×10^{-11}	4.15×10^{-11}	3.95×10^{-11}
^{46}Ti(n,p) rate (s^{-1})	Capsule	6.94×10^{-12}	6.76×10^{-12}	6.72×10^{-12}
^{63}Cu(n,α) rate (s^{-1})	Capsule	3.83×10^{-13}	3.63×10^{-13}	3.70×10^{-13}

F.7. Per-Assembly Distributions of νR_f

Figure F.11 shows the per-assembly distributions of νR_f obtained from the 1000 nuclear data samples. Most are close to Gaussian and centered on the unperturbed value, so the relative uncertainty used in Figure IV.15 captures the dominant spread; a few assemblies show mild departures from normality, for which a single relative-uncertainty value is only an approximate description of the spread. The separation between UOX and MOX assemblies, and the effect of burnup, is visible both in the central value and in the width of each histogram.

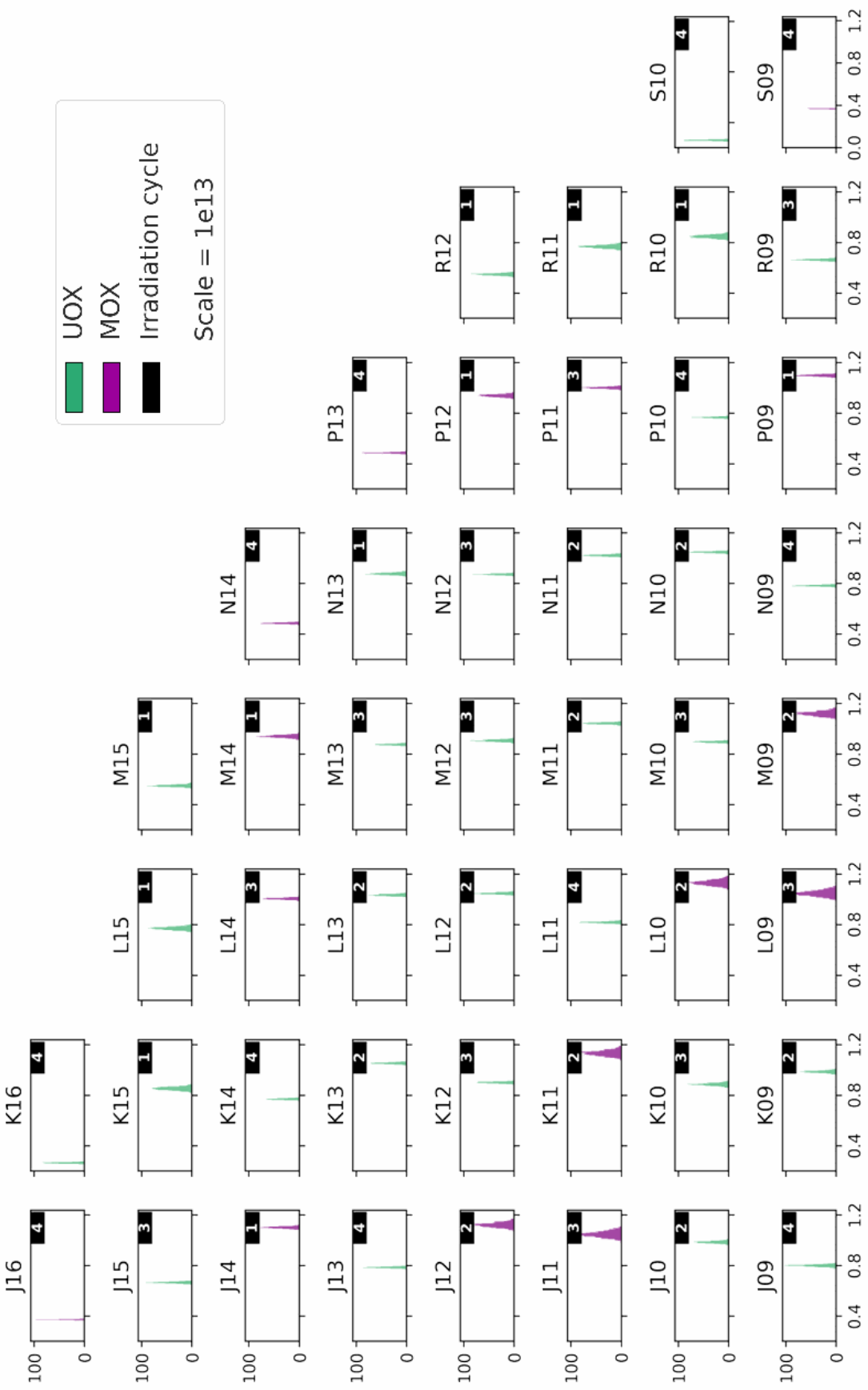
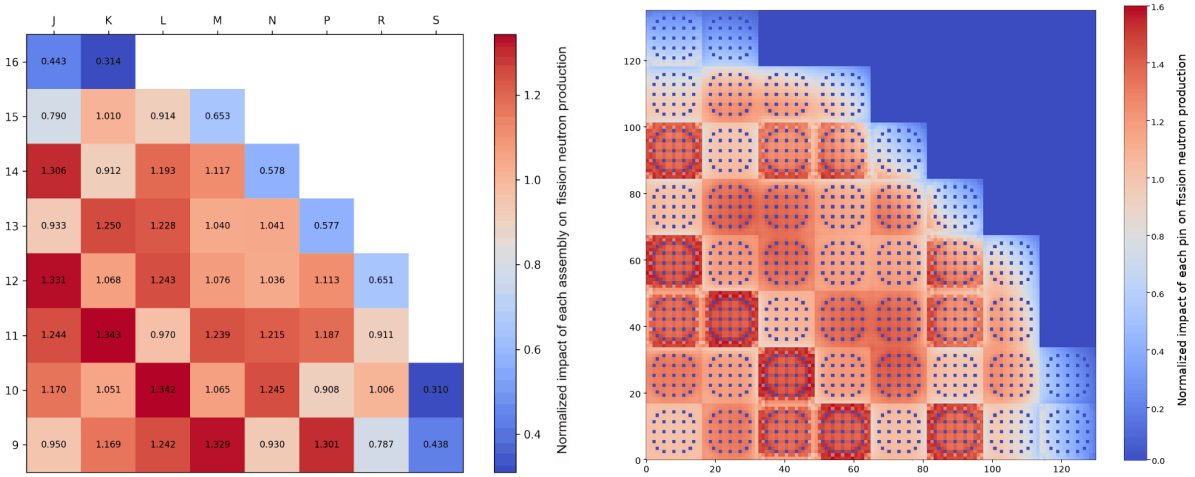


Figure F.11: Per-assembly distributions of the fission neutron production rate νR_f from 1000 nuclear data samples. The horizontal axis shows the production rate in units of 10^{13} n/s/cm^3, the vertical axis the frequency, and the numbers indicate the irradiation cycle.

F.8. Spatial Contribution to Fission Neutron Production

Figure F.12 shows the normalized contribution to the total fission neutron production in the core, resolved at the assembly level and at the pin level. The central MOX clusters dominate the production, in line with the loading strategy that places the most reactive assemblies away from the vessel. The peripheral assemblies, although individually small contributors to the total production, are the dominant contributors to the fast neutron flux on the vessel wall because of their geometric proximity to it.



(a) Assembly level. (b) Pin level.

Figure F.12: Normalized contribution to the total fission neutron production rate νR_f at the assembly level (a) and at the pin level (b).

F.9. Fast Neutron Flux Above 0.1 MeV

Figure F.13 reports the height-integrated combined uncertainty on the fast neutron flux above 0.1 MeV at the vessel inner surface. The pattern across azimuths and libraries follows the one observed on the flux above 1 MeV (Sec. 6.2), with values lower by 0.2% to 1.4% in absolute terms. The reduction is largest at $-20°$ and smallest at $-28°$, and it is largest for JENDL-5. The broader energy integral averages over part of the high-energy structure where the iron evaluations carry the widest covariances, so the reduction grows with the weight of iron along the path.

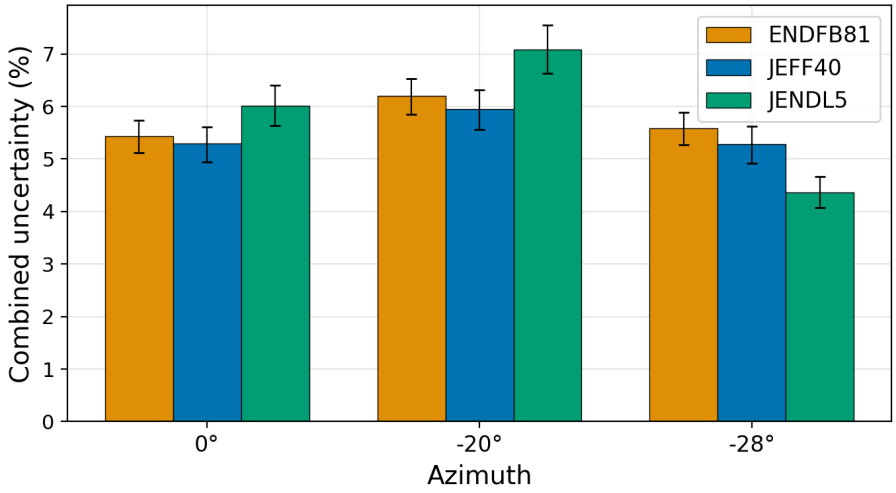
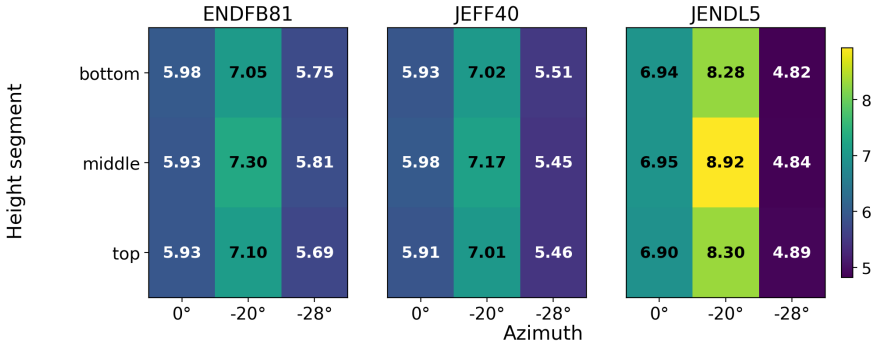


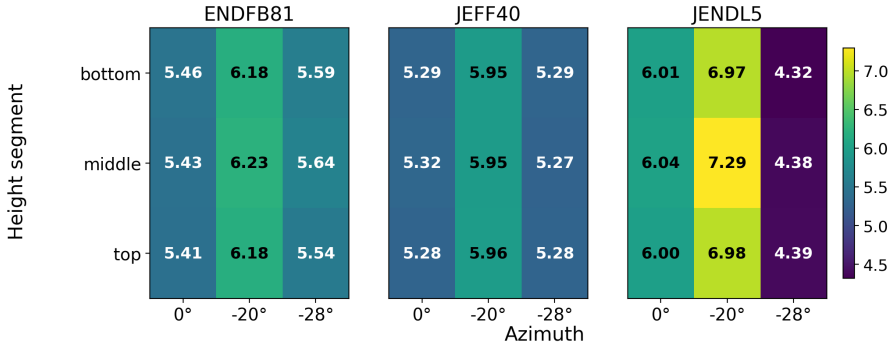
Figure F.13: Combined nuclear-data uncertainty on the fast neutron flux above 0.1 MeV at the vessel inner surface, height-integrated, for the three azimuths and the three libraries. Error bars are 95% bootstrap confidence intervals.

F.10. Azimuth-by-Height Uncertainty Maps

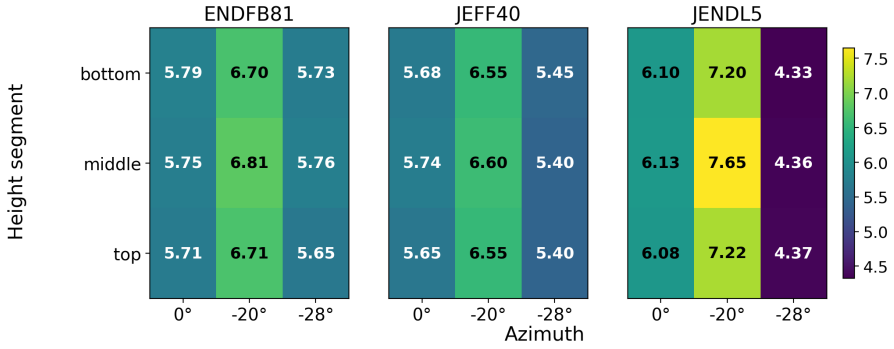
Figure F.14 displays, for each library and for each vessel response, the combined nuclear-data uncertainty resolved by azimuth (columns) and height segment (rows). The peak combined uncertainty on the fast flux above 1 MeV is located in the middle height segment of the $-20°$ azimuth for all three libraries, reaching 8.92% for JENDL-5 and 7.30% for ENDF/B-VIII.1 (Sec. 6.2). The dpa maps confirm the displacement-damage discussion of Sec. 6.3.



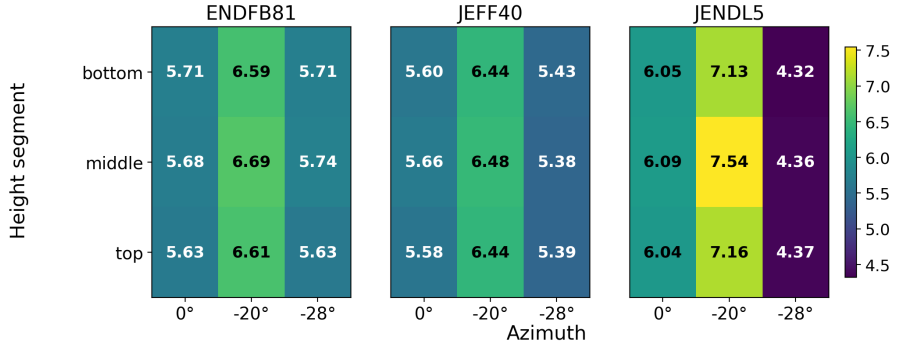
(a) Flux > 1 MeV.



(b) Flux > 0.1 MeV.



(c) NRT-dpa.



(d) ARC-dpa.

Figure F.14: Combined nuclear-data uncertainty (%) on the four vessel responses at the inner surface, resolved by azimuth and height segment, for the three libraries.

F.11. Highest-Uncertainty Tallies and Largest Library Spreads

Table F.8 lists the ten tallies with the highest combined nuclear-data uncertainty across the three libraries, and Table F.9 lists the ten tallies with the largest spread between libraries (maximum minus minimum combined value). The capsule dosimeters ^{63}Cu(n,α) and ^{46}Ti(n,p) dominate the highest-uncertainty ranking under ENDF/B-VIII.1 and JEFF-4.0 because they are the highest-threshold dosimeters and respond only to the fast tail of the capsule spectrum (Sec. 6.4 of Chapter IV). JENDL-5 places its highest values elsewhere, on the ^{238}U(n,f) rate and the fast flux at the capsule, which close the ranking. The largest cross-library disagreements all occur at the capsule, on the fast flux above 1 MeV and on the ^{238}U(n,f) and ^{237}Np(n,f) rates, where JENDL-5 exceeds the other two libraries by 2.3% to 4.0%. In the vessel structure the largest spread, 1.8%, is found in the middle height segment of the $-20°$ azimuth, followed by the height-integrated flux at the same azimuth and the displacement rates at $-28°$ (Sec. 6.2 of Chapter IV).

Table F.8: Ten tallies with the highest combined nuclear-data uncertainty across the three libraries. The position label refers to one of the three capsule positions or to the position-averaged response.

Response	Position	ENDF/B-VIII.1	JEFF-4.0	JENDL-5
^{63}Cu(n,α)	1	9.98	9.64	8.10
^{63}Cu(n,α)	2	9.96	9.53	8.13
^{63}Cu(n,α)	averaged	9.90	9.61	8.12
^{63}Cu(n,α)	3	9.81	9.72	8.12
^{46}Ti(n,p)	1	9.26	9.01	7.06
^{46}Ti(n,p)	2	9.26	9.00	7.07
^{46}Ti(n,p)	averaged	9.23	9.01	7.05
^{46}Ti(n,p)	3	9.18	9.07	7.01
^{238}U(n,f)	2	6.01	6.03	9.16
Flux $>$ 1 MeV	2	5.14	5.32	9.13

Table F.9: Ten tallies with the largest spread between libraries (maximum minus minimum combined value, in %). All ten lie at the surveillance capsule; the position label refers to one of the three capsule positions or to the position-averaged response.

Response	Position	ENDF/B-VIII.1	JEFF-4.0	JENDL-5	Spread
Flux $>$ 1 MeV	1	5.05	5.35	9.03	3.98
Flux $>$ 1 MeV	2	5.14	5.32	9.13	3.98
Flux $>$ 1 MeV	averaged	5.10	5.33	9.08	3.98
Flux $>$ 1 MeV	3	5.09	5.30	9.06	3.98
^{238}U(n,f)	1	5.88	6.06	9.08	3.19
^{238}U(n,f)	averaged	5.94	6.03	9.11	3.17
^{238}U(n,f)	3	5.93	6.00	9.08	3.16
^{238}U(n,f)	2	6.01	6.03	9.16	3.14
^{237}Np(n,f)	1	3.67	3.93	6.04	2.36
^{237}Np(n,f)	3	3.68	3.87	6.03	2.35

F.12. Nuclear-Data Budget at Every Location

Tables F.10 to F.13 extend the nuclear-data part of the budget of Table IV.12 to every response of the campaign, at the three vessel azimuths and at the capsule, with JEFF-4.0 as the transport library. The contributors are those of Sec. 7 of Chapter IV and combine in quadrature. The manufacturing and geometry term is left out. The spatial source term was estimated only for the fast flux at the 0° hot spot, and the same 1.5% is used for every response.

At the vessel, the total stays between 7.3% and 8.7% for every response and azimuth. The transport cross sections dominate at −20°, while at 0° and −28° they contribute about as much as the fission spectrum. At the capsule, both contributions grow with the threshold of the dosimeter, and the total rises from 5.0% on ^{237}Np(n,f) to 15.1% on ^{63}Cu(n,α).

Table F.10: Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, 0° azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ±60 cm central window.

	$\Phi_{E>1\,\mathrm{MeV}}$	$\Phi_{E>0.1\,\mathrm{MeV}}$	NRT-dpa	ARC-dpa
Transport cross sections (`MF33`)	5.9	5.3	5.7	5.6
Secondary angular distributions (`MF34`)	0.6	0.5	0.6	0.6
Spatial source term‡	1.5	1.5	1.5	1.5
Fission spectrum (PFNS shape)	6.1	5.0	6.1	6.0
Nuclear-data total (1σ)	**8.6**	**7.4**	**8.5**	**8.3**

‡ Estimated for $\Phi_{E>1\,\mathrm{MeV}}$ at the 0° hot spot from a three-case MCNP propagation (Sec. 5.4 of Chapter IV) and applied to every response as an approximation.

Table F.11: Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, −20° azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ±60 cm central window.

	$\Phi_{E>1\,\mathrm{MeV}}$	$\Phi_{E>0.1\,\mathrm{MeV}}$	NRT-dpa	ARC-dpa
Transport cross sections (`MF33`)	7.0	5.9	6.5	6.4
Secondary angular distributions (`MF34`)	0.8	0.6	0.7	0.7
Spatial source term‡	1.5	1.5	1.5	1.5
Fission spectrum (PFNS shape)	4.9	4.0	4.9	4.7
Nuclear-data total (1σ)	**8.7**	**7.3**	**8.3**	**8.1**

‡ Estimated for $\Phi_{E>1\,\mathrm{MeV}}$ at the 0° hot spot from a three-case MCNP propagation (Sec. 5.4 of Chapter IV) and applied to every response as an approximation.

Table F.12: Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, $-28°$ azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ± 60 cm central window.

	$\Phi_{E>1\,\mathrm{MeV}}$	$\Phi_{E>0.1\,\mathrm{MeV}}$	NRT-dpa	ARC-dpa
Transport cross sections (`MF33`)	5.5	5.3	5.4	5.4
Secondary angular distributions (`MF34`)	0.5	0.6	0.5	0.6
Spatial source term[‡]	1.5	1.5	1.5	1.5
Fission spectrum (PFNS shape)	5.6	5.0	5.8	5.7
Nuclear-data total (1σ)	**7.9**	**7.4**	**8.1**	**8.0**

[‡] Estimated for $\Phi_{E>1\,\mathrm{MeV}}$ at the $0°$ hot spot from a three-case MCNP propagation (Sec. 5.4 of Chapter IV) and applied to every response as an approximation.

Table F.13: Nuclear-data contributions to the uncertainty of the responses at the surveillance capsule, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are averaged over the three capsule positions.

Response	Cross sections (`MF33`)	Angular distr. (`MF34`)	Spatial source[‡]	Fission spectrum (PFNS shape)	**Nuclear-data total (1σ)**
$\Phi_{E>1\,\mathrm{MeV}}$	5.3	0.5	1.5	3.2	**6.4**
$\Phi_{E>0.1\,\mathrm{MeV}}$	2.9	0.3	1.5	2.5	**4.1**
^{237}Np(n,f)	3.9	0.5	1.5	2.8	**5.0**
^{238}U(n,f)	6.0	0.6	1.5	3.9	**7.3**
^{58}Ni(n,p)	7.3	1.2	1.5	5.7	**9.5**
^{54}Fe(n,p)	7.7	1.3	1.5	6.2	**10.0**
^{46}Ti(n,p)	8.8	1.8	1.5	9.2	**13.0**
^{63}Cu(n,α)	9.5	1.6	1.5	11.6	**15.1**

[‡] Estimated for $\Phi_{E>1\,\mathrm{MeV}}$ at the $0°$ hot spot from a three-case MCNP propagation (Sec. 5.4 of Chapter IV) and applied to every response as an approximation.

Appendix G

Synthèse en français

Cette annexe présente une synthèse en français du manuscrit, rédigé en anglais. Elle en reprend le fil complet, du contexte aux conclusions, et reproduit les principales figures et le tableau de synthèse du budget d'incertitude.

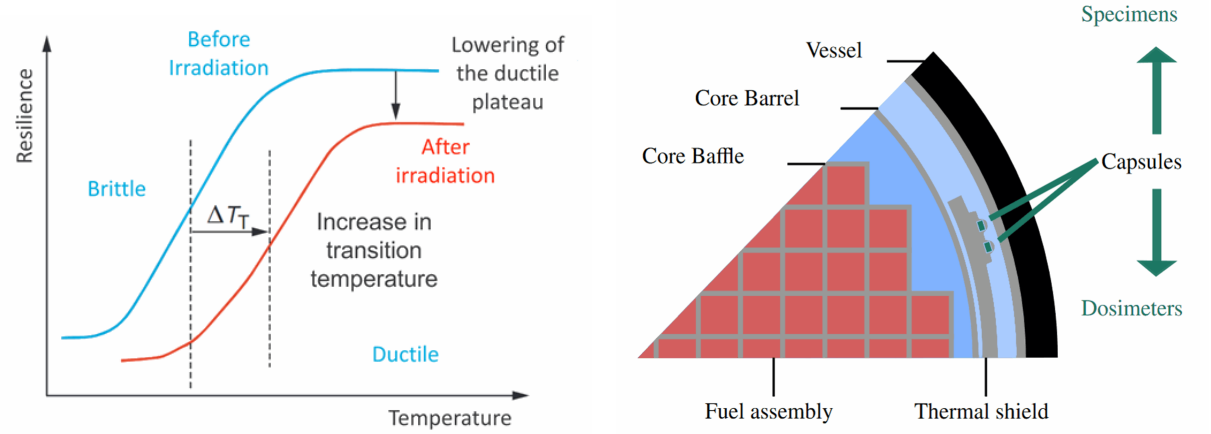
G.1. Contexte et problématique

La cuve d'un réacteur à eau sous pression est le seul composant majeur du réacteur qui ne peut pas être remplacé. Elle contient le cœur et le caloporteur primaire et constitue la principale barrière structurelle contre la dispersion de matières radioactives. Pendant toute la vie de l'installation, l'acier de la cuve est exposé aux neutrons rapides, d'énergie supérieure à 1 MeV, qui s'échappent du cœur. Ces neutrons déplacent les atomes du réseau cristallin de l'acier, et l'endommagement accumulé fragilise peu à peu le matériau : il élève la température à laquelle l'acier passe d'un comportement ductile à un comportement fragile. Cette fragilisation est irréversible, et comme la cuve ne peut être changée, elle fixe la limite ultime de la durée d'exploitation sûre d'un réacteur.

La question est centrale pour le parc français. La France tire près de 70% de son électricité du nucléaire, la part la plus élevée au monde, avec cinquante-huit réacteurs construits pour l'essentiel entre la fin des années 1970 et le début des années 1990 et autorisés initialement pour quarante ans. L'autorité de sûreté a autorisé les trente-deux réacteurs de 900 MWe à fonctionner jusqu'à cinquante ans, et prépare les conditions d'un fonctionnement au-delà. Le réacteur étudié dans cette thèse est l'une de ces tranches françaises de 900 MWe.

La grandeur qui pilote les corrélations de fragilisation est la fluence neutronique rapide à la paroi de la cuve, c'est-à-dire le flux au-dessus de 1 MeV intégré sur la durée de fonctionnement. Cette fluence ne peut pas être mesurée directement dans l'acier de la cuve. Elle est obtenue par le calcul de transport neutronique, calé sur les dosimètres d'activation de quelques capsules de surveillance placées entre l'enveloppe de cœur et la cuve (Figure G.1). Ces capsules n'échantillonnent qu'un petit nombre de positions et une partie seulement de la vie de la cuve. Le calcul porte donc la charge d'étendre une mesure éparse à toute la cuve, en espace, en énergie et dans le temps. Toute incertitude sur la fluence calculée se reporte alors en marge de sûreté : moins la prévision est précise, plus la marge à conserver est grande, avec un coût direct sur la souplesse d'exploitation. Quantifier cette incertitude, plutôt que de la masquer dans une marge prudente, est l'objet de ce travail.

L'incertitude d'un calcul de fluence a trois grandes origines : la méthode numérique, la description géométrique et technologique du réacteur, et les données nucléaires qui décrivent les interactions



(a) L'irradiation décale la transition ductile-fragile de l'acier vers les hautes températures et abaisse la ténacité.

(b) Les capsules de surveillance, porteuses d'éprouvettes d'acier et de dosimètres d'activation, sont placées entre l'enveloppe de cœur et la cuve, ici sur un huitième du cœur.

FIGURE G.1 : Fragilisation de la cuve et sa surveillance.

entre neutrons et matière. À mesure que les codes de transport et les modèles de réacteur ont mûri, les données nucléaires sont devenues le contributeur dominant, et parmi elles les covariances qui quantifient la connaissance de chaque section efficace. Les neutrons rapides atteignent la cuve après avoir traversé l'acier du baffle, de l'enveloppe de cœur et de la protection thermique, où le fer gouverne l'essentiel de leur diffusion et de leur ralentissement. Les sections efficaces du fer en sont donc le principal contributeur.

Un canal reste mal contraint par les bibliothèques évaluées : la distribution angulaire des neutrons diffusés élastiquement par le fer. L'angle de diffusion élastique du ^{56}Fe fixe la direction dans laquelle le neutron poursuit sa route vers la cuve, et l'incertitude associée est rangée dans une partie distincte des fichiers évalués. Cette information est éparse et incohérente : plusieurs bibliothèques modernes ne fournissent aucune covariance angulaire pour la diffusion élastique du ^{56}Fe, et celles qui en donnent sont en désaccord sur son amplitude et sa structure. Cette thèse réexamine donc directement les mesures différentielles compilées dans la base EXFOR, juge la qualité des covariances publiées au regard de ces mesures, et produit une évaluation indépendante de la distribution angulaire élastique du ^{56}Fe et de sa covariance, d'un type qu'aucune bibliothèque ne propose aujourd'hui. La propagation de cette évaluation répond à une question restée ouverte en dosimétrie de cuve : combien la distribution angulaire des neutrons diffusés ajoute-t-elle à l'incertitude de fluence neutronique rapide.

Ces éléments sont réunis dans un budget d'incertitude des données nucléaires pour la fluence neutronique rapide et la dosimétrie de surveillance d'une cuve française de 900 MWe, propagé à travers le schéma de calcul de vieillissement de cuve en deux étapes utilisé en France.

G.2. Méthodologie : analyse de sensibilité et quantification d'incertitudes

Le transport des neutrons dans le réacteur est décrit par l'équation de Boltzmann. Deux familles de méthodes la résolvent : les méthodes déterministes, qui discrétisent l'espace, l'énergie et la

direction, et les méthodes stochastiques de Monte-Carlo, qui simulent des histoires de particules sans erreur de discrétisation mais avec un bruit statistique. La fluence de cuve relève d'un problème d'atténuation profonde, où seule une fraction infime des particules atteint la cible ; les techniques de réduction de variance, fondées sur la théorie adjointe et les fenêtres de poids, y sont indispensables. Les calculs de cette thèse s'appuient principalement sur MCNP, avec des comparaisons par Serpent et un traitement des données nucléaires par NJOY.

Les incertitudes des données nucléaires sont portées par les covariances des fichiers évalués : la matrice `MF33` pour les sections efficaces, la matrice `MF34` pour les distributions angulaires. Pour propager ces covariances jusqu'à une réponse observable, deux stratégies complémentaires sont employées. La première est la propagation au premier ordre, ou règle « sandwich » : la variance de la réponse s'écrit $\sigma_R^2 \approx \boldsymbol{S}^\top \boldsymbol{C}\, \boldsymbol{S}$, où $\boldsymbol{S}$ est le vecteur des sensibilités de la réponse aux paramètres et $\boldsymbol{C}$ leur matrice de covariance relative. Cette voie est peu coûteuse et fournit une décomposition par paramètre, mais elle suppose la linéarité de la réponse. La seconde stratégie est l'échantillonnage de Monte-Carlo : N jeux de données nucléaires sont tirés de la distribution conjointe par décomposition de la matrice de covariance, N calculs de transport sont réalisés, et les statistiques de la réponse sont estimées. Cette voie capte les effets non linéaires, au prix d'une séparation soigneuse entre l'incertitude paramétrique et le bruit statistique du transport.

Ces deux voies, ainsi que le calcul des sensibilités par théorie des perturbations et par les estimateurs d'histoire de collision de Serpent, sont rassemblées dans un cadre logiciel développé pendant la thèse, la bibliothèque libre `kika-nd` et son application KIKA. Ce cadre forme une chaîne reproductible : lecture des fichiers ENDF-6, ACE et de covariance, décomposition matricielle et échantillonnage, génération des données nucléaires perturbées, et interfaçage avec MCNP, Serpent et NJOY. Deux chaînes de perturbation distinctes sont fournies, l'une au niveau ACE pour les sections efficaces, l'autre au niveau ENDF pour les distributions angulaires.

La méthodologie est vérifiée sur un modèle simplifié, une sphère représentative des matériaux de cuve : des coquilles concentriques d'acier et d'eau borée entourent une source ponctuelle de neutrons de 1 à 3 MeV, avec un détecteur en couronne dans l'air périphérique. Sur ce modèle, les profils de sensibilité du ^{56}Fe obtenus par MCNP et par Serpent concordent, la théorie des perturbations au premier ordre reste valide pour des amplitudes compatibles avec les covariances, et les incertitudes propagées par la règle sandwich et par l'échantillonnage de Monte-Carlo s'accordent à leurs barres statistiques près.

Cette vérification fait surtout apparaître le problème qui structure la suite de la thèse. Dans la gamme du mégaélectronvolt qui pilote l'endommagement, la distribution angulaire élastique du ^{56}Fe contribue à moins de 1% de la variance lorsque sa covariance est tirée de JEFF-4.0, et à près d'un cinquième lorsqu'elle est tirée de JENDL-5, avec une incertitude propagée d'environ 11,5% (Figure G.2). Les deux bibliothèques sont donc en désaccord de plus d'un ordre de grandeur sur une quantité qui n'avait jamais été confrontée aux données, et aucune ne peut être retenue en l'état. Ce constat motive l'évaluation détaillée du chapitre suivant.

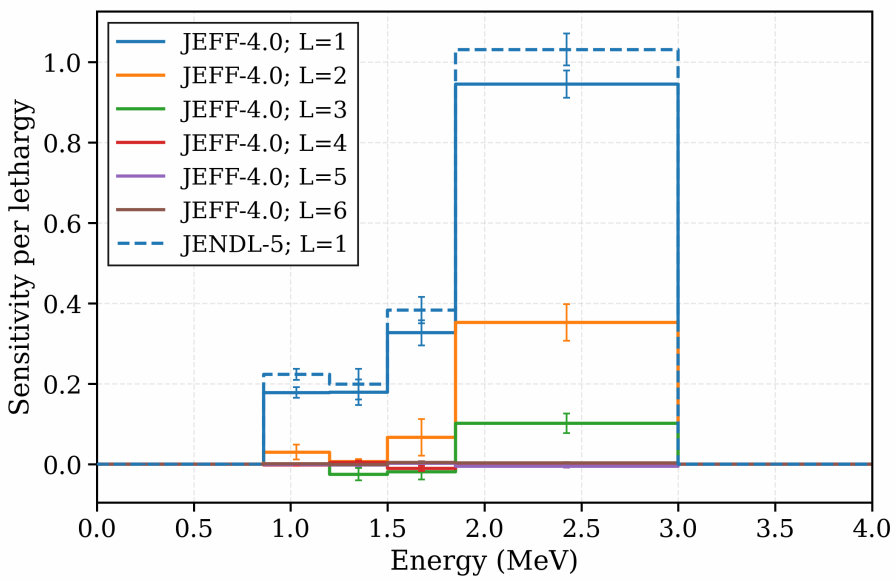


FIGURE G.2 : Sensibilité de la réponse du détecteur (1–3 MeV) aux moments de Legendre de la distribution angulaire élastique du ^{56}Fe. JEFF-4.0 (traits pleins, ordres 1 à 6) et JENDL-5 (tirets, ordre 1 seul, seul ordre porteur de covariance dans cette bibliothèque).

G.3. Évaluation des distributions angulaires du ^{56}Fe

Résoudre ce désaccord impose de revenir aux mesures différentielles sur la gamme 847 keV–4 MeV, celle qui pilote l'endommagement. Les bibliothèques décrivent presque la même distribution angulaire mais traitent son incertitude de façons différentes. Certaines ne fournissent aucune covariance pour la diffusion élastique du ^{56}Fe ; JENDL-5 place toute l'incertitude sur un unique ordre de Legendre ; JEFF-4.0 la répartit de façon inégale sur six ordres. Surtout, les deux le font sur des grilles en énergie bien plus grossières que la distribution elle-même : treize groupes pour JEFF-4.0, quatre pour JENDL-5.

Une évaluation indépendante de la distribution angulaire élastique du ^{56}Fe et de sa covariance a donc été construite à partir de la base EXFOR. Elle ingère soixante-sept expériences, soit environ 23 300 points de distribution angulaire après déduplication, sur la gamme 847 keV–4 MeV. Les incertitudes expérimentales sont décomposées en un bruit statistique par point et une composante systématique par expérience (normalisation, calibration des moniteurs), ce qui donne une structure de covariance de rang un par campagne. Les campagnes sont internes cohérentes mais décalées entre elles au-delà de leurs incertitudes déclarées, ce qui guide tout le traitement.

La chaîne d'évaluation procède en plusieurs étapes (Figure G.3). Un ajustement nominal par moindres carrés généralisés régularisés détermine, intervalle par intervalle, les coefficients de Legendre, l'ordre étant choisi automatiquement par le critère d'information d'Akaike pour équilibrer qualité et complexité de l'ajustement ; la régularisation supprime les oscillations parasites aux ordres élevés. Une première passe de Monte-Carlo estime les variances par intervalle, par rééchantillonnage des systématiques par expérience et du bruit par point. Une seconde passe estime les corrélations entre intervalles, en propageant les systématiques partagées d'une même expérience à tous les intervalles qu'elle recouvre, pondérées par le recouvrement en résolution énergétique. Les variances et la corrélation sont assemblées en une covariance complète, une projection garantissant sa positivité sur chaque échantillon. Le résultat est écrit en `MF4` et `MF34`,

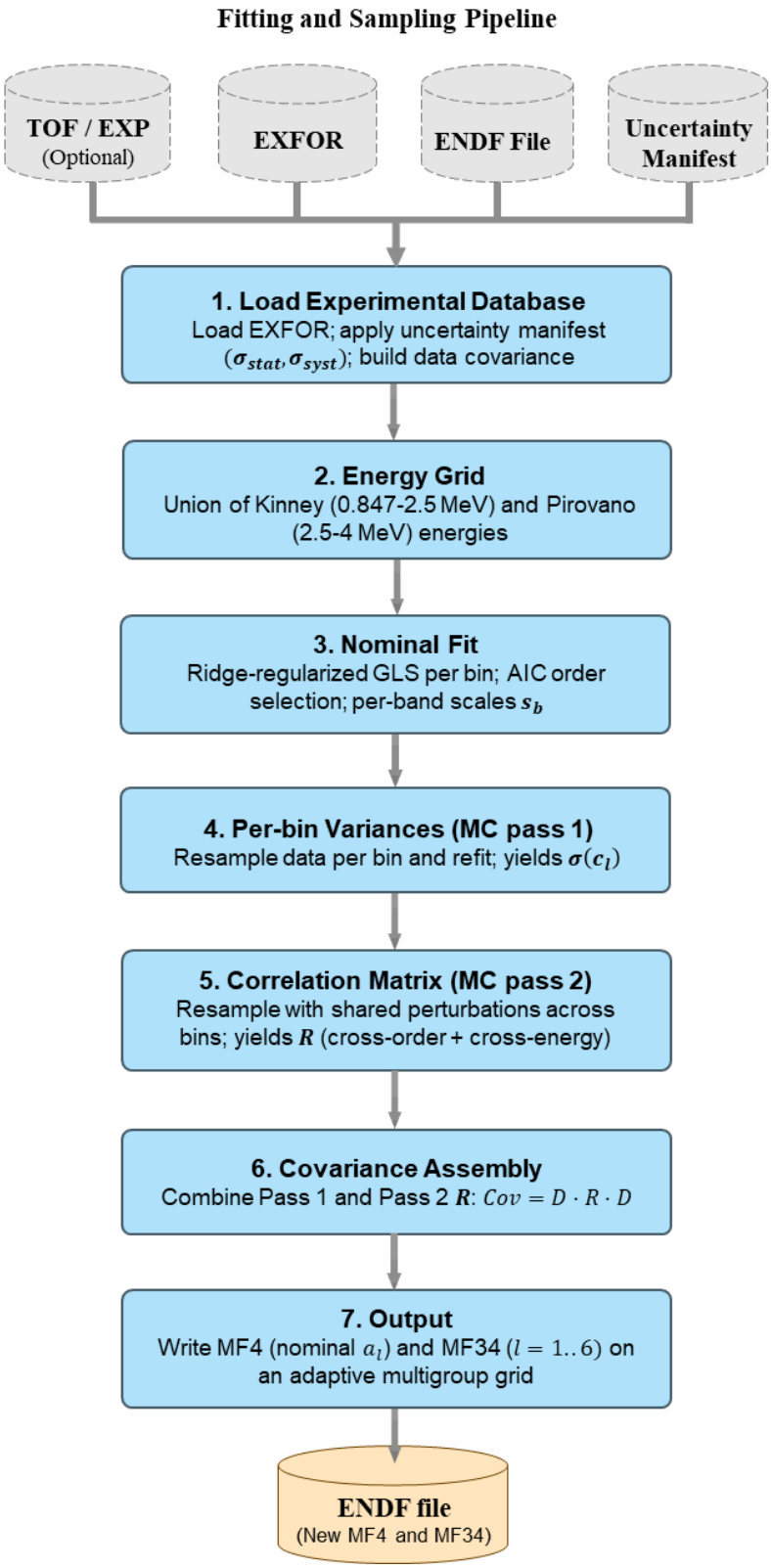


FIGURE G.3 : Chaîne d'évaluation. Les distributions angulaires EXFOR sont ajustées en coefficients de Legendre par intervalle d'énergie ; deux passes de Monte-Carlo estiment les variances par intervalle (passe 1) et la matrice de corrélation complète des paramètres (passe 2), assemblées dans la covariance écrite en `MF4` et `MF34`.

puis condensé sur des multigroupes par une fusion adaptative de groupes en énergie.

L'incertitude obtenue se répartit régulièrement sur les ordres de Legendre ajustés, là où les intervalles le justifient, et s'annule aux ordres que le critère d'Akaike n'a pas retenus. Elle produit ainsi une bande continue et physiquement interprétable sur la section efficace différentielle, qui suit la dispersion des données plutôt qu'une forme dictée par la paramétrisation. La covariance est par ailleurs portée par cinq-cent-vingt-six groupes en énergie, contre treize pour JEFF-4.0 et quatre pour JENDL-5. La Figure G.4 montre les sections efficaces différentielles reconstruites à quatre énergies représentatives : l'évaluation suit les données EXFOR sur toute la gamme et reste cohérente avec deux calculs indépendants par le modèle optique de TALYS, qui servent de contrôle physique.

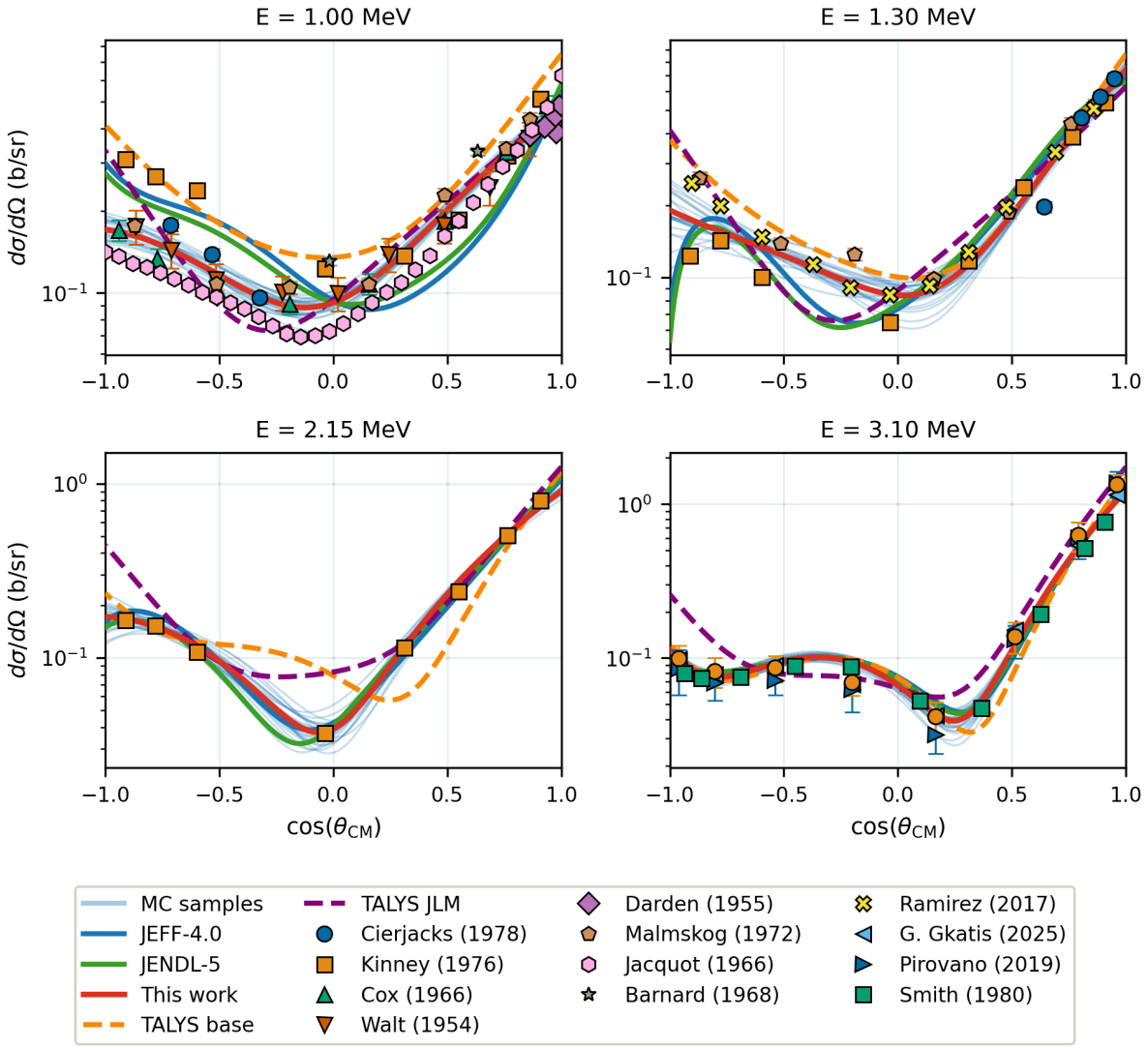


FIGURE G.4 : Sections efficaces différentielles élastiques du ^{56}Fe à quatre énergies incidentes (1,00, 1,30, 2,15 et 3,10 MeV) : ce travail (rouge), JEFF-4.0 (bleu), JENDL-5 (vert) et deux calculs TALYS. Les marqueurs sont les points expérimentaux de l'intervalle évalué ; les courbes fines sont des échantillons de Monte-Carlo illustrant la dispersion.

La qualité d'accord est mesurée par un χ^2 de Mahalanobis confronté à la base complète. Sous le budget de variance le plus complet, qui ajoute à l'incertitude expérimentale la covariance `MF34` propagée de chaque évaluation, le χ^2/N sur l'ensemble de la base vaut 7,5 pour ce travail, 9,6

pour JEFF-4.0 et 15,1 pour JENDL-5. La séparation est la plus nette sur le sous-ensemble qui exclut la campagne de calibration Kinney et le bloc dominant de Cierjacks : ce travail s'y tient à 4,7, tandis que JEFF-4.0 et JENDL-5 montent à 17,1 et 51,1. Sur les soixante-sept expériences prises une à une, ce travail obtient le plus faible χ^2/N pour soixante d'entre elles. Cette différence tient en partie à la méthode : sur Kinney seule, JENDL-5 descend à 0,53, sous les incertitudes que cette campagne déclare pour elle-même, alors que l'évaluation présentée ici s'accorde plus uniformément avec l'ensemble de la base. Ses valeurs centrales restent proches de JEFF-4.0 et JENDL-5, compatibles à leurs incertitudes près en dessous de 2,5 MeV et un peu plus piquées vers l'avant au-delà, de sorte que la section efficace ne bouge pas : ce qui change est surtout la covariance. Un contrôle sur quatre-vingt-huit benchmarks de criticité de l'ICSBEP confirme que ces faibles changements de valeurs centrales laissent la performance intégrale inchangée et améliorent légèrement les assemblages métalliques rapides les plus sensibles à la réflexion sur le fer.

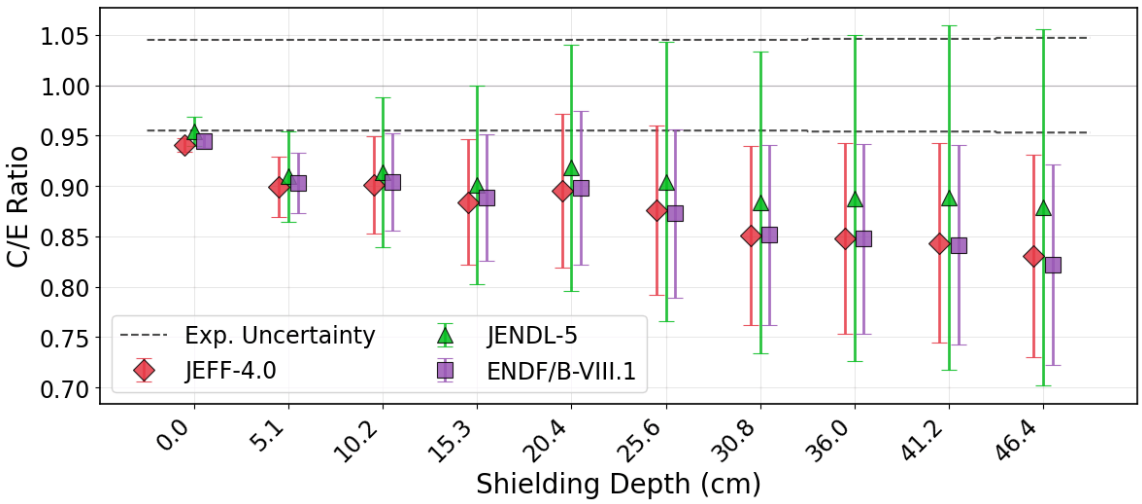
G.4. Confrontation aux benchmarks intégraux : ASPIS-88 et VENUS-3

L'évaluation a ensuite été confrontée à l'expérience dans des géométries où le fer gouverne le transport. Le benchmark ASPIS-88 envoie des neutrons à travers près de soixante-sept centimètres d'acier doux, un empilement où le ^{56}Fe est presque le seul acteur (Figure G.5). Des dosimètres d'activation insérés entre les plaques mesurent les taux de réaction le long de l'axe, leurs seuils couvrant le spectre de l'épithermique à plusieurs MeV.

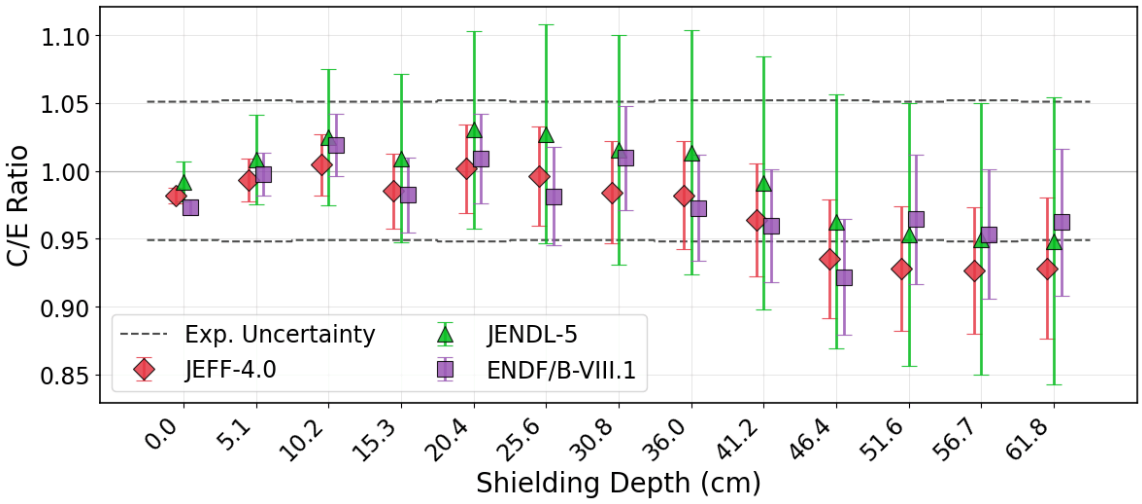
L'accord avec la mesure se sépare en deux effets. Certains dosimètres montrent un décalage constant dès la première position de mesure, présent dans toutes les bibliothèques ; ces biais permanents renvoient à la description de la source et aux données de dosimétrie, non à l'évaluation du fer. À ces décalages se superpose une tendance avec la profondeur, et c'est cette tendance que les données du fer contrôlent. L'échange de la seule évaluation du fer entre JEFF-4.0 et JENDL-5, à toutes choses égales par ailleurs, reproduit la quasi-totalité de l'écart entre leurs prévisions pour les dosimètres à seuil. Cet écart inter-bibliothèque dans la fenêtre de quelques centaines de keV au MeV se situe donc bien sur le ^{56}Fe.

L'incertitude de section efficace propagée croît avec la profondeur et avec le seuil du dosimètre, atteignant plusieurs dizaines de pour-cent aux positions les plus profondes et de plus haut seuil (jusqu'à 30 à 41% selon la bibliothèque sur le dosimètre soufre le plus profond). Le canal de distribution angulaire est plus faible et dépend du dosimètre ; pour ceux qui y sont les plus sensibles, la nouvelle évaluation améliore légèrement l'accord en pénétration profonde par rapport à JEFF-4.0 lorsqu'elle est utilisée comme valeurs centrales, et sa bande d'incertitude est plus étroite que celle, plus large, produite par la covariance à un seul ordre de JENDL-5.

Le benchmark VENUS-3, de type réacteur, place sur le trajet des neutrons la fission, l'acier et l'eau, plus proche de l'application. Les bibliothèques modernes y reproduisent les mesures à quelques pour-cent près, et les incertitudes de section efficace propagées y sont assez faibles pour rester sous les barres expérimentales. Cette marge réduite laisse peu de place pour resserrer les covariances évaluées par assimilation de ces mesures, ce qui explique que l'assimilation un temps envisagée pour la cuve n'ait pas été poursuivie. Deux anomalies logicielles ont été identifiées et signalées au cours de ce travail : un biais des estimations de sensibilité de MCNP lorsque les cartes de perturbation sont combinées aux fichiers de fenêtres de poids, et une covariance élastique non physique de l'^{16}O produite par le traitement NJOY de l'évaluation JEFF-4.0.



(a) ^{115}In.



(b) ^{103}Rh.

FIGURE G.5 : Rapports calcul/expérience le long de la profondeur d'acier pour les trois bibliothèques (ASPIS-88). Les barres d'erreur donnent l'incertitude des données nucléaires du ^{56}Fe (sections efficaces et distribution angulaire combinées en quadrature) ; les tirets marquent l'incertitude expérimentale.

G.5. Application : budget d'incertitude de la cuve REP900

La dernière étape propage les covariances de transport à travers le schéma de vieillissement de cuve jusqu'à la tranche française de 900 MWe et les combine avec les contributions du terme source en un budget unique de données nucléaires. Ce schéma comporte deux étapes : un calcul de source par le couple déterministe CASMO5/SIMULATE5, puis un calcul de transport Monte-Carlo par MCNP sur un modèle haute-fidélité du réacteur. Les données nucléaires sont propagées indépendamment pour ENDF/B-VIII.1, JEFF-4.0 et JENDL-5.

Le terme de source contribue pour environ 1,5% à la fluence de cuve par sa composante spatiale, propagée ici directement à travers l'échantillonnage CASMO5/SIMULATE5 sur tous les isotopes et réactions du cœur ; il est dominé par $\bar{\nu}$ et les sections efficaces de fission des actinides. La forme du spectre émis, perturbée ici à partir de la covariance `MF35` évaluée au moyen de 512 termes sources MCNP, apporte environ 6,1% au point chaud de la cuve et s'échelonne de 2,5% à 11,6% selon la réponse. Cette valeur se situe dans la fourchette publiée par Berge [183] pour un réacteur français du même type, de $\sim 1\%$ à 7% selon l'évaluation, avec 6,5% pour son ajustement Madland–Nix.

Au niveau du transport, les sections efficaces dominent. Leur incertitude sur le flux rapide atteint environ 5,9% au point chaud de la cuve et culmine à $-20°$, derrière la capsule de surveillance, où l'acier supplémentaire amplifie la contribution du fer. Une décomposition par isotope (Figure G.6) montre que le ^{56}Fe porte de l'ordre de 65% de la variance de section efficace à la cuve, l'^{16}O environ 18% et le ^{52}Cr environ 9%. La distribution angulaire reste mineure à la paroi de cuve : elle augmente l'incertitude totale de moins de 0,1% en valeur absolue pour l'évaluation partagée par JEFF-4.0 et ENDF/B-VIII.1 et pour la nouvelle évaluation ; la covariance à un seul coefficient de JENDL-5 en porte davantage, jusqu'à environ 0,5%, ce qui reste faible devant le total proche de 8,6% que se partagent la section efficace du fer et le spectre de fission.

En réunissant le terme de source et le terme de transport, le budget des données nucléaires atteint 8,6% à la cuve et 6,4% à la capsule. L'ajout d'un terme de fabrication et de géométrie, emprunté à une étude comparable, porte le total à environ 10% à la cuve et 8% à la capsule. Le Tableau G.1 place ce résultat face à une étude indépendante portant sur un réacteur comparable. Les totaux de données nucléaires à la cuve, 8,4% et 8,6%, sont proches, ce qui indique une certaine cohérence entre les deux budgets, construits par des chaînes différentes.

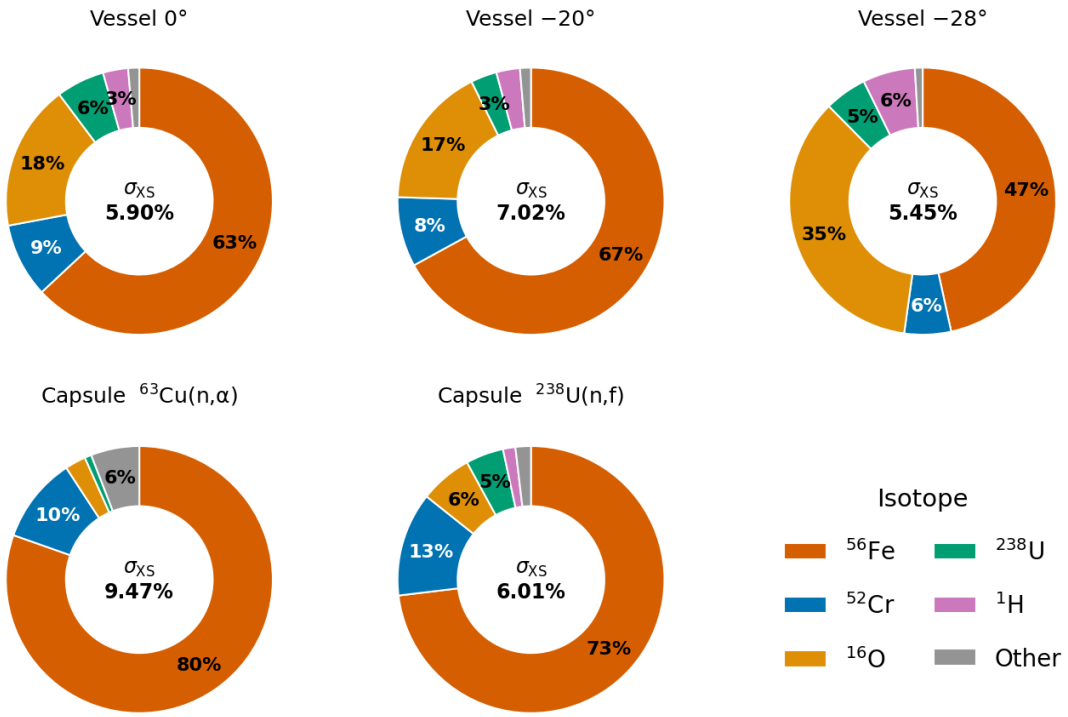


FIGURE G.6 : Décomposition par isotope de l'incertitude de section efficace JEFF-4.0 en cinq positions : les trois azimuts de cuve au-dessus de 1 MeV et deux taux de réaction de capsule. L'aire de chaque secteur vaut $\sigma^2_{\text{iso}}/\sigma^2_{\text{XS}}$; le σ_{XS} combiné est inscrit au centre.

TABLE G.1 : Budget d'incertitude du flux neutronique rapide $\Phi_{E>1\,\text{MeV}}$ pour une étude indépendante (Timpano) et le présent travail, séparé en contributions de données nucléaires et terme additionnel de fabrication et géométrie. Les contributions sont données au point chaud de cuve ; les totaux figurent à la cuve et à la capsule.

	Timpano [3]	Ce travail
Contributions des données nucléaires (%)		
Sections efficaces de transport (`MF33`)	7,3	5,9
Distributions angulaires (`MF34`)	4,1†	0,6‡
Terme de source spatial	0,3	1,5
Spectre de fission (forme)	—	6,1
Total données nucléaires, cuve (1σ)	**8,4**	**8,6**
Total données nucléaires, capsule (1σ)	**7,1**	**6,4**
Contribution additionnelle (%)		
Fabrication et géométrie	5,3	5,3⋆
Incertitude totale de fluence, cuve (1σ)	**9,9**	**10,1**
Incertitude totale de fluence, capsule (1σ)	**8,6**	**8,1**

† Issu d'un échantillonnage de modèle TALYS/TENDL (Total Monte-Carlo). ‡ Écart-type propre du canal angulaire ; combiné en quadrature avec les sections efficaces, il augmente l'incertitude totale de moins de 0,1% en valeur absolue. ⋆ Emprunté à Timpano, ce travail ne modélisant pas les tolérances de fabrication.

G.6. Conclusion et perspectives

Ce travail répond à la question autour de laquelle la thèse est construite. Propagée jusqu'à la paroi de la cuve, la distribution angulaire élastique du fer est un contributeur mineur : elle augmente l'incertitude de fluence de 0,1% au plus, en valeur absolue, avec la covariance partagée par JEFF-4.0 et ENDF/B-VIII.1 et avec la nouvelle évaluation, et d'environ 0,5% au plus avec la covariance à un seul coefficient de JENDL-5. Devant un total proche de 8,6% que se partagent les sections efficaces du fer et le spectre de fission, le canal angulaire reste faible pour chaque bibliothèque. Cette conclusion est propre à la paroi de cuve : les résultats ASPIS montrent que la contribution angulaire croît à mesure que les neutrons pénètrent l'acier, de sorte qu'un observable situé derrière la cuve en porterait davantage.

Les contributions principales sont les suivantes : la première évaluation guidée par les données de la distribution angulaire élastique du ^{56}Fe et de sa covariance sur 847 keV–4 MeV, construite à partir de la base EXFOR après filtrage et sélection en énergie documentés, livrée en `MF4` et `MF34` ; un budget d'incertitude des données nucléaires pour la fluence et la dosimétrie de surveillance de la cuve REP900, décomposé par isotope et par réaction et confronté à des études indépendantes ; le cadre logiciel `kika-nd` et l'application KIKA, chaîne reproductible de la lecture des covariances jusqu'au post-traitement du transport ; et le signalement de deux anomalies logicielles.

Plusieurs limites bornent ce qui peut être avancé. L'évaluation repose sur un enregistrement expérimental fini et inégal. Ses valeurs centrales sont stables, puisqu'elles s'accordent déjà à deux bibliothèques modernes indépendantes ; la covariance est le produit le plus fragile. Même en incluant la covariance de chaque bibliothèque, le χ^2/N reste supérieur à l'unité pour les trois évaluations, de sorte que les incertitudes rapportées ici, bien que plus larges et mieux réparties que celles des bibliothèques, pourraient encore sous-estimer la dispersion impliquée par les données. L'évaluation couvre 847 keV–4 MeV et traite la forme angulaire indépendamment de l'amplitude de la section efficace. Les campagnes diffèrent également par leur résolution en énergie, de sorte que chacune contraint une moyenne différente de la même distribution sous-jacente ; les combiner coûte soit la structure que seules les campagnes les plus fines résolvent, soit le moyennage que portent les plus grossières.

Plusieurs directions suivent naturellement. La même approche guidée par les données peut s'appliquer aux autres isotopes qui façonnent le spectre de cuve, à commencer par l'^{16}O, qui porte la part de variance de section efficace la plus importante après le fer. Une évaluation conjointe de l'amplitude et de la forme angulaire, portant leurs corrélations croisées, lèverait l'hypothèse d'indépendance. Une comparaison avec des covariances de ^{56}Fe issues des paramètres de modèle [110] séparerait l'information apportée par les mesures de celle imposée par le modèle nucléaire. Un estimateur portant la résolution en énergie de chaque expérience à l'intérieur du modèle, par exemple un ajustement par splines de la dépendance en énergie, permettrait d'utiliser conjointement les mesures à haute et à basse résolution sans dégrader les unes ni les autres. Le budget peut être porté vers des observables plus profonds que la paroi, où la contribution angulaire devrait croître. Un exercice de précisions cibles (TAR, *target accuracy requirements*) [187] traduirait la décomposition de la variance en la précision que chaque donnée nucléaire doit atteindre pour une incertitude visée sur la fluence en cuve.

Ces résultats aident à faire reposer sur la mesure une part de la marge de fluence de cuve, jusque-là portée par prudence. Les sources dominantes d'incertitude sur la fluence de cuve, les sections efficaces du fer et le spectre de fission, sont désormais identifiées et quantifiées, et le canal angulaire qui a motivé une grande partie du travail se révèle faible là où il est appliqué et

bien caractérisé là où il ne l'est pas. Pour un parc dont la poursuite de fonctionnement dépend de la connaissance de la durée de vie des cuves, une incertitude fondée sur la mesure est la base la plus solide pour ce jugement.

Bibliography

[1] Laura Clouvel. *Uncertainty quantification of the fast flux calculation for a PWR vessel.* Thèse de doctorat, Université Paris-Saclay (ComUE), November 2019.

[2] Romain Vuiart, Mariya Brovchenko, Julien Taforeau, Vaibhav Jaiswal, and Eric Dumonteil. A Versatile Methodology for Reactor Pressure Vessel Aging Assessments. *Nuclear Science and Engineering*, 196(4):455–477, April 2022.

[3] D. Timpano, A. Vasiliev, D. Rochman, and M. Hursin. Uncertainty quantification for a vessel fluence calculation: A PWR case study. *Progress in Nuclear Energy*, 193:106234, March 2026.

[4] Romain Vuiart. *Analyse du vieillissement sous irradiation neutronique des cuves des réacteurs nucléaires à eau sous pression.* Thèse de doctorat, université Paris-Saclay, November 2021.

[5] Dan Gabriel Cacuci. *Handbook of Nuclear Engineering*, volume 1-4. Springer, 2010.

[6] G. I. Bell and S. Glasstone. *Nuclear Reactor Theory.* Division of Technical Information, U.S. Atomic Energy Commission, Washington, DC, October 1970.

[7] Alain Hébert. *Applied Reactor Physics.* Presses internationales Polytechnique, 2 edition, 2016.

[8] Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith. OpenMC: A state-of-the-art Monte Carlo code for research and development. *Annals of Nuclear Energy*, 82:90–97, August 2015.

[9] Elmer Eugene Lewis and W. F. Miller. *Computational Methods of Neutron Transport.* Wiley, 1984. Google-Books-ID: hRxRAAAAMAAJ.

[10] J. Hykes, T. Simeonov, R. Ferrer, C. Jönsson, C. Wemple, V. P. Eronen, T. Ranta, A. Ranta-aho, V. Hynönen, J. Kumpula, and J. Huttunen. Nuclear data uncertainty quantification analysis at Studsvik Scandpower. *Annals of Nuclear Energy*, 208:110752, December 2024.

[11] Robert Macfarlane, Douglas W. Muir, R. M. Boicourt, I. I. I. Kahler, and Jeremy Lloyd Conlin. The NJOY Nuclear Data Processing System, Version 2016. Technical Report LA-UR-17-20093, Los Alamos National Laboratory (LANL), Los Alamos, NM (United States), January 2017.

[12] ORNL. SCALE 6.3.2 User Manual — SCALE 6.3.2 documentation, 2023.

[13] LANL. MCNP Code Version 6.3.0 Theory & User Manual. Technical Report 2022-09-28 (rev.1), Los Alamos National Laboratory, September 2022.

[14] Scott W Mosher, Aaron M Bevill, Seth R Johnson, Ahmad M Ibrahim, Charles R Daily, Thomas M Evans, John C Wagner, and Jeffrey O Johnson. ADVANTG An Automated Variance Reduction Parameter Generator. Technical Report ORNL/TM-2013/416, 1105937, Oak Ridge National Laboratory (ORNL), November 2013.

[15] Thomas Booth. Sample problem for variance reduction in MCNP. Technical Report LA-10363-MS, Los Alamos National Laboratory, October 1985.

[16] J. Hendricks and T. Booth. MCNP variance reduction overview. In *Monte-Carlo Methods and Applications in Neutronics, Photonics and Statistical Physics*, volume 240 of *Lecture Notes in Physics*, pages 83–92. Springer, Berlin, Heidelberg, January 1970. Journal Abbreviation: Lecture Notes Phys. (AXV64).

[17] B.C. Kiedrowski, F.B. Brown, and P.P.H. Wilson. Adjoint-weighted tallies for k-eigenvalue calculations with continuous-energy Monte Carlo. *Nuclear Science and Engineering*, 168(3):226–241, 2011.

[18] Mario Matijević, Dubravko Pevec, and Bojan Petrović. PCA Benchmark Analysis with ADVANTG3.0.1. and MCNP6.1.1b Codes. *Journal of Energy - Energija*, 68:171–183, July 2022.

[19] Davide Mancusi and Andrea Zoia. Zero-variance schemes for kinetic Monte Carlo simulations. *The European Physical Journal Plus*, 135(5):401, May 2020.

[20] John C. Wagner, Alireza Haghighat, and Bojan G. Petrovic. Monte Carlo Transport Calculations and Analysis for Reactor Pressure Vessel Neutron Fluence. *Nuclear Technology*, 114(3):373–398, June 1996. _eprint: https://doi.org/10.13182/NT96-A35241.

[21] United States: Nuclear Regulatory Commission, Brookhaven National Laboratory, and U.S. Nuclear Regulatory Commission: Office of Nuclear Regulatory Research. PWR and BWR Pressure Vessel Fluence Calculation Benchmark Problems and Solutions. Technical report, Nuclear Regulatory Commission, January 2001. Accession Number: GOVPUBY3_N88-PURL-gpo10401 Source: DGPO.

[22] Tanja Goričanec, Bor Kos, Klemen Ambrožič, Andrej Trkov, Luka Snoj, and Marjan Kromar. Determination of neutron flux redistribution factors for a typical pressurized water reactor ex-core measurements using Monte Carlo technique. *Frontiers in Energy Research*, 11, February 2023.

[23] Thomas M. Evans, Alissa S. Stafford, Rachel N. Slaybaugh, and Kevin T. Clarno. Denovo: A New Three-Dimensional Parallel Discrete Ordinates Code in SCALE. *Nuclear Technology*, 171(2):171–200, 2010.

[24] W. Engle, Jr. A Users Manual for ANISN: A one dimensional discrete ordinates transport code with anisotropic scattering. Technical Report K–1693, 4448708, Union Carbide Corpotation, Oak Ridge Gaseous Diffusion Plant, January 1967.

[25] Jaakko Leppänen, Ville Valtavirta, Antti Rintala, and Riku Tuominen. Status of Serpent Monte Carlo code in 2024. *EPJ Nuclear Sciences & Technologies*, 11:3, 2025.

[26] Tamer Bahadir. CMS-LINK Users Manual. Technical report, Studsvik Scandpower, Inc., 1999.

[27] Scott Vanevenhoven and Tamer Bahadir. Simulate 5 Users Manual. Technical report, Studsvik Scandpower, Inc., 2022.

[28] V. V. Zerkin and B. Pritychenko. The Experimental Nuclear Reaction Data (EXFOR): Extended Computer Database and Web Retrieval System. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 888:31–43, April 2018. arXiv:1802.05714 [physics].

[29] Pierre Solé. *Nuclear Data Uncertainties Processing and Propagation.* PhD thesis, Paris Saclay, April 2025.

[30] N. Otuka, E. Dupont, V. Semkova, B. Pritychenko, A. I. Blokhin, M. Aikawa, S. Babykina, M. Bossant, G. Chen, S. Dunaeva, R. A. Forrest, T. Fukahori, N. Furutachi, S. Ganesan, Z. Ge, O. O. Gritzay, M. Herman, S. Hlavač, K. Katō, B. Lalremruata, Y. O. Lee, A. Makinaga, K. Matsumoto, M. Mikhaylyukova, G. Pikulina, V. G. Pronyaev, A. Saxena, O. Schwerer, S. P. Simakov, N. Soppera, R. Suzuki, S. Takács, X. Tao, S. Taova, F. Tárkányi, V. V. Varlamov, J. Wang, S. C. Yang, V. Zerkin, and Y. Zhuang. Towards a More Complete and Accurate Experimental Nuclear Reaction Data Library (EXFOR): International Collaboration Between Nuclear Reaction Data Centres (NRDC). *Nuclear Data Sheets*, 120:272–276, June 2014.

[31] A. J.M. Plompen, T. Fukahori, H. Henriksson, A. Ignatyuk, T. Iwasaki, G. Manturov, R. D. McKnight, G. Rimpault, and D. L. Smith. The NEA High Priority Nuclear Data Request List for future needs. In *ND2007*, page 202, Nice, France, 2007. EDP Sciences.

[32] Joint Evaluated Fission and Fusion Project. JEFF-4.0 Evaluated Data: neutron induced fission products yields, June 2025.

[33] G. P. A. Nobre, R. Capote, M. T. Pigni, A. Trkov, C. M. Mattoon, D. Neudecker, D. A. Brown, M. B. Chadwick, A. C. Kahler, N. A. Kleedtke, M. Zerkle, A. I. Hawari, C. W. Chapman, N. C. Fleming, J. L. Wormald, K. Ramić, Y. Danon, N. A. Gibson, P. Brain, M. W. Paris, G. M. Hale, I. J. Thompson, D. P. Barry, I. Stetcu, W. Haeck, A. E. Lovell, M. R. Mumpower, G. Potel, K. Kravvaris, G. Noguere, J. D. McDonnell, A. D. Carlson, M. Dunn, T. Kawano, D. Wiarda, I. Al-Qasir, G. Arbanas, R. Arcilla, B. Beck, D. Bernard, R. Beyer, J. M. Brown, O. Cabellos, R. J. Casperson, Y. Cheng, E. V. Chimanski, R. Coles, M. Cornock, J. Cotchen, J. P. W. Crozier, D. E. Cullen, A. Daskalakis, M. A. Descalle, D. D. DiJulio, P. Dimitriou, A. C. Dreyfuss, I. Durán, R. Ferrer, T. Gaines, V. Gillette, G. Gert, K. H. Guber, J. D. Haverkamp, M. W. Herman, J. Holmes, M. Hursin, N. Jisrawi, A. R. Junghans, K. J. Kelly, H. I. Kim, K. S. Kim, A. J. Koning, M. Koštál, B. K. Laramee, A. Lauer-Coles, L. Leal, H. Y. Lee, A. M. Lewis, J. Malec, J. I. Márquez Damián, W. J. Marshall, A. Mattera, G. Muhrer, A. Ney, W. E. Ormand, D. K. Parsons, C. M. Percher, V. G. Pronyaev, A. Qteish, S. Quaglioni, M. Rapp, J. J. Ressler, M. Rising, D. Rochman, P. K. Romano, D. Roubtsov, G. Schnabel, M. Schulc, G. J. Siemers, A. A. Sonzogni, P. Talou, J. Thompson, T. H. Trumbull, S. C. van der Marck, M. Vorabbi, C. Wemple, K. A. Wendt, M. White, and R. Q. Wright. ENDF/B-VIII.1: Updated Nuclear Reaction Data Library for Science and Applications. *Nuclear Data Sheets*, 210:1–224, May 2026.

[34] Osamu Iwamoto, Nobuyuki Iwamoto, Satoshi Kunieda, Futoshi Minato, Shinsuke Nakayama, Yutaka Abe, Kohsuke Tsubakihara, Shin Okumura, Chikako Ishizuka, Tadashi Yoshida, Satoshi Chiba, Naohiko Otuka, Jean-Christophe Sublet, Hiroki Iwamoto,

Kazuyoshi Yamamoto, Yasunobu Nagaya, Kenichi Tada, Chikara Konno, Norihiro Matsuda, Kenji Yokoyama, Hiroshi Taninaka, Akito Oizumi, Masahiro Fukushima, Shoichiro Okita, Go Chiba, Satoshi Sato, Masayuki Ohta, and Saerom Kwon. Japanese evaluated nuclear data library version 5: JENDL-5. *Journal of Nuclear Science and Technology*, 60(1):1–60, January 2023.

[35] D. Rochman, A. Koning, S. Goriely, and S. Hilaire. TENDL-astro: A new nuclear data set for astrophysics interest. *Nuclear Physics A*, 1053:122951, January 2025.

[36] NEA. Specifications for the Generalised Nuclear Database Structure (GNDS): Version 2.0. Technical Report 7647, OECD, Paris, September 2023.

[37] Ivan A. Kodeli and Enrico Sartori. SINBAD – Radiation shielding benchmark experiments. *Annals of Nuclear Energy*, 159:108254, September 2021.

[38] OECD NEA. International Criticality Safety Benchmark Evaluation Project (ICSBEP), 2022.

[39] OECD NEA. International Handbook of Evaluated Reactor Physics Benchmark Experiments (IRPhE), 2023.

[40] Bor Kos and Ivan Kodeli. MCNP modelling of the ASPIS Iron88 SINBAD shielding benchmark. Technical report / INDC report INDC(NDS)-0771, IAEA Nuclear Data Section, Vienna, Austria, September 2018.

[41] David A. Brown. ENDF-6 Formats Manual - Data Formats and Procedures for the Evaluated Nuclear Data Files ENDF/B-VI, ENDF/B-VII and ENDF/B-VIII. Technical Report BNL–224854-2023-INRE, Brookhaven National Laboratory (BNL), Upton, NY (United States), September 2023.

[42] David Gottlieb and Chi-Wang Shu. On the Gibbs Phenomenon and Its Resolution. *SIAM Review*, 39(4):644–668, January 1997.

[43] Jeremy Lloyd Conlin and Paul Romano. A Compact ENDF (ACE) Format Specification. Technical Report LA-UR–19-29016, Los Alamos National Laboratory (LANL), Los Alamos, NM (United States), September 2019.

[44] Jaakko Leppänen. Serpent – a Continuous-energy Monte Carlo Reactor Physics Burnup Calculation Code. User's Manual, VTT, Espoo, Finland, June 2015.

[45] Tatsuhiko Sato, Yosuke Iwamoto, Shintaro Hashimoto, Tatsuhiko Ogawa, Takuya Furuta, Shin-Ichiro Abe, Takeshi Kai, Yusuke Matsuya, Norihiro Matsuda, Yuho Hirata, Takuya Sekikawa, Lan Yao, Pi-En Tsai, Hunter N. Ratliff, Hiroshi Iwase, Yasuhito Sakaki, Kenta Sugihara, Nobuhiro Shigyo, Lembit Sihver, and Koji Niita. Recent improvements of the particle and heavy ion transport code system – PHITS version 3.33. *Journal of Nuclear Science and Technology*, 61(1):127–135, January 2024. _eprint: https://doi.org/10.1080/00223131.2023.2275736.

[46] B. Cochet, A. Jinaphanh, L. Heulers, and O. Jacquet. Capabilities overview of the MORET 5 Monte Carlo code. *Annals of Nuclear Energy*, 82:74–84, August 2015.

[47] N. M. Greene, J. L. Lucius, L. M. Petrie, I. I. I. Ford, J. E. White, and R. Q. Wright. AMPX: a modular code system for generating coupled multigroup neutron-gamma libraries from ENDF/B. Technical Report ORNL/TM-3706, Oak Ridge National Lab., Tenn. (USA), February 1976.

[48] Caleb Mattoon, Bret Beck, and Godfree Gert. Managing and Processing Nuclear Data Libraries with FUDGE. *EPJ Web of Conferences*, 284:14010, 2023.

[49] G. Ferran, W. Haeck, and M. Gonin. Development Progress of the GAIA Nuclear Data Processing Software. *Nuclear Data Sheets*, 118:491–493, April 2014.

[50] Cédric Jouanne and Mireille Coste-Delclaux. Recent developments in the GALILÉE-1 processing code. *EPJ Web of Conferences*, 294:06003, 2024.

[51] Kenichi Tada, Akio Yamamoto, Satoshi Kunieda, Chikara Konno, Ryoichi Kondo, Tomohiro Endo, Go Chiba, Michitaka Ono, and Masayuki Tojo. Development of nuclear data processing code FRENDY version 2. *Journal of Nuclear Science and Technology*, 61(6):830–839, June 2024. _eprint: https://doi.org/10.1080/00223131.2023.2278600.

[52] V. V. Sinitsa and A. A. Rineiskij. GRUKON - A Package of appliced computer programs system input and operating procedures of functional modules. Technical Report INDC(CCP)-344, Institute of Physics and Power Engineering, Obninsk, April 1993.

[53] Dermott E. Cullen. PREPRO 2023: 2023 ENDF/B Pre-processing Codes. Technical Report IAEA-NDS-0241, Nuclear Data Section, International Atomic Energy Agency, Vienna, Austria, June 2023.

[54] R. E. MacFarlane and A. C. Kahler. Methods for Processing ENDF/B-VII with NJOY. *Nuclear Data Sheets*, 111(12):2739–2890, December 2010.

[55] I. Remec and F.B.K. Kam. H.B. Robinson-2 pressure vessel benchmark. Technical Report NUREG/CR–6453, ORNL/TM–13204, 576068, Oak Ridge National Laboratory (ORNL), February 1998.

[56] M. J. Norgett, M. T. Robinson, and I. M. Torrens. A proposed method of calculating displacement dose rates. *Nuclear Engineering and Design*, 33(1):50–54, August 1975.

[57] Kai Nordlund, S.J. Zinkle, Andrea Sand, Fredric Granberg, Robert Averback, R.E. Stoller, Tomoaki Suzudo, Lorenzo Malerba, Florian Banhart, William Weber, Francois Willaime, Sergei Dudarev, and D. Simeone. Improving atomic displacement and replacement calculations with physically realistic damage models. *Nature Communications*, 9, March 2018.

[58] J. Ch. Sublet, I. P. Bondarenko, G. Bonny, J. L. Conlin, M. R. Gilbert, L. R. Greenwood, P. J. Griffin, P. Helgesson, Y. Iwamoto, V. A. Khryachkov, T. A. Khromyleva, A. Yu. Konobeyev, N. Lazarev, L. Luneville, F. Mota, C. J. Ortiz, D. Rochman, S. P. Simakov, D. Simeone, H. Sjostrand, D. Terentyev, and R. Vila. Neutron-induced damage simulations: Beyond defect production cross-section, displacement per atom and iron-based metrics. *The European Physical Journal Plus*, 134(7):350, July 2019.

[59] Shengli Chen. *Maîtrise des biais et incertitudes des sections efficaces et de la modélisation de la cinématique associées aux réactions nucléaires conduisant aux dommages dans les matériaux sous irradiation.* These de doctorat, Université Grenoble Alpes, September 2020.

[60] Shengli Chen, David Bernard, Léonie Tamagno, Olivier Litaize, and Patrick Blaise. Uncertainty assessment for the displacement damage of a pressurized water reactor vessel. *Nuclear Materials and Energy*, 28:101017, September 2021.

[61] Romain Vuiart, Patrizio Console Camprini, Kenneth W. Burn, Mariya Brovchenko, Julien Taforeau, and Eric Dumonteil. Impact of Fission Spectrum Modeling on Aging Estimations of Pressurized Water Reactor Vessels. *2021 ANS Winter Meeting*, page 4, October 2021.

[62] JCGM. Guide to the expression of uncertainty in measurement — Part 1: Introduction. Technical report, Joint Committee for Guides in Metrology, January 2023.

[63] D. Rochman, S. C. van der Marck, A. J. Koning, H. Sjöstrand, and W. Zwermann. Uncertainty Propagation with Fast Monte Carlo Techniques. *Nuclear Data Sheets*, 118:367–369, April 2014.

[64] Ehud Greenspan. Sensitivity Functions for Uncertainty Analysis. In Jeffery Lewins and Martin Becker, editors, *Advances in Nuclear Science and Technology: Volume 14 Sensitivity and Uncertainty Analysis of Reactor Performance Parameters*, pages 193–246. Springer US, Boston, MA, 1982.

[65] H. Rief. Generalized Monte Carlo perturbation algorithms for correlated sampling and a second-order Taylor series approach. *Annals of Nuclear Energy*, 11(9):455–476, January 1984.

[66] Augusto Gandini. Generalized Perturbation Theory (GPT) Methods. A Heuristic Approach. In Jeffery Lewins and Martin Becker, editors, *Advances in Nuclear Science and Technology: Festschrift in Honor of Eugene P. Wigner*, Advances in Nuclear Science and Technology, pages 205–380. Springer US, Boston, MA, 1987.

[67] Micheal A. Smith, C. Adams, W. S. Yang, and E. E. Lewis. VARI3D & PERSENT: Perturbation and Sensitivity Analysis. Technical Report ANL/NE–13/8-Rev.6, Argonne National Laboratory (ANL), Argonne, IL (United States), April 2025.

[68] Brian C. Kiedrowski and Forrest B. Brown. Applications of Adjoint-Based Techniques in Continuous-Energy Monte Carlo Criticality Calculations. In D. Caruge, C. Calvin, C.M. Diop, F. Malvagi, and J.-C. Trama, editors, *SNA + MC 2013 - Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo*, page 03501, Paris, France, 2014. EDP Sciences.

[69] Juan A. Monleon de la Lluvia, Mariya Brovchenko, Dimitri Rochman, and Eric Dumonteil. Toward Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations. *Nuclear Science and Engineering*, pages 1–23, 2025.

[70] Augusto Gandini. The heuristically-based generalized perturbation theory. *EPJ Nuclear Sciences & Technologies*, 7:7, 2021.

[71] Manuele Aufiero, Adrien Bidaud, Mathieu Hursin, Jaakko Leppänen, Giuseppe Palmiotti, Sandro Pelloni, and Pablo Rubiolo. A collision history-based approach to sensitivity/perturbation calculations in the continuous energy Monte Carlo code SERPENT. *Annals of Nuclear Energy*, 85:245–258, 2015.

[72] J. D. Densmore, G. W. McKinney, J. S. Hendricks, and NM (United States) Los Alamos National Lab. Correction to the MCNP perturbation feature for cross-section dependent tallies. Technical Report LA–13374, Los Alamos National Lab., NM (United States), October 1997.

[73] Ville Valtavirta, Manuele Aufiero, and Jaakko Leppänen. Collision-history Based Sensitivity/Perturbation Calculation Capabilities in Serpent 2.1.30. In *Proceedings of the ANS International Conference on Best-Estimate Plus Uncertainties Methods (BEPU 2018)*, page 13, Lucca, Italy, 2018. American Nuclear Society.

[74] A. Jinaphanh, G. Valocchi, and A. Zoia. Implementation and testing of generalized perturbation theory capabilities in TRIPOLI-4®. *Nuclear Science and Engineering*, 199(sup1):S406–S434, 2025.

[75] VTT Technical Research Centre of Finland. Serpent Wiki.

[76] M. C. G. Hall. Monte Carlo perturbation theory in neutron transport calculations. In *A Review of the Theory and Application of Monte Carlo Methods*, pages 47–62, Oak Ridge, Tennessee, August 1980. Oak Ridge National Laboratory, Radiation Shielding Information Center. Number: ORNL/RSIC–44.

[77] Reuven Y. Rubinstein and Alexander Shapiro. Optimization of static simulation models by the score function method. *Mathematics and Computers in Simulation*, 32(4):373–392, October 1990.

[78] Peter W. Glynn. Likelihood ratio gradient estimation for stochastic systems. *Communications of the ACM*, 33(10):75–84, October 1990.

[79] G. W. McKinney and J. L. Iverson. Verification of the Monte Carlo differential operator technique for MCNP. Technical Report LA–13098, Los Alamos National Lab., NM (United States), January 1996.

[80] L. L. Carter and E. D. Cashwell. Particle-transport simulation with the Monte Carlo method. Technical Report TID–26607, Los Alamos Scientific Lab., N.Mex. (USA), January 1975.

[81] Jeffrey Favorite. Using the MCNP Taylor series perturbation feature (efficiently) for shielding problems. *EPJ Web of Conferences*, 153:06030, 2017.

[82] Alexander R Clark. Easy_pert: a Python tool for writting PERT cards and parsing PERT card results, October 2022.

[83] A. Jinaphanh and N. Leclaire. Continuous-energy perturbation methods in the MORET 5 code. *Annals of Nuclear Energy*, 114:395–406, April 2018.

[84] Odile Petit, François-Xavier Hugot, Yi-Kang Lee, Cédric Jouanne, and Alain Mazzolo. TRIPOLI-4 Version 4 User Guide. Technical report, CEA, 2008.

[85] Brian C. Kiedrowski. Review of Early 21st-Century Monte Carlo Perturbation and Sensitivity Techniques for k -Eigenvalue Radiation Transport Calculations. *Nuclear Science and Engineering*, 185(3), March 2017. Institution: North Carolina State University, Raleigh, NC (United States).

[86] Tao He and Bingjing Su. Comparison between correlated sampling and the perturbation technique of MCNP5 for fixed-source problems. *Annals of Nuclear Energy*, 38(6):1318–1326, June 2011.

[87] Jeffrey A. Favorite. A General-Purpose Code for Correlated Sampling Using Batch Statistics with MCNP6 for Fixed-Source Problems. Technical Report LA-UR–23-30650, Los Alamos National Laboratory (LANL), Los Alamos, NM (United States), September 2023.

[88] Achintya Haldar and Sankaran Mahadevan. *Probability, Reliability and Statistical Methods in Engineering Design*, volume 23. John Wiley & Sons, Inc., January 2001. Issue: 1 Journal Abbreviation: Structural Safety.

[89] William Wieselquist and Robert Alexander Lefebvre. SCALE 6.3.1 User Manual. Technical Report ORNL/TM-SCALE-6.3.1, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States), February 2023.

[90] A. J. Koning and D. Rochman. Towards sustainable nuclear energy: Putting nuclear physics to work. *Annals of Nuclear Energy*, 35(11):2024–2030, November 2008.

[91] M. D. McKay, R. J. Beckman, and W. J. Conover. A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code. *Technometrics*, 21(2):239–245, 1979.

[92] I. M Sobol'. On the distribution of points in a cube and the approximate evaluation of integrals. *USSR Computational Mathematics and Mathematical Physics*, 7(4):86–112, January 1967.

[93] Michael Stein. Large Sample Properties of Simulations Using Latin Hypercube Sampling. *Technometrics*, 29(2):143–151, 1987.

[94] Kenneth Lange. Eigenvalues and Eigenvectors. In Kenneth Lange, editor, *Numerical Analysis for Statisticians*, pages 113–128. Springer, New York, NY, 2010.

[95] Gašper Žerovnik. Random sampling and validation of covariance matrices of resonance parameters. *EPJ Web of Conferences*, 146:02014, January 2017.

[96] Kenneth Lange. Linear Regression and Matrix Inversion. In Kenneth Lange, editor, *Numerical Analysis for Statisticians*, pages 93–111. Springer, New York, NY, 2010.

[97] Kenneth Lange. Singular Value Decomposition. In Kenneth Lange, editor, *Numerical Analysis for Statisticians*, pages 129–142. Springer, New York, NY, 2010.

[98] T. Tony Cai, Cun-Hui Zhang, and Harrison H. Zhou. Optimal rates of convergence for covariance matrix estimation. *The Annals of Statistics*, 38(4), August 2010. arXiv:1010.3866 [math].

[99] Xiangzhao Cui, Chun Li, Jine Zhao, Li Zeng, Defei Zhang, and Jianxin Pan. Covariance structure regularization via Frobenius-norm discrepancy. *Linear Algebra and its Applications*, 510:124–145, December 2016.

[100] D. Rochman, W. Zwermann, S. C. van der Marck, A. J. Koning, H. Sjöstrand, P. Helgesson, and B. Krzykacz-Hausmann. Efficient Use of Monte Carlo: Uncertainty Propagation. *Nuclear Science and Engineering*, 177(3):337–349, July 2014. _eprint: https://doi.org/10.13182/NSE13-32.

[101] W Zwermann, B Krzykacz-Hausmann, L Gallner, M Klein, A Pautz, and K Velkov. Aleatoric and Epistemic Uncertainties in Sampling-Based Nuclear Data Uncertainty and Sensitivity Analyses. In *International Conference on the Physics of Reactors 2012 (PHYSOR 2012): Advances in Reactor Physics*, volume 4, pages 2816–2827, Knoxville, Tennessee, USA, April 2012. American Nuclear Society.

[102] Arnau Albà, Romana Boiger, Dimitri Rochman, and Andreas Adelmann. Lasso Monte Carlo, a variation on multi fidelity methods for high-dimensional uncertainty quantification. *Journal of Applied Statistics*, 52(15):2799–2835, November 2025. _eprint: https://doi.org/10.1080/02664763.2025.2487505.

[103] Juan A. Monleon de la Lluvia. KIKA (kika-nd), 2025.

[104] Kenichi Tada, Ryoichi Kondo, Tomohiro Endo, and Akio Yamamoto. Development of ACE file perturbation tool using FRENDY. *Journal of Nuclear Science and Technology*, 60(6):624–631, June 2023. _eprint: https://doi.org/10.1080/00223131.2022.2130463.

[105] Shuhei Maruyama, Akio Yamamoto, and Tomohiro Endo. Quantifying uncertainty induced by scattering angle distribution using maximum entropy method. *Annals of Nuclear Energy*, 205:110591, September 2024.

[106] L. Fiorito, G. Žerovnik, A. Stankovskiy, G. Van den Eynde, and P. E. Labeau. Nuclear data uncertainty propagation to integral responses using SANDY. *Annals of Nuclear Energy*, 101:359–366, March 2017.

[107] njoy/ACEtk, January 2026. original-date: 2017-06-22T21:41:07Z.

[108] W. Haeck, N. Gibson, and P. Talou. ENDFtk: A robust tool for reading and writing ENDF-formatted nuclear data. *Computer Physics Communications*, 303:109245, October 2024.

[109] Juan A. Monleon de la Lluvia. juanmonleon/kika-release, February 2026. original-date: 2026-01-27T17:53:09Z.

[110] Shengli Chen, David Bernard, Pascal Archier, Cyrille De Saint Jean, Gilles Noguere, and Pierre Tamagno. Covariance Matrices for Differential and Angle-Integrated Neutron-Induced Elastic and Inelastic Scattering Cross Sections of 56Fe. *EPJ Web of Conferences*, 211:07002, 2019.

[111] Alireza Haghighat and John Wagner. Monte Carlo variance reduction with deterministic importance functions. *Progress in Nuclear Energy*, 42:25–53, December 2003.

[112] Ivan A. Kodeli and Steven Van Der Marck. Consistency Among the Results of the ASPIS Iron88, PCA Replica, and PCA ORNL Benchmark Experiments. *Nuclear Science and Engineering*, 198(2):381–390, February 2024.

[113] M. Herman, A. Trkov, R. Capote, G. P. A. Nobre, D. A. Brown, R. Arcilla, Y. Danon, A. Plompen, S. F. Mughabghab, Q. Jing, G. Zhigang, L. Tingjin, L. Hanlin, R. Xichao, L. Leal, B. V. Carlson, T. Kawano, M. Sin, S. P. Simakov, and K. Guber. Evaluation of Neutron Reactions on Iron Isotopes for CIELO and ENDF/B-VIII.0. *Nuclear Data Sheets*, 148(C), February 2018. Number: BNL–203650-2018-JAAM.

[114] W. E. Kinney and J. W. McConnell. High resolution neutron scattering experiments at ORELA. In *Proceedings of the International Conference on the Interactions of Neutrons with Nuclei*, volume 2, page 1319, Lowell, Massachusetts, 1976. EXFOR entry 10571.002.

[115] A. Smith and P. Guenther. Scattering of MeV Neutrons from Elemental Iron. *Nuclear Science and Engineering*, 73(2):186–195, February 1980. _eprint: https://doi.org/10.13182/NSE80-A18698.

[116] M. Herman, R. Capote, M. Sin, A. Trkov, B. V. Carlson, P. Obložinsky, C. M. Mattoon, H. Wienke, S. Hoblit, C. Young-Sik, G. P. A. Nobre, V. A. Plujko, and V. Zerkin. EMPIRE-3.2 Malta: Modular System for Nuclear Reaction Calculations and Nuclear Data Evaluation. User's Manual / Technical report BNL-101378-2013, Brookhaven National Laboratory, Upton, NY, United States, August 2013.

[117] O. Iwamoto, N. Iwamoto, S. Kunieda, F. Minato, and K. Shibata. The CCONE Code System and its Application to Nuclear Data Evaluation for Fission and Other Reactions. *Nuclear Data Sheets*, 131:259–288, January 2016.

[118] Naoki Yamano, Tsunenori Inakura, Chikako Ishizuka, and Satoshi Chiba. Crucial importance of correlation between cross sections and angular distributions in nuclear data of Si28 on estimation of uncertainty of neutron dose penetrating a thick concrete. *Journal of Nuclear Science and Technology*, 59(5):641–646, May 2022.

[119] F. S. Dietrich. A New Method for Estimating Neutron Reaction Cross Sections Based on Wick's Limit. In *AIP Conference Proceedings*, volume 769, pages 300–303, Santa Fe, New Mexico (USA), 2005. AIP.

[120] International Conference on Neutron Physics and Nuclear Data for Reactors and Other Applied Purposes. *Proceedings of an International Conference on Neutron Physics and Nuclear Data for Reactors and Other Applied Purposes*. OECD Nuclear Energy Agency, Paris, 1978. Open Library ID: OL4025532M.

[121] E. Pirovano. Cross section and neutron angular distribution measurements of neutron scattering on natural iron. *Physical Review C*, 99(2), 2019.

[122] G. Gkatis, M. Diakaki, G. Noguere, A. Oprea, C. Paradela, E. Pirovano, and A. J. M. Plompen. Double-differential-cross-section measurements of neutron scattering on Fe56. *Physical Review C*, 112(4):044603, October 2025.

[123] R. Avery. Reactor Physics Division Annual Report. Technical Report ANL–7210, Argonne National Lab., Ill., January 1966.

[124] E. Barnard, J. A. M. De Villiers, C. A. Engelbrecht, D. Reitmann, and A. B. Smith. High-resolution fast-neutron cross sections of iron. *Nuclear Physics A*, 118(2):321–346, October 1968.

[125] Y. Tomita. Intermediate structure in the neutron scattering cross sections of iron. *Nuclear Physics A*, 210(1):51–59, August 1973.

[126] A. Jacquot and C. Rousseau. Fast neutron scattering cross sections of iron. *Nuclear Physics*, 84(1):239–240, August 1966.

[127] S. G. Malmskog. Differential neutron elastic scattering cross sections for Cr, Fe and Ni in the energy range 500 to 1400 keV. Technical Report AE-FN-25, Aktiebolaget Atomenergi, Stockholm, Sweden, 1972. Originally cited in EXFOR as (R,AE-FN-25,1972).

[128] J. T. Wells, A. B. Tucker, and W. E. Meyerhof. Threshold Effects in Neutron Elastic Scattering. *Physical Review*, 131(4):1644–1660, August 1963.

[129] A. P. D. Ramirez, J. R. Vanhoy, S. F. Hicks, M. T. McEllistrem, E. E. Peters, S. Mukhopadhyay, T. D. Harrison, T. J. Howard, D. T. Jackson, P. D. Lenzen, T. D. Nguyen, R. L. Pecha, B. G. Rice, B. K. Thompson, and S. W. Yates. Neutron scattering cross section measurements for Fe56. *Physical Review C*, 95(6):064605, June 2017.

[130] B. Holmqvist, T. Wiedling, V. Benzi, and L. Zuffi. Analysis of fast neutron elastic scattering from tantalum using a non-spherical optical potential. *Nuclear Physics A*, 150(1):105–113, July 1970.

[131] Kineo Tsukada, Shigeya Tanaka, Yoshiaki Tomita, and Michio Maruyama. Elastic and inelastic scattering of fast neutrons from iron, nickel and tungsten. *Nuclear Physics A*, 125(3):641–653, March 1969.

[132] C. O. Muehlhause, S. D. Bloom, H. E. Wegner, and G. N. Glasoe. Neutron Scattering from Iron and Carbon by Time-of-Flight. *Physical Review*, 103(3):720–726, August 1956.

[133] C. M. Perey, F. G. Perey, J. A. Harvey, N. W. Hill, and N. M. Larson. Fe56 and Ni60 Resonance Parameters. In Syed M. Qaim, editor, *Nuclear Data for Science and Technology*, pages 41–43, Berlin, Heidelberg, 1992. Springer.

[134] I. A. Korzh, V. A. Mishchenko, E. N. Mozhzhukhin, N. M. Pravdivyi, and I. E. Sanzhur. Differential neutron scattering cross-sections at 1.5 to 3.0 MeV for titanium, iron and bismuth nuclei. *Ukrainian Physical Journal*, 22, 1977. NSR: 1992PEZQ Institute: Inst. Yadernyh Doslidzhen, N.A.N. Ukraini, Kyiv Also indexed as: YFI-23 (1977).

[135] Lawrence Cranberg and Jules S. Levin. Neutron Scattering at 2.45 Mev by a Time-of-Flight Method. *Physical Review*, 103(2):343–352, July 1956.

[136] M. K. Machwe, D. W. Kent, and S. C. Snowdon. Elastic Scattering of 3.7-Mev Neutrons from S, Fe, Co, Ni, Cu, and Zn. *Physical Review*, 114(6):1563–1570, June 1959.

[137] V. M. Morozov, Yu G. Zubov, and N. S. Lebedeva. Two mechanisms of elastic scattering in the optical nuclear model. *Yadern. Fiz., v. 17, no. 4, pp. 734-741*, April 1973. Institution: Inst. of Atomic Energy, Moscow.

[138] J. R. Beyster, M. Walt, and E. W. Salmi. Interaction of 1.0-, 1.77-, 2.5-, 3.25-, and 7.0-Mev Neutrons with Nuclei. *Physical Review*, 104(5):1319–1331, December 1956.

[139] H. H. Landon, A. J. Elwyn, G. N. Glasoe, and S. Oleksa. Neutron Scattering at 2.2 MeV by Time of Flight. *Physical Review*, 112(4):1192–1200, November 1958.

[140] R. L. Becker, W. G. Guindon, and G. J. Smith. Elastic scattering of 3.2 MeV neutrons from many elements. *Nuclear Physics*, 89(1):154–164, December 1966.

[141] J. R. Vanhoy, S. H. Liu, S. F. Hicks, B. M. Combs, B. P. Crider, A. J. French, E. A. Garza, T. Harrison, S. L. Henderson, T. J. Howard, M. T. McEllistrem, S. Nigam, R. L. Pecha, E. E. Peters, F. M. Prados-Estévez, A. P. D. Ramirez, B. G. Rice, T. J. Ross, Z. C. Santonil, L. C. Sidwell, J. L. Steves, B. K. Thompson, and S. W. Yates. Fe54 neutron elastic and inelastic scattering differential cross sections from 2–6 MeV. *Nuclear Physics A*, 972:107–120, April 2018.

[142] Th. Schweitzer, D. Seeliger, and K. Seidel. Differentielle elastische und unelastische Streuquerschnitte für Magnesium, Silizium, Eisen und Wismut bei einer Neutroneneinschussenergie von 3,4 MeV. *Kernenergie*, 20:174, 1977.

[143] J. Cabé, M. Laurat, P. Yvon, and G. Bardolle. Recherche d'une structure intermédiaire dans les sections efficaces totales neutroniques. *Nuclear Physics A*, 102(1):92–104, September 1967.

[144] A. Begum and R. B. Galloway. Polarisation in 2.9 MeV neutron elastic energy scattering by W, Tl, Bi and U. *Journal of Physics G: Nuclear Physics*, 7(4):535, April 1981.

[145] N. A. Bostrom, I. L. Morgan, J. T. Prod'homme, P. L. Okhuysen, and Jr Hudson. Interaction of Fast Neutrons in Iron, Lead, Oxygen, and Nitrogen. Technical Report WADC-TR-59-31; AD-216652, Texas Nuclear Corp., Austin, Tex., February 1959.

[146] D. J. Bredin. Scattering and Polarization of Neutrons from Al, Si, Fe, and Co at 2 MeV. *Physical Review*, 135(2B):B412–B416, July 1964.

[147] M. Walt and H. H. Barschall. Scattering of 1-Mev Neutrons by Intermediate and Heavy Elements. *Physical Review*, 93(5):1062–1068, March 1954.

[148] R. B. Galloway and A. Waheed. The differential cross section and polarization for the elastic scattering of 2.9 MeV neutrons by Fe, Cu, I, Hg and Pb. *Nuclear Physics A*, 318(1):173–204, April 1979.

[149] S. A. Cox and E. E. Dowling Cox. Polarization in the Elastic Scattering of Neutrons from Medium- and Heavy-Weight Elements. Technical report ANL-7935, Argonne National Laboratory, Argonne, Illinois, USA, June 1972.

[150] I. A. Korzh, V. A. Mishchenko, M. V. Pasechnik, N. M. Pravdivyi, I. E. Sanzhur, and I. A. Tockii. Elastic scattering of 1.5 MeV neutrons by medium-atomic-weight nuclei. *Ukrainian Physics Journal*, 12, 1968. English translation of: Ukr. Fiz. Zh. 12, 1571 (1967). CFD optical model; data tables included. Also reported in YFI-4, p.36 (1967); INDC(CCP)-017.

[151] L. Ya. Kazakova, V. E. Kolesov, V. I. Popov, O. A. Salnikov, V. M. Sluchevskaja, and V. I. Trikova. Elastic scattering of neutrons with initial energy 2 MeV. Technical Report EANDC-50-S, Vol. 2, Paper 200, European-American Nuclear Data Committee, 1965. Abstract also in Nuclear Structure Study with Neutrons conference (Antwerp 1965) p. 576.

[152] M. V. Pasechnik, I. F. Barchuk, I. A. Totsky, V. I. Strizhak, A. M. Korolov, Y. V. Hofman, G. N. Lovchikova, E. A. Koltynin, and G. B. Yankov. Fast Neutron Scattering and Capture by Atomic Nuclei. In *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy*, volume 15. United Nations, Geneva, 1958. Meeting Name: UN International Conference on the Peaceful Uses of Atomic Energy (2nd : 1958 : Geneva).

[153] S. E. Darden, R. B. Perkins, and R. B. Walton. Small-Angle Scattering of Neutrons by Intermediate and Heavy Nuclei. *Physical Review*, 100(5):1315–1317, December 1955.

[154] V. I. Popov. Angular distribution of elastically and inelastically scattered 2.9-Mev neutrons. *The Soviet Journal of Atomic Energy*, 3(6):1379–1386, December 1957.

[155] G. N. Lovchikova. Measurement of the angular distribution of 0.9-Mev neutrons elastically scattered on Bi, Pb, Sn, Fe, and Al. *The Soviet Journal of Atomic Energy*, 2(2):197–201, March 1957.

[156] J. C. Hopkins and M. G. Silbert. Inelastic Scattering of 2- to 5-MeV Neutrons by Iron*. *Nuclear Science and Engineering*, 19(4):431–436, August 1964. _eprint: https://doi.org/10.13182/NSE64-A19000.

[157] W. L. Rodgers, E. F. Shrader, and J. T. Lindow. Neutron scattering from C-12, Fe-54, Fe-56, Cu-65, Ni-58 and Ni-60. Chicago Operations Office, U.S. Atomic Energy Commission Contract Reports COO-1573-33, Case Western Reserve University, 1967.

[158] O. A. Salnikov. Angular distribution of elastic and inelastic scattering of 2.34 MeV neutrons from chromium, iron, and lead. *The Soviet Journal of Atomic Energy*, 3(2):869–874, August 1957.

[159] Kineo Tsukada, Shigeya Tanaka, and Michio Maruyama. Inelastic Scattering of Neutrons by the Time-of-Flight Method. *Journal of the Physical Society of Japan*, 16(2):166–175, February 1961.

[160] P. H. Stelson and W. M. Preston. The Inelastic Scattering of Fast Neutrons from Iron. *Physical Review*, 86(1):132–133, April 1952.

[161] M.J. Poole. CXLVIT. The inelastic scattering of neutrons: II. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 44(359):1398–1402, December 1953. _eprint: https://doi.org/10.1080/14786441208520409.

[162] Denise Neudecker, Rudolf Frühwirth, and Helmut Leeb. Peelle's Pertinent Puzzle: A Fake Due to Improper Analysis. *Nuclear Science and Engineering*, 170(1):54–60, 2012.

[163] Satoshi Chiba and Donald L. Smith. A suggested procedure for resolving an anomaly in least-squares data analysis known as "Peelle's Pertinent Puzzle" and the general implications for nuclear data evaluation. Technical report ANL/NDM-121, Argonne National Laboratory, Argonne, Illinois, USA, 1991.

[164] Nancy M. Larson. Updated User's Guide for Sammy: Multilevel R-Matrix Fits to Neutron Data Using Bayes' Equations. Technical Report ORNL/TM-9179/R8, Oak Ridge National Laboratory (ORNL), October 2008.

[165] M. Moxon, T. C. Ware, and C. J. Dean. REFIT-2009, Multilevel Resonance Parameter Least Square Fit of N Transmission, Capture, Fission & Self Indication Data. User's Guide NEA-0914/09, OECD Nuclear Energy Agency Data Bank, Paris, France, 2010.

[166] Arthur E Hoerl and Robert W Kennard. Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1):55–67, February 1970.

[167] Clare Rutterford, Andrew Copas, and Sandra Eldridge. Methods for sample size determination in cluster randomized trials. *International Journal of Epidemiology*, 44(3):1051–1067, June 2015.

[168] Clifford M Hurvich and Chih-Ling Tsai. Regression and time series model selection in small samples. *Biometrika*, 76(2):297–307, June 1989.

[169] Raymond T. Birge. The Calculation of Errors by the Method of Least Squares. *Physical Review*, 40(2):207–227, April 1932.

[170] Peter J. Rousseeuw and Christophe Croux. Alternatives to the Median Absolute Deviation. *Journal of the American Statistical Association*, 88(424):1273–1283, December 1993. _eprint: https://www.tandfonline.com/doi/pdf/10.1080/01621459.1993.10476408.

[171] Kenneth P. Burnham and David R. Anderson, editors. *Model Selection and Multimodel Inference*. Springer, New York, NY, 2004.

[172] J A Dahl, B D Ganapol, and J E Morel. Positive Scattering Cross Section Using Constrained Least Squares. In *Proceedings of the ANS Mathematics and Computation Topical Meeting*, Madrid, Spain, 1999. American Nuclear Society.

[173] A J Koning, S Hilaire, and S Goriely. TALYS: modeling of nuclear reactions. *The European Physical Journal A*, 59(6):131, 2023.

[174] A. J. Koning and J. P. Delaroche. Local and global nucleon optical models from 1 keV to 200 MeV. *Nuclear Physics A*, 713(3):231–310, January 2003.

[175] E. Sheldon, G. P. Couchell, S. A. Goodwin, S. C. Mathur, and D. J. Pullen. International conference on the interactions of neutrons with nuclei. Volume II. Joint, parallel and papers sessions. Proceedings of a conference held at Lowell, Massachusetts, July 6–9, 1976. Technical Report CONF-760715-P2, Lowell Univ., MA (USA), January 1976.

[176] Barthélémi Petillon, Peron Arthur, Romain Vuiart, Vaibhav Jaiswal, and Mariya Brovchenko. CALINS: A New ASNR Software for Sensitivity and Uncertainty Analysis in Nuclear Criticality Safety Evaluations. In *Nuclear Criticality Safety Division 2025 Conference (NCSD 2025)*, pages 282–289, Austin, TX, United States, September 2025. American Nuclear Society.

[177] G. A. {Wright and M. J. Grimstone. Benchmark Testing of JEF-2.2 Data for Shielding Applications: Analysis of the Winfrith Iron 88 Benchmark Experiment. Technical Report AEA-RS-1231; EFF-Doc-229; JEF-Doc-421, AEA Technology, 1993.

[178] A. Trkov, P.J. Griffin, S.P. Simakov, L.R. Greenwood, K.I. Zolotarev, R. Capote, D.L. Aldama, V. Chechev, C. Destouches, A.C. Kahler, C. Konno, M. Koštál, M. Majerle, E. Malambu, M. Ohta, V.G. Pronyaev, V. Radulović, S. Sato, M. Schulc, E. Šimečková, I. Vavtar, J. Wagemans, M. White, and H. Yashima. IRDFF-II: A New Neutron Metrology Library. *Nuclear Data Sheets*, 163:1–108, January 2020.

[179] I. Kodeli and E. Sartori. Analysis of VENUS-3 benchmark experiment. In *Nuclear Energy in Central Europe 98, Proceedings*, pages 115–122, Terme Čatež, Slovenia, 1998. Nuclear Society of Slovenia. Number: INIS-SI–01-003.

[180] NEA. Prediction of Neutron Embrittlement in the Reactor Pressure Vessel. Venus-1 and Venus-3 benchmarks. Benchmark report NEA/NSC/DOC(2000)5, Nuclear Energy Agency, Organisation for Economic Co-operation and Development, Paris, France, March 2000.

[181] Romain Vuiart, Mariya Brovchenko, Julien Taforeau, and Eric Dumonteil. A Detailed Analysis of the H.B. Robinson-2 Reactor Pressure Vessel Dosimetry Benchmark. *Energies*, 15(14):5098, July 2022.

[182] D. A. Brown, M. B. Chadwick, R. Capote, A. C. Kahler, A. Trkov, M. W. Herman, A. A. Sonzogni, Y. Danon, A. D. Carlson, M. Dunn, D. L. Smith, G. M. Hale, G. Arbanas, R. Arcilla, C. R. Bates, B. Beck, B. Becker, F. Brown, R. J. Casperson, J. Conlin, D. E. Cullen, M.-A. Descalle, R. Firestone, T. Gaines, K. H. Guber, A. I. Hawari, J. Holmes, T. D. Johnson, T. Kawano, B. C. Kiedrowski, A. J. Koning, S. Kopecky, L. Leal, J. P. Lestone, C. Lubitz, J. I. Márquez Damián, C. M. Mattoon, E. A. McCutchan, S. Mughabghab, P. Navratil, D. Neudecker, G. P. A. Nobre, G. Noguere, M. Paris, M. T. Pigni, A. J. Plompen, B. Pritychenko, V. G. Pronyaev, D. Roubtsov, D. Rochman, P. Romano, P. Schillebeeckx, S. Simakov, M. Sin, I. Sirakov, B. Sleaford, V. Sobes, E. S. Soukhovitskii, I. Stetcu,

P. Talou, I. Thompson, S. van der Marck, L. Welser-Sherrill, D. Wiarda, M. White, J. L. Wormald, R. Q. Wright, M. Zerkle, G. Žerovnik, and Y. Zhu. ENDF/B-VIII.0: The 8th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data. *Nuclear Data Sheets*, 148(C), February 2018.

[183] Léonie Berge. *Contribution à la modélisation des spectres de neutrons prompts de fission. Propagation d'incertitudes sur un calul de fluence cuve.* Thèse de doctorat, Université Grenoble Alpes (ComUE), July 2015.

[184] D. Rochman, E. Bauge, A. Vasiliev, and H. Ferroukhi. Correlation $\overline{\nu}_p$–σ–χ in the fast neutron range via integral information. *EPJ Nuclear Sciences & Technologies*, 3:14, 2017.

[185] E. Bauge and D. A. Rochman. Cross-observables and cross-isotopes correlations in nuclear data from integral constraints. *EPJ Nuclear Sciences & Technologies*, 4:35, 2018.

[186] Thomas Ligonnet, Axel Laureau, Andreas Pautz, and Vincent Lamirand. High-Resolution Modeling Without Computation Slowdown for PETALE in CROCUS. *Nuclear Science and Engineering*, 199(sup1):33, April 2025.

[187] G. Aliberti, G. Palmiotti, M. Salvatores, T. K. Kim, T. A. Taiwo, M. Anitescu, I. Kodeli, et al. Nuclear data sensitivity, uncertainty and target accuracy assessment for future nuclear systems. *Annals of Nuclear Energy*, 33:700–733, 2006.

[188] I. M Sobol. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. *Mathematics and Computers in Simulation*, 55(1):271–280, February 2001.

[189] A. Saltelli, editor. *Global sensitivity analysis: the primer.* John Wiley, Chichester, England; Hoboken, NJ, 2008.

[190] Toshimitsu Homma and Andrea Saltelli. Importance measures in global sensitivity analysis of nonlinear models. *Reliability Engineering & System Safety*, 52(1):1–17, April 1996.

[191] Andrea Saltelli. Making best use of model evaluations to compute sensitivity indices. *Computer Physics Communications*, 145(2):280–297, May 2002.

[192] Bruno Sudret. Global sensitivity analysis using polynomial chaos expansions. *Reliability Engineering & System Safety*, 93(7):964–979, July 2008.

[193] Max D Morris. Factorial Sampling Plans for Preliminary Computational Experiments. *Technometrics*, 33(2):161–174, May 1991.

[194] S. Kucherenko, S. Tarantola, and P. Annoni. Estimation of global sensitivity indices for models with dependent variables. *Computer Physics Communications*, 183(4):937–946, April 2012.

[195] Thierry A. Mara and Stefano Tarantola. Variance-based sensitivity indices for models with dependent inputs. *Reliability Engineering & System Safety*, 107:115–121, November 2012.

[196] Bolade A. Adetula and Pavel M. Bokov. Computational Method for Global Sensitivity Analysis of Reactor Neutronic Parameters. *Science and Technology of Nuclear Installations*, 2012:1–11, 2012.

[197] Eric Cervi, Xuefei Lu, Antonio Cammi, Francesco Di Maio, and Enrico Zio. Sensitivity-Analysis-Driven Surrogate Model for Molten Salt Reactors Control. *Journal of Nuclear Engineering*, 3(4):277–294, December 2022.

[198] L. Gilli, D. Lathouwers, J. L. Kloosterman, T. H. J. J. van der Hagen, A. J. Koning, and D. Rochman. Uncertainty quantification for criticality problems using non-intrusive and adaptive Polynomial Chaos techniques. *Annals of Nuclear Energy*, 56:71–80, June 2013.

[199] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian processes for machine learning*. Adaptive computation and machine learning. MIT Press, Cambridge, Mass., 3. print edition, 2008.

[200] Xu Wu, Chen Wang, and Tomasz Kozlowski. Kriging-based Surrogate Models for Uncertainty Quantification and Sensitivity Analysis. In *International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2017)*, Jeju, Korea, April 2017. Korean Nuclear Society.

[201] N. Otuka, B. Pritychenko, M. Fleming, Y. Jin, G. Pikulina, R. Suzuki, V. Devi, M. Mikhailiukova, S. Okumura, N. Soppera, T. Tada, S. Takács, S. Taova, V. V. Varlamov, J. M. Wang, S. C. Yang, and V. Zerkin. Progress in international collaboration on EXFOR library. *EPJ Web of Conferences*, 239:15001, 2020.

[202] Viktor Zerkin. X4Pro – universal, fully relational EXFOR database. *EPJ Web of Conferences*, 284:14015, 2023.

[203] Stephen P. Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge University Press, Cambridge, UK ; New York, 2004.

[204] Dieter Kraft. A Software Package for Sequential Quadratic Programming. Technical Report DFVLR-FB 88-28, Deutsche Forschungs- und Versuchsanstalt für Luft- und Raumfahrt (DLR), Köln, Germany, 1988.

[205] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C. J. Carey, I. Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1 0 Contributors. SciPy 1.0 - Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17(3):261–272, March 2020. arXiv:1907.10121 [cs].

List of Figures

1 Location of the French nuclear power plants operated by EDF, colored by the age of each site. . . . 2
2 Embrittlement of the reactor pressure vessel and its surveillance. . . . 3

I.1 Comparison between accuracy and precision. . . . 14
I.2 Logic of the weight window technique showing the splitting and Russian Roulette regimes relative to the weight bounds [15] . . . 20
I.3 Schematic illustrating the duality between forward and adjoint transport. In the adjoint mode, particles effectively travel "backwards" from the detector to the source [18]. . . . 22
I.4 The Nuclear Data Cycle. This schematic illustrates the iterative process of data refinement, starting from application-driven needs (e.g., reactors, health, space) to the measurement, evaluation, validation, and eventual release of finalized libraries such as JEFF-4.0 and ENDF/B-VIII.1 [29]. . . . 29
I.5 Hierarchical structure of ENDF-6 `MT` reaction numbers. . . . 32
I.6 Scattering geometry in the laboratory (LAB) frame. An incident neutron scatters from a target initially at rest, emerging at the laboratory angle θ relative to the beam direction. . . . 35
I.7 The first six Legendre polynomials $P_n(x)$ in the interval $[-1, 1]$, where x here corresponds to the scattering cosine μ. All orders are normalized such that $P_n(1) = 1$. The zeroth order P_0 (green) represents an isotropic distribution, while higher orders introduce increasing angular complexity. . . . 36
I.8 Schematic overview of typical NJOY processing sequences: the generation of continuous-energy ACE libraries for Monte Carlo codes (left) and the production of multigroup covariance matrices (right). . . . 39
I.9 Correlation matrix and associated relative uncertainties for the ^{56}Fe elastic scattering cross section (`MT2`) in JENDL-5. The top panel displays the percentage uncertainty as a function of incident energy, while the main heatmap illustrates the correlation coefficients between energy bins. Grey cells mark energy ranges where the evaluation provides no covariance data. . . . 43

II.1 Illustration of representative particle tracks used in collision-history sensitivity analysis, highlighting accepted versus rejected interactions in the perturbed zone. 64
II.2 Matrix representation of the Sandwich Rule for the propagation of uncertainties to the response variance. . . . 74
II.3 Comparison of sampling techniques in 2D Space: Random, Sobol, LHS. A well-distributed sampling approach minimizes gaps and ensures better domain coverage. . . . 75

II.4 Module architecture of the `kika-nd` library. Fourteen modules are grouped into three categories: *Data* parsers for the nuclear data formats handled by the SA&UQ chain, *Tools* implementing the sensitivity, sampling, and UQ methods of this chapter, and *Codes* interfaces to the external processing and transport codes. The perturbation pipelines (Sec. 4.5) chain modules across all three columns. 82
II.5 Multigroup collapse of ^{56}Fe elastic scattering: (a) first-order Legendre coefficient a_1 and (b) first-order relative uncertainty. 84
II.6 Overview of the nuclear data perturbation pipelines implemented in `kika-nd`. The ENDF pipeline (left) perturbs angular distributions at the Legendre coefficient level; the ACE pipeline (right) perturbs cross sections directly. Three paths are color-coded: purple for angular distributions only, orange for cross sections only, and green for both combined. 85
II.7 Relative difference in the ACE ^{56}Fe elastic-scattering cross section (MT=2) for perturbed realizations obtained using multigroup scaling factors sampled from the covariance data. 86
II.8 ENDF angular distribution perturbation for ^{56}Fe elastic scattering. The nominal $a_1(E)$ (blue) with $\pm1\sigma$ bands from `MF34` covariance is shown together with perturbed realizations sampled from the covariance. 87
II.9 Schematic representation of the PWR sphere geometry (not to scale). Alternating layers of steel and borated water surround a central point source, with a thin spherical-shell detector located in the outermost air region. 88
II.10 Sensitivity profiles for ^{56}Fe in the 1–3 MeV detector response: total (MT 1), elastic (MT 2), and total inelastic (MT 4). Four configurations are compared: MCNP, Serpent, MCNP with weight windows, and MCNP with corrected weight windows, each with $\pm1\sigma$ statistical uncertainty bands. 90
II.11 Sensitivity profiles for ^{1}H reactions in the 1–3 MeV detector response: total (MT 1), elastic (MT 2), and radiative capture (MT 102). MCNP and Serpent results agree within statistical uncertainties. 91
II.12 Perturbed neutron flux as a function of the ^{56}Fe total cross section perturbation, computed with first-order only (blue), first-plus-second-order (red), and direct perturbation (black points). Bottom panels show the ratio of the approximate to the direct result with $\pm3\sigma$ error bars. (a) 0.1–1.0 MeV; (b) 1.0–3.0 MeV. . . 92
II.13 Second-to-first-order ratio $R_{2/1}$ for ^{56}Fe elastic scattering (MT 2) in the 1–3 MeV detector response, plotted as a function of perturbation magnitude for different energy bins. The steepest slope is approximately 1, confirming that first-order effects dominate for the uncertainty magnitudes present in the covariance data. 93
II.14 Energy-integrated sensitivity coefficients for the ^{56}Fe total cross section with 5% perturbation. Results from MCNP, MCNP with weight windows, and Serpent are compared with $\pm3\sigma$ error bars. 93
II.15 Distribution of neutron flux values in the 1–3 MeV detector group from 1024 Sobol–SVD samples (TMC, 10^8 histories per run). The fitted normal distribution (solid curve) and statistical metrics (mean, standard deviation, skewness, kurtosis, Shapiro–Wilk p-value) are overlaid. 97

II.16 Convergence of the nuclear data uncertainty estimate (1–3 MeV) as a function of sample size for three simulation strategies (TMC, fast TMC, fast GRS), each shown at two NPS levels. The shaded band spans the range of estimates from nested subsets of the 1024 samples for Sobol and RS; LHS points are independent runs at each sample size. The dashed band indicates ±5% of the reference value. 98
II.17 Convergence of the nuclear data uncertainty estimate with 95% confidence intervals (error bars), grouped by sampling technique and simulation strategy: (a, b) TMC; (c, d) fast TMC; (e, f) fast GRS. Results for Sobol and LHS sampling are shown at three different NPS levels. 99
II.18 Sensitivity of the 1–3 MeV detector response to the Legendre moments of ^{56}Fe elastic scattering angular distributions. JEFF-4.0 results (solid lines) are shown for orders 1–6; JENDL-5 (dashed) for order 1. 102

III.1 ^{56}Fe elastic-scattering cross section from JEFF-4.0 in the 847 keV–4 MeV range. 108
III.2 First Legendre coefficient $a_1(E)$ for ^{56}Fe elastic scattering in the 847 keV–4 MeV range, shown for JEFF-4.0 and JENDL-5. 108
III.3 Relative uncertainties of the Legendre coefficients a_ℓ for ^{56}Fe elastic scattering from the available angular-distribution covariances. 109
III.4 Contribution of a single Legendre order to the propagated $\pm 1\sigma$ uncertainty band on the ^{56}Fe elastic differential cross section. Black markers are EXFOR points within the corresponding energy bin. 109
III.5 Correlation matrices from the `MF34` sections of (a) JEFF-4.0 (orders 1–3) and (b) JENDL-5 (order 1) for ^{56}Fe elastic scattering in the 847 keV–4 MeV range. 110
III.6 Resolution folding for $^{56}\mathrm{Fe}(n, \mathrm{el})$ at a nominal 1.4 MeV (JEFF-4.0, center-of-mass frame; $\sigma_E \approx 17$ keV for a 27 m, 10 ns setup). Top: σ_{el} (solid) and a_1 (dashed) across the window, colored by energy. Bottom: $d\sigma/d\Omega$ at each energy in the window (faint), the nominal 1.4 MeV distribution (black dashed), the Gaussian window average (green), and EXFOR data within the window. 116
III.7 Elastic differential cross section of ^{56}Fe at $E_n = 1.4$ MeV (center-of-mass frame). Black dashed: nominal JEFF-4.0. Colored lines: same shape scaled by the folded total cross section (σ_f) for six TOF resolutions σ_E. Grey dashed: shape-normalized JEFF-4.0 with the isotropic component re-fitted to Kinney data [114]. 117
III.8 Overview of the evaluation pipeline. EXFOR angular distributions are fitted to Legendre coefficients per energy bin; two Monte Carlo passes estimate the per-bin variances (Pass 1) and the full parameter correlation matrix (Pass 2), assembled into the covariance written to `MF4` and `MF34`. 124
III.9 Legendre fits and AIC-based model selection for $^{56}\mathrm{Fe}(n, n)$ at $E = 1.200$ MeV. Left: ridge-regularized fits at orders $L = 0$ through 6 overlaid on the pooled experimental data (21 points from three experiments). Solid lines denote orders with Akaike weight $w_L > 0.5\%$; dashed lines denote orders with negligible weight. Right: Akaike weights w_L for each candidate order, with the selected degree L^* highlighted. 129

III.10 Effective per-point weights $w_{\mathrm{pp},i} = \omega_{g,i}\, g_{\mathrm{Kish}}$ for the kernel-weight Monte Carlo pass at $E = 1.30$ MeV. Each marker shape represents a different experiment; the color encodes the combined weight from the CDF-based overlap $\omega_{g,i}$ (Eq. (III.30)) and the Kish factor g_{Kish} (Eq. (III.24)). Red-edged markers denote the core data point of each experiment (highest overlap weight). Error bars are shown for core points only. 133
III.11 Differential elastic cross sections of ^{56}Fe at four representative energies. Each panel shows the evaluation of this work (red), JEFF-4.0 (blue), JENDL-5 (green), and two TALYS calculations using the default Koning–Delaroche OMP and the JLM model. Colored markers denote experimental data points whose nominal energy falls inside the energy bin being evaluated. Twenty-five Monte Carlo samples are drawn as thin curves to illustrate the spread of the sampled distributions. 138
III.12 First Legendre coefficient a_1 for ^{56}Fe elastic scattering comparing JEFF-4.0, JENDL-5, and this work. (a) Full energy range 847 keV–4 MeV. (b) Zoomed view of the resonance region 0.85–1.2 MeV. (c) Zoomed view of the 2.25–2.75 MeV region. 139
III.13 Zoomed comparison of a_2 and a_3 in the 0.85–1.2 MeV resonance region for JEFF-4.0, JENDL-5, and this work. 140
III.14 Relative uncertainties of the Legendre coefficients from the `MF34` covariance data of JEFF-4.0, JENDL-5, and this work: (a)–(c) the first three orders a_1, a_2, a_3, and (d) the higher orders a_4, a_5, a_6. 141
III.15 Energy correlation matrix of the first Legendre coefficient a_1 for ^{56}Fe elastic scattering from this work, across the 847 keV–4 MeV range. The full $\ell = 1$–6 matrix is shown in Figure E.4. 142
III.16 Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 1.0$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data. 143
III.17 Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 1.4$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data. 144
III.18 Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 2.38$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data. 144
III.19 Comparison of `MF34`-propagated $\pm 1\sigma$ uncertainty bands on the differential cross section at $E = 2.85$ MeV. Each panel shows a different evaluation (JEFF-4.0, JENDL-5, this work) together with the baseline angular distribution and EXFOR experimental data. 145
III.20 Reduced χ^2/N for JEFF-4.0, JENDL-5, and this work across the five experiment subsets under the EXFOR-only budget σ_{exp}. The statistic responds only to how well each evaluation's central values agree with the experimental data, without the cushion of the library's own covariance. 146
III.21 Same comparison as Figure III.20 under the full-covariance budget $\sigma_{\mathrm{exp}} \oplus \sigma_{\mathrm{eval}}$, which adds each library's propagated `MF34` block. 146

III.22 Per-experiment χ^2/N on all 67 experiments, ranked by ascending value for this work. The vertical axis is logarithmic; the dotted horizontal line marks $\chi^2/N = 1.148$
III.23 Calculated-minus-experimental bias $C-E$ on the fifteen benchmarks most sensitive to the ^{56}Fe elastic cross section, for JEFF-4.0 and after the ^{56}Fe swap, ordered by that sensitivity, whose value is given in brackets after each benchmark name. The bands mark each benchmark's experimental 1σ and 3σ. . . . 151
III.24 Shift $\Delta(C-E)$ produced by the ^{56}Fe swap against the EALF of each benchmark, colored by its ^{56}Fe elastic sensitivity over 850 keV–4 MeV. The band marks the Monte Carlo uncertainty on the difference. 151

IV.1 Side elevation of the ASPIS-88 single-material iron benchmark: fission-plate source, mild-steel slab stack, and the foil measurement positions along the central axis (not to scale). 158
IV.2 Activation cross sections of the five ASPIS-88 foil reactions. 158
IV.3 C/E ratios along the shield depth for the three libraries. Error bars show the propagated ^{56}Fe nuclear-data uncertainty, with the cross section and angular-distribution channels combined in quadrature; dashed lines mark the experimental uncertainty. 160
IV.4 Sensitivity of each foil response at its deepest measured position to the ^{56}Fe elastic (`MT` 2) and first two inelastic levels (`MT` 51+52) cross sections, as a function of incident neutron energy. 164
IV.5 Relative difference between the JEFF-4.0 and JENDL-5 ^{56}Fe cross sections as a function of incident neutron energy, for the elastic and first-level inelastic channels. The difference is averaged over 100 equal-lethargy groups. 165
IV.6 Radial cross section of the VENUS-3 configuration showing the main components and detector locations. Only some detector positions are shown for clarity. 167
IV.7 Nominal C/E for the four libraries, grouped by reaction and, for ^{58}Ni, by detector position a. Error bars in are the standard deviation across detectors. 168
IV.8 Change in the ^{58}Ni C/E between ENDF/B-VIII.0 and ENDF/B-VIII.1, by detector region ordered from the core to the vessel. The change grows with the steel thickness along the neutron path. Error bars are the standard error of the mean. 169
IV.9 C/E ratios with the propagated 1σ nuclear-data uncertainties for the four libraries. Detectors are grouped by reaction; dashed lines mark the experimental uncertainty. Pink points are detectors that fail the separation condition, for which the total 1σ band is shown. 170
IV.10 Cross-sectional views of the MCNP6 PWR900 model showing the tally positions. (1) fuel assemblies; (2) core baffle; (3) moderator bypass region; (4) core barrel; (5) asymmetric neutron pad; (6) annular moderator downcomer; (7) reactor pressure vessel; (8) surveillance capsule [4]. 172
IV.11 Evaluated cross sections of the six capsule dosimetry reactions as a function of incident neutron energy above 0.1 MeV. 173
IV.12 Horizontal cross-sectional view showing the ADVANTG-generated weight-window map used for detector-response variance reduction. The color scale represents the generated weight-window values, with the optimized detector regions in red (not to be confused with the highest WW values, also in red). . 174

IV.13 Neutron energy spectrum at the four detector positions of the PWR900 model, height-integrated over the central ±60 cm window. The spectra are scored from the JEFF-4.0 nominal run and lethargy-normalized. 175
IV.14 Fuel loading pattern for the PWR900 core at equilibrium cycle. UOX assemblies are shown in green and MOX assemblies in purple. Numbers indicate the irradiation cycle of each assembly, so that an assembly labeled "4" has already completed three irradiation cycles in the core [4]. 176
IV.15 Relative uncertainty (standard deviation over the mean) of the assembly-averaged fission neutron production rate νR_f on the PWR900 core in quarter-symmetry representation, propagated from 1000 ENDF/B-VIII.0 covariance samples through CASMO5/SIMULATE5. 177
IV.16 Prompt fission neutron spectra of the source isotopes from JEFF-4.0 and the spread of the 512 perturbed samples, per outgoing-energy bin. Each curve carries the `MF35` incident-energy band its source spectrum is taken from. 179
IV.17 Combined cross section and angular-distribution uncertainty on the fast neutron flux at the vessel inner surface, height-integrated over the ±60 cm central window, at each of the three vessel azimuths and for the three evaluated libraries. The left panel shows the > 1 MeV threshold and the right panel the > 0.1 MeV threshold. Error bars are 95% bootstrap confidence intervals. . . . 183
IV.18 Combined nuclear-data uncertainty on the NRT-dpa and ARC-dpa at the vessel inner surface, height-integrated, for the three azimuths and the three libraries. Error bars are 95% bootstrap confidence intervals. 185
IV.19 Nuclear-data uncertainty on the six capsule dosimetry reaction rates, with the cross section and angular-distribution channels combined in quadrature, averaged over the three capsule positions, for the three libraries. Error bars are 95% bootstrap confidence intervals. 185
IV.20 Per-isotope decomposition of the JEFF-4.0 cross section uncertainty at five representative locations: the three vessel azimuths above 1 MeV and two capsule reaction rates (^{63}Cu(n,α), ^{238}U(n,f)). Each wedge has area $\sigma^2_{\mathrm{iso}}/\sigma^2_{\mathrm{XS}}$, so the wedges add to 100%; the combined σ_{XS} for each location is printed at the center of the donut. Sub-1% isotopes (^{58}Ni, ^{54}Fe, ^{55}Mn, ^{28}Si, ^{10}B) and the bootstrap residual are pooled as "Other". 188
IV.21 Matched-scope cross-check and sandwich variance decomposition for the height-integrated 0° vessel fast flux from ^{56}Fe / JEFF-4.0. *Left*: FOSM versus MCS over the same reactions, with 95% confidence intervals. *Right*: the matched-scope sandwich variance split by reaction channel. 189
IV.22 Linearity check for the 512-sample ^{56}Fe / JEFF-4.0 run at the height-integrated 0° vessel flux. Each point is one Monte Carlo sample. 190

B.1 The KIKA home dashboard. Each tile links to one of the three sidebar sections (Data, Tools, Codes) and displays the number of files currently loaded in the workspace. 208
B.2 ACE file detail view showing interactive cross-section plots for ^{56}Fe. Reactions are selected by `MT` number from a side panel and overlaid on the same axes. . . 209
B.3 ENDF file detail view showing the `MF34` covariance tab. The color map displays the correlation matrix for a selected Legendre order pair, with energy on both axes. 209

B.4 Covariance file detail view showing a correlation heatmap for a selected reaction pair. The color scale represents the correlation coefficient between energy groups. 210
B.5 SDF file detail view showing the sandwich-formula UQ tab. A covariance file is selected and the application computes the per-reaction variance contributions and total relative uncertainty. 211
B.6 The interactive chart of nuclides in KIKA. Each tile represents a nuclide colored by decay mode. The chart supports zoom, pan, and click-to-detail navigation. . 212
B.7 The multi-tab plotter with an active cross-section comparison. Data series from different files are selected from a side panel and overlaid on the same axes. Each tab maintains an independent, fully customizable plot. 213
B.8 The SA&UQ hub showing the available analysis workflows: FOSM-based sensitivity and uncertainty analysis via PERT cards and sandwich formula, and Monte Carlo sampling for cross-section and angular distribution perturbation. 214
B.9 The NJOY pipeline builder. A processing chain is assembled from module blocks arranged in a vertical pipeline (left). Parameters are configured in the side panel (right), and the generated input deck is shown in the live preview pane (bottom). 215

E.1 Magnitude-only folding for $^{56}\mathrm{Fe}(n,\mathrm{el})$ at a nominal 1.4 MeV (JEFF-4.0, center-of-mass frame): σ_{el} varies across the window while the angular shape is fixed at its nominal value. Panels and curves as in Figure III.6. 226
E.2 Shape-only folding, plotted as in Figure E.1 with the roles reversed: the angular shape (all Legendre orders) varies across the window while σ_{el} is fixed. The top panel shows a_1 as a proxy for the full shape variation. 227
E.3 Conceptual overview of the positivity enforcement procedure. An initial Legendre expansion exhibiting non-physical negative values (left) is corrected by projecting its coefficient vector $\mathbf{c}$ to the nearest feasible point $\mathbf{c}^*$ satisfying the linear constraints (middle), yielding a strictly non-negative angular distribution (right). 234
E.4 Full correlation matrix of the Legendre coefficients ($\ell = 1$–6) of this work's evaluation for ^{56}Fe elastic scattering across the 847 keV–4 MeV range. The block structure groups parameters by Legendre order; within each block, rows and columns run over the 526 adaptive multigroups produced by the collapse step. The $\ell = 1$ sub-block on the diagonal is reproduced in Chapter III, Figure III.15. 238
E.5 Calculated k_{eff} for JEFF-4.0 and after the ^{56}Fe swap across all 88 ICSBEP benchmarks (top), with the experimental values and their 1σ and 3σ bands, and the per-benchmark difference $C-C$ between the two evaluations (bottom). Benchmarks are in alphabetical order. 240

F.1 C/E ratio for the ^{197}Au(n,γ) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3. 242
F.2 C/E ratio for the ^{32}S(n,p) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3. 242
F.3 C/E ratio for the ^{27}Al(n,α) reaction rate along the shield depth, with error bars and dashed lines as in Figure IV.3. 242
F.4 C/E ratio for the ^{197}Au(n,γ) reaction rate, nominal libraries and ^{56}Fe swap. . . 243
F.5 C/E ratio for the ^{103}Rh(n,n$'$) reaction rate, nominal libraries and ^{56}Fe swap. . 243
F.6 C/E ratio for the ^{115}In(n,n$'$) reaction rate, nominal libraries and ^{56}Fe swap. . . 244
F.7 C/E ratio for the ^{32}S(n,p) reaction rate, nominal libraries and ^{56}Fe swap. . . . 244

F.8 C/E ratio for the ^{27}Al(n,α) reaction rate, nominal libraries and ^{56}Fe swap. . . 244
F.9 C/E ratio for the ^{32}S(n,p) reaction rate, JEFF-4.0 angular distribution and Chapter III evaluation. 245
F.10 C/E ratio for the ^{115}In(n,n$'$) reaction rate, JEFF-4.0 angular distribution and Chapter III evaluation. 245
F.11 Per-assembly distributions of the fission neutron production rate νR_f from 1000 nuclear data samples. The horizontal axis shows the production rate in units of 10^{13} n/s/cm^3, the vertical axis the frequency, and the numbers indicate the irradiation cycle. 255
F.12 Normalized contribution to the total fission neutron production rate νR_f at the assembly level (a) and at the pin level (b). 256
F.13 Combined nuclear-data uncertainty on the fast neutron flux above 0.1 MeV at the vessel inner surface, height-integrated, for the three azimuths and the three libraries. Error bars are 95% bootstrap confidence intervals. 257
F.14 Combined nuclear-data uncertainty (%) on the four vessel responses at the inner surface, resolved by azimuth and height segment, for the three libraries. 258

G.1 Fragilisation de la cuve et sa surveillance. 264
G.2 Sensibilité de la réponse du détecteur (1–3 MeV) aux moments de Legendre de la distribution angulaire élastique du ^{56}Fe. JEFF-4.0 (traits pleins, ordres 1 à 6) et JENDL-5 (tirets, ordre 1 seul, seul ordre porteur de covariance dans cette bibliothèque). 266
G.3 Chaîne d'évaluation. Les distributions angulaires EXFOR sont ajustées en coefficients de Legendre par intervalle d'énergie ; deux passes de Monte-Carlo estiment les variances par intervalle (passe 1) et la matrice de corrélation complète des paramètres (passe 2), assemblées dans la covariance écrite en `MF4` et `MF34`. 267
G.4 Sections efficaces différentielles élastiques du ^{56}Fe à quatre énergies incidentes (1,00, 1,30, 2,15 et 3,10 MeV) : ce travail (rouge), JEFF-4.0 (bleu), JENDL-5 (vert) et deux calculs TALYS. Les marqueurs sont les points expérimentaux de l'intervalle évalué ; les courbes fines sont des échantillons de Monte-Carlo illustrant la dispersion. 268
G.5 Rapports calcul/expérience le long de la profondeur d'acier pour les trois bibliothèques (ASPIS-88). Les barres d'erreur donnent l'incertitude des données nucléaires du ^{56}Fe (sections efficaces et distribution angulaire combinées en quadrature) ; les tirets marquent l'incertitude expérimentale. 270
G.6 Décomposition par isotope de l'incertitude de section efficace JEFF-4.0 en cinq positions : les trois azimuts de cuve au-dessus de 1 MeV et deux taux de réaction de capsule. L'aire de chaque secteur vaut $\sigma^2_{\mathrm{iso}}/\sigma^2_{\mathrm{XS}}$; le σ_{XS} combiné est inscrit au centre. 272

List of Tables

I.1 Guidelines for interpreting the relative error (R) [13]. 15
I.2 Multigroup libraries available for Denovo calculations in ADVANTG [14]. . . . 26
I.3 Reaction types available for nuclear data sampling in CASMO5 [10]. Perturbations are generated from ENDF/B-VIII.0 covariance matrices via eigendecomposition. 27
I.4 Overview of major international nuclear data libraries. 30
I.5 Conceptual identifiers in the ENDF-6 hierarchy. 32
I.6 Mapping of ENDF-6 physics sections to their corresponding nominal and covariance file numbers (`MF`). 33
I.7 Fields in each line of the ENDF-6 file. 34
I.8 Overview of main nuclear data processing codes. 41

II.1 Collision-history example for a perturbed event type x. 64
II.2 DOS/`PERT` illustration for an `F4` flux tally with three particle histories. 70
II.3 Layer specifications of the spherical geometry. The detector shell is a 1 cm air region scoring the neutron flux in three energy groups: 0–0.1 MeV, 0.1–1.0 MeV, and 1.0–3.0 MeV. 88
II.4 Normalized isotopic compositions used in the model: (a) water, (b) air, and (c) steel. Isotopes used for uncertainty propagation are shown in bold. 89
II.5 Statistical checks for the first-order Taylor coefficient from the `PERT` card, for ^{56}Fe elastic scattering in the 2.479–3.0 MeV energy bin (10^7 histories). 91
II.6 FOSM-propagated nuclear data uncertainty for the neutron flux in each detector energy group. The 95% confidence interval is obtained by bootstrap resampling the Monte Carlo statistical uncertainty of the sensitivity coefficients through the sandwich rule. 94
II.7 Major contributors to the propagated variance of the neutron flux in the 1.0–3.0 MeV detector energy group. 94
II.8 Frobenius-norm relative error (%) between reconstructed and original covariance matrices for different sampling methods (RS, Sobol, LHS) and decomposition techniques (SVD, Cholesky, EVD), for ^{56}Fe and ^{1}H across varying sample sizes. Best values for each column are shown in bold. 95
II.9 Summary of MCS simulation sets. For each configuration, the table reports the sampling technique, strategy, number of particle histories per run, and the percentage of runs passing all ten MCNP statistical checks for the single spherical-shell flux detector. No variance reduction was applied. 96
II.10 Most efficient MCS configurations meeting the convergence criteria (reference within 95% CI, CI width $\leq \pm 8\%$) for the 1–3 MeV detector response. 100

II.11 Comparison of propagated nuclear data uncertainty from FOSM and MCS (fast TMC, Sobol, 10^6 histories) with the MCS reference value (average of IDs 1–5). All intervals are 95% confidence intervals obtained by bootstrap resampling: MCS resamples the nuclear-data sample ensemble; FOSM resamples the Monte Carlo statistical uncertainty of the sensitivity coefficients. 100
II.12 Propagated nuclear data uncertainty for the 1–3 MeV detector response from ^{56}Fe, decomposed into angular distribution (AD) and cross section (XS) contributions, for the FOSM and MCS methods. The last column shows the relative variance decomposition between the two sources. 102
II.13 FOSM variance decomposition of the angular distribution uncertainty by Legendre order for ^{56}Fe elastic scattering (1–3 MeV detector response). Negative contributions arise from opposite-sign sensitivities between Legendre orders. . . 103

III.1 EXFOR experimental datasets contributing data points in the neutron-energy interval $E_n = 0.847$–4.000 MeV. For datasets spanning a wider interval, the number of points listed corresponds only to this interval. The listed counts are those entering the fit after the per-bin closest-energy selection of Sec. 4.2. . . . 114
III.2 Reduced χ^2/N of JEFF-4.0 and JENDL-5 against the EXFOR experimental database for ^{56}Fe elastic scattering in the 847 keV–4 MeV range, under the two variance budgets defined in the text. Bold marks the lower χ^2/N between the two libraries in each column. 122
III.3 Fit statistics and Legendre coefficients c_ℓ for the order comparison at $E = 1.200$ MeV illustrated in Figure III.9. The information-criterion column uses the AIC form $\chi^2 + 2k$ of Eq. (III.25). The ridge penalty uses $\lambda = 10^{-4}$ with power $p = 4$. Dashes indicate coefficients not present at a given order. 128
III.4 The seven experiments (of 67) on which JEFF-4.0 or JENDL-5 reaches a χ^2/N at least as low as this work, under the full-covariance budget $\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$, ranked by this work's value. N is the number of angular points entering the metric; bold marks the lowest χ^2/N in each row. EXFOR identifiers and references are given in Table III.1. 148
III.5 Reduced χ^2/N on the three campaigns published after the evaluated angular distributions of JEFF-4.0 and JENDL-5, under the full-covariance budget $\sigma_{\text{exp}} \oplus \sigma_{\text{eval}}$. Bold marks the lowest value in each row. 149
III.6 Agreement with the ICSBEP criticality benchmarks before and after replacing the ^{56}Fe evaluation in JEFF-4.0 with the present one, for the full suite, for the 56 benchmarks that combine the highest ^{56}Fe elastic sensitivities with an experimental k_{eff} known to better than 300 pcm, and for the remainder. $\overline{C-E}$ is the mean calculated-minus-experimental bias and RMS $(C-E)$ its root-mean-square, both in pcm. 150

IV.1 Activation foils of the ASPIS-88 benchmark, ordered by effective energy threshold. The threshold is the approximate neutron energy above which each reaction responds; the number of measured axial positions and the deepest position are taken from the benchmark dataset. 158
IV.2 Propagated ^{56}Fe nuclear-data uncertainty at the deepest measured position of each foil: cross section (XS) channel for JEFF-4.0 and JENDL-5, and angular-distribution (AD) channel for the three `MF34` evaluations. Values in percent at 1σ. 161

IV.3 Agreement with the ASPIS-88 foil measurements before and after replacing the JEFF-4.0 elastic angular distribution with the Chapter III evaluation, with the ^{56}Fe cross sections held fixed. χ^2/N is the reduced chi-square of the C/E against unity, weighted by the experimental uncertainty, and $\overline{C/E}$ the mean ratio, each over the N measured positions of the foil. The foils are ordered by increasing effective threshold. 162
IV.4 Mean C/E values, root-mean-square error (RMSE), and percentage of detectors within 1σ experimental uncertainty for each evaluated library. 167
IV.5 Reference central values of the vessel and capsule tallies for JEFF-4.0, in physical units, from the nominal run; the last column is its MCNP statistical error. Vessel quantities are volume-averaged over the ±60 cm central window, capsule quantities over the three axial detectors at −17°. Reaction rates are per target atom ($s^{-1}\,atom^{-1}$), the dosimeters being scored as fictitious materials. 173
IV.6 Fast neutron flux (E = 1–20 MeV) at the RPV inner surface, 0° vessel azimuth (the hot spot), for three representative source term cases, in MCNP normalization (per source neutron). The relative deviation is computed with respect to the nominal case. 179
IV.7 Emitted-spectrum contribution to the vessel and capsule responses, propagated from the JEFF-4.0 `MF35` covariance through 512 MCNP calculations. Values are relative 1σ uncertainties in percent, with their 95% bootstrap confidence interval (CI). Vessel responses are height-integrated over the ±60 cm central window, capsule responses averaged over the three capsule positions. 181
IV.8 Isotopes carrying covariance data in the three evaluations propagated in this chapter. A check mark indicates an isotope is perturbed; `MF33` covers cross sections, `MF34` covers elastic-scattering Legendre coefficients. 183
IV.9 Central values of the height-integrated fast neutron flux ($E > 1$ MeV), in $n\,cm^{-2}\,s^{-1}$, at the three vessel azimuths for the three libraries, with the cross-library spread (maximum minus minimum over the mean) and the range of the propagated nuclear-data uncertainty across libraries. Central values from the nominal run of each library. 184
IV.10 Propagated ^{56}Fe angular-distribution (`MF34`) relative uncertainty at the three vessel azimuths (height-integrated > 1 MeV flux) and the six capsule reaction rates, for the published JEFF-4.0 and JENDL-5 evaluations and the EXFOR-based evaluation of Chapter III. Each value is the isolated ^{56}Fe angular channel (cross section and ^{16}O channels excluded), reported as a relative 1σ uncertainty in percent with its 95% bootstrap confidence interval in brackets. 187
IV.11 Propagated relative uncertainty σ_R at the height-integrated 0° vessel flux for ^{56}Fe / JEFF-4.0: FOSM against MCS over the matched reaction scope (first two rows), and MCS extended to all inelastic levels (third row). 191
IV.12 Fast neutron flux uncertainty budget $\Phi_{E>1\,\mathrm{MeV}}$ for Timpano and the present work, separated into nuclear-data contributions and an additional manufacturing-and-geometry term. The per-contributor rows are quoted at the vessel hot spot (0° for the present work, the RPV tally for Timpano). Totals are given at the vessel and at the surveillance capsule, first for the nuclear-data contributions alone and then including the manufacturing term. 193

D.1 Placeholders in the NJOY perturbation template. Each placeholder is substituted at runtime with sample-specific values. 223

E.1 Per-experiment uncertainty resolution for the largest contributors. "Case" refers to the classification in the text. σ_g^{sys} is the per-experiment relative systematic (rank-one normalization). The "$\sigma_{g,E}^{\mathrm{sys},E}$ range" summarizes any documented additional contribution before routing; for Cierjacks this is folded into σ_{stat} as diagonal residual (see Sec. E.2.2). "Median $\sigma_{\mathrm{stat}}/|y|$" is the resolved per-point statistical fraction. 231
E.2 Number of bins selecting each Legendre degree, per criterion, across the 1 735 fitted bins of the 0.847–4 MeV range. 232
E.3 Per-bin reduced chi-squared $\tilde{\chi}^2_{\mathrm{red}}$ of the nominal fit against the EXFOR angular data, summarized over the fitted bins of the 0.847–4 MeV range, for the three model-selection criteria. Lower values indicates a closer fit; the last column is the fraction of bins fitted within the data uncertainties ($\tilde{\chi}^2_{\mathrm{red}} < 1$). 233

F.1 Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{197}Au(n,γ) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. 247
F.2 Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{103}Rh(n,n$'$) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. 248
F.3 Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{115}In(n,n$'$) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. 249
F.4 Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{32}S(n,p) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. 250
F.5 Per-position C/E and propagated ^{56}Fe nuclear-data uncertainties for the ^{27}Al(n,α) foil of the ASPIS-88 iron benchmark. Uncertainties in percent at 1σ, with the 95% bootstrap confidence interval in parentheses. 251
F.6 Isotopes perturbed in the VENUS-3 nuclear-data uncertainty propagation for each evaluated library. A check mark indicates the isotope carries covariance data in that library and is perturbed; a dash indicates it is absent. 253
F.7 Reference central values of the vessel and capsule tallies, by library. Vessel quantities are height-integrated over the ±60 cm central window; capsule reaction rates are averaged over the three capsule positions. 254
F.8 Ten tallies with the highest combined nuclear-data uncertainty across the three libraries. The position label refers to one of the three capsule positions or to the position-averaged response. 259
F.9 Ten tallies with the largest spread between libraries (maximum minus minimum combined value, in %). All ten lie at the surveillance capsule; the position label refers to one of the three capsule positions or to the position-averaged response. 259
F.10 Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, 0° azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ±60 cm central window. 260
F.11 Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, −20° azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ±60 cm central window. 260

F.12 Nuclear-data contributions to the uncertainty of the responses at the vessel inner surface, $-28°$ azimuth, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are height-integrated over the ± 60 cm central window. 261

F.13 Nuclear-data contributions to the uncertainty of the responses at the surveillance capsule, as relative 1σ uncertainties in percent, with JEFF-4.0 as the transport library. Values are averaged over the three capsule positions. 261

G.1 Budget d'incertitude du flux neutronique rapide $\Phi_{E>1\,\mathrm{MeV}}$ pour une étude indépendante (Timpano) et le présent travail, séparé en contributions de données nucléaires et terme additionnel de fabrication et géométrie. Les contributions sont données au point chaud de cuve ; les totaux figurent à la cuve et à la capsule. 272